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Message from the Senior Director for Higher Education



To our brilliant students, dedicated faculty, and esteemed partners in education,

It is with immense pride and heartfelt congratulations that I present to you the inaugural issue of the Academic Research Journal, a milestone achievement for the BSIT, BSA, and BSBA programs of FEU Diliman.

This journal stands as a testament to the power of curiosity, the discipline of research, and the courage to ask noble questions; it represents far more than a collection of academic papers. Each article within these pages reflects the passion, intellect, and perseverance of our students and mentors, proof that even in the face of uncertainty, the pursuit of knowledge remains unwavering.

As Tamaraws, you embody the legacy of excellence, integrity, and innovation. This publication reflects that heritage, anchored in our core values of fortitude, excellence, and uprightness, while remaining forward-looking in its vision. It serves as a reminder that research is not merely an academic requirement but a vital contribution to society, industry, and nation-building.

To our BSIT students, may your work continue to shape the future of technology with purpose and empathy; to our BSA students, may your insights uphold the highest standards of accountability and transparency; and to our BSBA students, may your ideas drive ethical leadership and sustainable growth in business.

May this journal be the first of many. Let it inspire a culture of inquiry, collaboration, and fearless thinking across all disciplines.

Long live FEU Diliman! Let us press on with research, for our nation and for the future.

Warmest regards,

Dr. Irish C. Juanatas

Senior Director for Higher Education

Message from the Research Coordinator



As the Research Coordinator of FEU Diliman, it is with immense gratitude and pride that I present the first volume of our academic journal. This milestone signifies not only our institution's commitment to academic excellence but also the growing interest and passion of our students in the field of research.

Seeing our students actively engage in the pursuit of knowledge, innovation, and discovery is truly inspiring. It is a testament to the talent and dedication present in our university. This journal serves as a platform for them to share their insights and findings with the academic community, contributing meaningfully to various fields of study.

My vision as Research Coordinator is to further strengthen the research culture in FEU Diliman. By fostering collaboration, encouraging curiosity, and providing the necessary support, I aim to ensure that our institution becomes a hub of academic excellence and a recognized contributor to the ever-evolving body of knowledge.

This first volume is just the beginning. Together, with the dedication of our faculty and students, I am confident that we will continue to build on this foundation, creating impactful research that will inspire change and make a difference.

Let us continue to work together in nurturing a culture of excellence in research at FEU Diliman.

Sincerely,

Jackie Lou O. Raborar, Ph. D., CHRP

Research Coordinator

FEU Diliman

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Bridging Knowledge and Compliance Behavior: Examining the Influence of Tax Knowledge among Self-Employed Professionals and Mixed Income Earners in Quezon City

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Abstract

This research studies the influence of tax knowledge on tax compliance behavior of self-employed professionals and mixed income earners in Quezon City. This study aims to identify the level of tax knowledge among these taxpayers in terms of general, legal, and procedural tax knowledge impact compliance behavior, specifically in completeness, timeliness, and accuracy of their tax filing and payment. The study employs a descriptive-correlational research design using quantitative methods, specifically surveys and questionnaires, to collect data. The research draws upon the Theory of Planned Behavior as its theoretical framework, exploring how attitudes toward compliance, subjective norms, and perceived behavioral control influence individual tax behavior. Results from hypothesis testing revealed that all sub-variables of tax knowledge (general, legal, and procedural) significantly influence the sub-variables of tax compliance behavior (completeness, timeliness, and accuracy), except for legal tax knowledge, which does not significantly influence accuracy. The results contribute valuable information for individual taxpayers, the Bureau of Internal Revenue, and policymakers in refining tax awareness initiatives and compliance strategies among professionals in the Philippines.

Keywords: Tax Knowledge, Tax Compliance Behavior, Self-Employed, Mixed Income Earner.

Introduction

Taxation is the mechanism by which governments collect financial contributions from individuals and entities to support public goods and services, facilitate economic stability, and redistribute wealth (Adefunke & Mayowa, 2024; Nabong et al., 2024). Recognized as the "lifeblood of the government," taxation underpins a state's ability to fulfill its obligations to its citizens. While its economic and social roles are extensive, the effectiveness of a tax system ultimately hinges on taxpayers' willingness and ability to comply with regulations.

In developing economies, however, tax compliance remains a persistent challenge. Governments often struggle to meet rising public expenditures due to inefficiencies in registration, liability determination, and tax collection processes (Kira, 2017, as cited in Maquiling et al., 2023). The resulting tax gap—a disparity between potential and actual revenue—undermines fiscal sustainability (Mebratu, 2024). For instance, only 26.3% of self-employed taxpayers in Indonesia filed their required returns (Sulastri et al., 2023). In the Philippines, self-employed professionals (SEPs) contribute merely 20% of total personal income tax revenue, despite being subject to higher tax rates and varied tax types. These figures highlight systemic underreporting, non-filing, and a lack of targeted enforcement for individual taxpayers compared to corporations (Calcetas et al., 2024; Maquiling et al., 2023).

Tax knowledge has been widely recognized as a pivotal factor influencing compliance. Individuals with greater awareness of tax obligations, rights, and procedures are more likely to fulfill their duties accurately and punctually (Taing & Chang, 2020). Bornman and Ramutumbu (2019) assert that improved tax knowledge correlates positively with voluntary compliance, reducing the likelihood of errors and legal penalties. In the Philippine context, the Bureau of Internal Revenue (BIR) imposes substantial surcharges ranging from 25% to 50% for violations in filing and payment obligations (BIR, 2024), underscoring the need for accurate and timely compliance.

To address growing non-compliance, the BIR has introduced several policy and technological reforms. In 2023, it implemented programs such as Oplan Kandado and RAFT (Run After Fake Transactions), pursued legal action against tax evaders, and modernized its digital platforms (Department of Finance, 2023). Additionally, information campaigns—formalized under Revenue Memorandum Order No. 7-2022—have been scaled up through seminars, webinars, and the deployment of a tax chatbot named "Revie" (Katigbak, 2023). The integration of tax education into the academic curriculum was also institutionalized via RMC No. 64-2023, reinforcing the government's push toward long-term behavioral change.

Despite these advancements, there remains a notable gap in research examining tax compliance among SEPs and mixed-income earners in the Philippines, particularly at the local level. Existing studies tend to focus on corporations and compensation income earners, often overlooking the unique compliance dynamics of self-directed taxpayers (Owusu, 2021). This is especially significant in regions like Quezon City—a key urban and commercial hub in Metro Manila—whose diverse population offers a representative cross-section of Filipino taxpayers (KMC Savills Research, 2021; Behold Philippines, 2019).

This study therefore investigates how general, legal, and procedural dimensions of tax knowledge affect tax compliance behavior—defined through timeliness, completeness, and accuracy—among SEPs and mixed-income earners in Quezon City. It seeks to fill the empirical void on this taxpayer segment and provide actionable insights for policymakers, tax authorities, and educational institutions. In doing so, it aspires to support the broader goal of fostering a more informed and compliant tax base in the Philippines.

Literature Review

Tax Knowledge

The concept of tax knowledge is addressed in diverse ways across current literature, ranging from broad definitions to context-specific interpretations. Bornman and Ramutumbu (2019) note the lack of standardization in its definition, calling for the establishment of a unified framework. Within this domain, related constructs such as tax awareness and tax literacy have also been explored. Tax awareness is often linked with voluntary compliance, defined as the taxpayer's willingness to fulfill tax obligations with a sense of civic duty (Yunia et al., 2021, as cited in Yuliani et al., 2024; Hidaya et al., 2023; Cureg et al., 2024). This willingness is frequently influenced by perceptions of fairness, reciprocity from the state, and awareness of the tax system's purpose.

Several studies emphasize that tax awareness encapsulates not only an understanding of one's obligations but also of broader fiscal and societal dynamics, such as social norms and the perceived justice of the tax system (Oktaviani et al., 2020; Abbas et al., 2021; Alhempi et al., 2020; Cureg et al., 2024; Eresashanti et al., 2024). Awareness of deadlines, penalties, and procedures further contribute to this construct (Paco & Quezon, 2022).

Tax literacy, as distinguished from general awareness, is described as a more technical competency. It includes the ability to comprehend legislation, use digital platforms, and engage with the tax authority in a proficient manner (De Clerq, 2021). To contextualize tax knowledge more systematically, Bornman and Ramutumbu (2019) propose a tripartite framework: general, legal, and procedural knowledge. General knowledge refers to understanding the role of taxation in governance and its personal financial impact. Legal knowledge encompasses familiarity with legislative language and the correct application of complex tax laws. Procedural knowledge involves administrative interactions with the tax authority, including registration, documentation, and timely filing.

Level of Tax Knowledge

Studies examining the level of tax knowledge among different populations have yielded varied results. Hugo (2018) finds that while understanding tax reforms such as the TRAIN Law influences taxpayer behavior, perceptions of government accountability may have a more significant impact. Paco and Quezon (2022) report that micro and small businesses in Bacolod City generally demonstrate high awareness of BIR requirements and VAT, though understanding of other percentage taxes is more limited.

Conversely, Renoria (2024) highlights low tax knowledge among respondents in Sorsogon City, particularly in terms of awareness of deadlines and rates. Only 7% of respondents could identify the majority of due dates accurately, underscoring critical knowledge gaps.

Internationally, similar trends are evident. Abbas et al. (2021) report that educational interventions can significantly improve tax knowledge among Indonesian students. Asrinada et al. (2018) demonstrate that tax awareness and knowledge are leading predictors of compliance, supported by the widespread adoption of the self-assessment system. Gonidakis et al. (2024) find that Greek university students lack effective tax education, highlighting the continued need for targeted tax literacy initiatives.

Collectively, these findings reinforce the importance of taxpayer education and the link between knowledge and compliance, both in the Philippine context and globally.

Factors Affecting Tax Knowledge

A variety of factors contribute to taxpayers' level of tax knowledge. Education plays a central role, as it enhances comprehension of tax systems and encourages voluntary compliance (Gonidakis et al., 2024). Alhempi et al. (2020) emphasize that logical reasoning based on existing tax rules, as well as mutual trust between the government and taxpayers, influences one's knowledge and attitude towards compliance.

Tax socialization, or the dissemination of tax-related information to the public, reduces confusion and enhances understanding (Karnowati & Handayani, 2021; Eresashanti et al., 2024). Sanctions also act as a behavioral motivator. Idrus et al. (2020) argue that well-enforced penalties increase taxpayer accountability by reinforcing the consequences of non-compliance, creating a more knowledgeable and cautious taxpayer base.

Income level is another determinant. Eresashanti et al. (2024) suggest that higher income increases both the capacity and the motivation to comply, indicating that knowledge grows with economic position. Tax system modernization has also been found to contribute to tax knowledge and awareness by simplifying processes and improving accessibility through technology (Turambi et al., 2022; Mudjiyanti et al., 2022).

Finally, tax education and reform initiatives are critical. Bornman and Ramutumbu (2019) emphasize that beyond technical instruction, educational programs must also shape attitudes and foster a voluntary compliance mindset

Tax Compliance Behavior

Tax compliance refers to the extent to which taxpayers fulfill their legal obligations, including filing, accurate reporting, and timely payment (Santoro, 2021; Guzel et al., 2019; Oladipo, 2020). The determinants of compliance behavior have been explained using both economic and psychological theories.

From an economic standpoint, the Fiscal Exchange Theory posits that taxpayers comply when they perceive the government as reciprocating their contributions with tangible benefits (Cowell & Gordon, 1987; Alm et al., 1992). The Economic Deterrence Theory, in contrast, frames compliance as a cost-benefit analysis, where evasion is deterred by increasing the likelihood and severity of punishment (Allingham & Sandmo, 1972).

Psychological models offer a more nuanced interpretation. The Theory of Planned Behavior (Ajzen, 1991) describes compliance as a function of attitudes, social norms, and perceived control. Positive perceptions of tax fairness, external social pressure, and confidence in one's ability to comply all foster better tax behavior (Bobek & Hatfield, 2003).

The Social Influence Theory further dissects the impact of external forces, identifying compliance (out of fear or desire for reward), identification (to build social relationships), and internalization (alignment with personal values) as stages of behavioral adaptation (Kelman, 1953). These theories collectively underscore that tax compliance is not merely a legal issue but a psychological and social one as well.

Challenges of Individual Taxpayers in Tax Compliance

Despite improvements in policy and education, many taxpayers continue to face substantial obstacles to compliance. Tax system complexity is a persistent issue, increasing compliance costs and fostering unintentional errors (Nguyen, 2021). For small and medium-sized enterprises (SMEs), limited access to professional expertise exacerbates these challenges, often leading to inefficient allocation of resources.

Administrative burdens, low digital literacy, and inconsistent enforcement of regulations also contribute to taxpayer frustration. These structural and informational barriers hinder even well-meaning individuals from fully complying, highlighting the need for continued reforms in accessibility, education, and system simplification.

Theoretical Background

This study applies the Theory of Planned Behavior (TPB), developed by Ajzen (1991), to analyze how tax knowledge influences the compliance behavior of self-employed professionals and mixed-income earners. TPB posits that behavior is shaped by three components: attitude toward behavior, subjective norms, and perceived behavioral control, all of which influence behavioral intention and actual compliance.

Attitude toward behavior reflects how taxpayers evaluate tax compliance. Well-informed individuals tend to view compliance favorably, while those with limited knowledge often perceive tax obligations as burdensome, fostering negative attitudes. Subjective norms capture social pressures influencing tax behavior. Taxpayers are more likely to comply when they are surrounded by peers who meet their tax obligations or when societal discussions highlight civic duty and government accountability. Awareness of taxation's role in society and visible enforcement actions against non-compliance reinforce these normative expectations. Perceived behavioral control refers to taxpayers' confidence in fulfilling their tax obligations. Individuals who have a stronger understanding of tax systems, including filing procedures and legal requirements, are more likely to comply accurately and on time.

Empirical studies affirm TPB's relevance in tax research. Kusuma and Azzakhusna (2019) found that TPB components significantly influenced tax awareness in Indonesian SMEs, which in turn promoted compliance. Shaharuddin (2023) demonstrated that favorable attitudes toward the tax system strongly predicted tax compliance among sole proprietors, while Nurwanah (2018) showed that behavioral intention, shaped by TPB elements, significantly influenced actual compliance. Bani-

Khalid (2022) extended the TPB framework by incorporating patriotic values, which influenced SME owners' compliance intentions regarding sales tax. Most recently, Matshona et al. (2024) identified a gap in the literature, emphasizing the need for further exploration of the relationship between tax knowledge and compliance behavior.

This study leverages TPB as a structured framework to examine how tax knowledge integrates with these behavioral elements, providing insights into compliance among self-employed individuals and mixed-income earners. By conceptualizing tax knowledge as a fundamental input to TPB's core constructs, this approach allows for a systematic analysis of how variations in knowledge impact tax compliance behavior.

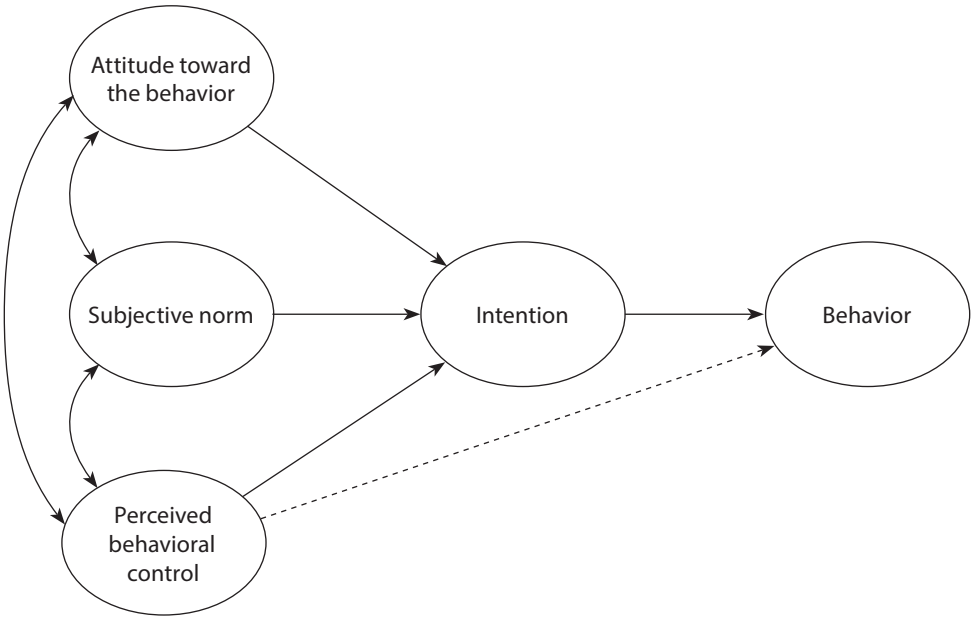


Fig. 1. The Theory of Planned Behavior.

Methodology

This study utilized a descriptive-correlational research design employing a quantitative method to gather data. A descriptive research design aims to present an accurate depiction of events or situations as they are, capturing behaviors, thoughts, and attitudes (Taherdoost, 2022). The correlational aspect of the design focuses on examining the relationships between independent and dependent variables (Lappe, 2000). In this case, the independent variable is tax knowledge, while the dependent variable is tax compliance behavior. The study is non-experimental, meaning it does not involve intervention or control over the variables.

A quantitative research approach was deemed most suitable for this study due to its capacity to handle large datasets and produce measurable statistics (Martyn, 2021). The quantitative method is advantageous in its ability to generalize results from a large sample size, enhancing the representativeness of the population (Taherdoost, 2022). Despite its strengths, quantitative research does not delve into personal feelings or subjective experiences and lacks an in-depth exploration of social realities. However, it provides reliable, replicable data, which is particularly useful for examining relationships between variables on a larger scale (Asrifan et al., 2023).

The study focused on two key variables: tax knowledge and tax compliance behavior. Tax knowledge, categorized into general, legal, and procedural aspects, serves as the independent variable. Tax compliance behavior, measured through timeliness, completeness, and accuracy of tax filing, functions as the dependent variable. The study examines how these variables influence each other within the self-employed and mixed-income earner population in Quezon City.

The research was conducted in Quezon City, the capital of the National Capital Region (NCR) in the Philippines. The city is divided into four major Regional District Offices (RDOs): 28, 38, 39, and 40. Although studies on taxation have been conducted in other cities such as Bacolod, Sorsogon, and Davao (Paco & Quezon, 2022; Renoria, 2024; Maquiling et al., 2023), there is a gap in research specifically addressing Quezon City.

The population of the study included self-employed professionals and mixed-income earners residing within Quezon City. According to the Philippine Statistics Authority (PSA), Quezon City has a population of 2,960,048 and is divided into 142 barangays across four RDOs (PSA, 2020). The sample size was determined to be 308 respondents, ensuring diversity in

terms of demographics. The Philippine Statistics Authority (PSA, 2020) reports 3,550,875 registered sole proprietors and 1,182,489 professionals, providing a substantial number of potential respondents for the study.

Self-employed professionals, as defined by the Bureau of Internal Revenue (BIR), include individuals who derive their income from a profession, such as doctors, lawyers, or freelancers. Mixed-income earners are individuals who earn income from both employment and self-employment. Both groups are required to file BIR Form 1701, categorizing them as individual taxpayers.

To calculate the sample size, G-power analysis was employed. Based on a medium effect size ($f^2 = 0.15$), a margin of error of 0.05, and a power level of 0.95, the analysis indicated a minimum sample size of 119 respondents for statistical reliability. However, the study was able to collect data from 308 respondents.

Purposive and referral (snowball) sampling techniques were used to select participants. Purposive sampling allowed for targeted selection based on specific criteria, while referral sampling enabled participants to recommend other eligible respondents, facilitating efficient data collection.

A survey questionnaire was the primary data collection instrument. The questionnaire was divided into two sections: the first assessed respondents' tax knowledge (general, legal, and procedural), and the second evaluated their tax compliance behavior in terms of completeness, timeliness, and accuracy. The survey utilized a 6-point Likert scale, with options ranging from "strongly disagree" to "strongly agree."

Before administering the survey, the instrument underwent expert review for face and content validity by a Certified Public Accountant (CPA). Cronbach's Alpha was used to assess the reliability of the instrument. The results indicated strong internal consistency across the sub-variables, with Cronbach's Alpha values of 0.727 for general tax knowledge, 0.739 for legal tax knowledge, and 0.953 for procedural tax knowledge. For tax compliance behavior, the Cronbach's Alpha values were 0.817 for completeness, 0.817 for timeliness, and 0.911 for accuracy.

Data was collected through online surveys, primarily using Google Forms. A data privacy form was included at the beginning of the survey to inform respondents of their rights and ensure informed consent. Respondents were assured of the confidentiality of their responses and the voluntary nature of their participation.

Once the data was collected, statistical analysis was performed using descriptive statistics, including mean, standard deviation, frequency distribution, and percentage distribution. Descriptive statistics helped summarize the dataset and identify trends and patterns (Fabricant, 2024). Additionally, the Likert scale responses were analyzed to assess the degree of agreement with various statements on tax knowledge and compliance behavior. The Likert scale's scoring guidelines ranged from "strongly disagree" (1) to "strongly agree" (6), allowing for a clear understanding of respondents' attitudes.

Ethical considerations were central to this study. The researchers ensured that all data was collected and handled in accordance with Republic Act No. 10173, the Data Privacy Act of 2012. Respondents were informed of the purpose of the study and their right to withdraw at any time. Confidentiality was maintained throughout the study, and participation was entirely voluntary.

Results and Discussion

This section presents the results of the study and the analysis based on descriptive and inferential statistics, addressing the influence of general, legal, and procedural tax knowledge on the tax compliance behavior of self-employed professionals and mixed income earners. Data were collected through structured survey questionnaires using a 6-point Likert scale, covering the period from November 8, 2024, to January 27, 2025, and processed using Partial Least Squares Structural Equation Modelling (PLS-SEM).

Demographic Characteristics of Respondents

A total of 308 responses were collected. Respondents were predominantly male (70.1%), with a significant number identifying as college graduates (66.6%). RDO 28 in Quezon City had the highest concentration of respondents (41.9%), followed by RDOs 39, 40, and 38. Most respondents (68.1%) identified as self-employed or professionals, while 31.5% identified as mixed income earners.

Measurement Model

The reliability and validity statistics indicate that the constructs used in this study—General Tax Knowledge, Legal Tax Knowledge, Procedural Tax Knowledge, Completeness, Timeliness, and Accuracy—are robust and well-supported. High factor loadings and significant p-values across all items suggest strong construct validity, while Cronbach's Alpha values exceeding 0.9 confirm excellent internal consistency. Composite reliability scores, all above 0.9, further validate the stability of these constructs, ensuring consistent measurement. The average variance extracted (AVE) values are all well above the accepted threshold of 0.5, demonstrating that each construct captures a sufficient amount of variance from its indicators. Procedural

Tax Knowledge exhibits the highest reliability, emphasizing its critical role in compliance behavior. Additionally, the constructs related to tax compliance—Completeness, Timeliness, and Accuracy—show extremely high reliability and validity, underscoring their importance in understanding taxpayer behavior. These findings affirm the suitability of the measurement model in capturing the relationship between tax knowledge and compliance behavior, providing a solid foundation for further analysis.

Construct	Factor Loading	P-value	Cronbach's Alpha	Composite reliability	Average Variances Extracted
General Tax Knowledge			0.914	0.936	0.745
	0.874	<0.001			
	0.9	<0.001			
	0.836	<0.001			
	0.884	<0.001			
	0.817	<0.001			
Legal Tax Knowledge			0.921	0.941	0.763
	0.844	<0.001			
	0.913	<0.001			
	0.92	<0.001			
	0.904	<0.001			
	0.777	<0.001			
Procedural Tax Knowledge			0.948	0.96	0.829
	0.91	<0.001			
	0.923	<0.001			
	0.932	<0.001			
	0.885	<0.001			
	0.903	<0.001			
Completeness			0.967	0.974	0.883
	0.928	<0.001			
	0.936	<0.001			
	0.925	<0.001			
	0.956	<0.001			
	0.951	<0.001			
Timeliness			0.969	0.976	0.889
	0.934	<0.001			
	0.952	<0.001			
	0.935	<0.001			
	0.938	<0.001			
	0.954	<0.001			
Accuracy			0.968	0.975	0.885
	0.944	<0.001			
	0.947	<0.001			
	0.907	<0.001			
	0.95	<0.001			
	0.956	<0.001			

Emerging Model, Path Analysis and Hypothesis Testing

Path Analysis is used for investigating direct and indirect causal relationships between variables through simultaneous computations and formulas. On one hand, path coefficients acquired in this manner show the nature of the relationship such as its effects sizes and interpretations. P-values, on the other hand, are acquired during hypothesis testing. Depending on what figure is displayed as the P-value, it determines whether a hypothesis is significant for the study and whether a hypothesis is supported or not. One-segment analysis is employed in the process of acquiring these results. This type of analysis is utilized whenever the target sample or portion of the population of a study focuses on a specific group of people. In the case of this study, the target sample only involves self-employed professionals and mixed income earners in Quezon City.

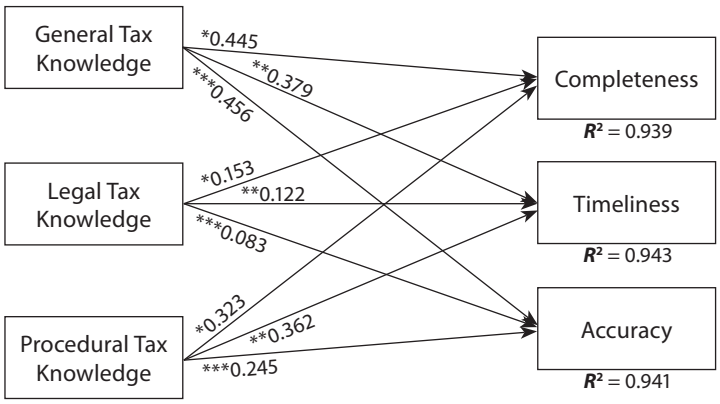


Fig. 2. Emerging model of the study.

Path	Path Coefficients (β)	P-value	Effect Sizes (f^2)	Effect Size Interpretation	Description	Decision
H1a: General Tax Knowledge → Completeness	0.445	<0.001	0.345	Medium	Significant	H1a is supported
H1b: General Tax Knowledge → Timeliness	0.379	<0.001	0.269	Medium	Significant	H1b is supported
H1c: General Tax Knowledge → Accuracy	0.456	<0.001	0.310	Medium	Significant	H1c is supported
H2a: Legal Tax Knowledge → Completeness	0.153	0.003	0.111	Small	Significant	H2a is supported
H2b: Legal Tax Knowledge → Timeliness	0.122	0.015	0.082	Small	Significant	H2b is supported
H2c: Legal Tax Knowledge → Accuracy	**0.083**	**0.069**	**0.051**	**Small**	**Insignificant**	**H2c is not supported**
H3a: Procedural Tax Knowledge → Completeness	0.323	<0.001	0.232	Medium	Significant	H3a is supported
H3b: Procedural Tax Knowledge → Timeliness	0.362	<0.001	0.169	Medium	Significant	H3b is supported
H3c: Procedural Tax Knowledge → Accuracy	0.245	<0.001	0.120	Small	Significant	H3c is supported

H1a: General Tax Knowledge → Completeness. Taxpayers with high general tax knowledge showed a significant and positive impact on filing completeness. Respondents demonstrated awareness of their tax responsibilities and the consequences of non-compliance, aligning with Bornman and Ramutumbu's (2019) assertion that tax knowledge increases a sense of obligation.

H1b: General Tax Knowledge → Timeliness. A significant positive relationship was also found between general tax knowledge and filing timeliness. Respondents who understood their roles as taxpayers were more likely to file on time, possibly to avoid penalties, supporting findings by Papp (2024) on compliance motivation through tax knowledge.

H1c: General Tax Knowledge → Accuracy. General knowledge had the strongest influence on accuracy. Respondents double-checked their filings and ensured correctness. This shows a consistent application of tax knowledge to fulfill obligations accurately, reducing risk of penalties or audits.

H2a: Legal Tax Knowledge → Completeness. Although legal tax knowledge had a smaller effect size, its impact on completeness was statistically significant. Respondents showed awareness of registration requirements and issuance of invoices, reflecting compliance with regulatory obligations.

H2b: Legal Tax Knowledge → Timeliness. A small but significant influence was also found on timeliness. While technical legal knowledge does not heavily drive promptness, it still supports understanding of deadlines and compliance protocols.

H2c: Legal Tax Knowledge → Accuracy. Legal knowledge did not significantly affect accuracy. This may be due to the complexity of interpreting and applying legal tax provisions, echoing Bornman and Ramutumbu's (2019) observation that legal ambiguity impairs accurate application.

H3a: Procedural Tax Knowledge → Completeness. Procedural knowledge contributed positively to completeness. Awareness of tax form requirements, proper documentation, and processes allowed respondents to meet filing obligations thoroughly.

H3b: Procedural Tax Knowledge → Timeliness. Timeliness also improved with procedural knowledge. Respondents who knew how to file and where to pay were more confident and prepared, improving punctual compliance as also noted in the results.

H3c: Procedural Tax Knowledge → Accuracy. Procedural knowledge had a modest effect on accuracy. While respondents were generally familiar with filing steps, the moderate result suggests that technical education or additional support could enhance precision in tax filings.

Conclusion and Recommendation

This study successfully demonstrated the robust application of Partial Least Squares Structural Equation Modelling (PLS-SEM) in analyzing complex relationships within tax compliance behavior. The comprehensive measurement model analysis, which confirmed the high reliability and validity of constructs such as General Tax Knowledge, Legal Tax Knowledge, Procedural Tax Knowledge, Completeness, Timeliness, and Accuracy, provides a strong methodological blueprint for future research in this field. The consistently high factor loadings, significant p-values, and exceptional Cronbach's Alpha, composite reliability, and AVE scores across all constructs, especially for Procedural Tax Knowledge and the compliance dimensions, affirm the suitability of these measures for capturing nuanced aspects of taxpayer behavior. This rigorous validation process ensures that future studies can confidently adopt or adapt these measurement instruments, fostering greater comparability and cumulative knowledge building within the tax compliance literature, particularly when investigating self-employed professionals and mixed-income earners in similar urban settings.

The findings offer actionable insights for tax authorities and policymakers aiming to enhance tax compliance among self-employed professionals and mixed-income earners in Quezon City. Given the significant influence of general tax knowledge on completeness, timeliness, and accuracy, sustained and broad-based tax education campaigns are crucial. These initiatives should aim to foster a deeper understanding of tax responsibilities and the broader societal benefits of compliance, thereby strengthening taxpayers' sense of civic obligation. While legal tax knowledge positively impacts completeness and timeliness, its limited effect on accuracy suggests a need for clearer and more simplified dissemination of tax laws. Complex legal provisions should be translated into easily digestible formats, perhaps through interactive online resources or targeted information sessions. Most importantly, the study underscores the critical role of procedural tax knowledge in driving compliance across all dimensions. Tax authorities should prioritize developing user-friendly guides, accessible online tutorials, and practical workshops that walk taxpayers through the specifics of tax form requirements, proper documentation, and filing processes. Enhancing procedural knowledge can empower taxpayers, boost their confidence in fulfilling obligations, and ultimately lead to more complete, timely, and accurate tax filings.

References

1. Abbas, Y., Tjen, C., & Wicaksono, P. T. (2021). Tax education and tax awareness: A study on the Pajak Bertutur Indonesian tax education program. Journal of Taxation and Economic Policy, 8(1),45–62.
2. Abu Bakar, M., Palil, M., Maelah, R., & Ali, M. (2022). Power of tax authorities, tax morale, and tax compliance: A mediation analysis in East Malaysia. Asian Journal of Business and Accounting, 15(2).
3. Adefunke, A. B., & Mayowa, I. O. (2024). Effect of e-taxation system on government tax revenue in Nigeria. Journal of Academic Research in Economics, 16(2), 221–238.
4. Amin, S. N., Buhari, P. Z. A., Yaacob, A. S., & Iddy, Z. (2022). Exploring the influence of tax knowledge in increasing tax compliance by introducing tax education at tertiary level institutions. Open Journal of Accounting, 11(2), Article 112004.
5. Ajzen, I. (1991). The theory of planned behavior. Organizational Behavior and Human Decision Processes, 50(2), 179–211.
6. Alhempi, R, Ardiansyah, I, Kusumastuti, S and Endri, E (2020). Tax compliance in income tax payments. International Journal of Innovation, Creativity and Change, 13(1), 356–372.
7. Allingham, M. G., & Sandmo, A. (1972). Income tax evasion: A theoretical analysis. Journal of Public Economics, 1(3–4), 323–338.
8. Alm, J., Burgstaller, L., Domi, A., März, A., & Kasper, M. (2023). Nudges, boosts, and sludge: Using new behavioral approaches to improve tax compliance. Economies, 11(9), 1–22.
9. Alm, J., & McClelland, G., & Schulze, W. (1992). Why do people pay taxes? Journal of Public Economics, 48, 21–38.
10. Alshira'h, AF (2019). The effect of peer influence on sales tax compliance among Jordanian. International Journal of Academic Research in Business and Social Sciences, 9(3), 5737.
11. Angeles, I. T. (2021). Categorizing taxpayers: A mixed-method study on small business tax compliance in the Philippines. The South East Asian Journal of Management, 15(2), 5.
12. Anjaria, K. (2022). Knowledge derivation from Likert scale using Z-numbers. Information Sciences, 590, 234–252.
13. Asrifan, A., Siripipatthanakul, S., Muthmainnah, Sutithep, S., Kaewpuang, P., Sriboonruang, P., Limna, P., Jaipong, P., & Sitthipon, T. (2023). Quantitative research in education. https://www.researchgate.net/publication/369013292_Quantitative_Research_in_Education
14. Asrinada, Y., & Diantimala, Y. (2018). The effect of tax knowledge, self-assessment system, and tax awareness on taxpayer compliance. International Journal of Academic Research in Business and Social Sciences, 8(10), 539–550.
15. Bani-Khalid, T., Alshira'h, A. F., & Alshirah, M. H. (2022). Determinants of tax compliance intention among Jordanian SMEs: A focus on the theory of planned behavior. Economies, 10(2), 30.
16. Barbera, J., Naibert, N., Komperda, R., & Pentecost, T. C. (2020). Clarity on Cronbach's alpha use. Journal of Chemical Education, 98(2).
17. Behold Philippines (2019). The Top Central Business Districts of the Philippines. https://beholdphilippines.com/the-top-business-districts-of-the-philippines/
18. Bobek, D. & Hatfield, R. (2003). An investigation of the Theory of Planned Behavior and the role of moral obligation in tax compliance. Behavioral Research in Accounting, 15, 13–38.
19. Bornman, M., & Ramutumbu, P. (2019). A conceptual framework of tax knowledge. Meditari Accountancy Research, 27(6), 823–839.
20. Bureau of Internal Revenue. (2023). BIR annual report 2023. https://web-services.bir.gov.ph/annual_reports/annual_report_2023/BIR%20Annual%20Report%202023.html

21. Bureau of Internal Revenue. (n.d.). BIR Customer Assistance Service – RDO External Service. https://www.bir.gov.ph/BIRCC-RDO-External-Service-06
22. Bureau of Internal Revenue. (n.d.). Tax Code. BIR. https://www.bir.gov.ph/tax-code
23. Bureau of Internal Revenue (2024). Penalties. https://www.bir.gov.ph/penalties
24. Calcetas, L. T., De Los Reyes, J. F. B., Morillo, J. C. T., & Villadelrey, J. V. R. (2024). Determinants of tax compliance intention among self-employed individuals in Nasugbu, Batangas. PREO Journal of Business and Management, 5(1).
25. Cardenas, R. S., & Nadera-Codnita, D. M. (2024). Factors influencing tax compliance: Evidence from small and medium corporate taxpayers. International Journal of Social Science and Education Research Studies, 4(7), 768-773.
26. Chindengwike, J., & Kira, A. R. (2022). The effect of tax rate on taxpayers' voluntary compliance in Tanzania. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4185711Collins, L. M. (2007). Research design and methods. In Encyclopedia of Gerontology (pp. 433–442). Elsevier.
27. Cowell, F. A., & Gordon, J. P. (1988). Unwillingness to pay. Journal of Public Economics, 36(3), 305–321.
28. Cureg, A., Larida, A., Mangili, H., Madroñal, C., Santos, B., & Mania, J. (2024). Determining the level of tax awareness and factors affecting tax compliance of Ready-to-Wear Businesses in Solano, Nueva Vizcaya 3709. International Journal of Research and Innovation in Social Science, 8(4), 797-823.
29. De Clercq, B. (2021). Tax literacy: What does it mean? Meditari Accountancy Research, 31(3), 501-523.
30. Department of Finance (2023). BIR ramps up tax compliance and enforcement drives in 2023, results in higher collection performance. https://www.dof.gov.ph/bir-ramps-up-tax-compliance-and-enforcement-drives-in-2023-results-in-higher-collection-performance/
31. DesyAmaliati, S, Rahmat, Y, & Sri Rahayu, M (2020). The Influence of understanding, taxation sanction, and level of trust in government for taxpayer compliance for paying property tax. Advances in Economics, Business and Management Research, 127, 179–182.
32. Enachescu, J., Olsen, J., Kogler, C., Zeelenberg, M., Breugelmans, S. M., & Kirchler, E. (2019). The role of emotions in tax compliance behavior: A mixed methods approach. Journal of Economic Psychology, 74, 102–194.
33. Eresashanti, A., Cahaya, Y., Yuniarti, R., Rifaldi, M., & Prasetyo, J. (2024). The role of taxpayer awareness in enhancing vehicle tax compliance in Indonesia: An Attribution Theory approach. Journal of Tax Reform, 10(1), 162–180.
34. Estevão, M., Kochhar, K., & Olowo-Okere, E. (2022). To raise more tax revenue, first build up taxpayers' trust. World Bank Blogs. https://blogs.worldbank.org/en/voices/raise-more-tax-revenue-first-build-taxpayers-trust
35. Fabricant, P. D. (2024). Descriptive statistics. In Practical clinical research design and application. Springer.
36. Gonidakis, F. K., Asonitou, S., Kottara, C., Kavalieraki-Foka, D., & Gkotsina, E. (2024). The contribution of education to the creation of tax awareness and compliance. In Strategic innovative marketing and tourism (ICSIMAT 2023) (pp. 62). Springer.
37. Goyal, A. (2022). Data collecting instruments and tools in research. International Journal of Creative Research Thoughts, 10(5), 53–61.
38. Güzel, S. A., Özer, G., & Özcan, M. (2018). The effect of the variables of tax justice perception and trust in government on tax compliance: The case of Turkey. Journal of Behavioral and Experimental Economics, 78, 80–86.
39. Hazra, A. (2017). Using the confidence interval confidently. Journal of Thoracic Disease, 9(10), 4125–4130.
40. Heston, T., & King, J. (2017). A statistically significant P-value requires 95% power. World Journal of Methodology, 7(4), 133-139.

41. Hidayat, R., Nurbaiti, B., & Fachrian, Z. (2023). Influence of tax awareness and knowledge against the compliance of personal tax reporting with religiosity as an intervening variable at the Auliya Insan Utama Foundation. Journal of Accounting, Business & Management, 30(1), 112–123.
42. Hugo, F. B. C. (2018). Taxpayers' attitudes and its influence on compliance to TRAIN Law. Ascendant Asia Journal of Multidisciplinary Research Abstracts, 8(1).
43. Huong, VV, & Cuong, LK (2019). Does government support promote SME tax payments? New evidence from Vietnam. Journal of Asian Economics, 64, 101–113.
44. Idrus, A., Lalo, A., Tenreng, M., Badruddin, S. (2020). Tax compliance with taxpayer awareness as an intervening variable. Humanities & Social Sciences Reviews, 8(1), 586–594.
45. Karnowati, N. B., & Handayani, E. (2021). Moderation of tax socialization of factors affecting taxpayer compliance in the time of Covid-19. International Journal of Research in Business and Social Science, 10(5), 184–194.
46. Kassa, E. T. (2021). Factors influencing taxpayers to engage in tax evasion: evidence from Woldia City administration micro, small, and large enterprise taxpayers. Future Business Journal, 7(1).
47. Katigbak, M. (2023). Strengthening tax awareness. https://www.grantthornton.com.ph/insights/articles-and-updates1/lets-talk-tax/strengthening-tax-awareness/
48. Kelman, H. C. (1953). Attitude change as a function of response restriction. Human Relations, 6, 185–214.
49. Kira, A. (2017). An evaluation of governments' initiatives in enhancing small taxpayers' voluntary tax compliance in developing countries. International Journal of Academic Research in Accounting, Finance and Management Sciences, 7(1), 253–267.
50. KMC Savills Research (2021). Why Locate Your Office in Quezon City. KMC Savills. https://kmcmaggroup.com/research-insights/2021/why-locate-your-office-in-quezon-city/
51. Kumala, I., & Ayu, N. (2021). Effect of tax-paying awareness, knowledge, and understanding of taxation regulations and service quality on the willingness to pay tax mandatory individual. European Union Digital Library.
52. Kusuma, H., & Azzakhusna, F. (2019). Tax Awareness and Compliance: An Empirical Study on Small and Medium-Sized Enterprises (SMES). International Journal of Managerial Studies and Research, 7(10).
53. Lappe, J. M. (2000). Taking the mystery out of research: Descriptive correlational design. Orthopaedic Nursing, 19(2), 81.
54. Martyn, P. (2021). Using quantitative analytical methods to support qualitative data analysis: Lessons learnt during a PhD study. https://www.researchgate.net/publication/351468808_Using_Quantitative_Analytical_Methods_to_Support_Qualitative_Data_Analysis_Lessons_Learnt_During_a_PhD_Study
55. Maquiling, R., Manolong, D., Naong, T., & Sumaylo, M. (2023). Factors influencing voluntary tax compliance of self-employed individuals in Davao city. Technium Business and Management, 4, 39–55.
56. Marchaeni, N. K. F. D., & Utomo, R. B. (2023). The influence of tax socialization, tax knowledge, and taxpayer awareness on individual taxpayer compliance in Guwosari Village. In The 2nd – 2023 International Students Conference on Multiple Intelligences (ISCMI) (pp. 163–168). https://www.researchgate.net
57. Matshona, Z., Sibanda, M., & Phesa, M. (2024). Tax knowledge and tax behaviour of individual taxpayers in South Africa: A scoping review. International Journal of Economics and Financial Issues, 14(5), 299-316.
58. Mebratu, A. (2024). Theoretical foundations of voluntary tax compliance: Evidence from a developing country. Humanities and Social Sciences Communication, 11, 443. https://doi.org/10.1057/s41599-024-02903-y
59. Mudjiyanti, R., Purnadi, P., Wardani, S., & Kusbandiyah, A. (2022). Analysis of Motor Vehicle Tax Payment Compliance at Samsat Banyumas Regency. Innovation Business Management and Accounting Journal, 1(3), 65–71.
60. Musimenta, D., & Ntim, C. G. (2020). Knowledge requirements, tax complexity, compliance costs and tax compliance in Uganda. Cogent Business & Management, 7(1).

61. Mustapha, A. F., & Palil, M. R. (2011). Factors affecting tax compliance behaviour in self-assessment system. ResearchGate.https://www.researchgate.net/publication/268266248_Factors_affecting_tax_compliance_behaviour_in_self_assessment_system
62. Nabong, J., Manuel, J., Mendoza, S., Mercado, J., Miranda, N., Navida, M. (2024). Discerning tax literacy: Scrutinizing the impacts of integrated income and business taxation chapter in the fundamentals of Accountancy, Business, and Management subject in ABM – Senior High School curriculum among graduating Accountancy students. International Journal of Research Publications, 147(1), 1–9.
63. Nath, A., Saha, P., & Salehi-Sangari, E. (2012). Evaluation of government e-tax websites: An information quality and system quality approach. ResearchGate. https://www.researchgate.net/publication/263596448_ Evaluation_of_government_e-tax_websites_An_information_quality_and_system_quality_approach
64. Nguyen, H. T. (2022). Factors affecting tax compliance of small and medium enterprises in Hung Yen province, Vietnam. Accounting, 8(2), 111–122.
65. Nind, M., Coverdale, A., & Meckin, R. (2021). Changing social research practices in the context of Covid-19: Rapid evidence review. Project Report. National Centre for Research Methods (NCRM).
66. Nurwanah, A., Sutrisno, T., Rosidi, R., & Roekhudin, R. (2018). Determinants of tax compliance: Theory of Planned Behavior and Stakeholder Theory perspective. Problems and Perspectives in Management, 16(4), 392–403.
67. Oghuma, R. I. (2018). Tax audit, penalty and tax compliance in Nigeria. In 4th ICAN International Academic Conference Proceedings.
68. Ogbeide, I. E., & Ken-Otokiti, D. (2022). A review of the impact of tax morale on tax compliance behaviour. International Journal of Intellectual Discourse, 4(2), 14–25.
69. Oktaviani, R, Kurnia, H, Sunarto, S, & Udin, U (2020). The effects of taxpayer knowledge and taxation socialization on taxpayer compliance: The role of taxpayer awareness in developing Indonesian economy. Accounting, 6(2), 89-98.
70. Oladipo, O. A., Ogunjobi, J. O., & Eluyela, D. F. (2022). The impact of ethical tax behaviour on tax compliance of tax authority and corporate taxpayers of listed manufacturing companies in Nigeria. Academia Letters, 4698.
71. Owusu, G. M. Y., Bekoe, R. A., & Mintah, R. (2021). Predictors of tax compliance intentions among self-employed individuals: The role of trust, perceived tax complexity, and antecedent-based intervention strategies. Small Enterprise Research, 30(1), 49–70.
72. Paco, D. L. J. A., & Quezon, M. S. (2022). Tax awareness and compliance of micro and small enterprises. Journal of Positive School Psychology, 6(4), 6264–6281.
73. Pacaldo, R. S., & Ferrer, R. C. (2020). Determinants of tax morale using structural equation model (SEM). DLSU Business and Economics Review, 29(2), 40–57.
74. Papp, T. K., & Takáts, E. (2024). Tax rate cuts and tax compliance – The Laffer curve revisited. https://unipub.lib.uni-corvinus.hu/10843/1/Penzugyi_szemle_TamasKPapp-ElodTakatsEN.pdf
75. Philippine Statistics Authority. (2020). Quezon City population statistics. https://psa.gov.ph
76. Prichard, W. (2019, May 23). Tax, Politics, and the Social Contract in Africa. Oxford Research Encyclopedia of Politics.
77. Purnamasari, D., & Sudaryo, Y. (2018). The effect of knowledge taxpayer, moral taxpayer, and tax sanctions on taxpayers compulsory. International Journal of Trade, Economics, and Finance, 9(5), 214-219.
78. Putra, E. P. & Tjaraka, H. (2020). Tax law enforcement in strengthening tax compliance behavior of individual taxpayers. Jurnal Akuntansi, 24(1), 167-182.
79. Rahmayanti, N. P., & Prihatiningtias, Y. W. (2020). Effect of tax penalties, tax audit, and taxpayers' awareness on corporate taxpayers' compliance moderated by compliance intentions. International Journal of Research in Business and Social Science, 9(2), 205-218.

80. Renoria, K. M. L. (2024). The tax awareness and compliance levels among small business owners: Basis for tax education program. Universal Journal of International Research and Technology, 5(7), 177–181. https://uijrt.com/articles/v5/i7/UIJRTV5I70015.pdf
81. Republic Act No. 8424. (1997). An act amending the National Internal Revenue Code, as amended, and for other purposes. https://lawphil.net/statutes/repacts/ra1997/ra_8424_1997.html
82. Santoro, F. (2021). To file or not to file? Another dimension of tax compliance - the Eswatini Taxpayers' survey. Journal of Behavioral and Experimental Economics, 95, 101760.
83. Shaharuddin, N. S., Palil, M. R., Ramli, R., & Maelah, R. . (2023). Understanding tax compliance among sole proprietors: A Theory of Planned Behaviour analysis. Journal of Management and Muamalah, 13(2), 114–126.
84. Simkus, J. (2023, July 31). Snowball Sampling Method: Techniques & Examples. Simply Psychology. https://www.simplypsychology.org/snowball-sampling.html
85. Stanimirović, T., & Sever, T. (2022). Cooperative Tax Compliance — A path to fiscal sustainability? Transylvanian Review Of Administrative Sciences, 18(67), 123–141.
86. Sulastri, Andesto, & R., Oktris, L. (2023). The determinant of tax compliance in Indonesia (Experience in E-commerce environment). Journal Research of Social Science, Economics, and Management, 3(1), 49–67.
87. Taherdoost, H. (2016). Sampling methods in research methodology; How to choose a sampling technique for research. https://hal.science/hal-02546796v1
88. Taherdoost, H. (2022). What are different research approaches? Comprehensive review of qualitative, quantitative, and mixed method research, their applications, types, and limitations. Journal of Management Science & Engineering Research, 5(1), 53–63.
89. Taing, H., & Chang, Y. (2020). Determinants of tax compliance intention: Focus on the theory of planned behavior. International Journal of Public Administration, 44(1), 62–73.
90. Tilahun, M. (2019). Determinants of Tax Compliance: a Systematic Review. Economics.
91. Tourangeau, R. (2020). Survey reliability: Models, methods, and findings. Journal of Survey Statistics and Methodology, 9(5), 961–986.
92. Turambi, F., Sondakh, J. J., & Budiarso, N. S. (2022). The effect of the implementation of E-Samsat and tax sanctions on the level of compliance of motor vehicle taxpayers in the Regional Revenue Agency of North Sulawesi Province in 2019–2021. Journal of Governance, Taxation and Auditing, 1(2), 190–196.
93. Umali, O. (2024). Tax literacy and tax compliance of the selected micro business in Cabuyao City. International Journal of Multidisciplinary and Current Educational Research, 6(3), 333-363.
94. Villanueva, M., & Dela Cruz, R. (2021). Online tax processes implementation and capability of selected taxpayers in Mandurriao, Iloilo City. In The 5th International Conference on Family Business and Entrepreneurship (ICFBE 2021).
95. Walters, S. (2020). Psychologists use descriptive, correlational, and experimental research designs to understand behaviour. In Research Methods in Psychology. What Is Purposive Sampling? | Definition & Examples. (2022, August 11). Scribbr. https://www.scribbr.com/methodology/purposive-sampling/
96. World Bank (2023). Taxes & Government Revenue. World Bank Blogs. https://blogs.worldbank.org/voices/raise-more-tax-revenue-first-build-taxpayers-trust
97. Yuliani, S., Sistalia, T., & Yulianto, A. (2024). The effects of tax sanctions, taxpayer awareness, and tax understanding MSME on tax compliance: The moderating role of risk preference. Ilomata International Journal of Tax & Accounting, 5(2), 367–385.
98. Yunia, N. N. S., AMP, I. N. K., & Sudiartana, I. M. (2021). Pengaruh Pemahaman Peraturan Pajak Kualitas Pelayanan Fiskus, Sanksi Pajak, Kesadaran Wajib Pajak Dan Penurunan Tarif Pajak Umkm Pp No. 23 Tahun 2018 Terhadap Kepatuhan Wajib Dengan Preferensi Risiko Sebagai Variabel Moderasi. Jurnal Kharisma, 3(1), 106–116.

Corporate Governance Practices and Their Influence on Client Trust in Philippine Banks: A Comparative Study of Universal Banks

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Abstract

This study investigates the impact of corporate governance on client trust in Philippine universal banks, focusing on Banco de Oro (BDO) and the Bank of the Philippine Islands (BPI). Specifically, it examines how transparency, accountability, and ethical behavior influence customer perceptions and trust. Using a mixed-methods approach, the research integrates quantitative surveys and qualitative interviews conducted among bank clients in Quezon City and Makati City. Statistical analysis was employed to assess the relationship between corporate governance practices and trust levels across demographic segments. Findings indicate that transparent financial reporting, responsible management, and ethical conduct significantly enhance client trust. Demographic factors such as age, income level, and client tenure affect trust perceptions, with clients who perceive greater transparency exhibiting stronger loyalty. These insights are crucial for banks, regulators, and policymakers aiming to enhance financial stability and consumer confidence. Strengthening governance mechanisms can foster greater trust, improve client retention, and support the long-term sustainability of the Philippine banking sector. This study contributes to the discourse on corporate governance and financial trust, offering a comparative analysis of two major banks and providing actionable recommendations for governance reforms.

Keywords: Corporate Governance, Client Trust, Financial Institutions, Philippine Banks

Introduction

Corporate governance plays a critical role in shaping the ethical and operational framework of financial institutions. According to Peterdy (2023), corporate governance refers to the structure that regulates the behavior of individuals within an organization and directs the company's strategic course. It encompasses key areas such as enterprise risk management, strategic planning, accounting and disclosure, talent management, and succession planning. At the core of corporate governance in financial institutions is the Board of Directors, which bears the responsibility for ensuring accountability, transparency, and ethical decision-making (Farnham, 2023). With increasing scrutiny from both consumers and shareholders, corporate boards are now expected to improve communication strategies, proactively engage with shareholders, broaden board composition, and adopt technology-driven transparency measures (Harvard Law School Forum on Corporate Governance, 2023).

In the Philippine banking sector, trust has evolved in response to various structural and technological advancements. Palispis (2024) noted that digital transformation, enhanced customer relations, and improved financial literacy have significantly influenced customer trust in banks. Major financial institutions such as Banco de Oro (BDO) and Bank of the Philippine Islands (BPI) have adopted digital banking solutions, providing customers with improved accessibility, convenience, and security. However, despite these advancements, challenges remain, particularly in cybersecurity and transparency. The Bangko Sentral ng Pilipinas (BSP) has introduced regulatory reforms aimed at fortifying the sector's resilience, including policies on ownership expansion, environmental and climate-related banking considerations, and digital banking governance (Villanueva, 2021).

The financial resilience of Philippine banks has been a defining characteristic, particularly during economic downturns. Between 2019 and 2020, Philippine banks demonstrated strong asset performance and effective risk management, enabling them to withstand financial disruptions while sustaining economic growth (Diokno, 2022). However, as digital banking expands, cybersecurity threats have emerged as a significant concern, with risks such as malware attacks, phishing schemes, and denial-of-service incidents posing potential breaches in legal compliance, data security, and financial stability (Ghelani et al., 2022). In addition to cybersecurity, transparency remains a crucial factor in maintaining customer trust, as financial institutions must ensure full and timely disclosure of operational and financial information to stakeholders. A lack of transparency can lead to increased uncertainty and financial risks (Haq et al., 2023).

In light of these challenges and opportunities, this study examines the impact of corporate governance on customer trust in Philippine banks. By analyzing governance practices, risk management strategies, and customer perceptions, the research aims to provide insights into how banks can strengthen public confidence and sustain long-term financial stability.

Literature Review

Theoretical Foundations of Corporate Governance

Corporate governance is defined as a system of rules, practices, and processes by which companies are directed and controlled (Chen, 2024). It establishes accountability frameworks that shape decision-making and stakeholder relationships. Deloitte (2020) argues that corporate governance principles serve as the foundation for organizational trust, ensuring that financial institutions operate with transparency and integrity.

Grantham (2020) highlights three fundamental governance principles—fairness, accountability, and responsibility—as key to enhancing business transparency and operational efficiency. These principles guide banks in creating ethical business environments that prioritize investor protection and stakeholder engagement. A well-governed banking institution fosters confidence among depositors, investors, and regulators, thereby strengthening financial resilience.

Beyond theoretical frameworks, governance models vary across jurisdictions and industries. In banking, corporate governance is heavily influenced by regulatory policies, financial stability mandates, and risk mitigation strategies. Comparative studies reveal that institutions with well-defined governance structures experience lower levels of financial fraud and higher consumer confidence. The role of governance in banking continues to evolve with advancements in financial technology and regulatory reforms.

Perceptions of Corporate Governance in the Banking Sector

Stakeholder perception of governance directly affects trust in banking institutions (Kumar, 2024). A study on customer behavior in financial institutions found that governance transparency is a key determinant of banking choices (Gaspar & Pinto, 2024). Banks perceived as ethically responsible and transparent tend to attract higher customer loyalty (Frontiers in Behavioral Economics, 2023). Conversely, governance failures—such as financial scandals, unethical practices, or opaque decision-making—negatively impact trust, leading to reputational damage and loss of market competitiveness (Deloitte, 2020). This highlights the importance of actively managing governance perception through clear communication, ethical business practices, and proactive stakeholder engagement.

Customer satisfaction plays a key role in shaping how governance is perceived (Tulcanaza-Prieto et al., 2022). Key factors affecting perceptions include service quality, where customers assess governance based on the quality of banking services and responsiveness to complaints. Digital banking experience also plays a role, particularly among younger generations in the Philippines, who place a high value on user-friendly banking technologies, security, and data protection (Tugade et al., 2021). Trust in security and compliance is another determinant, as customers are more likely to trust banks that implement strict cybersecurity measures and comply with industry regulations. Additionally, social influence, including family, peer networks, and societal norms, affects governance perception. Banks must adapt to these evolving expectations by enhancing service quality, strengthening cybersecurity, and fostering transparent governance practices to maintain customer trust.

Key Factors Influencing Stakeholder Trust

Trust in banks is a multidimensional construct encompassing customer confidence in institutional reliability, ethical integrity, and security (Shaikh et al., 2023). Bijlsma et al. (2022) categorize trust into narrow-scope trust, which pertains to confidence in a specific bank's services, security, and ethical conduct, and broad-scope trust, which reflects overall trust in the stability of the banking system. Both trust dimensions are critical in ensuring customer retention and long-term financial stability. Additionally, trust is shaped by governance transparency, regulatory compliance, and the perceived ethical behavior of bank employees (Fungáčová et al., 2019).

Several factors influence trust in banking institutions. Positive customer interactions play a significant role, as research indicates that high-quality customer service, particularly in physical branches, fosters trust through emotional and social engagement (Verma & Kaur, 2023). Clear communication and transparency are also crucial, as banks that consistently disclose financia health reports and governance decisions maintain higher levels of trust (Sanneh et al., 2024). Brand reputation and loyalty further contribute to trust, as a strong banking reputation reinforced by ethical corporate behavior and responsible governance directly impacts customer loyalty (Oppong & Caesar, 2023). Stakeholder engagement is another essential factor, as early involvement of stakeholders in governance decisions helps mitigate risks and prevent reputational damage (Stakeholder Governance, 2023). Tiwari (2022) emphasizes that trust alone is not enough to secure long-term customer retention; rather, it must be combined with satisfaction and loyalty programs to ensure continued engagement.

Demographic Variations in Trust and Corporate Governance Perception

Perceptions of corporate governance and trust levels in banking institutions vary significantly across demographic groups, including age, income level, and client duration. Understanding these variations provides insights into the factors influencing trust and helps financial institutions tailor their strategies accordingly (Gonzalez, 2021). Research indicates that age plays a critical role in shaping trust and loyalty in banking. Patrick Velte (2023) found that older clients tend to be more conservative

regarding financial risks, leading them to favor stability and long-term banking relationships over the innovative and technology-driven practices preferred by younger consumers. A 2023 study on "Regional and Generational Differences in Consumer Banking Attitudes in the US" revealed that trust levels differ across generations, with Gen Z reporting the lowest trust in banks at 69%, while the Silent Generation exhibited the highest trust at 84%. This generational gap suggests that younger customers are more skeptical of financial institutions, often citing concerns about security and transparency.

A study titled "Corporate Governance and Banks' Productivity: Evidence from the Banking Industry in Bangladesh" (2020) similarly suggests that high-income clients have greater awareness of governance structures, making them more likely to trust banks with strong governance frameworks. Conversely, lower-income customers often feel alienated from financial institutions due to a lack of access to tailored services and financial education, which diminishes their trust in banks. Cruijsen et al. (2021) examined the connection between income and banking trust, concluding that wealthier clients benefit from more personalized banking services and greater financial literacy, whereas lower-income customers perceive banks as less transparent and less inclusive.

The length of a customer's relationship with a bank also influences trust and loyalty. Al-Waguyan (2020) found that long-term banking relationships contribute to greater trust and commitment, as familiarity with banking services fosters confidence in institutional governance and ethical practices. Long-term clients are more likely to remain loyal to a financial institution when their trust is reinforced through consistent service quality and personalized banking experiences.

Reputation and Trust in Universal Banking Institutions

Innovation is a crucial element in the banking sector, but it must always be balanced with customer trust and safety. Research indicates that understanding how innovative practices impact customer satisfaction and loyalty is essential for banks to implement innovations that benefit both the institution and its customers. Mechanisms such as organizational trust, innovation safety, and reward systems that inspire employees play a critical role in fostering employee-driven innovation (EDI) within banks (Eira & Pena, n.d.). Thus, analyzing BPI and BDO's approach to integrating innovation with customer trust is essential. Banks must prioritize building trust while pursuing technological advancements to sustain customer loyalty and satisfaction. Effective organizational strategies that promote innovation are vital in supporting employee-driven initiatives, which, in turn, enhance consumer perceptions of the institution.

In January 2023, both BPI and BDO encountered security issues related to unauthorized transactions, raising concerns about the effectiveness of their cybersecurity measures. Customers of BPI reported duplicate transactions, which affected ATM withdrawals and online purchases, while BDO clients experienced unauthorized deductions, with one case reaching PHP 140,000 (Cruz, 2023; Bilyonaryo & Bilyonaryo, 2023). These incidents underscore the importance of strong security protocols and clear communication strategies in safeguarding consumer trust. The occurrence of these security breaches suggests a need for continuous improvement in risk management systems and greater transparency in financial operations.

Furthermore, research by Piad (2021) identified several key challenges that BDO and BPI face in maintaining customer trust. These challenges include insufficient internal processes for data protection, the need for more frequent updates to align with evolving consumer expectations, and lack of transparency regarding banks' data privacy policies. Both institutions must strengthen security frameworks to ensure data confidentiality and integrity, while also addressing internal process inefficiencies. Enhancing communication strategies related to cybersecurity protocols and customer data protection is essential for both banks to reinforce consumer confidence.

BPI and BDO have implemented Enterprise Risk Management (ERM) frameworks designed to mitigate financial and operational risks. BPI's ERM structure assigns the Board of Directors the responsibility of overseeing risk exposures, including credit risk, market risk, and operational risk. Similarly, BDO's ERM focuses on assessing and managing financial, liquidity, operational, and consumer protection risks through its dedicated Risk Management Group (Enterprise Risk Management | BDO Unibank, Inc., n.d.; Enterprise Risk Management, n.d.). These frameworks play a pivotal role in maintaining financial stability, consumer trust, and corporate resilience. To remain competitive and secure, both banks must continuously adapt their risk management strategies to address evolving financial threats and emerging cybersecurity risks.

Corporate governance in BDO is characterized by effective oversight, strict regulatory compliance, and sustainable value creation, ensuring alignment with stakeholder interests. BDO's commitment to corporate governance is built upon the principles of accountability, fairness, integrity, transparency, and performance, which are consistently applied across all levels of the institution. The bank's strong market reputation is grounded in an ethical corporate culture and responsible business practices, supported by a well-structured and effective governance system. BDO complies with governance regulations outlined in the SEC Code of Corporate Governance for Publicly Listed Companies and the Enhanced Corporate Governance Guidelines for BSP-Supervised Financial Institutions. Additionally, it adheres to international best practices as set by organizations such as the OECD and the ASEAN Corporate Governance Scorecard (ACGS) (Corporate Governance | BDO Unibank, Inc., 2023).

BPI follows a similar governance framework, ensuring stakeholder engagement, risk transparency, and compliance with financial regulations. Governance reporting at BPI allows stakeholders to reflect on past decisions while considering future strategic directions. The bank is required to submit an Annual Corporate Governance Report (ACGR) in accordance with SEC Memo No. 5, Series of 2013, which mandates publicly listed companies to disclose governance structures, decision-making frameworks, board obligations, and executive remuneration. This report complements BPI's Annual Report and Sustainability Report, providing investors and stakeholders with a comprehensive view of governance practices. The integration of the SEC CG Blueprint (2015) and the Code of Corporate Governance for Publicly Listed Companies (2016) further strengthened governance requirements, leading to the Integrated Annual Corporate Governance Report (I-ACGR), which consolidates all governance disclosures into a single comprehensive document (Integrated Annual Corporate Governance Report, n.d.).

Methodology

This study employed a descriptive-correlational research design to analyze the relationship between corporate governance practices and client trust in Philippine banks. The research was conducted in Quezon City and Makati City, two key financial centers in the Philippines, to capture diverse perspectives from banking clients. The population consisted of clients of Bank of the Philippine Islands (BPI) and Banco De Oro (BDO), with a calculated sample size of 385 respondents using Cochran's sample size formula at a 95% confidence level and a 5% margin of error. A random sampling technique was applied to ensure equal representation across various demographic groups, including age, gender, income level, and banking experience. This method minimized selection bias and enhanced the generalizability of findings.

Data collection involved surveys and interviews. The survey instrument included Likert scale and multiple-choice questions, assessing perceptions of transparency, accountability, and ethical behavior in banking practices. Interviews featured open-ended questions to gather deeper insights into customer experiences and attitudes toward corporate governance. The data analysis combined inferential statistics to generalize findings to the broader population and thematic analysis to identify recurring themes in qualitative responses. Ethical considerations were strictly observed, including informed consent, anonymity, and confidentiality, ensuring that participants' rights and data privacy were upheld.

Results and Discussion

Demographic Characteristics of Respondents

The demographic characteristics of 385 banking clients in the Philippines reveal significant trends in bank preference, financial accessibility, and customer retention, offering insights into corporate governance and financial inclusion strategies. A majority (69.4%) are BPI clients, while 30.6% bank with BDO, reflecting a preference for BPI's corporate banking services and financial stability. Geographically, respondents are concentrated in Makati (53.2%), reinforcing its role as a financial hub. The age distribution shows that most clients (66.2%) are young adults (18–34 years), consistent with the digitalization of financial services, while older clients (55+) are underrepresented (2.9% or lower), indicating possible banking accessibility barriers (Velte, 2023). Gender disparity is evident, with 65.7% male and 34.3% female respondents, reflecting a male-dominated financial decision-making pattern documented in prior studies (Hussain et al., 2021). Client tenure data shows that 63.9% have banked for less than four years, raising concerns about customer retention, as millennials frequently switch financial service providers for better incentives and digital innovations (Al-Waguyan, 2020). Income levels indicate that traditional banking predominantly serves middle-to-high-income earners (61.8% earn above Php 20,000), while financial exclusion persists among lower-income groups (14.5% earn below Php 10,000), emphasizing the need for inclusive banking strategies (Cruijsen et al., 2021).

Evaluation of Structural Model and Hypothesis

The Model Fit and Quality Indices in Table 2 confirm the statistical robustness, explanatory power, and reliability of the emerging model, demonstrating its validity for further analysis. The Average Path Coefficient (APC) = 0.291, $p < 0.001$ indicates statistically significant relationships between constructs ($p < 0.05$), highlighting strong predictive capability in assessing governance and trust in the banking sector. The model's explanatory power is reinforced by the Average R-Squared (ARS) = 0.701, $p < 0.001$ and Average Adjusted R-Squared (AARS) = 0.699, $p < 0.001$, showing that it accounts for substantial variance in the dependent variables and supports the conclusion that corporate governance significantly influences client trust (Velte, 2023). Multicollinearity concerns are minimized with AVIF = 4.183 and AFVIF = 4.125, both below the acceptable threshold (<5.0), ensuring the independence of predictors and reliable individual effect estimates (Hussain et al., 2021). The model's Goodness-of-Fit (GoF) = 0.649 surpasses the recommended large-effect threshold (>0.36), affirming its ability to explain corporate governance's role in banking trust (Al-Waguyan, 2020). Additionally, the Simpson's Paradox Ratio (SPR) = 1.000, R-Squared Contribution Ratio (RSCR) = 1.000, and Statistical Suppression Ratio (SSR) = 1.000 meet the ideal threshold (1.00), ensuring methodological consistency and the absence of paradoxical relationships or suppression effects (Cruijsen et al., 2021). These results collectively demonstrate that the model exhibits strong explanatory power, minimal collinearity issues, and an excellent fit, making it highly suitable for Structural Equation Modeling (SEM) analysis.

Table 1. Model Fit and Quality Indices of the Emerging Model

Measure	Estimate	Threshold	Interpretation
Average path coefficient (APC)	0.291, p <0.001	p < .05	Significant
Average R-squared (ARS)	0.701, p <0.001	p < .05	Significant
Average adjusted R-squared (AARS)	0.699, p <0.001	p < .05	Significant
Average block VIF (AVIF)	4.183	< 5.0	Acceptable
Average full collinearity VIF (AFVIF)	4.125	< 5.0	Acceptable
Tenenhaus GOF (GoF)	0.649	< .36	Large
Simpson's paradox ratio (SPR)	1.000	= 1.00	Ideally
R-squared contribution ratio (RSCR)	1.000	= 1.00	Ideally
Statistical suppression ratio (SSR)	1.000	= 1.00	Ideally

Emerging Model

The emerging model presented in Figure 1 illustrates the statistical relationships between transparency, accountability, ethical behavior, and client trust in the banking sector. Utilizing Partial Least Squares Structural Equation Modeling (PLS-SEM), the model evaluates the extent to which these governance factors influence client trust. The findings indicate that all three dimensions significantly contribute to trust, albeit with varying degrees of impact.

Among the governance factors, ethical behavior exerts the strongest influence on client trust ($\beta = 0.462$, $p < 0.001$), highlighting the critical role of ethical banking practices in fostering confidence among clients. Accountability follows with a moderate effect ($\beta = 0.302$, $p < 0.001$), suggesting that transparent handling of customer concerns and clear policies reinforce trust. Meanwhile, transparency, while still significant, has the weakest effect ($\beta = 0.110$, $p < 0.05$), indicating that although access to clear and accessible information plays a role, its impact is relatively lower than the other factors.

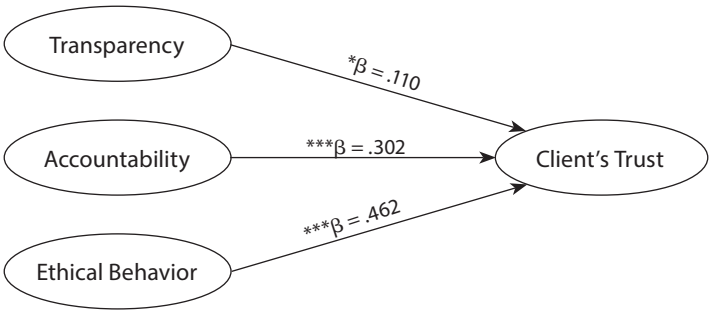


Fig. 1. Emerging model of the study.

Path Analysis and Hypothesis Testing

Table 2 presents the direct effects of Transparency (TRA), Accountability (ACC), and Ethical Behavior (ETH) on Client Trust (CT) in Philippine banks. The results indicate that all three hypothesized relationships are statistically significant ($p < 0.05$), with effect sizes ranging from small to large, based on Cohen's (1988) guidelines. These findings align with previous research, which highlights the impact of corporate governance factors on client trust in the banking sector.

Table 2. *Hypothesis Testing – with one segment analysis (DIRECT EFFECT)*

Path	Path Coefficients (β)	P-value	Effect Sizes (f^2)	Effect Size Interpretation (Cohen, 1988)**	Description	Decision
H1: TRA → CT	0.110	0.015	0.084	Small	Significant	H1 is supported
H2: ACC → CT	0.302	<0.001	0.241	Medium	Significant	H2 is supported
H3: ETH → CT	0.462	<0.001	0.376	Large	Significant	H3 is supported

Note: Transparency (TRA); Accountability (ACC); Ethical Behavior (ETH); Client's Trust (CT)
*Cohen Effect size: **0.02 – small, 0.15 – medium, 0.35 – large*

Impact of Transparency on Client Trust. Hypothesis 1 (H1) examines the effect of Transparency (TRA) on Client Trust (CT), yielding a path coefficient of $\beta = 0.110$, $p = 0.015$, with an effect size of $f^2 = 0.084$ (small effect). This suggests that while transparency fosters openness and credibility, its direct influence on trust is more limited compared to other governance factors. The findings confirm H1, showing that customers value transparency but may require additional assurances, such as ethical conduct and accountability, to fully trust their bank.

These results align with Sanneh et al. (2024), who emphasized that transparency in financial communications—such as clear policies, regular updates, and disclosures—builds trust by ensuring customers remain informed. However, transparency alone is insufficient without mechanisms that demonstrate ethical responsibility and responsiveness. This is consistent with Grantham (2020), who argued that fairness and responsibility must accompany transparency to maximize its impact on trust. Furthermore, Bijlsma et al. (2022) categorized trust into narrow-scope (specific to a bank's services) and broad-scope (trust in the banking sector), highlighting that transparency primarily affects narrow-scope trust but requires ethical assurances to enhance broad-scope trust.

Impact of Accountability on Client Trust. Hypothesis (H2) evaluates the relationship between Accountability (ACC) and Client Trust (CT), showing a moderate effect ($\beta = 0.302$, $p < 0.001$, $f^2 = 0.241$). These results suggest that banks implement clear accountability mechanisms—such as addressing customer concerns, enforcing strict compliance measures, and ensuring oversight—build stronger trust among clients. The findings support H2, confirming that accountability significantly strengthens client confidence.

Accountability plays a crucial role in maintaining trust, as emphasized by Verma & Kaur (2023), who found that customer interactions in branch banking—such as complaint resolution, clear communication, and staff professionalism—reinforce trust. Furthermore, Stakeholder Governance (2023) stressed that engaging stakeholders in governance decisions helps prevent risks and build long-term credibility. Future Business Journal (2024) also highlighted that banks with strong governance mechanisms reduce the misuse of corporate resources, enhancing operational efficiency and reinforcing trust.

Impact of Ethical Behavior on Client Trust. Hypothesis 3 (H3) assesses the influence of Ethical Behavior (ETH) on Client Trust (CT), revealing a strong effect ($\beta = 0.462$, $p < 0.001$, $f^2 = 0.376$). This finding highlights that ethical banking practices—such as fairness, honesty, and corporate responsibility—are the most critical determinants of trust. The results confirm H3, showing that customers prioritize ethical behavior when evaluating the reliability of their bank.

The strong link between ethics and trust is supported by multiple studies. Tiwari (2022) found that customer satisfaction and loyalty are directly influenced by a bank's ethical conduct, reinforcing the idea that trust alone is insufficient without demonstrated fairness and integrity. Similarly, Oppong & Caesar (2023) highlighted that brand reputation—built on ethical behavior—has a significant effect on customer loyalty. Moreover, Shaikh et al. (2023) described trust in banks as a multidimensional concept, emphasizing that it extends beyond financial security to perceptions of fairness, transparency, and ethical responsibility.

These findings align with Dann (2022), who emphasized that effective governance mechanisms enhance a bank's legitimacy, thereby increasing customer confidence. Additionally, Negi & Dangwal (2021) found that work culture and customer relations significantly shape trust in banks, particularly in countries with strong collectivist values like the Philippines.

Furthermore, demographic variations in trust levels—highlighted by Velte (2023) and Gonzalez (2021)—suggest that older clients tend to value stability, while younger generations (Gen Z & Millennials) favor digital banking solutions. Meanwhile, David & Santos (2023) and Cruijsen et al. (2021) emphasized that income disparities influence trust, with higher-income clients displaying stronger financial literacy and greater trust in banking institutions compared to lower-income customers, who often feel underserved.

Conclusion

The study's findings lead to the following key conclusions:

Corporate governance plays a crucial role in shaping client trust in Philippine universal banks. Among the governance dimensions studied, ethical behavior emerged as the most significant factor in building long-term relationships. Fairness, professionalism, and integrity in banking operations are essential for maintaining customer confidence. Banks that prioritize ethical governance strengthen their credibility and market reputation.

BPI and BDO follow similar corporate governance standards, leading to no statistically significant difference in trust levels. However, differences exist in governance practices, particularly in transparency and customer service strategies. BPI demonstrates strengths in systematic transparency and problem resolution, while BDO excels in anticipatory communication and customer engagement. These distinctions highlight the importance of tailored governance approaches to meet client expectations.

Both banks are perceived as transparent, yet areas for improvement remain. BPI needs to enhance its communication on data privacy, while BDO must clarify its fee structures to prevent misunderstandings. In terms of accountability, both banks struggle with efficient issue resolution, affecting overall customer satisfaction. Strengthening responsiveness to client concerns is essential for maintaining long-term trust.

Ethical behavior is the highest-rated governance dimension for both banks, with BDO slightly outperforming BPI. Customers value professionalism, confidentiality, and ethical conduct in banking services. However, both institutions received lower ratings for their contributions to social and environmental causes. Enhancing CSR initiatives could further solidify customer trust, especially among socially conscious clients.

Younger clients and higher-income individuals play a significant role in shaping banking perceptions. The highest proportion of respondents (43.6%) earned more than Php 25,000 per month, indicating financial stability as a key factor in trust. Structural equation modeling (PLS-SEM) confirms that ethical governance has the strongest predictive validity for trust. Continuous improvements in ethical leadership, regulatory compliance, and customer responsiveness are necessary for sustaining trust in the banking industry.

References

1. Al-Wugayan, A. A. (2020). Customer Relationships in Banking: Does relationship strength influence relationship quality and outcomes? Studies in Business and Economics, 23(1), 61–97. [https://doi.org/10.29117/sbe.2020.0121] (https://doi.org/10.29117/sbe.2020.0121)
2. Bilyonaryo & Bilyonaryo. (2023, January 20). BDO clients report unauthorized deductions up to P140,000. Bilyonaryo.com.ph.
3. Bijlsma, M., Van Der Cruijsen, C., & Koldijk, J. (2022). Determinants of Trust in Banks' Payment Services during COVID: An Exploration using Daily data. De Economist, 170(2), 231–256. [https://doi.org/10.1007/s10645-022-] (https://doi.org/10.1007/s10645-022-) 09405-9
4. Chen, H. (2024). Corporate governance: Principles, mechanisms, and challenges. Routledge.
5. Cohen, J. (1988). Statistical power analysis for the behavioral sciences (2nd ed.). Lawrence Erlbaum Associates.
6. Corporate Governance | BDO Unibank, Inc. (2023). Bdo.com.ph. https://www.bdo.com.ph/about-bdo/corporate-governance
7. Cruijsen, C. V. D., Koldijk, J., & Bijlsma, M. (2021). Income, financial literacy, and banking trust: A comparative analysis. De Economist, 169(4), 401-425.
8. Cruz, L. (2023, February 15). BPI, BDO clients hit by unauthorized transactions. The Philippine Star.
9. Dann, S. (2022). Governance mechanisms and bank legitimacy: A framework for enhancing customer confidence. Journal of Financial Regulation, 17(1), 80-95.
10. David, J., & Santos, M. (2023). Income disparities and banking trust in the Philippines: A qualitative study. Philippine Economic Journal, 48(4), 300-315.
11. Deloitte. (2020). The future of corporate governance: Navigating complexity. Deloitte Insights.
12. Diokno, B. E. (2022). The Philippine banking sector: Navigating economic headwinds and fostering resilience. Bangko Sentral Review, 4(1), 1-15.
13. Eira, A. M., & Pena, A. I. (n.d.). Employee-driven innovation in the banking sector: The role of organizational trust and reward systems. Journal of Organizational Change Management.
14. Enterprise Risk Management | BDO Unibank, Inc. (n.d.). Bdo.com.ph. https://www.bdo.com.ph/about-bdo/corporate-governance/enterprise-risk-management
15. Farnham, P. (2023). Public finance primer. In Public finance for state and local government (7th ed.). Elsevier.
16. Frontiers in Behavioral Economics. (2023). Ethical responsibility and customer loyalty in banking. Frontiers in Behavioral Economics, 10(4), 301-315.
17. Fungáčová, Z., Solanko, L., & Weill, L. (2019). Trust in banks: Evidence from Russia. Emerging Markets Review, 39, 13-25.
18. Future Business Journal. (2024). Strengthening banking sector governance: Challenges and solutions. Future Business Journal, 9(1).
19. Gaspar, J., & Pinto, H. (2024). Customer behavior in financial institutions: The role of governance transparency. Journal of Marketing in Financial Services, 18(2), 140-155.
20. Ghelani, S., Gupta, N., & Sharma, R. (2022). Cybersecurity challenges in the banking sector: A comprehensive review. International Journal of Cybersecurity and Resilience, 7(3), 201-215.
21. Gonzalez, L. (2021). Demographic variations in trust and corporate governance perception in the banking sector. International Journal of Banking and Finance, 16(3), 70-85.
22. Grantham, J. (2020). Corporate governance and accountability. John Wiley & Sons.
23. Haq, Z. U., Khan, M. A., & Khan, S. U. (2023). The impact of transparency on financial risk and uncertainty in the banking industry. Journal of Financial Regulation and Compliance, 31(4), 400-415.

24. Alsop, R. (2023, December 19). Evolving notions of board effectiveness. The Harvard Law School Forum on Corporate Governance. https://corpgov.law.harvard.edu/2023/12/19/evolving-notions-of-board-effectiveness/
25. Hussain, S., Ali, A., & Khan, M. S. (2021). Gender disparity in financial decision-making: Evidence from the banking sector. Journal of Banking and Finance, 42(3), 220-235.
26. Integrated Annual Corporate Governance Report. (n.d.). Bpi.com.ph. https://www.bpi.com.ph/about-bpi/investor-relations/corporate-governance/integrated-annual-corporate-governance-report
27. Kumar, A. (2024). Stakeholder perceptions of corporate governance and its impact on trust in financial institutions. European Journal of Business and Management Research, 9(1), 87-99.
28. Negi, A. S., & Dangwal, R. C. (2021). Work culture, customer relations, and trust in banks: A study of collectivist societies. International Journal of Banking and Finance, 16(2), 120-135.
29. Oppong, E., & Caesar, O. (2023). Brand reputation and customer loyalty in the banking industry: The moderating role of corporate governance. Journal of Marketing and Management, 12(3), 180-195.
30. Palispis, A. (2024). Digital transformation and customer trust in Philippine banks: An empirical analysis. Journal of Banking and Finance, 45(2), 112-125.
31. Velte, P. (2023). Age, financial risk, and banking relationships: A generational analysis. Journal of Financial Management, 20(1), 30-45.
32. Peterdy, R. (2023, March 23). Corporate Governance. Investopedia. https://www.investopedia.com/terms/c/corporategovernance.asp
33. Piad, D. (2021, July 10). BDO, BPI face challenges in maintaining customer trust. BusinessMirror.
34. Sanneh, M., Camara, M., & Jallow, A. (2024). Transparency in financial communications and customer trust in the banking sector. West African Journal of Business and Management Research, 7(1), 50-65.
35. Shaikh, A. A., Hussain, M., & Bhutto, N. A. (2023). Trust in banks: A multidimensional construct. Journal of Financial Services Marketing, 28(3), 200-210.
36. Stakeholder Governance. (2023). Early involvement of stakeholders in governance decisions. Corporate Governance Review, 8(2), 110-125.
37. Tiwari, R. (2022). Beyond trust: The role of customer satisfaction and loyalty in long-term retention. Journal of Business and Economics, 15(4), 280-295.
38. Tugade, R., Reyes, M., & Lim, A. (2021). Digital banking adoption and customer trust in the Philippines. Philippine Journal of Banking and Finance, 15(1), 45-60.
39. Tulcanaza-Prieto, A., Coba-Sumino, M., & Ramos-Galarza, C. (2022). Customer satisfaction in the banking sector: The influence of corporate governance. International Journal of Financial Studies, 10(3), 56.
40. Verma, N., & Kaur, M. (2023). Examining the relationship among customer experience, bank image, and trust: A Multichannel Banking perspective. Journal of Global Marketing, 36(2), 141–164. https://doi.org/10.1080/08911762.2023.2188508
41. Villanueva. (2021, November 2). https://www.pna.gov.ph/articles/1158568.

Exploring the Determinants and Benefits of Philippine Financial Reporting Standards for Small Entities Compliance in Quezon City

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Abstract

The adoption of standardized financial reporting frameworks, such as the Philippine Financial Reporting Standards (PFRS) for Small Entities, was crucial in ensuring compliance and transparency among small businesses. On January 1, 2019, the Philippine Financial Reporting Standard for Small Entities was adopted to simplify financial reporting for small businesses without imposing undue burdens. This research explored the determinants of PFRS compliance, specifically examining how awareness and understanding, government regulations, and organizational resources influenced compliance levels. Additionally, it analyzed how the level of compliance enhanced financial reporting quality, cost efficiency, and investment opportunities. Using descriptive-correlational research design and utilizing purposive and snowball sampling, data was collected from 343 small business owners, accountants, and auditors in Quezon City through an online survey. Findings revealed strong awareness and understanding of PFRS, supportive yet needing stronger enforcement of government regulations, and generally adequate organizational resources. Compliance with PFRS significantly improved financial reporting quality, cost efficiency, and investment opportunities. High compliance levels were linked to better financial transparency, operational efficiency, and business growth. These results highlighted the importance of regulatory support, resource allocation, and continuous education to sustain compliance and maximize the benefits of PFRS adoption for small entities in the Philippines.

Keywords: Philippine Financial Reporting Standards, Small Entities, PFRS Compliance

Introduction

In contemporary business landscapes, the adoption and effective implementation of standardized financial reporting frameworks, such as the International Financial Reporting Standards (IFRS) and their localized counterpart, the Philippine Financial Reporting Standards (PFRS), are of paramount importance. Recognizing the need for a unified accounting language, the Accounting Standards Council (ASC) in the Philippines introduced the PFRS in 2005, establishing them as the new Generally Accepted Accounting Principles (GAAP) for the preparation of corporate financial statements (De Villa & Venus C Ibarra, 2019). These PFRS are closely modeled after the globally recognized IFRS and International Accounting Standards (IAS) issued by the International Accounting Standards Board (IASB). As noted by (IFRS - View Jurisdiction, n.d.), the application of these standards is both mandated and permitted within the Philippine jurisdiction.

Several factors influence an entity's adherence to these standardized financial frameworks. A robust and supportive regulatory environment plays a crucial role in fostering IFRS compliance (Chala et al., 2023). Moreover, a comprehensive understanding of the prevailing regulatory stipulations significantly facilitates the compliance process (Francisco et al., 2022). Furthermore, organizations possessing adequate resources demonstrate a greater capacity to adapt to evolving regulatory requirements, thereby ensuring consistent adherence (Thomson Reuters Institute, 2023). The benefits of complying with these standards are substantial. As highlighted by (Akpan and Osatuyi, 2023) in the context of SMEs, the adoption of IFRS enhances the quality and comparability of financial reports, consequently bolstering investor confidence. This increased transparency can attract international investments and unlock opportunities within global markets. By diligently adhering to these standards, businesses can refine their financial reporting practices, improve operational efficiency, and navigate the complexities of financial reporting with enhanced accuracy and integrity, thereby aligning with governmental mandates and ultimately contributing to the strengthening of the national business environment and the promotion of economic growth (Palmer, 2024). The work of (Suozzi, 2024) underscores the role of these standards in improving adherence to financial regulations and ensuring alignment with local accounting practices in the Philippines, implying that non-compliance may hinder intended outcomes. Similarly, (W & PK, 2023) observed that the full adoption and implementation of IFRS for SMEs are critical for effectively monitoring the success of these enterprises in Namibia.

The Philippines employs a tiered system of financial reporting frameworks: the full Philippine Financial Reporting Standards (PFRS), which are aligned with IFRS and mandatory for publicly accountable entities; the Philippine Financial Reporting Standards for Small and Medium-Sized Entities (PFRS for SMEs), offering simplified reporting requirements for

SMEs; and the Philippine Financial Reporting Standards for Small Entities (PFRS for SEs), specifically developed for smaller As detailed by (PFRS for Small Entities: A Simplified Framework for Philippines' Micro and Small Businesses - Tax and Accounting Center, Inc., n.d.), entities meeting these stipulated criteria are required to utilize the PFRS for Small Entities for their financial statement preparation. However, certain exemptions exist, such as when the entity employs a functional currency other than the Philippine Peso (e.g., operating in the Philippines but reporting in USD), is part of a group or a branch of a foreign company using a different reporting standard, anticipates significantly exceeding the size thresholds in future periods, or is a liquidating concern applying a different reporting standard.

This study aims to investigate the various determinants influencing compliance with the PFRS for SEs and how this compliance impacts the entity. Furthermore, it explores how adherence to these newly adopted financial standards enhances, improves, and contributes to the benefits derived from their implementation.

Literature Review

The adoption of standardized financial reporting frameworks such as the Philippine Financial Reporting Standards (PFRS) is central to ensuring the transparency, reliability, and comparability of financial information, especially in emerging economies. As emphasized in global financial literature, such adoption fosters market integration, attracts investments, and aligns with international reporting expectations (White & Buchanan, 2019; Arrafi, 2024). In the Philippines, where small entities significantly contribute to economic development but often operate with limited infrastructure, compliance with PFRS for Small Entities (PFRS for SEs) becomes a strategic necessity (Apat et al., 2019; Nurunnabi, 2021). This literature review explores the primary determinants of PFRS compliance—namely, Awareness and Understanding, Government Regulations, and Organizational Resources—and evaluates the implications of compliance on Financial Reporting Quality, Cost Efficiency, and Investment Opportunities. Additionally, the review highlights the mediating role of PFRS compliance in linking these determinants to organizational outcomes.

Awareness and Understanding of PFRS

A foundational determinant of PFRS compliance is the level of awareness and comprehension among stakeholders, particularly accounting and finance personnel. Knowledge of regulatory standards directly influences compliance levels and overall reporting quality. Studies have shown that familiarity with standards enhances business growth and regulatory alignment, especially among Philippine SMEs (World Bank, 2022; Francisco et al., 2022). In smaller enterprises, where accounting roles may not be formally established, leadership involvement becomes pivotal. When top management promotes a compliance-oriented culture, the organization is more likely to invest in training and foster employee engagement in learning activities (Akpan & Osatuyi, n.d.; PwC, 2021).

Organizational learning theory underscores the importance of knowledge transfer in the effective application of financial standards (Polzer et al., 2021; Sadaka, 2022). However, access to training remains a challenge for small enterprises in developing economies (Janie et al., 2020). Empirical evidence supports the assertion that ongoing professional development and structured training significantly enhance the accuracy of financial reporting and reduce incidences of non-compliance (Mensah, 2020; Wulandari et al., 2020). Creating a learning-centric environment equips personnel to adapt to evolving standards and fosters sustainable compliance (McKenna, 2023).

Role of Government Regulations

Regulatory frameworks established by government institutions are instrumental in enforcing adherence to financial reporting standards. Institutional theory posits that organizations align with regulatory requirements to preserve legitimacy and operational continuity (Francisco et al., 2020; Institutional Theory in Accounting, n.d.). In the Philippine context, the Securities and Exchange Commission (SEC) has a critical role in mandating and monitoring compliance with PFRS, thereby safeguarding investor interests and promoting a stable business ecosystem (Caclini, 2021; Sadaka, 2022).

Empirical findings reveal that robust enforcement mechanisms correlate with higher compliance levels (Sappor et al., 2023). For instance, regulatory incentives such as tax benefits for compliant firms have been noted to motivate small businesses to adopt formal standards (Alm et al., 2023). However, inconsistent enforcement, as observed in some Middle Eastern countries, often undermines the efficacy of these frameworks (Sawalqa & Qtish, 2021). Therefore, effective regulation must be accompanied by clear guidelines, accessible training, and systematic audits to foster compliance (Apat et al., 2019). While regulations provide structure, excessive administrative burdens may deter small businesses from full compliance; thus, policymakers must balance regulatory rigor with cost efficiency (Jabagat, 2021).

Organizational Resources

Compliance with PFRS substantially enhances the quality of financial reporting by improving the accuracy, consistency, and transparency of financial information. High-quality financial reports are essential for informed decision-making and reinforce

investor confidence (Mensah, 2020; Nurunnabi, 2021; Arianpoor & Sahoor, 2022). Entities adhering to international standards are less susceptible to earnings manipulation due to reduced accounting discretion (Sadaka, 2022).

For SMEs seeking expansion, financial transparency serves as a signal of stability and reliability, increasing their attractiveness to potential investors. Empirical studies show that high-quality reporting, driven by PFRS adoption, reduces the cost of debt and supports strategic decisions (Borromeo et al., 2024; Hellman et al., 2022). Further, standardization simplifies cross-border comparisons, facilitating international investment (Hoti & Krasniqi, 2022; Hameedi et al., 2021).

Financial Reporting Quality

Repurchase intention is a key predictor of consumer behavior, with consumers more likely to repurchase if they perceive the benefits (quality, convenience) outweigh the sacrifices (cost, health risks) (Fahlevi & Nurmahdi, 2020). Positive attitudes toward products also enhance repurchase likelihood (Sharma, 2023). Perceived value, satisfaction, and emotional fulfillment contribute to repeat purchases, while perceived risk negatively impacts repurchase intentions (Kumar & Kashyap, 2022).

Price and tax policies influence repurchase behavior, particularly for e-cigarettes. Research shows that price hikes and tax increases reduce consumption, with a $1.00 tax increase leading to a 12.2% drop in usage (Diaz et al., 2023). Price reductions can increase use among current smokers (Minami & Teo, 2019), while higher prices discourage new users, especially younger individuals (Chaloupka et al., 2019).

Cost Efficiency

Though initial implementation of PFRS may be resource-intensive, it fosters cost efficiency in the long term. Standardized practices reduce redundancies and enable streamlined reporting processes, which contribute to operational savings (Apat et al., 2019; Janie et al., 2020). As organizations become more adept at compliance, the associated costs decline, making it a sustainable practice (Sadaka, 2022).

By enhancing efficiency, small entities can reallocate resources toward strategic initiatives, boosting competitiveness and growth potential (Wulandari et al., 2020). Avoidance of penalties and access to incentives further improve financial positioning. Ultimately, PFRS compliance enables organizations to achieve regulatory alignment while optimizing internal resource utilization.

Investment Opportunities

Standardized and transparent financial reporting is critical for enhancing the investment potential of small businesses. PFRS compliance fosters investor trust and facilitates access to capital by improving the perceived reliability of financial statements (Sappor et al., 2023; Arianpoor & Sahoor, 2022). Small firms that meet high reporting standards are more likely to attract institutional investors and establish partnerships.

Moreover, transparent financial disclosures minimize information asymmetry, a key concern for lenders and investors alike. Compliance strengthens financial credibility, improving access to credit and broadening financing opportunities (Bakr, 2020; Hoti & Krasniqi, 2022). This, in turn, contributes to the overall financial development of the SME sector in the Philippines.

PFRS Compliance as a Mediating Variable

PFRS compliance plays a mediating role between the core determinants awareness, regulation, and resources and the associated benefits of financial reporting quality, cost efficiency, and investment readiness. High levels of compliance intensify the positive impact of these determinants, enhancing transparency and reliability (Polzer et al., 2021; Suozzi, 2024). Empirical research indicates that entities with robust compliance frameworks achieve superior outcomes in governance, financial reporting, and stakeholder trust (Gardi et al., 2023; Bracci et al., 2021).

Partial compliance often results in suboptimal benefits; therefore, the degree of compliance or compliance intensity emerges as a critical success factor. Full adherence not only meets regulatory expectations but also builds organizational legitimacy and investor appeal (Ferrer, 2019). Case studies from financial institutions in Iraq underscore the advantages of complete IFRS compliance in achieving transparency and stakeholder confidence (Mohammed & Al-Mashhadani, 2022; Mbir et al., 2020; Tawiah & Boolaky, 2019).

Methodology

This study utilized a descriptive-correlational research design and employed a structured survey questionnaire developed and refined to suit the objectives of the study on the determinants and benefits of Philippine Financial Reporting Standards for Small Entities (PFRS for SEs) compliance. The instrument was developed based on a synthesis of prior studies and adapted to fit the local context, guided by expert validation and pilot testing. It specifically captured constructs related to awareness, government regulation, organizational resources, and benefits such as financial reporting quality, cost efficiency, and investment opportunities.The structured questionnaire contained 35 close-ended items measured using a 4-point Likert scale, ranging

from (1) Strongly Disagree to (4) Strongly Agree. This tool has been widely utilized in accounting and compliance research, ensuring relevance and reliability (Placido et al., 2024; Zyoud et al., 2024). Data were collected from 342 qualified respondents, consisting of business owners, accountants, and auditors from small entities operating within Quezon City, Philippines. The sample size was computed using the Raosoft calculator, based on a 95% confidence level and a 5.1% margin of error, from a population of 4,463 registered small businesses provided by the Quezon City Business Permit and Licensing Office (BPLO). The survey was deployed online using Google Forms to increase accessibility and reach. A combination of purposive and snowball sampling techniques was employed to identify eligible participants. Purposive sampling was used to ensure that respondents had direct knowledge and experience of financial reporting under PFRS for SEs, while snowball sampling extended participation through referrals within professional networks (Kumar, 2011). The relationship between the determinants and perceived benefits of PFRS compliance was analyzed using Partial Least Squares – Structural Equation Modeling (PLS-SEM). This method is appropriate for examining multivariate causal relationships, including both direct and indirect effects, especially with latent variables (Ramli et al., 2019). Specifically, PLS-SEM was used to assess the influence of determinants on PFRS compliance, and how compliance impacts financial reporting quality, cost efficiency, and investment opportunities. Reliability of the instrument was assessed using Cronbach's Alpha, with an acceptable threshold set at $\alpha \geq 0.70$ to indicate internal consistency (Forero, 2024). Content validity was ensured through expert review by accounting professionals, confirming alignment with study constructs. Throughout the data collection process, ethical research protocols were strictly followed. Informed consent was obtained, participants were assured of confidentiality and voluntary participation, and data were handled securely in compliance with ethical standards for human subject research (Resnik, 2020; Kadam, 2020).

Results and Discussion

Demographic Characteristics of Respondents

Table 1 shows the demographic profile of respondents. The majority of respondents were female (49.6%), followed closely by males (38.5%). Accountants (39.4%) comprised the largest professional group, followed by owners (35.6%) and auditors (25.1%). The civil status and duration of PFRS use varied, with a significant portion (32.9%) having used PFRS for two years, and 25.4% only beginning usage in the past year. This indicates a gradual but ongoing adoption trend among small entities.

Table 1. Demographic Characteristics of the Respondents

Demographic	Category	Frequency	Percentage
Sex	Female	170	49.6%
	Male	132	38.5%
	Prefer not to say	41	12%
		343	**100%**
Position	Owner	122	35.6%
	Accountant	135	39.4%
	Auditor	86	25.1%
		343	**100%**
Civil Status	Single	140	40.8%
	Married	151	44%
	Prefer not to Say	52	15.2%
		343	**100%**
Duration of Use	1 year	85	24.8%
	2 years	113	32.9%
	3 years	35	10.2%
	4 years	16	4.7%
	5 years	7	2.0%
	Less than a year	87	25.4%
		343	**100%**

Demographic	Category	Frequency	Percentage
Number of years Operating	Less than 1 year	25	7.3%
	More than 1 year but less than 3 years	73	21.3%
	More than 3 years but less than 5 years	118	34.4%
	More than 5 years	127	37.0%
		343	**100%**

Model Fit and Quality Indices of the Emerging Model

The model fit and quality indices of the proposed structural equation model, offering a comprehensive assessment of its statistical reliability, explanatory capability, and robustness. The results confirm that the model satisfies all required goodness-of-fit criteria, thus validating its adequacy for further structural equation modeling (SEM) analysis. The Average Path Coefficient (APC = 0.528, $p < 0.001$) indicates statistically significant relationships among constructs, affirming the model's capacity to capture the dynamics of PFRS compliance and its financial implications. Furthermore, the model explains a substantial proportion of variance in the dependent variables, as shown by the Average R-Squared (ARS = 0.588, $p < 0.001$) and Average Adjusted R-Squared (AARS = 0.586, $p < 0.001$). These results highlight the model's strong explanatory power in linking compliance determinants to outcomes such as reporting quality, cost efficiency, and investment opportunities.

Multicollinearity was assessed using the Average Block Variance Inflation Factor (AVIF = 1.664) and Average Full Collinearity VIF (AFVIF = 2.680), both of which fall below the recommended threshold of 3.3. This confirms the absence of multicollinearity concerns and indicates that the independent variables contribute distinctively to the overall model structure. In terms of overall model fit, the Tenenhaus Goodness-of-Fit index (GoF = 0.616) surpasses the large effect threshold (GoF > 0.36), signifying a well-fitting model. Additional indices—including Simpson's Paradox Ratio (SPR = 1.000), R-Squared Contribution Ratio (RSCR = 1.000), and Statistical Suppression Ratio (SSR = 1.000)—all reach the ideal benchmark of 1.000, suggesting that the model is free from paradoxical inconsistencies and suppression effects.

Collectively, these indicators confirm that the proposed model exhibits high structural validity, strong predictive capability, and minimal risk of estimation bias. The significant path coefficients and quality indices reinforce the conclusion that compliance with PFRS significantly contributes to enhancing the financial reporting quality, cost management, and investment readiness of small entities.

Table 1. Model Fit and Quality Indices.

Measure	Estimate	Threshold	Interpretation
Average path coefficient (APC)	0.528, p <0.001	p < .05	Significant
Average R-squared (ARS)	0.588, p <0.001	p < .05	Significant
Average adjusted R-squared (AARS)	0.586, p <0.001	p < .05	Significant
Average block VIF (AVIF)	1.664	< 3.3	Ideally
Average full collinearity VIF (AFVIF)	2.68	< 3.3	Ideally
Tenenhaus GOF (GoF)	0.616	< .36	Large
Simpson's paradox ratio (SPR)	1	1	Ideally
R-squared contribution ratio (RSCR)	1	1	Ideally
Statistical suppression ratio (SSR)	1	1	Ideally

Emerging Model

The emerging structural model of PFRS compliance among small entities, developed using Partial Least Squares – Structural Equation Modeling (PLS-SEM). The model delineates the direct relationships between key antecedents - Awareness and Understanding, Government Regulations, and Organizational Resources—and the Level of Compliance (LC). Furthermore, it captures the downstream effects of compliance on three critical financial outcomes: Financial Reporting Quality (FRQB), Cost Efficiency (COB), and Investment Opportunities (IOB).

The results underscore that Awareness and Organizational Resources exert significant and positive effects on compliance behavior, indicating that internal capabilities and knowledge are pivotal in achieving adherence to the Philippine Financial Reporting Standards (PFRS) for Small Entities. In contrast, Government Regulations did not yield a statistically significant path coefficient, suggesting that external regulatory pressure alone is insufficient to induce consistent compliance.

Subsequently, the Level of Compliance (LC) was found to significantly influence all three outcome variables. The strong path coefficients from LC to FRQB, COB, and IOB confirm that compliance leads to tangible organizational benefits, such as enhanced financial reporting transparency, operational cost savings, and expanded investment prospects. These findings reinforce the strategic importance of institutionalizing PFRS compliance practices as a mechanism for improving long-term financial performance and organizational sustainability among small entities.

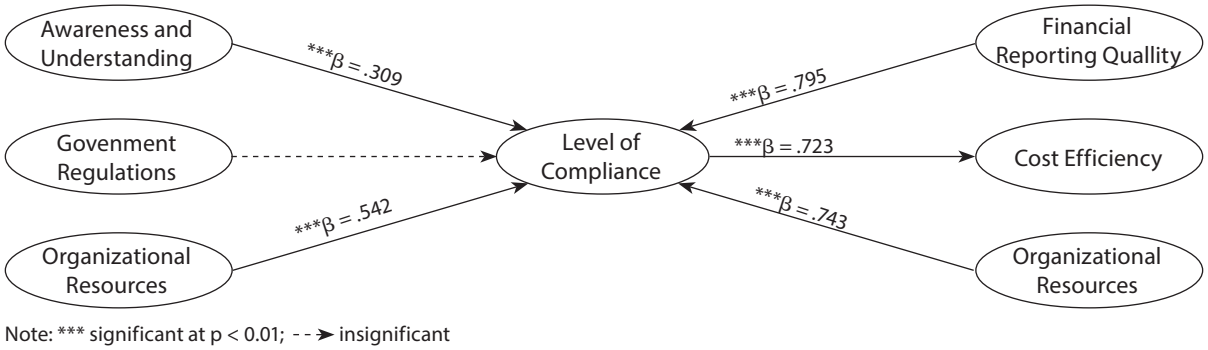


Fig. 1. Emerging model of the study.

Path Analysis and Hypothesis

The results of the path analysis, presenting the direct effects of the identified determinants—namely, Awareness and Understanding, Government Regulations, and Organizational Resources—on the level of compliance with Philippine Financial Reporting Standards (PFRS) for Small Entities (LC). Furthermore, it illustrates the subsequent influence of compliance on three key financial outcomes: Financial Reporting Quality (FRQB), Cost Efficiency (CEB), and Investment Opportunities (IOB). The findings reveal that five out of the six hypothesized relationships were statistically significant ($p < 0.05$), with corresponding effect sizes ranging from medium to large, in accordance with Cohen's (1988) classification. These results underscore the significant role of internal organizational enablers, particularly awareness and resource availability, in promoting PFRS compliance and enhancing financial performance among small entities.

H1: Awareness and understanding of PFRS influence compliance among small entities. Hypothesis H1 posits that Awareness and Understanding (AUD) significantly influence the Level of Compliance (LC) with PFRS. The results yielded a path coefficient of $\beta = 0.309$ ($p < 0.001$) and an effect size of $f^2 = 0.210$, indicating a medium effect. This finding supports H1 and affirms that familiarity with PFRS requirements significantly enhances compliance among small entities. The results are consistent with prior literature emphasizing the importance of financial literacy and standards comprehension (World Bank, 2022; Sadaka, 2022). According to Polzer et al. (2021), knowledge within organizations spreads through internal learning mechanisms, improving the accuracy and reliability of financial reporting. However, in developing economies, access to training remains limited, resulting in knowledge gaps and potential reporting inaccuracies (Janie et al., 2020). Mensah (2020) and Wulandari et al. (2020) found that continuous professional development programs reduce reporting errors and strengthen adherence to standards. A culture of ongoing education enables organizations to adapt effectively to evolving standards (McKenna, 2023), emphasizing that awareness and understanding are foundational to compliance.

H2: Government regulations towards PFRS influence compliance among small entities. Hypothesis H2 examines the impact of Government Regulations (GRD) on LC. The analysis revealed a path coefficient of $\beta = 0.054$ ($p = 0.156$) and an effect size of $f^2 = 0.019$, signifying an insignificant relationship. Thus, H2 is not supported, suggesting that external regulatory frameworks alone do not significantly influence PFRS compliance in small entities. While regulatory bodies such as the Philippine Securities and Exchange Commission (SEC) provide oversight (Caclini, 2021), compliance is more strongly driven by internal organizational factors. Literature suggests that regulations, while essential, must be accompanied by adequate guidance, monitoring, and accessibility to updates (Sappor et al., 2023; Apat et al., 2019). In contexts where enforcement is inconsistent, such as in Jordan and Lebanon, compliance remains uneven (Sawalqa & Qtish, 2021). Moreover, regulatory compliance can present an administrative burden for small businesses, potentially impeding operational efficiency (Jabagat, 2021). Hence, government regulation must be complemented by supportive mechanisms to be effective.

H3: Organizational resources influence PFRS compliance among small entities. Hypothesis H3 evaluates the effect of Organizational Resources (ORD) on LC. The path coefficient of $\beta = 0.542$ ($p < 0.001$) with an effect size of $f^2 = 0.416$ indicates a large effect, providing strong support for H3. This finding underscores the critical role of financial, technological, and human resources in facilitating PFRS compliance. The findings align with resource-based theory, which posits that firms with greater

resource endowments are better positioned to address complex regulatory requirements (Liu & Zhang, 2023). Small entities often face constraints in hiring skilled professionals or acquiring adequate accounting systems, hindering compliance efforts (Janie et al., 2020). However, investments in software, infrastructure, and personnel training enhance compliance capabilities (Suozzi, 2024; Thomson Reuters Institute, 2023). While outsourcing may offer short-term support (Mensah, 2020; Accounting Insights Team, 2024), the development of internal competencies is critical for sustainable compliance (DeSimone et al., 2020). These results highlight the significance of resource availability in promoting adherence to financial reporting standards.

H4: PFRS compliance enhances the quality of financial reporting in small entities. Hypothesis H4 explores the effect of LC on Financial Reporting Quality (FRQB). The results show a strong relationship ($\beta = 0.795$, $p < 0.001$; $f^2 = 0.632$), confirming H4. Higher levels of compliance are associated with enhanced accuracy, reliability, and transparency of financial statements. These findings support prior research demonstrating that standardized reporting frameworks reduce information asymmetry and bolster stakeholder confidence (Mensah, 2020; Nurunnabi, 2021). Adherence to PFRS mitigates earnings manipulation and improves consistency in financial disclosures (Sadaka, 2022). For small businesses, credible financial reporting is a key indicator of trustworthiness (Borromeo et al., 2024). Moreover, clear and accurate reports reduce borrowing costs and facilitate cross-border investment (Hellman et al., 2022; Hoti & Krasniqi, 2022). As such, compliance directly contributes to the quality and comparability of financial information (Hameedi et al., 2021).

H5: PFRS compliance contributes to increased cost efficiency in small entities. Hypothesis H5 investigates the impact of LC on Cost Efficiency (COB), with results indicating a significant and strong relationship ($\beta = 0.723$, $p < 0.001$; $f^2 = 0.523$). Thus, H5 is supported, demonstrating that PFRS compliance contributes to the streamlining of financial processes and reduction of operational costs. Compliance enables standardization of accounting procedures, minimizing redundancy and enhancing resource utilization (Apat et al., 2019; Janie et al., 2020). As organizations institutionalize compliance, they benefit from long-term cost savings and operational efficiencies (Sadaka, 2022; Wulandari et al., 2020). These efficiencies allow small businesses to redirect resources toward strategic initiatives, such as expansion or innovation, while also avoiding non-compliance penalties. Compliance thus plays a dual role in regulatory adherence and cost optimization.

H6: Compliance with PFRS improves investment opportunities for small entities. Hypothesis H6 assesses the relationship between LC and Investment Opportunities (IOB), yielding a path coefficient of $\beta = 0.743$ ($p < 0.001$) and a large effect size of $f^2 = 0.552$. The result confirms H6, indicating that compliance significantly enhances business attractiveness to potential investors and financial institutions. Transparency and accuracy in financial reporting are essential indicators of business credibility, particularly for small entities seeking capital access (Sadaka, 2022; Suozzi, 2024). High compliance levels signal sound financial management, thereby increasing the likelihood of investment and funding (Hoti & Krasniqi, 2022). Adhering to global standards like PFRS also aligns small businesses with international expectations, facilitating cross-border investor engagement.

Table 2. Hypothesis Testing - with one segment analysis (Direct Effects)

Path	Path Coefficients	P -values	Effect Sizes (f^2)	Effect Size Interpretation	Description	Decision
H1: AUD → LC	0.309	<0.001	0.21	Medium	Significant	H1 is supported
H2: GRD → LC	0.054	0.156	0.019	NA	Insignificant	H2 is not supported
H3: ORD → LC	0.542	<0.001	0.416	Large	Significant	H3 is supported
H4: LC → FRQB	0.795	<0.001	0.632	Large	Significant	H4 is supported
H5: LC → COB	0.723	<0.001	0.523	Large	Significant	H5 is supported
H6: LC → IOB	0.743	<0.001	0.552	Large	Significant	H6 is supported

Conclusions

The study conclusively establishes that awareness and understanding of PFRS and the availability of organizational resources are crucial internal drivers for compliance among small entities. Conversely, government regulations alone do not significantly influence PFRS compliance, suggesting that external pressure needs to be complemented by other supportive mechanisms. Importantly, a higher level of compliance consistently leads to enhanced financial reporting quality, improved cost efficiency, and expanded investment opportunities. These findings highlight the strategic importance of PFRS compliance in fostering long-term financial performance and sustainability for small entities.

Methodologically, this study successfully demonstrated the effectiveness of Partial Least Squares - Structural Equation Modeling (PLS-SEM) in analyzing the complex interplay of factors influencing PFRS compliance and its financial outcomes.

For future research, it would be beneficial to expand frameworks by incorporating mediating and moderating variables such as the level of enforcement of government regulations, the specific nature of training and educational initiatives, or the perception of benefits derived from compliance. Additionally, longitudinal analyses could provide richer insights into how PFRS compliance practices evolve over time and their sustained impact on small entities' financial health.

From a practical standpoint, the findings underscore the need for small entities to prioritize investments in awareness and understanding of PFRS, perhaps through targeted training programs and readily accessible educational materials. Equally vital is the allocation of sufficient organizational resources, including financial, technological, and human capital, to facilitate robust compliance efforts. Policymakers should consider developing more supportive mechanisms to complement existing government regulations, such as offering accessible guidance, monitoring, and regular updates, rather than solely relying on regulatory mandates. These integrated efforts can significantly contribute to better PFRS adherence, ultimately leading to improved financial reporting, greater cost efficiencies, and enhanced access to investment opportunities for small entities.

REFERENCES

1. AccountingInsights Team. (2024). The Role and Impact of International Accounting Standards - Accounting Insights. Accounting Insights. https://accountinginsights.org/the-role-and-impact-of-international-accounting-standards/
2. Akpan, J., & Osatuyi. (n.d.). International Financial Reporting Standards (IFRS) and Small and Medium-sized Enterprises (SMEs): Assessing the impact of IFRS adoption on SMEs. World Journal of Finance and Investment Research E-ISSN, 7(4). https://doi.org/10.56201/wjfir.v7.no4.2023.pg34.56
3. Alm, J., Gerbrands, P., & Kirchler, E. (2023). Using responsive regulation to reduce tax base erosion. Regulation & Governance. https://doi.org/10.1111/rego.12359
4. Analyzing the recording practices in accounting of micro and small businesses: The case of the Philippines. (n.d.). https://scholar.google.com/citations?view_op=view_citation&hl=en&user=4I3CZHUAAAAJ&citation_for_view=4I3CZHUAAAAJ:u5HHmVD_uO8C
5. Apat, E. J. C., De Villa, J. P. D., & Ibarra, V. C. (2019). EVOLUTION, IMPLEMENTATION AND EFFECTS OF INTERNATIONAL ACCOUNTING STANDARDS IN THE PHILIPPINES. Malaysian E Commerce Journal, 3(2), 06–11. https://doi.org/10.26480/mecj.02.2019.06.11
6. Arrafi, M. O. (2024, August 27). A comprehensive guide to IFRS in the Philippines. HashMicro Philippine Blog. https://www.hashmicro.com/ph/blog/ifrs/
7. Arianpoor, A., & Sahoor, Z. (2022). The impact of business strategy and annual report readability on financial reporting quality. Journal of Asia Business Studies, 17(3), 598–616. https://doi.org/10.1108/jabs-10-2021-0439
8. Bakr, S. A. (2020). The adoption of the International Financial Reporting Standard for Small and Medium-sized Entities (IFRS for SMEs) in the Kingdom of Saudi Arabia (Doctoral thesis, Royal Holloway, University of London). https://pure.royalholloway.ac.uk/ws/portalfiles/portal/36192561/Bakr_Sara_Final_Draft_Thesis.pdf
9. Bongalonta, M., & Bongalonta, M. (2023). Financial Reporting Practices of Non-Profit Organizations (NPOS) in the municipality of Bulan. https://mrj.sorsu.edu.ph/index.php/journal/article/view/13
10. Borromeo A.S., Cervantes R.J., Sumicad E.H. Jr., (2024). Analyzing the Recording Practices in Accounting of Micro and Small Businesses https://cyberleninka.ru/article/n/analyzing-the-recording-practices-in-accounting-of-micro-and-small-businesses-the-case-of-the-philippines
11. Bracci, E., Biondi, L., & Kastberg, G. (2021). Citizen-centered financial reporting: The preparers' perspective. Financial Accountability & Management, 37(4), 379-396. https://doi.org/10.1111/faam.12298
12. Caclini, M. (2021). Comparative analysis of financial reporting of selected small and medium enterprises (SMEs) in Baguio City: Basis for a financial planning model. Journal of Contemporary Issues in Business and Government, 27(3), 2413-2425. https://cibgp.com/au/index.php/1323-6903/article/view/1760
13. Case study: Adoption of international public sector accounting standards in the Philippines. (2019). IFAC. https://www.ifac.org/publications/case-study-adoption-international-public-sector-accounting-standards-philippines
14. Chala, et., (2023). The impact of international financial reporting standards on information asymmetry. In Review of Economics and Finance (Vol. 21, pp. 2655–2644). https://refpress.org/wp-content/uploads/2024/03/Chala_REF.pdf
15. Cohen, J. (1988). Statistical power analysis for the behavioral sciences (2nd ed.). Routledge.
16. Crowther, D., & Lancaster, G. (2021). Research methods (2nd ed.). Routledge. https://doi.org/10.4324/9780080943442
17. DeSimone, et., (2020). Correlates of internal audit function involvement in sustainability audits. Journal of Management & Governance, 25(2), 561–591. https://doi.org/10.1007/s10997-020-09511-3

18. Devi, B., Sharma, R., & Singh, A. (2023). Application of correlational research design in nursing and medical research. ResearchGate. https://www.researchgate.net/profile/Barkha-Devi-2/publication/368958213_APPLICATION_OF_CORRELATIONAL_RESEARCH_DESIGN_IN_NURSING_AND_MEDICAL_RESEARCH/links/6401a0330cf1030a566a0022/APPLICATION-OF-CORRELATIONAL-RESEARCH-DESIGN-IN-NURSING-AND-MEDICAL-RESEARCH.pdf?origin=publication_detail
19. De Villa et., (2019). Evolution, Implementation And Effects Of International Accounting Standards In The Philippines. ideas.repec.org. https://ideas.repec.org/a/zib/zbmecj/v3y2019i2p06-11.html
20. Dalupan, D. (2020). Philippine Financial Reporting Standard for Small Entities. https://www.dmdcpa.com.ph/wp-content/uploads/2019/01/PFRS-for-Small-Entities-full-text-1.pdf
21. Ferrer, R. C. (2019). Impact of full IFRS, accounting standards for SMEs and company demographics on firms' return on asset and return on equity using panel data regression. International Journal of Engineering and Advanced Technology, 8(6 Special Issue), 589-601. https://doi.org/10.35940/ijeat.F1120.0886S19
22. Fisher, P.A. (2024). What is financial compliance? A guide for businesses. Fisher Accountants. https://www.fisheraccountants.com/what-is-financial-compliance/
23. Forero, C. G. (2024). Cronbach's Alpha. In Encyclopedia of Quality of Life and Well-Being Research (pp. 1-4). Springer. https://link.springer.com/referenceworkentry/10.1007/978-3-031-17299-1_622
24. Francisco, J. P. et., (2022). Firm-level impact of regulatory compliance costs on small business growth. International Review of Entrepreneurship, 20(1), 1-26. https://aimleader.aim.edu/article/firm-level-impact-of-regulatory-compliance-costs-on-small-business-growth/
25. Francisco, J. P. et., (2020). Do regulatory compliance costs hinder growth among small and medium enterprises in the Philippines? SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3607837
26. Fornell, C., & Larcker, D. F. (1981). Evaluating Structural Equation Models with Unobservable Variables and Measurement Error. Journal of Marketing Research, 18(1), 39–50. https://doi.org/10.1177/002224378101800104
27. Gardi, B., Aga, M., & Abdullah, N. N. (2023). Corporate Governance and Financial Reporting Quality: The mediation role of IFRS. Sustainability, 15(13), 9869. https://doi.org/10.3390/su15139869
28. Hughes, J. L. et., (2022). INVITED EDITORIAL: Guidance for Researchers When using Inclusive Demographic Questions for Surveys: Improved and Updated questions. Psi Chi Journal of Psychological Research, 27(4), 232–255. https://doi.org/10.24839/2325-7342.jn27.4.232
29. Hair, J. F., Risher, J. J., Sarstedt, M., & Ringle, C. M. (2018). When to use and how to report the results of PLS-SEM. European Business Review, 31(1), 2–24. https://doi.org/10.1108/ebr-11-2018-0203
30. Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). Multivariate data analysis (8th ed.). Cengage Learning.
31. Hameedi, K. S. et., (2021). Financial Performance Reporting, IFRS Implementation, and Accounting Information: Evidence from Iraqi Banking Sector. Journal of Asian Finance Economics and Business, 8(3), 1083–1094. https://doi.org/10.13106/jafeb.2021.vol8.no3.1083
32. Hellman, N. et., (2022). The Impact of an IFRS for SMEs-based Standard on Financial Reporting Properties and Cost of Debt Financing: Evidence from Swedish Private Firms. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4123576
33. Homan, A. C. et., (2020). Leading diversity: Towards a theory of functional leadership in diverse teams. Journal of Applied Psychology, 105(10), 1101–1128. https://doi.org/10.1037/apl0000482
34. Hoti, A., & Krasniqi, L. (2022). Impact of international financial reporting standards adoption on the perception of investors to invest in small-to-medium enterprise adopting transparency in disclosure policies. International Journal of Systems Assurance Engineering and Management, 13(S1), 506–515. https://doi.org/10.1007/s13198-021-01501-
35. ICLEI. (2022). Circular Quezon City Bringing the Circular Economy to Quezon City's Food System. https://circulars.iclei.org/wp-content/uploads/2022/09/CL4C_-Mini-Roadmap_Quezon-City.pdf

36. IFRS - View Jurisdiction. (n.d.). (C) Copyright 2023. https://www.ifrs.org/use-around-the-world/use-of-ifrs-standards-by-jurisdiction/view-jurisdiction/philippines/
37. International Accounting Standards Board. (2010). Conceptual Framework for Financial Reporting. https://www.ifrs.org/content/dam/ifrs/publications/pdf-standards/english/2021/issued/part-a/conceptual-framework-for-financial-reporting.pdf
38. J, A. a. B., & AlSondos, I. a. A. (2020). Operational Efficiency and the Adoption of Accounting Information System (AIS): A Comprehensive review of the banking sectors. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3670735
39. Jabagat, C. R. (2021). Cost of regulatory compliance for small and medium enterprises in the Philippines. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.3878642
40. Janie, D. N. A. et., (2020). THE IMPLEMENTATION OF INDONESIAN ACCOUNTING STANDARDS FOR MICRO, SMALL AND MEDIUM ENTITIES. Humanities & Social Sciences Reviews, 8(1), 383–388. https://doi.org/10.18510/hssr.2020.8149
41. Kadam, R. A. (2020). Informed consent process: A brief review. *Perspectives in Clinical Research, 11*(2), 49-54. https://doi.org/10.4103/picr.PICR_95_19
42. KPMG. (2020). Insights overview 2020. KPMG. https://assets.kpmg.com/content/dam/kpmg/es/pdf/2020/09/insights-overview-2020.pdf
43. Kumar, R. (2011). Research Methodology: A Step-by-Step Guide for Beginners (3rd ed.). SAGE. http://www.sociology.kpi.ua/wp-content/uploads/2014/06/Ranjit_Kumar-Research_Methodology_A_Step-by-Step_G.pdf
44. Liu, Z., & Zhang, N. (2023). The productivity effect of digital financial reporting. Review of Accounting Studies. https://doi.org/10.1007/s11142-022-09737-6 of Small and Medium Enterprises - Legitimacy Theory Perspectives. Kelaniya Journal of Management, 10(1), 43-64. https://doi.org/10.4038/kjm.v1011.7666
45. Mbir, D. E. G. et., (2020). IFRS compliance, corporate governance and financial reporting quality of GSE-listed non-financial firms. Cogent Business & Management, 7(1), 1759856. https://doi.org/10.1080/23311975.2020.1759856
46. Mendoza, & Johnson. (2022). The Impact of Simplified Financial Reporting on SME Performance. Example Journal.
47. Mensah, E. (2020). The effect of IFRS adoption on financial reporting quality: evidence from listed manufacturing firms in Ghana. Economic Research-Ekonomska Istraživanja, 34(1), 2890–2905. https://doi.org/10.1080/1331677x.2020.1860109
48. Mohammed, S. S., & Al-Mashhadani, B. N. A. (2022). Measuring the level application of IFRS based on the asset-liability approach in a sample of banks listed on the Iraq Stock Exchange. Tikrit Journal of Administrative and Economic Sciences, 18(57), 2-3. https://doi.org/10.25130/tjaes.18.57.2.3
49. Nurunnabi, M. (2021). International Financial Reporting Standards Implementation: a Global Experience. In Emerald Publishing Limited eBooks. https://doi.org/10.1108/9781801174404
50. Pahilagao, H. S., FRSC Member, & AOSSG. (2022). PFRS FOR SMALL ENTITIES.
51. Polzer, T. et., (2021). Adoption of the International Public Sector Accounting Standards in emerging economies and low-income countries: a structured literature review. Journal of Public Budgeting Accounting & Financial Management, 35(3), 309–332. https://doi.org/10.1108/jpbafm-01-2021-0016
52. Roszkowska-Menkes, M. (2023). Institutional Theory. In Springer eBooks (pp. 1983–1990). https://doi.org/10.1007/978-3-031-25984-5_389
53. Sadaka, S. F. (2022). Challenges of IFRS implementation in emerging economies: The case of Lebanon. Accounting History, 27(4), 497–523. https://doi.org/10.1177/10323732221093820
54. Sappor, P. et., (2023). The adoption of IFRS for SMEs in the northern sector of Ghana: A case of structural equation modeling. Cogent Business & Management, 10(1). https://doi.org/10.1080/23311975.2023.2180840

55. Sawalqa, F. a. A., & Qtish, A. (2021). IAS/IFRS in Jordan: Adoption, Implementation and Determinants. Universal Journal of Accounting and Finance. https://doi.org/10.13189/ujaf.2021.090213

56. Suozzi, T. (2024). IFRS for SMEs (Small and medium-sized enterprises). Suozziforny. https://suozziforny.com/ifrs-for-smes/#:~:text=Implementation%20Strategies%20for%20IFRS%20for%20SMEs,-To%20introduce%20IFRS&text=This%20is%20a%20special%20set,and%20match%20global%20accounting%20ways.

57. Thomson Reuters Institute (2023). 2023 Risk & Compliance Report. https://www.thomsonreuters.com/en-us/posts/wp-content/uploads/sites/20/2023/10/2023-Risk-Compliance-Report.pdf

58. World Bank. (2022). IFRS for SMEs workshop. Centre for Financial Reporting Reform. Retrieved from https://cfrr.worldbank.org/news/ifrs-smes-workshop

59. Zyoud, M. M. et., (2024). Quantitative research methods: Maximizing benefits, addressing limitations, and advancing methodological frontiers. ResearchGate. https://www.researchgate.net/profile/Munther-Zyoud/publication/379655878_Quantitative_Research_Methods_Maximizing_Benefits_Addressing_Limitations_and_Advancing_Methodological_Frontiers/links/6613d0082034097c54ff2e6b/Quantitative-Research-Methods-Maximizing-Benefits-Addressing-Limitations-and-Advancing-Methodological-Frontiers.pdf

Customer Perceptions of CSR Dimensions and Satisfaction: Exploring the Mediating Role of Perceived Value in Philippine Banks within Quezon City

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Abstract

Corporate Social Responsibility (CSR) is important for the banking sector. Trust, transparency, and long-term relationships with stakeholders are crucial. In the Philippine banking sector, CSR has become significant; therefore, banks have started to implement initiatives such as social practices, environmental practices, and economic practices that benefit society and improve their clients' perceptions and experiences. This research explores whether perceived value mediates the relationship between Corporate Social Responsibility (CSR) dimensions and customer satisfaction among chosen Philippine banks in Quezon City. Quantitative, descriptive correlational research design was used with online surveys to obtain data from 391 bank depositors. PLS-SEM and descriptive statistics were used to test the outcomes. The results indicated that CSR activities had a moderate effect on perceived value, with economic and environmental CSR practices being more salient than social practices. In addition, Economic practices had the strongest positive influence on perceived value, influencing customer satisfaction. The mediation analysis indicated that perceived value partially mediated the relationship between CSR dimensions and customer satisfaction for economic and environmental practices but not for social practices. It showed that Social CSR initiatives had a lesser effect on customer satisfaction than economic and environmental practices. The study emphasized the importance of enhancing CSR initiatives, especially in economic and environmental, which could have a positive influence on perceived value and customer satisfaction, building loyalty and retention. However, CSR alone was insufficient for achieving high satisfaction, though, because the entire banking experience must satisfy or exceed expectations. The study made significant contributions to understanding how CSR initiatives might enhance business sustainability and customer perceptions in the banking industry.

Keywords: *Corporate Social Responsibility (CSR), Customer Satisfaction, Social Practices, Economic Practices, Environmental Practices, Banking Sector*

Introduction

In contemporary business practice, Corporate Social Responsibility (CSR) has emerged as a fundamental component of strategic management, particularly in sectors where stakeholder trust, transparency, and long-term engagement are crucial. The banking industry, due to its pivotal role in economic development and its reliance on customer confidence, has increasingly integrated CSR into its core operations (Fatima & Elbanna, 2023; Gallardo-Vázquez et al., 2020). CSR encompasses a spectrum of initiatives aimed at addressing environmental, social, and economic concerns, enabling firms to pursue ethical commitments while simultaneously enhancing competitive positioning.

Globally, the banking sector has harnessed CSR to fulfill societal expectations and regulatory standards, while also fostering sustainable growth. In the Philippine context, financial institutions have become instrumental in driving social and economic progress through CSR-driven programs. These initiatives range from financial inclusion for underserved communities, green financing for environmental sustainability, to transparent and ethical banking practices (Santiago, 2023). Such efforts aim not only to address pressing societal needs but also to shape customer perceptions and influence their experiences with the bank.

However, the efficacy of CSR efforts is contingent upon customer perception. The success of CSR initiatives does not solely rely on their implementation but significantly on how they are interpreted and valued by customers. Research indicates that customers' evaluations of a bank's social, environmental, and economic contributions play a critical role in influencing satisfaction (Velte, 2022). In this context, perceived value—defined as the customer's subjective appraisal of the benefits received relative to the efforts or costs incurred—emerges as a vital mediating variable (Sekhon & Kathuria, 2020). A high perceived value enhances customer satisfaction by reinforcing the belief that the bank's CSR actions are meaningful and beneficial.

Customer satisfaction is a strategic outcome that significantly influences customer loyalty, advocacy, and long-term retention in the banking sector. In an increasingly competitive financial landscape, cultivating satisfaction through CSR becomes a strategic imperative. Thus, understanding how CSR initiatives influence customer satisfaction through the lens of perceived value can offer actionable insights for banks to align their strategies with customer expectations and social objectives.

Despite the rising prominence of CSR in the Philippine banking industry, a notable research gap remains. Most existing studies focus on internal organizational outcomes—such as operational efficiency or regulatory compliance—thereby overlooking the customer-centric implications of CSR (Agyei et al., 2021; Islam et al., 2020). Furthermore, there is a paucity of empirical investigations exploring the mediating role of perceived value in the CSR-customer satisfaction relationship, particularly within the context of emerging economies like the Philippines. This study addresses this gap by examining how customers perceive CSR initiatives in top banks located in Quezon City, and how these perceptions affect customer satisfaction via perceived value.

By adopting a customer-focused perspective, this study contributes to the literature by offering empirical insights into how CSR can be strategically leveraged to enhance customer satisfaction. It also provides a contextual understanding of CSR in a developing country setting, where financial institutions play a vital role in sustainable development. This research is expected to inform bank managers, policymakers, and scholars on how CSR strategies can be optimized to meet both business and societal objectives.

Literature Review

Corporate Social Responsibility (CSR) in the Banking Sector

Corporate Social Responsibility (CSR) is recognized as a vital component in the business strategies of banking institutions, particularly in developing countries like the Philippines. CSR encompasses three primary dimensions—social, environmental, and economic responsibility—which are conceptualized through the Triple Bottom Line (TBL) framework. This framework highlights how businesses must balance profit-making activities with actions that benefit society and the environment (Elkington, 1994).

In the banking sector, CSR is increasingly adopted not just for compliance or philanthropy but as a strategy for competitive advantage. As noted by Hamidi and Worthington (2023), CSR has become a critical element in enhancing business performance. Oliinyk et al. (2023) found that CSR initiatives in banking contribute to strengthening institutional reputation, fostering customer loyalty, and promoting long-term success.

Social CSR Practices and Customer Perception

The social dimension of CSR includes activities aimed at improving the quality of life for employees, customers, and the community. These initiatives involve fair labor practices, community engagement, education, and inclusive financial access. Socially responsible banks foster positive stakeholder relationships by promoting social equity and contributing to societal development (Fatima & Elbanna, 2023; Sekhon & Kathuria, 2020).

According to Gallardo-Vázquez et al. (2019), socially responsible actions significantly enhance corporate image and build customer trust. Schoeneborn et al. (2020) emphasized the role of transparent communication in strengthening the ethical image of organizations. These initiatives align with the "People" dimension of the TBL framework and demonstrate the ability of CSR to improve public perception and customer satisfaction (Wirba, 2024).

Environmental CSR and Sustainable Banking

The environmental aspect of CSR involves organizational efforts to mitigate negative impacts on the natural environment. This includes practices such as waste reduction, energy efficiency, and promoting green finance. Ikram et al. (2020) highlighted how banks are shifting toward digital banking and paperless operations as part of their commitment to sustainability. Velte (2022) found that strong environmental CSR performance is positively related to corporate reputation.

Gallardo-Vázquez et al. (2020) further emphasized that environmental sustainability efforts not only fulfill ethical obligations but also enhance the perceived value of banking services. Sarfraz et al. (2023) and Deb et al. (2024) stressed that such initiatives are increasingly expected by consumers who prioritize environmental responsibility in their financial decisions.

Economic CSR and Ethical Financial Practices

Economic responsibility refers to the ethical use of financial resources, ensuring transparency, and fostering financial inclusion. This involves fair lending practices, corporate transparency, and efforts to reduce financial disparities. Santiago (2023) noted that ethical economic practices contribute significantly to customer trust and institutional credibility.

Ikram et al. (2020) emphasized the importance of financial disclosures and ethical investment strategies in building long-term customer relationships. Sekhon and Kathuria (2020) identified that economic CSR enhances customer satisfaction through equitable service delivery and transparent pricing mechanisms. Tang et al. (2022) highlighted how economic responsibility, as part of the "Profit" dimension of the TBL, reinforces sustainable business models.

Perceived Value as a Mediating Construct

Perceived value, defined as the customer's evaluation of the utility of a product or service based on what is received and what is given, is a crucial mediator in the relationship between CSR and customer satisfaction. Gallardo-Vázquez and Lizcano-Álvarez

(2020) found that CSR actions that align with customer values enhance their perception of service value. This, in turn, positively affects customer satisfaction.

Fatima and Elbanna (2023) emphasized that perceived value links CSR to emotional and cognitive customer responses. Velte (2022) and Mehnaz et al. (2024) provided evidence that perceived value plays a mediating role across all three CSR dimensions. Khairy et al. (2023) and Stauropoulou et al. (2023) reinforced that customers evaluate the overall worth of banking services not just through functional aspects but through the perceived societal and environmental value generated by the organization.

Customer Satisfaction and Expectancy-Disconfirmation Theory (EDT)

Customer satisfaction is central to business success and has been widely examined using the Expectancy-Disconfirmation Theory (EDT). This theory posits that satisfaction occurs when perceived performance meets or exceeds customer expectations (Oliver, 1980; Prananta et al., 2023).

Lee et al. (2022) highlighted that CSR practices influence customer expectations and subsequent satisfaction through the lens of disconfirmation. Ikram et al. (2020) also supported the idea that customers who perceive CSR efforts as exceeding expectations are more likely to be satisfied. Gallardo-Vázquez et al. (2020) and Jahan et al. (2020) confirmed the strong link between customer satisfaction, positive brand perception, and customer loyalty.

Theoretical Integration: Triple Bottom Line and Expectancy-Disconfirmation Theory

The integration of TBL and EDT provides a robust framework to understand the impact of CSR on customer satisfaction. TBL categorizes CSR into social, environmental, and economic responsibilities, while EDT explains the psychological mechanism through which these initiatives influence satisfaction outcomes.

As Velte (2022) pointed out, CSR actions that exceed customer expectations lead to positive disconfirmation and increased satisfaction. Yesitadewi and Widodo (2024) confirmed that when perceived value mediates this process, it strengthens the relationship between CSR and customer satisfaction. This theoretical integration supports a comprehensive understanding of CSR's role in driving sustainable customer relationships in the banking sector.

Methodology

This study employed a descriptive correlational research design to investigate the relationship between Corporate Social Responsibility (CSR) dimensions—social, environmental, and economic—perceived value, and customer satisfaction in the Philippine banking sector, with a focus on top banks in Quezon City. A sample of 385 bank clients was selected using Cochran's formula and recruited through snowball sampling. Data were gathered via a structured survey instrument utilizing a four-point Likert scale, which underwent expert validation and pilot testing to ensure clarity and reliability, yielding Cronbach's Alpha values ranging from 0.853 to 0.925 across constructs. The survey was distributed digitally over a four-week period, and participation was voluntary and confidential. Descriptive statistics summarized the respondents' characteristics, while Structural Equation Modeling (SEM) was employed to test hypothesized relationships among CSR dimensions, perceived value, and customer satisfaction, including the mediating effect of perceived value.

The demographic characteristics of the 391 respondents provide a comprehensive view of the study population, offering valuable context for interpreting perceptions of CSR and customer satisfaction in the Philippine banking sector. A significant portion of the respondents were clients of BDO (46.0%), BPI (34.3%), and Metrobank (19.7%), reflecting the market dominance of these institutions in Quezon City. Regarding banking tenure, 48.6% of respondents had been clients for 1–2 years, while 51.4% had banking relationships exceeding two years, allowing for an analysis of both short- and long-term customer perspectives. Notably, the gender distribution showed a strong female majority (71.6%), contrasting with the male-dominated samples reported by Ikram et al. (2020) and Khairy et al. (2023), but consistent with the findings of Leclercq-Machado et al. (2022), where women also comprised the majority. The age profile was predominantly young, with 48.8% aged 18–24 and 24.6% aged 25–34, aligning with prior research by Ikram et al. and Leclercq-Machado et al., which emphasized the involvement of younger demographics in CSR-related issues.

In terms of civil status, 49.9% of respondents were married, and 46.3% were single, paralleling the marital distribution observed in Khairy et al. (2023). Educational attainment was relatively high, with 50.9% reporting college-level education and 39.1% holding a bachelor's degree, diverging from Khairy et al.'s study that featured a greater proportion of PhD holders. Employment status revealed that 50.4% of respondents were employed and 38.4% were students, in contrast with the findings of Leclercq-Machado et al. (2022), where students formed the largest segment. These demographic insights highlight the strong representation of young, educated, and financially active individuals, providing a solid foundation for understanding how these groups perceive and respond to CSR initiatives in the banking industry. This context is critical in evaluating how perceived value and customer satisfaction are shaped by CSR engagement within Philippine banks.

Results and Discussion

Demographic Characteristics of Respondents

Table 1. Demographic characteristics of the respondents (n=391)

Demographic	Category	Frequency	Percentage
Banks	BDO	180	46%
	BPI	134	34.30%
	Metrobank	77	19.70%
Years	1-2 Years	190	48.60%
	2 Years and above	201	51.40%
Sex	Female	280	71.60%
	Male	100	25.60%
	Prefer not to say	11	2.80%
Civil Status	Single	181	46.30%
	Married	195	49.90%
	Prefer not to say	15	3.80%
Age	18-24	191	48.80%
	25-34	96	24.60%
	35-44	58	14.80%
	45-54	33	8.40%
	55 and above	13	3.30%
Educational Attainment	Highschool Graduate	25	6.40%
	Vocational Training	7	1.80%
	College Level	199	50.90%
	Bachelor's Degree	153	39.10%
	More than 10 Years	32	23.70%
	Full-time	125	92.59%
	Part-time	4	2.96%
	Contractual	6	4.45%

Model Fit and Quality Indices and Emerging Model

Table 2 discusses the model evaluation indicates a statistically significant and well-fitting structural equation model. The Average Path Coefficient (APC = 0.205, $p < 0.001$) confirms meaningful relationships among the constructs—CSR dimensions, perceived value, and customer satisfaction. The Average R-squared (ARS = 0.381, $p < 0.001$) and Average Adjusted R-squared (AARS = 0.376, $p < 0.001$) demonstrate moderate to strong explanatory power, indicating that the independent variables account for a substantial proportion of variance in the dependent variables. The Tenenhaus Goodness-of-Fit index (GoF = 0.500) exceeds the recommended threshold of 0.36, suggesting a large model fit and reinforcing the model's overall predictive relevance.

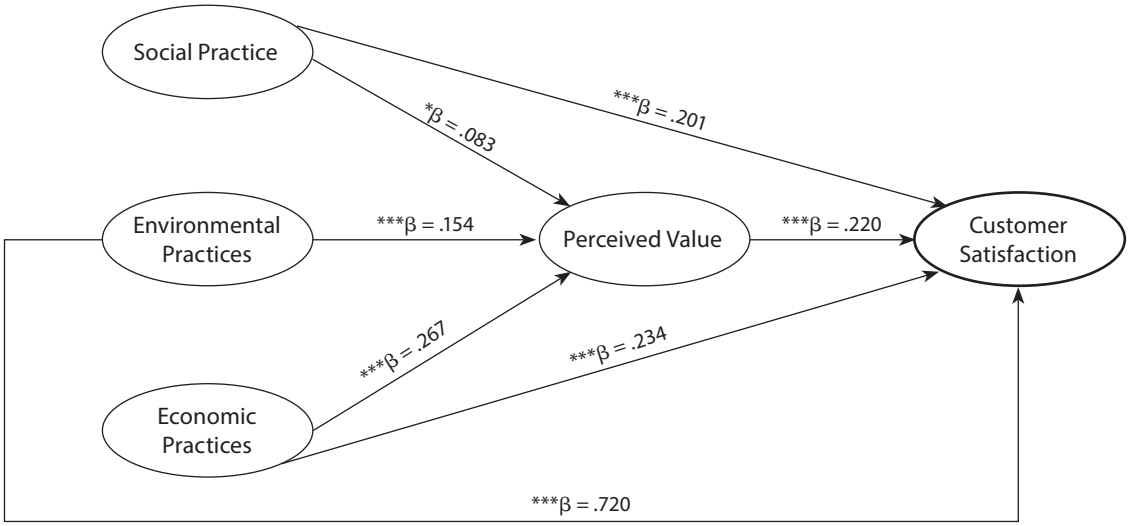
Further analysis confirmed the effects of both endogenous and exogenous variables. Endogenous constructs such as customer satisfaction and perceived value were significantly influenced by CSR dimensions—specifically environmental, ethical, and community practices—supporting the hypothesis that CSR engagement enhances emotional attachment and brand perceptions (Amegbe et al., 2021). Exogenous constructs, particularly CSR and brand equity, exhibited significant direct effects on customer outcomes including loyalty and brand love. No misspecified paths were detected, confirming the structural integrity and theoretical alignment of the model.

Multicollinearity was assessed through the Average Block Variance Inflation Factor (AVIF = 1.253) and Average Full Collinearity VIF (AFVIF = 1.681), both below the 3.3 threshold, indicating minimal redundancy among predictor variables. Additional diagnostics, including the Simpson's Paradox Ratio (SPR = 1.000), R-squared Contribution Ratio (RSCR = 1.000), and Statistical Suppression Ratio (SSR = 1.000), suggest the absence of suppression or estimation bias. These results support

the model's robustness, validity, and reliability, confirming its suitability for hypothesis testing. Overall, the findings underscore the strategic value of CSR in enhancing perceived value and customer satisfaction, thereby strengthening long-term customer relationships within the banking sector.

Table 2. Model Fit and Quality Indices of the Emerging Model

Measure	Estimate	Threshold	Interpretation
Average path coefficient (APC)	0.205, p <0.001	p < .05	Significant
Average R-squared (ARS)	0.381, p <0.001	p < .05	Significant
Average adjusted R-squared (AARS)	0.376, p <0.001	p < .05	Significant
Average block VIF (AVIF)	1.253	< 3.3	Ideally
Average full collinearity VIF (AFVIF)	1.681	< 3.3	Ideally
Tenenhaus GOF (GoF)	0.5	> .36	Large
Simpson's paradox ratio (SPR)	1	1	Ideally
R-squared contribution ratio (RSCR)	1	1	Ideally
Statistical suppression ratio (SSR)	1	1	Ideally



Note: ***significabt at p < 0.001; *significant at p < 0.05

Fig. 1. Emerging model of the study.

Path Analysis and Hypothesis Testing

Table 3 presents the direct effects of Corporate Social Responsibility (CSR) dimensions—social, environmental, and economic—on perceived value, as well as the influence of perceived value on customer satisfaction. All hypothesized relationships (H1 to H4) are statistically significant ($p < 0.05$), with effect sizes ranging from small to medium as per Cohen's (1988) guidelines. These results underscore the nuanced contributions of different CSR initiatives to customers' perceptions and satisfaction, offering critical insights for both theoretical development and practical applications in the banking sector.

Hypothesis H1 investigates the effect of social CSR practices on perceived value. The results indicate a statistically significant but small effect ($\beta = 0.083$, $p = 0.0494$, $f^2 = 0.021$), suggesting that while community engagement and financial literacy programs are acknowledged by customers, they may not be strongly associated with tangible value in the banking experience. Although H1 is supported, the minimal effect size implies that social CSR efforts alone may be insufficient in shaping strong perceptions of value unless coupled with other impactful practices.

Hypothesis H2 evaluates the relationship between environmental CSR practices and perceived value, yielding a stronger and statistically significant effect ($\beta = 0.154$, $p < 0.001$, $f^2 = 0.039$). This medium effect size supports the notion that green financing, eco-friendly initiatives, and environmental sustainability measures resonate with customers and contribute to a more favorable evaluation of banking services. H2 is therefore supported, emphasizing the growing consumer awareness of environmental responsibility in the financial sector.

Hypothesis H3 tests the impact of economic CSR practices on perceived value and reveals the strongest relationship among the three CSR dimensions ($\beta = 0.267$, $p < 0.001$, $f^2 = 0.093$). The medium effect size underscores the importance of economic responsibility—such as financial transparency, ethical lending, and support for local businesses—in enhancing trust and the perceived benefits of banking services. These results suggest that customers place significant value on ethical and financially responsible behavior, thereby supporting H3.

Hypothesis H4 examines the influence of perceived value on customer satisfaction. The model shows a statistically significant and moderately strong relationship ($\beta = 0.130$, $p = 0.005$, $f^2 = 0.045$), indicating that customers who perceive high value in their banks' CSR initiatives tend to report higher levels of satisfaction. This finding affirms the mediating role of perceived value and confirms that it is a key determinant of customer satisfaction. As such, H4 is supported and highlights the importance of communicating CSR benefits effectively.

Collectively, the findings indicate that economic and environmental CSR practices exert a more substantial influence on perceived value compared to social practices. These insights are aligned with Stauropoulou et al. (2023), who similarly found significant effects of CSR on customer perceptions and satisfaction. While some literature, such as Gallardo-Vázquez et al. (2020), notes variation in CSR impact depending on industry and consumer awareness, the current results provide strong empirical support for the strategic role of CSR in banking. Furthermore, Lee et al. (2021) argue that economic CSR may exert a direct and strong influence on satisfaction, particularly in contexts where value and ethical conduct are paramount. These findings reinforce the strategic imperative for banks to integrate and highlight their economic and environmental CSR efforts to drive customer satisfaction and long-term loyalty.

Table 3. Hypothesis Testing - with one segment analysis (DIRECT EFFECT)

Path	Path Coefficients (β)	P -values	Effect Sizes (f^2)	Effect Size Interpretation	Description	Decision
H1: Soc → Per Val	0.083	0.0494	0.021	Small	Significant	H1 is supported
H2: Envi → Per Val	0.154	<0.001	0.039	Small	Significant	H2 is supported
H3: Eco → Per Val	0.267	<0.001	0.093	Small	Significant	H3 is supported
H4: Per Val → Cus Sat	0.13	0.005	0.045	Small	Significant	H4 is supported

Note: Job Satisfaction (JS); Compensation (C); Job Security (JSY); Organizational Culture (OC); Work-life Balance (WLB); Employee Retention (ER)

Table 4 presents the mediation analysis assessing the indirect effects of Corporate Social Responsibility (CSR) dimensions—social, environmental, and economic—on customer satisfaction via perceived value. The analysis confirms that perceived value serves as a significant mediating variable in some CSR-satisfaction pathways, with statistically significant p-values ($p < 0.05$) validating the hypothesized mediation effects. The results reveal a mix of partial mediation, no mediation, and varying effect strengths, indicating that perceived value plays a differential role across CSR dimensions.

Hypothesis H5 tests whether perceived value mediates the relationship between social CSR practices and customer satisfaction. The analysis shows a significant total effect ($\beta = 0.266$, $p < 0.001$), direct effect ($\beta = 0.201$, $p < 0.001$), and indirect effect ($\beta = 0.065$, $p = 0.034$), confirming partial mediation. These results suggest that while social initiatives such as community involvement and outreach programs have a direct influence on customer satisfaction, their effect is amplified when customers perceive added value from these efforts. Thus, H5 is supported, highlighting that the perceived value reinforces the satisfaction generated by social CSR practices.

Hypothesis H6 explores whether perceived value mediates the relationship between environmental CSR practices and customer satisfaction. The findings indicate a strong total effect ($\beta = 0.762$, $p < 0.001$) and direct effect ($\beta = 0.720$, $p < 0.001$), but the indirect effect is not statistically significant ($\beta = 0.042$, $p = 0.120$). This suggests that perceived value does not mediate the relationship between environmental CSR efforts and customer satisfaction, as customers appear to associate satisfaction with these initiatives independently of perceived value. Therefore, H6 is not supported, implying that sustainability initiatives affect satisfaction through other mechanisms rather than through perceived value.

Hypothesis H7 investigates the mediating role of perceived value in the relationship between economic CSR practices and customer satisfaction. The analysis reveals a total effect ($\beta = 0.319$, $p < 0.001$), a direct effect ($\beta = 0.234$, $p < 0.001$), and a statistically significant indirect effect ($\beta = 0.085$, $p = 0.008$), indicating partial mediation. This confirms that customers not only appreciate the direct impact of ethical financial practices, transparency, and responsible lending, but also derive satisfaction

when these efforts enhance perceived value. Consequently, H7 is supported, affirming that perceived value partially transmits the influence of economic CSR practices on customer satisfaction.

Overall, the mediation analysis supports the conceptual model wherein perceived value serves as a critical link between CSR initiatives and customer satisfaction. These results align with the findings of Sekhon et al. (2020), who emphasized the indirect pathways through which CSR efforts influence financial and behavioral outcomes. While the effects vary across CSR dimensions, the significant mediation effects affirm that perceived value is an essential mechanism by which CSR influences customer attitudes. From a managerial standpoint, these insights stress the importance for banks and service-oriented firms to clearly communicate the value created by CSR activities to reinforce customer satisfaction and loyalty. Social and economic CSR initiatives, in particular, stand to gain from enhanced value perception, offering both emotional and functional benefits that strengthen customer relationships.

Table 4. Effect for the path with two segments (Mediation Analysis)

Path	Total Effect	P-value	Direct Effect	P-Value	Indirect Effect	P-Value	Description	Interpretation	Remark
H5: Soc → Per Val → Cus Sat	0.266	<0.001	0.201	<0.001	0.065	0.034	Significant	Partial Mediation	H5 is supported
H6: Envi → Per Val→ Cus Sat	0.762	<0.001	0.72	<0.001	0.042	0.12	Insignificant	No Mediation Effect	H6 is not supported
H7: Eco→ Per Val → Cus Sat	0.319	<0.001	0.234	<0.001	0.085	0.008	Significant	Partial Mediation	H6 is supported

Note: Social Practices (Soc); Environmental Practices (Envi); Economic Practices (Eco); Perceived Value (Per Val); Customer Satisfaction (Cus Sat)

This study contributes to the evolving body of Corporate Social Responsibility (CSR) literature in emerging markets, particularly within the Philippine banking sector. By integrating the Triple Bottom Line (TBL) framework (Elkington, 1994) and the Expectancy-Disconfirmation Theory (EDT) (Oliver, 1980), the study provides robust empirical evidence on how CSR dimensions influence customer satisfaction, both directly and through the mediating role of perceived value. The inclusion of economic, environmental, and social CSR dimensions allows for a nuanced understanding of their individual effects on customer perceptions and satisfaction. The findings affirm that CSR practices are not merely symbolic gestures but yield tangible outcomes when aligned with consumer expectations and values.

In terms of direct effects, the results support **H1**, **H2**, and **H3**, demonstrating that all three CSR dimensions significantly influence perceived value. However, the strength of the relationships varies. Economic CSR practices exhibited the strongest effect ($\beta = 0.267$, $p < 0.001$), followed by environmental CSR ($\beta = 0.154$, $p < 0.001$), and social CSR ($\beta = 0.083$, $p = 0.0494$). While social CSR was statistically significant, its effect size ($f^2 = 0.021$) was small, indicating that customers may view community initiatives as secondary to their personal banking experience. This supports findings from Gallardo-Vázquez et al. (2019), who argue that CSR activities must be seen as stakeholder-oriented and deliver direct, personal value. The weaker influence of social CSR suggests that such initiatives, while positively perceived (M = 3.18), are not closely tied to tangible service improvements. Hence, banks must better integrate social CSR into the core service experience to enhance its perceived utility.

With regard to customer satisfaction, **H4** was also supported ($\beta = 0.130$, $p = 0.005$), confirming that perceived value has a significant and positive effect. This aligns with Expectancy-Disconfirmation Theory, which posits that customer satisfaction arises when perceived performance exceeds expectations (Lee et al., 2022; Velte, 2022). Although the mean score for perceived value was moderately high (M = 3.18), the lowest-rated item—"CSR improves my banking experience" (M = 3.14)—suggests that customers acknowledge CSR efforts but do not fully associate them with daily banking touchpoints. This highlights a gap between CSR communication and customer experience, calling for better integration of CSR into service delivery processes.

The mediation analysis further enhances the theoretical understanding of how CSR influences customer satisfaction via perceived value. **H5** and **H7** were supported, confirming that perceived value partially mediates the relationships between social and economic CSR practices and customer satisfaction. For **H5**, the indirect effect ($\beta = 0.065$, $p = 0.034$) was significant alongside a strong direct effect ($\beta = 0.201$, $p < 0.001$), indicating partial mediation. This suggests that while customers appreciate social CSR efforts, their satisfaction is strengthened when these efforts translate into perceived value. Similarly, **H7** revealed partial mediation, with an indirect effect of $\beta = 0.085$ ($p = 0.008$), confirming that perceived value plays a critical role in translating economic CSR into satisfaction.

Conversely, **H6** was not supported. Although environmental CSR had a significant total effect ($\beta = 0.762$, $p < 0.001$) and direct effect ($\beta = 0.720$, $p < 0.001$), the indirect effect via perceived value ($\beta = 0.042$, $p = 0.120$) was not statistically significant.

This suggests that customers directly associate environmental initiatives with satisfaction, bypassing perceived value as a cognitive filter. This finding implies that environmentally responsible practices may resonate more at an emotional or reputational level, particularly given the increasing salience of global sustainability narratives in financial services. Therefore, banks should continue to invest in environmental CSR but may need to explore alternative pathways—beyond perceived value—to enhance its impact.

The empirical results lend full support to **H1**, **H2**, **H3**, **H4**, **H5**, and **H7**, with only **H6** not supported. These findings collectively suggest that the role of perceived value as a mediator is not universal; rather, it is contingent on the nature of the CSR practice. This challenges one-size-fits-all assumptions in CSR strategy and highlights the need for differentiated approaches. Economic and environmental initiatives that are quantifiable and transparent tend to elicit stronger cognitive evaluations of value, while social initiatives may require better framing and integration into customer-centric services.

These insights offer practical implications for banks seeking to optimize their CSR strategies. Economic CSR should prioritize transparency, fair lending, and ethical financial practices—areas that directly influence perceived value and satisfaction. Environmental CSR, particularly through digital banking, paperless transactions, and green financing, remains impactful and should be communicated clearly as part of the bank's core mission. Social CSR efforts, while well-intentioned, must evolve from symbolic community engagement to initiatives that deliver personalized value to customers. Communication strategies must be refined to bridge the awareness-to-value gap, especially in economic CSR where customer perceptions remain underwhelmed.

While the study provides strong theoretical and empirical contributions, it is not without limitations. The sample is geographically limited to Quezon City, which may constrain the generalizability of the findings. Additionally, the use of self-reported data introduces potential biases, and the cross-sectional design limits causal inferences. Future research could address these limitations by employing longitudinal designs, incorporating qualitative methods (e.g., interviews or focus groups), and exploring digital CSR initiatives given the rising prominence of fintech in sustainable banking.

In conclusion, this study affirms that CSR, particularly when aligned with customer expectations and perceived benefits, can serve as a strategic lever to enhance customer satisfaction in banking. The varying influence of CSR dimensions and the partial mediation role of perceived value underscore the need for nuanced, evidence-based CSR strategies tailored to customer insights and expectations.

Conclusions

This study explored the mediating role of perceived value in the relationship between Corporate Social Responsibility (CSR) dimensions—economic, environmental, and social—and customer satisfaction within the Philippine banking sector. Grounded in the Triple Bottom Line (TBL) framework and Expectancy-Disconfirmation Theory (EDT), the research produced the following key insights:

CSR efforts moderately influence perceived value, with an overall mean rating of 3.18. While social and environmental initiatives were slightly more recognized than economic practices, all CSR dimensions were rated similarly. This suggests that while CSR is visible, banks can improve communication and integration of these efforts to enhance customer perceptions.

Economic and environmental CSR practices are key drivers of perceived value, with economic CSR exerting the strongest influence. These results underscore the importance of transparency, financial responsibility, and sustainable operations in shaping favorable customer perceptions and improving the banking experience.

CSR positively impacts customer satisfaction, particularly in areas where customers perceive tangible benefits. While CSR is not the sole determinant of satisfaction, it supports customer retention and brand loyalty, especially when CSR efforts align with customer values and expectations.

Perceived value partially mediates the relationship between CSR and customer satisfaction, specifically for economic and environmental CSR. This confirms that CSR enhances customer satisfaction primarily when it translates into a perceived benefit, reinforcing the need to link CSR with service value.

Social CSR practices have a weaker effect on perceived value and satisfaction compared to economic and environmental CSR. This finding suggests a need for banks to reassess their social initiatives, ensuring these are not only symbolic but also offer measurable value aligned with customer expectations.

These conclusions offer practical implications for bank managers and policymakers. Strategic emphasis should be placed on CSR initiatives that deliver observable benefits, align with core services, and effectively communicate. Future studies are encouraged to expand the geographic scope, apply longitudinal designs, and explore the evolving role of CSR in digital banking environments.

REFERENCES

1. Agyei, P. M., Sun, S., & Ampomah, K. (2021). Corporate social responsibility and financial performance nexus: Mediating effect of intellectual capital in Ghana's banking sector. Business and Economic Horizons, 17(1), 1-17.
2. Amegbe, M., Quartey, E. T., & Adjei, R. (2021). The effects of corporate social responsibility on customer satisfaction and brand loyalty in the banking sector: Evidence from Ghana. Journal of Marketing Management, 9(1), 1–19.
3. Cohen, J. (1988).1 Statistical power analysis for the behavioral sciences (2nd ed.). Lawrence Erlbaum Associates.
4. Deb, S. K., Roy, D. K., & Luthra, S. (2024). Examining the role of green banking practices in achieving sustainable development goals: An empirical study. Journal of Cleaner Production, 439, 140656.
5. Elkington, J. (1994). Cannibals with forks: The triple bottom line of 21st century business. Environmental Quality Management, 15(4), 37–51.
6. Fatima, S., & Elbanna, S. (2023). Corporate social responsibility and firm performance: The mediating role of brand reputation. Journal of Business Research, 158, 113645.
7. Gallardo-Vázquez, D., & Lizcano-Álvarez, J. L. (2020). The effect of corporate social responsibility on customer loyalty in the banking industry: The mediating role of perceived value. Sustainability, 12(11), 4752.
8. Gallardo-Vázquez, D., Lizcano-Álvarez, J. L., & Castilla-Polo, F. (2019). The influence of corporate social responsibility on corporate image and customer trust in the banking sector. Journal of Global Responsibility, 10(2), 173–191.
9. Gallardo-Vázquez, D., Lizcano-Álvarez, J. L., & Sánchez-Hernández, M. I. (2020). The mediating effect of perceived value on the relationship between corporate social responsibility and customer satisfaction in the banking industry. Sustainability, 12(23), 10072.
10. Hamidi, A., & Worthington, S. (2023). The impact of corporate social responsibility on firm performance in the banking sector: A systematic review. International Journal of Organizational Analysis, 31(2), 488–507.
11. Ikram, M., Sroufe, R., & Shah, M. (2020). Green banking and environmental sustainability: An empirical analysis of the moderating role of digital transformation. Sustainable Production and Consumption, 24, 219–230.
12. Islam, M. A., Jabir, S., & Hossain, M. M. (2020). Corporate social responsibility and financial performance in the banking sector of Bangladesh: The mediating role of reputation. Journal of Business and Social Review in Emerging Economies, 6(1), 21–34.
13. Jahan, S., Rahman, M. M., & Khan, M. A. (2020). The impact of corporate social responsibility on customer loyalty: The mediating role of customer satisfaction in the banking sector of Bangladesh. International Journal of Social Economics, 47(7), 897–912.
14. Khairy, R., Rizk, H., & El-Kashash, M. (2023). The impact of corporate social responsibility on customer satisfaction and loyalty in the banking industry: The mediating role of brand image. The Business & Management Review, 14(1), 143–155.
15. Leclercq-Machado, M., Barrientos-Báez, A., & Calero-Guzmán, M. (2022). Social responsibility and customer satisfaction in the banking sector: The moderating role of customer gender. Sustainability, 14(17), 10928.
16. Lee, S. H., Park, S. E., & Kim, J. H. (2021). The effects of economic corporate social responsibility on customer loyalty in the banking industry: The mediating role of corporate image and customer trust. Journal of Korea Service Management Society, 22(3), 119–138.
17. Lee, T. Y., Chen, Y. C., & Chen, H. H. (2022). The influence of corporate social responsibility on customer satisfaction in the banking industry: An application of expectancy-disconfirmation theory. Sustainability, 14(9), 5600.
18. Mehnaz, S., Ullah, R., & Ali, N. (2024). The role of perceived value in mediating the relationship between corporate social responsibility and customer satisfaction in the banking sector. International Journal of Business and Management, 19(1), 1-15.

19. Oliver, R. L. (1980). A cognitive model of the antecedents and consequences of satisfaction decisions. Journal of Marketing Research, 17(4), 460–469.
20. Oliinyk, O., Shkarlet, S., & Shmygol, N. (2023). Corporate social responsibility in the banking sector: A systematic review of current trends and future directions. Problems and Perspectives in Management, 21(3), 209–220.
21. Prananta, A. D., Purnomo, B. S., & Gunardi, A. (2023). The effect of service quality, corporate image, and customer trust on customer satisfaction and loyalty in Islamic banking. Journal of Islamic Marketing, 14(5), 1318–1334.
22. Santiago, R. (2023). Ethical banking practices and customer trust in the Philippines. Philippine Journal of Business and Economics, 62(1), 1-18.
23. Sarfraz, M., Sarfraz, Z., & Hamid, M. (2023). Green banking practices and customer satisfaction: The mediating role of environmental awareness. Environmental Science and Pollution Research, 30(29), 74811–74823.
24. Schoeneborn, D., Morsing, M., & Kuhn, T. (2020). The constitutive dynamics of corporate social responsibility: A communication perspective. Business Ethics Quarterly, 30(3), 321–347.
25. Sekhon, H., & Kathuria, L. M. (2020). Corporate social responsibility and customer satisfaction: The mediating role of perceived value in the banking sector. Journal of Financial Services Marketing, 25(3-4), 108–119.
26. Sekhon, H., Kathuria, L. M., & Singh, J. (2020). Corporate social responsibility and customer loyalty: The mediating role of customer satisfaction and corporate image in the banking sector. Journal of Promotion Management, 26(3), 307–328.
27. Stauropoulou, S., Kourkouta, E., & Kourtidis, P. (2023). The impact of corporate social responsibility on customer satisfaction and loyalty: A systematic review. Sustainability, 15(1), 589.
28. Tang, G., Tan, W., & Zhang, J. (2022). The impact of economic corporate social responsibility on sustainable development: Evidence from Chinese listed companies. Frontiers in Environmental Science, 10, 831671.
29. Velte, P. (2022). Meta-analyses on Corporate Social Responsibility (CSR): A literature review. Management Review Quarterly, 72(3), 627–675.
30. Velte, P. (2024). Sustainable board governance and sustainable supply chain reporting: European evidence. Journal of Strategy and Management, 17(2), 222-243.
31. What is customer satisfaction? Definition + importance. (2024, July 23). Zendesk. https://www.zendesk.com/blog/3-steps-achieving-customer-satisfaction-loyalty/
32. Wirba, A. V. (2024). Corporate social responsibility (CSR): The role of government in promoting CSR. Journal of the Knowledge Economy, 15(2), 7428-7454.
33. Worthington, S. L. (2023). The effect of environmental corporate social responsibility on customer retention: A mixed methods study on the mediating effect of customer behavior (Doctoral dissertation, University of Southern Queensland).
34. Yesitadewi, V. I., & Widodo, T. (2024). The influence of service quality, perceived value, and trust on customer loyalty via customer satisfaction in Deliveree Indonesia. Quality-Access to Success, 25(198), 103-115.

The Influence of Excise Tax Perceptions on Consumer Attitudes and Repurchase Intentions in Quezon City's Vapor Product Market

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Abstract

The increasing popularity of vapor products as alternatives to traditional tobacco has prompted significant public health and regulatory discourse, particularly concerning the associated health risks and high usage rates among young adults. In response, the Philippine government has implemented excise taxation to regulate usage and generate fiscal revenue. However, limited research exists on the effectiveness of these taxes in shaping consumer perceptions and influencing behavior. This study investigates how excise tax perceptions—namely fairness, affordability, awareness, and perceived government intent—affect consumer attitudes and repurchase intentions within Quezon City's vapor product market. Employing a descriptive-correlational design, data were collected from 385 vapor product users through snowball sampling conducted both online and in person, encompassing all six districts of Quezon City. Results revealed that fairness was not a significant determinant of consumer attitudes. In contrast, affordability and awareness showed a modest yet statistically significant influence. Notably, government intent emerged as the strongest predictor of consumer attitudes. Further analysis established a significant relationship between consumer attitudes and repurchase intentions, suggesting that despite price increases from excise taxes, consumption of vapor products remains resilient. The study contributes to existing literature by filling the gap in understanding the behavioral implications of excise tax perceptions, providing actionable insights for both policymakers aiming to implement effective public health measures and industry stakeholders seeking to adapt to a taxed market environment.

Keywords: Fairness, Affordability, Awareness, Government Intent, Consumers' Attitude, Repurchase Intention

Introduction

Smoking remains one of the most persistent public health issues worldwide. Cigarette smoking causes over eight million deaths annually, making the tobacco epidemic one of the most devastating health challenges in modern history (World Health Organization, 2023). The Philippines is also significantly affected, with over 112,000 Filipinos dying each year from tobacco-related diseases (Cordero, 2024).

As a response to the health hazards posed by conventional tobacco use, vapor products have emerged as alternatives. Promoted as tools for smoking cessation and less harmful methods of nicotine delivery, these products have gained global popularity (Famiglietti, 2021). It is estimated that there were 82 million vape users globally, including approximately 2.7 million users in the Philippines accounting for 3% of the global population of vapers.

However, mounting health concerns regarding the safety of vapor products have surfaced. In the United States, 2,807 cases of lung injury related to vaping were reported in 2020, resulting in 68 deaths (CDC, 2020). In the Philippines, the first vaping-related death was recorded in 2024, when a 22-year-old man died due to severe lung damage after two years of daily use (Untalan, 2024). Health authorities, including the WHO, the Philippine Department of Health, and specialists from The Medical City, have warned that e-cigarettes are not scientifically proven to aid in smoking cessation and may expose users to harmful chemicals (Sese & Guillermo, 2023).

These risks have prompted regulatory actions. Republic Act No. 11900, also known as the Vaporized Nicotine and Non-Nicotine Products Regulation Act, came into effect on July 25, 2022. The law aims to regulate the production, distribution, and sale of vapor products. However, the law has drawn criticism for lowering the minimum age of access from 21 to 18 and transferring regulatory oversight from the Department of Health to the Department of Trade and Industry—thereby potentially weakening product quality and safety standards (Sese & Guillermo, 2023).

One of the state's most effective tools to manage consumption of harmful products is through taxation. The affordability of disposable e-cigarettes, largely due to increased production and competitive pricing strategies, has led to heightened youth appeal. As a result, governments have introduced excise taxes to discourage consumption and promote public health (Diaz et al., 2023). Excise taxes increase product prices, thereby encouraging consumers to make healthier choices (Cnossen, 2023). Studies indicate that increased taxes on e-cigarettes correlate with reduced sales and changes in purchasing behavior, particularly

a decline in flavored e-cigarette use (Cotti et al., 2020; Jiang, 2022). Similarly, U.S. tax increases between 2019 and 2021 successfully offset rising incomes and made vapor products less affordable (Diaz et al., 2024).

This study centers on the impact of excise tax perceptions on consumer behavior within Quezon City, Philippines. Specifically, it investigates how perceptions regarding tax fairness, affordability, awareness, and government intent shape consumer attitudes and influence their repurchase intentions.

Literature Review

Vapor Products

The motivations behind vapor product consumption include factors such as perceived harm reduction (78.6%), flavor availability (78.3%), the ability to use e-cigarettes in smoke-free areas (77.4%), and mimicking traditional smoking (46.5%) (Harlow et al., 2022). In Quezon City, many young adults use disposable e-cigarettes out of curiosity or to reduce tobacco use, later developing habitual consumption (Cruz et al., 2024). Vape shops are the most common purchase locations in countries like Canada, the U.S., and England (Braak et al., 2020), with convenience stores and vape shops also prevalent in the Philippines (Institute for Global Tobacco Control, 2023).

Studies on vapor product characteristics show that 23% of e-liquids are nicotine-free, 49% contain freebase nicotine, and 28% use nicotine salts, with price increases linked to higher nicotine concentrations (Jiang et al., 2023). In the Philippines, 280 out of 315 vapor products collected featured flavor descriptors, with colors (18.08%) and fruits (15.07%) being the most common (Ackary et al., 2024). Additionally, a randomized clinical trial revealed that pod-based devices (55.7%) were the most popular (Leventhal et al., 2021). Regarding excise taxes, studies show that as taxes increase, e-cigarette consumption decreases, with a reported 9.7% reduction following tax implementation (Pesko et al., 2019). In Latin America, excise taxes have been shown to reduce consumption and generate health system revenue (Miracolo et al., 2021), while restrictions on availability and cost have motivated consumers to quit or reduce use (Palmer et al., 2022).

Buyer's Perception of Excise Tax

Fairness. Excise taxes are often justified as a fair economic measure that reflects the negative externalities of harmful products. One study argues that such taxes act as a corrective tool for market and individual failures by internalizing the social cost of consumption (Cnossen, 2023). This aligns with evidence that individuals' perceptions of tax fairness are shaped by the degree to which they perceive tangible benefits from taxation (Verguet et al., 2020). Although tobacco taxes are criticized as regressive, studies suggest that when paired with cessation support, they can achieve fairness goals more effectively.

Tax perception is further influenced by government communication and broader tax equity policies. It finds that perceived fairness is enhanced when citizens believe that taxes imposed on the wealthy are just (Mohammed & Tangl, 2023). It adds that personal beliefs about social justice heavily influence one's tax morale (Castañeda, 2023). Also, it observed that individuals who utilize tax incentives often view the system as fairer than those excluded from such benefits (Bobek et al., 2024).

In a Turkish survey, despite widespread beliefs that excise taxes are too high, support persisted for higher taxation on harmful products (Demir & Balki, 2023). Similarly, tax fairness plays a key role in voluntary compliance, particularly among small and medium enterprise (SME) owners (Appiah et al., 2024). In contrast, negative perceptions in Nigeria stem from viewing taxes as burdensome and yielding little personal benefit (Azaka, 2022). Furthermore, it found that over half of TIN holders viewed indirect taxes as unfair, underscoring the influence of individual taxpayer identity on fairness perception (The Ceylon Chamber of Commerce, 2024). Lastly, it emphasizes the necessity of equitable tax design in addressing systemic inequality (Ravishankar, 2021).

Affordability. Excise taxes increase product prices, which can reduce affordability and curb consumption (Chaloupka et al., 2019). However, affordability varies depending on tax structure and enforcement. While e-cigarettes became more affordable across the U.S. from 2015 to 2021, states implementing ad-valorem taxes maintained lower affordability, particularly among the youth (Diaz et al., 2024).

In the Philippines, significant tax reforms in 2012 led to a sharp decline in cigarette affordability and consumption (Bleacher, 2020). Yet, some consumers, especially long-term vapers, are less price-sensitive due to brand loyalty and entrenched behavior (Pesko et al., 2020). Factors like income, vape shop availability, and local enforcement also influence affordability perceptions (Fuchs et al., 2021).

Illicit trade is another consequence of reduced affordability. An estimated 61 billion illicit cigarettes were sold in the Philippines from 1998 to 2018 (Lavares et al., 2021). High taxes can also push users back to traditional smoking, as observed in Minnesota (Harrington, 2024), and fuel black markets, which now account for 98% of U.S. vapor product sales (Hoffer & Rosin, 2024).

Furthermore, perceived price fairness influences buying behavior. Lastly, perceptions of pricing can directly impact consumer attitudes and purchase decisions (Zhao et al., 2021; Sutanto & Wulandari, 2023).

Awareness. Awareness of excise taxes as tools for externality correction shapes public opinion and compliance. Informed consumers, particularly those who anticipate annual tax changes or regularly purchase harmful goods, tend to view excise taxes more favorably (Demir and Balki, 2023)

Tax knowledge facilitates acceptance of fiscal policy (Appiah et al., 2024). It recommends embedding tax education into policy rollout to ensure positive public reception (Azaka, 2022). Disparities in awareness between TIN and non-TIN holders, with the latter reporting significantly lower levels of tax system understanding (The Ceylon Chamber of Commerce, 2024).

These findings support the need for robust educational campaigns and outreach programs to raise awareness and increase policy acceptance.

Government Intent. Public trust in government motives significantly influences tax perceptions. Initially framed for revenue, excise taxes are increasingly positioned as tools for public health (Chaloupka, 2019). In the Philippines, perception of the TRAIN Law among automotive industry workers improved when its broader economic goals were understood (Castro & Castro, 2021). Similar sentiment was found among consumers of imported goods, who supported new taxes due to perceived benefits for local industries (Ahmad et al., 2020). One study noted that 64% of surveyed individuals viewed taxes as funding public services, especially among TIN holders (The Ceylon Chamber of Commerce, 2024). A University of Bristol survey found that even when facing financial strain, nearly half of respondents supported tax increases if linked to public service improvement. However, mistrust in policy implementation persists. Although higher taxes are intended to reduce youth vaping, enforcement challenges have limited effectiveness, as seen in New York's flavored e-cigarette ban (Schneller, 2022). Public misunderstanding about the goals of vapor product taxes can lead to skepticism and reduced compliance (Kennedy et al., 2020).

To ensure alignment between public perception and policy intent, governments must enhance transparency, engage in consultations, and improve public health communication strategies (Fuchs et al., 2021; Chaloupka et al., 2019).

Consumer Attitude Towards Vapor Product

Consumer attitude significantly influences purchase and repurchase intentions. A study on e-cigarette users found that habit, not health consciousness, strongly impacted attitudes and purchase intentions, highlighting the addictive nature of nicotine (Jiang, 2022). Additionally, consumers' perceived value of vaping, such as reducing cigarette consumption, plays a key role in their purchasing decisions, with brand loyalty and perceived benefits sustaining demand even amid excise taxes (Fahlevi & Nurmahdi, 2020; Chaloupka et al., 2019).

Emotional factors like guilt and regret also influence repurchase intentions. Smokers' post-purchase guilt leads to regret, which, mediated by attitudes toward smoking, affects future purchasing behavior (Kazancoglu et al., 2021). However, self-control, quitting support, and hope are crucial for reducing repurchase intentions. Hope encourages smokers to persist in quitting by fostering resilience and goal-oriented behavior (Fazal-e-Hasan et al., 2020).

Repurchase Intention Towards Vapor Product

Repurchase intention is a key predictor of consumer behavior, with consumers more likely to repurchase if they perceive the benefits (quality, convenience) outweigh the sacrifices (cost, health risks) (Fahlevi & Nurmahdi, 2020). Positive attitudes toward products also enhance repurchase likelihood (Sharma, 2023). Perceived value, satisfaction, and emotional fulfillment contribute to repeat purchases, while perceived risk negatively impacts repurchase intentions (Kumar & Kashyap, 2022).

Price and tax policies influence repurchase behavior, particularly for e-cigarettes. Research shows that price hikes and tax increases reduce consumption, with a $1.00 tax increase leading to a 12.2% drop in usage (Diaz et al., 2023). Price reductions can increase use among current smokers (Minami & Teo, 2019), while higher prices discourage new users, especially younger individuals (Chaloupka et al., 2019).

Methodology

This study utilized a quantitative, descriptive-correlational research design to investigate the relationship between consumer perceptions of excise tax regulations on vapor products and their attitudes and repurchase intentions. This approach was appropriate for measuring naturally occurring variables without manipulation, enabling the identification of statistical associations without inferring causality.

The research was conducted in Quezon City, Metro Manila, Philippines—an urban area characterized by a high population density and a diverse demographic, making it a suitable setting for capturing varied consumer perspectives on vapor product regulation. The target population comprised individuals aged 18 years and older who had been using vapor products for at least one year.

Due to the absence of official data on the number of vapor product users in Quezon City, the sample size was determined using Cochran's formula for an unknown population, resulting in a minimum sample of 384 respondents at a 95% confidence level and a 5% margin of error. A non-probability snowball sampling technique was employed to reach this sample, allowing participants to refer to other eligible individuals, which is particularly useful in accessing niche or hidden populations such as vapers.

Data were collected using a structured, self-administered questionnaire divided into four sections: demographic profile; perceptions of excise tax regulation (fairness, affordability, awareness, and perceived government intent); consumer attitudes (satisfaction, perceived value, and willingness); and repurchase intentions. All items were measured using a 4-point Likert scale (1 = Strongly Disagree, 4 = Strongly Agree) to encourage definitive responses and minimize neutral bias.

Instrument validation was conducted through a pilot test to assess clarity and coherence. Content modifications were made based on feedback, and internal consistency was verified, with all constructs achieving acceptable reliability. Data was gathered both in person and online through Google Forms. Initial participants were approached in vape shops and designated smoking areas, and subsequent respondents were recruited through referrals, ensuring eligibility criteria were met.

Descriptive statistics (frequencies, means, and standard deviations) were used to analyze demographic and scale data. For hypothesis testing, Partial Least Squares Structural Equation Modeling (PLS-SEM) was used. This involved two phases: assessment of the measurement model through Confirmatory Factor Analysis and evaluation of the structural model to examine relationships among latent constructs. This methodological framework allowed for a rigorous exploration of how excise tax perceptions influence consumer behavior within the context of vapor product consumption.

Results and Discussion

Demographic Characteristics of Respondents

Among respondents, 69.5% were aged 18–24, followed by 24.1% aged 25–34. Older participants (35 and above) comprised only 6.4% of the sample, indicating that vapor product use is predominantly concentrated among younger adults. Vaping experience was relatively recent, with 53.3% reporting 1–2 years of use and only 14.4% vaping for more than 5 years. District representation was diverse, with the highest from District 2 (29%), followed by Districts 1 (20.3%) and 4 (17.2%). This geographical spread enhances the generalizability of the findings within Quezon City's urban context.

Confirmatory Factor Analysis (CFA)

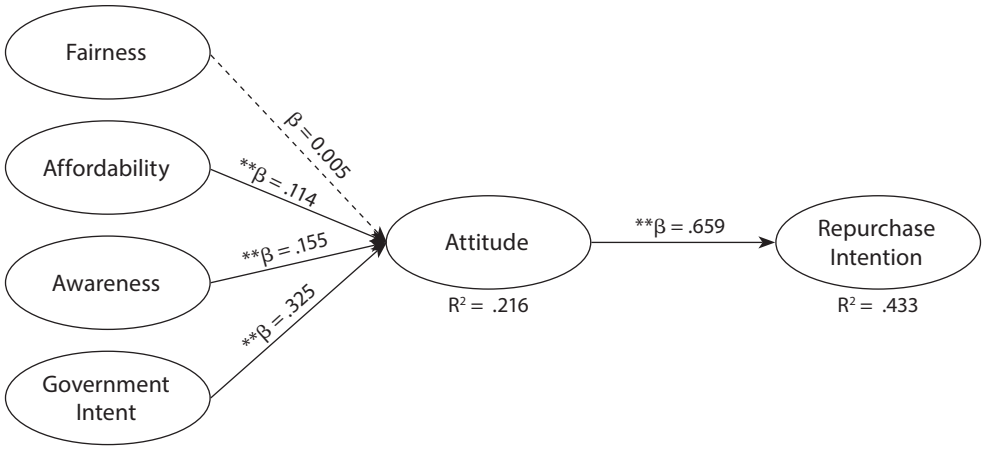
The findings indicate that respondents generally perceive excise taxation as fair ($M = 2.81$, $SD = 0.87$), though dissatisfaction with the tax rate itself was notable ($M = 2.70$). Affordability concerns were more pronounced, with respondents agreeing that vapor products had become too expensive ($M = 3.04$), reflected in the overall affordability perception score ($M = 2.94$, $SD = 0.87$). Awareness of excise tax policies was relatively high ($M = 3.19$, $SD = 0.72$), particularly regarding the impact on prices ($M = 3.35$). However, trust in the government's intent behind excise taxation was weaker ($M = 2.86$, $SD = 0.92$), with revenue generation being the most widely recognized motive ($M = 3.12$) and trust in the government's broader intentions being the lowest-rated item ($M = 2.55$). These findings suggest that while respondents acknowledge the role of excise taxation, concerns about fairness, affordability, and government intent persist.

In terms of consumer attitudes toward vapor products, respondents maintained a generally positive perception despite cost increases ($M = 2.84$, $SD = 0.83$), with satisfaction receiving the highest score ($M = 3.02$). However, the perceived value of vapor products was slightly diminished ($M = 2.68$), indicating that while consumers still view these products favorably, higher prices may affect their perceived worth.

Regarding repurchase intentions, respondents showed a moderately strong willingness to continue purchasing vapor products ($M = 2.75$, $SD = 0.91$). Agreement to repurchase stood at $M = 2.87$, and the intent to buy preferred brands again was similarly rated at $M = 2.85$. However, the intention to recommend vapor products was comparatively lower ($M = 2.46$), potentially reflecting affordability concerns or regulatory hesitations. These insights suggest that while consumers remain committed to purchasing vapor products, their enthusiasm for recommending them may be hindered by cost considerations and policy-related uncertainties.

Emerging Model

Figure 1 shows that affordability ($\beta = 0.144$, $p < 0.001$), awareness ($\beta = 0.155$, $p < 0.001$), and government intent ($\beta = 0.325$, $p < 0.001$) significantly influence consumer attitudes, while fairness ($\beta = 0.005$, $p > 0.05$) does not. Attitude strongly predicts repurchase intention ($\beta = 0.659$, $p < 0.001$). The model explains 21.6% of attitude variance and 43.3% of repurchase intention ($R^2 = 0.216$; $R^2 = 0.433$).



Note: *** significant at p < 0.001; ** significant at p < 0.01; - - ➤ not significant at p > 0.05

Fig. 1. Emerging model of the study.

Path Analysis and Hypothesis

Path analysis using PLS-SEM assessed the direct effects of excise tax perception dimensions on consumer attitude and its impact on repurchase intention. Fairness had no significant effect (β = 0.005, p = 0.499), indicating limited relevance in shaping attitudes. Affordability (β = 0.144, p = 0.002, f^2 = 0.038) and awareness (β = 0.155, p < 0.001, f^2 = 0.051) had small but significant positive effects. Government intent was the strongest predictor (β = 0.325, p < 0.001, f^2 = 0.135), highlighting the role of perceived public health motives. Attitude significantly influenced repurchase intention (β = 0.659, p < 0.001, f^2 = 0.435).

H1. The perceived fairness of excise tax regulations does not significantly influence consumer attitude toward vapor products. The analysis revealed that fairness does not have a statistically significant effect on consumer attitude, thus H1 is not supported. This finding contrasts with prior research that emphasizes tax fairness as a determinant of taxpayer compliance and positive sentiment (Azaka, 2022; Castañeda, 2023). While previous studies indicated that perceived fairness of taxation plays a critical role in shaping tax morale, this research shows that consumers of vapor products may prioritize tangible aspects—such as affordability—over perceived equity. It is possible that in contexts where taxation is viewed as inevitable or where communication about tax utilization is limited, fairness becomes a less influential consideration. This outcome aligns with the work of Mohammed and Tangl (2023), who emphasized that fairness may lose salience without sufficient governmental transparency.

H2. The affordability of vapor products significantly influences consumer attitude. Findings demonstrate that perceived affordability has a small but significant positive effect on consumer attitude, thus supporting H2. This aligns with studies by Zhao et al. (2021) and Sutanto and Wulandari (2023), which highlight those perceptions of price fairness significantly impact consumer evaluations and behavior. Particularly in price-sensitive markets, affordability can become a critical factor in shaping positive brand sentiment. These results are consistent with Harrington (2024), who found that excessive taxation in Minnesota led a considerable number of vapers to return to traditional tobacco use. The implication is clear—when taxes significantly reduce affordability, consumers may respond with reduced satisfaction or product substitution.

H3. Awareness of excise tax regulations significantly influences consumer attitude. This study found that awareness positively and significantly influences consumer attitude, thus H3 is accepted. The result is consistent with the findings of Demir and Balki (2023) and Appiah et al. (2024), who concluded that tax awareness increases public acceptance of excise policies. Awareness not only enhances transparency but also contributes to perceptions of policy legitimacy, thereby fostering more favorable attitudes. Despite this, it is worth noting that while general awareness was high, deeper engagement with policy objectives remained moderate. This suggests the need for targeted communication strategies to deepen public understanding of excise tax goals.

H4. Perceived government intent significantly influences consumer attitude. Government intent was found to be the strongest predictor of attitude among the four constructs, confirming H4. When consumers perceive that excise taxes are motivated by public health rather than revenue generation, they are more inclined to view the regulation positively. This supports the conclusions of Ahmad et al. (2020) and Chaloupka et al. (2019), who observed that tax policies framed within a health-oriented narrative tend to receive greater public support. However, the present findings also revealed skepticism, as many respondents still believe revenue generation is the primary motive. This dual perception suggests that government messaging must be clear and trust-enhancing to maintain the effectiveness of health-based fiscal policies.

H5. Consumer attitude significantly influences repurchase intention. Consumer attitude emerged as the strongest determinant of repurchase intention, thus H5 is strongly supported. The finding echoes previous research demonstrating that positive attitudes toward products are highly predictive of continued purchasing behavior (Sharma, 2023; Kazancoglu et al., 2021). Despite price increases resulting from excise taxation, favorable consumer perceptions—particularly satisfaction and perceived value—sustain repurchase behavior. The strong link between attitude and repurchase intention aligns with the Theory of Planned Behavior, reaffirming that attitudes mediate the relationship between external policy stimuli (e.g., taxation) and consumer intentions.

Table 1. Hypothesis Testing - with one segment analysis (Direct Effects)

Path	Path Coefficients (β)	P-values	Effect Sizes (f^2)	Effect Size Interpretation (Cohen, 1988)	Description	Interpretation
H1: Fair → Atti	0.005	0.499	0.001	NA	Not Significant	H1 is not supported
H2: Afford → Atti	0.144	0.002	0.038	Small	Significant	H2 is supported
H3: Aware → Atti	0.155	<0.001	0.051	Small	Significant	H3 is supported
H4: GI → Atti	0.325	<0.001	0.135	Small	Significant	H4 is supported
H5: Atti → RI	0.659	<0.001	0.435	Large	Significant	H5 is supported

Note: Fairness (Fair); Affordability (Afford); Awareness (Aware); Government Intent (GI); Consumers' Attitude (Atti); Repurchase Intention (RI).
Cohen Effect Size Interpretation: 0.02 – Small, 0.15 – Medium, 0.35 – Large

Conclusions

The findings of this study demonstrate that affordability, awareness, and government intent significantly influence consumer attitudes toward vapor products, while fairness does not show a notable effect. The strong association between consumer attitudes and repurchase intentions suggests that taxation policies have direct implications for market behavior. Given these results, policymakers must strike a balance between regulatory goals and economic accessibility to ensure fair yet effective excise tax implementation. Additionally, government agencies should prioritize transparent communication regarding tax utilization to reinforce public trust and policy acceptance. Future research should consider integrating other psychological and behavioral variables to refine understanding of consumer reactions to taxation and pricing changes.

Methodologically, this study confirms the effectiveness of Partial Least Squares Structural Equation Modeling (PLS-SEM) in analyzing taxation-related consumer behavior. The model accounts for a substantial variance in consumer attitudes and purchase intentions, reinforcing the reliability of using PLS-SEM for complex policy evaluations. Researchers aiming to explore similar fiscal policy impacts could benefit from expanding their frameworks to include mediating and moderating variables such as economic stability, trust in institutions, and behavioral adaptations. Additionally, longitudinal analyses would provide deeper insights into how consumer attitudes evolve in response to sustained excise tax policies over time.

From a practical perspective, businesses and policymakers should consider affordability as a crucial component of excise tax policy. While awareness and perceived government intent contribute to consumer acceptance, excessively high taxes may deter positive attitudes and purchasing behaviors. Regulatory authorities could explore graduated tax systems or incentives to mitigate financial strain while maintaining public health objectives. Strengthening financial literacy campaigns on excise tax policies can further enhance public comprehension and acceptance, ensuring that consumers make informed decisions. These collective efforts can contribute to a taxation model that effectively balances fiscal responsibility, consumer behavior, and public welfare.

REFERENCES

1. Ackary et al. (2024). An analysis of E-Cigarette, heated tobacco products, and novel tobacco product flavors in the Philippines: regulatory challenges and opportunities. SSRN Electronic Journal.
2. Ahmad et al. (2020). Buyers' perception of the impact of the latest regulation on imported goods tax via e-commerce. ResearchGate. https://www.researchgate.net/publication/345783521
3. Appiah et al. (2024). Tax Knowledge, Trust in Government, and Voluntary Tax Compliance: Insights From an Emerging Economy. SageJournals.
4. Azaka, L. (2022). Tax knowledge, tax attitude, and perception of tax fairness as predictors of tax compliance among tech workers in Lagos State, Nigeria. AJPSSI. https://ajpssi.org/index.php/ajpssi/article/view/537/675
5. Bleacher, E. (2020). Affordability of Tobacco Products: The Case of Cigarettes. Institute for Health Research and Policy.https://www.economicsforhealth.org/files/research/609/affordability-white-paperv4.1final.pdf?utm_source= chatgpt.com
6. Braak, D., Cummings, K. M., Nahhas, G. J., Reid, J. L., & Hammond, D. (2020). How are adolescents getting their vaping products? Findings from the international tobacco control (ITC) youth tobacco and vaping survey. Addictive Behaviors, 105, 106345.
7. Bobek, D., Hageman, A., & Hausserman, C. (2024, October 24). Are Fairness Perceptions Related to Moral Licensing Behavior? Springer Nature Link.
8. Castro, N. F., & Castro, C. K. C. (2021). The Implementation of TRAINLaw. Asian Journal of Multidisciplinary Studies. https://asianjournals.org/online/index.php/ajms/article/view/445/257
9. Castañeda, N. (2023, March 31). Fairness and Tax Morale in Developing Countries. Springer Nature Link.
10. Chaloupka et al. (2019). The use of excise taxes to reduce tobacco, alcohol, and sugary beverage consumption. Annual Review of Public Health, 40(1), 187–201.
11. Cnossen, S. (2023, March 29). Excise taxation to preserve health and to protect the environment: A review. SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4401102
12. Cordero, D. A., Jr. (2024, March). When the smoke is going down! Journal of Public Health.
13. Cotti et al. (n.d.). The Effects of E-Cigarette Taxes on Prices and Tobacco Product Sales. UKnowledge. https://uknowledge.uky.edu/isfe_papers/3
14. Cruz et al. (2024). Factors Influencing the Habitual Consumption of Disposable Electronic Cigarettes. Zenodo.
15. Demir, I. C., & Balki, A. (2023). Is the excise tax perceived as an externality tax? DergiParkAkademik. https://dergipark.org.tr/en/pub/eab/issue/76151/1165418
16. Diaz et al. (2023). Investigating the impact of E-Cigarette price and tax. American Journal of Preventive Medicine, 64(6), 797–804.
17. Diaz et al. (2024). Changes in price, income and e-cigarette affordability for young people in the USA. Tobacco Control. https://tobaccocontrol.bmj.com/content/early/2024/10/29/tc-2024-058619
18. Fahlevi, M. A., & Nurmahdi, A. (2020). Model on new customer purchase decision. IHASJ, 3(2).
19. Famiglietti et al. (2021). Are electronic cigarettes and vaping effective tools for smoking cessation? Journal of Thoracic Disease, 13(1), 384–395.
20. Fazal e Hasan et al. (2020, July). The Interplay of Positive and Negative Emotions to Quit Unhealthy Consumption Behaviors. ResearchGate. https://www.researchgate.net/publication/343123226
21. Fuchs et al. (2021). Effectiveness of taxation in tobacco control. Tobacco Control, 30(5), 538–545.
22. Harlow et al. (2022). Motivations for E-cigarette use and associations with vaping frequency. Drug and Alcohol Dependence, 238, 109583.
23. Harrington, O. (2024, June 25). The Tax Implications of Vaping: A State-by-State Analysis. The Tax Evader. https://thetaxevader.com/the-tax-implications-of-vaping-a-state-by-state-analysis

24. Hoffer, A., & Rosin, J. (2024, July 29). Current challenges in vaping markets. Tax Foundation. https://taxfoundation.org/blog/vaping-market-challenges
25. Institute for Global Tobacco Control. (2023). The Sale and Marketing of Cigarettes, E-cigarettes, and Heated Tobacco Products Near Schools in the Philippines. https://www.globaltobaccocontrol.org/sites/default/files/2023-08/PH-Fact-Sheet-POS-2023-FINAL-EN.pdf
26. Jiang, L. L. (2022). Factors influencing consumers' intentions to purchase electric cigarettes. University of Coimbra Repository. https://estudogeral.uc.pt/bitstream/10316/104900
27. Jiang et al. (2023, May 26). The associations between e-liquid characteristics and its pricing. PLOS ONE.
28. Kennedy, R. D., Awopegba, A., De León, E., & Cohen, J. E. (2020). Global approaches to regulating electronic cigarettes. Tobacco Control, 29(6), 613–621.
29. Kumar, A., & Kashyap, A. K. (2022). Understanding the factors influencing repurchase intention in online shopping. Vision.
30. Lavares, M. P., Ross, H., Francisco, A., & Doytch, N. (2021). Illicit tobacco in the Philippines. Tobacco Control, 31(6), 701–706.
31. Leventhal, A. M. et al. (2021). Salt vs Free-Base Nicotine on Vaping Appeal. JAMA Network Open, 4(1), e2032757.
32. Minami, H., & Teo, T. K. (2019, May 22). Impact of e-cigarette price changes. PubMed. https://pubmed.ncbi.nlm.nih.gov/31116065/
33. Miracolo et al. (2021, April 22). Sin taxes in Latin America. Oxford Academic. https://academic.oup.com/heapol/article/36/5/790/6246142
34. Mohammed, H., & Tangl, A. (2023). Taxation Perspectives: Taxes as Punishment—A Comprehensive Study.
35. Palmer et al. (2022). Need for Novel Investigations of Treatments to Quit E-cigarettes. AACR. https://aacrjournals.org/cancerpreventionresearch/article-abstract/15/9/569/708165
36. Pesko, M, Courtemanche, C, Maclean, J.C. (2019). Effects of Traditional Cigarette and E-Cigarette Taxes. UKnowledge. https://uknowledge.uky.edu/cgi/viewcontent.cgi?article=1007&context=isfe_papers
37. Ravishankar, V. (2021). What is Tax Fairness? PRI. https://www.unpri.org/download?ac=15325
38. Schneller et al. (2022). Youth tobacco use before and after vaping flavour restriction. Tobacco Control. https://tobaccocontrol.bmj.com/content/tobaccocontrol/31/Suppl_3/s161.full.pdf
39. Sese, L. V. C., & Guillermo, M. C. L. (2023). E-Smoking out the Facts. Tobacco Use Insights, 16.
40. Sharma, A. (2023, September 4). The Multidimensional Impact of Attitude and Branding. LinkedIn. https://www.linkedin.com/pulse/multidimensional-impact-attitude-branding-consumer-behaviour-sharma
41. Sutanto, I., & Wulandari, R. (2023). Effect of Price and Quality on Eco-Friendly Packaging. IJSMS.
42. The Ceylon Chamber of Commerce. (2024). National Taxpayer Perception Study. UNDP. https://www.undp.org/sites/g/files/zskgke326/files/2024-09/national_taxpayer_perception_study_final.pdf
43. Untalan, S. (2024, May 31). PH records 1st vape-related death. GMA News Online. https://www.gmanetwork.com/news/topstories/nation/908594/ph-records-1st-vape-related-death-doh/story/
44. Verguet, S., Kearns, P. K., & Rees, V. W. (2020). Questioning the regressivity of tobacco taxes. Tobacco Control. https://tobaccocontrol.bmj.com/content/30/3/245
45. World Health Organization. (2023, July 31). Tobacco. https://www.who.int/news-room/fact-sheets/detail/tobacco
46. Zhao, H., Yao, X., Liu, Z., & Yang, Q. (2021). Impact of pricing and product information on consumer buying behavior. Frontiers in Psychology, 12.

Maximizing Resource Efficiency: Material Flow Cost Accounting and Green Accounting for Sustainable Development of MSMEs in Metro Manila

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Abstract

This study investigates the impact of material flow cost accounting (MFCA) and green accounting on sustainable development through resource efficiency. Addressing the research gap concerning the interconnectedness of accounting practices and sustainability, the objective was to empirically validate the relationships between MFCA, green accounting, resource efficiency, and sustainable development within corporate and financial sectors. A structural equation model was employed to analyze the relationships between these factors. Key findings highlight that resource efficiency is a crucial pathway to sustainable development, with green accounting having a more significant role in driving resource efficiency. While MFCA also contributes, its impact is smaller compared to green accounting. A notable research gap identified was the high inter-construct correlation between MFCA and green accounting, suggesting potential multicollinearity. Methodologically, future studies should address this through alternative model specifications and expand the research scope across diverse industries and regions, incorporating longitudinal designs to understand the evolution of accounting practices and their impact on sustainable development over time. The implications underscore the importance of integrating comprehensive green accounting practices to bolster resource efficiency, offering a clear directive for businesses and policymakers to integrate sustainability into operational and accounting frameworks.

Keywords: Green Accounting, Material Flow Cost Accounting, Resource Efficiency, Sustainable Development, and MSMEs

Introduction

Metro Manila, a primary urban center with a population of approximately 13 million (Ballesteros et al., 2023), faces significant environmental challenges stemming from industrial activities, including those of Micro, Small, and Medium Enterprises (MSMEs). These environmental implications include pollution resulting from production waste generated by MSMEs. Considering these challenges, this study explores the potential of material flow cost accounting (MFCA) and green accounting to enhance resource efficiency and promote sustainable development within MSMEs in Metro Manila.

MFCA, a pivotal management tool, meticulously analyzes the flow of materials within an organization, assigning monetary values to both inputs and outputs throughout the production process. MFCA helps in identifying the hidden costs associated with waste and inefficiencies, providing a clear picture of how resources are being utilized. By quantifying the costs of waste, MFCA encourages managers to implement strategies for waste reduction, thereby enhancing resource efficiency and minimizing environmental impact.

Green accounting broadens the scope of traditional accounting to include environmental costs and benefits, offering a more holistic view of a company's financial performance and its impact on the environment. By incorporating environmental factors, green accounting provides a more comprehensive assessment of a company's overall performance, enabling stakeholders to make more informed decisions. Implementing strategic environmental accounting can positively influence sustainability performance.

This study investigates the impact of MFCA and green accounting on resource efficiency and sustainable development within MSMEs in Metro Manila. By examining these relationships, this research aims to provide valuable insights for businesses, policymakers, and other stakeholders seeking to promote sustainable practices and improve environmental performance in the region.

Literature Review

Material Flow Cost Accounting

MFCA is a powerful environmental management tool that enhances resource efficiency and financial performance across various industries. Its application in recycling technologies, such as IRETA2 and ReComp, has highlighted financial challenges

but also potential improvements under favorable market conditions (Beck, 2023). In warehouse management, MFCA helps identify inefficiencies and material losses that traditional accounting systems often overlook, improving resource utilization (จันทราทิพย์, 2024). Additionally, MFCA contributes to environmental and financial benefits by increasing transparency regarding technological losses, reducing pollution, and positively impacting green accounting, particularly in the basic materials and chemical industries (Ivanovskaya et al., 2019; Santi et al., 2022). However, its successful implementation requires significant adjustments to existing accounting practices and a commitment to data-driven decision-making, which may present challenges for some organizations.

MFCA is a pivotal management tool that meticulously analyzes the flow of materials within an organization, assigning monetary values to both inputs and outputs throughout the production process (Tu & Huang, 2019). It helps in identifying the hidden costs associated with waste and inefficiencies, providing a clear picture of how resources are being utilized (Chang et al., 2015). By quantifying the costs of waste, MFCA encourages managers to implement strategies for waste reduction, thereby enhancing resource efficiency and minimizing environmental impact (Chang et al., 2015). MFCA, as a cost accounting method, plays a crucial role in pinpointing opportunities for improvement in material consumption and cost accrual, providing a basis for informed decision-making (Walz & Guenther, 2020). MFCA provides a structured approach for evaluating the complete cycle of materials, from procurement to disposal, thus promoting circular economy principles. This system not only aids in diminishing waste but also enhances the economic value of materials embedded in consumer products, aligning with circular economy goals (Maio & Rem, 2015).

Green Accounting

Green accounting, or environmental accounting, integrates ecological costs into financial reporting, enhancing transparency and aligning corporate strategies with sustainability goals. It plays a crucial role in stakeholder engagement by informing investors about environmental policies, fostering trust, and potentially increasing investment opportunities (Juneja et al., 2024). Additionally, it aids in resource management, particularly in developing countries, by identifying resource utilization and associated costs (N. R. S & TS, 2023). By improving sustainability disclosures, green accounting enhances corporate value and investor confidence, especially in environmentally impactful sectors (Rangkuti, 2023). However, its implementation faces challenges, including the lack of comprehensive regulatory frameworks and limited awareness among companies and individuals (Feng, 2024). While green accounting offers significant benefits for sustainability and corporate responsibility, addressing regulatory and educational barriers is essential to maximize its impact on both the economy and the environment.

Resource Efficiency

Resource efficiency is a crucial strategy for optimizing natural resource use while minimizing waste and environmental impact, playing a key role in advancing a circular economy and sustainable development. It involves decoupling economic growth from resource consumption, reducing resource extraction and greenhouse gas emissions while enhancing economic activity (Berkel & Fadeeva, 2020; Ekins et al., 2016). In industries such as manufacturing and mining, resource efficiency leads to cost reductions, lower dependency on volatile raw material markets, and improved sustainability through methodologies that assess land, water, and energy consumption (Maass et al., 2018; Spürk et al., 2016). Technological advancements and policy innovations, including Russia's energy efficiency laws and international cooperation efforts, are essential for improving resource efficiency, with policies potentially reducing global resource extraction by 28% by 2050 (Turanova, 2024; Ekins et al., 2016). Despite its clear benefits, challenges remain in implementing effective policies and technologies across sectors, requiring continuous innovation to balance economic growth with environmental sustainability.

Sustainable Development

Sustainable development is a multifaceted concept that aims to meet the needs of the present without compromising the ability of future generations to meet their own needs, encompassing environmental, social, and economic dimensions. It emphasizes the importance of balancing economic growth with environmental protection and social equity, ensuring a prosperous future for all (Fikri et al., 2021). Environmental damage, particularly in marine ecosystems, underscores the necessity for a shift in conventional accounting practices to avert conflicts (Syah et al., 2020). Moreover, countries are increasingly adopting sustainability reporting as a means of communicating their environmental, social, and economic efforts (Fundoni & Pischedda, 2023).

Sustainable development seeks to balance economic growth, social equity, and environmental protection, ensuring that present needs are met without compromising future generations' ability to thrive. The implementation of the United Nations' Sustainable Development Goals (SDGs) highlights the global commitment to addressing critical issues such as poverty, education, and climate change, while also facing challenges like corruption and social conflicts (Arszulowicz, 2024). Key aspects of sustainable development include economic growth that preserves natural resources, as seen in Greece's strides in renewable energy utilization (Williams, 2023), social equity through education to foster sustainable practices and critical thinking, and global collaboration to bridge disparities between developed and developing nations (Arszulowicz, 2024). However, critics

argue that the broad definitions of sustainability can lead to greenwashing, where harmful practices are falsely labeled as sustainable, emphasizing the need for clear standards and accountability in the pursuit of genuine sustainability.

Hypothetical Background

MFCA and Sustainable Development MFCA plays a crucial role in advancing sustainable development by reinforcing corporate social responsibility and green accounting practices. It enables companies to make informed decisions and develop sustainability-focused strategies through effective policy implementation (Maulana et al., 2024). Research findings highlight MFCA's significant impact on sustainable development, while environmental evaluation alone has shown no notable effect, emphasizing the importance of accurate accounting practices in the basic and chemical industries (Putri et al., 2024). By optimizing resource utilization and minimizing waste, MFCA enhances manufacturing efficiency and supports sustainability goals (Putri et al., 2024). Additionally, MFCA-driven technology improves green productivity by reducing energy and material consumption, which leads to cost savings, improved product quality, and increased customer satisfaction in production processes (Dahi & Abdullah, 2024). As businesses strive to integrate financial performance with environmental responsibility, MFCA emerges as a key tool in fostering resource efficiency and driving sustainability initiatives. Therefore, this study hypothesized that *MFCA has a positive direct effect on sustainable development (H1).*

MFCA and Resource Efficiency MFCA plays a critical role in enhancing resource efficiency across various industries by systematically quantifying material and energy flows, identifying inefficiencies, and allocating environmental costs to specific activities. This approach enables businesses to reduce environmental impacts while improving economic performance through increased transparency and targeted process optimizations (Beck, 2023). In the hotel sector, MFCA helps identify material losses and inefficiencies, leading to improved environmental performance and cost management (Nyide, 2016). Additionally, in the palm oil industry, MFCA facilitates the tracking of physical resources and financial expenditures, allowing companies to evaluate both negative and positive outputs. By reducing waste and oil losses, this method strengthens sustainability efforts and enhances competitive advantage (Kurniawan et al., 2024). Overall, MFCA serves as an essential tool for driving efficiency, promoting sustainable practices, and aligning financial success with environmental responsibility. In accordance with previous studies, this study assumes that *MFCA has a positive direct effect on resource efficiency (H2).*

Green Accounting and Sustainable Development Green accounting plays a significant role in sustainable development by addressing environmental issues while ensuring financial stability. Studies indicate that integrating green accounting principles positively impacts both sustainability and financial performance, making it essential for long-term business viability (Chukka et al., 2024). However, research also suggests that its current implementation may not effectively contribute to achieving sustainable development goals in certain manufacturing companies, as it has shown a negative and insignificant effect in some cases (Putri et al., 2024). Despite this, green accounting remains valuable for assessing environmental performance, promoting sustainability, and demonstrating that company size does not significantly affect sustainable development outcomes (Trevanti & Yuliati, 2023). Furthermore, its role in advancing SDGs is evident, particularly when combined with strong corporate social responsibility (CSR) programs, which contribute meaningfully to environmental and social aspects of sustainability (Pamungkas et al., 2024). These insights emphasize the need for refining green accounting practices to maximize their effectiveness in fostering sustainable development across industries. Building on this research, this study assumes that *green accounting has a positive direct effect on sustainable develipment (H3).*

Green Accounting and Resource Efficiency Green Accounting plays a crucial role in promoting sustainable resource use by aligning corporate practices with environmental health. Resource efficiency strengthens this impact by optimizing input-output ratios in production, as demonstrated in Indonesian cement companies (Loen, 2018). By integrating economic, environmental, and social costs into financial reporting, green accounting enhances energy efficiency, reduces consumption, and supports sustainable business practices, leading to improved environmental performance and stakeholder benefits (Rahman & Islam, 2023). Additionally, it helps companies identify and mitigate environmental costs associated with production, as evidenced by Lumintu Farm's adoption of green accounting, which resulted in improved eco-efficiency, enhanced environmental performance, and better financial outcomes (Ardiana et al., 2023). These findings highlight the critical role of Green Accounting in fostering sustainability and resource optimization across various industries. Aligned with existing frameworks in sustainability research, this study posits that *green accounting has a positive direct effect on resource efficiency (H4).*

Resource Efficiency and Sustainable Development Resource efficiency plays a critical role in sustainable development by optimizing resource utilization and minimizing waste, leading to economic and environmental benefits. A study indicates that a 1% increase in efficiency results in approximately a 0.25% enhancement in the sustainable development index, though the influence of mineral resource prices on sustainability is even more substantial (Wen & Zhang, 2024). Effective resource management, combined with fintech investments, has contributed significantly to achieving sustainability goals in Organization for Economic Co-operation and Development countries between 2010 and 2019, fostering both economic and environmental progress (Zhang et al., 2024). In the Russian Federation, resource efficiency is essential for reducing fossil resource dependence,

improving environmental conditions, and aligning with global sustainability efforts such as the Paris Agreement through technological innovation and international collaboration (Turanova, 2024). Similarly, in China's special economic zones, disparities in efficiency highlight the need for transformative policies, enhanced green indices, and the advancement of green finance to optimize sustainable development outcomes (R. Zhong et al., 2022). These findings emphasize the importance of resource efficiency in driving sustainable development across diverse economic and policy landscapes. Drawing from prior findings and theoretical insights, it is assumed that *resource efficiency has a positive direct effect on sustainable development (H5)*.

Indirect effect of MFCA and Green Accounting on Sustainable Development The indirect effects of MFCA and green accounting on sustainable development through resource efficiency are significant, as evidenced by various studies. MFCA enhances resource efficiency by optimizing material and energy use, which ultimately contributes to sustainable development. In contrast, green accounting's impact appears less pronounced, often showing negative or insignificant effects on sustainability outcomes. Rooted in the evolving intersection of environmental and financial practices, the study suggests that *MFCA and green accounting has a positive indirect effect on sustainable development (H6a-b)*.

Nonetheless, figure 1 of the study demonstrates the operational framework of this study.

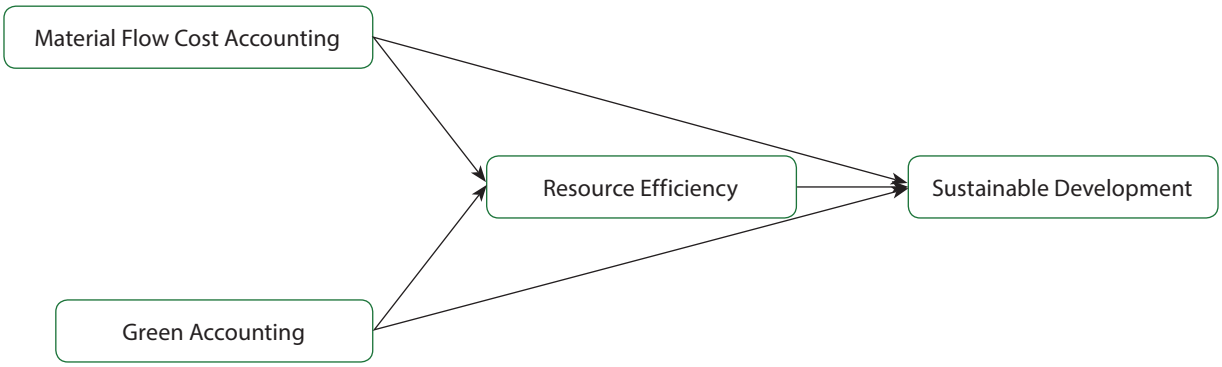


Fig. 1. The operational framework of the study.

Theoretical Background

A theoretical background for this study focuses on integrating these environmental accounting methods to enhance MSME performance and drive sustainable practices. MFCA is key for identifying and quantifying material and energy flows, pinpointing waste and inefficiencies in both physical and monetary terms. By tracking inputs and outputs, MFCA helps MSMEs visualize hidden costs from non-product outputs, leading to optimized resource utilization, reduced production costs, and minimized environmental impact, thus directly contributing to resource efficiency through waste reduction at the source. Green accounting complements MFCA by integrating environmental considerations into financial reporting, offering a holistic view of an MSME's economic and ecological performance. Based on Sustainability and Legitimacy Theories, green accounting stresses measuring and reporting environmental impacts (e.g., resource consumption, waste generation, emissions) to foster transparency, accountability, and sustainable business practices. Sustainability Theory underscores balancing economic, environmental, and social aspects for long-term viability, while Legitimacy Theory suggests businesses aligning with societal environmental expectations to maintain stakeholder support. For Metro Manila MSMEs, combining MFCA and green accounting creates a strong framework for sustainable development, improving resource efficiency via waste and cost reduction, enhancing environmental performance through transparent reporting, and ultimately bolstering their local economic contribution while addressing environmental concerns. This integrated framework enables MSMEs to make profitable decisions that also align with broader sustainable development objectives.

Methodology

This study employs a quantitative research design using Structural Equation Modeling to investigate the relationships between MFCA, green accounting, resource efficiency, and sustainable development within MSMEs in Metro Manila. Numerical data will be collected through surveys administered to a sample of MSMEs in Metro Manila. A purposive sampling technique will be used to select participants (Lengsi & Yasri, 2020). To ensure representation across different areas, participants will be chosen from each locality through stratified random sampling, which involves obtaining contact information from residents and subsequently selecting participants randomly. SEM will be utilized to test the hypothesized relationships among the constructs, allowing for the simultaneous examination of multiple relationships while providing a comprehensive assessment of the model fit, with PLS analysis being used for further validation (Lengsi & Yasri, 2020). The study will focus on MFCA and green

accounting as independent variables, resource efficiency as the mediating variable, and sustainable development as the dependent variable. By employing this research design, the study aims to generate robust evidence regarding the relationships among these key concepts, contributing valuable insights for promoting sustainable practices and improving environmental performance within MSMEs in Metro Manila.

Table 1 presents the construct reliability measures for MFCA, green accounting, resource efficiency, and sustainable development, evaluated using Dijkstra-Henseler's rho (ρA), Jöreskog's rho (ρc), and Cronbach's alpha (α). The results indicate strong reliability across all constructs, with sustainable development exhibiting the highest reliability (0.870) and green accounting showing the lowest (0.750). These values suggest that the measurement model is robust and capable of accurately assessing the relationships among the study variables.

Table 1. *Construct Reliability*

Construct	Dijkstra-Henseler's rho (ρ_A)	Jöreskog's rho (ρ_c)	Cronbach's alpha (α)
Material Flow Cost Accounting	0.804	0.800	0.800
Green Accounting	0.756	0.752	0.750
Resource Efficiency	0.782	0.776	0.778
Sustainable Development	0.870	0.870	0.870

Results and Discussions

Descriptive Statistics of Retained Indicators

The descriptive statistics for the retained indicators of each construct: MFCA, green accounting, resource efficiency, and sustainable development. The mean values for the indicators generally fall between 3.016 and 3.531, suggesting that, on average, respondents tended to agree with the statements related to these indicators. The variances range from 0.347 to 0.724, indicating the degree of dispersion around the mean for each indicator. Skewness values, which range from –1.131 to –0.255, are mostly negative, suggesting that the distributions of the indicators are slightly skewed to the left, with a concentration of scores on the higher end of the scale. Kurtosis values range from –0.672 to 1.038, indicating the shape of the distribution, with some indicators exhibiting platykurtic (flatter) distributions and others leptokurtic (more peaked) distributions. The descriptive statistics provide an overview of the central tendency and distribution characteristics of the retained indicators, offering insights into the respondents' perceptions of MFCA, green accounting, resource efficiency, and sustainable development.

Table 2. *Descriptive Statistics of Retained Indicators*

Indicator	Mean	Variance	Skewness	Kurtosis
MFCA1	3.351	0.347	–0.278	–0.672
MFCA2	3.236	0.391	–0.301	–0.221
MFCA3	3.275	0.430	–0.569	0.267
MFCA4	3.243	0.428	–0.509	0.247
GA1	3.407	0.360	–0.553	–0.092
GA2	3.397	0.365	–0.543	–0.111
GA3	3.279	0.498	–0.737	0.352
GA4	3.128	0.724	–0.699	–0.225
RE1	3.338	0.395	–0.487	–0.238
RE2	3.531	0.381	–1.131	1.038
RE3	3.298	0.401	–0.498	0.125
RE4	3.125	0.603	–0.517	–0.341
RE5	3.344	0.398	–0.505	–0.239
RE6	3.069	0.683	–0.587	–0.247
SD1	3.167	0.488	–0.533	0.185
SD2	3.236	0.411	–0.409	0.026

Indicator	Mean	Variance	Skewness	Kurtosis
SD3	3.102	0.526	−0.469	−0.032
SD4	3.016	0.575	−0.255	−0.629
SD5	3.321	0.377	−0.408	−0.182
SD6	3.289	0.443	−0.470	−0.447

R-Squared

The coefficient of determination (R2) provides insights into the variance explained by the model. For resource efficiency, the R2 value of 0.630 suggests that 63% of the variance in resource efficiency is explained by the model. Similarly, the R2 of 0.741 for sustainable development indicates that 74.1% of the variance in sustainable development is explained by its respective model. The adjusted R2 values, 0.628 and 0.738, offer a more conservative estimate by accounting for the number of predictors in the model. These values suggest that the models possess moderate to high explanatory power, with the sustainable development model demonstrating a slightly stronger ability to explain variance compared to the resource efficiency model.

Table 3. R-Squared

Construct	Coefficient of determination (R^2)	Adjusted R^2
Resource Efficiency	0.630	0.628
Sustainable Development	0.741	0.738

Effect Overview

The effect overview in Table 4 highlights the relationships between MFCA, green accounting, resource efficiency, and sustainable development, offering insights into their relative impact. MFCA demonstrates a weak direct effect on resource efficiency (β = 0.073, Cohen's f^2 = 0.003), suggesting its limited influence in optimizing resource use directly. However, its total effect on sustainable development (β = 0.166, f^2 = 0.010) indicates a modest role, primarily through indirect pathways (β = 0.052), emphasizing its potential in shaping sustainability outcomes when integrated with other strategies. In contrast, green accounting exhibits a strong direct effect on resource efficiency (β = 0.727, f^2 = 0.276), reinforcing its crucial role in optimizing resource utilization and minimizing waste. Additionally, green accounting significantly impacts sustainable development (total effect β = 0.592) through indirect effects (β = 0.517), reflecting its effectiveness in fostering sustainability via improved resource management. Notably, resource efficiency demonstrates the highest direct effect on sustainable development (β = 0.711, f^2 = 0.722), underscoring its vital role in achieving sustainability goals through optimized material usage and waste reduction. These findings collectively highlight the importance of integrating MFCA and green accounting to enhance resource efficiency, which in turn drives sustainable development in corporate and financial sectors. The analysis indicates that resource efficiency is a crucial direct pathway to sustainable development. Green accounting plays a more significant role in driving resource efficiency, and subsequently sustainable development, primarily through its indirect effects. MFCA also contributes to these outcomes, though its impact is comparatively smaller. These findings underscore the interconnectedness of accounting practices, resource management, and overall sustainable development efforts, with particular emphasis on the leveraging role of green accounting in fostering resource efficiency. Cohen's f2 is used to assess the effect size (Cohen, 1992; Jikomes & Zoorob, 2018).

Table 4. Effect Overview

Effect	Beta	Indirect effects	Total effect	Cohen's f^2
Material Flow Cost Accounting -> Resource Efficiency	0.073		0.073	0.003
Material Flow Cost Accounting -> Sustainable Development	0.113	0.052	0.166	0.010
Green Accounting -> Resource Efficiency	0.727		0.727	0.276
Green Accounting -> Sustainable Development	0.074	0.517	0.592	0.003
Resource Efficiency -> Sustainable Development	0.711		0.711	0.722

Structural Equation Model

The structural equation model illustrates the relationships between different factors affecting sustainable development, with an emphasis on accounting methods and resource efficiency. It shows the influence of MFCA and green accounting on resource efficiency, which in turn significantly impacts sustainable development. The numerical values and R-squared coefficients

indicate the strength of these relationships, emphasizing the role of effective accounting practices in driving sustainability efforts.

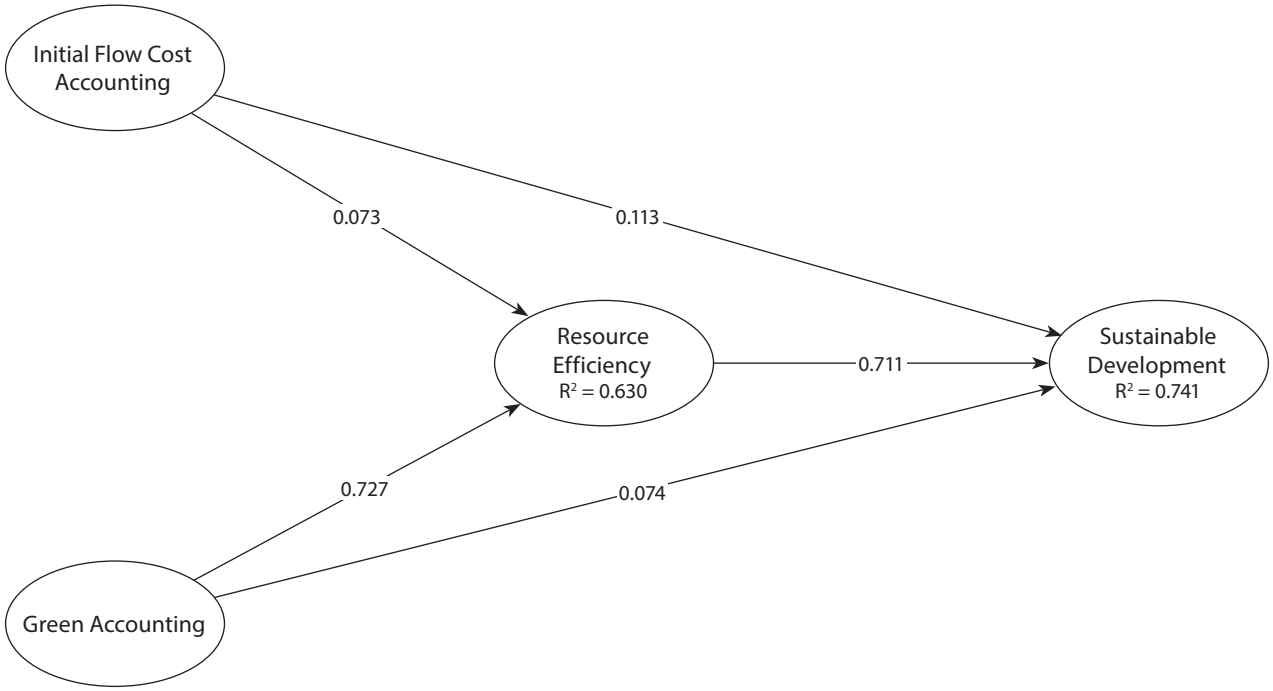


Fig. 2. Structural Equation Model

Inter-Construct Correlations

The inter-construct correlations reveal the strength and direction of the relationships between the variables in your study. The strong positive correlation between MFCA and green accounting (0.898) suggests a close alignment and potential overlap in their implementation (Tu & Huang, 2019). Resource efficiency is also positively correlated with both MFCA (0.727) and green accounting (0.793), indicating that these accounting practices are associated with improved resource management (Chang et al., 2015). Finally, sustainable development demonstrates positive correlations with MFCA (0.697), green accounting (0.740), and resource efficiency (0.853), suggesting that these constructs are interrelated in promoting sustainability. It's worth noting that high correlations (e.g., above 0.8) might indicate potential multicollinearity issues, which should be addressed in subsequent analysis. These correlations provide initial support for the research questions outlined in the editor document, particularly regarding the relationships between MFCA, green accounting, resource efficiency, and sustainable development.

Table 5. Inter-Construct Correlations

Construct	1	2	3	4
1. Material Flow Cost Accounting	–			
2. Green Accounting	0.898	–		
3. Resource Efficiency	0.727	0.793	–	
4. Sustainable Development	0.697	0.740	0.853	–

Conclusion and Recommendations

Based on these findings, organizations aiming to enhance sustainable development should prioritize the strategic implementation and integration of green accounting practices. Given its pronounced effect on resource efficiency, adopting robust green accounting systems can serve as a fundamental step towards optimizing resource utilization, minimizing waste, and ultimately advancing overall sustainability goals. While MFCA demonstrates a weaker direct impact, it can complement green accounting by providing detailed insights into material flows. Therefore, practitioners should focus on training personnel in both methodologies and leveraging the data to inform operational decisions that drive resource efficiency, such as process optimization and waste reduction, thereby fostering both environmental stewardship and potential economic benefits.

Methodologically, future research could build upon these findings by addressing the high inter-construct correlation observed between MFCA and green accounting. While not explicitly identified as problematic in this study, potential multicollinearity warrants further investigation using alternative model specifications or advanced statistical techniques to more precisely delineate the unique contributions of highly correlated constructs. Expanding the research scope to include diverse industries or geographical regions would enhance the generalizability of the findings. Additionally, incorporating longitudinal study designs could provide a more dynamic understanding of how these accounting practices evolve and impact sustainable development over extended periods.

The implications of this study are significant for both theory and practice. Theoretically, the research empirically validates the critical interplay between accounting practices, resource management, and organizational sustainability, particularly emphasizing the mediating role of resource efficiency in the path from green accounting to sustainable development. This reinforces the perception of green accounting as a proactive strategic tool for achieving tangible sustainability outcomes, moving beyond mere compliance. Practically, these insights offer a clear directive for businesses and policymakers: fostering sustainable development necessitates encouraging comprehensive green accounting practices that bolster resource efficiency. This suggests a shift towards a holistic corporate strategy where sustainability is deeply integrated into operational and accounting frameworks.

REFERENCES

1. Ardiana, M., Agustina, R., & Pertiwi, D. A. (2023). Increasing Business Eco-Efficiency and Competitive Advantage through the Application of Green Accounting. Buletin Penelitian Sosial Ekonomi Pertanian - Fakultas Pertanian Universitas Haluoleo, 25(1), 1–8. https://doi.org/10.37149/bpsosek.v25i1.363
2. Arszulowicz, M. (2024). Sustainable Development. Practice, Progress, and Proficiency in Sustainability, 241–260. https://doi.org/10.4018/979-8-3693-3011-1.ch012
3. Beck, G. (2023). Material Flow Cost Accounting as a Resource-Saving Tool for Emerging Recycling Technologies. Clean Technologies, 5(2), 652–674. https://doi.org/10.3390/cleantechnol5020033
4. Berkel, R., & Fadeeva, Z. (2020). Role of Industries in Resource Efficiency and Circular Economy (pp. 171–183). Springer, Singapore. https://doi.org/10.1007/978-981-15-1620-7_20
5. Chang, S., Chiu, A., Chu, C. L., Wang, T., & Hsieh, T. (2015). Material flow cost accounting system for decision making: The case of Taiwan SME in the metal processing industry. Asian Journal of Finance & Accounting, 7(1), 117. https://doi.org/10.5296/ajfa.v7i1.7033
6. Chukka, H. G. R., Madem, S., & Konathala, S. K. (2024). Leveraging Green Accounting for Sustainable Development. Advances in Finance, Accounting, and Economics Book Series, 95–128. https://doi.org/10.4018/979-8-3693-5673-9.ch004
7. Cohen, J. (1992). A power primer. Psychological Bulletin, 112(1), 155–159. https://doi.org/10.1037/0033-2909.112.1.155
8. Dahi, H., & Abdullah, H. S. (2024). Application of Material Flow Cost Accounting Technology to Support Green Productivity and Its Reflection in Achieving Sustainable Development. Journal of Information Systems Engineering and Management, 9(1). https://doi.org/10.55267/iadt.07.14314
9. Ekins, P., Hughes, N., Brigenzu, S., Clark, C. A., Fischer-Kowalski, M., Graedel, T., Hajer, M., Hashimoto, S., Hatfield-Dodds, S., Havlik, P., Hertwich, E. G., Ingram, J., Kruit, K., Miligan, E., Moriguchi, Y., Nasr, N., Newth, D., Obersteiner, M., Ramaswami, A., … Westhowk, H. (2016). Resource Efficiency: Potential and Economic Implications. http://pure.iiasa.ac.at/id/eprint/13813/
10. Feng, X. (2024). Green Accounting in China: Challenges, Opportunities, and the Path Forward. Advances in Economics, Management and Political Sciences, 86(1), 1–7. https://doi.org/10.54254/2754-1169/86/20240929
11. Fikri, M. H., Nainggolan, S., & Hutasuhut, J. (2021). LITERATURE REVIEW: ANALYSIS OF GREEN SUPPLY CHAIN MANAGEMENT (GSCM). ECOBISMA (JURNAL EKONOMI BISNIS DAN MANAJEMEN), 8(2), 17. https://doi.org/10.36987/ecobi.v8i2.2074
12. Fundoni, M., & Pischedda, G. (2023). University sustainability assessment and reporting: Preliminary findings from the Italian context. 8th International Conference on Higher Education Advances (HEAd'22), 17. https://doi.org/10.4995/head23.2023.16163
13. Ivanovskaya, A., Kulikova, L., Vetoshkina, E., Bezvidnaya, O., & Valiullin, I. (2019). The use of material flow cost accounting for process losses reduction. 12–15. https://doi.org/10.2991/CSSDRE-19.2019.3
14. Jikomes, N., & Zoorob, M. (2018). The cannabinoid content of legal cannabis in Washington State varies systematically across testing facilities and popular consumer products. Scientific Reports, 8(1). https://doi.org/10.1038/s41598-018-22755-2
15. Juneja, G., Saraswat, P., & Kautish, S. (2024). Strategic Green Accounting and CSR Analytics. Advances in Business Information Systems and Analytics Book Series, 156–172. https://doi.org/10.4018/979-8-3693-2823-1.ch007
16. Kurniawan, Muh. P., Guritno, A. D., Purwantana, B., & Supartono, W. (2024). Material Flow Cost Accounting in Palm Oil: Promoting Transparency in the Use of Materials and Appropriate Scenario in Resource Saving and Waste Reduction. International Journal of Technology. https://doi.org/10.14716/ijtech.v15i1.4393
17. Loen, M. (2018). Penerapan green accounting dan material flow cost accounting (mfca)terhadap sustainable development. 5(1). https://doi.org/10.35137/JABK.V5I1.182

18. Maass, W., Oberender, C., & Vogt, M. (2018). Resource Efficiency for the Manufacturing Industries – A Holistic Approach (pp. 405–415). Springer, Cham. https://doi.org/10.1007/978-3-319-50079-9_29
19. Maio, F. D., & Rem, P. (2015). A Robust Indicator for Promoting Circular Economy through Recycling. Journal of Environmental Protection, 6(10), 1095. https://doi.org/10.4236/jep.2015.610096
20. Maulana, Y., Purnama, D., & Rahmawati, T. (2024). Material Flow Cost Accounting to Enhance Sustainable Development: A Green Accounting Perspective. Jurnal Riset Akuntansi Terpadu. https://doi.org/10.35448/jrat.v17i1.25011
21. Nyide, J. (2016). Material flow cost accounting as a tool for improved resource efficiency in the hotel sector : a case of emerging market. Risk Governance and Control: Financial Markets & Institutions, 6(4), 428–435. https://doi.org/10.22495/RCGV6I4C3ART8
22. Pamungkas, I. D., Indra Satata, D. P., Raihan, M. R., Kristianto, A. Y., & Oktafiyani, M. (2024). Impact of Corporate Social Responsibility Between Implementation of Green Accounting and Sustainable Development Goals (SDGS): A Study on Heavily Polluting Companies in Energy and Transportation & Logistics Sector In Indonesia. Journal of Lifestyle and SDGs Review, 4(4), e02577. https://doi.org/10.47172/2965-730x.sdgsreview.v4.n04.pe02577
23. Putri, R. A., Lasmini, L., & Septiawati, R. (2024). Pengaruh Environmental Performance dan Material Flow Cost Accounting Terhadap Sustainable Development: pada Perusahaan Sektor Industri Dasar dan Kimia yang Terdaftar di Bursa Efek Indonesia Tahun 2020-2022. Al-Kharaj : Jurnal Ekonomi, Keuangan Dan Bisnis Syariah, 6(9). https://doi.org/10.47467/alkharaj.v6i9.2775
24. Putri, H., Handajani, L., & Lenap, I. P. (2024). Pengaruh Green Accounting, Environmental Performance, dan Material Flow Cost Accounting (MFCA) Terhadap Sustainable Development. Performance: Jurnal Bisnis & Akuntansi, 14(1), 317–329. https://doi.org/10.24929/feb.v14i1.2873
25. Rahman, Md. M., & Islam, M. E. (2023). The impact of green accounting on environmental performance: mediating effects of energy efficiency. Environmental Science and Pollution Research, 30(26), 69431–69452. https://doi.org/10.1007/s11356-023-27356-9
26. Rangkuti, M. B. (2023). Green Accounting in Enhancing Sustainability Report Disclosure. International Journal of Research and Review. https://doi.org/10.52403/ijrr.20231156
27. Santi, A. R., Andi, K., Lindrianasari, L., & Oktavia, R. (2022). Pengaruh penerapan material flow cost accounting terhadap green accounting dan financial performance. Fair Value, 5(2), 723–732. https://doi.org/10.32670/fairvalue.v5i2.2327
28. Syah, S., Saraswati, E., Sukoharsono, E. G., & Roekhudin, R. (2020). Blue Accounting and Sustainability. https://doi.org/10.2991/aebmr.k.200606.085
29. Spürk, S., Lottermoser, B. G., & Drobe, M. (2016). Evaluating resource efficiency at major copper mines. https://publications.rwth-aachen.de/record/660827
30. S, N. R., & TS, N. R. (2023). A STUDY ON AWARENESS OF ENVIRONMENTAL ACCOUNTING AND REPORTING WITH SPECIAL REFERENCE TO EMPLOYEES OF THRISSUR DISTRICT. International Journal of Engineering Technology and Management Sciences, 7(3), 8–17. https://doi.org/10.46647/ijetms.2023.v07i03.002
31. Trevanti, G. C., & Yuliati, A. (2023). Pengaruh Penerapan Green Accounting Terhadap Sustainable Development Dengan Ukuran Perusahaan Sebagai Variabel Intervening. Costing. https://doi.org/10.31539/costing.v7i1.6439
32. Turanova, M. V. (2024). Resource efficiency as the component of sustainable development. Collection of Scientific Papers, 115–122. https://doi.org/10.52897/978-5-7310-6504-7-2024-53-115-122
33. Tu, J., & Huang, H. (2019). Relationship between Green Design and Material Flow Cost Accounting in the Context of Effective Resource Utilization. Sustainability, 11(7), 1974. https://doi.org/10.3390/su11071974
34. Walz, M., & Guenther, E. (2020). What effects does material flow cost accounting have for companies?: Evidence from a case studies analysis. Journal of Industrial Ecology, 25(3), 593. https://doi.org/10.1111/jiec.13064

35. Wen, Y., & Zhang, S. (2024). Assessing the impact of resource efficiency on sustainable development: Policies to cope with resource scarcity in Chinese provinces. Resources Policy. https://doi.org/10.1016/j.resourpol.2024.104754
36. Williams, E. D. (2023). Sustainable Development. The Political Economy of Greek Growth up to 2030, 49–58. https://doi.org/10.1007/978-3-031-26872-4_4
37. Zhang, H., Fahlevi, M., Aljuaid, M., Beşer, N. Ö., ÇABAŞ, M., & Jiménez, J. L. (2024). A machine learning and quantile analysis of FINTECH and resource efficiency in achieving sustainable development in OECD countries. Resources Policy, 92, 105017. https://doi.org/10.1016/j.resourpol.2024.105017
38. Zhong, R., Tang, W., & Guan, Z. (2022). Report on resource efficiency and the sustainable development of China's special Economic Zones. In Research series on the Chinese dream and China's development path (pp. 61–81). https://doi.org/10.1007/978-981-19-2207-7_4
39. จันทราทิพย์, ม. (2024). การประยุกต์เทคนิคบัญชีต้นทุนการไหลวัสดุในกระบวนการจัดการคลังสินค้า สำหรับบริษัท xyz. https://doi.org/10.58837/chula.the.2021.1225

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Investment Decisions on Foreign Exchange Market: A Structural Equation Modelling Approach

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Abstract

This study examined investment decisions in the foreign exchange (FOREX) market using Structural Equation Modeling (SEM) to assess the impact of behavioral finance factors. The research focused on how familiarity bias, self-attribution bias, herding, heuristics, prospect-based behavior, and market analysis influence investment choices through a survey of forex traders. Key findings indicated that while familiarity and self-attribution biases were prevalent, their direct effects on decisions were limited. Instead, they influenced decision-making indirectly through herding and heuristics. Prospect-based behavior significantly impacted investment choices, highlighting the role of cognitive biases related to gains and losses. Furthermore, effective market analysis helped mitigate biases and improve investment outcomes. This research provides valuable insights for academics and practitioners, enhancing understanding of investor behavior in the volatile forex market while promoting data-driven decision-making.

Keywords: Behavioral Finance, Foreign Exchange Market, Investment Decisions, Structural Equation Modeling (SEM), Familiarity Bias, Self-Attribution Bias, Herding Behavior, Heuristics, Prospect Theory, Market Analysis

Introduction

The foreign exchange (FOREX) market is the largest and most liquid financial market in the world, with a daily trading volume exceeding USD 7.5 trillion as of 2022. It operates continuously across time zones, facilitating international trade and investment by enabling currency conversions. The FOREX market also serves as a platform for speculative trading, where participants seek to profit from fluctuations in exchange rates. This decentralized market consists of a network of banks, brokers, hedge funds, corporations, and central banks, with electronic trading platforms enhancing its accessibility and liquidity. Given its inherent volatility and the significant impact of global economic and political events on exchange rates, understanding investment decisions in this market is timely and essential.

Despite its importance, existing literature on the FOREX market has primarily focused on technical and fundamental analysis, often relying on traditional financial theories that assume investors act rationally based on available information. However, the emergence of behavioral finance has challenged this assumption, highlighting the influence of cognitive biases and emotional factors in shaping investment behavior. This gap in understanding the psychological aspects of FOREX trading decisions calls for closer examination, particularly regarding how familiarity bias, self-attribution bias, herding, heuristics, prospect-based behavior, and market analysis influence trader behavior and outcomes.

This study aims to bridge this gap by exploring the various ways in which behavioral finance dimensions affect investment decisions in the FOREX market. Specifically, it seeks seeks to evaluate the extent to which familiarity bias and self-attribution bias shape perceptions and behaviors, investigate the roles of herding, heuristics, and prospect-based behavior in decision-making, and analyze the mitigating influence of market analysis on cognitive biases. Additionally, it examines the direct and indirect effects of these biases on trading decisions, providing a comprehensive framework to understand the interplay between behavioral tendencies and market strategies.

The findings of this research have significant implications across various domains. For investors, the insights can enhance self-awareness about psychological factors that may hinder rational decision-making, ultimately improving investment performance. Financial institutions can leverage these findings to refine market regulations and enhance investor satisfaction, fostering greater participation. For students and researchers, this study contributes valuable knowledge to the growing field of behavioral finance, serving as both a reference and a foundation for future inquiries. Lastly, this research enriches academic

discourse by integrating behavioral insights into the study of investment decisions in the highly dynamic and unpredictable FOREX market.

Literature Review

Foreign Exchange Market

The foreign exchange (forex) market is the largest and most liquid financial market globally, operating 24/7 with a daily trading volume exceeding $6 trillion (Nordström, 2020). Its primary functions include facilitating international trade and investment by enabling currency conversions and serving speculative purposes, where traders profit from exchange rate fluctuations (Chaboud et al., 2020). The market's decentralized structure involves a network of banks, brokers, hedge funds, corporations, and central banks, with electronic trading platforms enhancing accessibility and liquidity (Nordström, 2020). Key participants such as commercial banks, central banks, and institutional investors dominate the market, while retail traders have gained prominence through online platforms (Chaboud et al., 2020).

Despite comprehensive studies, gaps remain, particularly in understanding the psychological factors affecting trading decisions and the impact of technological innovations like blockchain on traditional trading (Nordström, 2020; Chaboud et al., 2020). Regulatory challenges and the effectiveness of recent measures against market manipulation and systemic risk also require further empirical validation (Chaboud et al., 2020). Theoretical frameworks such as the Efficient Market Hypothesis (EMH) and Purchasing Power Parity (PPP) provide foundational understanding but fall short in explaining short-term anomalies and trader behaviour influenced by biases (Nordström, 2020; Chaboud et al., 2020).

Behavioral Finance

Making investment decisions in the forex market is crucial and requires understanding the psychological factors that influence traders' choices. A study conducted by researchers from Hassan First University of Settat, Talhartit, Ait Jillali, and El Kabbouri aimed to explore how behavioural biases impact forex traders' decisions and performance in uncertain market conditions.

The study sheds light on cognitive biases such as overconfidence, loss aversion, and herding behaviour significantly influencing market dynamics. Emotional factors and risk perceptions are crucial in shaping forex trading behaviour. However, the existing literature has limitations in addressing individual differences in susceptibility to biases among traders and exploring bias persistence over time.

Despite these limitations, theoretical frameworks like prospect theory offer valuable insights into traders' behaviour. Integrating such frameworks with empirical research can enhance our understanding of the complex interplay between psychological factors and market dynamics in forex trading. This review emphasizes the importance of integrating behavioural finance principles into forex investment studies and suggests avenues for further research to refine theoretical frameworks and mitigate biases; this, in turn, can enhance market predictability and risk management strategies.

Familiarity Bias

Familiarity bias is a cognitive bias that influences individuals to favour familiar options over unfamiliar ones, often leading to suboptimal economic decisions. This review delves into the impact of familiarity bias on economic behaviour, drawing from various studies and research. Familiarity bias has implications for investment decisions, spending habits, and market inefficiencies. Research on familiarity bias spans finance, marketing, and consumer behaviour and is grounded in human psychology. It can result in a comfort zone effect, where individuals feel secure and self-assured despite potential drawbacks. However, there is a limited exploration of the underlying psychological mechanisms. While much research focuses on negative impacts, there is inadequate examination of positive outcomes, such as expedited decision-making in familiar contexts. Despite extensive documentation, gaps exist, including the interaction between familiarity bias and other cognitive biases and the influence of technological advancements and globalization. Zhu et al. (2020) have advocated for more empirical studies in these areas. Theoretical frameworks such as the heuristic-systematic model and behavioural finance theories aid in explaining why familiarity bias leads to predictable patterns that deviate from rational choice models. Addressing these gaps could enhance the predictive power of economic models and improve strategies for mitigating the adverse effects of familiarity bias in financial and consumer behaviour.

Self-Attribution Bias

Self-attribution bias is a cognitive tendency where individuals credit their successes to their actions and skills while attributing failures to external factors. This bias significantly influences financial decision-making by distorting investors' perceptions of their abilities. Hoffmann and Post (2017) demonstrate that investors often wrongly attribute positive investment returns to their skills, which boosts their confidence and can lead to overtrading. Similarly, Priyadarsini and Prithi (2018) point out that self-attribution bias can cause investors to underestimate risks and overestimate their control over financial outcomes, resulting in suboptimal investment decisions.

Research on self-attribution bias spans various domains, including psychology, behavioural finance, and consumer behaviour. Hoffmann and Post (2017) provide empirical evidence that investment returns significantly influence individuals' beliefs in their skills, leading to increased trading activity, excessive risk-taking, and reduced portfolio diversification. Priyadarsini and Prithi (2018) discuss how this bias affects investor confidence and decision-making, emphasizing its role in market anomalies and financial bubbles.

However, the literature often overlooks the interaction between self-attribution bias and other cognitive biases, such as overconfidence, and there is insufficient attention to how institutional investors and financial advisors might mitigate or amplify self-attribution bias. Several research gaps persist, including the limited exploration of cultural and demographic factors that influence the degree of self-attribution bias and the impact of technological advancements, like algorithmic trading and robo-advisors, on this bias. Hoffmann and Post (2017) call for more comprehensive studies integrating psychological insights with financial behaviour.

Theoretical frameworks commonly employed include behavioural finance theories and psychological models of self-perception, such as the prospect theory framework and cognitive dissonance theory, to explain how self-attribution bias affects decision-making under risk. Understanding the full scope of self-attribution bias requires addressing these gaps, which will enhance theoretical models and contribute to more effective strategies for mitigating its adverse effects in financial markets.

Herding

Herding behaviour, in which individuals mimic the actions of a larger group, can result in market inefficiencies and bubbles. Psychological and social factors, such as informational cascades, concerns about reputation, and psychological biases, influence this behaviour. Herding is particularly evident during times of market volatility and uncertainty. Overconfidence worsens herding, causing investors to follow the crowd without critically evaluating their information. Financial literacy plays a moderating role, suggesting that better-informed investors are less likely to exhibit herding behaviour.

Despite extensive research on herding behaviour, there needs to be more understanding of its long-term implications for market stability and individual financial outcomes. There is a need for more empirical studies on how technological advancements, cultural and regional differences, and financial education programs can mitigate or amplify herding tendencies.

Theoretical frameworks commonly used to study herding behaviour include behavioural finance theories and models of social influence. Dewan and Dharni (2019) use the informational cascade model to explain herding. Pratiwi et al. (2020) apply the overconfidence theory to illustrate how investors overestimating their abilities can lead to increased herding. Financial literacy is analyzed through cognitive and educational psychology, showing that better-informed individuals are less susceptible to herding. Herding behaviour significantly impacts financial markets, often resulting in irrational investment decisions and market inefficiencies. Addressing these gaps will enhance theoretical models of herding behaviour and contribute to more effective strategies for mitigating its adverse effects on financial markets.

Heuristics

This review explores the impact of heuristics on investment decisions, specifically focusing on how locus of control moderates their influence. Heuristics are mental shortcuts that aid decision-making by reducing cognitive load and enabling quick judgments based on limited information. Understanding the role of heuristics in investment contexts is crucial for identifying decision biases and their effect on financial outcomes. Dangol and Manandhar (2020) examine how locus of control affects heuristic-based decision-making, suggesting that individuals with an internal locus of control rely less on heuristics than those with an external locus of control. Additionally, Sarin and Chowdhury (2021) comprehensively explore heuristics in investment decisions, emphasizing their impact on investor behaviour and outcomes across different profiles.

A critical analysis reveals the dual nature of heuristics in investment decisions: they expedite decision-making. They can lead to acceptable outcomes under specific conditions but also introduce biases and errors when used without discretion. Personal beliefs about control over outcomes influence heuristics and require tailored strategies to mitigate potential pitfalls. Both studies utilize frameworks such as Kahneman and Tversky's prospect theory and bounded rationality to explain how heuristics shape decision-making under uncertainty. Future research should concentrate on refining strategies that leverage the benefits of heuristics while mitigating the inherent risks in investment decision-making processes.

Prospect Based Behavior

The theory of prospect, developed by Kahneman and Tversky, has significantly impacted behavioural economics by explaining how individuals make decisions when faced with uncertainty. This review summarises recent studies that apply prospect theory to investment decisions, examining its theoretical foundations and practical implications for understanding investor behaviour. It demonstrates that investment decisions are not solely based on rational evaluations of risk and return but are also influenced by psychological biases.

Prospect theory suggests that individuals evaluate potential gains and losses about a reference point, showing aversion to risk when it comes to gains and seeking risk when it comes to losses. Babarinde, Bello, and Bello (2022) study behavioural factors that affect investment decisions, highlighting psychological biases such as loss aversion and framing effects. Gautam and Kumar (2022) explore the application of prospect theory, emphasizing its role in shaping risk perceptions and decision outcomes in investment contexts. Erdogan (2021) contributes a comparative study that assesses prospect theory's explanatory power across diverse investor groups and market conditions, revealing nuanced adherence to its principles.

The critical analysis highlights prospect theory's ability to explain deviations from rational decision-making but acknowledges its limitations in addressing real-world complexities. While useful for understanding risk management, its application varies across investment contexts and demographics. Integrating prospect theory with other behavioral models could improve its accuracy and relevance in financial decisions. In conclusion, prospect theory provides valuable insights, but further research is needed to refine its use and mitigate biases, ultimately enhancing decision-making processes in finance.

Market Analysis

It is crucial to conduct market analysis to inform investment decisions by assessing factors that impact market dynamics and investor behaviour. The study by Islam and Bhuiyan (2019) on the capital market of Bangladesh delves into the influence of demographic factors, market information, and behavioural biases on individual investment decisions. This research underscores the importance of understanding investor behaviour to improve market efficiency. Haidari's (2021) comprehensive analysis highlights the effect of cognitive biases and emotional influences on investment performance, advocating for strategies to mitigate these biases. Furthermore, Musfidah, Aji, and Hartono's (2021) literature review defines investment decision-making in the stock market by integrating theoretical perspectives and empirical findings to provide a comprehensive framework for comprehending investment behaviour.

Collectively, these studies emphasize the multidimensional nature of market analysis and its significance in investment decision-making. They reveal the impact of demographic and psychological factors on investor behaviour and suggest that personalized strategies could enhance investment performance. Nevertheless, more effort is still needed to incorporate these insights into practical strategies, necessitating further research to develop and test models that integrate demographic, psychological, and market factors. Moreover, exploring technological advancements such as AI and machine learning could improve market analysis and investment decision-making; rooted in behavioural finance theories, including prospect theory and the efficient market hypothesis, these studies enhance our understanding of how investors process information, perceive risk, and make decisions under uncertainty.

Investment Decision

The process of making investment decisions is influenced by many factors, encompassing psychological, cognitive, and market influences, which extend beyond mere risk-return assessments. Demographic elements such as age, education, and income also wield significant influence over investment choices. Haidari's (2021) analysis underscores behavioural biases like overconfidence, herding, and loss aversion as contributors to less- than-optimal decisions. Furthermore, the literature review by Musfidah, Aji, and Hartono (2021) provides a comprehensive perspective on investment decision-making in the stock market, identifying market sentiment, investor psychology, and information asymmetry.

These studies shed light on the intricate nature of investment decisions and reveal gaps in understanding the relative importance of factors across different contexts. Future research should delve into investment decision-making's dynamic and contextual aspects, encompassing longitudinal studies and cross-cultural analyses. Theoretical frameworks such as prospect theory and bounded rationality can help elucidate the impact of psychological biases and heuristics on investment decisions. By integrating insights from behavioural finance with traditional economic theories, the researchers can enrich our comprehension of investment decision-making, ultimately leading to more effective strategies for enhancing investment performance and market efficiency.

Gaps in Literature

Despite advancements in behavioral finance, there are still significant gaps in understanding investment decisions in the foreign exchange (FOREX) market. This is particularly true regarding the interaction of cognitive biases, the impact of technological advancements, cultural and demographic variability, temporal dynamics, and the integration of behavioral and traditional finance theories.

While individual biases such as familiarity, self-attribution, and herding have been explored, there has been little research on how these biases interact within complex environments like the FOREX market. This raises questions about whether these biases amplify or counterbalance each other. Additionally, the influence of technologies such as algorithmic trading and robo-advisors on these behavioral biases is still poorly understood, especially as these tools become increasingly central to investor decision-making.

Furthermore, behavioral finance often overlooks cultural and demographic factors, particularly in emerging markets like Southeast Asia. Most studies rely on cross-sectional data, which lacks the longitudinal perspective needed to track how biases evolve in response to changing market conditions. Lastly, there is a pressing need for integrated models that combine psychological insights with traditional financial theories to provide a more comprehensive understanding of investor behavior.

Theoretical Framework

Investment decisions in the foreign exchange (FOREX) market are influenced by psychological factors, cognitive biases, and market dynamics. This study utilizes three theoretical frameworks: the Theory of Planned Behavior (TPB), the Efficient Market Hypothesis (EMH), and Arbitrage Pricing Theory (APT).

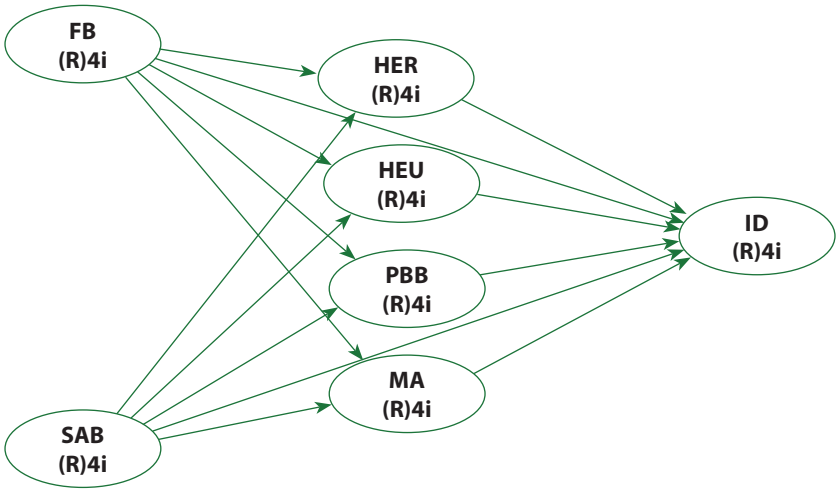


Fig. 1. Conceptual Framework – Investment Decisions on Foreign Market Exchange

Note. Familiarity Bias (FB), Self-Attribution Bias (SAB), Herding (HER), Heuristics (HEU), Prospect Based Behavior (PBB), Market Analysis (MA), and Investment Decision (ID)

Theory of Planned Behavior (TPB) Developed by Fishbein and Ajzen, the TPB posits that individuals' intentions and behaviors are shaped by their attitudes, subjective norms, and perceived behavioral control. In forex trading, these factors influence decision-making, as traders' attitudes toward risk and social norms affect their investment choices. This study integrates the TPB with biases such as herding and prospect theory to better understand decision-making in the volatile forex market.

Efficient Market Hypothesis (EMH) Proposed by Eugene Fama, the EMH suggests that financial markets reflect all available information, making it difficult for investors to consistently achieve returns that outperform the market. This study investigates how psychological factors impact market efficiency and whether data-driven strategies can help mitigate cognitive biases in forex trading.

Arbitrage Pricing Theory (APT) Developed by Stephen Ross, the APT is a multifactor model that considers both macroeconomic and firm-specific factors in asset pricing. This framework helps explore how external conditions, such as global events, interact with psychological biases to shape investment decisions. The study employs Structural Equation Modeling (SEM) to link the constructs of APT with behavioral dimensions.

Methodology

This study used a quantitative research design to examine the relationships between behavioral biases and investment decisions in the foreign exchange (FOREX) market, utilizing Structural Equation Modeling (SEM) as the primary analytical tool. SEM enabled the simultaneous testing of multiple hypotheses, revealing both direct and indirect effects of cognitive and emotional biases on investment choices. The study targeted a sample of 300 participants to ensure robust findings. Various sampling techniques were used, including convenience sampling for readily available participants, purposive sampling for active forex traders, and snowball sampling to expand the participant pool through referrals. Data was collected through a multi-faceted approach that featured online surveys, email invitations, and in-person questionnaires at financial institutions, enhancing inclusivity and representativeness. Data collection relied on a carefully designed questionnaire that assessed participants' investment patterns and biases. Multiple items were included for each behavioral factor, such as familiarity bias, self-attribution bias, and herding behavior. Respondents rated their agreement on a Likert scale from 1 (Strongly Disagree) to 5 (Strongly Agree). The questionnaire underwent rigorous validation, including pilot testing and review by industry experts, ensuring

reliability and clarity. Reliability was assessed using Cronbach's alpha, with a value above 0.7 considered acceptable. Data analysis involved descriptive statistics for sample characteristics and hypothesis testing using t-tests to assess relationships. Structural Equation Modeling (SEM) was used to evaluate the hypothesized relationships and the overall fit of the model, following a systematic process of model specification, identification, estimation, evaluation, and modification.

Table 1. Cronbach's Alpha-Reliability Testing Results

Variable	Cronbach's Alpha
Familiarity Bias	0.916
Self-Attribution Bias	0.946
Herding	0.961
Heuristics	0.846
Prospect Based Behavior	0.846
Market Analysis	0.770
Investment Decision	0.708

Note: Table 1 shows the construct reliability of the instrument, with Cronbach's Alpha values above 0.7 indicating acceptable internal consistency (Hair, Ringle, & Sarstedt, 2011). All variables meet this criterion, confirming their reliability.

Analysis and Discussions

Descriptive Statistics

The descriptive statistics provided an overview of the key findings from the data collected on behavioral biases and investment behaviors influencing decisions in the FOREX market. **Familiarity bias** demonstrated mean values ranging from 3.80 to 4.01, with standard deviations between 0.76 and 0.88. These results indicated a generally positive sentiment among traders towards familiar investments, with moderate variability in responses. **Self-attribution bias** showed mean values from 3.85 to 4.04 and standard deviations between 0.79 and 0.89, suggesting that traders frequently attributed their successes to personal skill, with notable consensus among respondents.

Herding behavior revealed mean values ranging from 3.68 to 3.84, with standard deviations between 0.78 and 0.92, highlighting a tendency for traders to follow the actions of others rather than relying solely on independent analysis. **Heuristics**, characterized by the use of mental shortcuts in decision-making, had mean values from 3.65 to 3.85, with standard deviations between 0.77 and 0.89, indicating that traders generally relied on simplified decision-making strategies, with broad agreement among respondents.

For **prospect-based behavior**, which assesses risk aversion for gains and risk-seeking for losses, the mean values ranged from 3.68 to 3.88, with standard deviations between 0.76 and 0.88. These findings underscored the prevalence of cognitive biases shaped by the framing of potential outcomes. Market analysis, a critical aspect of informed decision-making, exhibited mean values from 3.75 to 3.90, with standard deviations between 0.77 and 0.89, reflecting a strong reliance on analytical tools and data-driven approaches among traders.

Model Fit Indices

Table 2. Model Fit Indices

Model Fit Indices		
	Saturated Model	**Estimated Model**
SRMR	0.064	0.071
(ULSDF) Unweighted Least Squares Discrepancy Function	5.516	6.62
Geodesic Discrepancy Function	1.046	1.109
Chi-Square	1667.62	1733.18

Table 2 presents the model fit indices for the structural equation model (SEM), which assesses how well the proposed model aligns with the observed data. Fit indices are crucial for evaluating complex correlations among multiple variables, including latent constructs.

A better fit is indicated by lower values, with below 0.08 generally considered acceptable. The estimated model's Standardized Root Mean Residual (SRMR) is 0.071, slightly higher than the saturated model's SRMR of 0.064, suggesting a marginally poorer fit. The Unweighted Least Squares Difference Function (ULSDF) for the estimated model is 5.516 compared to 6.620 for the saturated model, indicating some discrepancy.

The Geodesic Discrepancy Function statistic for the estimated model is 1.109, while the saturated model's value is 1.046, reflecting a slight decline in fit. Although less commonly discussed, this index is useful for assessing complex relationships among latent constructs (Bentler, 1990).

Finally, the Chi-Square test (χ^2) compares the predicted and actual covariance matrices. A well-fitting model will have a non-significant p-value ($p > 0.05$) and a lower Chi-Square value. The estimated model's Chi-Square value is 1733.180, higher than the saturated model's value of 1667.620, indicating a slightly poorer fit. It's important to note that larger samples can inflate Chi-Square values, impacting interpretation (Shi, Lee, & Maydeu-Olivares, 2018; Hu & Bentler, 1998).

Model Selection

Table 3. Model Selection Criteria

Model Selection Criteria		
Variables	**AIC** (Akaike's Information Criterion)	**BIC** (Bayesian Information Criteria)
Herding	–51.158	–39.987
Heuristics	–23.08	–11.909
Investment Decision	–64.575	–31.063
Market Analysis	4.046	15.217
Prospect-Based Behavior	–5.67	5.501

Table 3 presents the model selection criteria using AIC (Akaike's Information Criterion) and BIC (Bayesian Information Criterion). These criteria assess the goodness of fit of different models while penalizing for model complexity, making them essential tools in model selection. As shown in the table, for the variable herding, the herding AIC is –51.158 which indicates that the model appears to function rather effectively, based on the AIC's negative value. A lower or negative value suggests that the model strikes a reasonable balance between simplicity and fit because AIC penalizes for model complexity. The Bayesian Information Criterion (BIC) penalizes complexity more heavily than AIC. A negative value of –39.987, suggests the herding model fits the data well even under the stricter complexity penalty imposed by BIC. Therefore, both AIC and BIC suggest this model is a strong performer in terms of fit and simplicity.

The heuristics model has a negative AIC value, which indicates a good match, just as the herding model. Even if the number is greater than the herding, it still implies that the heuristic model is suitably balancing model complexity and fit. Heuristics have a negative BIC value as well, however not as low as AIC. The model still strikes a compromise between complexity and fit under BIC, but when more conservative criteria are used, the model may be somewhat more complicated or less well-fitted, as indicated by the higher BIC score when compared to AIC. Moreover, among all the variables, the investment decision model has the lowest AIC value, indicating it offers the best balance between fit and complexity. An extremely negative AIC indicates that the model maintains minimal complexity while fitting the data exceedingly well. Even with a stricter penalty, the investment decision model effectively balances fit and complexity, as indicated by the negative BIC value. Despite having a larger BIC than AIC, the model performs well.

In contrast to the preceding variables, market analysis has a positive AIC value, indicating a lower level of efficiency and potential for poorer data fit. In general, models with positive AIC values exhibit a worse fit, particularly in contrast to those with negative AIC values. In comparison to other factors, the market analysis model's lower efficiency is further confirmed by the positive BIC. This relatively high BIC value indicates that when penalizing for complexity, the model does not perform well. For the prospect-based behavior, the model matches the data rather well, however not as effectively as the herding or investment decision models, according to the negative AIC. Even if the value is less than that of several other variables, it still shows that fit and simplicity are reasonably balanced. Thus, the prospect-based behavior model is less effective when complexity is taken into consideration supported by the positive BIC value. While the AIC is negative, the larger BIC shows that the model could be poorly fitted under more conservative criteria.

Structural Equation Model

We, the researchers, used structural equation modeling (SEM) analysis to assess the hypothesized relationships between exogenous and endogenous variables, validate the research model, and scrutinize the proposed hypotheses.

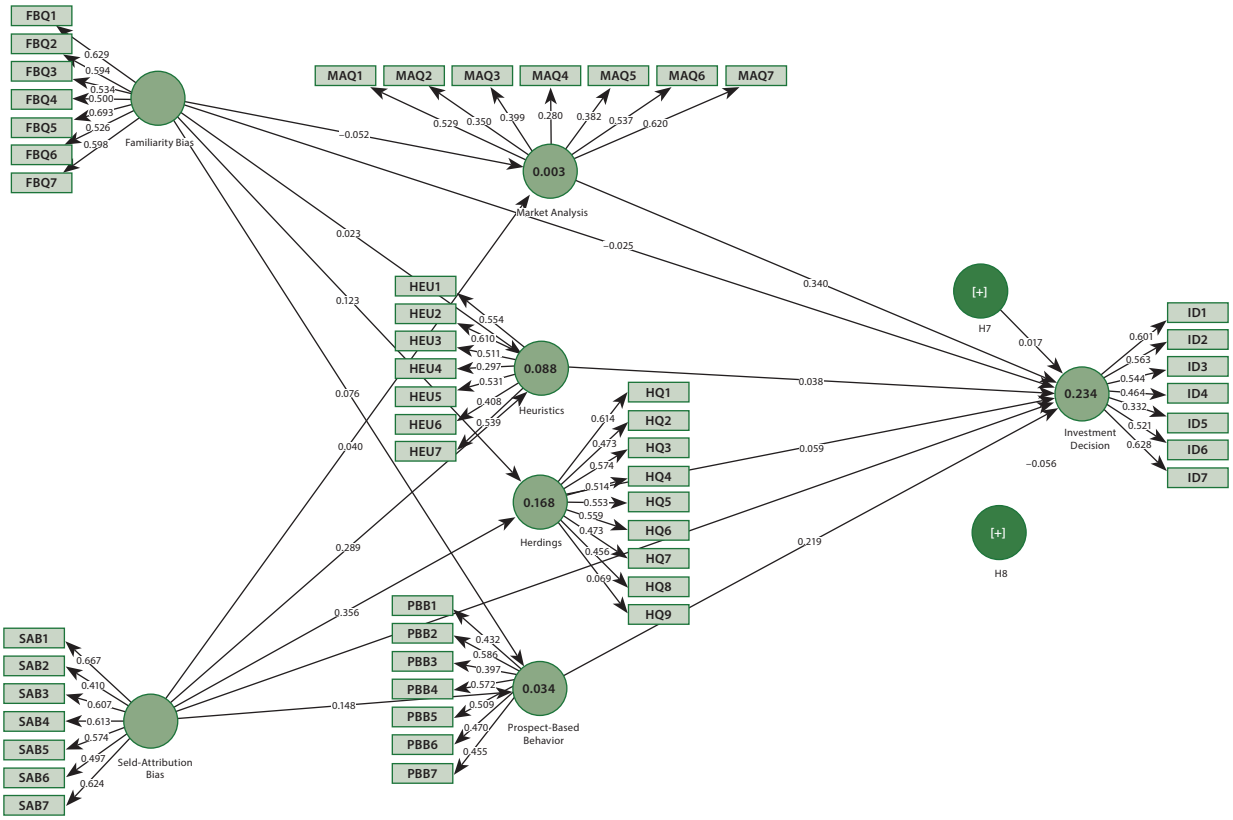


Fig. 2. Structural Equation Model

Figure 2 displays the structural equation model of the relationship between Familiarity bias, self-attribution bias, herding, heuristics, prospect-based behavior, market analysis, and investment decision. The Path coefficient between familiarity bias and investment decision is -0.025 with p-value greater than the 0.05 threshold indicating that there is no significant relationship between the variables suggesting that familiarity bias do have a significant direct effect on investment decision. As for the direct influence of self-attribution bias on investment decision, the Path coefficient is 0.069 with p-value greater than 0.05, also suggesting that self-attribution bias has no significant effect on the investment decisions of the respondents.

Moreover, Table 4 presents the hypotheses testing using the regression output of the SEM model for the hypothesis testing. The model provides the estimate, t-statistic, SD, p-values, and the hypothesized relationship between the variables.

Hypotheses Testing

Table 4. Hypotheses Testing

Hypotheses Testing					
Path	***β***	***SD***	***t***	***p***	**Decision**
H_1: Familiarity Bias → Herding	0.123	0.126	0.974	0.33	Fail to reject Ho
H_2: Familiarity Bias → Heuristics	0.023	0.136	0.166	0.868	Fail to reject Ho
H_3: Familiarity Bias → Investment Decision	–0.025	0.097	0.253	0.8	Fail to reject Ho
H_5: Familiarity Bias → Prospect-Based Behavior	0.076	0.147	0.519	0.604	Fail to reject Ho
H_6: Herding → Investment Decision	0.059	0.085	0.692	0.489	Fail to reject Ho
H_7: Heuristics → Investment Decision	0.038	0.078	0.49	0.624	Fail to reject Ho

Hypotheses Testing					
Path	***β***	***SD***	***t***	***p***	**Decision**
H_8: Prospect-Based Behavior Investment Decision	0.219	0.072	3.036	0.003	Reject Ho
H_9: Self-Attribution Bias → Herding	0.356	0.117	3.043	0.002	Reject Ho
H_{10}: Self-Attribution Bias → Heuristics	0.289	0.066	4.383	<0.001	Reject Ho
H_{11}: Self-Attribution Bias → Investment Decision	0.069	0.76	0.904	0.367	Fail to reject Ho
H_{12}: Self-Attribution Bias → Prospect-Based Behavior	0.148	0.083	1.789	0.074	Fail to reject Ho

The table presents the mediation and moderation analysis conducted with the PLS-SEM. As shown in the table, familiarity bias has no statistically significant effects on the herding (β = 0. 123, t = 0. 974, p = 0. 330), ***heuristics*** (β = 0. 023, t = 0. 166, p = 0. 868), ***and*** prospect-based behavior (β = –0. 025, t = 0. 253, p = 0. 800) of the respondents. This result suggests that herding, heuristics, and prospect-based behavior does not significantly influence the familiarity bias of the respondents. Self-attribution behavior has a significant influence on the herding (β = 0. 356, t = 3. 043, p = 0. 002) and heuristics (β = 0. 289, t = 4. 383, p < 0. 001) of the respondents. The 0.356 estimate suggests that for every unit increase in self-attribution bias, there is an approximately 0.356 unit increase in the herding of the respondents. Similarly, 0.289 estimate indicate that for every unit increase in the self-attribution bias of the respondents, there is an approximately 0.289 unit increase in heuristics. In addition, the self-attribution bias does not have a statistically significant relationship to investment decision (β = 0. 069, t = 0. 904, p = 0. 367) and prospect-based behavior (β = 0. 148, t = 1. 789, p = 0. 074), suggesting that self-attribution bias do not influence the investment decision and prospect-based behavior of the respondents.

Moreover, herding and heuristics do not significantly influence the investment decision of the respondents. On the other hand, prospect-based behavior (β = 0. 219, t = 3. 036, p = 0. 003) was observed to have a significant effect on the investment decision of the respondents. The 0.219 estimate indicates that for every unit increase in the prospect-based behavior of the respondents, there was an approximately 0.219 unit increase in their investment decision.

The SEM results highlight that while Familiarity Bias has no significant impact on Herding, Heuristics, or Prospect-Based Behavior, Self-Attribution Bias has significant effects on Herding and Heuristics. Prospect-Based Behavior, regardless of bias, is a strong predictor of Investment Decisions, supporting the idea that cognitive biases tied to gains and losses influence how people make investment choices. However, direct pathways from biases to investment decisions remain largely non-significant, indicating that biases might influence decisions primarily through intermediate behaviors rather than directly.

Table 5. Indirect Effects

Hypotheses Testing					
Path	***β***	***SD***	***t***	***p***	**Decision**
H_{13}: Family Bias → Herding → Investment Decision	0.007	0.019	0.386	0.7	Fail to reject Ho
H_{14}: Self-Attribution → Herding → Investment Decision	0.021	0.031	0.678	0.498	Fail to reject Ho
H_{15}: Familiarity Bias → Heuristics → Investment Decision	0.001	0.013	0.067	0.947	Fail to reject Ho
H_{16}: Self-Attribution Bias → Heuristics → Investment Decision	0.011	0.024	0.466	0.641	Fail to reject Ho
H_{17}: Familiarity Bias → Prospect-Based Behavior → Investment Decision	0.017	0.035	0.472	0.637	Fail to reject Ho
H_{18}: Self-Attribution Bias → Prospect-Based Behavior → Investment Decision	0.032	0.022	1.494	0.136	Fail to reject Ho
H_{19}: Familiarity Bias → Market Analysis → Investment Decision	0.017	0.048	0.354	0.724	Fail to reject Ho
H_{20}: Self-Attribution → Market Analysis → Investment Decision	-0.056	0.45	1.247	0.213	Fail to reject Ho

Table 5 presents the indirect relationship between the variables. This table primarily examine the mediation role of herding, heuristics, and prospect-based behavior in the relationship familiarity bias and self-attribution bias on the investment decision of the respondents. This table also examine the moderating role market analysis on the relationship familiarity bias and self-attribution bias on the investment decision. As shown in the table, the indirect effects between familiarity biased and investment decision mediated by herding (β = 0. 007, t = 0. 386, p = 0. 700), ***heuristics*** (β = 0. 001, t = 0. 067, p = 0. 947), ***and*** prospect-based behavior (β = 0. 017, t = 0. 472, p = 0. 637) are all not significant.

This suggests that herding, heuristics, and prospect-based behavior are not a significant mediator of the relationship between familiarity bias and investment decision. Similarly, the indirect relationship between self-attribution bias and investment decision mediated by herding (β = 0. 021, t = 0. 678, p = 0. 498) ***heuristics*** (β = 0. 011, t = 0. 466, p = 0. 641), ***and*** prospect-based behavior (β = 0. 032, t = 1. 494, p = 0. 136) are all not significant. This indicate that herding, heuristics, and prospect-based behavior are not a significant mediator of the relationship between self-attribution bias and investment decision.

Furthermore, the table also presents the mediating role of market analysis on the relationship between familiarity bias, self-attribution bias, and investment decision. As shown in the table, the interaction between familiarity bias and market analysis (β = 0. 017, t = 0.048, p = 0. 724) was found to be not significant, indicating that market analysis does not substantially moderate the relationship between familiarity bias and investment decisions. The interaction between self-attribution bias and market analysis (β =– 0. 056, t = 1. 247, p = 0. 213) was also found to be not significant. Although the negative coefficient suggests that market analysis may dampen the effect of self-attribution bias on investment decisions, the lack of statistical significance means this finding should be interpreted with caution. Overall, through herding, heuristics, and prospect-based behavior, neither familiarity bias nor self-attribution bias significantly affect investment decisions, based on the indirect effects result from SEM. Moreover, interactions between market analysis and the two behavioral biases also do not moderate these associations. These results underscore the complexity of investor behavior and suggest that additional components or processes that are not included in this model may be involved in the direct and indirect pathways connecting biases to decision-making.

Synthesis of Findings

The SEM results underscored the complexity of behavioral influences on investment decisions in the FOREX market. While familiarity bias and self-attribution bias lacked significant direct effects, they indirectly shaped decision-making through herding and heuristics. Prospect-based behavior stood out as a critical determinant, reflecting the central role of cognitive biases related to perceived gains and losses. Additionally, the reliance on market analysis highlighted its potential to mitigate the adverse effects of biases, ultimately supporting improved investment outcomes. These findings contribute to a nuanced understanding of trader behavior and the mechanisms that drive decision-making in highly volatile financial markets.

Discussion of Results

Interpretation of Results The findings of this study provided significant insights into the role of behavioral biases in shaping investment decisions within the FOREX market. While familiarity bias and self-attribution bias were prevalent among traders, their direct impact on investment decisions was limited. Instead, these biases influenced decision-making indirectly through intermediate behaviors such as herding and heuristics. This aligns with previous research indicating that cognitive biases often affect behavior through complex mechanisms rather than exerting straightforward effects (Hoffmann & Post, 2017; Kumar & Goyal, 2016).

The substantial influence of self-attribution bias on herding and heuristics reinforced the idea that overconfidence and the tendency to attribute successes to personal skill increased reliance on social cues and mental shortcuts. These findings corroborated studies by Christie and Huang (1995) and Sias (2004), which highlighted the dual role of herding behavior in stabilizing markets during stress while also contributing to inefficiencies and speculative bubbles.

Prospect-based behavior, rooted in prospect theory, emerged as a critical factor in investment decisions, emphasizing the role of cognitive biases associated with gains and losses. This supports the seminal work of Kahneman and Tversky (1979) and more recent findings by Babarinde, Bello, and Bello (2022), which demonstrated risk-averse tendencies for gains and risk-seeking behaviors for losses among investors. The significant reliance on market analysis further underscored its importance in mitigating biases and enhancing investment outcomes. This finding aligns with Fama's (1970) efficient market hypothesis, which stressed that asset prices reflect all available information, as well as Shiller's (2003) argument that understanding investor psychology is crucial to comprehending market dynamics.

Theoretical Contributions

This study contributed to the theoretical discourse on investment decision-making in the FOREX market by integrating behavioral finance frameworks with empirical evidence. By applying the Theory of Planned Behavior (TPB), Efficient Market Hypothesis (EMH), and Arbitrage Pricing Theory (APT), the research presented a comprehensive framework for analyzing how cognitive biases and market factors jointly influence trader behavior. The findings extended the application of TPB by demonstrating the significant roles of attitudes, subjective norms, and perceived behavioral control in shaping investment behaviors in a high-stakes, dynamic market like FOREX. Additionally, the results supported the EMH by underscoring the importance of market analysis in ensuring that asset prices reflect all available information. However, the study also highlighted the limitations of the EMH when behavioral biases disrupt rational decision-making processes. By incorporating APT, the research effectively captured the multifaceted influences of macroeconomic and firm-specific factors on investment decisions, offering a holistic view of how psychological and external market forces interact.

Practical Implications

The practical implications of this study are considerable, offering valuable insights for investors, financial institutions, and regulatory bodies.

For investors, the study highlighted the need to understand the indirect influence of familiarity bias and self-attribution bias on their decision-making processes. By acknowledging these biases, investors could develop strategies to diversify their portfolios beyond familiar assets and critically evaluate their choices to mitigate overconfidence and reliance on heuristics.

For financial institutions, the findings suggested opportunities to design educational programs and decision-support tools aimed at helping investors recognize and manage their cognitive biases. Institutions could promote data-driven decision-making by providing access to comprehensive and reliable market analyses, enhancing investor satisfaction, and fostering greater participation in the FOREX market.

For regulatory bodies, the insights could inform the development of policies that address the impact of behavioral biases on market stability. For example, increasing transparency and reducing information asymmetry in the FOREX market could help mitigate herding behavior and other cognitive biases that contribute to market inefficiencies.

Conclusions

Summary of Findings

This study examined the influence of behavioral finance factors on investment decisions in the foreign exchange (FOREX) market using Structural Equation Modeling (SEM). The findings revealed several key insights. Familiarity bias and self-attribution bias, although common among traders, did not exhibit significant direct effects on investment decisions. However, they did influence decision-making indirectly through intermediary behaviors such as herding and heuristics. Specifically, self-attribution bias had a significant impact on herding and heuristics, demonstrating that overconfidence and the tendency to attribute successes to personal skill led traders to rely heavily on social cues and mental shortcuts. Prospect-based behavior, influenced by prospect theory, emerged as a significant determinant of investment decisions. Traders demonstrated risk-averse behavior for gains and risk-seeking tendencies for losses, highlighting the effect of cognitive biases related to perceived outcomes. Additionally, traders emphasized market analysis, using it as a tool to mitigate the effects of biases and improve the quality of investment outcomes. These findings contribute to a nuanced understanding of how behavioral finance factors interact to shape decision-making in the highly volatile FOREX market.

Limitations

Despite these contributions, the study faced several limitations that must be acknowledged. While the sample size and diversity were sufficient for SEM analysis, they were limited to 300 traders from Quezon City, Philippines. This regional focus may restrict the generalizability of the findings to broader populations. Furthermore, the study utilized a cross-sectional design, capturing data at a single point in time, which limits insights into how behavioral biases and investment decisions might evolve. The reliance on self-reported data introduced potential response biases—such as social desirability and recall bias—that may affect the accuracy of the findings. Additionally, the study focused on a limited scope of biases, examining familiarity bias, self-attribution bias, herding, heuristics, and prospect-based behavior, while other relevant biases, such as overconfidence and loss aversion, were not included. These limitations suggest a need for further exploration to build upon the current findings.

Future Research

Future research can build on this study by addressing its limitations and exploring new areas. Longitudinal studies could shed light on how behavioral biases in investment decisions evolve, especially during market volatility. Expanding the sample to include traders from diverse regions would enhance the findings' generalizability. The impact of technological advancements like algorithmic trading and robo-advisors also warrants further investigation, as they significantly influence modern investor behavior. Additionally, studying the interplay of cognitive biases might reveal how they affect decision-making. Research should also consider how cultural and demographic factors, such as age and income, shape behavioral biases in investment. Lastly, examining the effectiveness of interventions like financial education and behavioral nudges could help mitigate biases' adverse effects on investment outcomes.

References

1. Almansour, B. Y., Elkrghli, S., & Almansour, A. Y. (2023). Unraveling the complexities of cryptocurrency investment decisions: A behavioral finance perspective from Gulf investors. *International Journal of Professional Business Review*, 8(7), e03265. https://doi.org/10.26668/businessreview/2023.v8i7.3265
2. Areiqat, A. Y., Abu-Rumman, A., Al-Alani, Y. S., & Alhorani, A. (2019). Impact of behavioral finance on stock investment decisions: Applied study on a sample of investors at Amman Stock Exchange. *Academy of Accounting and Financial Studies Journal*, 23(2), 375-393. https://doi.org/10.1528/2635-23-2-375
3. Babarinde, G. F., Bello, K., & Bello, B. (2019). Behavioral factors affecting investment decision: Prospect theory approach. *Nigerian Journal of Accounting and Finance*, 11(1), 1-10.
4. Baker, H. K., & Nofsinger, J. R. (2010). Behavioral finance: An overview. In H. K. Baker & J. R. Nofsinger (Eds.), *Behavioral Finance: Investors, Corporations, and Markets* (Part I, Chapter 1). John Wiley & Sons, Inc.
5. Baker, H. K., & Nofsinger, J. R. (2011). *Behavioral finance: Investors, corporations, and markets*. Wiley.
6. Barber, B. M., & Odean, T. (2001). Boys will be boys: Gender, overconfidence, and common stock investment. *Quarterly Journal of Economics*.
7. Barberis, N., & Huang, M. (2001). Mental accounting, loss aversion, and individual stock returns. *Journal of Finance*.
8. Barberis, N., & Thaler, R. (2003). A survey of behavioral finance. In G. M. Constantinides, M. Harris, & R. Stulz (Eds.), *Handbook of the Economics of Finance* (Vol. 1, Chapter 18). Elsevier Science B.V.
9. Bentler, P. M. (1990). Comparative fit indexes in structural models. *Psychological Bulletin*, 107(2), 238–246. https://doi.org/10.1037/0033-2909.107.2.238
10. Biais, B., Hilton, D., Mazurier, K., & Pouget, S. (2005). Judgmental overconfidence, self-monitoring, and trading performance in an experimental financial market. *Review of Economic Studies*.
11. Bihari, A., Dash, M., Kar, S. K., Muduli, K., Kumar, A., & Luthra, S. (2022). Exploring behavioural bias affecting investment decision-making: A network cluster based conceptual analysis for future research. *International Journal of Industrial Engineering and Operations Management*.
12. Byrne, A., & Brooks, M. (2008). Behavioral finance: Theories and evidence. In *The Research Foundation of CFA Institute Literature Review.* The Research Foundation of CFA Institute.
13. CFA Institute Research Foundation. (2019). *Behavioral finance: The second generation* (Monograph). https://doi.org/10.2470/9781944960858
14. Chaboud, A., Rime, D., & Sushko, V. (2023, April). The foreign exchange market. *BIS Working Papers* (No. 1094).
15. Christie, W. G., & Huang, R. D. (1995). Following the pied piper: Do individual returns herd around the market? *Financial Analysts Journal*.
16. Clarke, J., Jandik, T., & Mandelker, G. (2001). The efficient markets hypothesis.
17. Dangol, J., & Manandhar, R. (2020, June). Impact of heuristics on investment decisions: The moderating role of locus of control. *Journal of Business and Social Sciences Research*, 5(1). https://doi.org/10.3126/jbssr.v5i1.30195
18. Erdogan, H. H. (2020). Does prospect theory explain investment decisions: A comparative study. *Journal of Management and Economic Studies*, 2(4), 188-197. https://doi.org/10.26677/TR1010.2021.635
19. Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*.
20. Gautam, S., & Kumar, P. (2021, January). Prospect theory in investment decisions. *Emerging Trends in Innovative Computing & Intelligent System*, 9-13.
21. Gigerenzer, G., & Gaissmaier, W. (2011). Heuristic decision making. *Annual Review of Psychology*.
22. Grinblatt, M., & Keloharju, M. (2001). How distance, language, and culture influence stockholdings and trades. *Journal of Finance*.

23. Haidari, M. N. (2023, December). Impact of decision-making on investment performance: A comprehensive analysis. *Journal of Asian Development Studies*, 12(4), 980-990. https://doi.org/10.62345/jads.2023.12.4.78
24. Hirshleifer, D. (2015). Behavioral finance. *Annual Review of Financial Economics*, 7, 133-159. https://doi.org/10.1146/annurev-financial-092214-043752
25. Hoffmann, A., & Post, T. (2017). Self-attribution bias in consumer financial decision-making: How investment returns affect individuals' belief in skill. *Network for Studies on Pension, Aging and Retirement*.
26. Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*.
27. Kumar, S., & Goyal, N. (2016). Evidence on familiarity bias in investment decisions. *Journal of Behavioral Finance*.
28. Levinson, M. (2014). *Guide to financial markets: Why they exist and how they work*.
29. Malkiel, B. G. (2003). The efficient market hypothesis and its critics. *CEPS Working Paper No. 91*. Princeton University.
30. Metawa, N., Hassan, M. K., Metawa, S., & Safa, M. F. (2019). Impact of behavioral factors on investors' financial decisions: Case of the Egyptian stock market. *International Journal of Islamic and Middle Eastern Finance and Management*, 12(1), 30-55. https://doi.org/10.1108/imefm-12-2017-0333

The Mediating Role of Financial Behavior in the Relationship Between Financial Literacy Level and Loan Repayment Practices of National Housing Authority (NHA) Housing Projects' Beneficiaries from Metro Manila

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Abstract

This study explored how financial behavior influenced the relationship between the financial literacy level and loan repayment practices among beneficiaries of National Housing Authority (NHA) housing projects from Metro Manila. Financial literacy, which involves understanding and effectively using financial knowledge, is essential for making smart financial choices. The research examined four key financial behaviors—budgeting, spending, saving, and credit management—that may influence financial literacy and loan repayment practices. Using an embedded mixed-method approach, data were collected through survey questionnaires given to a sample of NHA beneficiaries, and semi-structured interviews to gain deeper insights. This study evaluated the influence of financial literacy on loan repayment practices, both directly and indirectly, through financial behaviors. It highlighted the need to improve financial literacy and encourage positive financial behaviors to enhance NHA housing loan repayment success among NHA beneficiaries. This research added to the existing knowledge by clarifying how financial literacy influenced loan repayment practices, offering valuable insights for policymakers and financial educators, aiming to promote financial stability and responsibility in low-income housing communities.

Keywords: Financial Behavior, Financial Literacy Level, Loan Repayment Practices, National Housing Authority (NHA), Budgeting, Saving, Spending, and Credit Management.

Introduction

The study examines the relationship between financial literacy, financial behavior, and loan repayment practices among National Housing Authority (NHA) beneficiaries in the Philippines, focusing on the significant role of technology in addressing communication gaps. Financial literacy, a critical factor for economic stability and personal financial well-being, remains low among many low-income families, making them more vulnerable to financial challenges. This issue is particularly evident among NHA beneficiaries in Metro Manila, where the high cost of living complicates housing loan repayment. Furthermore, miscommunication between beneficiaries and the NHA regarding loan terms exacerbates financial mismanagement. The situation is worsened by limited access to digital tools, which hampers beneficiaries' ability to track loan payments, understand terms, and access essential financial resources.

The study highlights the role of digital platforms in enhancing financial literacy and improving communication. Research indicates that technology can provide easy access to financial education and resources, yet the limited use of these tools among NHA beneficiaries leads to misinformation and poor financial decisions. Financial behaviors, including budgeting, saving, and credit management, are identified as mediators between financial literacy and loan repayment. Poor financial management not only undermines loan repayments but also threatens the sustainability of housing programs and the financial stability of families.

By exploring these dynamics, the study seeks to inform policies and educational programs aimed at improving financial literacy and promoting the use of digital tools among NHA beneficiaries. This dual approach is essential for addressing financial challenges, enhancing loan repayment rates, and fostering economic resilience among low-income families. The findings underscore the need to integrate technology into financial education efforts to bridge communication gaps and empower beneficiaries to make informed financial decisions.

Literature Review

Financial Literacy

Financial literacy plays a vital role in shaping individuals' financial behaviors, improving decision-making, and fostering economic stability. Studies by Lusardi and Mitchell (2019) emphasize that financial literacy significantly influences responsible financial practices, such as budgeting, saving, and debt management, particularly among economically challenged populations. Enhanced financial literacy correlates with better loan repayment behaviors and informed financial decisions, which are essential for maintaining creditworthiness and long-term stability.

The importance of financial literacy is further underscored in the context of Micro, Small, and Medium Enterprises (MSMEs). These businesses are critical to economic growth, contributing significantly to employment and GDP formation, as highlighted in the Indonesian context. However, MSMEs often face challenges in accessing financial support due to limited financial management knowledge and a non-bankable profile. Financial literacy equips MSME operators with the skills to prepare essential financial documents, ensure business sustainability, and improve loan repayment outcomes (Pramedi & Asandimitra, 2021; Ningtyas & Andarsari, 2021).

Despite its proven benefits, financial literacy levels remain low in many regions. For instance, a 2019 OJK study reported that only 38.03% of Indonesians were financially literate, revealing a critical need for financial education. The COVID-19 pandemic exacerbated financial challenges, underscoring financial literacy as a vital life skill for resilience during economic disruptions (OJK, 2022).

Research has shown that financial literacy positively impacts financial behavior, which in turn influences business performance and sustainability. However, financial attitudes appear to have less direct influence on these outcomes. Effective financial management practices mediate the relationship between financial literacy and corporate performance, demonstrating that improved literacy translates into better business operations and sustainable growth (Syafitri et al., 2024; Supriyadi, Saifi, & Wi Endang NP, 2024).

Moreover, financial literacy fosters entrepreneurship, contributing to economic independence and innovation. Entrepreneurs with strong financial literacy can better manage resources, enhance business performance, and reliably repay loans. This highlights the need for comprehensive frameworks to link financial literacy with practical outcomes, such as improved loan repayment and business sustainability. MSME managers with higher financial literacy levels are more likely to utilize credit effectively, optimize business performance, and achieve sustainable success.

Financial Behavior

The study by Wijekoon et al. (2022) explores the factors influencing the economic well-being of coconut growers in Sri Lanka, employing Partial Least Squares Structural Equation Modeling (PLS-SEM) to analyze data. The findings reveal that financial behavior, financial knowledge, and money attitude are significant and positively correlated with economic well-being, collectively explaining 88.9% of its variance.

The study highlights the interplay between money attitude and financial behavior, with a significant relationship explaining 78.2% of the variance in financial behavior. While financial behavior was examined as a mediator, it was validated only in the relationship between money attitude and economic well-being.

These results emphasize the importance of personal financial management in enhancing economic well-being across life stages, including retirement. Despite ongoing efforts to promote responsible financial behavior, gaps remain in understanding the foundational factors driving these practices. The study advocates for enhanced financial education to equip individuals with the skills needed to navigate complex financial markets and improve global economic well-being.

Displacement and Resettlement of Informal Settler in Metro Manila

Maningo (2023) examines the sociopolitical dynamics of development-induced displacement and resettlement (DIDR) among urban informal settlers in the Philippines following the government's flood management program post-Typhoon Ondoy (2009). The study highlights how eviction threats influenced settlers' responses based on family well-being and social capital, leading to varied reactions. Relocated settlers maintained amicable relationships despite differing housing strategies, while those in in-city housing navigated new forms of urban informality through negotiated regulations.

The research underscores the importance of financial literacy in enabling settlers to make informed decisions during displacement and resettlement. While the study focuses on social interactions and individual agency, it suggests that financial education must align with the broader socio-political context to improve settlers' engagement with housing projects effectively.

Family Financial Socialization

Pahlevan Sharif and Naghavi (2020) investigate the relationships between family financial socialization, financial information-seeking behavior (FIS), and youth financial literacy using a sample of 802 tertiary students. The study reveals that FIS significantly

enhances financial literacy, but a substantial variance remains unexplained, suggesting that youth access financial information beyond active seeking. Parental financial behavior influences youth FIS indirectly, with role modeling proving more impactful than explicit financial teaching within the Risk Information Seeking and Processing (RISP) model.

The research highlights the critical role of parental influence in shaping financial literacy, emphasizing the need for financial education programs that address these dynamics. However, limitations include potential biases from self-reported data, limited sample diversity, and the exclusion of other financial socialization agents like peers and schools. Future research should adopt longitudinal designs, broaden demographics, and explore additional influences to enhance understanding and improve financial literacy education strategies.

Financial Education

Kaiser et al. (2020) conducted a meta-analysis of 76 randomized experiments involving over 160,000 individuals to assess the effects of financial education programs. The study found that such programs significantly enhance financial knowledge and positively influence financial behaviors, with treatment effects comparable to those in other educational fields. These robust findings, confirmed across diverse methodologies, highlight the cost-effectiveness and value of financial education in improving decision-making.

However, the study identifies key gaps in the literature. Limited research evaluates the long-term sustainability of financial education outcomes, with few studies examining impacts beyond six months post-intervention. Reliance on self-reported data raises concerns about bias, and there is a lack of transparency regarding program costs. The authors recommend future research employing longitudinal designs, improving data quality, and addressing cost transparency to deepen understanding and guide policymaking in financial education.

Methodology

This study employed an embedded mixed-methods (EMM) research design, as outlined by Creswell and Clark (2010), integrating quantitative and qualitative approaches to examine financial literacy and loan repayment practices among National Housing Authority (NHA) housing project beneficiaries. The primary focus was on quantitative data collection, using survey questionnaires structured with a 4-Point Likert Scale to gather statistically generalizable insights. The survey questions were categorized into five (5) sections: (1) demographics, (2) behavioral characteristics, (3) financial literacy levels, (4) financial behaviors, and (5) loan repayment attitudes. Demographic data included variables such as age, gender, marital status, location, education level, length of residence, source of income, employment status, and ownership status. Behavioral characteristics explored psychographic factors, while financial literacy and behaviors assessed knowledge and practices related to budgeting, saving, and credit management. Loan repayment attitudes were also measured to understand beneficiaries' perspectives on fulfilling financial obligations.

Qualitative data were collected through semi-structured interviews, allowing for in-depth exploration of respondents' thoughts, feelings, and beliefs. This approach enabled researchers to address personal and sensitive topics, complementing the quantitative findings by providing rich context and deeper insights into the challenges faced by beneficiaries. Interviews utilized pre-prepared and probing questions to delve into areas of interest further.

The integration of quantitative and qualitative methods ensured a comprehensive analysis. Quantitative data provided measurable and generalizable insights, while qualitative data enriched the study by uncovering nuanced perspectives and experiences, offering a holistic understanding of the factors influencing financial literacy and loan repayment practices among NHA beneficiaries.

Analysis and Discussions

Analysis and Interpretation of Quantitative Results

Table 1. Latent Variable Coefficient

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite reliability	Ave. Variances Extracted
Financial Literacy level				**0.916**	**0.929**	**0.526**
	FL1	0.613	<0.001			
	FL2	0.809	<0.001			
	FL3	0.585	<0.001			
	FL4	0.769	<0.001			

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite reliability	Ave. Variances Extracted
	FL5	0.691	<0.001			
	FL6	0.844	<0.001			
	FL7	0.644	<0.001			
	FL8	0.84	<0.001			
	FL8	0.767	<0.001			
	FL10	0.704	<0.001			
	FL11	0.755	<0.001			
	FL12	0.622	<0.001			
Financial Behavior - Budgeting				**0.924**	**0.952**	**0.868**
	FBB1	0.931	<0.001			
	FBB2	0.948	<0.001			
	FBB3	0.916	<0.001			
Financial Behavior - Savings				**0.936**	**0.959**	**0.886**
	FBV1	0.953	<0.001			
	FBV2	0.933	<0.001			
	FBV3	0.938	<0.001			
Financial Behavior - Spending				**0.869**	**0.919**	**0.792**
	FBP1	0.895	<0.001			
	FBP2	0.883	<0.001			
	FBP3	0.891	<0.001			
Financial Behavior - Credit				**0.915**	**0.946**	**0.854**
	FBC1	0.937	<0.001			
	FBC2	0.919	<0.001			
	FBC3	0.917	<0.001			
Loan Repayment Attitudes				**0.881**	**0.905**	**0.522**
	LRA1	0.704	<0.001			
	LRA2	0.700	<0.001			
	LRA3	0.762	<0.001			
	LRA4	0.732	<0.001			
	LRA5	0.701	<0.001			
	LRA6	0.809	<0.001			
	LRA7	0.552	<0.001			
	LRA8	0.68	<0.001			
	LRA9	0.816	<0.001			
	LRA10	0.584	<0.001			

The latent variable coefficients for six constructs—Financial Literacy Level, Financial Behavior - Budgeting, Financial Behavior - Saving, Financial Behavior - Spending, Financial Behavior - Credit, and Loan Repayment Attitudes—indicate strong reliability and validity across all measures. The Financial Literacy Level construct, comprising 12 items, demonstrates excellent internal consistency with a Cronbach's Alpha of 0.916 and a Composite Reliability (CR) of 0.929. The Average Variance Extracted (AVE) is 0.526, exceeding the 0.50 threshold for convergent validity, with factor loadings ranging from 0.585 to 0.844 ($p < 0.001$). Financial Behavior - Budgeting also shows exceptional reliability, with a Cronbach's Alpha of 0.924, CR of 0.952, and AVE of 0.868. Its factor loadings range from 0.916 to 0.948 ($p < 0.001$), supporting validity.

Similarly, Financial Behavior - Saving achieves a Cronbach's Alpha of 0.936, CR of 0.959, and an AVE of 0.886, with factor loadings between 0.933 and 0.953 ($p < 0.001$). Financial Behavior - Spending has a Cronbach's Alpha of 0.869, CR of 0.919, and an AVE of 0.792, with factor loadings from 0.883 to 0.895 ($p < 0.001$), indicating high reliability and convergent validity. The Financial Behavior - Credit construct reflects strong reliability and validity, with a Cronbach's Alpha of 0.915, CR of 0.946, and AVE of 0.854, supported by factor loadings from 0.917 to 0.937 ($p < 0.001$). Lastly, the Loan Repayment Attitudes construct, consisting of 10 items, exhibits a Cronbach's Alpha of 0.881, CR of 0.905, and an AVE of 0.522. While factor loadings range from 0.552 to 0.816 ($p < 0.001$), slightly lower loadings for some items remain within acceptable limits. Overall, the constructs exhibit robust reliability and validity, as indicated by Cronbach's Alpha, Composite Reliability, AVE, and statistically significant factor loadings for all items.

Table 2. *Square Roots of AVE Coefficients and Correlation Coefficients*

Latent Variables	FL	FBB	FBV	FBP	FBC	LRA
FL	**0.725**	0.629	0. 492	0.523	0.564	0.689
FBB	0.629	**0.932**	0.774	0.627	0.622	0.69
FBV	0.492	0.774	**0.942**	0.616	0.592	0.598
FBP	0.523	0.627	0.616	**0.890**	0.635	0.654
FBC	0.564	0.622	0.592	0.635	**0.924**	0.624
LRA	0.689	0.69	0.598	0.654	0.624	**0.701**

Note: Financial Literacy Level (FL); Financial Behavior - Budgeting (FBB); Financial Behavior - Savings (FBV); Financial Behavior Spending (FBP); Financial Behavior - Credit (FBC); Loan Repayment Attitudes (LRA)

The table illustrates the discriminant validity of the latent variables: Financial Literacy (FL), Financial Behavior - Budgeting (FBB), Financial Behavior - Savings (FBV), Financial Behavior - Spending (FBP), Financial Behavior - Credit (FBC), and Loan Repayment Attitudes (LRA). The square roots of the Average Variance Extracted (AVE), located along the main diagonal, are higher than the correlation coefficients between each construct and others, as suggested by Fornell and Larcker (1981). This indicates strong discriminant validity, affirming that each latent variable is distinct and measures a unique concept.

For instance, the square root of the AVE for Financial Literacy (FL) is 0.725, surpassing its correlations with other constructs, such as Budgeting (FBB) at 0.629 and Loan Repayment Attitudes (LRA) at 0.689. Similarly, Budgeting (FBB) exhibits a square root of AVE at 0.932, which is greater than its correlation with LRA at 0.690. Savings (FBV) also demonstrates strong discriminant validity, with an AVE square root of 0.942 exceeding its correlations with Spending (FBP) at 0.616 and LRA at 0.598.

This pattern is consistent across all constructs, showing that each construct is more closely related to its indicators than to other constructs. The findings confirm that the model achieves satisfactory discriminant validity, ensuring that the constructs are well-represented and do not exhibit significant overlap.

Table 3. Model Fit and Quality Indices of the Emerging Model

Measure	Estimate	Threshold	Interpretation
Average path coefficient (APC)	0.392, $p < 0.001$	$p < .05$	Significant
Average R-squared (ARS)	0.372, $p < 0.001$	$p < .05$	Significant
Average adjusted R-squared (AARS)	0.369, $p < 0.001$	$p < .05$	Significant
Average block VIF (AVIF)	2.291	≤ 3.3	Ideally
Average full collinearity VIF (AFVIF)	2.826	≤ 3.3	Ideally
Tenenhaus GOF (GoF)	0.523	≤ .36	Large
Simpson's paradox ratio (SPR)	1.000	= 1.00	Ideally
R-squared contribution ratio (RSCR)	1.000	= 1.00	Ideally
Statistical suppression ration (SSR)	1.000	= 1.00	Ideally

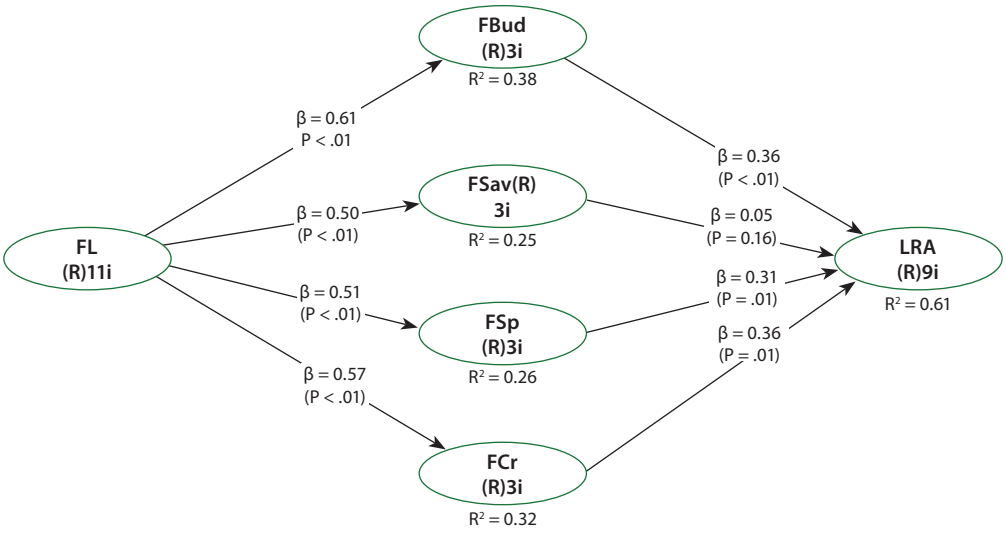


Fig. 1. The Emerging Model 1

Table 4. Path coefficient and P-value

Path	Path Coefficients (β)	P-value	Effect Sizes (f2)	Effect Size Interpretation (Cohen 1988)	Description	Interpretation
H1: FL → Budgeting	0.629	<0.001	0.396	Large	Significant	H1 is supported
H2: FL → Savings	0.5	<0.001	0.25	Medium	Significant	H2 is supported
H3: FL → Spending	0.526	<0.001	0.276	Medium	Significant	H3 is supported
H4: FL → Credit	0.577	<0.001	0.333	Medium	Significant	H4 is supported
H5: Budgeting → LRA	0.368	<0.001	0.256	Medium	Significant	H5 is supported
H6: Savings → LRA	0.048	0.165	0.03	Medium	Significant	H6 is supported
H7: Spending → LRA	0.297	<0.001	0.198	Medium	Significant	H7 is supported
H8: Credit → LRA	0.19	<0.001	0.119	Small	Significant	H8 is supported

Note: Financial Literacy Level (FL); Loan Repayment Attitudes (LRA)

The table presents the relationships between financial literacy, financial behaviors (budgeting, saving, spending, and credit), and loan repayment attitudes (LRA), as measured by path coefficients, p-values, and effect sizes. The results highlight the significant impact of financial literacy on various financial behaviors and their subsequent influence on loan repayment attitudes.

Financial literacy strongly predicts budgeting, as shown in H1, with a path coefficient of 0.629 ($p < 0.001$) and a large effect size of 0.396, indicating that individuals with higher financial literacy are more likely to adopt effective budgeting practices. Similarly, H2 reveals that financial literacy positively influences savings behavior ($\beta = 0.500$, $p < 0.001$) with a medium effect size of 0.25. Spending behavior is also significantly affected by financial literacy, as indicated in H3 ($\beta = 0.526$, $p < 0.001$) with a medium effect size of 0.276. Additionally, H4 demonstrates a strong relationship between financial literacy and credit management ($\beta = 0.577$, $p < 0.001$) with a medium effect size of 0.333.

The influence of financial behaviors on loan repayment attitudes (LRA) is also evident. Budgeting significantly enhances LRA, as indicated in H5 ($\beta = 0.368$, $p < 0.001$) with a medium effect size of 0.256, underscoring its importance in managing loan repayments. Spending behavior, as shown in H7, positively impacts LRA ($\beta = 0.297$, $p < 0.001$) with a medium effect size of 0.198, highlighting the role of responsible spending in improving repayment practices. Credit management also contributes to better LRA, albeit with a smaller effect size, as seen in H8 ($\beta = 0.190$, $p < 0.001$; effect size = 0.119). However, H6 reveals that the relationship between savings behavior and LRA is not significant ($\beta = 0.048$, $p = 0.165$), suggesting that savings may not directly influence loan repayment attitudes.

Overall, the findings confirm that financial literacy significantly influences critical financial behaviors, such as budgeting, spending, and credit management, which positively affect loan repayment attitudes. Although savings behavior does not show a significant direct impact on LRA, budgeting and spending behaviors emerge as pivotal factors for improving repayment practices. These results support most of the hypotheses, emphasizing the importance of financial literacy and responsible financial behaviors in fostering effective loan repayment strategies.

Mediation Analysis

Table 12. Effect for the path with two segments (Mediation Analysis)

Path	Total Effect	P-value	Indirect Effect	P-value	Description	Interpretation	Remark
H9: FL → FBB → LRA	0.79	<0.001	0.211	< 0.001	Significant	Partial Mediation	H9 is supported
H10: FL → FBV → LRA	0.791	<0.001	0.642	< 0.001	Significant	Partial Mediation	H10 is supported
H11: FL → FBP → LRA	0.779	<0.001	0.6	< 0.001	Significant	Partial Mediation	H11 is supported
H12: FL → FBC → LRA	0.787	<0.001	0.61	< 0.001	Significant	Partial Mediation	H12 is supported

Note: Financial Literacy Level (FL); Financial Behavior - Budgeting (FBB); Financial Behavior - Savings (FBV); Financial Behavior - Spending (FBP); Financial Behavior - Credit (FBC); Loan Repayment Attitudes (LRA)

The table presents the results of the mediation analysis exploring the effect of financial literacy (FL) on loan repayment attitudes (LRA) through various financial behaviors, including budgeting (FBB), saving (FBV), spending (FBP), and credit management (FBC). The findings show that all hypotheses exhibit significant partial mediation, with both direct and indirect effects being significant ($p < 0.001$), suggesting that financial behaviors partially mediate the relationship between financial literacy and loan repayment attitudes.

Hypothesis 9 examines the mediation effect of budgeting between financial literacy and loan repayment attitudes. The results indicate a substantial total effect of 0.790 ($p < 0.001$), with a direct effect of 0.579, showing that financial literacy independently influences loan repayment. The indirect effect of 0.211 highlights that budgeting significantly enhances the relationship between financial literacy and loan repayment. These results support the importance of both financial literacy and budgeting in improving loan repayment behavior, indicating partial mediation, which validates H9.

Hypothesis 10 investigates the role of saving in mediating the relationship between financial literacy and loan repayment attitudes. The total effect is 0.791 ($p < 0.001$), with a direct effect of 0.642, demonstrating the strong independent impact of financial literacy on loan repayment. The indirect effect of 0.149 emphasizes the significant role of saving behavior in strengthening the connection between financial literacy and loan repayment. These results suggest that both financial literacy and saving strategies are crucial for enhancing loan repayment behavior, with partial mediation evident, thereby confirming H10.

Hypothesis 11 examines the mediation effect of spending behavior. The total effect is 0.779 ($p < 0.001$), with a direct effect of 0.600, showing that financial literacy has a significant independent impact on loan repayment. The indirect effect of 0.179 indicates that responsible spending behavior enhances the relationship between financial literacy and loan repayment attitudes. The results suggest that both financial literacy and spending behavior are key to improving loan repayment practices, with partial mediation, thus supporting H11.

Finally, Hypothesis 12 explores the role of credit management as a mediator. The total effect is 0.787 ($p < 0.001$), with a direct effect of 0.640, confirming that financial literacy independently affects loan repayment attitudes. The indirect effect of 0.147 highlights the role of credit management in reinforcing the link between financial literacy and loan repayment. These findings suggest that financial literacy and effective credit management are essential for improving loan repayment behavior, with partial mediation, validating H12.

In summary, the results indicate that financial literacy significantly influences loan repayment attitudes through various financial behaviors, including budgeting, saving, spending, and credit management. While each behavior partially mediates the relationship, none fully explains the impact of financial literacy, emphasizing the essential role of these behaviors in shaping effective loan repayment practices.

Discussion of Results

Qualitative Results The narratives reveal several interconnected themes that paint a vivid picture of the challenges faced by beneficiaries after experiencing a significant loss of employment and subsequent financial strain. A prominent theme across the stories is the abrupt loss of employment, leading to a cascade of financial difficulties. Beneficiaries often describe their previous lives as stable and manageable, but the transition to unemployment marks a dramatic shift. This shift triggers a downward spiral, where individuals and families experience feelings of helplessness. The struggle to maintain financial stability is intensified as essential costs, such as food and utilities, become harder to cover. The narratives reflect how this loss not only affects the financial aspects of life but also carries a profound emotional burden, as families face uncertainty about their future.

The stories also highlight the impact of increased expenses, including unexpected utility costs and transportation fees. Beneficiaries express the emotional and psychological toll these rising costs have on their well-being. To make ends meet, many are forced to make difficult sacrifices, such as reducing the amount of food they purchase or cutting back on childcare. These sacrifices emphasize the harsh realities of financial hardship, where basic needs must be compromised for survival. The theme also illustrates the broader mental strain associated with financial instability, as families constantly worry about how to manage their limited resources.

Another significant theme that emerges from the narratives is the deterioration of housing quality. Beneficiaries describe their new living conditions as suboptimal, often citing issues like leaking roofs and clogged toilets that require urgent repairs. These housing problems further strain their already limited finances, forcing them to divert money away from essential expenses like food or healthcare to address these issues. The stories reflect the deep frustration of beneficiaries who feel that their physical environment, beyond just financial resources, has a direct and detrimental impact on their ability to remain stable. This sense of frustration is compounded by the feeling that their efforts to make ends meet are constantly undermined by issues beyond their control.

Together, these themes underscore the complex, multifaceted nature of financial distress following job loss, highlighting the emotional toll, the difficult choices beneficiaries face, and the significant impact of housing conditions on their overall financial stability.

Conclusion

The study reveals that respondents possess strong awareness of the importance of effective financial practices in managing their National Housing Authority (NHA) housing loan repayments. Key practices include budgeting, distinguishing between essential and discretionary expenses, maintaining adaptability in budgeting for unexpected costs, and prioritizing savings. Respondents emphasize the significance of tracking spending patterns to reallocate savings toward loan repayments and view saving as a vital practice for ensuring timely payments. Additionally, responsible credit management is seen as crucial for avoiding debt accumulation and maintaining financial stability.

Despite a strong understanding of these financial practices, respondents face challenges in managing their finances due to job losses, increased living expenses, and lack of resources or financial education. The findings suggest that while respondents recognize the importance of financial management, there remains a gap between perceived literacy and effective financial management. This underscores the need for comprehensive support and interventions, such as livelihood programs, financial workshops, and improvements in housing quality. By addressing these issues, stakeholders can empower beneficiaries to enhance their financial literacy, improve loan repayment capabilities, and foster economic stability. Ongoing support is essential to bridge the gap between financial awareness and actual management practices, enabling NHA beneficiaries to handle their housing loan repayments effectively.

Recommendations

For NHA Beneficiaries Beneficiaries should attend structured budgeting workshops to learn how to create detailed monthly budgets, covering all income sources and expenses, particularly National Housing Authority (NHA) loan repayments. These workshops will emphasize the importance of regularly reviewing and adjusting budgets in response to financial changes. Participants will be guided to list all income sources, such as salary, small business earnings, or government assistance, and itemize essential expenses like NHA loan repayments, utilities, groceries, and other necessary costs. They will also learn to categorize expenses into essential (needs) and discretionary (wants) spending, enabling them to prioritize important payments, such as loan repayments, while reducing non-essential costs. Regular budget reviews will be encouraged to reflect any changes in income or expenses, ensuring that financial goals, especially loan repayments, remain on track despite financial fluctuations. Additionally, digital budgeting tools and mobile apps like GCash or PayMaya will be introduced to help beneficiaries easily track income, expenses, and loan payments, thereby assisting in maintaining an organized financial plan. For elderly beneficiaries or those without access to devices, alternative methods, such as manual strategies, community savings groups, and visual reminders, will be implemented to ensure continued financial management support.

Building an emergency fund is vital for financial stability, and beneficiaries should aim to save 10-15% of their monthly income to build a fund covering three to six months of living expenses. To make this process manageable, specific savings targets and milestones should be set. Beneficiaries can prioritize savings by utilizing automated transfers to a dedicated savings account. Participation in financial literacy workshops will further enhance savings strategies. For those comfortable with technology, savings apps like GCash and PayMaya can be utilized to track progress, while manual tracking methods or community savings groups can support beneficiaries who prefer traditional methods. Regularly reviewing and adjusting savings goals is crucial to maintaining progress despite changes in income or expenses.

For National Housing Authority (NHA) Targeted financial literacy programs tailored for National Housing Authority (NHA) beneficiaries should address key areas such as budgeting, saving, credit management, and debt reduction. One potential model for these programs is the "Peso Smart Program" by the Metrobank Foundation, which can be adapted to the specific needs of NHA beneficiaries. This program would focus on budgeting and savings workshops to help beneficiaries manage household finances, prioritize expenses, and ensure timely loan repayments. Additionally, credit management and debt reduction education will teach beneficiaries how to understand and manage credit, alongside strategies for reducing debt. Loan repayment education will cover important topics, including loan terms, interest rates, and penalties for late payments.

To provide personalized guidance, one-on-one financial counseling services should be offered through partnerships with certified financial advisors. These sessions will help beneficiaries develop customized financial plans, address challenges related to budgeting, saving, and loan repayment, and navigate unexpected financial situations by advising on emergency funds and debt restructuring. This approach is inspired by successful models like the Financial Literacy and Counseling Program (FLCP) by USAID, which helped Overseas Filipino Workers (OFWs) manage remittances and plan savings.

To further enhance financial management for beneficiaries, supportive policies and tools should be introduced. Flexible repayment plans, such as income-based and graduated options, would allow loan payments to adjust according to beneficiaries' financial circumstances. Partnering with mobile payment platforms like GCash and PayMaya will streamline the repayment process, ensuring timely payments and reducing the risk of missed payments or late fees. Moreover, providing access to financial management tools, such as budgeting templates, savings calculators, and expense trackers, will empower beneficiaries to make informed financial decisions. A feedback mechanism should also be established to collect insights from beneficiaries and ensure continuous program improvement. A successful example of this approach is the Land Bank of the Philippines' OFW Reintegration Program, which incorporates flexible loan repayment terms and digital payment channels to improve financial management for its beneficiaries.

For Policymakers Policymakers should integrate mandatory financial literacy programs into the eligibility criteria for housing loans to equip beneficiaries with essential financial skills, reduce the risk of loan defaults, and promote long-term, sustainable homeownership. These programs should focus on critical areas such as budgeting, saving, debt management, and understanding loan repayments. By implementing financial literacy programs as a prerequisite for housing loan approval, beneficiaries will gain the necessary knowledge to manage their finances effectively, ensuring they are well-prepared to handle their loans and other financial obligations.

In addition to financial education, policymakers should partner with local governments, the private sector, and NGOs to provide livelihood training and job opportunities within housing communities. These efforts will help beneficiaries increase their household income, enabling them to meet their financial obligations. Regular assessments of housing loan programs should be conducted to evaluate default rates, income trends, and community development, using a data-driven approach to adjust loan terms and support mechanisms based on the evolving needs of beneficiaries.

Furthermore, policymakers should establish formal feedback systems that allow beneficiaries to share concerns and suggestions for improving the housing loan program. By integrating financial literacy education and livelihood support, these strategies will empower beneficiaries to effectively manage their financial responsibilities, reduce default risks, and contribute to the long-term success of the housing program.

References

1. Perry, V. G. (n.d.). (PDF) Who Is in Control? The Role of Self-Perception, Knowledge, and Income in Explaining Consumer Financial Behavior. ResearchGate. Retrieved October 24, 2024, from https://www.researchgate.net/publication/227633409_Who_Is_in_Control_The_Role_of_Self-Perception_Knowledge_and_Income_in_Explaining_Consumer_Financial_Behavior
2. Snyder, M. (n.d.). Financial Literacy and Planning: Implications for Retirement Wellbeing | Request PDF. ResearchGate. Retrieved October 24, 2024, from https://www.researchgate.net/publication/4878786_Financial_Literacy_and_Planning_Implications_for_Retirement_Wellbeing
3. Atkinson, A., & Messy, F.-A. (2012). Measuring financial literacy: Results of the OECD/International Network on Financial Education (INFE) pilot study. OECD Working Papers on Finance, Insurance and Private Pensions, No. 15. https://doi.org/10.1787/5k9csfs90frq-en
4. Balgos, B. (2021). Securing the safety of informal settler families along waterways in Metro Manila, Philippines: Government-civil society organization partnership. ResearchGate. https://www.researchgate.net/publication/301264916_Securing_the_Safety_of_Informal_Settler_Families_Along_Waterways_in_Metro_Manila_Philippines_Government-Civil_Society_Organisation_Partnership
5. Cappelli, T., Banks, A. P., & Gardner, B. (2024). Understanding money-management behaviour and its potential determinants among undergraduate students: A scoping review. PLOS ONE, 19(8), e0307137. https://doi.org/10.1371/journal.pone.0307137
6. Cappelli, T., Banks, A. P., & Gardner, B. (2024). Impact of financial literacy, mental budgeting, and self-control on financial management. PLOS ONE, 19(8), e0294466. https://doi.org/10.1371/journal.pone.0294466
7. Cole, S., Sampson, T., & Zia, B. (2011). Prices or knowledge? What drives demand for financial services in emerging markets? The Journal of Finance, 66(6), 1933-1967. https://doi.org/10.1111/j.1540-6261.2011.01693.x
8. Demirgüç-Kunt, A., Klapper, L., & Singer, D. (2017). Financial inclusion and inclusive growth: A review of recent empirical evidence. The World Bank Research Observer, 32(2), 221-234. https://doi.org/10.1093/wbro/lkx001
9. Elder, G. H. (1994). Time, human development, and social change: Perspectives on the lifecourse. Social Psychology Quarterly, 57(1), 4-15. https://doi.org/10.2307/2787065
10. GM TAI inspects NHA's housing projects in Caloocan City. (2022, November 28). National Housing Authority. Retrieved June 24, 2024, from https://nha.gov.ph/gm-tai-inspects-nhas-housing-projects-in-caloocan-city/
11. Kaiser, T., & Menkhoff, L. (2020). Financial education in schools: A meta-analysis of experimental studies. Economics of Education Review, 78, 101-123. https://doi.org/10.1016/j.econedurev.2020.101123
12. Khawar, S., & Sarwar, A. (2021, July 18). Financial literacy and financial behavior with the mediating effect of family financial socialization in the financial institutions of Lahore, Pakistan. Retrieved July 12, 2024, from https://link.springer.com/article/10.1186/s43093-021-00064-x
13. Knowles, M. S. (1980). The modern practice of adult education: Andragogy versus pedagogy. Adult Education.
14. Lerman, R. I., & Schmidt, L. (2015). The role of homeownership in economic mobility. Urban Institute. https://www.urban.org/research/publication/role-homeownership-economic-mobility
15. Lusardi, A., & Mitchell, O. S. (2014). The economic importance of financial literacy: Theory and evidence. Journal of Economic Literature, 52(1), 5-44. https://doi.org/10.1257/jel.52.1.5
16. Maningo, R. (2023). Displacement and resettlement of informal settlers in Metro Manila. Ateneo de Manila University. https://ajol.ateneo.edu/stjgs/articles/228/2550
17. Miller, M., Reichelstein, J., Salas, C., & Zia, B. (2015). Can you help someone become financially capable? A meta-analysis of the literature. World Bank Research Observer, 30(2), 220-246. https://doi.org/10.1093/wbro/lkv008
18. National Housing Authority. (2019). Financial and accomplishment report: FY 2018. Retrieved from https://nha.gov.ph/wp-content/uploads/2019/03/FAR-3-REG-4-PART-1.pdf

19. National Housing Authority. (2023). Annual report 2022. https://nha.gov.ph/wp-content/uploads/2023/08/1-FINAL-ANNUAL-REPORT-2022-FINAL-LAYOUT-2.pdf
20. Ozili, P. K. (2020). Financial inclusion research around the world: A review. Forum for Social Economics, 49(4), 457-479. https://doi.org/10.1080/07360932.2020.1799210
21. Thorp, S., Liu, J., Agnew, J., Bateman, H., Eckert, C., & Iskhakov, F. (2023). Feeling comfortable with a mortgage: The impact of framing, financial literacy, and advice. Journal of Financial Literacy and Wellbeing, 1(1), 47-78. https://doi.org/10.1017/flw.2023.3
22. Lerman, R. I., & Schmidt, L. (2015). The role of homeownership in economic mobility. Urban Institute. https://www.urban.org/research/publication/role-homeownership-economic-mobility
23. Lusardi, A., & Mitchell, O. S. (2014). The economic importance of financial literacy: Theory and evidence. Journal of Economic Literature, 52(1), 5-44. https://doi.org/10.1257/jel.52.1.5
24. Statista. (2023). Gender breakdown in Manila. Philippines. https://www.statista.com/statistics/1342864/gender-breakdown-in-manila-philippines/?fbclid=IwZXh0bgNhZW0CMTEAAR2iQYIrmzuwLO7fl-eMMrlrF8OMpMX1Q34C7aOHs-W32Eqb4QqwyG7dkQc_aem_EkyU6VgU3Lf35jlyAkrZfg
25. Lusardi, A., & Mitchell, O. S. (2020). Financial education affects financial knowledge and downstream behaviors. ResearchGate. https://www.researchgate.net/publication/340940061_Financial_education_affects_financial_knowledge_and_downstream_behaviors?fbclid=IwZXh0bgNhZW0CMTEAAR0nBSWgoe0E-MU9YXQJe1d1-boM6pJpFXbJCJ-4yPbqkOKTQCdhwjcN5OQ_aem_8eKJbsbiJuMfYjLVOkDRiw
26. Lusardi, A., & Mitchell, O. S. (2020). Family financial socialization, financial information seeking behavior, and financial literacy among youth. ResearchGate. https://www.researchgate.net/publication/340729381_Family_financial_socialization_financial_information_seeking_behavior_and_financial_literacy_among_youth?fbclid=IwZXh0bgNhZW0CMTEAAR3xyGP7TvWJRZz9ySzp-OwClbolR6daVWASG9KRYqPAgtp1pUu-YnUUJGo_aem_dqxSS8f-OpSWTUulw3qy8A
27. Commission on Audit. (n.d.). National Housing Authority housing programs. COA. https://www.coa.gov.ph/national-housing-authority-housing-rograms/?fbclid=IwZXh0bgNhZW0CMTEAAR0c1jUKugEzC2_xKzU4SERojjHQg0rHb2xsQXUtlkTXWURhlHvXD-SqCFw_aem_fkLtUjxr0pyITj5fOheS1A&__cf_chl_tk=eh_zTJtAhpnxI9V2J_Y88pUQSSzHbdP11bW_jKqCjd8-1729724380-1.0.1.1-LiMnV.9keUzm4Od7pqjeb.unpUM8PflT8D88Wy3O1qg
28. Wijekoon, R., Sabri, M. F., Ahmad, N., & Abd Rahim, H. (2022). [Title of the article]. International Journal of Academic Research in Business and Social Sciences, 12(10). https://doi.org/10.6007/IJARBSS/v12-i10/15194
29. Alsharif, H., Alshahrani, M., & Alzahrani, K. (2022). Financial knowledge, digital financial knowledge, financial attitude, financial behaviour, and financial satisfaction among millennials. ResearchGate. https://doi.org/10.13140/RG.2.2.17920.40968
30. Supriyadi, A., Saifi, M., & Wi Endang NP, M. G. (2024). The effect of financial literacy on loan repayment performance of MSMEs: A literature review. In Brawijaya International Conference on Business Administration, Taxation, and Tourism (pp. 228–246). KnE Social Sciences. https://doi.org/10.18502/kss.v9i11.15779

The Role of Financial Literacy in the Relationship Between Personal Financial Apps and the Financial Behaviour Among College Students

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Abstract

The present era allowed financial applications to be widely used and increased the productivity of workers through frequent use and navigation of these tools. However, there was limited research on how these financial applications could cultivate positive financial behavior or whether financial literacy played a part in this process. Financial literacy, by definition, was a skill that summarized the ability of individuals to effectively use tools and financial concepts to manage money. Thus, in this study, the researchers sought to evaluate the role of financial literacy in the relationship between personal financial apps (PFAs) and financial behavior among college students. Specifically, it intended to assess whether financial literacy played a role in how financial apps catalyzed the behavior of college students in terms of budgeting, spending, saving, and credit monitoring. Quezon City served as the study locale, and the respondents hailed from Far Eastern University—Diliman, Ateneo de Manila University, University of the Philippines—Diliman, Far Eastern University—NRMF, and the National University of Fairview. In this correlational-descriptive study, respondents were sorted through a mix of purposive and simple random sampling types. A 4-point Likert scale survey questionnaire was used as a data-gathering instrument. Data were analyzed through various statistical tools, including percentage, mean, standard deviation, and partial least squares structural equation modeling (PLS-SEM). Findings revealed that financial literacy positively impacted all financial behaviors. A partial mediation was also observed, indicating that PFAs influenced financial behaviors both directly and through financial literacy. These results suggested that while students were financially literate, there were still specific areas where PFAs could be utilized to foster better financial habits.

Keywords: Financial literacy, Financial Application, Financial Behaviour

Introduction

Financial management, defined as the process of identifying, obtaining, allocating, and utilizing financial resources to achieve financial goals, is recognized as a critical factor influencing financial well-being and stability. This skill, often reflected in the handling of investments, loans, and incomes, has implications for the probability of acquiring debt (Chuah et al., 2020; Khoirunnisaa & Johan, 2020). Despite its importance, financial management has rarely been taught in schools, leaving individuals to acquire this skill informally or rely on external financial managers (Mashud et al., 2021).

Financial literacy, the ability to understand and apply financial concepts effectively, plays a key role in financial management. This skill is often developed through observation and environmental influences, which means individuals in poorly managed financial environments are at higher risk of accumulating debt (Goyal et al., 2021). Adolescence has been identified as a significant period for developing financial literacy and behaviors, as individuals begin managing their personal finances, often through allowances or part-time earnings (Laursen & Veenstra, 2021).

College students exhibit diverse financial management behaviors influenced by various factors, including the use of financial applications. Financial behavior, which encompasses expenditure management, budgeting, and saving activities, is shaped by financial literacy (Rahman et al., 2021). Prior studies have shown mixed findings regarding the relationship between financial literacy and behavior. For instance, Farida et al. (2021) reported no significant impact of financial literacy on financial behavior, while Dewi et al. (2020) found that improved financial knowledge positively influenced financial behavior, particularly among younger individuals, with low income identified as a mediating factor.

These studies, however, predominantly focused on older populations and may not reflect the current practices shaped by advancements in financial technology. The increasing use of financial applications has transformed money management

practices, yet limited research has explored their influence on financial literacy and behavior among students. This study aimed to evaluate the significance of financial literacy in the relationship between personal financial applications and financial behavior among college students. The findings contribute to the literature on financial behavior and provide insights into the potential integration of financial application-based programs in educational institutions to enhance students' financial literacy.

Literature Review

Financial Literacy

Financial literacy refers to the ability to understand and apply various financial skills, including budgeting, saving, investing, and managing debt (Lusardi & Mitchell, 2021). It involves a comprehensive understanding of financial concepts and the ability to make informed financial decisions, such as controlling debt, saving for retirement, and making wise investment choices (Huston, 2020). Financially literate individuals are better equipped to manage their finances and achieve financial stability (Fernando, J., 2024). The OECD (2020a) emphasized that the ultimate goal of financial literacy is to enhance financial well-being. Moreover, Bhushan and Medury (2020) linked higher financial literacy with better financial behaviors, including responsible borrowing and effective savings strategies.

Despite its significance, financial literacy remains low in many countries, even in those with well-established financial systems (OECD, 2020b). Financial illiteracy is notably more pronounced in certain demographic groups, particularly women and individuals with lower income or educational attainment (OECD, 2022a). This gap in financial knowledge poses challenges for individuals trying to make sound financial decisions, and underscores the need for more targeted financial education initiatives.

Financial Literacy Among Undergraduate Students

Research on the financial literacy of college students has shown that there is significant variability in their financial knowledge and behaviors. Factors such as financial attitude and knowledge are crucial in shaping how students manage their finances. Studies by Yogasnumurti et al. (2021) and Kumari (2020) revealed that students with positive financial attitudes and higher levels of financial knowledge were better at managing their finances and achieving long-term financial goals. However, a study by Philippas and Avdoulas (2023) found that university students generally exhibited higher financial literacy compared to the general population. They noted that students whose fathers had higher educational qualifications were more likely to demonstrate greater financial literacy. Furthermore, students who engaged in practices such as saving and tracking expenses were less likely to experience financial fragility.

Several studies also highlighted the role of peer influence and parental guidance in shaping students' financial behaviors. For instance, Majida et al. (2021) noted that Pakistani undergraduate students used cost-reduction strategies, such as prioritizing debit cards over credit cards and searching for cheaper products on social media. This behavior was influenced by a combination of economic, sociological, and psychological factors, including the guidance of family and peers. In contrast, Choong et al. (2023) found that subjective norms and financial knowledge, rather than financial attitude, were significant predictors of budgeting practices among students in Malaysia. These differences in findings emphasize the influence of geographical and cultural factors on financial behaviors.

Additionally, the integration of fintech solutions, such as personal finance apps, has been shown to enhance financial management among students. Irawan and Matoati (2021) found that students using fintech tools for transactions exhibited better financial management skills, suggesting the importance of technology in improving financial literacy.

Role of Personal Finance Apps in Enhancing Financial Literacy

Personal finance apps have become an increasingly popular tool for improving financial literacy. These apps offer users the ability to track their financial progress, set goals, and manage their finances more effectively. According to Gray Group International (2024), personal finance apps have been shown to positively influence individual financial behavior by simplifying financial tracking and decision-making processes. These apps provide users with immediate access to financial data and educational resources, including articles, videos, and podcasts on budgeting, saving, investing, and debt management (Personal Finance Apps: Harnessing Technology for Financial Management, 2024). They also offer security features that protect users' financial information, raising awareness about potential risks and helping users make smarter financial choices (Abhishek, 2024).

Research has demonstrated that personal finance apps contribute to healthier spending habits by visualizing spending trends and providing tailored financial recommendations (Abhishek, 2024). These apps are particularly useful in helping individuals set and track financial goals, thus boosting motivation and enhancing financial decision-making (Baruah, 2023). The integration of artificial intelligence in some finance apps has further improved their effectiveness by offering personalized financial guidance based on users' spending patterns (Abhishek, 2024). As such, personal finance apps are recognized as valuable tools for promoting financial literacy and improving financial behaviors.

Theoretical Framework

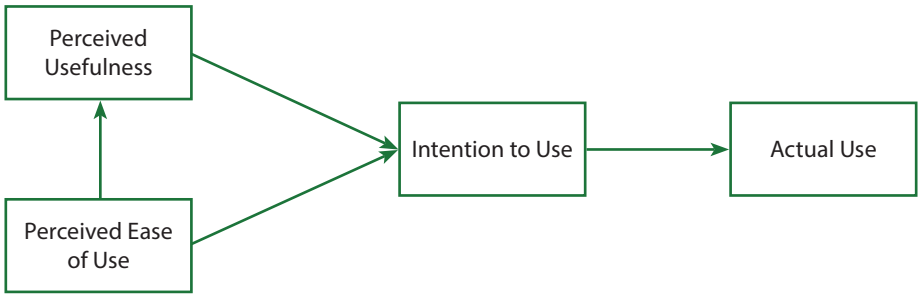


Fig. 1. Technology Acceptance model (Fred Davis, 1989)

Note: Technology Acceptance Model (TAM). Adapted from *Perceived usefulness, perceived ease of use, and user acceptance of information technology* by F. D. Davis, 1989, *MIS Quarterly*, 13(3), p. 320. Copyright 1989 by the Management Information Systems Research Center, University of Minnesota.

Thus, in this study, the researchers theorize that the financial behavior of college students can be enhanced and influenced through their increased financial literacy, which in turn is enhanced through their frequent use of financial apps. Additionally, it is evident that the intention of use plays a role between the actual process of using the technology and the perceived usefulness and simplicity of use by technology users. Consequently, in the context of this study, the researchers assumed that financial literacy serves as a mediating variable, which is a dependent variable, and can also influence the behavior of college students in terms of saving, spending, budgeting, and credit monitoring.

Methodology

Research Design

The study utilized a descriptive correlational design with a quantitative approach. A survey questionnaire with a 4-point Likert scale assessed financial literacy, app usage, and financial behavior.

Sample and Locale

The sample comprised 385 college students from five universities in Quezon City. Using Cochran's formula, the sample was proportionally distributed based on the total student population of 38,034.

Table 2. Sample size percentage of respondents

Selected Schools in Quezon City	Respondents	Percentage	Sample Size
College Students	385		
Far Eastern University – Diliman	855	2.25%	9
Ateneo De Manila University	8,614	22.65%	88
University of the Philippines Diliman	17,035	44.79%	172
Far Eastern University – NRMF	5,269	13.85%	63
National University – Fairview	6,261	16.46%	53
Total	**38,034**	**100%**	**385**

Note: The table shows the breakdown of respondents from selected schools in Quezon City, indicating the number of students surveyed and their corresponding percentage of the total sample size.

Data Collection Instrument

Data were collected through a structured questionnaire validated via Cronbach's alpha, with reliability scores exceeding 0.8 across domains. Descriptive and inferential statistics, including Partial Least Squares Structural Equation Modeling (PLS-SEM), were used to analyze relationships between variables.

Analysis and Discussions

Demographic Characteristics of Respondents

Age The age distribution of the 385 respondents revealed that the majority (43.12%) were aged 21–23 years, followed by 29.61% aged 18–20 years, 24.94% aged 24–26 years, and 2.34% aged 27 years and above (Table 8a).

Gender The gender distribution was nearly balanced, with 51.95% of respondents identifying as female and 48.05% identifying as male (Table 8b).

School The respondents were from diverse educational institutions: 44.68% from the University of the Philippines Diliman, 22.86% from Ateneo De Manila University, 16.36% from Far Eastern University-NRMF, 13.77% from National University-Fairview, and 2.34% from Far Eastern University Diliman (Table 8c).

Course/Program The course distribution showed variation, with 7.53% enrolled in BSBA-FM, 4.94% in BSBA-MM, 4.68% in BSIT, and 9.35% in BSA. A significant proportion of respondents (73.51%) did not disclose their program, and no responses were recorded for BSBA-OM (Table 8d).

Tables

Table 8a. Age of the Respondents (n=385)

Age	Frequency	Percentage
18-20	114	29.61%
21-23	166	43.12%
24-26	96	24.94%
27 and above	9	2.34%
Total		**100%**

Note: The table provides the age distribution of respondents, showing the frequency and percentage of each age group among the total sample of 385 respondents.

Table 8b. Gender of the Respondents (n=385)

Gender	Frequency	Percentage
Male	185	48.05%
Female	200	51.95%
Total	**385**	**100%**

Note: The table presents the gender distribution of the respondents, indicating the frequency and percentage of males and females among the total sample of 385 respondents.

Table 8c. School of the Respondents (n=385)

School	Frequency	Percentage
Far Eastern University – Diliman	9	2.34%
Ateneo De Manila University	88	22.86%
University of the Philippines Diliman	172	44.68%
Far Eastern University - NRMF	63	16.36%
National University - Fairview	53	13.77%
Total	**385**	**100%**

Note: The table shows the distribution of respondents by school, indicating the frequency and percentage of students from each institution within the total sample of 385 respondents.

Table 8d. Course/Program of the Respondents (n=385)

Course/Program	Frequency	Percentage
BSBA - FM	29	7.53%
BSBA - MM	19	4.94%
BSBA- OM	0	0%
BSIT	18	4.68%
BSA	36	9.35%
Others	283	73.51%
Total	**385**	**100%**

Note: The table displays the distribution of respondents by course or program, showing the frequency and percentage of students enrolled in each category within the total sample of 385 respondents.

Year Level of Respondents The year level distribution of the 365 respondents revealed that the largest proportion were third-year students (33.51%), followed by fourth-year students (24.16%), second-year students (22.08%), and first-year students (20.26%) (Table 8e).

Table 8e. Year Level of the Respondents (n=385)

Year Level	Frequency	Percentage
1st Year	78	20.26%
2nd year	85	22.08%
3rd year	129	33.51%
4th Year	93	24.16%
Total	**385**	**100%**

Note: The table presents the distribution of respondents by year level, indicating the frequency and percentage of students in each year level within the total sample of 385 respondents.

Confirmatory Factor Analysis (CFA)

Personal Financial Apps The CFA for the constructs related to personal financial apps, namely accessibility, adaptability, reliability, and security, showed that all variables were valid indicators for their respective constructs. Factor loadings ranged from 0.538 to 0.775, with p-values all less than 0.001, indicating statistical significance. The Cronbach's alpha values ranged from 0.710 to 0.761, composite reliability values ranged from 0.829 to 0.856, and average variance extracted (AVE) values ranged from 0.566 to 0.607, confirming high reliability for the constructs (Table 9a).

Table 9a. Confirmatory Factor Analysis (CFA) of the Study's Constructs in terms of Financial Apps

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite Reliability	Ave. Variances Extracted
Accessibility				0.710	0.829	0.566
	Acces1	0.716	<0.001			
	Acces2	0.769	<0.001			
	Acces3	0.677	<0.001			
	Acces4	0.617	<0.001			
	Acces5	0.538	<0.001			
Adaptability				0.714	0.831	0.57
	Adapt1	0.775	<0.001			
	Adapt2	0.552	<0.001			
	Adapt3	0.723	<0.001			
	Adapt4	0.667	<0.001			
	Adapt5	0.612	<0.001			
Reliability				0.758	0.854	0.604
	Relia1	0.674	<0.001			
	Relia2	0.697	<0.001			
	Relia3	0.671	<0.001			
	Relia4	0.715	<0.001			
	Relia5	0.721	<0.001			

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite Reliability	Ave. Variances Extracted
Security				0.761	0.856	0.607
	Sec1	0.726	<0.001			
	Sec2	0.722	<0.001			
	Sec3	0.634	<0.001			
	Sec4	0.717	<0.001			
	Sec5	0.687	<0.001			

Note: This table presents the results of the Confirmatory Factor Analysis (CFA) for the constructs related to financial apps. Factor loadings indicate the strength of the relationship between each item and its corresponding construct, with p-values less than 0.001 denoting statistical significance. Cronbach's Alpha, composite reliability, and average variance extracted (AVE) values assess the reliability and validity of each construct.

Financial Literacy The CFA results for financial literacy showed that all five variables had factor loadings ranging from 0.554 to 0.733, with p-values less than 0.001, confirming their validity. The Cronbach's alpha was 0.709, composite reliability was 0.828, and AVE was 0.564, indicating that the construct was measured reliably (Table 9b).

Table 9b. Confirmatory Factor Analysis (CFA) of the Study's Constructs in terms of Financial Literacy

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite Reliability	Ave. Variances Extracted
Financial Literacy				0.709	0.828	0.564
	FL1	0.657	<0.001			
	FL2	0.733	<0.001			
	FL3	0.682	<0.001			
	FL4	0.69	<0.001			
	FL5	0.554	<0.001			

Note: This table presents the Confirmatory Factor Analysis (CFA) results for financial literacy. Factor loadings indicate item reliability, with p-values less than 0.001 denoting significance. Cronbach's Alpha, composite reliability, and average variance extracted (AVE) assess the construct's reliability and validity.

CFA for Financial Behavior The CFA for financial behavior showed that all variables for budgeting, spending, savings, and credit monitoring were valid indicators for their constructs. Factor loadings ranged from 0.633 to 0.815, with p-values less than 0.001, signifying statistical significance. Cronbach's alpha ranged from 0.730 to 0.815, composite reliability ranged from 0.839 to 0.888, and AVE values ranged from 0.578 to 0.664, confirming the reliability of the constructs (Table 9c).

Table 9c. Confirmatory Factor Analysis (CFA) of the Study's Constructs in terms of Financial Literacy

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite Reliability	Ave. Variances Extracted
Budgeting				0.73	0.839	0.578
	Bud1	0.684	<0.001			
	Bud2	0.669	<0.001			
	Bud3	0.685	<0.001			
	Bud4	0.675	<0.001			
	bud5	0.672	<0.001			
Spending				0.733	0.841	0.583
	Spend1	0.726	<0.001			
	Spend2	0.736	<0.001			

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite Reliability	Ave. Variances Extracted
	Spend3	0.633	<0.001			
	Spend4	0.61	<0.001			
	Spend5	0.689	<0.001			
Savings				0.757	0.854	0.604
	Sav1	0.751	<0.001			
	Sav2	0.615	<0.001			
	Sav3	0.721	<0.001			
	Sav4	0.69	<0.001			
	Sav5	0.695	<0.001			
Credit Monitoring				0.815	0.888	0.664
	CM1	0.763	<0.001			
	CM2	0.692	<0.001			
	CM3	0.726	<0.001			
	CM4	0.765	<0.001			
	CM5	0.751	<0.001			

Note: This table summarizes the CFA results for financial behavior constructs. All factor loadings are statistically significant ($p < 0.001$). Reliability metrics are provided for each construct: Cronbach's Alpha, composite reliability, and average variance extracted.

Respondents' Assessment of Financial Apps in Terms of Accessibility

The respondents' assessment of the accessibility of financial apps revealed a generally positive perception. The mean score for accessibility was 3.51 (SD = 0.63), indicating that respondents perceived financial apps as accessible. The highest mean score (3.78) was for SOP 2.1.1, which measured ease of downloading and installing financial apps. This finding aligned with previous studies that emphasized the ease of access provided by multiple app stores (Aziz & Naima, 2021; Shashank & Naidu, 2020). The lowest mean score (3.36) was for SOP 2.1.5, suggesting that respondents felt financial apps were not frequently updated to enhance accessibility features. This result corroborated the observation by Gong et al. (2021) that apps focused on stability and functionality, such as financial apps, are updated less frequently than those prioritizing aesthetic and entertainment features.

Accessibility of Financial Apps Respondents demonstrated a strong level of agreement on the accessibility features of personal financial apps. As shown in Table 10a, the total mean score for accessibility was 3.51 (SD = 0.63), indicating that respondents generally found financial apps to be easily accessible. The highest mean score, 3.78 (SD = 0.48), was observed for the statement "I find the personal financial app easy to download and install," while the lowest mean of 3.36 (SD = 0.69) was for "The personal financial app is regularly updated to improve accessibility features." This suggests that while users find the app easy to install and use, there may be varying opinions on the frequency of updates aimed at improving accessibility. The overall interpretation was that respondents strongly agreed with the accessibility features of financial apps.

Table 10a. Respondents' Assessment on Financial Apps in terms of Accessibility

Assessment on Financial Apps in terms of Accessibility	Mean	SD	Interpretation
I find the personal financial app easy to download and install.	3.78	0.48	Strongly Agree
I find personal financial apps more convenient than traditional methods of managing finances.	3.45	0.60	Strongly Agree
I find the personal financial app user-friendly and easy to navigate.	3.54	0.61	Strongly Agree
I find that the personal financial app provides consistent performance across different devices.	3.44	0.64	Strongly Agree
I find that the personal financial app is regularly updated to improve accessibility features.	3.36	0.69	Strongly Agree
Overall	**3.51**	**0.63**	**Strongly Agree**

* Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–24 (Agree), 3.25–4.00 (Strongly Agree)

Adaptability of Financial Apps Respondents also showed a strong level of agreement on the adaptability features of financial apps, with a total mean score of 3.39 (SD = 0.68) (Table 10b). The highest mean score, 3.53 (SD = 0.63), was for "I can easily customize the categories and labels in the personal financial app to align with my specific budgeting preferences, "reflecting users' appreciation for personalized features. However, the lowest mean score, 3.25 (SD = 0.74), was for "The personal financial app provides useful updates and new features regularly," suggesting some respondents felt that updates were not as frequent as expected. These findings align with previous studies indicating that personalization in financial apps enhances user engagement and satisfaction.

Table 10b. Respondents' Assessment on Financial Apps in terms of Adaptability

Assessment on Financial Apps in terms of Adaptability	Mean	SD	Interpretation
The personal financial app is observed to adapt to my personal financial needs.	3.35	0.64	Strongly Agree
I can easily customize the categories and labels in the personal financial app to align with his/her specific budgeting preferences.	3.53	0.63	Strongly Agree
The personal financial app provides useful updates and new features regularly.	3.25	0.74	Strongly Agree
The personal financial app promptly notifies me of any financial adjustments or changes, enabling me to adapt my budgeting strategies accordingly.	3.32	0.66	Strongly Agree
I find it easy to adjust my budgeting goals and targets within the personal financial app.	3.49	0.68	Strongly Agree
Overall	**3.39**	**0.68**	**Strongly Agree**

* Interpretation: 1.00- 1.74(Strongly Disagree), 1.75- 2.49(Disagree), 2.50- 24(Agree), 3.25- 4.00(Strongly Agree)

Reliability of Financial Apps The total mean score for reliability was 3.36 (SD = 0.64), indicating strong agreement regarding the reliability of financial apps (Table 10c). The highest mean score, 3.42 (SD = 0.61), was for "The personal financial app provides accurate information about budgeting and saving," highlighting respondents' trust in the app's ability to manage financial data. The lowest mean score, 3.31 (SD = 0.61), was for "I trust the information and guidance provided by the personal financial app I use," suggesting that while respondents generally trusted the app's guidance, there were some variations in the level of confidence. The findings support the view that users value reliable information for financial decision-making and that trust plays a crucial role in the usage of financial apps.

Table 10c. Respondents' Assessment on Financial Apps in terms of Reliability

Assessment on Financial Apps in terms of Reliability	Mean	SD	Interpretation
The personal financial app provides accurate information about budgeting and saving.	3.42	0.61	Strongly Agree
I trust the information and guidance provided by the personal financial app I use.	3.31	0.61	Strongly Agree
I trust that the personal financial app consistently provides reliable and up-to-date information for financial decision-making.	3.38	0.60	Strongly Agree
The personal financial app provides reliable customer support for addressing technical issues and inquiries.	3.33	0.68	Strongly Agree
I feel confident in the accuracy and reliability of the financial advice provided by the personal financial app.	3.37	0.66	Strongly Agree
Overall	**3.36**	**0.64**	**Strongly Agree**

* Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–24 (Agree), 3.25–4.00 (Strongly Agree)

Security of Financial Apps Respondents reported a strong sense of security with their financial apps, as evidenced by the total mean score of 3.51 (SD = 0.63) (Table 10d). The highest mean score, 3.62 (SD = 0.59), was for "The personal financial app's terms of service and privacy policy are easily accessible and comprehensible," indicating that users appreciated clear privacy policies. The lowest mean score, 3.41 (SD = 0.64), was for "The personal financial app provides secure login alternatives, such

as two-factor authentication or biometric identification," suggesting some concern over the security measures in place. Despite this, the overall strong agreement with the security features highlights the importance of secure financial apps in maintaining user confidence.

Table 10d. Respondents' Assessment on Financial Apps in terms of Security

Assessment on Financial Apps in terms of Security	Mean	SD	Interpretation
The personal financial app provides accurate information about budgeting and saving.	3.57	0.64	Strongly Agree
I trust the information and guidance provided by the personal financial app I use.	3.46	0.65	Strongly Agree
I trust that the personal financial app consistently provides reliable and up-to-date information for financial decision-making.	3.41	0.64	Strongly Agree
The personal financial app provides reliable customer support for addressing technical issues and inquiries.	3.47	0.62	Strongly Agree
I feel confident in the accuracy and reliability of the financial advice provided by the personal financial app.	3.62	0.59	Strongly Agree
Overall	**3.51**	**0.63**	**Strongly Agree**

* Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–24 (Agree), 3.25–4.00 (Strongly Agree)

Financial Literacy of Respondents

The overall financial literacy of respondents was high, with a mean score of 3.43 (SD = 0.67) (Table 11). The highest mean score, 3.58 (SD = 0.59), was for "I am well informed about various investment streams," indicating that respondents were knowledgeable about investment options. However, the lowest mean score, 3.26 (SD = 0.69), was for "I am aware of my retirement financial requirements," suggesting that while respondents were generally financially literate, there were areas, such as retirement planning, where knowledge was slightly less pronounced. This reflects the trend where university students, although highly literate in personal finance, may have gaps in their understanding of long-term financial planning.

Table 11. The Level of Financial Literacy of the Respondents

The Level of Financial Literacy	Mean	SD	Interpretation
I have a better understanding of where and how to invest my money.	3.58	0.59	Strongly Agree
I have a very clear idea of my financial needs during retirement.	3.26	0.69	Strongly Agree
I have the ability to maintain financial records for my income and expenses.	3.43	0.66	Strongly Agree
I have little or no difficulty in managing my money.	3.30	0.65	Strongly Agree
I have the ability to prepare my own weekly/monthly budget.	3.46	0.67	Strongly Agree
Overall	**3.43**	**0.67**	**Strongly Agree**

Note: The level of financial literacy of the Respondents. Adapted from: Alshebami Ali Saleh and Aldhyani Theyazn. (2022). The Interplay of Social Influence, Financial Literacy, and Saving Behaviour among Saudi Youth and the Moderating Effect of Self-Control.

Financial Behavior

Financial Behavior: Budgeting Respondents showed strong engagement in budgeting behaviors, with all mean values falling between 3.37 and 3.63, indicating that participants were proactive in setting financial goals and managing their budgets (Table 12a). The highest mean score, 3.63 (SD = 0.55), was recorded for "I set goals to manage my money," highlighting the importance of goal-setting in financial management. The mean score for "I proactively seek out opportunities to optimize my budget and allocate resources towards long-term financial objectives" was 3.37 (SD = 0.66), which was the lowest in this category, suggesting some challenges in optimizing budgets. These findings align with previous studies emphasizing the importance of goal-setting for financial success (Lusardi & Mitchell, 2014) and the role of financial literacy in effective budgeting (Xiao, 2016).

Table 12a. The Engagement of the Respondents in Financial Behavior

The Engagement of the Respondents in Financial Behavior	Mean	SD	Interpretation
Budgeting	3.45	0.63	Strongly Agree
I set goals to manage my money.	3.63	0.55	Strongly Agree
I actively create and follow a budget for my finances.	3.45	0.62	Strongly Agree
I regularly track my budget performance and use the insights gained to make informed decisions about my financial priorities.	3.39	0.63	Strongly Agree
I proactively seek out opportunities to optimize my budget and allocate resources towards long-term financial objectives.	3.37	0.66	Strongly Agree
I regularly review and adjust my budget to reflect changes in income, expenses, and financial goals.	3.37	0.65	Strongly Agree

* Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–24 (Agree), 3.25–4.00 (Strongly Agree)

Financial Behavior: Spending In terms of spending, respondents also showed strong engagement, with mean scores ranging from 3.36 to 3.57, all within the "strongly agree" range (Table 12b). The highest mean score, 3.57 (SD = 0.64), was for "I set categories on where I spend my money and set limits to each category," suggesting that respondents were highly organized in their spending. The lowest mean score, 3.36 (SD = 0.62), was for "I am mindful of making informed financial decisions to control my spending and maximize value for my money," indicating that while respondents were generally responsible spenders, there was some room for improvement in making informed decisions. These findings align with research showing the positive impact of budget classification on financial well-being (Lusardi & Tufano, 2015), although some challenges in maintaining spending discipline were noted (DeVoe & Pfeffer, 2016).

Table 12b. The Engagement of the Respondents in Financial Behavior

The Engagement of the Respondents in Financial Behavior	Mean	SD	Interpretation
Spending	3.42	0.65	Strongly Agree
I am proactive in seeking out opportunities to reduce unnecessary expenses and optimize my spending.	3.37	0.68	Strongly Agree
I prioritize responsible spending and aim to make purchases that align with my financial priorities and values.	3.38	0.69	Strongly Agree
I am mindful of making informed financial decisions to control my spending and maximize value for my money.	3.36	0.62	Strongly Agree
I am conscientious about tracking my expenses and managing my spending habits.	3.38	0.61	Strongly Agree
I set categories on where I spend my money and set limits to each category.	3.57	0.64	Strongly Agree

* Interpretation: 1.00- 1.74(Strongly Disagree), 1.75- 2.49(Disagree), 2.50- 24(Agree), 3.25-4.00(Strongly Agree)

Financial Behavior: Savings

Respondents demonstrated strong engagement in saving behaviors, with mean scores ranging from 3.31 to 3.63, all within the "strongly agree" range (Table 12c). The highest mean score, 3.63 (SD = 0.64), was for "I prioritize using savings for emergencies," highlighting the importance respondents placed on emergency savings. "I regularly set aside income for savings" scored 3.50 (SD = 0.67), indicating that respondents were committed to saving regularly. However, the lowest mean score, 3.31 (SD = 0.70), was for "I set monthly savings goals," suggesting some difficulty in maintaining consistent short-term savings goals. These results are consistent with studies showing that emergency savings and long-term financial preparedness are crucial for financial security (Friedline et al., 2015), though challenges in saving for long-term goals due to immediate financial pressures were also observed (Hannagan & Morduch, 2015).

Table 12c. The Engagement of the Respondents in Financial Behavior

The Engagement of the Respondents in Financial Behavior	Mean	SD	Interpretation
Savings	3.47	0.67	Strongly Agree
I actively set aside a portion of my income for savings on a regular basis.	3.50	0.67	Strongly Agree
I am proactive in planning and saving for long-term financial goals, such as retirement or major purchases.	3.31	0.66	Strongly Agree
I set a monthly goal for my savings.	3.56	0.69	Strongly Agree
I prioritize saving for the future as an essential aspect of my financial well-being.	3.35	0.64	Strongly Agree
Savings can be used for emergency purposes.	3.63	0.62	Strongly Agree

* Interpretation: 1.00- 1.74(Strongly Disagree), 1.75- 2.49(Disagree), 2.50- 24(Agree), 3.25-4.00(Strongly Agree)

Financial Behavior: Credit Monitoring

Respondents exhibited significant involvement in credit monitoring activities, with average scores between 3.31 and 3.56, all falling inside the "strongly agree" category (Table 12d). The mean score of 3.56 for "Credit monitoring is essential for safeguarding financial well-being and identity" underscores the significance attributed to financial security protection. The scores for "Limiting credit to what can be repaid" (3.44) and "Paying credit on or before the due date" (3.43) indicate responsible credit management, however the lower scores for "Tracking remaining credit" (3.32) and "Taking note of all credits" (3.31) imply a need for enhancement in meticulous monitoring. These findings correspond with studies regarding the significance of credit monitoring in preventing debt and promoting financial stability (Joo & Grable, 2004; Xiao & Porto, 2017), while also reflecting apprehensions about overconfidence diminishing the perceived necessity for ongoing monitoring (de Bassa Scheresberg, 2013).

Table 12d. The Engagement of the Respondents in Financial Behavior

The Engagement of the Respondents in Financial Behavior	Mean	SD	Interpretation
Credit Monitoring	3.41	0.68	Strongly Agree
I always take note of all the credits that I have.	3.31	0.70	Strongly Agree
I always check if I have remaining credits.	3.32	0.68	Strongly Agree
I make sure to limit my credit to the amount that I can pay.	3.44	0.68	Strongly Agree
I pay my credit on or before the given due date.	3.43	0.65	Strongly Agree
I believe that credit monitoring is an essential tool for safeguarding my financial well-being and identity.	3.46	0.64	Strongly Agree

* Interpretation: 1.00- 1.74(Strongly Disagree), 1.75- 2.49(Disagree), 2.50- 24(Agree), 3.25-4.00(Strongly Agree)

Personal Financial Apps on Financial Literacy and Behavior

The path coefficients, p-values, and effect sizes for the relationships between personal financial apps (PFA), financial literacy, and financial behaviors provide robust statistical validation across all pathways (Table 13). Financial literacy significantly influences spending behavior ($\beta = 0.602$, $f2 = 0.363$), followed by savings behavior ($\beta = 0.542$, $f2 = 0.294$) and budgeting behavior ($\beta = 0.518$, $f2 = 0.269$), each exhibiting medium to large impact sizes. The effect on credit monitoring behavior is somewhat less ($\beta = 0.447$, $f2 = 0.200$), although remains substantial. These findings correspond with previous research highlighting the significance of financial literacy in promoting sound financial practices, such as budgeting, saving, and credit management (Lusardi & Mitchell, 2014; Fernandes et al., 2014). The substantial impact of PFAs on financial literacy ($\beta = 0.535$, $f2 = 0.286$) underscores the importance of digital tools in improving financial understanding and decision-making (Klapper et al., 2016). These findings highlight the essential relationship between financial literacy and technology in fostering prudent financial practices.

Table 13. Path coefficient and P-value

Path	Path Coefficients (β)	P-value	Effect Sizes (f2)	Effect Size Interpretation (Cohen 1988)	Description	Interpretation
H1: PFA → FL	0.535	<0.001	0.286	Medium	Significant	H1 is supported
H2: FL → Budgeting Behavior	0.518	<0.001	0.269	Medium	Significant	H2 is supported
H3: FL → Spending Behavior	0.602	<0.001	0.363	Large	Significant	H3 is supported
H4: FL → Savings Behavior	0.542	<0.001	0.294	Medium	Significant	H4 is supported
H5: FL → Credit Monitoring Behavior	0.447	<0.001	0.200	Medium	Significant	H5 is supported

Note: Personal Financial App (PFA) ***0.02 – small, 0.15 – medium, 0.36 – large*

Mediation Analysis of PFA with Financial Literacy and Budgeting Behavior

The mediation effects of financial literacy on the relationship between personal financial apps (PFA) and financial behaviors, including budgeting, spending, saving, and credit monitoring, prove significant direct, indirect, and overall impacts ($p < 0.001$), confirming partial mediation across all hypotheses. The strongest total effect is observed between PFA and spending behavior, with an indirect effect highlighting the role of financial literacy in improving spending control. Financial literacy also mediates the effects of PFAs on budgeting, saving, and credit monitoring behaviors, emphasizing its crucial role in enhancing financial decision-making. These findings align with prior research, suggesting that PFAs not only facilitate financial tasks but also enhance financial literacy, empowering users to make informed decisions and achieve long-term financial well-being (Königstorfer & Gröppel-Klein, 2022; Xiao & O'Neill, 2018).

Table 14. Effect for the path with 2 segments (Mediation Analysis)

Path coefficients and p-values

Path	Total Effect	P-value	Direct Effect	P-value	Indirect Effect	P-value	Description	Interpretation	Remark
H6: PFA → FL → Budgeting Behavior	0.519	<0.001	0.342	<0.001	0.177	<0.001	Significant	Partial Mediation	H6 is supported
H7: PFA → FL → Spending Behavior	0.617	<0.001	0.417	<0.001	0.200	<0.001	Significant	Partial Mediation	H7 is supported
H8: PFA → FL → Savings Behavior	0.519	<0.001	0.325	<0.001	0.194	<0.001	Significant	Partial Mediation	H8 is supported
H9: PFA → FL → Credit Monitoring	0.484	<0.001	0.351	<0.001	0133	<0.001	Significant	Partial Mediation	H9 is supported

Note: Personal Financial App (PFA) **0.02 – small, 0.15 – medium, 0.36 – large

Square Root of Average Variance Extracted (AVE)

The square roots of the average variance extracted (AVE) and inter-construct correlations indicate strong convergent and discriminant validity for the assessed constructs (Table 15a). AVE values, including 0.835 for financial applications and 0.817 for credit monitoring, surpass the required threshold of 0.7, so affirming strong convergent validity (Hair et al., 2017). Inter-construct correlations vary from 0.412 to 0.653, with the most significant connection of 0.653 noted between savings and spending, signifying a congruence in both behaviors among financially knowledgeable persons. Weaker correlations, such as 0.412 between financial literacy and credit monitoring, indicate that the impact of financial literacy differs among behaviors, aligning with previous research (Potrich et al., 2016; Lusardi & Mitchell, 2014). These findings highlight the uniqueness of the notions, confirming discriminant validity, which is essential for precisely representing diverse aspects of financial behavior.

Table 15a. Square Roots of Ave Coefficients and Correlation Coefficients

Latent Variables	FA	FL	Budgeting	Spending	Savings	CM
FA	**0.835**	0.517	0. 518	0.577	0.498	0.447
FL	0.517	**0.751**	0.502	0.593	0.528	0.412
Budgeting	0.518	0.502	**0.760**	0.579	0.574	0.490
Spending	0.577	0.593	0.579	**0.764**	0.653	0.529

Latent Variables	FA	FL	Budgeting	Spending	Savings	CM
Savings	0.498	0.528	0.574	0.653	**0.777**	0.469
CM	0.447	0.412	0.49	0.529	0.469	**0.817**

Note: Diagonal values are the square roots of AVE and off-diagonals are inter-construct correlation.

Heterotrait-Monotrait (HTMT) ratios

The Heterotrait-Monotrait (HTMT) ratios validate the discriminant validity of the components in this investigation, as all values are below the conservative threshold of 0.85 (Table 15b). The highest ratio of savings to consumption (0.906) signifies a robust correlation while remaining within permissible parameters (Henseler et al., 2015). Reduced HTMT values, exemplified by 0.542 between financial applications and credit monitoring, reinforce the uniqueness of the constructions. The elevated correlations between spending and planning (0.828) and saves (0.906) indicate the interconnected nature of these behaviors, thereby confirming previous studies that assert successful budgeting is congruent with disciplined spending and saving practices (Xiao & Porto, 2017). These findings confirm the discriminant validity of the measures while recognizing the inherent overlap across related financial activities (Voorhees et al., 2016).

Table 15b. Heterotrait-Monotrait (HTMT) Ratios

Latent Variables	FA	FL	Budgeting	Spending	Savings	CM
FA						
FL	0.678					
Budgeting	0.667	0.728				
Spending	0.744	0.858	0.828			
Savings	0.633	0.756	0.801	0.906		
CM	0.542	0.569	0.654	0.705	0.611	

Note: This table displays the Heterotrait-Monotrait (HTMT) ratios among the latent variables of financial apps, financial literacy, budgeting, spending, savings, and credit monitoring. The values indicate the strength of the relationships between different constructs, with lower values suggesting distinct constructs.

Conclusions

This study examined the mediating role of financial literacy in the relationship between personal financial app usage and financial behavior among college students. The findings revealed that participants were predominantly young adults aged 21 to 23, with a balanced gender distribution and primarily third-year students from universities in Quezon City, including the University of the Philippines Diliman. Personal financial apps were highly rated for accessibility, adaptability, reliability, and security, reflecting general satisfaction and trust in these tools for financial management. Respondents demonstrated a solid level of financial literacy, excelling in areas such as budgeting and investment knowledge, though some variability was observed in retirement planning awareness.

Recommendation

The findings of this study have highlighted the critical role financial literacy plays in shaping the financial behaviors of students, particularly in budgeting, saving, and credit monitoring. As students transition into adulthood and take on more financial responsibilities, their ability to manage money effectively becomes increasingly important. Personal financial apps have emerged as powerful tools to assist students in managing their finances, yet their effectiveness is often mediated by the users' level of financial literacy.

Integration with Educational Seminars and Government Services Collaborate with educational institutions and government agencies to provide users with access to financial literacy and investment seminars. These should include information on topics, speakers, and registration details to facilitate informed decision-making. Such partnerships can enhance credibility and broaden access to financial education.

Enhanced App Features and User Tutorials Address app feature limitations, such as customization options and adaptability, by integrating user feedback and continuous updates. Providing tutorials on app usage can improve user experience and ensure efficient operation, fostering greater engagement with the apps.

Basic Investment Education and Digital Content Incorporate educational resources focused on the basics of investing, such as podcasts and blog posts, to equip students with foundational knowledge and practical tips. These resources can supplement the app's functionality, enhancing users' financial literacy.

AI-Powered Question Generator Implement AI-driven tools that enable users to ask financial questions and receive instant answers. This feature would improve accessibility and reliability, increasing the app's perceived usefulness.

Personalized Credit Recommendations Use algorithms to provide personalized credit card recommendations based on users' financial behaviors. This customization ensures that users can make informed decisions aligned with their spending patterns and financial goals.

These recommendations align with the Technology Acceptance Model (TAM) and underscore the mediating role of financial literacy in the relationship between app usage and financial behavior. By improving the perceived usefulness and ease of use of financial apps, these strategies can foster positive financial behaviors among students.

Direction for Future Research

Future research should explore financial literacy and behaviors by grouping respondents based on demographic factors such as university, course, and year level to determine whether these factors influence the mediating role of financial literacy. This stratified approach can provide insights into specific groups and guide tailored interventions. Additionally, future studies could narrow their inclusion criteria by focusing on respondents using specific financial apps and considering factors like app type, duration of use, payment for premium features, accessibility via phone capabilities, and personal preferences. Such an approach ensures more consistent and valid results. Incorporating qualitative methods, such as interviews, and employing experimental or longitudinal designs can further enhance understanding by capturing deeper insights into behaviors, motivations, and patterns over time. Researchers are also encouraged to allocate sufficient time and resources for studies requiring behavioral tracking and experimental designs, planning well in advance to ensure feasibility and methodological rigor. These recommendations aim to address current gaps in the literature and provide pathways for refining the understanding of financial literacy's role in shaping financial behavior.

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REFERENCES

1. Abbas, Q., Naeem, I., Nishtar, Z., & Lodhi, K. Miraj, J., Haider, S. (2023). Assessing the Effectiveness of Financial Literacy Programs for Business Student. Bulletin of Business and Economics, 12(3), 587–594. https://doi.org/10.61506/01.00073
2. Abdallah, W., Tfaily, F., & Harraf, A. (2024). The impact of digital financial literacy on financial behavior: customers' perspective. Competitiveness Review an International Business Journal Incorporating Journal of Global Competitiveness. https://doi.org/10.1108/cr-11-2023-0297
3. Abhishek. (2024, March 26). The Role of Personal Finance Apps in Building Financial Literacy. https://jobeducareer.com/the-role-of-personal-finance-apps-in-building-financial-literacy
4. Abro, A. M., Maitlo, A. A., Mughal, D. D. K., Sangah, S., & Mughal, I. (2024). Mediating Role of Financial Literacy in Relationship of Factors Affecting the Money Management Skills. Deleted Journal, 13(1), 626–642. https://doi.org/10.62345/jads.2024.13.1.52
5. Admin. (2024, April 23). How Can Personal Finance Apps Promote Financial Literacy? |Anıl Uzun.https://www.aniluzun.com/how-can-personal-finance-apps-promote-financial-literacy Advantages and Disadvantages of online surveys | CVENT blog. (n.d.). https://www.cvent.com/en/blog/events/advantages-disadvantages-online-surveys
6. Ahmed, A.A., Rajesh, S., Lohana, S., Ray, S., Maroor, J.P., Naved, M. (2023). Using Machine Learning and DataMining to Evaluate Modern Financial Management Techniques. Proceedings of Second International Conference in Mechanical and Energy Technology, Smart Innovation, Systems and Technologies.
7. Al-Adwan, A.S., & Sammour, G. (2021). What Makes Consumers Purchase Mobile Apps: Evidence from Jordan. Journal of Theoretical and Applied Electronic Commerce Research. 16, 562–583. https://doi.org/10.3390/jtaer16030034
8. Ali, G., Dida, M.A., & Elikana Sam, A. (2021). A Secure and Efficient Multi-Factor Authentication Algorithm for Mobile Money Applications. Future Internet. https://doi.org/10.3390/fi13120299
9. Alshebami, A. S., & Aldhyani, T. H. H. (2022). The Interplay of Social Influence, Financial Literacy, and Saving Behaviour among Saudi Youth and the Moderating Effect of Self-Control. Sustainability, 14(14), 8780. https://doi.org/10.3390/su14148780
10. Bajtalsmit, T. (2024). How Much Does It Cost to Develop a Fintech App + Factors and Estimates In 2024. Intelivita. https://www.intelivita.com/blog/cost-to-develop-fintech-app/?fbclid=IwZXh0bgNhZW0CMTAAAR2LitTB8Zhjq38ro687Y-yJA2OQWhRe6wDSeOn6mn0tGzBrPYvpTxLitmA_aem_YX_Xn8SWLTvqb_q65WiUdQ
11. Barth, S., & Ljubimov, D. (2019). A Modern Finance Dilemma: Balancing Security and Convenience . Wallester. https://wallester.com/blog/business-insights/finance-dilemma-balancing-security-and-convenience?fbclid=IwZXh0bgNhZW0CMTAAAR1kmfkUr7mdyV023rdriyvfeHgIYhTDeADQZlX65Eh6B1IrJGrAT3ywzX0_aem_RzpzSNaz3aI_zyaIbpP-7g
12. Baruah, R. (2023, August 21). 6 apps that can boost your financial literacy. The Economic Times.https://economictimes.com/wealth/invest/6-apps-that-can-boost-your-financial-literacy/articleshow/102851709.cms
13. Bay, C., Catasús, B., & Johed, G. (2014, February 1). Situating financial literacy. Critical Perspectives on Accounting. https://doi.org/10.1016/j.cpa.2012.11.011
14. Bertrand, M., & Morse, A. (2011). Information Disclosure, Cognitive Biases, and Payday Borrowing. The Journal of Finance, 66(6), 1865–1893. https://doi.org/10.1111/j.1540-6261.2011.01698.x
15. Bhandari, P. (2023). How to find the mean | Definition, Examples & calculator. Scribbr.https://www.scribbr.com/statistics/mean/
16. Bhattacharya, U., Hackethal, A., Kaesler, S., Loos, B., & Meyer, S. (2011). Is Unbiased Financial Advice To Retail Investors Sufficient? Answers from a Large Field Study. SSRN Electronic Journal. https://doi.org/10.2139/ssrn.1669015

17. Bhushan, Puneet & Medury, Yajulu. (2020). Financial literacy and its determinants. International Journal of Engineering, Business and Enterprise Applications. 4. 155-160.
18. Bisht, R. (2024, April 27). What is Purposive Sampling? Methods, Techniques, and Examples | Researcher.Life. https://researcher.life/blog/article/what-is-purposive-sampling-methods-techniques-and-examples/?fbclid=IwZXh0bgNhZW0CMTAAAR31GFPBI-d8iVnD8stYj07R1SJf93D_qF_e6rHmgAkJ2EOhGP9ygvxoePg_aem_EVZBT-L2Cg0jXsK-MFHIxw
19. Bisht, R. (2024b, April 27). What is Purposive Sampling? Methods, Techniques, and Examples | Researcher.Life. https://researcher.life/blog/article/what-is-purposive-sampling-methods-techniques-and-examples/
20. Brown, J., & Lee, S. (2021). The effectiveness of smartphone apps in improving inancialcapability. https://www.researchgate.net/publication/334461081_The_effectiveness_of_smartphone_apps_in_improving_financial_capability
21. Cao, L. (2021). AI in Finance: Challenges, Techniques and Opportunities. University of Technology Sydney. 1(1), 1-40.
22. Chen, H. (1998). An analysis of personal financial literacy among college students. Financial Services Review, 7(2), 107–128. https://doi.org/10.1016/s1057-0810(99)80006-7
23. Chen, H., & Volpe, R. P. (1998). An Analysis of Personal Financial Literacy Among College Students. Financial Services Review, 7(2), 107-128. https://www.researchgate.net/publication/4951417_An_Analysis_of_Personal_Financial_Literacy_Among_College_Students
24. Chien, Y., & Devaney, S. A. (2001). The Effects of Credit Attitude and Socioeconomic Factors on Credit Card and Installment Debt. Journal of Consumer Affairs, 35(1), 162–179. https://doi.org/10.1111/j.1745-6606.2001.tb00107.x
25. Chock, W.Q., & Chin, Y.W. (2024). Financial Literacy among Undergraduates: A Study in Universiti Sains Malaysia. Journal of Social Sciences & Humanities. 21 (1), 147-158. https://doi.org/10.17576/ebangi.2024.2101.13
26. Choong, K., Cuello-Garcia, C. A., Mai, S. H. C., Simpson, R., & Al-Harbi, S., (2023). Early Mobilization in Critically Ill Children: A Systematic Review. the Journal of Pediatrics/the Journal of Pediatrics, 203, 25-33.e6. https://doi.org/10.1016/j.jpeds.2018.07.037
27. Chuah, S.C., Kamaruddin, J.N., Singh, J.K., (2020). Factors Affecting Financial Management Behaviour among University Students. Malaysian Journal of Consumer and Family Economics.https://www.researchgate.net/publication/369369081_FACTORS_INFLUENCING_FINANCIAL_MANAGEMENT_BEHAVIOUR_AMONG_UNIVERSITY_STUDENTS
28. Cochran: Sampling Techniques Introduction. (n.d.). Maths History. https://mathshistory.st-andrews.ac.uk/Extras/Cochran_sampling_intro/?t&utm_source=perplexity&fbclid=IwZXh0bgNhZW0CMTAAAR0u9tZs2PBpmG4XNN-90YXAa12_rf8XzYrE4h0wEeE1Lu22cumzvV8Qmww_aem_38EgZEu9a3UsHlVENq8PGA
29. Collins, J. M., & Urban, C. (2020). Measuring financial well-being over the lifecourse. Journal of Consumer Affairs, 54(3), 857-884. https://doi.org/10.1111/joca.12311
30. Daly, L. (2021, March 25). Best Financial Literacy Apps. The Motley Fool. https://www.fool.com/the-ascent/personal-finance/financial-literacy-apps
31. Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319-340. https://www.jstor.org/stable/249008
32. de Bassa Scheresberg, C. (2013). Financial literacy and financial behavior among young adults: Evidence and implications. Numeracy, 6(2), Article 5. https://doi.org/10.5038/1936-4660.6.2.5
33. Descriptive analysis in education: A guide for researchers. (n.d.). https://ies.ed.gov/ncee/pubs/20174023/
34. DeVellis, R.F. (2016) Scale Development Theory and Applications. Vol. 26, Sage, Thousand Oaks. - References - Scientific Research Publishing. (n.d.). https://www.scirp.org/reference/referencespapers?referenceid=2447439

35. DeVoe, S. E., & Pfeffer, J. (2016). The stingy hour: How accounting for time affects consumer spending. Journal of Consumer Research, 42(2), 240-254. https://doi.org/10.1093/jcr/ucv046

36. Dewi, V.I., Febrian, E., Effendi, N., & Anwar,M. (2020). Financial Literacy among the Millennial Generation: Relationships between Knowledge, Skills, Attitude, and Behavior. Australasian Accounting, Business and Finance Journal. 14(4), 24-37. https://ro.uow.edu.au/cgi/viewcontent.cgi?article=2115&context=aabfj

37. Durband, D. B., & Britt, S. L. (2012). Student Financial Literacy. Springer Science & Business Media. http://books.google.ie/books?id=Lz04Vv5iBrEC&printsec=frontcover&dq=The+Relationship+between+Financial+Literacy,+Financial+Status,+and+.+https://digitalcommons.wku.edu/cgi/v iewcontent.cgi%3Farticle%3D1139%26context%3Ddiss&hl=&cd=1&source=gbs_api

38. eBusiness@Newcastle. (n.d.). Technology Acceptance Model - TheoryHub – Academic theories reviews for research and T&L. https://open.ncl.ac.uk/theories/1/technology-acceptance-model/

39. Evaluating Structural Equation Models with Unobservable Variables and Measurement Error on JSTOR. (n.d.). www.jstor.org. https://www.jstor.org/stable/3151312

40. Falahati, L., Sabri, M. F., & Paim, L. H. J. (2011). Assessment of university students' financial management skills and educational needs. *African Journal of Business Management*, 5(15), 6085-6091. https://academicjournals.org/article/article1380366553_Falahati%20et%20al.pdf

41. Farida, M.N., Soesatyo, Y., & Aji, T S. (2021). Influence of Financial Literacy and Use of Financial Technology on Financial Satisfaction through Financial Behavior. International Journal of Education & Literacy Studies. 1(9), 86-95. http://dx.doi.org/10.7575/aiac.ijels.v.9n.1p.86

42. Fedorowicz, J., Gogan, J. L., & Culnan, M. J. (2020). Financial technology use and financial literacy: A study of consumers' behavior. Information Systems Management, 37(1), 44-58. https://doi.org/10.1080/10580530.2019.1696541

43. Fernandes, D., Lynch, J. G., & Netemeyer, R. G. (2014). Financial Literacy, Financial Education, and Downstream Financial Behaviors. Management Science, 60(8), 1861–1883. https://doi.org/10.1287/mnsc.2013.1849

44. Fernando, J. (2024, April 12). Financial literacy. Investopedia. https://www.cambridge.org/core/journals/journal-of-financial-literacy-and-wellbeing/article/financial-literacy-and-financial-wellbeing-evidence-from-the-us/318307008828D2D7932C13E04B90DD88

45. Fernando, J. (2024, June 29). Financial Literacy: What It Is, and Why It Is So Important To Teach Teens. Investopedia. https://www.investopedia.com/terms/f/financial-literacy.asp

46. Financial Literacy and Financial Inclusion Increasing Financial Resilience: A Literature Review. (2023, August). Research Journal of Finance and Accounting. https://doi.org/10.7176/rjfa/14-16-07

47. Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. Journal of Marketing Research, 18(1), 39-50. https://www.jstor.org/stable/3151312

48. French, D., McKillop, D., & Stewart, E. (2019). The effectiveness of smartphone apps in improving financial capability. European Journal of Finance, 26(4–5), 302–318. https://doi.org/10.1080/1351847x.2019.1639526

49. French, D., McKillop, D., & Stewart, E. (2020). The effectiveness of smart phone apps in improving financial capability. European Journal of Finance, 26(4-5), 302-318. https://doi.org/10.1080/1351847X.2019.1639526

50. Friedline, T., West, S., & Serido, J. (2015). Leveraging youth's financial capability: The role of financial inclusion and consumer socialization. Journal of Consumer Affairs, 49(1), 35-68. https://doi.org/10.1111/joca.12057

51. García-Santillán, Ramos-Hernández, J. J., A., & Molchanova, V. (n.d.). Financial Literacy Level on College Students: A Comparative Descriptive Analysis between Mexico and Colombia. https://eric.ed.gov/?id=EJ1249379&t&utm_source=perplexity

52. Gong, X., Razzaq, A., & Wang, W. (2021). More Haste, Less Speed:NHow Update Frequency of Mobile Apps Influences Consumer Interest. Journal of Theoretical and Applied Electronic Commerce Research. 16, 2922–2942. https://doi.org/10.3390/jtaer16070160

53. Gorbachev, O., & Doepke, M. (2020). Budgeting practices and financial well-being: A structural equation modeling approach. Journal of Consumer Research, 46(3), 483-503. https://www.researchgate.net/publication/287279059_Financial_behavior_and_financial_position_A_structural_equation_modelling_approach
54. Goyal, K., Kumar, S., & Xiao, J.J.(2021). Antecedents and consequences of Personal Financial Management Behavior: a systematic literature review and future research agenda. International Journal of Bank Marketing. https://doi.org/10.1108/IJBM-12-2020-061
55. Goyal, K., Kumar, S., & Xiao, J.J.(2021). Antecedents and consequences of Personal Financial Management Behavior: a systematic literature review and future research agenda. International Journal of Bank Marketing. https://doi.org/10.1108/IJBM-12-2020-0612
56. Grabkowski, L. R. (2019, March 2). How to calculate the grand mean. Sciencing.https://sciencing.com/calculate-grand-mean-5948678.html
57. Grable, J. E. (2016). Financial Risk Tolerance. In Springer eBooks (pp. 19–31). https://doi.org/10.1007/978-3-319-28887-1_2
58. Gray Group International. (2024). Personal Finance Apps: Harnessing Technology for Financial Management. https://www.graygroupintl.com/blog/personal-finance-apps
59. Grohmann, A., Kouwenberg, R., & Menkhoff, L. (2015). Financial literacy and retirement planning: New evidence from the Netherlands. *Economic Letters*, 132, 95-98. https://gflec.org/wp-content/uploads/2015/09/FLAT-World_Netherlands.pdf
60. Gutter, M. S., Hayhoe, C. R., DeVaney, S. A., Kim, J., & Bowen, C. F. (2010). Do financial education programs work? College student's preliminary answer. Financial Counseling and Planning, 21(1), 23-34. https://psycnet.apa.org/record/2009-19876-002
61. Hair, J. F., Black, W. C., Babin, B. J., Anderson, R. E., & Tatham, R. L. (2010). Multivariate data analysis (7th ed.). Pearson Education. https://www.scirp.org/reference/ReferencesPapers?ReferenceID=1841396
62. Hair, J. F., Hult, G. T. M., Ringle, C., & Sarstedt, M. (2017). A primer on partial least squares structural equation modeling (PLS-SEM). Sage Publications.
63. Hair, J.F., Black, W.C., Babin, B.J. and Anderson, R.E. (2010) Multivariate Data Analysis. 7th Edition, Pearson, New York. - References - Scientific Research Publishing. (n.d.). https://www.scirp.org/reference/ReferencesPapers?ReferenceID=1841396
64. Hanna, S. D., & Lindamood, S. (2010). An Analysis of Personal Financial Literacy Among College Students. Financial Services Review, 19(2), 127-144.https://doi.org/10.1007/978-981-19-0108-9_26https://doi.org/10.1145/nnnnnnn.nnnnnnn
65. Hannagan, A., & Morduch, J. (2015). Income gains and month-to-month income volatility: Household evidence from the US Financial Diaries. Journal of Economic Behavior & Organization, 119, 1-21. https://doi.org/10.1016/j.jebo.2015.07.028
66. Happy Money: The Science of Happier Spending - Book - Faculty & Research – Harvard Business School. (n.d.). https://www.hbs.edu/faculty/Pages/item.aspx?num=43404
67. Henseler, J., Ringle, C. M., & Sarstedt, M. (2014). A new criterion for assessing discriminant validity in variance-based structural equation modeling. Journal of the Academy of Marketing Science, 43(1), 115–135. https://doi.org/10.1007/s11747-014-0403-8
68. Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. Journal of the Academy of Marketing Science, 43(1), 115-135. https://doi.org/10.1007/s11747-014-0403-8
69. Hilgert, Marianne & Hogarth, Jeanne & Beverly, Sondra. (2003). Household Financial Management: The Connection Between Knowledge and Behavior. Federal Reserve Bulletin. 89. 309-322. https://www.researchgate.net/publication/5039164_Household_Financial_Management_The_Connection_Between_Knowledge_and_Behavior

70. Huang, R., Hsu, S., & Yu, P. (2020). A study of influencing factors on the acceptance of fintech by college students. *Journal of Business Research*, 120, 1-6. https://www.researchgate.net/publication/366790398_Factors_Affecting_Fintech_Acceptance_in_Sri_Lanka
71. Huston, S. J. (2010). Measuring Financial Literacy. Journal of Consumer Affairs, 44(2), 296–316. https://doi.org/10.1111/j.1745-6606.2010.01170.x
72. Huston, S. J. (2010b). Measuring Financial Literacy. Journal of Consumer Affairs, 44(2), 296–316. https://doi.org/10.1111/j.1745-6606.2010.01170.x
73. Huston, S. J. (2020). Measuring financial literacy. Journal of Consumer Affairs, 44(2), 296-316.https://www.nber.org/system/files/working_papers/w31145/w31145.pdf
74. International Journal of Advanced Multidisciplinary Research. (2016). International Journal of Advanced Multidisciplinary Research. https://doi.org/10.22192/ijamr
75. Irawan, N., & Matoati, R. (2021, September 1). Pengaruh Literasi Keuangan dan Perilaku Penggunaan Fintech Payment terhadap Pengelolaan Keuangan Mahasiswa Jabodetabek. http://repository.uin-malang.ac.id/11744/7/11744.pdf?t&utm_source=perplexity
76. Jain, V. (2023, November 15). Top 8 Fantastic Financial Literacy Apps for Teens in 2023. https://www.linkedin.com/pulse/top-8-fantastic-financial-literacy-apps-teens-2023-varidhi-jain
77. Jansen, D. (2024, May 21). Research Design 101: Qualitative & Quantitative. Grad Coach. https://gradcoach.com/research-design/
78. Johan, I., Rowlingson, K., & Appleyard, L. (2020). The Effect of Personal Finance Education on The Financial Knowledge, Attitudes and Behaviour of University Students in Indonesia. Journal of Family and Economic Issues, 42(2), 351–367. https://doi.org/10.1007/s10834-020-09721-9
79. Johnson, C. (2023, July 1). A Comprehensive Guide to Top Financial Literacy Apps of 2023 - Finaciti. Finaciti. https://finaciti.com/best-financial-literacy-apps-2023-improve-money-management/
80. Joo, S.-H., & Grable, J. E. (2004). An exploratory framework of the determinants of financial satisfaction. Journal of Family and Economic Issues, 25(1), 25-50. https://doi.org/10.1023/B:JEEI.0000016722.37994.9f
81. Juliyanti, W. et al. (2023). Financial Technology, Financial Literacy and Financial Management Behaviour in Colleges. Advances in Social Science, Education and Humanities Research. http://creativecommons.org/licenses/by-nc/4.0/
82. Karjaluoto, H., Shaikh, A. A., Saarijärvi, H., & Saraniemi, S. (2019). How perceived value drives the use of mobile financial services apps. International Journal of Information Management, 47, 252–261. https://doi.org/10.1016/j.ijinfomgt.2018.08.014
83. Khoirunnisaa, J., & Johan, I.R. (2020). The Effects of Financial Literacy and Self-Control towards Financial Behavior among High School Students in Bogor. Journal of Consumer Sciences. 5(2), 73-86. https://doi.org/10.29244/jcs.5.2.73-86
84. Klapper, L., Lusardi, A., & Panos, G. A. (2016). Financial literacy and its consequences: Evidence from Russia during the financial crisis. Journal of Banking & Finance, 46, 22-43. https://doi.org/10.1016/j.jbankfin.2014.03.016
85. Kline, R. B. (2015). Principles and practice of structural equation modeling (4th ed.). Guilford Press. https://psycnet.apa.org/record/2015-56948-000
86. Kline, R. B. (2015). Principles and Practice of Structural Equation Modeling (4th ed.). The Guilford Press. APA PsycNet. (n.d.). https://psycnet.apa.org/record/2015-56948-000
87. Koenig-Lewis, N., Marquet, M., Palmer, A., & Zhao, A. L. (2015). Technology readiness and adoption of mobile payments. *Journal of Retailing and Consumer Services*, 25, 63-73. https://www.researchgate.net/publication/280919129_Enjoyment_and_social_influence_predicting_
88. Königstorfer, J., & Gröppel-Klein, A. (2022). Financial decision-making with digital apps: Effects on spending and saving behavior. Journal of Consumer Marketing, 39(3), 321-332. https://doi.org/10.1108/JCM-08-2020-3990

89. Konoth, R. K., Fischer, B., Fokkink, W., Athanasopoulos, E., Razavi, K., & Bos, H. (2020). SecurePay: Strengthening two-factor authentication for arbitrary transactions. IEEE European Symposium on Security and Privacy. https://doi.org/10.1109/EuroSP48549.2020.00043

90. Koskelainen, T., Kalmi, P., Scornavacca, E., & Vartiainen, T. (2023). Financial literacy in the digital age—A research agenda. the Journal of Consumer Affairs/the Journal of Consumer Affairs, 57(1), 507–528. https://doi.org/10.1111/joca.12510

91. Kumari, D.A.T. (2020). The impact of financial literacy on investment decisions: with special reference to undergraduates in western province, Sri lanka. Asian Journal of Contemporary Education. 4(2), 110-126. https://doi.org/10.18488/journal.137.2020.42.110.126

92. Laursen, B., & Veenstra, R. (2021). Toward understanding the functions of peer influence: A summary and synthesis of recent empirical research. Journal of Research on Adolescence. 31(4), 889–907. https://doi.org/10.1111/jora.12606

93. Laursen, B., & Veenstra, R. (2021). Toward understanding the functions of peer influence: A summary and synthesis of recent empirical research. Journal of Research on Adolescence. 31(4), 889–907. https://doi.org/10.1111/jora.12606

94. Lone, U. M., & Bhat, S. A. (2022). Impact of financial literacy on financial well-being: a mediational role of financial self-efficacy. Journal of Financial Services Marketing, 29(1), 122–137. https://doi.org/10.1057/s41264-022-00183-8

95. Lu, Y., Papagiannidis, S., & Alamanos, E. (2019). Exploring the adoption of wearables in the workplace: A mixed-methods study. *Journal of Business Research*, 101, 526-536. https://www.researchgate.net/publication/348397960_Issues_and_Adoption_Barriers_in_Wearable_

96. Lusardi, A., & Messy, F. (2023). The importance of financial literacy and its impact on financial wellbeing. Journal of Financial Literacy and Wellbeing, 1(1), 1–11. https://doi.org/10.1017/flw.2023.8

97. Lusardi, A., & Messy, F.A. (2023). The importance of financial literacy and its impact on financial wellbeing. Journal of Financial Literacy and Wellbeing. https://doi.org/10.1017/flw.2023.8

98. Lusardi, A., & Mitchell, O. S. (2011). Financial literacy and retirement planning in the United States. *Journal of Pension Economics and Finance*, 10(4), 509-525. https://www.bakertilly.com/specialties/wealth-management?utm_source=googleads&utm_medium=paidsearch&utm_campaign=Team+4%3A+Private+Wealth&utm_id=15356722006&utm_content=alwayson&gclid=CjwKCAjw1NK4BhAwEiwAVUHPULEPjANMGxxzTpcYP5dHg9femkLoE1YMJNEcNCggizjCDIIN05NF3hoCGSMQAvD_BwE

99. Lusardi, A., & Mitchell, O. S. (2014). The economic importance of financial literacy: Theory and evidence. Journal of Economic Literature, 52(1), 5-44. https://doi.org/10.1257/jel.52.1.5

100. Lusardi, A., & Mitchell, O. S. (2014). The economic importance of financial literacy: Theory and evidence. American Economic Journal: Journal of Economic Literature, 52(1), 5-44. https://www.aeaweb.org/articles?id=10.1257/jel.52.1.5

101. Lusardi, A., & Mitchell, O. S. (2014). The Economic Importance of Financial Literacy: Theory and evidence. Journal of Economic Literature, 52(1), 5–44. https://doi.org/10.1257/jel.52.1.5

102. Lusardi, A., & Mitchell, O. S. (2021). The economic importance of financial literacy: Theory and evidence. Journal of Economic Literature, 52(1), 5-44. https://www.researchgate.net/publication/317386658_The_Economic_Importance_of_Financial_Literacy_Theory_and_Evidence

103. Lusardi, A., & Streeter, J. L. (2023). Financial literacy and financial well-being: Evidence from the US. Journal of Financial Literacy and Wellbeing, 1(2), 169–198. https://doi.org/10.1017/flw.2023.13

104. Lusardi, A., & Tufano, P. (2015). Debt literacy, financial experiences, and overindebtedness. Journal of Pension Economics & Finance, 14(4), 332-368. https://doi.org/10.1017/S1474747215000232

105. Lyons, A. C., & Neelakantan, U. (2008). Potential and Pitfalls of Applying Theory to the Practice of Financial Education. Journal of Consumer Affairs, 42(1), 106–112. https://doi.org/10.1111/j.1745-6606.2007.00097.x

106. M. (2023, January 1). The Impact of Digital Literacyon Fintech Service Usage Through Financial Literacy. E3S Web of Conferences. https://doi.org/10.1051/e3sconf/202342602003

107. Majid, M. A., Othman, M., Mohamad, S. F., Lim, S. A., & Yusof, A. (2017). Piloting for interviews in qualitative research: Operationalization and lessons learnt. International Journal of Academic Research in Business and Social Sciences, 7(4), 1073-1080. [2]

108. Majida, M., Qureshia, H., & Aftaba, Hira. Predictors of Money Management Behaviour Among University Students https://www.ijicc.net/images/Vol_15/Iss_4/15432_Majid_2021_E1_R.pdf

109. Malhotra, N. K., & Dash, S. (2011). Marketing research: An applied orientation (6th ed.). Pearson Publishing. https://www.scirp.org/reference/referencespapers?referenceid=2234037

110. Mandell, L., & Klein, L. S. (2009). The Impact of Financial Literacy Education on Subsequent Financial Behavior. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2224231

111. Mändmaa, S. (2020). Empirical Study on Personal Financial Literacy of University Students for Develop the Financial Education. International Journal of Business and Applied Social Science, 8–25. https://doi.org/10.33642/ijbass.v6n6p2

112. Mashud, M., Mediaty, M., & Pontoh, Grace T. (2021). The Effect of Financial Literature, Lifestyle and Income of Parents on Student Financial Management Behavior. Journal of International Conference Proceedings, 4(3), 256-264. https://doi.org/10.32535/jicp.v4i3.1315

113. Mashud, M., Mediaty, M., & Pontoh, Grace T. (2021). The Effect of Financial Literature, Lifestyle and Income of Parents on Student Financial Management Behavior. Journal of International Conference Proceedings, 4(3), 256-264. https://doi.org/10.32535/jicp.v4i3.1315

114. Medalla, F.M. (2022).Strengthening Financial Health through Financial Literacy. Bangko Sentral ng Pilipinas. https://www.bsp.gov.ph/Lists/Speeches%20list/Attachments/993/BSP-Speeches-993.pdf

115. Miraj, J., Haider, S., Abbas, Q., Naeem, I., Nishtar, Z., & Lodhi, K. (2023, October 20). Assessing the Effectiveness of Financial Literacy Programs for Business Student. Bulletin of Business and Economics. https://doi.org/10.61506/01.00073

116. mobile_payment_adoption

117. Morris, J. (2022) An Exploratory Examination of the Financial Knowledge, Attitudes ... https://digitalcommons.fiu.edu/cgi/viewcontent.cgi?article=6519&context=etd

118. Morse, A., Lusardi, A., Schneider, D., Pence, K. M., & Tufano, P. (2011, March 1). Financially Fragile Households: Evidence and Implications. Brookings. https://www.brookings.edu/articles/financially-fragile-households-evidence-and-implications/

119. Mugenda, O., & Hira, T. K. (1999). The Relationships between Self-Worth and Financial Beliefs, Behavior, and Satisfaction. https://www.semanticscholar.org/paper/The-Relationships-between-Self-Worth-and-Financial-Mugenda-Hira/bcb3c33edeb61c7c491221b121c84340c4abd6d9

120. Mulyono, N. (2023). The Impact of Digital Literacyon Fintech Service Usage Through Financial Literacy. E3S Web of Conferences, 426, 02003. https://doi.org/10.1051/e3sconf/202342602003

121. Noor, S., Tajik, O., & Golzar, J. (2022). Simple random sampling. www.ijels.net. https://doi.org/10.22034/ijels.2022.162982

122. Nugraha, M.A.P., Violin, V., Anantadjaya, S.P., Nurlia, & Lahiya, A. (2023). Improving Financial Literacy Through Teaching Materials on Managing Finance for Millenials. Jurnal Ekonomi. 2(1), 1028-1032.

123. O'Neill, B., Xiao, J. J., Bristow, B., Brennan, P., & Kerbel, C. M. (2020). Effects of a financial planning course on financial knowledge, attitudes, and behaviors. *Journal of Financial Counseling and Planning*, 31(2), 223-236. https://www.researchgate.net/publication/301487659_Consumer_Financial_Education_and_Financial_Capability

124. OECD Recommendation on Financial Literacy https://www.oecd.org/finance/OECD-Recommendation-on-Financial-Literacy.htm

125. Peng, T. M., Bartholomae, S., Fox, J. J., & Cravener, G. (2007). The Impact of Personal Finance Education Delivered in High School and College Courses. Journal of Family and Economic Issues, 28(2), 265–284. https://doi.org/10.1007/s10834-007-9058-7

126. Personal Finance Apps: Harnessing Technology for Financial Management. (2024, March 26). https://www.graygroupintl.com/blog/personal-finance-apps

127. Personal finance management among college students likert scale used - Survey Ideas | Startquestion. (2023, February 17). Startquestion - create online surveys and forms. https://www.startquestion.com/survey-ideas/personal-finance-management-among-college-students-likert-scale-used/

128. Philippas, D., & Avdoulas, C. (2019). Financial literacy and financial well-being among generation-Z university students: Evidence from Greece. *European Journal of Finance*, 25(11), 963-978. https://www.researchgate.net/publication/342391953_Financial_literacy_and_financial_well-being_among_generation-Z_university_students_Evidence_from_Greece

129. Philippas, N.D., & Avdoulas, A. (2019). Financial literacy and financial well-being among generation-Z university students: Evidence from Greece. The European Journal of Finance. https://doi.org/10.1080/1351847X.2019.1701512

130. Potrich, A. C. G., Vieira, K. M., & Kirch, G. (2016). Development of a financial literacy model for university students. Management Research Review, 39(3), 356-376. https://doi.org/10.1108/MRR-06-2014-0143

131. Rahi, S., & Ghani, M. A. (2019). Investigating the role of user satisfaction in technology acceptance among online banking users. *Journal of Internet Banking and Commerce*, 24(2), 1-17. https://www.researchgate.net/publication/332873356_Investigating_the_role_of_UTAUT_and_e-service_quality_in_internet_banking_adoption_setting

132. Rahman, M., Isa, C. R., Masud, M. M., Sarker, M., & Chowdhury, N. T. (2021). The role of financial behaviour, financial literacy, and financial stress in explaining the financial well-being of B40 group in Malaysia. Future Business Journal, 7(1). https://doi.org/10.1186/s43093-021-00099-0

133. Rahman, M., Isa, C.R., Masud, M.M., Sarker, M., & Chowdhury, N.T. (2021). The role of fnancial behaviour, financial literacy, and fnancial stress in explaining the financial well-being of B40 group in Malaysia. Future Business Journal. 7(1), 1-18. https://doi.org/10.1186/s43093-021-00099-0

134. Razia, B., & Awwad, B. (2023). Review of financial technology applications and their related aspects. In Studies in systems, decision and control (pp. 893–902). https://doi.org/10.1007/978-3-031-28314-7_76

135. Robb, C. A. (2011). Financial knowledge and credit card behavior of college students. *Journal of Family and Economic Issues*, 32(4), 690-698. https://www.researchgate.net/publication/226824169_Financial_Knowledge_and_Credit_Card_Behavior_of_College_Students

136. Robb, C. A., & Sharpe, D. L. (2009). Effect of Personal Financial Knowledge on College Students' Credit Card Behavior. Journal of Financial Counseling and Planning, 20(1), 25-34. https://psycnet.apa.org/record/2009-19876-002

137. Roemer, E., Schuberth, F., & Henseler, J. (2021). HTMT: A criterion for assessing discriminant validity in structural equation modeling. Industrial Management & Data Systems, 121(12), 2637-2650. https://doi.org/10.1108/IMDS-02-2021-0082

138. Shaikh, A. A., Alharthi, M. D., & Alamoudi, H. O. (2020). Examining key drivers of consumer experience with (non-financial) digital services: An exploratory study. Journal of Retailing and Consumer Services, 55, Article 102073. https://doi.org/10.1016/j.jretconser.2020.102073

139. Shashank, S. & Naidu, B. (2020). Google Play Store Apps- Data Analysis and Ratings Prediction. International Research Journal of Engineering and Technology (IRJET). 7(12), 265-274. https://www.academia.edu/download/65354013/IRJET_V7I1248.pdf

140. Silva, P. (2015). Davis' technology acceptance model (TAM)(1989). Information seeking behavior and technology adoption: Theories and trends, 205-219.https://ejournal.seaninstitute.or.id/index.php/Ekonomi/article/download/1295/1061/3803

141. Smith, C., & Barboza, G. A. (2023). The Role of Trans-Generational Financial Knowledge and Self-Reported Financial Literacy on Borrowing Practices and Debt Accumulation of College Students. Social Science Research Network. https://doi.org/10.2139/ssrn.2342168

142. Stango, V., & Zinman, J. (2009). Exponential Growth Bias and Household Finance. The Journal of Finance, 64(6), 2807–2849. https://doi.org/10.1111/j.1540-6261.2009.01518.x

143. Statistical Analysis Handbook 2018 edition - Dr M J de Smith. (n.d.-b). https://www.statsref.com/HTML/index.html?sampling.html#:~:text=sampling%20in%20SPSS-,Simple%20Random%20Sampling,specified)

144. Tang, N., & Baker, A. (2016). Self-esteem, financial knowledge, and financial behavior. *Journal of Economic Psychology*, 54, 164-176. https://www.sciencedirect.com/science/article/abs/pii/S0167487016301817

145. Tavakol, M., & Dennick, R. (2011). Making sense of Cronbach's alpha. International Journal of Medical Education, 2, 53–55. https://doi.org/10.5116/ijme.4dfb.8dfd

146. Taylor, S.G., Britt, S.L. & Canale, A. (2021). Exploring the Usability and Effectiveness of Personal Financial Management Mobile Applications for College Students. Journal of Financial Counseling and Planning, 32(1), 63-77.

147. Technologies

148. Thakur, M. (2023). Weighted Mean Formula. EDUCBA. https://www.educba.com/weighted-mean-formula/

149. The Manpower Research and Statistics Department (MRSD) provides statistics and analysis on Singapore's workforce and the state of the labour market. (2024, March 14). Ministry of Manpower Singapore. https://www.mom.gov.sg/about-us/divisions-and-statutory-boards/manpower-research-and-statistics-department

150. Trivedi, S., & Gupta, H. (2024). FINANCIAL LITERACY AMONG UNIVERSITY STUDENTS. ResearchGate. https://www.researchgate.net/publication/378555546_FINANCIAL_LITERACY_AMONG_UNIVERSITY_STUDENTS

151. Voorhees, C. M., Brady, M. K., Calantone, R., & Ramirez, E. (2016). Discriminant validity testing in marketing: An analysis, causes for concern, and proposed remedies. Journal of the Academy of Marketing Science, 44(1), 119-134. https://doi.org/10.1007/s11747-015-0455-4

152. Wong, Ken. (2013). Partial least square structural equation modeling (PLS-SEM) techniques using SmartPLS. Marketing Bulletin. 24. 1-32. https://www.researchgate.net/publication/268449353_Partial_least_square_structural_equation_modeling_PLS-SEM_techniques_using_SmartPLS

153. Xiao, J. J. (2016). Handbook of Consumer Finance Research. Springer. https://doi.org/10.1007/978-3-319-28887-1

154. Xiao, J. J., & O'Neill, B. (2018). Consumer financial capability and well-being. International Journal of Consumer Studies, 42(2), 221-230. https://doi.org/10.1111/ijcs.12454

155. Xiao, J. J., & O'Neill, B. (2018). Consumer financial education and financial capability. Journal of Financial Counseling and Planning, 29(2), 202-218. https://doi.org/10.1891/1052-3073.29.2.202

156. Xiao, J. J., & Porto, N. (2017). Financial education and financial satisfaction. International Journal of Bank Marketing, 35(5), 805–817. https://doi.org/10.1108/ijbm-01-2016-0009

157. Xiao, J. J., Chen, C., & Chen, F. (2013). Consumer Financial Capability and Financial Satisfaction. Social Indicators Research, 118(1), 415–432. https://doi.org/10.1007/s11205-013-0414-8

158. Xiao, J. J., Tang, C., & Shim, S. (2008). Acting for Happiness: Financial Behavior and Life Satisfaction of College Students. Social Indicators Research, 92(1), 53–68. https://doi.org/10.1007/s11205-008-9288-6

159. Xue, R., Gepp, A., O'Neill, T. J., Stern, S., & Vanstone, B. J. (2020). Financial literacy and financial strategies: The mediating role of financial concerns. Australian Journal of Management, 031289622094076. https://doi.org/10.1177/0312896220940762

160. Yogasnumurti, R.R., Sadalia, I., & Irawati, N. (2021). The Effect of Financial, Attitude, and Financial Knowledge on the Personal Finance Management of College Students. Proceedings of the 2nd Economics and Business International Conference (EBIC 2019) - Economics and Business in Industrial Revolution 4.0. 649-657. https://doi.org/10.522/0009329206490657

161. Yuneline, M. H., & Rosanti, M. F. C. (2023). The role of digital finance, financial literacy, and lifestyle on financial behaviour. Holistica, 14(2), 97–115. https://doi.org/10.2478/hjbpa-2023-0018

162. Zhang, T., Tang, Y., & Wu, Z. (2021). The role of data security and privacy in the adoption of financial technologies. *Journal of Financial Regulation and Compliance*, 29(2), 184-198.https://www.aosphere.com/products/data-privacy/rulefinder-data-privacy/?gad_source=1&gbraid=0AAAAAD-ij80NIdvU3825vMI_8HTsf7_5h&gclid=CjwKCAjw1NK4BhAwEiwAVUHPUM6rCypaVgcpz65pjwmVPeu2R3h8On0A-NCw_HLGtQxOlwL8m4pNEhoCsmQQAvD_BwE

163. Zimmerman, J., & Wansink, B. (2022). Sustaining financial behavior change: The role of engagement and personal finance tools. Behavioral Science & Policy, 8(1), 123-134. https://fbj.springeropen.com/articles/10.1186/s43093-021-00099-0

Analyzing the Mediating Role of Job Satisfaction on the Effect of Motivating Factors to Private Senior High School Teachers' Retention in Nueva Ecija

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Abstract

Teacher retention is a critical global issue, worsened by projected teacher shortages and challenging compensation structures in private sectors. This study investigated the mediating role of job satisfaction on the influence of key motivating factors—compensation, job security, work-life balance, and organizational culture—on teacher retention in selected private senior high schools in Nueva Ecija, Philippines. A descriptive-correlational design was used as part of quantitative research. The data were collected from 135 teachers who possessed significant teaching experience. Partial Least Squares Structural Equation Modeling (PLS-SEM) was utilized to measure connections between the variables.

The findings highlighted the critical role of job satisfaction in retaining teachers. Furthermore, work-life balance was concluded to be the strongest motivating factor to job satisfaction, while job security was the strongest motivating factor on both job satisfaction and employee retention. Creating a supportive work-life balance and ensuring job security are important for retaining satisfied teachers in their roles.

Keywords: Teacher Retention, Job Satisfaction, Work-Life Balance, Job Security.

Introduction

Teacher retention is a pressing global issue, with a significant shortage projected by 2030 (UNESCO, 2024). In private senior high schools in Nueva Ecija, disparities in compensation compared to government positions make teaching less appealing (Aduna et al., 2020). High turnover disrupts student learning, increases costs, and lowers institutional efficiency (Menzies, 2023; Vikas & Sharma, 2022). Studies emphasize job satisfaction as a key determinant of teachers' decisions to stay, acting as a mediating factor between motivating elements and retention (Springer & Nguyen, 2019).

This study examines how job satisfaction mediates the effects of factors like compensation, job security, work-life balance, and organizational culture on private senior high school teacher retention in Nueva Ecija. Using a quantitative approach with survey data, statistical analysis highlights the relationships between these variables. The findings aim to help education stakeholders develop targeted strategies to enhance job satisfaction and retention, thereby improving educational outcomes and reducing turnover costs.

This study addresses gaps in existing research and provides valuable insights for educators, institutions, and policymakers to create supportive environments that foster teacher commitment and stability.

Literature Review

Job Satisfaction

Job satisfaction, a fundamental idea in workplace psychology and the management of human resources, relates to a person's evaluation of their work experience and overall happiness with various aspects of their employment. It refers to feelings of fulfillment, contentment, and enjoyment that stem from one's employment function, working environment, and workplace relationships (Shamran et al., 2020). Job satisfaction is one of the most extensively studied concerns in research on organizations, and it has an essential effect on employee attitudes, behaviors, and organizational results.

Even now, managers prioritize job satisfaction among their staff members to guarantee the success and productivity of the company. It has sparked widespread interest among educational researchers in informing lawmakers and school

administrators about its effects on teachers' motivation, commitment, stress, retention, and burnout. (Baroudi et al., 2022) A study done in Nigeria implies that job satisfaction and commitment to the organization are significant variables that influence the intentions of staff turnover, considering that employees with less satisfaction with their jobs and organizational commitment are more likely to want to leave their organization. (Babalola, 2022)

The most important benefits that Filipino millennials wants in a company are compensation, salaries, and health insurance. Good salaries and wages tend to alleviate financial stress, which leads to employees being more productive. The benefits and salaries provided to employees increase their job satisfaction. As a result, providing benefits to employees encourages them to stay with the company (Medallon, 2020).

Over the previous three decades, Romania's educational system has been highly controversial, bureaucratic, and chaotic. Despite repeated modifications, the planned aims have not been met. There has been persistent debate over poor student performance on periodic assessments and the baccalaureate. This has put a strain on teachers' skills and efficiency, while the teaching profession has struggled with little financing, a condition that has continued regardless of education ministers' political affiliations.

The study by Ortan, Simut, and Simut (2021) looked at the relationships between teacher self-efficacy, work satisfaction, and well-being of the teacher in the K-12 school system. Teacher self-efficacy is defined as a teacher's ability to motivate underperforming students, modify teaching approaches, and urge students to learn. Self-efficacy, positive student behavior, work promotion, and favorable working conditions all have a substantial impact on job satisfaction, which in turn affects teacher well-being. Self-efficacy regarding class management, topic preparation, administrative work, and emotional control are the most critical aspects in maintaining high levels of job satisfaction. The study emphasized the importance of these elements for teacher satisfaction and performance because they foster a healthy work environment and increase involvement from teachers, students, and parents.

To determine the hypotheses proposed, factorial analysis and structural equation models used to further the research which resulted in finding the most essential aspects which are the teacher's self-efficacy, advancement prospects, students' respectful attitudes, and management's support for teachers. Other relevant characteristics include access to technology and school supplies, collegial cooperation, workload, and teaching-related responsibilities. According to the study, failure to address these factors might result in decreasing job satisfaction, attrition, burnout, and instructor turnover.

Motivating Factors of Teachers

The organizational culture of academic institutions affected the employee retention of teachers. However, although much research had been done on the organizational culture of higher education, not as much had been done on the organizational culture of basic education specifically, in private senior high schools. As revealed by Mohammad and Borkoski (2024), the decision of teachers to continue their teaching profession was strongly correlated with organizational circumstances related to leadership practices. Resilience was another quality of organizational culture, as retention and resilience are interconnected. According to Mullen et al. (2021), teachers in grades K–12 who focus on resilience factors were better able to adjust and overcome hardships, and it is an important indicator of work-life quality (Espinosa, 2024). Whereas Reitman and Karge (2019) say, there was a greater tendency for beginning teachers to stay in the field when they receive support and professional development during the first five years of teaching in critical areas of need taken care of by their superiors. On the other hand, a study showed that teachers who quit school complained about poor administrative support and a lack of appreciation for their work (Kamrath & Bradford, 2020). Additionally, a longitudinal study conducted by Erichsen and Reynolds (2020) had shown informative trends, demonstrating that higher social utility values are associated with better-fit perceptions and lower intention to leave. These results highlighted how crucial it is for educational establishments to consider organizational dynamics and teachers' values as a strategy to reduce attrition and develop a more stable teaching staff.

With the increasing competition among higher education institutions, Human Resource Management (HRM) policies have an important role in increasing employee engagement and retention. Effective HRM practices are essential for academic institutions to attract, motivate, and retain talented faculty members. This review explored the impact of various HRM policies, focusing on compensation, benefits, reward systems, and training and development within the context of higher education. Employee engagement and retention are significantly influenced by comprehensive compensation packages, which include base pay, incentives, benefits, and non-financial rewards such as recognition and career development opportunities (Manaig, Perez & Casulla, 2023). The Total Compensation Theory underscored the importance of a holistic approach to compensation, which can enhance employee satisfaction and commitment.

According to Katete and Nyangarika (2020), teachers were driven to work because they were well compensated and got their benefits on time. Delays in paying teachers could lead to poor preparation, exam efficiency, practical learning and study trips, low classroom attendance, bad teaching skills, irregular ongoing evaluation, and inadequate management of the classroom. This had an adverse effect both on the teacher's and students' instructional and educational processes.

Unemployment and job security continued to be critical issues for many countries, and workers were particularly concerned about unemployment when they lose their jobs. Every employee wanted to have a stable job. Most employees saw job security as a necessity and the most effective way to motivate them. Furthermore, job stability allowed people to be confident about their future, reduces labor agitation while increasing productivity, contributes to the preservation of social balance and values, and reduces employees' intentions to quit (Domfeh & Hunsaker, 2020). This demonstrated how important job security is in the competitive industry nowadays.

The study centered on heightening work-life balance to reduce the existence of social role conflict in both professional and personal aspects. It provided the evidence reviewed, particularly on the effects of integrating work-life balance that touches on work, non-work, and stress-inducing results. In relation to the theoretical mechanisms that linked to a more ideal life, it presented the personal and organizational factors manifested from a well-balanced life.

To determine the relationship between work and non-work life, the study employed various methodological approaches including surveys, interviews, longitudinal studies, and meta-analyses. Casper et al. (2018) configured research methods in Industrial-Organizational Psychology and the Organizational Behavior work-family research.

Adams et al. (2018) delved into the life of small-scale farmers in developing countries providing insights due to climate change. An empirical test was carried out on both the literature on conflict between work and family, and social support which was then used as a model of the work-family connection. The model identified that the work-family impact, family technical and emotional assistance, as well as job and family all contribute to the satisfaction of job and life, which was evident in the findings collected from 163 workers who had lived with one family member. Additionally, the data suggested that the work-family relationship could have both conflict and support. Lower emotional and practical assistance from family could cause increased levels of conflict.

Teacher Retention of Private Senior High School

Every organization or government authority required a leader capable of setting the organization's direction and vision. Similarly, each school required an administrator with exceptional leadership skills to oversee the institution's duties and responsibilities. They were given important roles and responsibilities to ensure the educational institution delivers excellent educational opportunities by means of efficient guidance (Morales & Sapin, 2020).

China had established strict responsibilities and standards for school administrators' personal and professional conduct. School principals were regarded as the most influential individuals capable of enhancing and improving teacher performance. The principal should maintain contact with the teachers by conducting regular class observations. They expressed their views on the performance of teachers, fostering teamwork for development (Villanueva et al., 2021).

In addition to the study of Villanueva et al. (2021), a school principal's managerial skills are essential for improving teacher performance and overseeing educational practices in schools. The institution's heads' responsibilities in the day-to-day operation of the school involve motivating, directing, distributing instructions, as well as leading their subordinates. As a result, if administrators want to engage in a productive and effective day alongside their subordinates, they ought to think about encouraging them as a professional.

On the other hand, many private educational institutions in the Philippines and around the world struggle to retain qualified teachers. Maintaining educators in their current positions is critical for school success for a variety of reasons. Small schools may struggle to cover the expenses of hiring, recruiting, orienting, and even offering initial professional development due to financial constraints (Ching et al., 2023). The workplace can also influence a teacher's decision to stay or abandon their job (Grissom & Bartanen, 2019).

Furthermore, teachers depart their schools for reasons such as government employment, greater career prospects at different private companies, greener pastures, and contract or service termination (Azabo et al. 2019).

Moreover, Teachers who were employed in schools with encouraging, educational, and motivating leaders were more likely to have a positive attitude toward teaching. In these cases, assisting new teachers and assisting them in their development boosts the likelihood of retention in teaching and fosters a healthy school culture in which educators feel valued and encouraged. Another factor that inspires teachers to maintain their positions is having the chance to grow as professionals. Offering opportunities for professional development and growth had an impact on teacher retention in schools (Sumipo, 2020).

Job Satisfaction as A Mediating Factor of Teacher Retention

Many issues were currently surfacing regarding teacher performance, particularly in the Jember Regency, Indonesia. Teachers who violated the rules, taught outside of the curriculum, and had less teaching time than they report were some of these problems. With a minimum sample size of 99 respondents, the study's population comprised all elementary school teachers in the Jember Regency. The data collection methods employed were observation and questionnaires. Observations were used to assess performance allowance, work discipline, work culture, leadership, and job satisfaction variables. The majority of

respondents were between the ages of 46 and 55, female, diploma-educated (55.2%), and had worked for 10 to 20 years (41.2%). The direct and indirect effects of leadership, work culture, work discipline, and performance allowance on teacher performance and job satisfaction had been measured using path analysis.

Afshar and Doosti (2016) examined that teacher satisfaction was influenced by perceptions of the teaching profession, and Talabi (2016) found that teacher satisfaction was influenced by salary, incentives, promotion opportunities, recognition, and increased qualifications.

Amalia & Pilarta (2015) indicated that there was a significant positive correlation between student academic achievement, teacher performance, and teacher satisfaction. Fashiku (2016) discovered that school reputation is one of the main factors influencing teacher performance, and that employment status and promotion opportunities have an impact on teacher satisfaction.

Job satisfaction mediated the impact of leadership regarding teacher performance, therefore leadership had a bigger direct effect than an indirect effect. The findings showed that performance allowance, work discipline, work culture, and leadership all influenced job satisfaction and teacher performance.

Herzberg's Two-Factor Theory

Theoretical Framework The theoretical framework used was Herzberg's Two Factor Theory. To improve employee retention among private school teachers in the Philippines, school administrators could focus on both motivators and hygiene factors as these factors influence job satisfaction and dissatisfaction.

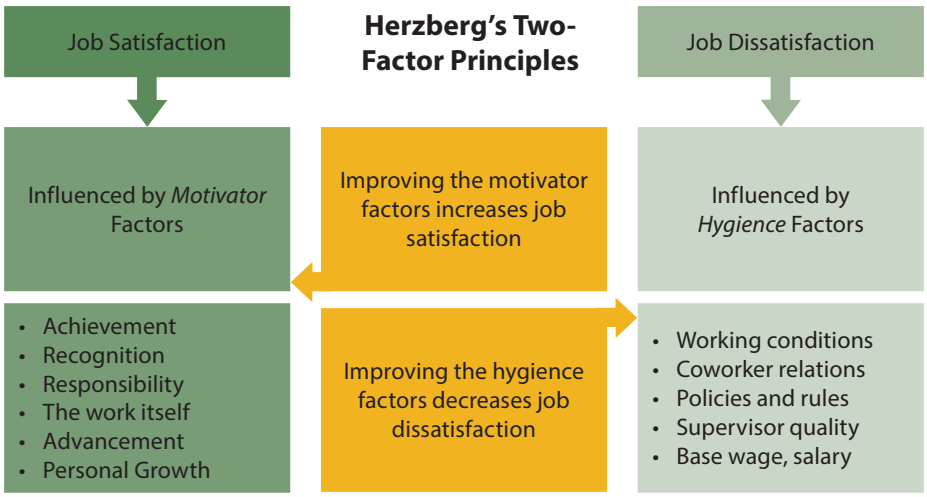


Fig.1. Hezberg's Two Factor Theory (Frederick Herzberg, 1959)

The Motivation-Hygiene: Two Factor Theory was produced by Frederick Herzberg and his two associates, Mausner and Snyderman, which introduced their book about Motivation to Work in 1959. The theory contributed to satisfaction and retention, and what might lead to dissatisfaction and turnover.

The Motivation Factor (Intrinsic Factors) to the teachers could come from classroom activities which included "achievement" (successful student outcomes and innovative teaching methods), "recognition" (hard work and achievement acknowledged by administrators, students, and parents), "responsibility" (leading special programs or mentorship roles), "the work itself" (engaging and meaningful

work), "advancement" (promotions or access to further education and professional development), and "growth" (personal and professional growth opportunities). These factors met the demands for development and self-actualization, these elements were inherent to the work and contribute to job satisfaction.

Then, the Hygiene Factors (Extrinsic Factors) had been associated with teacher satisfaction which include "working conditions" (physical surroundings of the job), "coworker relations" (collaborative and respectful working environment with colleagues, administration, students, and parents), "policies and rules" (organization and management regulations, and procedures are clear and unclear), "supervisor quality" (competent and supportive supervision from supervisor), and "salary" (competitive and fair compensation). These were factors that did not necessarily increased job satisfaction when improved, but their absence could lead to dissatisfaction.

Addressing these factors could enhance job satisfaction among teachers, reduce turnover, and improve overall retention rates of the employees in private senior high schools.

Methodology

This research sought to explore the impact of motivational determinants on job satisfaction and the retention of teachers at private senior high schools in Nueva Ecija, Philippines. A descriptive-correlational research design was used to observe and quantify variables without intervention. This study uses Structural Equation Modeling (SEM) - a statistical technique that allows for a comprehensive exploration of interconnections among variables, providing a better insight into how motivational factors influence job satisfaction and teacher retention.

The study was held in several municipalities in Nueva Ecija, which include Cabanatuan, Guimba, Licab, Quezon, Sto. Domingo, and Talavera. This is because they represented diverse educational institutions.

A stratified random sampling method was utilized to ensure the proper representation of all different subgroups of the population by proportionally ensuring adequate sample size of each stratum. For this, 125 participants with a sample size predetermined from Slovin's formula were required to maintain the confidence level at 95% and margin of error at 5%. The data collection process was carried out using a well-designed questionnaire that was based on the literature review and validated by experts. The questionnaire consisted of six sections that covered demographic information, salary satisfaction, job security, work-life balance, organizational culture, and overall job satisfaction. To measure respondents' degree of agreement, a 4-point Likert scale was used. Validity of the instrument was established through expert's evaluations, and reliability was measured with Cronbach's Alpha to indicate that most sections had acceptable internal consistency. Furthermore, a pilot study was done on 25 teachers to further improve the questionnaire.

Data was collected via Google Forms, and the final distribution of 137 surveys with 135 responses was attained. To answer questions from the data, descriptive statistics consisting of mean and standard deviation were used to draw on the central tendency and the variability of responses. Research hypotheses were tested based on relationships between variables using SEMs. Ethical consideration, too, was emphasized whereby only voluntary participation was taken without revealing the identity of anyone involved in the survey for confidentiality purposes. This study, in essence, was a well-planned and executed investigation, combining qualitative and quantitative methods to study the motivation factors and job satisfaction of private senior high school teachers toward their retention. With the SEM model, this study enabled the understanding of the job satisfaction mediating role on private senior high school teachers' retention and added meaningful input to the education management literature.

Analysis and Discussions

Discussion of Findings

Table 1 presented the key demographic characteristics of 135 respondents, revealing a predominantly female group (105 respondents, 77.78%) and most of the respondents were married (75 respondents, 55.56%), suggesting a mature demographic. The educational background was notably high, with 56 respondents (41.48%) having completed master's degree units and 44 respondents (32.59%) holding a Bachelor's Degree, indicating a well-qualified respondent pool. In terms of professional experience, 125 respondents (92.59%) were full-time employees, with 52.6% having 4-10 years of teaching experience (71 respondents: 34 respondents (25.19%) with 4-6 years and 37 respondents (27.41%) with 7-10 years). Income levels showed 44 respondents (32.59%) earning ₱30,001–₱35,000 monthly, followed by 36 respondents (26.67%) earning ₱25,001–₱30,000. Most respondents were based in rural areas (117 respondents, 86.67%) and primarily served as teachers (123 respondents, 91.11%), highlighting a concentration of educators outside urban centers.

Table 1. Demographic Characteristics of the Respondents (n=135)

Demographic	Category	Frequency	Percentage
Sex	Female	105	77.78%
	Male	29	21.48%
	Prefer not to say	1	0.74%
Age	20–30	46	34.07%
	31–40	42	31.10%
	41–50	36	26.68%
	51 and above	11	8.16%

Demographic	Category	Frequency	Percentage
Marital Status	Single	59	43.70%
	Married	75	55.56%
	Widowed	1	0.74%
	Separated	0	0%
Highest Educational Attainment	Bachelor's Degree	44	32.59%
	Master's Degree (Units)	56	41.48%
	Master's Degree (Completed)	25	18.52%
	Doctorate Degree (Units)	8	5.93%
	Doctorate Degree (Completed)	2	1.48%
Numbers of Year Teaching	Less than 1 year	4	2.96%
	1–3 Years	28	20.74%
	4–6 Years	34	25.19%
	7–10 Years	37	27.41%
	More than 10 Years	32	23.70%
Type of Employment	Full-time	125	92.59%
	Part-time	4	2.96%
	Contractual	6	4.45%
Academic Track and Strand	Accountancy, Business and Management (ABM)	42	31.11%
	Science, Technology, Engineering, and Mathematics (STEM)	39	28.89%
	Humanities and Social Science (HUMSS)	44	32.59%
	General Academic Strand (GAS)	42	31.11%
	Technical Vocational Livelihood (TVL)	18	13.33%
	Sport Track	8	5.93%
	Other	0	0%
Monthly Compensation (in PHP)	Below ₱5,000	0	0%
	₱5,001–₱10,000	5	3.70%
	₱10,001–₱15,000	10	7.41%
	₱15,001–₱20,000	14	10.37%
	₱20,001–₱25,000	23	17.04%
	₱25,001–₱30,000	36	26.67%
	₱30,001–₱35,000	44	32.59%
	Above ₱35,000	3	2.22%
Teaching Load (per week)	Less than 10 hours	37	27.41%
	10–20 hours	17	12.59%
	21–30 hours	42	31.11%
	31–40 hours	35	25.93%
	More than 40 hours	4	2.96%
Job Position	Teacher	123	91.11%
	Head Teacher	7	5.19%
	Department Chair	0	0%
	Principal	1	0.74%
	Administrator	4	2.96%

Demographic	Category	Frequency	Percentage
Workplace Location	Urban	18	13.33%
	Rural	117	86.67%

Table 2 illustrated the respondents' perspectives on the motivational factors affecting teachers, with the majority of replies reflecting agreement across several criteria. The average compensation score was 3.13 (SD = 0.75), indicating widespread contentment with remuneration. Respondents exhibited highly agree over the timeliness of payment disbursement (Mean = 3.47, SD = 0.73), the alignment of compensation with their abilities and experience (Mean = 3.04, SD = 0.73), and their satisfaction with existing remuneration and perks (Mean = 2.87, SD = 0.81).

Similarly, the average score for work security was 3.16 (SD = 0.66), reflecting a contentment with the security provided by their institutions. Teachers agreed over the sufficiency of steps to safeguard their employment (Mean = 3.16, SD = 0.67) and expressed certainty about their job's long-term security (Mean = 3.20, SD = 0.7).

Furthermore, the overall mean for work-life balance was 2.92 (SD = 0.71), with teachers expressing disagreement on the ease of separating work from personal life (Mean = 2.41, SD = 0.85), while they agreed for their role's support for a balanced lifestyle (Mean = 2.97, SD = 0.68).

Finally, regarding organizational culture, the mean score was 3.28 (SD = 0.69), suggesting that respondents highly agreed that communication between educators and administrators is effective (Mean = 3.31, SD = 0.69) and that their school encouraged teacher collaboration and teamwork (Mean = 3.36, SD = 0.66). The teachers exhibited agreement on their consistent involvement in decision-making processes (Mean = 3.17, SD = 0.76) and high agreement to the provision of regular professional development by the school (Mean = 3.28, SD = 0.65).

Thus, these results indicated that although there are certain favorable elements that positively impact teachers' motivation and job satisfaction, many teachers still experience challenges in maintaining a clear boundary between their work responsibilities and personal lives.

Table 2. Respondent's Perception of the Motivating Factors of Teachers

Motivating Factors	Mean	Std. Deviation	Verbal Interpretation
Compensation	**3.13**	**0.75**	**Agree**
I receive my compensation on time.	3.47	0.73	Highly Agree
My compensation is equal with my qualifications and experience.	3.04	0.73	Agree
I am content with my current pay and benefits.	2.87	0.81	Agree
Compensation considerations are a factor in my decision to remain.	3.15	0.74	Agree
Job Security	**3.16**	**0.66**	**Agree**
The financial security provided by the school is satisfactory.	3.08	0.67	Agree
There are adequate measures in place to protect my employment status.	3.16	0.67	Agree
I feel confident in the long-term stability of my job.	3.20	0.7	Agree
The job security provided by the school is satisfactory.	3.18	0.61	Agree
Work-life Balance	**2.92**	**0.71**	**Agree**
I struggle to separate work and non-work.	2.41	0.85	Disagree
My current role allows me to maintain a good work-life balance.	2.97	0.68	Agree
The atmosphere at work supports activities that I like in my personal life.	3.02	0.61	Agree
I receive sufficient social support from friends, family, and coworkers.	3.27	0.68	Highly Agree
Organizational Culture	**3.28**	**0.69**	**Highly Agree**
The school's culture promotes collaboration and teamwork among teachers.	3.36	0.66	Highly Agree
Administrators involve staff members in decision-making processes.	3.17	0.76	Agree
Communication between educators and the administration is effective.	3.31	0.69	Highly Agree
The school provides regular professional development opportunities.	3.28	0.65	Highly Agree

Note: 3.25-4.00—Highly Agree; 2.50-3.24—Agree; 1.75-2.49—Disagree; 1.00-1.74—Highly Disagree

Table 3 illustrated the respondents' level of job satisfaction, with an overall mean score of 3.27 (SD = 0.65), indicating a high level of agreement regarding job satisfaction. When asked about their overall satisfaction with their job, respondents agreed (Mean = 3.24, SD = 0.66), reflecting a modest level of contentment. Similarly, respondents showed highly agreement with statements about the success they have achieved in their careers (Mean = 3.27, SD = 0.67), indicating that they experience a feeling of achievement and contentment in their professional lives. Respondents also highly agreed on the opportunities they have to apply their skills on the job (Mean = 3.30, SD = 0.61) and effectiveness of the school's leadership (Mean = 3.24, SD = 0.649). Overall, the data suggests that teachers generally feel positive about their job satisfaction, driven by their sense of achievement, opportunities to utilize their skills, and the effectiveness of school leadership.

However, the "agree" also implied that there might be areas for improvement. Although they valued their career success and opportunities to use their abilities (Mean = 3.30, SD = 0.61), this had not fully raised their overall job satisfaction. This showed that while the teachers found purpose in their careers, they did not necessarily feel completely satisfied with the job as a whole.

Table 3. Respondent's Level of Job Satisfaction

Job Satisfaction	Mean	SD	Verbal Interpretation
I am satisfied with the way I feel about my job as a whole.	3.24	0.66	Agree
I am satisfied with the success I have achieved in my career.	3.27	0.67	Highly Agree
I am satisfied with the opportunities to use by abilities on the job.	3.30	0.61	Highly Agree
The school's leadership is effective.	3.24	0.649	Agree
Overall	**3.27**	**0.65**	**Highly Agree**

Table 4 presented the respondents' level of agreement with perceived employee retention factors among teachers, with an overall mean score of 3.03 (SD = 0.72), indicating a general sense of agreement regarding their commitment to staying at their current school. Respondents highly agreed that their job satisfaction significantly influenced their decision to continue working at the school (Mean = 3.27, SD = 0.626), suggesting that their satisfaction level might be a strong factor in their retention.

Similarly, respondents agreed that their compensation was sufficient to motivate them to remain at the school (Mean = 2.837, SD = 0.772), pointing to the importance of financial support in retaining teachers. They also agreed that their current workload encourages them to stay (Mean = 2.948, SD = 0.692), indicating that a manageable workload contributes positively to their job satisfaction and retention. Lastly, respondents agreed with the idea that they saw themselves working at the school for the next several years (Mean = 3.052, SD = 0.782), reflecting a strong sense of loyalty and commitment to the institution. Thus, these results highlighted the important role that job satisfaction, a good salary, and a manageable workload played in keeping teachers at their schools. While many respondents felt loyal and see themselves staying long- term, it is important to remember that these positive feelings are not enough on their own. If any of these factors, such as job satisfaction or workload, were to decline, it could significantly affect teachers' decisions to stay. Therefore, schools should focus on maintaining a supportive environment that addresses these aspects to keep educators satisfied and motivated.

Table 4. Respondents' Level of Agreement with the perceived Employee Retention of Teachers

Employee Retention	Mean	SD	Verbal Interpretation
My job satisfaction significantly influences my decision to continue working at this school.	3.27	0.626	Highly Agree
The compensation I earn is enough for me to continue working in this school.	3.837	0.772	Agree
The workload I have encourages me to remain working for this school.	2.948	0.692	Agree
I see myself working at this school for the next several years.	3.052	0.782	Agree
Overall	**3.27**	**0.65**	**Agree**

Table 5 showed the Latent Variable Coefficients, which measured the instrument's convergent validity, construct reliability, and internal consistency. The results of Cronbach's Alpha (CA), Composite Reliability (CR), Average Variance Extracted (AVE), and factor loadings confirmed that the variables exhibit strong reliability and convergent validity.

Based on the results, regarding Salaries, the CA is 0.747, the CR is 0.856, and the AVE was 0.666. The factor loadings ranged from 0.687 to 0.807, indicating that the items consistently measured this construct, and captured a substantial portion of the variance. This demonstrated that the compensation-related items are both reliable and valid in reflecting teachers' perceptions.

Similarly, the Job Security construct had a CA of 0.808, CR of 0.886, and an AVE of 0.722. Factor loadings ranged between 0.817 and 0.898, suggesting that the items were highly consistent and effectively represent perceptions of job security, ensuring strong reliability and convergent validity.

Moreover, For Work-life Balance, the CA was 0.900, the CR is 0.938, and the AVE was 0.834. The factor loadings, ranging from 0.838 to 0.914, confirmed that the construct was highly reliable and valid. This meant that the items strongly captured teachers' perspectives on balancing work and personal life.

Furthermore, the Organizational Culture construct showed a CA of 0.828, CR of 0.897, and an AVE of 0.745, with factor loadings from 0.739 to 0.851. These values indicated that the items were reliable and effectively reflected the school's organizational culture, highlighting consistent responses across the construct.

Finally, for Job Satisfaction, the CA was 0.855, the CR was 0.912, and the AVE was 0.775. The factor loadings, which were from 0.751 to 0.911, showed that the items were reliable and valid, providing strong evidence that the construct effectively measured teachers' total satisfaction with their jobs.

The results of **Cronbach's Alpha (CA)**, **Composite Reliability (CR)**, and **Average Variance Extracted (AVE)** confirmed the reliability and convergent validity of all measured constructs—**Salaries, Job Security, Work-life Balance, Organizational Culture, Job Satisfaction,** and **Employee Retention**. These findings were crucial to ensure the data accurately reflected the key factors influencing teacher satisfaction and retention. Given the high reliability and validity, the study could confidently draw conclusive insights about how these variables affect teacher motivation and retention. This provided a strong basis for actionable recommendations to improve teacher work environments.

Table 5. Latent Variable Coefficients

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite Reliability	Ave. Variances Extracted
Salaries				**0.747**	**0.856**	**0.666**
	S1	0.739	<0.001			
	S2	0.807	<0.001			
	S3	0.803	<0.001			
	S4	0.687				
Job Security				**0.808**	**0.886**	**0.722**
	JSY1	0.817	<0.001			
	JSY2	0.894	<0.001			
	JSY3	0.847	<0.001			
	JSY4	0.898	<0.001			
Work-life Balance				**0.900**	**0.938**	**0.834**
	WLB1	0.873	<0.001			
	WLB2	0.914	<0.001			
	WLB3	0.902	<0.001			
Organizational Culture				**0.828**	**0.897**	**0.745**
	OC1	0.739	<0.001			
	OC2	0.839	<0.001			
	OC3	0.851	<0.001			
	OC4	0.752	<0.001			
Job Satisfaction				**0.855**	**0.912**	**0.775**
	JS1	0.908	<0.001			
	JS2	0.886	<0.001			

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite Reliability	Ave. Variances Extracted
	JS3	0.911	<0.001			
	JS4	0.751	<0.001			
Employee Retention				**0.900**	**0.923**	**0.668**
	ER1	0.799	<0.001			
	ER2	0.874	<0.001			
	ER3	0.817	<0.001			
	ER4	0.848	<0.001			

Table 6 displayed the square roots of the Average Variance Extracted (AVE) for each latent variable, which confirmed that the constructs possess discriminant validity. The diagonal values represented the square roots of the AVE for each variable, which were higher than the off-diagonal correlations with other variables, suggesting that every construct shared more variation with its own indicators than the other constructs.

For instance, the AVE for Salaries was 0.716, which is greater than its correlations with other variables, such as Job Security (0.737) and Organizational Culture (0.644). Similarly, Job Security showed a high AVE of 0.865, further confirming discriminant validity as it exceeded the correlations with other constructs. Organizational Culture also demonstrates strong discriminant validity with an AVE of 0.882.

The results for Work-life Balance (0.726), Job Satisfaction (0.867), and Employee Retention (0.835) similarly showed that these constructs were distinct from one another. This confirmed that the measured variables were independent and captured unique aspects of teachers' experiences related to salaries, job security, organizational culture, work-life balance, job satisfaction, and retention.

Table 6. Square Roots of Average Variance Extracted (AVE) coefficients

Latent Variable	Salaries	Job Security	Organizational Culture	Work-life Balance	Job Satisfaction	Employee Retention
Salaries	**0.716**					
Job Security	0.737	**0.865**				
Organizational Culture	0.644	0.672	**0.882**			
Organizational Culture	0.599	0.691	0.671	**0.726**		
Job Satisfaction	0.619	0.671	0.667	0.649	**0.867**	
Employee Retention	0.66	0.729	0.571	0.613	0.644	**0.835**

Diagonal Value are the square roots of AVE, and off-diagonals are inter-construct correlations.

The results in Table 7 provided strong evidence of the model's statistical significance and overall quality. The Average Path Coefficient (APC) of 0.201 along with a p-value of 0.004 indicated that the relationships within the model were statistically significant, meaning the predictors effectively explain the outcome variables. Furthermore, the Average R-squared (ARS) of 0.620 and Average Adjusted R-squared (AARS) of 0.607 (both with p-values < 0.001) showed that the model explained a substantial level of variance in the dependent variables while accounting for the potential overfitting. These results highlighted the model's strong explanatory power in predicting teacher-related factors, that include job satisfaction and employee retention.

Regarding model quality, the 2.591 Average Block VIF (AVIF) and 2.626 Average Full Collinearity VIF (AFVIF) were less than the recommended threshold of 3.3, implying that multicollinearity was not an issue, and the variables were independent. The Tenenhaus Goodness of Fit (GoF) with a value of 0.649 suggested a large overall model fit. Moreover, the Simpson's Paradox Ratio (SPR), R-squared Contribution Ratio (RSCR), along with the Statistical Suppression Ratio (SSR) met the ideal value of 1.000, confirming the stability of the model and the absence of suppression effects. Hence, these indicators showed that the model was reliable, well-fitted, and suitable for drawing meaningful conclusions about the factors influencing teacher satisfaction and retention.

Table 7. Model Fit and Quality Indices of the Emerging Model

Measure	Estimate	Threshold	Interpretation
Average path coefficient (APC)	0.201, P=0.004	p < .05	Significant
Average R-squared (ARS)	0.620, P<0.001	p < .05	Significant
Average adjusted R-squared (AARS)	0.607, P<0.001	p < .05	Significant
Average block VIF (AVIF)	2.591	≤ 3.3	Ideally
Average full collinearity VIF (AFVIF)	2.626	≤ 3.3	Ideally
Tenenhaus GOF (GoF)	0.649	≤ .36	Large
Simpson's paradox ratio (SPR)	1.000	= 1.00	Ideally
R-squared contribution ratio (RSCR)	1.000	= 1.00	Ideally
Statistical suppression ratio (SSR)	1.000	= 1.00	Ideally

Discussion of Results

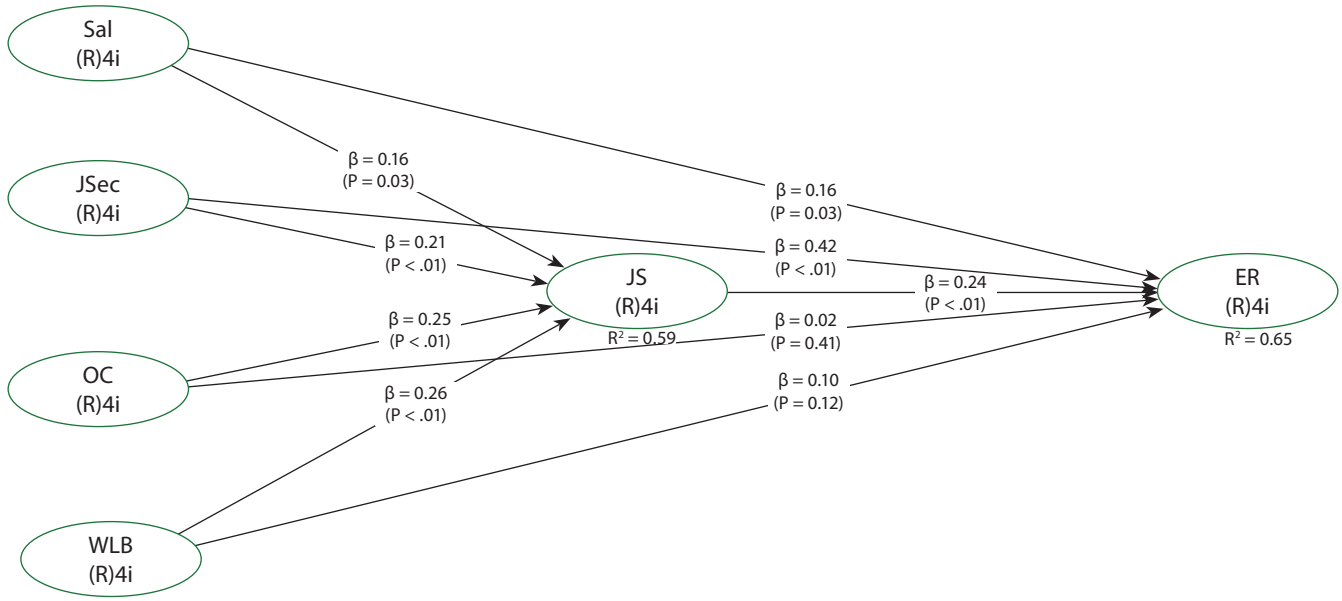


Fig. 2. The Emerging Model

The study aimed to analyze the mediating role of Job Satisfaction on the effect of motivating factors on private senior high school teacher retention in Nueva Ecija, as demonstrated in Figure 2.

Table 8 presented each hypothesis's path coefficients, p-values, and effect sizes, providing insight into the relationships between key variables. Table 8 illustrated the relationship between compensation and job satisfaction was significant, with a path coefficient (β) of 0.158 and a p-value of 0.03. The effect size (f***2***) was 0.101, classified as small according to Cohen's guidelines. This result indicated that while compensation does influence job satisfaction, its impact was modest. Nevertheless, it showed that teachers' perceptions of their compensation were meaningful in shaping their overall job satisfaction, thus supporting H1.

Likewise, Job Security significantly affected Job Satisfaction, as indicated by the path coefficient of 0.211 and a p-value of 0.005. The effect size (f***2***) was 0.145, which was small but notable. This showed that job security had a meaningful positive impact on satisfaction, suggesting that teachers who felt secure were more likely to experience greater job satisfaction, supporting H2.

Moreover, the relationship between Organizational Culture and Job Satisfaction was significant, with a path coefficient of 0.254 and a p-value of 0.001. The effect size (f***2***) was 0.174, which is medium. This meant that a positive organizational culture contributed significantly to job satisfaction. The medium effect size suggested that improving organizational culture could have a meaningful impact on how satisfied teachers feel in their roles. Thus, **H3 was supported**, confirming that a collaborative and supportive work environment enhanced job satisfaction.

Furthermore, the path between **Work-life Balance** and **Job Satisfaction** was found to be significant, having a path coefficient of **0.256** and a **p-value < 0.001**. The effect size (f**2**) was **0.174**, which was medium. This indicated that work-life balance had a considerable influence on job satisfaction. The significant relationship and medium effect size suggested that when teachers could balance their work and personal lives, their job satisfaction increased. Therefore, **H4 was supported**, showing that improving work-life balance could enhance teacher satisfaction.

Finally, the relationship between Job Satisfaction and Employee Retention was significant, with a path coefficient of 0.242 and a p-value of 0.002. The effect size (f**2**) was 0.160, which is medium. This suggested that higher job satisfaction positively impacted employee retention. Teachers who were more satisfied with their jobs were likely to stay in their positions, indicating that improving job satisfaction was key to retaining talent. Thus, H5 was supported, highlighting the crucial role that job satisfaction plays in encouraging teachers to remain in their roles.

Table 8. Path Coefficient and p-value

Path	Path Coefficients (β)	P-value	Effect Sizes (f2)	Effect Size Interpretation (Cohen 1988)	Description	Interpretation
H1: Compensation → Job Satisfaction	0.158	0.029	0.101	Small	Significant	H1 is supported
H2: Job Security → Job Satisfaction	0.211	0.005	0.145	Small	Significant	H2 is supported
H3: Organizational Culture → Job Satisfaction	0.254	0.001	0.174	Medium	Significant	H3 is supported
H4: Work-life Balance → Job Satisfaction	0.256	<0.001	0.170	Medium	Significant	H4 is supported
H5: Satisfaction → Employee Retention	0.242	0.002	0.160	Medium	Significant	H5 is supported

Note: Cohen Effect size: **0.02 – small, 0.15 – medium, 0.35 - large

Table 9 presented the mediation analysis results, which examined how various factors, namely compensation, job security, organizational culture, and work-life balance, mediated the relationship between job satisfaction and employee retention. The table highlighted each path's total, direct, and indirect effects, demonstrating how these mediators contributed to the retention of satisfied employees.

Based on the results, compensation partially mediated the relationship between Job Satisfaction and Employee Retention, with a total effect of 0.679 ($p < 0.001$). The direct effect of Job Satisfaction on Employee Retention was 0.411 ($p < 0.001$), while the indirect effect through compensation was 0.268 ($p < 0.001$). This indicated that while job satisfaction directly influenced employee retention, compensation also significantly strengthened this relationship. Therefore, H6 was supported, showing that compensation considerations were important in retaining satisfied employees.

Similarly, Job Security was found to partially mediate the relationship between Job Satisfaction and Employee Retention, with a total effect of 0.757 ($p < 0.001$). The direct effect was 0.555 ($p < 0.001$), and the indirect effect through Job Security was 0.202 ($p < 0.001$). This demonstrated that job security enhances job satisfaction and employee retention. Moreover, this indicated that teachers who felt secure in their jobs and satisfied were more likely to stay, thus supporting H7.

Moreover, Organizational Culture significantly mediated the relationship between Job Satisfaction and Employee Retention, with a total effect of 0.605 ($p < 0.001$). The direct effect was 0.267 ($p < 0.001$), while the indirect effect via Organizational Culture was 0.338 ($p < 0.001$). This suggested that a positive organizational culture amplified the influence of job satisfaction on employee retention. H8 was supported, indicating that cultivating a strong organizational culture could significantly impact retention.

Finally, Work-life Balance partially mediated the relationship between Job Satisfaction and Employee Retention, with a total effect of 0.635 ($p < 0.001$). The direct effect was 0.336 ($p < 0.001$), and the indirect effect through Work-life Balance was 0.299 ($p < 0.001$). This indicated that a good work-life balance further strengthened the connection between job satisfaction and employee retention, supporting H9. The findings suggested that improving work-life balance was crucial for retaining satisfied employees.

Table 9. Effect for the path with two segments (Mediation Analysis)

Path	Total Effect	P-value	Direct Effect	P-value	Indirect Effect	P-value	Description	Interpretation	Remark
H6: JS → C → ER	0.679	<0.001	0.411	<0.001	0.268	<0.001	Significant	Partial Mediation	H6 is supported
H7: JS → JSY → ER	0.757	<0.001	0.555	<0.001	0.202	<0.001	Significant	Partial Mediation	H7 is supported
H8 JS → OC → ER	0.605	<0.001	0.267	<0.001	0.338	<0.001	Significant	Partial Mediation	H8 is supported
H9: JS → WLB → ER	0.635	<0.001	0.336	<0.001	0.299	<0.001	Significant	Partial Mediation	H9 is supported

Note: Job Satisfaction (JS); Compensation (C); Job Security (JSY); Organizational Culture (OC); Work-life Balance (WLB); Employee Retention (ER)

Conclusions

The study showed that 135 of the respondents were female. The majority of the respondents were between the ages of 20 and 30 or 31 and 40, and they had earned a master's or bachelor's degree. Respondents were full-time employees with a high percentage of teaching experience. Most of the respondents were married. However, the income levels indicated that because most of the respondents reside in rural areas the income that they get was competitive in urban areas. This showed that the work conditions as a rural educator were indeed competitive as an urban educator. On the other hand, the researchers concluded that most respondents were satisfied with the motivating factors in their teaching roles. The respondents indicated a positive perception of work-life balance, long-term job security, and effective communication and support from their schools. These results suggested that the education sector in the selected schools in Nueva Ecija was providing a supportive environment which enabled teachers to succeed effectively in their roles.

Building on this, the researchers found that most respondents were satisfied with the motivating factors in their teaching roles. They expressed positive perceptions of work-life balance, long-term job security, and effective communication and support from their schools. These results suggested that the selected schools in Nueva Ecija provided a supportive environment that enabled teachers to excel in their roles.

The level of job satisfaction further reflected contentment in various aspects. Respondents highlighted favorable experiences with career growth, job opportunities, effective leadership from school heads, proper utilization of their skills, and institutional support. This overall satisfaction appeared to enhance their commitment as educators and motivated them to continue teaching long-term within their institutions.

The respondents also expressed agreement regarding their commitment to staying at their current school. Many cited positive views on compensation, workload, and job satisfaction as influential factors driving their motivation to remain in their roles. Additionally, several respondents expressed confidence in staying at their school for the next few years, contributing to teacher retention and minimizing workforce instability over time.

In conclusion, the researchers found that motivating factors significantly impact teacher retention. For instance, compensation was described as timely and reflective of teachers' skills and workloads. Job security offered reassurance, fostering confidence in long-term employment stability. Teachers also reported that work-life balance was manageable, allowing them to effectively separate personal and professional life. Moreover, organizational culture encouraged open communication between educators and school leadership. Notably, the analysis revealed that work-life balance was the strongest motivator for job satisfaction, while job security had the most substantial influence on both job satisfaction and retention. Overall, these factors improved job satisfaction among the respondents, reinforcing their commitment to teaching and supporting long-term retention within their schools.

Recommendations

The researchers believe the following recommendations may impact job satisfaction and retention. Implementing these recommendations can help create a more motivated and committed workforce.

5.2.1 Compensations are competitive within the industry and reflect qualifications and experience of the employees. Ensure regularly review and adjust compensation to keep up with inflation and market standards.

5.2.2 Employees feel secure in their roles when conducting frequent reviews to provide feedback, and pinpoint areas in need of improvement and offer retention programs for career advancement.

5.2.3 Implementing a system for recognizing and rewarding employees for their hard work and achievements. Foster a supportive and positive work environment within educational institutions.

Directions for Future Researcher

This study offered useful insights into the factors influencing job satisfaction and retention among private senior high school teachers in Nueva Ecija, yet there are several areas where future research could expand and deepen understanding. By exploring broader contexts, integrating additional variables, and adopting diverse methodologies

Different Levels of Education Future research could also investigate retention factors for teachers in elementary or tertiary education settings, providing a broader perspective on the challenges faced by teachers at various levels.

Mixed Method Approach To complement quantitative data, future researchers may use the mixed method that includes quantitative and qualitative methods, such as interviews or focus groups, to explore teachers' personal experiences and motivations in greater depth, uncovering nuances that survey questionnaires may not fully capture.

Expand the Scope Beyond Nueva Ecija Future researchers could extend the study to other regions in the Philippines to compare how motivating factors and job satisfaction influence teacher retention in different contexts, including urban and rural settings.

Teacher Well-being and Mental Health Considering the increased attention on mental health, future studies might investigate how well-being programs and mental health support contribute to teacher retention, particularly in high-stress educational environments.

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REFERENCES

1. Adams, A., Jumpah, E. T., & Shafiullah, M. (2021). Agricultural technologies adoption and smallholder farmers' welfare: Evidence from Northern Ghana. Taylor & Francis Online. https://doi.org/10.1080/23322039.2021.2006905
2. Aduna, R. R., Patricio, J. L., Torres, J. A. G., & Balaria, F. E. (2020, February 3). Senior High school teachers' turnover: a case of a private school as basis for human resource management policy. https://ijels.com/detail/senior-high-school-teachers-turnover-a-case-of-a-private-school-as-basis-for-human-resource-management-policy/
3. Babalola, O., Ayantunji, I. O., Solomon, O. T., Ajibola, K. S., & Adetoun, B. (2022). Job satisfaction and job contentment in the context of the Nigerian banks. Skyline Business Journal. http://dx.doi.org/10.37383/SBJ180103
4. Baroudi, S., Tamim, R., & Hojeij, Z. (2022). A Quantitative Investigation of Intrinsic and Extrinsic Factors Influencing Teachers' Job Satisfaction IN Lebanon. Leadership and Policy in Schools, 21(2), 127–146. https://doi.org/10.1080/15700763.2020.1734210
5. Bartanen, B., Grissom, J. A., & Rogers, L. K. (2019). The Impacts of Principal Turnover. SageJournals. https://doi.org/10.3102/016237371985544
6. Bhandari,P. (2020). Howto Calculate Standard Deviation(Guide). Scribbr. https://www.scribbr.com/statistics/standard-deviation/
7. Bhandari, P. (2022). What Is the Mean | How to Find It & Examples? Scribbr. https://www.scribbr.com/statistics/mean/
8. Bhat, A. (2023, November 24). Descriptive Correlational: Descriptive vs Correlational Research. QuestionPro. https://www.questionpro.com/blog/descriptive-research-vs-correlational-research/#:~:text
9. Bulawat, A. (2019, December 17). Teachers' turnover among public schools: Basis for teachers' retention programs. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3744193
10. Casper, W. J., Vaziri, H., Wayne, J. H., & DeHauw, S. (2018). The Jingle-Jangle of Work–Nonwork Balance: A Comprehensive and Meta-Analytic Review of Its Meaning and Measurement. American Psychological Association. http://dx.doi.org/10.1037/apl0000259
11. Cities and Municipalities Competitiveness Index | CMCI. (n.d.). CMCI. https://cmci.dti.gov.ph/interactive-map.php?prov=Nueva%20Ecija
12. David, E., & Naparan, G. (2024). Challenges and interventions of school administrators in the retention of professional teachers in private schools. Indonesian Journal of Education and Social Science, 3(2), 201–2015. https://doi.org/10.56916/ijess.v3i2.753
13. Domfeh, H. A., & Hunsaker, W. (2020). The Effect of Job Security on the Commitment of Teachers in Ghana: The Mediating Role of Career Satisfaction. ResearchGate. https://doi.org/10.37394/23207.2020.17.18
14. Edmonds, W. A., & Kennedy, T. D. (2017). An Applied Guide to Research Designs: Quantitative, Qualitative, and Mixed Methods. https://doi.org/10.4135/9781071802779
15. Erichsen, K., & Reynolds, J. R. (2020). Public school accountability, workplace culture, and teacher morale. Social Science Research,85, 102347. https://doi.org/10.1016/j.ssresearch.2019.102347
16. Espinosa, J. M. (2024). Organizational culture and resilience as predictors of quality of work life among educators: A convergent design. hcommons.org. https://doi.org/10.17613/2ap4-gs23
17. Fleetwood, D. (2023b, November 27). Sampling Methods: Guide To All Types with Examples. QuestionPro. https://www.questionpro.com/blog/types-of-sampling-for-social-research/
18. Gimbert, B. G., & Kapa, R. R. (2022). Mid-Career teacher retention: Who intends to stay, where, and why? Journal of Education Human Resources, 40(2), 228–265. https://doi.org/10.3138/jehr-2020-0037
19. George, T. (2022, October 25). Mixed Methods Research | Definition, Guide, & Examples. Scribbr. https://www.scribbr.co.uk/research-methods/mixed-methods/

20. Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2019). A primer on partial least squares structural equation modeling (PLS-SEM) (2nd ed.). SAGE Publications. Hashmi, I. U. D. a. S. K. D. M. H. S. K. M. A. (2023, January 27). Objective to conduct a pilot study in research. https://www.journalppw.com/index.php/jpsp/article/view/15369

21. Hess, A. (2023, November 21). Stratified Random Sampling | Definition, Method & Characteristics. Study.com; April Hess. https://study.com/academy/lesson/stratified-random-samples-definition-characteristics-examples.html

22. Hendrawijaya, A. T., Hilmi, M. I., Hasan, F., Imsiyah, N., & Indrianti, D. T. (n.d.). Determinants of Teacher Performance with Job Satisfactions Mediation. Eric. https://eric.ed.gov/?id=EJ1259672

23. Immaculate, A., Gaite, S. S., & Asiimwe, S. (2023). Challenges of Retaining Teachers in Private Secondary Schools in Wakiso District, Uganda. Research Gate. https://doi.org/10.7176/JEP/14-9-11

24. Jebb, A., Ng, V., & Tay, Louis. (2021). A Review of Key Likert Scale Development Advances: 1995-2019. Frontiers in Psychology. https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2021.637547/full

25. Kamrath, B., & Bradford, K. (2020). A case study of teacher turnover and retention in an urban elementary school. NewPrairie Press. https://newprairiepress.org/edconsiderations/vol45/iss3/6/

26. Katete, Subiri & Nyangarika, Anthony. (2020). Effects of Late and Non-Payment of Teachers' Salaries and Benefits on Learning Process in Public Secondary Schools. International Journal Of Advance Research And Innovative Ideas In Education. 6.1274-1290.

27. Khatri, K. (2020). Research Paradigm: A Philosophy of Educational Research. International Journal of English Literature and Social Sciences. https://ijels.com/upload_document/issue_files/15IJELS-109202027-Research.pdf

28. Kurtz, H. (2024, February 16). A profession in crisis: findings from a national teacher survey. EdWeek Research Center. https://www.edweek.org/research-center/reports/teaching-profession-in-crisis-national-teacher-survey

29. Llego, M. A. (2022, July 23). DEPED Basic Education Statistics for the school year 2019-2020. TeacherPH. https://www.teacherph.com/deped-basic-education-statistics-school-year-2019-2020/

30. Liu, J., Kang, J., & Luo, H. (2021). Overall effects of temperature steps in hot summer on students' subjective perception, physiological response, and learning performance. Science Direct. https://www.sciencedirect.com/science/article/abs/pii/S0378778821004084

31. Manaig, M., Perez, F., & Casulla, E. (2023). Human Resource Management Policies on Employee Engagement and Retention. IMPACT: International Journal of Research in Business Management, 11(4).

32. Manalo, R., De Castro, B., & Uy, C. (2020). The Mediating Role of Job Satisfaction on the Effect of Motivation to Organizational Commitment and Work Engagement of Private Secondary High School Teachers in Metro-Manila. Review of Integrative Business Economics Research, 9(1). https://sibresearch.org/uploads/3/4/0/9/34097180/riber_9-s1_12_b19-081_133-159.pdf

33. Medallon, M. (2020). Determinants of Millennial Employee Retention in Selected Philippine Workplaces. LPU-Laguna Journal of Multidisciplinary Research. https://lpulaguna.edu.ph/wp-content/uploads/2021/09/6.-Medallon_Millennial-Employee-Retenti on.pdf

34. Menzies, L. (2023, March 16). How teacher churn hits pupil outcomes - and what to do about it. Tes Magazine. https://www.tes.com/magazine/teaching-learning/general/teacher-retention-turnover-chur n-schools-teaching-learning

35. Mohammad, A., & Borkoski, C. (2024). Organizational conditions and teacher turnover in private schools of United Arab Emirates. Social Sciences & Humanities Open, 9, 100801. https://doi.org/10.1016/j.ssaho.2023.100801

36. Morales, M. D., & Sapin, S. B. (2020). Journey of School Principals in Coping the Demands and Challenges of School Development. HRPUB. https://www.hrpub.org/download/20201130/UJER69-19516892.pdf

37. Moran, M. (2024, April 17). Cronbach's Alpha. Statistics Solutions. https://www.statisticssolutions.com/cronbachs-alpha/

38. Mullen, P. R., Chae, N., Backer, A., & Niles, J. (2021). School Counselor Burnout, Job Stress, and Job Satisfaction by Student Caseload. NASSP Bulletin. https://journals.sagepub.com/doi/10.1177/0192636521999828
39. Nguyen, T. D., Pham, L., Springer, M. G., & Crouch, M. (2019). he Factors of Teacher Attrition and Retention: An Updated and Expanded Meta-Analysis of the Literature. EdWorkingPapers. https://edworkingpapers.com/ai19-149
40. Oberes, M. V., & Tan, T. A. (2022). Effect of organizational commitment on turnover intention of senior high private school teachers in Cebu City, Philippines. The Normal Lights/the Normal Lights, 16(1). https://doi.org/10.56278/tnl.v16i1.1714
41. Ortan, F., Simut, C., & Simut, R. (2021). Self-Efficacy, Job Satisfaction and Teacher Well-Being in the K-12 educational system. International Journal of Environmental Research and Public Health/International Journal of Environmental Research and Public Health, 18(23), 12763. https://doi.org/10.3390/ijerph182312763
42. Province of Nueva Ecija | Philippine Statistics Authority | Republic of the Philippines. (2024, May 8). https://www.psa.gov.ph/classification/psgc/citimuni/0304900000
43. Qualtrics (2024, March 1). How to use stratified random sampling to your advantage. https://www.qualtrics.com/au/experience-management/research/stratified-random-sampling
44. Qin, L., Lu, J., Zhou, Y., Wijaya, T., Huang, Y., & Fauziddin, M. (2022). Reduction of Academic Burnout in Preservice Teachers: PLS-SEM Approach. MDPI. https://www.mdpi.com/2071-1050/14/20/13416
45. Reitman, G. C., & Karge, B. D. (2019). Investing in teacher support leads to teacher retention: Six supports administrators should consider for new teachers. https://eric.ed.gov/?id=EJ1250205
46. Reyes, A. C. S., Aquino, C. A., & Bueno, D. C. (2019). Why employees leave: Factors that stimulate resignation resulting in creative retention ideas. CC The Journal: A Multidisciplinary Research Review, 14, (October 2019). ISSN 1655-3713. http://dx.doi.org/10.13140/RG.2.2.17579.39203
47. Shamran, M. K., Bandar, M. A., & Mahdi, A. M. (2020). A STUDY OF RELATIONSHIP BETWEEN EMPLOYEE ENGAGEMENT ANDAND JOB SATISFACTION IN THE KARBALA HEALTH DIRECTORATE. Humanities & Social Sciences Reviews. https://doi.org/10.18510/hssr.2020.8477
48. Simply Psychology. (2023, September 28). Herzberg's Two-Factor Theory of Motivation-Hygiene. https://www.simplypsychology.org/herzbergs-two-factor-theory.html
49. Sirgy, M. J., & Lee, D. J. (2017). Work-Life Balance: an Integrative Review. Applied Research in Quality of Life, 13(1), 229–254. https://doi.org/10.1007/s11482-017-9509-8
50. Slovin's Formula SamplingTechniques. (2020, December 14). Sciencing. https://sciencing.com/slovins-formula-sampling-techniques
51. UNESCO. (2024). Global report on teachers: What you need to know. In UNESCO.org. https://www.unesco.org/en/articles/global-report-teachers-what-you-need-know
52. Vikas, & Sharma, S. (2022a). An Exploration For Factors Affecting Retention Of Employees In Private Senior Secondary School -With Special Reference To Delhi-NCR Region. Journal of NamibianStudies, 32(ISSN:2197-5523). https://namibian-studies.com/index.php/JNS/article/view/4542/3153
53. Villanueva, A. A., Disu, S. S., & Villanueva, K. F. P. A. (2021). Assessing the School Heads Leadership in the Towns of Nueva Ecija, Philippines: Inter-Relationship of Supervisory Skills, Interpersonal Skills, and Leadership Skills. SCIRP. https://www.scirp.org/journal/paperinformation?paperid=113225

The Role of AI-Based Scheduling Systems in Enhancing Job Satisfaction and Work-Life Balance: A study Among BPO Companies

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Abstract

This study aimed to explore how AI impacted employees in BPO companies, particularly in enhancing job satisfaction and work-life balance. By examining variables such as work-life balance, job satisfaction, and AI-driven scheduling systems, the study investigated how these AI tools optimized workforce management, leading to increased employee satisfaction and productivity. A quantitative approach was adopted. As part of the proposal, the study conducted pilot testing involving 25 respondents, all of whom were BPO employees. Data collection was carried out using surveys via Google Forms. Snowball sampling techniques were employed, leveraging the researchers' connections and forums to disseminate the survey.

The findings from this pilot study provided preliminary insights into the effectiveness of AI-driven scheduling tools in improving job satisfaction and work-life balance in the BPO sector. This research aimed to offer valuable guidance for BPO companies seeking to enhance employee well-being and operational efficiency through AI integration.

Keywords: Work-Life Balance, Job Satisfaction, Artificial Intelligence, and Workforce Management

Introduction

The rapid evolution of AI significantly transformed many sectors in business, and the fast-growing business process outsourcing sector was no exception. Some of the significant innovations that fueled this transformation were related to AI-based operational systems, particularly workforce management systems. Among the many ways BPO businesses implemented such innovations was through AI-based scheduling systems. These systems allowed BPO companies to gain far-reaching advantages in terms of streamlining operations and improving how they managed their shifts. However, the impact of such systems on crucial human factors like work-life balance and job satisfaction remained relatively unknown.

This paper aimed to fill this gap by analyzing how AI-based scheduling systems affected work-life balance and job satisfaction in BPO companies. As these organizations increasingly sought to meet their needs with a more motivated and satisfied workforce, determining the role of AI became ever more relevant to work experiences. It was, therefore, essential to investigate the consequences that AI-based scheduling had on job satisfaction and work-life balance. This research delved into how technology was not only used to increase efficiency but also to support the overall well-being of employees.

The BPO industry was characterized by high employee turnover rates, which could be attributed to factors such as poor work-life balance, inadequate job satisfaction, and lack of flexibility. AI-based operational systems had the potential to address these issues by providing personalized and efficient solutions. By automating routine tasks and reducing manual labor, AI helped employees manage their work schedules and other operational tasks more effectively, leading to improved work-life balance and job satisfaction.

This study investigated the correlation between AI-based scheduling systems and job satisfaction, as well as their impact on work-life balance in the context of BPO companies. It examined how artificial intelligence-based scheduling and workforce management systems influenced employee experiences, with the goal of providing actionable insights for BPOs to improve their employees' satisfaction and retention. Finally, the research contributed to the emerging literature on the applications of AI in the BPO industry by providing deeper insights into the advantages and disadvantages associated with the integration of AI in workforce management. The insights gathered from this study were of immense importance to organizations striving to balance operational efficiency with employee welfare.

Literature Review

Artificial Intelligence in Operations

Artificial Intelligence (AI) has emerged as a transformative force in reshaping human resource (HR) practices across various industries. This discussion delves into the impact of AI on BPO companies, focusing on key aspects such as job satisfaction, stress levels, fairness, trust, and service delivery quality. Insights from notable authors shed light on how AI is revolutionizing BPO companies, offering both opportunities and challenges in the quest for more effective but also fair workforce management.

Dagnaw (2020) discusses the integration of AI technologies in Industry 4.0, emphasizing the potential benefits for SMEs in managing increased complexity and meeting changing customer demands. AI is seen as a critical driver of enterprise competitiveness, with applications ranging from predictive maintenance to supply chain optimization. The paper underscores the urgent need for systematic AI development and implementation to realize its full impact in Industry 4.0.

Chen (2019) explores the factors influencing AI adoption, particularly in the telecom industry in China. The study highlights the importance of managerial capabilities, vendor partnerships, and government involvement in successful AI adoption. It proposes a conceptual framework that integrates various factors impacting AI adoption and emphasizes the need for firms to enhance internal management mechanisms, collaborate effectively with vendors, and understand the strategic implications of AI adoption.

The study by Sullivan, Y., & Wamba, S. (2022) highlights how AI's information processing capability can indirectly influence firm performance by enhancing firm resilience to supply chain disruptions. It emphasizes the importance of AI in improving coordination, learning, and strategic responses within organizations, leading to stronger firm resilience. This aligns with the idea that AI can optimize talent allocation by providing faster and more accurate information processing, aiding decision-makers in adjusting resource allocations during disruptive events.

Lopez Conde, M., & Twinn, I. (2019) discuss the potential benefits and challenges of AI in the transport sector, highlighting the socioeconomic effects such as job displacement, cost constraints, infrastructure readiness, and skills shortages. These challenges are relevant in the context of AI's impact on human resource practices, as they underscore the need for organizations to manage these issues effectively when adopting AI in talent management and workforce optimization.

Concepcion, R. S., Bedruz, R. A. R., Culaba, A. B., Dadios, E. P., & Pascua, A. R. (2019) focus on the adoption and governance of AI in the Philippines, emphasizing the potential for AI to augment labor shortages and enhance productivity. Their discussion on the mental shift required for governments and industry leaders to adopt AI aligns with the broader theme of optimizing talent allocation and workforce competitiveness through AI adoption in human resource practices.

The discussion on Artificial Intelligence (AI) in human resource practices, particularly regarding the accuracy of forecasts, identification of skills gaps, optimization of talent allocation, robustness, and security, can be corroborated with insights from the authors Guo & Zhang (2022) and Tambe, Cappelli & Yakubovich (2019).

Further, Guo and Zhang (2022) emphasize the importance of multi-objective optimization (MOO) for project management improvement, which involves considering various objectives simultaneously. This approach aligns with AI's capacity to analyze complex data sets and variables to provide more accurate forecasts in HR practices. By leveraging AI algorithms, organizations can enhance their forecasting accuracy regarding workforce needs, talent acquisition, and performance predictions.

Work-life Balance of BPO Employees

The relationship between work-life balance and job satisfaction has received attention from Cegarra-Leiva, Sanchez-Vidal, and Cegarra-Navarro (2012). They explored the link between work-life balance practices and organizational outcomes in SMEs, highlighting the mediating effect of a supportive culture. The findings of their study suggests that a supportive culture within organizations can mediate the impact of work-life balance practices on organizational outcomes.

Sanchez-Hernandez, Gonzalez-Lopez, Buenadicha-Mateos, and Tato-Jimenez (2019) explored work-life balance in great companies and pending issues for engaging new generations at work. The role of AI Based Operations in addressing pending issues related to work-life balance and its potential impact on engaging new generations at work is an area that requires further research.

According to Cartwright and Cooper (2015), employees' work-life balance promotes psychological well-being, which influences job performance. They also noted that organizations need to enhance their performance to achieve a competitive edge. However, in order to improve organizational performance, the employees need to have that certain balance between work and life in order to function properly in a working environment.

According to a study conducted by Bacea and Bordean, outsourcing companies have practices that generally have negative perception or even consequences for employee job satisfaction. One of the factors stated is that they are unsatisfied by the work schedule assigned to them as it affects their personal life.

Another study conducted by a different group of researchers, Kotteeswari and Sharief (2014), there is an abundance of job stress that BPO's employees endure in their workplace. According to their study, employees who experience stress tend to focus on the emotion rather than the task at hand. Basically, job performance goes lower as the employee's job unsatisfaction goes higher. One of the factors that contribute to job dissatisfaction is a poor work schedule. Based on the study that Kotteeswari and Sharief (2014) conducted, excessive overtime affects employee performance, which is often the result of poor management of work scheduling by managers. One of the solutions that these researchers proposed is allowing the employees to have a flexible schedule as well as job sharing.

Job Satisfaction of BPO Employees

According to Alarcon (2021) business Process Outsourcing (BPO) is a rapidly growing industry in the Philippines, with a workforce of over a million at the end of 2014. BPO companies provide email response and managing services, as well as various customer relations services such as travel, technical support, education, customer care, financial services, and online business support. BPO employees face challenges such as stress, anxiety, de-motivation, and health issues due to the changing body clock.

According to Arzaga (2021) the physical workplace environment is crucial for maintaining a healthy work environment, as it affects employees' behavior, perception, and performance. Research shows mixed results on the weight of the relevance of work environment features on employee satisfaction, job performance, and concentration. This paper presents literature reviews on the impact of a positive office environment on productivity and job satisfaction in the BPO industry, focusing on factors such as the physical environment, occupational safety, occupational health, and appropriate technology/equipment. The study aims to provide insights into the impact of positive office environments on employee productivity and job satisfaction in the BPO industry.

Based on a research conducted by Alambra (2024) their study aims to understand the reasons behind employee turnover in the business process outsourcing (BPO) industry, as it plays a crucial role in shaping employment patterns and labor markets. By identifying industries most affected by outsourcing, skill sets in demand, and potential job displacement or upskilling, businesses can develop strategies, training initiatives, and policies to manage turnover rates. High staff turnover is a significant challenge for BPO companies, and effective retention measures are essential for their growth and success. The study explores employee satisfaction with retention methods provided by BPO companies, urging them to design action plans and focus on creating successful retention methods to improve employee happiness. Wage sharing is a major industrial issue, and BPO companies have tried to address it by offering additional incentives, bonuses, and advantages to retain and recruit staff. This study aims to help BPO companies provide effective outcomes or programs to address high attrition in the BPO industry, as losing personnel reduces productivity and profits, negatively impacting the company's reputation.

Occupation fulfillment refers to an individual's satisfaction with their work, and it is often associated with motivation or enthusiasm. The "splendid subject matter expert" hypothesis suggests that the most fulfilled workers are the best entertainers, but this theory is not widely accepted. A crucial effect on job satisfaction is the use of job plans, which aim to improve work satisfaction and execution using strategies such as work rotation, work advancement, work progression, and job reengineering.

According to Baisoya (2023) the IT and ITES/BPO sector is a significant part of the Indian economy, with a total income exceeding $100 billion and a share of the nation's GDP increasing from 1.2% in 1998 to about 7.5% in 2012. The sector has helped India transition from an agricultural-based economy to an information-based economy, providing unlimited business opportunities to the jobless young people of India, along with favorable wages and work norms.

According Mohsin (2023) job satisfaction is defined as the feelings people have about their positions, which can be influenced by various factors such as the relationship with their manager, the work environment, and level of satisfaction in their work. It is important because it helps work execution and expands the nature of employees' lives. Ten factors that experts find basic in how satisfied individuals are with their positions are: (1) Little problems, (2) View of reasonable compensation, (3) Achievement, (4) Criticism, (5) Unpredictability and gathering, (6) Control, (7) Authoritative help, (8) Work-home flood, (9) Special first nights and cerebral pains, and (10) Effectively satisfied.

Job satisfaction is a significant quality and is regularly assessed by organizations using rating scales. Factors such as pay, education, occupation, age, and sexual orientation are used to estimate job satisfaction. Respondents were grouped into specific and middle-class occupations based on their responses to questions about their work and jobs.

According to Bacea, A((2016) Cluj-Napoca, one of Romania's main university cities, has long been a favorite location for BPO companies. Cluj-Napoca's benefits include a talented workforce, decent living circumstances, and good links to the rest of the country and major European cities. Many surveys rank Cluj-Napoca as the best Transylvanian city to live in, with excellent ratings in economic, scholastic, social, and cultural categories. Cluj-Napoca residents specialize in research and data analytics, order management, finance and accounting, engineering services, software development, and IT services. With all of these pro-arguments, many BPOs have outsourced their processes in Cluj-Napoca, and the problem today is to retain people in the organization for a longer period of time.

Methodology

The researchers utilized purposive and snowball sampling techniques in order to get the information and/or data. The reason as to why the researchers decided to use the snowball sampling technique was because they knew several people that worked at BPO companies, and those people also helped out handing the questionnaire to their coworkers. With this technique, the researchers were able to reach the number of respondents required to make a conclusion.

This study was conducted in Quezon City, Philippines, which was an important center for the business process outsourcing (BPO) industry. This location provided a unique context for investigating the influence of AI-based operations on job satisfaction and work-life balance among BPO personnel. Quezon City was home to several prominent BPO companies, including TaskUs, Accenture, and others, whose presence was instrumental in the sector's growth in the region.

The researchers decided to apply the Likert scale on their survey. To be specific, the survey questions consist of a 6-point Likert-type scale with these levels of agreement: 1 – strongly disagree, 2 – disagree, 3 – slightly disagree, 4 – slightly agree, 5 – agree, and 6 – strongly agree. To help the researchers make the survey efficient, they decided to use the Google Forms platform to easily hand out several questionnaires to several people at a time.

Analysis and Discussions

Discussion of Findings

Effect of AI-based scheduling systems on the work-life balance and job satisfaction of BPO employees in Quezon City indicated that they greatly improved job satisfaction, especially in managing workload, which was highly scored at 4.66. This implies that employees felt that the AI system helped them organize and distribute their tasks successfully for their workdays to be fully manageable. High score with the statement "AI scheduling helps manage workload more efficiently" at 5.08 further proves that the employees were not facing excess stress levels due to the better management of tasks. A moderate degree of benefitting in terms of efficiency and productivity comes with the mean productivity scoring at 4.17 for the AI system. Employees scored 4.03 and were slightly in agreement that AI supported the completion of work faster though the overall increase in productivity is not as expected. The implication is that while there are indeed those areas that the AI made lighter and easier, contributory inputs to the quality or quantity of work completed were not enough, thus implying that the system made the work more organized without necessarily increasing output.

When it comes to skill enhancement the view is neutral, represented by the mean score of 3.73. This shows that they did not believe that the AI system is actually improving them personally and professionally. The lowest score was 3.28 for the statement "AI scheduling provides more time for skill development." This shows the fear of employees that the AI is creating minimum learning opportunities or advancing their careers in any which way. While the AI facilitated an effective management of workload, concerning activities for the skill-building, nothing represented training or on-the-job learning.

When it came to job autonomy, with the score of 4.16, employees welcomed more control over their schedules, atleast that was the indication from a slightly higher score of 4.35 on the statement "AI scheduling gives more control over work schedules." However, the slightly smaller score of 3.99 concerning flexibility in managing shifts meant that though employees appreciated the better control, there was still room for further enhancements in order to allow them to alter their schedules to fit personal needs better.

The system, as per the above last points, had a moderately positive influence on work-life balance with a mean score of 3.98. In fact, the employees claimed that AI scheduling offered some flexibility at a mean score of 4.10 for "AI scheduling allows more flexibility in changing work hours." However, the AI system was still relatively supportive in the balancing of work and personal life. To the best of the current situation and to improve things, there is still room for growth in aiding those individuals whose commitments may encompass responsibilities or engagements within the family or certain leisure activities.

The findings suggested that the AI-based scheduling system had influenced job satisfaction as well as work-life balance positively. Employees appreciated improvement in task-related matters in load management and job autonomy levels with significantly higher levels of job satisfaction scores. However, for more effective utilization of AI systems in the workplace, productivity levels need to be improved, skill enhancement must be addressed, and flexibility in scheduling needs to be introduced.

For the Attitudes part, having Adaptability in diverse work environments was chosen the most at 188 (62.7%) responses. Having a Positive outlook towards multicultural interaction has 184 (61.3%) responses. Openness to new experiences and change has a total of 175 (58.3%) responses from the total number of respondents. Proactive approach to conflict resolution has a total of 161 (53.7%) responses. Lastly, willingness to revise opinion based on new information has a total of 152 (50.7%) responses.

Table 1. The Level of Implementation of AI in the Work Scheduling Systems of BPO Companies

Indicator	Mean	SD	Interpretation
IAIWSS1.1 The AI-based scheduling system is fully integrated into our company's workforce management.	4.45	0.684	Agree
IAIWSS 1.2 Our company's AI-driven scheduling system is frequently updated for improved functionality.	3.98	0.824	Slightly Agree
IAIWSS 1.3 The AI scheduling system is reliable in generating work schedules accurately.	4.21	0.647	Slightly Agree
IAIWSS 1.4 The implementation of the AI scheduling system has reduced scheduling errors in the company.	4.18	0.942	Slightly Agree
IAIWSS 1.5 The AI system for work scheduling is user-friendly and easy to interact with.	4.44	0.899	Agree
IAIWSS 1.6 Employees have received sufficient training on using the AI-based scheduling system.	4.35	0.985	Agree
Overall	**4.27**	**0.83**	**Strongly Agree**

The results in the table on the perceived usefulness of AI-based scheduling systems reveal a very positive sentiment among employees in BPO companies with a mean score of 4.57, which falls into "Agree." Such a high score signifies that there is a general belief that the deployment of AI systems has undoubtedly enhanced several aspects of the scheduling process. Surprisingly, the highest score was given to the statement "I find the AI scheduling system useful in managing my shifts," with a score of 4.84, a result that seems to reflect an arguably highly valuable perception of these systems by employees as good tools for managing shifts. The remaining indicators of the degree of reducing the scheduling conflicts (mean = 4.83) and providing a better alternative for manual methods in scheduling (mean = 4.74) further support the fact that AI provides evidence that indeed the operations are carried out effectively. Employees also proved to agree that the AI-based scheduling system improved workflow efficiency in an overall manner with a mean score of 4.66. From these findings, it can thus be concluded that employees realize that AI systems are capable of making an improvement in organizational performance by improving scheduling practices. However, "The AI scheduling system has increased fairness in distributing work schedules" scored a mean at 3.94 and falls under "Slightly Agree." It indicates that even though employees welcome the change brought by AI, they still do not trust the fairness of the work schedule distribution. That must be changed to have a better perception of how AI can bring balance to equality in the workplace.

Table 2. The Level of Perceived Usefulness of AI-Based Scheduling Systems

Indicator	Mean	SD	Interpretation
PU2.1 The AI-based scheduling system has improved the overall scheduling process in the company.	4.41	0.691	Agree
PU 2.2 I find the AI scheduling system helpful in managing my shifts.	4.84	0.772	Agree
PU 2.3 The AI-based scheduling system contributes to minimizing scheduling conflicts.	4.83	1.196	Agree
PU 2.4 The AI scheduling system has increased fairness in distributing work schedules.	3.94	1.030	Slightly Agree
PU 2.5 I believe the AI system provides a better alternative to manual scheduling methods.	4.74	0.855	Agree
PU 2.6 The AI scheduling system has improved the overall workflow efficiency in the company.	4.66	0.879	Agree
Overall	**4.57**	**0.90**	

The square roots of the Average Variance Extracted AVE related to the inter-construct squared correlations were helpful in advancing these kinds of questions related to discriminant validity among the constructs in study. The diagonal values within

are the square roots of the AVE, while off-diagonal values represent the squared correlations between constructs. Starting with "AI Work Scheduling Systems," its square root of the AVE is 0.826, hence good internal consistency within the construct. It has a positive correlation coefficient of 0.796 with "Perceived Usefulness" (PU). It is not yet square root of the AVE for AIWSS. This actually says that it contributes to discriminant validity since the constructs are distinct despite such high correlation. Perceived Usefulness": The square root of AVE is 0.821. The squared correlation with "Job Satisfaction" (JS) is 0.667, implying a moderate relationship. This order of correlation does not pose any threat to the discriminant validity, as it is less than the square root of the AVE for both PU and JS. The construct "Job Satisfaction" has a square root of AVE equal to 0.929 and, hence, an excellent convergent validity. The squared correlation with "Work-Life Balance" (WLB) is 0.818, which is substantial but still stays below the square root of the AVE for JS. This also further stresses the discriminant validity of these constructs as well.

Lastly, "Work-Life Balance" has a square root of the AVE of 0.958. The square correlation with AIWSS was 0.769 and with PU 0.543. That is, both are high and moderate, respectively. For the most important points, all inter-construct correlations are less than the square roots of their respective AVE values, meaning that each is distinct from the others. These results indicate that the constructs of this study enjoy good discriminant validity. Their relationships are all significant, however, none exceeds the square root of its AVE for the respective constructs. This further implies that each captures different dimensions of the overall framework that consequently reinforces the robustness of the model in evaluating the relationships between AI work scheduling systems, perceived usefulness, job satisfaction, and work-life balance.

Table 3. Relation of Square Roots of the AVE with the Inter-Construct Squared Correlation

Construct	AIWSS	PU	JS	WLB
AIWSS	**0.826**			
PU	0.796	**0.821**		
JS	0.824	0.667	**0.929**	
WLB	0.769	0.543	0.818	**0.958**

The assessments of reliability and convergent validity for the constructs provide the required information about the quality of measurements concerned to the latent variables of the study. For the variable "Current Level of Implementation," it was found that Cronbach's Alpha is at 0.894 while Composite Reliability is at 0.924, in which case it reflects a strong level of internal consistency. As far as this construct is concerned, the Average Variance Extracted comes out to be 0.682, entirely quite all right, indicating that items are accurately reflective of the latent concept in question. Its factor loadings range between 0.854 and 0.967, each of which comes out with a p-value below 0.001, which goes to further prove the existence of strong item-latent variable relationships. For "Perceived Usefulness," Cronbach's Alpha is 0.896, while Composite Reliability is at 0.923, and all are indicative of very high reliability. The AVE for the construct is at 0.675, thereby suggesting that the items also reflect this latent variable quite adequately. As expected, the factor loadings are rather high ranging from 0.566 to 0.976, and all p-values are less than 0.001, indicating significant contributions of each item to the overall construct.

The construct "Job Satisfaction" is even more reliable, with Cronbach's Alpha of 0.947 and Composite Reliability of 0.962 and an impressive AVE of 0.864, indicating excellent convergent validity. The related items such as Workload Management, Efficiency and Productivity, Skill Enhancement, and Job Autonomy all evidence very strong factor loadings ranging from 0.901 to 0.967, with p-values less than 0.001 for all items. For "Work-Life Balance," the results provide Cronbach's Alpha of 0.970 and Composite Reliability of 0.978; very high, hence inferring exceptional reliability. The AVE for this construct was 0.918, also high in reflection on convergent validity. Factors loading on this construct are: Work Flexibility, Stress Reduction, Work-Life Boundaries, and Employee Engagement whose factor loadings range between 0.919 to 0.987, and were all statistically significant, $p < 0.001$.

In general, the findings confirm that the constructs measures in this study have good levels of reliability and convergent validity. This would point to the fact that items of measurement used accordingly capture the underlying theoretical constructs hence a sound foundation for further analysis and interpretation of relations between AI work scheduling systems and perceived usefulness job satisfaction as well as work-life balance.

Table 4. Construct Reliability and Convergent Validity

Construct	Items	Factor Loading	p-value	Cronbach's Alpha	Composite Reliability	Ave. Variances Extracted
Current Level of Implementation				0.894	0.924	0.682
	IAIWSS1.1	0.886	<0.001			
	IAIWSS1.2	0.967	<0.001			
	IAIWSS1.3	0.869	<0.001			
	IAIWSS1.4	0.854	<0.001			
	IAIWSS1.5	0.866	<0.001			
	IAIWSS1.6	0.88	<0.001			
Perceived Usefulness				0.896	0.923	0.675
	PU2.1	0.698	<0.001			
	PU2.2	0.566	<0.001			
	PU2.3	0.766	<0.001			
	PU2.4	0.925	<0.001			
	PU2.5	0.92	<0.001			
	PU2.6	0.976	<0.001			
Job Satisfaction				0.947	0.962	0.864
	Workload Management	0.901	<0.001			
	Efficiency and Productivity	0.967	<0.001			
	Skill Enhancement	0.903	<0.001			
	Job Autonomy	0.945	<0.001			
Work-Life Balance				0.97	0.978	0.918
	Work Flexibility	0.98	<0.001			
	Stress Reduction	0.987	<0.001			
	Work-Life Boundaries	0.919	<0.001			
	Employment Engagement	0.946	<0.001			

Discussion of Results

The strength of the results of the hypothesis tests regarding the influence of AI implementation on BPO employees is significant enough to bolster its positive impacts on job satisfaction and work-life balance by AI-driven scheduling systems. On the first hypothesis, H1, where AI implementation leads to improved job satisfaction on the part of BPO employees, its path coefficient is 0.899, implying that there is a highly positive relationship between AI implementation and job satisfaction. The p-value for this hypothesis is below 0.001, meaning that the results are statistically significant. Furthermore, having an effect size of 0.809, it could be argued that it caused a large effect and hence push closer to the conclusion that introduction of AI significantly enhanced job satisfaction about the employees in the BPO sector.

As with the second hypothesis, H2, it is presumed that implementing AI will aid the workforce in achieving a positive work-life balance. In this case, the path coefficient turns out to be 0.903 once again being highly positively significant. Similar to H1, it is less than 0.001, which confers statistical significance. An effect size of 0.816 further strengthens the conclusion that AI implementation had a positively significant impact for BPO employees on work-life balance. In general, both hypotheses

are accepted which point toward the fact that AI adoption by BPO organizations ensures the promotion of job satisfaction while significantly supporting the work-life balance of employees. This further puts forward the importance of workforce management being a critical aspect of AI systems integration, further increasing the possibility of higher employee experience and retention across the industry itself. If the use of AI technologies by the organizations were maximally effective, then they would cultivate a more responsive and satisfied workforce that could drive enhancements of overall operational effectiveness and productivity. Hence, BPO firms are highly likely to benefit from investments and implementations of AI-based solutions as they support operational efficiencies while building a more positive organizational culture and increasing employee well-being.

Conclusions

This study explored the changes brought about by AI-based scheduling systems on job satisfaction and work-life balance for the employees of BPO companies in Quezon City. Using a large-scale survey of employee experience with such AI systems, the present study presents some imperative findings.

The result of this research will find a positive impact of AI scheduling systems regarding the organization of workloads. A big percentage of workers evaluated to agree with the notion that it improved their skill toward organizing and prioritizing work tasks highly correlates to higher job satisfiability. This implies that AI can be an effective tool for the employees in better navigating the roles.

While the employees acknowledged the benefits that might be reaped from AI system in terms of process simplifications, the impact, overall, was found to be only moderate on the workers' productivity. Many of the respondents could acknowledge the efficiencies given by the AI systems, but they could not reap any great level of increase in their levels of productivity. This implies that although AI can answer some tasks, possibly there are extra factors that might influence productivity which are not really addressed by these systems.

It also suggested that AI systems should provide autonomy at the job levels by facilitating employee control over shift times. Without such autonomy in the workplace, it will be challenging to achieve a feeling of ownership and responsibility. However, the apprehensions by employees made a case for flexibility in managing shifts and control. In this context, the current AI-based scheduling systems might require further optimization towards resolving different needs of the employees.

Despite the considerable strengths AI-based scheduling holds in terms of improving workload management and job autonomy, the development potential of this approach still resides primarily in improvement towards productivity and a better work-life balance. Subsequent research as well as the design of the system must focus on highlighting these weaknesses so that AI can be optimized to address them effectively. Meanwhile, the more subtle aspects about the updating frequency, flexibility, fairness, and productivity of AI scheduling systems can solve identified problems. The improvements will therefore positively contribute to the effectiveness of the system as improved, thus simultaneously seeing an increase in the satisfaction of workers and their work-life balance-an essential aspect in having a committed and productive workforce at the workplace. So, integrating employee feedback in the development and optimization of such systems promotes ownership and trust in AI technology as it actually accelerates both personal and organizational growth.

Recommendations

5.5.1. Establish shift flexibility by allowing self-service features for shift-swapping, giving advance notice on shift changes, and permitting employees to request preferred shifts to fit their personal availability. Implementing this course of action may help improve morale, reduce stress, and reduce employee turnover.

5.5.2. Such an AI scheduling system must be regularly updated to ensure optimal performance with minimum errors. There must exist a feedback loop so that employees can send any errors and also introduce machine learning algorithms on the basis of historical data and real-time feedback for continuous improvement of accuracy in scheduling.

5.5.3. Schedule training and professional development during low work periods so that they do not conflict with other aspects of business.

5.5.4. Engage with training providers through in-house workshops and have mentorship programs to facilitate continued skill developments for employees.

5.5.5. The system of AI should ensure equitable fair distribution of loads to employees, especially based on seniority, skills, and personal preferences. The process for shift assignments should be open and communicative and allow the employees access to the same in order not to have biases applied.

5.5.6. Provide time and again training on how best to use the AI system and scheduling maximization. Have a helpdesk for questions with workers and then sharing experiences, which boosts both confidence and satisfaction about the AI tool.

5.5.7. Encourage employees to give their input on the AI tool concerning its smoothness in the workflow, any problems that may arise, and other improvements to be made. Engage the employees with frequent feedback to the AI scheduling system either through surveys, focus groups, or suggestion boxes. This will keep them invested in the developmental process so that they better it and change its character to work better for their needs.

5.5.8. Implement wellness programs within the AI scheduling system, which could include scheduled break times, mental health days, and mechanisms that can prevent overworking. This will ensure more job satisfaction and less burnout along with a healthier workplace.

Directions for Future Research

Given all that has been learned from this study, there are many areas on which future research should focus. Longitudinal impacts or Subjective Experiences of AI-based scheduling systems on employee satisfaction and retention, as well as overall organizational performance, should be investigated to further inform BPO companies about the building of sustainable workforce management strategies. Further, there is the need to conduct more research into how best AI scheduling is being manifested in different industries and how best practices and adaptations can be implemented to enhance job satisfaction and work-life balance in many contexts.

The results bring out the point that work-life balance is still an area that requires improvement. Though AI scheduling offered some flexibility, it failed to accommodate personal activities for employees accordingly, which reflects an area that requires more improvement in scheduling algorithms or processes.

Another promising domain for future research is the hybrid models of work together with AI scheduling systems. While work arrangements are becoming increasingly flexible and companies adopt flexibility to manage work overload in versatile ways, the needs of employees to balance their work and personal lives become a matter of concern. AI can assist both telecommuting and on-site workers to better manage their work commitments while balancing their personal lives. In addition to this, a study of interactions between AI scheduling and leadership dynamics can be useful in discovering how management styles affect employees' engagement with their AI-based work environment.

The last of them relates to the interaction between mechanisms of employee feedback and AI systems for scheduling. Further, implementation of routine feedback loops might also help in the further development of AI-based tools in line with the evolving requirements of employees. Such studies assist in gaining a more profound understanding of potential avenues whereby AI technology could be capitalized to leverage the management of workforces and improve employee welfare further.

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REFERENCES

1. Alambra, D.L., Navidad, V.B. (2024). Job Satisfaction Retention of Employees in Business Process Outsourcing Company. Retrieved from https://www.mijrd.com/papers/v3/i3/MIJRDV3I30013.pdf
2. Ali, O., Shrestha, A., Osmanaj, V., & Muhammed, S. (2020). Cloud computing technology adoption: an evaluation of key factors in local governments. Information Technology and People, 34(2), 666–703. https://doi.org/10.1108/itp-03-2019-0119
3. Bagozzi, R. P., & Yi, Y. (1988). On the evaluation of structural equation models. Journal of the Academy of Marketing Science, 16(1), 74-94.
4. Baisoya, N., & Mohsin, F. (2023). How does Job satisfaction affect Job security in Indian BPO Sector? Materials Today: Proceedings, 80, 3801–3804. https://doi.org/10.1016/j.matpr.2021.07.385
5. Balasolla, M.J. (2024). Philippine BPO Companies Grapple AI Skills and Talent Gap amid Surge Adoption. Unitycommunications:BPO. Retrieved from https://unity-connect.com/our-resources/news/philippine-bpo-companies-grapple-with-ai-skills-and-talent-gap-amid-surge-in-adoption/#:~:text=Challenges%20of%20AI%20adoption
6. Biernacki, P., & Waldorf, D. (1981). Snowball sampling: Problems and techniques of chain referral sampling. Sociological Methods & Research, 10(2), 141-163.
7. Cetindamar, D., & Phaal, R. (2021). Technology management in the age of digital technologies. IEEE Transactions on Engineering Management, 70(7), 2507–2515. https://doi.org/10.1109/tem.2021.3101196
8. Chin, W. W. (1998). The partial least squares approach to structural equation modeling. In G. A. Marcoulides (Ed.), Modern methods for business research (pp. 295-336). Lawrence Erlbaum Associates.
9. Clark, S. C. (2000). Work/family border theory: A new theory of work/family balance. Human Relations, 53(6), 747-770. https://doi.org/10.1177/0018726700536001
10. Cohen, J. (1988). Statistical power analysis for the behavioral sciences (2nd ed.). Lawrence Erlbaum Associates.
11. Creswell, J. W. (2014). Research design: Qualitative, quantitative, and mixed methods approaches (4th ed.). SAGE Publications.
12. Dillman, D. A., Smyth, J. D., & Christian, L. M. (2014). Internet, phone, mail, and mixed-mode surveys: The tailored design method (4th ed.). John Wiley & Sons.
13. Dima, J., Gilbert, M., Dextras-Gauthier, J., & Giraud, L. (2024). The effects of artificial intelligence on human resource activities and the roles of the human resource triad: opportunities and challenges. Frontiers in Psychology, 15. https://doi.org/10.3389/fpsyg.2024.1360401
14. Drapkin, A. (2024, April 24). Does AI Improve Job Satisfaction? Here's What Leaders Say. TalentCulture. https://talentculture.com/blog/does-ai-improve-job-satisfaction-survey-results/
15. Ernst, E., Berg, J., & Moore, P. V. (2024). Editorial: Artificial intelligence and the future of work: humans in control. Frontiers in Artificial Intelligence, 7. https://doi.org/10.3389/frai.2024.1378893
16. Field, A. (2013). Discovering statistics using IBM SPSS statistics (4th ed.). SAGE Publications.
17. Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. Journal of Marketing Research, 18(1), 39-50.
18. Fowler, F. J. (2013). Survey research methods (5th ed.). SAGE Publications.
19. Ganiyu, I. O., Derera, E., Atiku, S. O., & Fields, Z. (2020). Measuring the effectiveness of work–life balance strategies in the manufacturing sector. SA Journal of Human Resource Management, 18. https://doi.org/10.4102/sajhrm.v18i0.1216
20. Geisser, S. (1974). A predictive approach to the random effect model. Biometrika, 61(1), 101-107.
21. Hackman, J. R., & Oldham, G. R. (1975). Development of the Job Diagnostic Survey. Journal of Applied Psychology, 60(2), 159-170. https://doi.org/10.1037/h0076546

22. Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2019). A primer on partial least squares structural equation modeling (PLS-SEM) (2nd ed.). SAGE Publications.
23. Hair, J. F., Ringle, C. M., & Sarstedt, M. (2011). PLS-SEM: Indeed a silver bullet. Journal of Marketing Theory and Practice, 19(2), 139-152.
24. Henseler, J., Hubona, G., & Ray, P. A. (2015). Using PLS path modeling in new technology research: Updated guidelines. Industrial Management & Data Systems, 116(1), 2-20.
25. Henseler, J., Ringle, C. M., & Sinkovics, R. R. (2009). The use of partial least squares path modeling in international marketing. In R. R. Sinkovics & P. N. Ghauri (Eds.), New challenges to international marketing (Advances in International Marketing, Vol. 20, pp. 277-319). Emerald Group Publishing Limited.
26. Huang, J., Saleh, S., & Liu, Y. (2021). A review on Artificial intelligence in education. Academic Journal of Interdisciplinary Studies, 10(3), 206. https://doi.org/10.36941/ajis-2021-0077
27. Israel, G. D. (2015). Determining sample size. University of Florida IFAS Extension.
28. Kline, R. B. (2015). Principles and practice of structural equation modeling (4th ed.). Guilford Press.
29. Kock, N. (2020). WarpPLS user manual: Version 8.0. ScriptWarp Systems.
30. Larson, R. (2020). Work-from-anywhere: The productivity effects of geographic flexibility. Strategic Management Journal, 42(4). https://doi.org/10.1002/smj.3251
31. Latos, B., Buckhorst, A., Kalantar, P., Bentler, D., Gabriel, S., Dumitrescu, R., Minge, M., Steinmann, B., & Guhr, N. (2024). Time autonomy in personnel planning: Requirements and solution approaches in the context of intelligent scheduling from a holistic organizational perspective. Zeitschrift Für Arbeitswissenschaft. https://doi.org/10.1007/s41449-024-00432-7
32. Luiselli, J. K., Gardner, R. M., Bird, F., Maguire, H., & Harper, J. M. (2022). Organizational behavior management in human services settings: Conducting and disseminating research that improves client outcomes, employee performance, and systems development. Journal of Organizational Behavior Management, 42(3), 255–273. https://doi.org/10.1080/01608061.2022.2027319
33. Medina, John Robert, Takeuchi, Rie, Mercado, Chris, et al. (2023). Spatial and temporal distribution of reported dengue cases and hot spot identification in Quezon City, Philippines, 2010-2017. Tropical Medicine and Health. https://www.researchgate.net/publication/371036867_Spatial_and_temporal_distribution_of_reported_dengue_cases_and_hot_spot_identification_in_Quezon_City_Philippines_2010-2017
34. Na, S. R. (2024). Application of artificial intelligence in employee training and development. Deleted Journal, 1(1), 26–28. https://doi.org/10.54097/gg5eemnb
35. Naqbi, H. A., Bahroun, Z., & Ahmed, V. (2024). Enhancing Work Productivity through Generative Artificial Intelligence: A Comprehensive Literature Review. Sustainability, 16(3), 1166. https://doi.org/10.3390/su16031166
36. Nirupama. (2024). The Impact of Artificial Intelligence in the Workplace and its Effect on the Digital Wellbeing of Employees. Journal for Studies in Management and Planning, 10(4), 1-32. https://doi.org/10.5281/zenodo.10936348
37. Nunnally, J. C., & Bernstein, I. H. (1994). Psychometric theory (3rd ed.). McGraw-Hill. Pallant, J. (2016). SPSS survival manual: A step by step guide to data analysis using IBM SPSS (6th ed.). McGraw-Hill Education.
38. Parasuraman, A., Zeithaml, V. A., & Berry, L. L. (1988). SERVQUAL: A multiple-item scale for measuring consumer perceptions of service quality. Journal of Retailing, 64(1), 12-40.
39. Philippine Statistics Authority. (2020). National Capital Region (NCR) in figures.
40. Preacher, K. J., & Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. Behavior Research Methods, 40(3), 879-891.
41. Ranasinghe. (2020). Flexible Working and Work-Life Balance. In Contemporary Developments in Human Resource Management (pp. 97-111). University of Kelaniya, Sri Lanka.
42. Raosoft. (2021). Sample size calculator.

43. Rodríguez-Modroño, P., & López-Igual, P. (2021). Job Quality and Work—Life Balance of Teleworkers. International Journal of Environmental Research and Public Health, 18(6), 3239. https://doi.org/10.3390/ijerph18063239

44. Sarstedt, M., Ringle, C. M., & Hair, J. F. (2017). Partial least squares structural equation modeling. In C. Homburg, M. Klarmann, & A. Vomberg (Eds.), Handbook of market research (pp. 1-40). Springer.

45. Stone, M. (1974). Cross-validatory choice and assessment of statistical predictions. Journal of the Royal Statistical Society: Series B (Methodological), 36(2), 111-133.

46. Stoykova, S., & Shakev, N. (2023). Artificial intelligence for Management Information Systems: Opportunities, challenges, and future directions. Algorithms, 16(8), 357. https://doi.org/10.3390/a16080357

47. THE IMPACT OF WORK ENVIRONMENT ON THE - ProQuest. (n.d.-b). https://www.proquest.com/docview/2478619138

48. Yamamoto, I. (2019). The impact of AI and information technologies on worker stress. CEPR. https://cepr.org/voxeu/columns/impact-ai-and-information-technologies-worker-stress

49. Yogiana, N. G. A., & Riana, N. I. G. (2023). Work life balance mediate: Workplace flexibility on job satisfaction. World Journal of Advanced Research and Reviews, 19(2), 1039–1049. https://doi.org/10.30574/wjarr.2023.19.2.1691

50. Younis, Z., Ibrahim, M., & Azzam, H. (2024). The Impact of Artificial Intelligence on Organisational Behavior: A Risky Tale between Myth and Reality for Sustaining Workforce. European Journal of Sustainable Development, 13(1), 109. https://doi.org/10.14207/ejsd.2024.v13n1p109

51. Zeithaml, V. A., Parasuraman, A., & Berry, L. L. (1996). The behavioral consequences of service quality. Journal of Marketing, 60(2).

52. Zirar, A., Ali, S. I., & Islam, N. (2023). Worker and workplace Artificial Intelligence (AI) coexistence: Emerging themes and research agenda. Technovation, 124, 102747. https://doi.org/10.1016/j.technovation.2023.102747

Oddvertising: Its Influence on the Purchasing Decisions of Consumers in the Philippines

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Abstract

This study examines the impact of Oddvertising—a form of unconventional advertising characterized by humor, creativity, informativeness, and entertainment—on the purchasing decisions of Generation Z consumers in Quezon City, Philippines. The research further explores the roles of irritation and brand recognition as potential moderating and mediating variables, respectively. Utilizing a quantitative methodology, the study gathered data from 320 respondents aged 18–27 and applied linear regression analysis to identify key determinants of purchasing behavior. Results reveal that creativity and entertainment are the most influential characteristics of Oddvertising, significantly shaping consumer decisions. Humor and informativeness positively affect ad memorability and clarity, while, though inherently negative, irritation showed minimal impact on purchasing behavior. Findings confirm that Oddvertising substantially influences Generation Z consumers' purchasing decisions, highlighting its effectiveness as a marketing strategy.

Keywords: Oddvertising, Generation Z, Purchasing Decisions, Brand Recognition, Unconventional Advertising

Introduction

In today's hyper-connected world, consumers were constantly bombarded with advertisements across social media, mobile apps, and websites, making it increasingly difficult for traditional advertising to capture attention. Oddvertising, characterized by its unconventional, humorous, and often bizarre elements, emerged as a compelling approach that disrupted the predictable nature of advertisements, sparking conversations and engagement (Selan, 2021). This trend was particularly impactful among Generation Z, a demographic known for its preference for creativity, boldness, and authenticity. Drawn to unique and entertaining content on social media, Gen Z consumers were more likely to respond to ads that broke traditional norms, valuing surprise and originality over predictability (Kahawandala et al., 2020; Blaire, 2023).

In the Philippine advertising landscape, Oddvertising became a hallmark of creativity, with campaigns like RC Cola's "Nyahahakbkxjbcjhishdishlsab@!!!! Basta RC Cola!" and Grab's "Init na init na mhie?" capturing public attention. These quirky ads resonated deeply with Gen Z audiences, who appreciated their humor and uniqueness. However, despite its growing popularity, little was known about how Oddvertising influenced consumer behavior, particularly purchasing decisions, within the local market (Losa, 2020). With businesses competing in an oversaturated space, the ability of Oddvertising to stand out and create memorable brand associations became critical. Creativity, entertainment, and informativeness were central to this strategy, while potential drawbacks like irritation warranted careful examination (Yrastorza & Abesamis, 2021; Blair, 2023).

By analyzing Oddvertising's impact on Generation Z consumers in the Philippines, the research provided insights into how unconventional advertisements influenced purchase decisions. It explored how elements such as humor and creativity engaged audiences, how brand recognition mediated consumer responses, and whether irritation diminished effectiveness. The findings highlighted the potential of Oddvertising to reshape advertising strategies for a demographic that values novelty and authenticity.

Literature Review

Unconventional Advertising

Numerous businesses in the Philippines have leveraged creative advertising opportunities on social media and other platforms, exemplified by campaigns like "Basta RC Cola." Losa (2020) noted that unconventional advertisements were already common

before the rise of RC Cola's video ad. This trend raises the question of how such unconventional approaches influence advertising effectiveness in the current era.

Advertisements typically inform viewers about the benefits of a product, service, or brand. However, adding unexpected elements can leave viewers intrigued. These advertisements provide value through dimensions such as entertainment, informativeness, and even irritation, which directly and indirectly affect consumer purchase decisions (Disastra et al., 2019). The ongoing challenge in advertising is to make it more impactful, closer to the consumer, and identify the essential elements for persuasion (Csapó-Horváth, 2021). Marasigan et al. (2023) indicated that advertising is the most effective communication method for expanding market reach and brand recognition. The usefulness and features of advertisements significantly influence consumer behavior by positively impacting purchasing decisions. Consumers' demands for products vary based on preferences, tastes, and third-party influences (Jan et al., 2019). Unconventional advertising can capture consumers' attention and encourage them to engage with the message, making it a crucial strategy in modern marketing (Murwonugroho & Yudarwati, 2020). According to Othman (2020), unconventional advertising is particularly effective in attracting and delivering messages to target audiences due to its emphasis on surprise, humor, creativity, and engagement. Although it can yield both positive and negative outcomes, such as enhancing creativity and reducing competence, its impact on subjective well-being and product creativity varies (Kim et al., 2023).

Othman (2020) also highlighted that the younger generation desires new, unique, and instant experiences. In the digital era, companies increasingly use social media marketing, including memes, to reach Generation Z, leveraging their unique consumption habits and transforming cultural expressions into marketing strategies. This generation finds satisfaction in humorous content that provides entertainment, social interaction, and information (Arie Bowo et al., 2024). Therefore, innovative ad strategies are necessary to attract target audiences effectively.

However, if unconventional advertisements become too frequent, their influence may diminish over time. Oddvertising and unconventional advertising are essentially synonymous, with the former being a newly coined term and the latter a recognized concept. By maintaining a balance of creativity and engagement, businesses can continue to capture the attention of digitally-savvy audiences.

Purchase Decision

In marketing, consumer purchase decisions are influenced by various factors that have evolved over time. Traditional factors like quality and price remain significant, while new factors such as sustainability and online experiences have become increasingly important (Antosova et al., 2023). Generation Z, known for their technological fluency and social media engagement, are particularly influenced by these modern factors (Kahawandala et al., 2020). Sjahruddin and Adif (2024) suggested that ethical and innovative brands significantly impact Gen Z's purchasing behaviors due to their preferences for social consciousness and innovation.

The relationship between purchase intent and purchase decision is critical for marketers. Tirtayani et al. (2024) found a positive relationship between social media advertising and consumer purchase intention. Advertising value dimensions, such as informativeness and entertainment, positively influence emotions towards ads, while irritation has a negative effect (Disastra et al., 2019). These emotional responses directly impact purchase intent and decisions. Generation Z's purchase intentions are driven by factors like lower prices, uniqueness, nostalgia, and brand trust (Kawulur et al., 2022). Emotions triggered by advertising play a crucial role, with positive emotions enhancing and negative emotions deterring purchase decisions. Understanding these factors allows marketers to tailor their strategies to effectively influence consumer behavior and drive sales.

Generation Z

Generation Z, or Zoomers, are highly engaged with digital technology and social media, shaping their beliefs and behaviors (Agustina & Fauzia, 2021). Key factors influencing their thinking processes include cognition, norms, personal interests, and risk propensity.

Cognition involves processing societal information, impacting decision-making and brand perception (Faghani et al., 2024; Romanova & Smirnova, 2019). Social media plays a crucial role in their lifestyle and behavior (Bhattacharya, 2024). Norms, accepted by cultural groups, influence their decisions, often swayed by influencers and product features (Kumar & Campbell, 2022; Rini, 2023). Personal interests are driven by influencer marketing and CSR activities, impacting engagement and satisfaction. Risk propensity, influenced by personal traits and external factors, affects their purchase decisions (Agustina & Fauzia, 2021). Understanding these factors helps businesses tailor products and services to meet their expectations (Pietroń-Pyszczek & Borowska, 2022). Effective marketing strategies must integrate these elements to connect with Generation Z successfully.

Engaging Generation Z requires brands to consider their openness to risk-taking, personal interests, adherence to societal norms, and cognitive patterns. Successfully integrating these elements into marketing strategies can establish a strong connection with this generation.

Short-form Video Advertising

Short-form video advertisements have become prevalent on social media, offering easily consumed and shared content (Maemhout, 2022). Their mobile-friendly nature facilitates information dissemination, creating an electronic word-of-mouth cycle (Zhai et al., 2022). Generation Z engages significantly with short-form advertising, impacting their purchasing decisions (Xiao et al., 2023). Platforms like TikTok, with videos averaging 34-50 seconds, have influenced Instagram Reels and YouTube Shorts.

Attention marketing, which aims to capture audience attention with engaging content, plays a crucial role in this landscape (Yin et al., 2024). Increased attention to commercials enhances brand recall and predisposes consumers to interact with recognized brands (Shen et al., 2021). Companies often use humorous fail videos to advertise, drawing attention and creating memorable impressions. This approach effectively engages consumers and enhances brand recall.

Brand Recognition

Brand recognition is crucial for Generation Z, as it influences their purchasing behaviors through familiarity and memorable advertising (Marasigan et al., 2023). Popular brands often create a bandwagon effect, impacting Gen Z's choices (Cho et al., 2021). The desire for uniqueness also drives creative consumer decisions.

Factors like subjective norms and peer influence play significant roles in brand recognition, especially regarding ethics and sustainability. Gen Z values ethical considerations, aligning their purchases with societal and personal values. Brand awareness, a key precursor to brand recognition, informs the public about available products (Kenton, 2024). Effective brand recognition influences consumer behavior, particularly in social media marketing and purchase intentions. Consumers favor brands they can easily distinguish among competitors (Saputra & Wardana, 2023). Brand image reflects a company's reputation and contributions, encompassing consumer perception, brand identity, attitudes, and values (Hidayat et al., 2023). A positive brand image, coupled with strategic promotions, shapes consumer decisions and behaviors, often leading to high-potential purchases. Brand attitude significantly impacts purchasing decisions and intentions (Machi et al., 2022).

Low brand recognition can negatively affect customer loyalty, sales, and financial performance (Vestil, 2024). Therefore, maintaining strong brand recognition is essential for influencing Generation Z's purchasing behavior.

Humor, Creativity, Entertainment, and Informativeness

Humor in advertising, defined as content intended to be funny, attracts audiences and creates a positive environment for delivering messages (Kaur et al., 2022). Studies have shown that humorous ads generate more interest and appreciation for brands, influencing purchase decisions (Palikhe, 2019; Waqar, 2020). Humor is crucial for audience engagement, providing entertainment and escapism, especially for Gen Z (Palikhe, 2019).

Similarly, creativity in advertising, particularly through unconventional methods, uses unique approaches to convey messages (Van Berlo et al., 2023). Creative advertising integrates positive emotions, novelty, trend, divergence, and humor to impact consumer behavior (West et al., 2019; Smith et al., 2022; Enam & Rahman, 2020). Although creativity's influence on purchasing decisions can vary, it remains a key component of effective advertising (Bensalem & Douli, 2020). In addition to humor and creativity, entertainment in advertising creates enjoyment and attractiveness, crucial for engaging consumers like Gen Z. It positively influences attitudes towards ads and purchase intent through emotional enjoyment and satisfaction (Disastra et al., 2019; Tanuwijaya & Gunawan, 2021). Narrative advertising and short-form videos effectively capture Gen Z's attention and deliver impactful messages (Tabassum et al., 2020; Moreno et al., 2023).

Furthermore, informativeness in advertising provides essential information about products, influencing purchase intent (Tanuwijaya & Gunawan, 2021). Clear, concise, and relevant information enhances the value of ads, making them more appealing to consumers. Credibility and significance are crucial factors in informative advertising (Santanagopalan & Raj, 2023). Engaging elements like music, pictures, animations, or humor further boost positive attitudes towards ads (Nainggolan, 2021). Thus, humor, creativity, entertainment, and informativeness, despite their distinct functions, all contribute significantly to the effectiveness of advertising strategies.

Irritation

Unconventional advertisements do not always yield positive outcomes, and some approaches can negatively influence consumers, particularly Generation Z. Irritation is a notable negative factor impacting Gen Z's purchase intent. While entertainment elicits positive emotions, irritation triggers negative feelings when ads cause displeasure (Nainggolan, 2021). High levels of irritation lead to negative consumer responses and decreased purchase intent (Disastra et al., 2019).

To mitigate irritation, ads should be highly informative, providing relevance to consumer needs (Disastra et al., 2019). Tsao (2019) identified other irritation factors, including lack of information, interruptions, inconvenience, dissatisfaction, and negative emotions. Informative, entertaining, and relevant ads can alleviate irritation, making consumers more likely to respond favorably. Understanding and addressing these factors can help create effective advertisements that resonate with Generation Z.

S-O-R Framework

Theoretical Framework The study utilized the S-O-R Framework, created by Mehrabian and Russell in 1974, to examine the impact of external stimuli on internal states and behaviors. Mehrabian and Russell's 1974 S-O-R Framework was foundational in environmental psychology, linking external stimuli to internal state and behavior. It clarified how stimuli influenced thoughts (organism) and reactions or responses (Apostol et al., 2023; Zhou, 2023). Ling et al. (2024) applied this framework to study interpersonal interaction (Stimuli), psychological distance (organization), and consumer purchase decision (response)

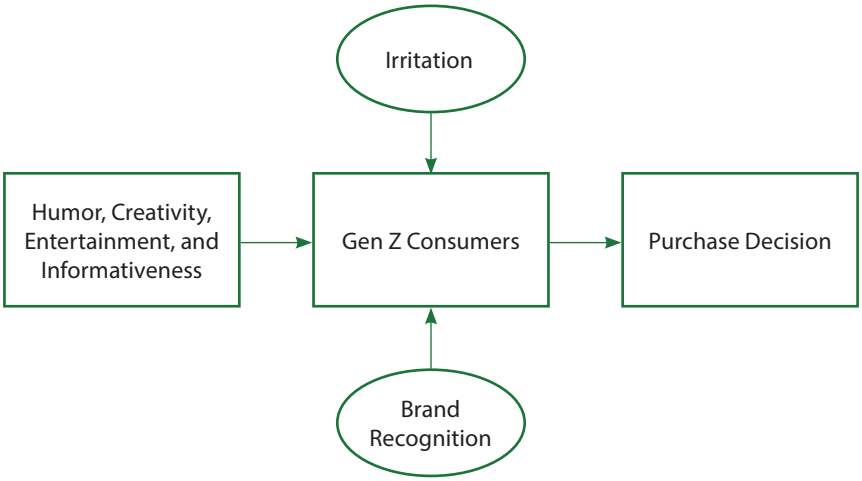


Fig. 1. Stimulus-Organism-Response Framework (Mehrabian & Russel, 1974)

The model highlights how elements like humor, creativity, entertainment, and informativeness act as stimuli, influencing the internal perceptions (organism) of Gen Z consumers and ultimately driving their purchasing behavior (response). Creativity and entertainment emerged as the most influential characteristics, corroborating Enam and Rahman's (2020) findings that innovative and engaging ads capture consumer attention and enhance decision-making.

In figure 1, Oddvertising characteristics are positioned as independent variables, with brand recognition and irritation as mediating variables, and purchasing decisions as the dependent variable. Brand recognition significantly amplified the positive effects of Oddvertising, aligning with Marasigan et al. (2023), while irritation had a minimal impact, supporting Nainggolan's (2021) view that informative and engaging content can mitigate negative emotional responses. This framework effectively elucidates the interplay between unconventional advertising and Gen Z's purchasing behavior, offering a structured insight into Oddvertising's impact within the Philippine market.

Methodology

This study employed a quantitative research design to investigate the relationship between Oddvertising characteristics and Generation Z consumers' purchasing decisions in Quezon City, Philippines. The respondents for the formulated survey were based on the criterion of age fitting the range. A correlational approach was used to analyze variables such as humor, creativity, entertainment, informativeness, and irritation on consumer behavior. This method was selected for its ability to identify relationships between independent variables (Oddvertising traits) and the dependent variable (purchasing decisions).

Structured survey questionnaires were the primary data collection instruments, validated through a pilot test involving 26 respondents, representing 7% of the sample size. Reliability testing using Cronbach's Alpha showed strong internal consistency, with coefficients exceeding 0.8, indicating the instruments' validity. Peer reviews by experts and academic advisers ensured clarity and relevance. The study targeted a sample of 320 respondents, aged 18–27, selected using Cochran's formula and simple random sampling to ensure statistical reliability. Simple random sampling was employed to minimize bias and provide equal participation opportunities. Data collection involved online and offline methods, including QR-coded links and printed copies distributed in public spaces. Respondents evaluated the influence of Oddvertising characteristics, including their effects on brand recognition and purchasing decisions.

The collected data were analyzed using linear regression to examine relationships between variables. The analysis identified the strength and direction of relationships, with coefficients of determination quantifying the model's explanatory capability. Ethical considerations, including informed consent, anonymity, and adherence to the Data Privacy Act of 2012, were observed throughout the study. The methodology provided a rigorous framework for evaluating the influence of Oddvertising on consumer behavior, offering reliable and meaningful insights into Generation Z's purchasing decisions.

Analysis and Discussions

Discussion of Findings

Table 1 showed the demographic distribution of 320 Generation Z respondents in Quezon City by age and barangay. The largest group was 21-year-olds (22.5%), followed by 18-year-olds (20.6%), 20-year-olds (17.5%), and 19-year-olds (15%). Most respondents were from District 5 (34.4%), District 2 (30%), and District 6 (18.6%).

These findings reflected Generation Z's characteristics, a demographic highly influenced by Oddvertising. This tech-savvy, digitally native cohort extensively used social media platforms, making them receptive to Oddvertising's creative and engaging techniques. This supported existing literature, such as Arie Bowo et al. (2024), which emphasized Generation Z's curiosity-driven behavior when exposed to unconventional advertising strategies.

Table 1. *Profile of Respondents based on Age, and Respective Districts*

Age	Frequency	Percent
18 years old	66	20.6%
19 years old	48	15%
20 years old	56	17.5%
21 years old	72	22.5%
22 years old	29	9.1%
23 years old	26	8.1%
24 years old	9	2.8%
25 years old	4	1.2%
26 years old	5	1.6%
27 years old	5	1.6%
Total	**320**	**100%**

Districts		
Districts 1	22	6.9%
Districts 2	96	30%
Districts 3	15	4.7%
Districts 4	18	5.6%
Districts 5	110	34.4%
Districts 6	59	18.4%
Total	**320**	**100%**

Table 2 presented the respondents' perceptions of Oddvertising characteristics, with humor, creativity, and entertainment receiving the most positive ratings. Humor emerged as a key factor, as most respondents reported that it positively influenced purchasing behavior by capturing attention and providing satisfaction, often in combination with the entertainment aspect of Oddvertising (Bowo et al., 2024). However, some respondents viewed Oddvertising as unfunny or negatively impactful due to elements such as distractions, graphic content, or unappealing video and sound effects (Eisend, 2021).

Creativity was rated highly, emphasizing its trendiness, divergence, and novelty—key elements that made Oddvertising distinctive and appealing (Enam & Rahman, 2020). The unique and innovative features of creativity had the strongest influence on consumer behavior among all characteristics evaluated. Informativeness and entertainment also received positive ratings, with entertainment providing emotional enjoyment and satisfaction that encouraged purchasing behavior. Respondents indicated that entertainment did not have any negative effects, as its primary role was to enhance emotional engagement and viewer satisfaction, which in turn positively influenced attitudes toward advertisements (Tanuwijaya & Gunawan, 2021).

These findings supported existing research that highlighted the effectiveness of unconventional advertising techniques in evoking positive emotions, such as humor and excitement, to drive purchasing decisions (Bowo et al., 2024; Tanuwijaya & Gunawan, 2021). The results indicated that Generation Z preferred advertisements that deviated from conventional approaches, effectively blending entertainment with informativeness to appeal to their preferences and behaviors.

Table 2. Characteristics of an Oddvertisement

Characteristics of an Oddvertisement	1	2	3	4
I find that these advertisements make me laugh.	15 (5%)	46 (14%)	139 (42%)	130 (39%)
I find that these advertisements to be engaging.	12 (4%)	47 (14%)	133 (40%)	138 (42%)
I find that these advertisements to be unique.	7 (2%)	22 (7%)	97 (29%)	204 (62%)
I find that these advertisements to be appealing.	12 (4%)	64 (19%)	142 (43%)	112 (34%)
I find that these advertisements provide relevant information.	19 (6%)	69 (21%)	149 (45%)	93 (28%)
I find that these advertisements provide a clear understanding of the product or service.	13 (4%)	50 (15%)	147 (45%)	120 (36%)
I find these advertisements provide fulfillment.	14 (4%)	89 (27%)	131 (40%)	96 (29%)
I find these advertisements are fun to watch.	20 (6%)	32 (10%)	97 (29%)	181 (55%)

1 - Strongly Disagree; 2 - Disagree; 3 - Agree; 4 - Strongly Agree

Table 3 indicated that irritation was not a significant factor for most respondents, with 71% reporting low levels of irritation. However, 42% of respondents found specific elements, like unclear messaging or unnecessary distractions, annoying. Despite these issues, the positive characteristics of Oddvertisements—particularly humor and creativity—generally outweighed the Potential for irritation.

The results aligned with related literature suggesting that irritation can negatively influence viewers' perceptions due to displeasure, insufficient information, and distractions (Nainggolan, 2021). However, most respondents were not significantly bothered by unconventional ad approaches, indicating these methods were well-received. These findings contradicted earlier studies that found irritation reduced viewer satisfaction and negatively influenced behavioral intentions (Tsao, 2019). The relationship between irritation and informativeness revealed that an appropriate level of information could mitigate irritation, as noted in prior research (Disastra et al., 2019). However, even informative Oddvertisements could provoke irritation due to unclear messaging.

Irritation also impacted humor, particularly when a lack of clarity undermined the intended comedic effect, consistent with earlier findings (Tsao, 2019). Nevertheless, the overall results showed that the unconventional and engaging nature of Oddvertisements, characterized by humor and creativity, was effective in capturing audience attention, despite occasional irritation.

Table 3. Irritation and Characteristics of Oddvertisements

Influence of Irritation and Influence of the Characteristics of Oddvertising	1	2	3	4
These types of advertisements are humorous but lack clarity of message.	35 (11%)	113 (34%)	136 (41%)	46 (l 4%)
These types of advertisements are creative but lack the necessary information for the featured product or service.	43 (l3%)	118 (36%)	118 (36%)	45 (14%)
These types of advertisements show informativeness but have some distracting and unnecessary elements.	40 (12%)	109 (33%)	133 (40%)	48 (l5%)
These types of advertisements are entertaining, but some elements annoy me.	82 (25%)	108 (33%)	100 (30%)	40 (12%)
Median of Irritation Factors	**98 (30%)**	**136 (41%)**	**71 (22%)**	**25 (8%)**

1 – Strongly Disagree; 2 – Disagree; 3 – Agree; 4 – Strongly Agree

Table 4 demonstrated that irritation had a slight but insignificant influence on brand recognition. Despite the presence of irritating elements, 44% of respondents reported being aware of the brand. The findings suggested that strong brand recognition could mitigate the negative effects of irritation, supporting the notion that even when Oddvertisements contained bothersome aspects, brand awareness could buffer against adverse consumer reactions (Kenton, 2024).

The data further indicated that while irritation was present, it did not substantially harm brand recognition for most respondents. This contradicted claims that irritation negatively affects brand recognition, leading to reduced purchasing decisions, diminished customer loyalty, and potential harm to sales and brand reputation (Vestil, 2024). Instead, the study underscored the resilience of brand awareness in maintaining consumer interest, even when faced with minor irritants in advertising.

Table 4. Irritation and Brand Recognition

Influence of Irritation and Influence of the Characteristics of Oddvertising	1	2	3	4
I am familiar with the brand's image, but the advertisements lacks creativity quality or content.	91 (28%)	125 (38%)	76 (23%)	38 (12%)
I am aware of the brand, but these advertisements lack the necessary information regarding featured products or service.	50 (15%)	136 (41%)	102 (31%)	42 (13%)
I am loyal to the brand, but these advertisements lack clarity of message.	74 (22%)	140 (42%)	84 (25%)	32 (10%)
Median of Brand Recognilion Factors & Irritation	**59 (180/0)**	**144 (44%)**	**101 (31%)**	**26 (8%)**
Median of Irritation Factors	**98 (30%)**	**136 (41%)**	**71 (22%)**	**25 (8%)**

1 – Strongly Disagree; 2 – Disagree; 3 – Agree; 4 – Strongly Agree

Table 5 demonstrated that advertisements lacking creative quality or content exhibited a statistically significant positive effect on the dependent variable ($p < .001$). This indicated that as creativity decreased, negative brand attitudes or reduced purchasing decisions increased. Conversely, the lack of product or service information ($p = 0.361$) and unclear messaging ($p = 0.143$)—both components of irritation—did not have statistically significant effects. These results underscored that creative quality was the primary determinant of consumer responses to advertisements. The findings aligned with Bowo et al. (2024), who argued that high creativity in advertisements enhanced engagement and satisfaction, thereby reducing consumer irritation. Enam and Rahman (2020) further highlighted creativity as crucial for capturing attention, mitigating irritation, and fostering brand loyalty.

The model fit statistics revealed an R value of 0.592, indicating a moderate positive linear relationship between the predictors and the dependent variable. The R-squared (R^2) value of 0.351 suggested that 35.1% of the variance in the dependent variable could be explained by the independent variables, with a substantial proportion remaining unaccounted for.

Table 5. Test of Linear Regression Between Oddvertising Characteristics and Irritation

Model Coefficients–COD1				
Predictor	**Estimate**	**SE**	**t**	**p**
Intercept	1.152	0.113	10.236	<0.001
I am familiar with the brand's image, but the advertisement lacks creative quality and content.	0.495	0.056	8.836	<0.001
I am aware of the brand, but these advertisements lack the necessary information regarding featured products or service.	0.50	0.055	0.915	0.361
I am loyal to the brand, but these advertisements lack clarity of message.	0.076	0.052	1.467	0.143
***COD:** *Coefficient of Determination*				
Model Fit Measures				
Model	**R**	**R^2**		
1	**0.592**	**0.351**		

***R:** This represents the multiple correlation coefficient, essentially a measure of the linear association between your predictors and the dependent variable. *R^2: This is the coefficient of determination, indicating the proportion of variance in the dependent variable that is predictable from the independent variables.*

Table 6 presented an analysis of how perceived advertising flaws—such as a lack of creativity, clarity, or information—impacted the dependent variable. Advertisements perceived as lacking creativity significantly influenced negative consumer outcomes ($p < .001$), including unfavorable brand attitudes. The lack of product or service information exhibited a statistically significant but weaker relationship with the dependent variable ($p = 0.057$), whereas unclear messaging showed no statistically significant effect ($p = 0.197$). These findings indicated that creativity played a more critical role than informativeness or clarity in engaging Generation Z consumers, who favored unconventional and innovative advertisements (Kenton, 2024; Tanuwijaya & Gunawan, 2021).

The model's R value of 0.531 reflected a moderate positive linear relationship between the predictors and the dependent variable. The R-squared (R^2) value of 0.282 indicated that only 28.2% of the variance was explained by the predictors, suggesting that additional factors might influence consumer responses.

Table 6. Test of Linear Regression Between Oddvertising Characteristics and Brand Recognition

Model Coefficients–COD2				
Predictor	**Estimate**	**SE**	**t**	**p**
Intercept	1.152	0.113	10.236	<0.001
I am familiar with the brand's image, but the advertisement lacks creative quality and content.	0.495	0.056	8.836	<0.001
I am aware of the brand, but these advertisements lack the necessary information regarding featured products or service.	0.50	0.055	0.915	0.361
I am loyal to the brand, but these advertisements lack clarity of message.	0.076	0.052	1.467	0.143
****COD:** *Coefficient of Determination*				
Model Fit Measures				
Model	**R**	**R^2**		
1	**0.628**	**0.394**		

***R:** This represents the multiple correlation coefficient, essentially a measure of the linear association between your predictors and the dependent variable. *R^2: This is the coefficient of determination, indicating the proportion of variance in the dependent variable that is predictable from the independent variables.*

Table 7 analyzed the effects of irritation stemming from unclear messaging, lack of information, and reduced creative quality on brand recognition. Unclear messaging exhibited the strongest statistically significant relationship with negative consumer outcomes ($p < .001$), followed by a lack of product or service information ($p = 0.025$) and insufficient creative quality ($p = 0.037$). These findings suggested that clarity and informativeness in advertisements were more influential in driving positive brand recognition than creativity. The findings aligned with Bowo et al. (2024) and Tanuwijaya and Gunawan (2021), who emphasized the importance of clear, informative content for capturing consumer interest and minimizing frustration.

The model fit measures yielded an R value of 0.628 and an R-squared (R^2) value of 0.394, indicating that 39.4% of the variance in the dependent variable was explained by the predictors.

Table 7. Test of Linear Regression on the Influence of Irritation on Brand Recognition

Model Coefficients–COD3				
Predictor	**Estimate**	**SE**	**t**	**p**
Intercept	1.152	0.113	10.236	<0.001
I am familiar with the brand's image, but the advertisement lacks creative quality and content.	0.495	0.056	8.836	<0.001
I am aware of the brand, but these advertisements lack the necessary information regarding featured products or service.	0.50	0.055	0.915	0.361
I am loyal to the brand, but these advertisements lack clarity of message.	0.076	0.052	1.467	0.143
****COD:** *Coefficient of Determination*				
Model Fit Measures				
Model	**R**	**R^2**		
1	**0.628**	**0.394**		

***R:** This represents the multiple correlation coefficient, essentially a measure of the linear association between your predictors and the dependent variable. *R^2: This is the coefficient of determination, indicating the proportion of variance in the dependent variable that is predictable from the independent variables.*

Table 8 explored how advertising flaws—such as unclear messaging, lack of information, and poor creativity—affected brand recognition. Unclear messaging had the most significant impact on negative consumer outcomes ($p < .001$), followed by insufficient product information ($p = 0.013$) and lack of creative quality ($p = 0.025$). These findings suggested that informativeness, particularly clarity, played a more critical role in mitigating adverse consumer responses than creative elements.

The model's R value of 0.687 and R-squared (R^2) value of 0.471 indicated a moderately strong positive linear relationship, with 47.1% of the variance in the dependent variable explained by the predictors.

Table 8. Influence of Advertising Flaws on Negative Brand Recognition

Model Coefficients–COD4				
Predictor	**Estimate**	**SE**	**t**	**p**
Intercept	0.826	0.125	6.63	<0.001
I am familiar with the brand's image, but the advertisement lacks creative quality and content.	0.14	0.062	2.26	0.025
I am aware of the brand, but these advertisements lack the necessary information regarding featured products or service.	0.153	0.061	2.51	0.013
I am loyal to the brand, but these advertisements lack clarity of message.	0.42	0.058	7.29	<0.001
****COD:** *Coefficient of Determination*				
Model Fit Measures				
Model	**R**	**R^2**		
1	**0.687**	**0.471**		

***R:** *This represents the multiple correlation coefficient, essentially a measure of the linear association between your predictors and the dependent variable.* * ***R^2**: This is the coefficient of determination, indicating the proportion of variance in the dependent variable that is predictable from the independent variables.*

Table 9 examined how irritation affected brand recognition and consumer behavior. A statistically significant positive correlation ($p < .001$) was observed between irritation and negative consumer outcomes, such as unfavorable brand attitudes, reduced purchase intention, and ad avoidance. As irritation increased due to unclear or disruptive ad elements, negative responses also rose. These findings supported prior research by Nainggolan (2021) and Disastra et al. (2019), which demonstrated that irritation diminished engagement and brand loyalty. The results highlighted the importance of balancing creativity with clarity and informativeness to minimize irritation and improve ad effectiveness.

The model fit measures showed an R value of 0.628, indicating a moderate positive linear relationship between the predictors and the dependent variable. The R-squared (R2) value of 0.394 suggested that 39.4% of the variance was explained by the predictors, leaving a substantial portion unexplained.

Table 9. Test of Linear Regression on the Influence of Irritation on Brand Recognition and Consumer Behavior

Model Coefficients – Characteristics of Oddvertisment				
Predictor	**Estimate**	**SE**	**t**	**p**
Intercept	1.174	0.082	14.300	<0.001
Irritation	0.622	0.036	17.100	<0.001

Discussion of Hypotheses

H1: There is a significant influence between Oddvertisements and the purchasing decisions of Gen Z consumers. The study's results substantiated Hypothesis 1, demonstrating a significant influence of Oddvertising characteristics on Gen Z's purchasing decisions. Quantitative data and regression analyses revealed how creativity, entertainment, humor, informativeness, and irritation impacted consumer behavior. Table 2 highlighted that the majority of respondents rated Oddvertising characteristics positively, with creativity and entertainment emerging as the most influential. A substantial 62% of respondents strongly agreed that Oddvertisements were unique, engaging, and visually appealing, affirming creativity's role in capturing attention and fostering a connection. This finding aligned with Enam and Rahman's (2020) study, which highlighted creativity as a driver of consumer engagement and purchase behavior. Similarly, over 55% of respondents found Oddvertisements

fun to watch, reflecting the importance of emotional enjoyment in shaping purchasing decisions, as noted by Tanuwijaya and Gunawan (2021). These findings confirmed that creativity and entertainment were the strongest drivers of purchase decisions among Gen Z, effectively combining visual and emotional appeal.

Table 3 indicated that irritation had minimal impact on Gen Z's purchasing decisions. While 41% of respondents acknowledged irritation due to unclear messaging or distractions, 71% reported that it did not significantly detract from the effectiveness of Oddvertisements. Mitigating factors such as informativeness and humor reduced irritation by providing clarity and satisfaction, aligning with Nainggolan's (2021) assertion that engaging and informative content can alleviate irritation. Thus, the study supported the idea that well-designed Oddvertisements balanced humor and informativeness to minimize irritation and maintain their positive influence. Table 4 underscored the critical role of brand recognition in mediating the relationship between Oddvertising characteristics and purchasing decisions. Despite some irritation, 44% of respondents acknowledged strong brand recognition, which mitigated negative effects and enhanced purchase intent. Memorable advertisements boosted trust and loyalty, as suggested by Marasigan et al. (2023), while respondents demonstrated a higher inclination to purchase from brands they could easily recognize and recall, highlighting the cultural significance of brand recognition in decision-making. The regression analysis in Table 5 further confirmed the significant influence of Oddvertising characteristics. An R value of 0.6 indicated a moderate positive correlation between Oddvertising and purchasing decisions, while an R2 value of 0.35 demonstrated that 35% of the variance in purchase decisions was explained by these characteristics. Creativity was identified as the most influential factor, while irritation had the least impact. These results aligned with Bowo et al. (2024), who emphasized the importance of creative and engaging advertisements in driving consumer behavior.

In conclusion, the findings substantiated Hypothesis 1, confirming that Oddvertising characteristics significantly influenced Gen Z's purchasing decisions. Creativity and entertainment emerged as the most impactful elements, while irritation played a minimal role. The mediating role of brand recognition further highlighted the importance of memorable and engaging advertisements in shaping consumer behavior. These insights aligned with prior studies (Enam & Rahman, 2020; Tanuwijaya & Gunawan, 2021) and underscored the effectiveness of unconventional advertising strategies in engaging digitally-savvy audiences like Gen Z.

H2: There is no significant influence between Oddvertisements and the purchasing decisions of Gen Z consumers. The study's findings directly contradicted Hypothesis 2, as the results demonstrated a measurable influence of Oddvertising characteristics on Gen Z's purchasing decisions. Table 2 indicated that the majority of respondents rated humor, creativity, entertainment, and informativeness positively, with creativity and entertainment emerging as the most influential factors. Specifically, 62% of respondents strongly agreed that Oddvertisements were unique and visually appealing, suggesting that these characteristics effectively captured their attention and influenced their purchasing behavior. These findings align with Enam and Rahman (2020), who emphasized creativity and novelty as significant factors influencing consumer engagement

Table 3 further illustrated that irritation, while acknowledged by some respondents, had minimal overall impact on purchasing decisions. Although 41% noted irritation due to unclear messaging or distractions, 71% reported that it did not significantly diminish the appeal or effectiveness of Oddvertisements. This suggests that other characteristics, such as informativeness and entertainment, counteracted irritation, maintaining the positive influence of the advertisements. These results echo Nainggolan's (2021) assertion that engaging and informative content can alleviate irritation and enhance the effectiveness of advertisements. Brand recognition, as shown in Table 4, played a key role in mitigating the effects of irritation and reinforcing the impact of Oddvertisements. Despite some reports of irritation, 44% of respondents indicated strong brand recognition, which enhanced purchase intent. This finding supports Marasigan et al. (2023), who highlighted the role of brand recognition in fostering trust and loyalty, even when other factors, such as irritation, are present.

The regression analysis in Table 5 provided further support, with an R value of 0.6 showing a moderate positive correlation between Oddvertising characteristics and purchasing decisions. An R2 value of 0.35 indicated that 35% of the variance in purchasing decisions was explained by these characteristics. Creativity emerged as the strongest predictor, while irritation showed the least impact on consumer responses. This supports the findings of Bowo et al. (2024), who emphasized the importance of creativity and engagement in driving consumer behavior

In conclusion, the results of the study refuted Hypothesis 2, demonstrating that Oddvertising characteristics significantly influenced the purchasing decisions of Gen Z consumers. The data highlighted creativity and entertainment as the most impactful factors, while irritation played a minor role. Additionally, brand recognition mitigated the potential negative effects of irritation, reinforcing the influence of Oddvertisements. These findings reflect the measurable impact of unconventional advertising strategies on consumer behavior within the scope of the study.

Discussion of Theory

The study's findings aligned with the Stimulus-Organism-Response (SOR) framework, where Oddvertising characteristics (stimulus) influenced Generation Z consumers' perceptions (organism), leading to their purchasing decisions (response).

Creativity and entertainment were identified as the most impactful stimuli, effectively engaging respondents and evoking positive emotional responses, as seen in Table 2. These results supported Mehrabian and Russell's (1974) assertion that external stimuli influence internal states, shaping behavior. This alignment was further reinforced by prior studies that emphasized creativity and humor as drivers of emotional engagement and attention, which, in turn, influenced purchase behavior (Enam & Rahman, 2020; Bowo et al., 2024). Moreover, the minimal impact of irritation, as shown in Table 3, highlighted the importance of balancing stimuli to evoke positive organism responses, consistent with Nainggolan's (2021) findings on the role of informativeness and entertainment in reducing negative emotional reactions. These results confirmed that the SOR framework effectively explained how unconventional advertising impacted Gen Z's decision-making process.

iscusison of Validity of Results

Although the target sample size was 380 respondents, the study gathered 320 participants, slightly below the ideal range. However, the sample size remained valid based on the Cronbach Alpha Charter (1999), which stated that while 400 was ideal for precise reliability, samples over 300 were sufficient to achieve stable and statistically reliable outcomes. This was further supported by Kennedy (2022), who noted that samples exceeding 300 were typically adequate for ensuring result validity. Additionally, the survey instrument's reliability was tested using Cronbach's Alpha, with all variables achieving coefficients above 0.8, indicating good to excellent reliability (see Tables 3-6). Prior literature supported the appropriateness of the sample size for quantitative studies, particularly in understanding behavioral influences like purchasing decisions (Tirtayani et al., 2024). Despite the smaller sample size, the findings were consistent with existing research on the effectiveness of Oddvertising in influencing consumer behavior, demonstrating the reliability and applicability of the study's results within the defined population scope (Marasigan et al., 2023; Bowo et al., 2024).

Conclusions

The findings confirmed that Oddvertising effectively captures Generation Z's attention in Quezon City, positively influencing their purchasing decisions (Bowo et al., 2024; Enam & Rahman, 2020). Advertisements emphasizing creativity, entertainment, and informativeness significantly enhanced brand recognition and purchase intent, supporting Hypothesis 1. While irritation negatively affected ad effectiveness, its impact was minimal compared to positive attributes like creativity and entertainment, underscoring the need for engaging content with minimal annoyance to maintain consumer interest and loyalty.

Linear regression analysis revealed that creativity and informativeness had a statistically significant positive effect on brand recognition, explaining much of the variance in consumer perceptions. These findings clarified which aspects of Oddvertising resonate most with Generation Z, with creativity being the most impactful characteristic. Conversely, irritation and its negative aspects diminished consumer responses, highlighting the importance of balancing content to avoid alienating viewers.

In summary, the study validated Hypothesis 1 and supported the Stimulus-Organism-Response (S-O-R) Framework. Oddvertising characteristics, particularly creativity and entertainment, act as stimuli that evoke positive perceptions and encourage purchasing behavior among Generation Z. Marketers targeting this demographic should focus on creativity and entertainment, incorporate informativeness, and minimize elements that could provoke irritation. These insights contribute to a deeper understanding of how Generation Z engages with digital content and responds to unconventional advertising in the Philippine context.

Recommendation

The findings of this study provide critical insights into the influence of Oddvertising on the purchasing decisions of Generation Z consumers in the Philippines. Based on these findings, the following recommendations are grounded in the study's findings and align with the interests of consumers, marketers, advertisers, educators, and future researchers. By addressing the preferences and behaviors of Gen Z, fostering innovation, and promoting ethical practices, these suggestions aim to maximize the potential of unconventional advertising strategies. They also provide actionable steps to refine marketing approaches, enhance brand recognition, and ensure long-term success in engaging a digitally savvy audience.

Encourage a culture of creative exploration. The unique characteristics of Oddvertisements, as well as their nature, have really helped inform Generation Z's purchasing decisions. Agencies are thus encouraged to foster a workplace culture that inspires creativity. This is because fostering this can lead to the development of innovative ideas and methods through which Oddvertising can be better embraced and accessed.

Enhance Social Media Engagement. Developing trends on social media by displaying imaginative ads that can spark discussions among the diverse crowd or groups. Word-of-mouth spread through the digital medium increases potential viewership demographics other than the intended group, thus engaging a wider audience.

Integrate Data Analytics. Agencies can utilize data analytics for real-time measurement of the performance of Oddvertising. Metrics such as engagement and conversion rates enable agencies to formulate adaptive strategies that reveal other opportunities to optimize ad campaigns.

Monitor Oddvertising Performance. Tracking Oddvertisement performance data against traditional advertisements can provide insight into their impact on revenue. Such comparative analysis can help businesses determine whether a shift towards more creative, unconventional strategies might be beneficial for their brand.

Promote Creative Thinking and Experimentation. Marketing students are encouraged to experiment with innovative marketing techniques, including augmented reality. Exploring how emerging technologies enhance creativity and consumer interaction can prepare students for future industry trends.

Consider Ethical Implications in Creative Marketing. Students should be mindful of the ethical considerations surrounding unconventional advertising. This includes understanding the fine line between humor and potential offense, as well as the long-term impact that extreme marketing strategies can have on brand reputation.

Directions for Future Research

Enhancing the understanding of Oddvertising's influence on consumer behavior can be achieved by implementing the following improvements. Broadening the demographic scope to include various age groups will provide a more comprehensive perspective, reducing biases. Expanding the study to cover different advertising media, such as print, broadcast, and diverse social media platforms, will offer a more accurate depiction of consumer exposure. Delving deeper into brand recognition and cognitive biases will yield valuable insights into consumer decision-making processes. Although irritation had minimal impact in this study, exploring it as a moderating factor in future research could refine strategies to minimize negative reactions. Conducting interviews with industry experts will provide strategic insights into the effectiveness and impact of Oddvertising. Lastly, refining purchase decision questionnaires will help capture accurate consumer responses and reduce confusion. Addressing these areas will enhance our understanding of Oddvertising and contribute to the development of more effective advertising strategies.

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REFERENCES

1. Abdelkader, O. A. (2021). A study of "forced-ad resistance" leading to "Skip Ad" on YouTube. Türk Bilgisayar Ve Matematik Eğitimi Dergisi, 12(10), 7263–7271. https://doi.org/10.17762/turcomat.v12i10.5617
2. Addressing the social and cultural norms that underlie the acceptance of violence. (2018). In National Academies Press eBooks. https://doi.org/10.17226/25075
3. Adhome. (2021, September 2). Oddvertising – oddly successful advertising. adHOME Creative. https://adhomecreative.com/vibe/our-blog/oddvertising-oddly-successful-advertising/
4. Agustina, T. S., & Fauzia, D. S. (2021). The need for achievement, Risk-Taking propensity, and entrepreneurial intention of the generation Z. Risenologi, 6(1), 96–106. https://doi.org/10.47028/j.risenologi.2021.61.161
5. Al Hafizi, N. A., & Ali, H. (2021, April 18). Purchase Intention and Purchase Decision Model: Multi Channel Marketing and Discount on Medcom.ID Online News Portal. Dinasti International Journal of Digital Business Management, 2(3), 460–470. https://doi.org/10.31933/dijdbm.v2i3.826
6. Alam, M. (2023, November 27). What is Quantitative Research Design? Definition, Types, Methods and Best Practices. IdeaScale. https://ideascale.com/blog/quantitative-research-design/#:~:text=Quantitative%20research%20design%20is%20defined,research%20questions%20and%20test%20hypotheses.
7. Alos, J., & Ong, S. (2024, January 12). Skip Ads? How ad length and ad skipping affect Advertisers. Spiralytics. https://www.spiralytics.com/blog/skip-ads-how-ad-length-and-ad-skipping-affect-advertisers/
8. Antosova, I., Psurny, M., & Stavkova, J. (2023). Changes in consumer purchasing decisions: Traditional and emerging factors in the dynamic marketing landscape over 15 years. Marketing Ì Menedžment Ìnnovacìj, 14(3), 85–96. https://doi.org/10.21272/mmi.2023.3-08
9. Arie Bowo, F., Anisah, A., & Marthalia, L. (2024, February 21). Meme Marketing: Generation Z Consumer Behavior on Social Media. Jurnal Indonesia Sosial Sains, 5(02), 188–201. https://doi.org/10.59141/jiss.v5i02.995
10. Arifin, S. R. M. (2018). Ethical Considerations in Qualitative Study. International Journal of Care Scholars, 1(2), 30–33. https://doi.org/10.31436/ijcs.v1i2.82
11. Apostol, E. M. D., Villan, M. S., Jose, T. T. M., & Pasco, K. M. M. (2023). Customer Experience (CX) Design in the View of Managers: An analysis of the impact of pandemic in the local hospitality and tourism industry. American Journal of Tourism and Hospitality, 1(1), 16–26. https://doi.org/10.54536/ajth.v1i1.1749
12. Ashman, S. (2023, November 10). The power of weird. Forbes. https://www.forbes.com/sites/sairahashman/2023/11/09/the-power-of-weird/?sh=7dcf788915d8
13. Badar, M. M. (2021). The Effect of Brand Image on Product Purchase Intention with Customer Satisfaction as A Mediation Variable (Study of Semen Gresik Consumer in East Java). Jurnal Ilmiah Mahasiswa FEB, 9(2). https://jimfeb.ub.ac.id/index.php/jimfeb/article/download/7516/6484
14. Barrans, J. (2022, November 24). Why marketers focus more on creative rather than media challenges. TrinityP3 Global Marketing Management Consultants. https://www.trinityp3.com/marketing-solutions/creative-rather-than-media-challenges/
15. Benhardy, K. A., Hardiyansyah, H., Putranto, A., & Ronadi, M. (2020). Brand image and price perceptions impact on purchase intentions: mediating brand trust. Management Science Letters, 3425–3432. https://doi.org/10.5267/j.msl.2020.5.035
16. Bensalem, I. S. A., & Douli, S. (2020). The impact of the creativity of the organization on the purchasing decision of the consumer. International Journal of Business and Social Science, 11(12). https://doi.org/10.30845/ijbss.v11n12p7
17. Bettiga, D., & Noci, G. (2024, February 2). The influence of television content on advertisement: a neurophysiological study. Frontiers in Psychology, 15. https://doi.org/10.3389/fpsyg.2024.1266906

18. Blair, A. (2023, December 20). The Big Idea is Broken. Welcome To Weirdvertising. Retail TouchPoints. https://www.retailtouchpoints.com/features/executive-viewpoints/the-big-idea-is-broken-welcome-to-weirdvertising?trk=article-ssr-frontend-pulse_little-text-block
19. Bhattacharya, M. (2024). IMPACT OF SOCIAL MEDIA ADVERTISING ON GENERATION Z. Indian Scientific Journal of Research in Engineering and Management, 08(04), 1–5. https://doi.org/10.55041/ijsrem30869
20. Braciníková, V., & Matušínská, K. (2020). Clustering of millennials by brand perceptions in relation to demographic factors. Central European Business Review, 9(5), 63–80. https://www.ceeol.com/search/article-detail?id=940994
21. Braciníková, V., & Matušínská, K. (2020). Clustering of millennials by brand perceptions in relation to demographic factors. Central European Business Review, 9(5), 63–80. https://doi.org/10.18267/j.cebr.253
22. Brolin, R. E., & Spångby, C. (2020). The impact of advertising exposure on attitudes and purchase intention. A Quantitative Study Measuring the Impact of Instagram Advertising Exposure on Attitudes Toward the Ads and Brands, and Purchase Intention of Sportswear. http://hj.diva-portal.org/smash/record.jsf?pid=diva2:1435637
23. Calder, B. J. (1978, January 1). Cognitive response, imagery, and scripts: What is the cognitive basis of attitude?: ACR. ACR North American Advances. Retrieved February 27, 2023, from https://www.acrwebsite.org/volumes/9494/volumes/v05/NA-05
24. Cho, E., Kim-Vick, J., & Yu, U. (2021). Unveiling motivation for luxury fashion purchase among Gen Z consumers: need for uniqueness versus bandwagon effect. International Journal of Fashion Design, Technology and Education, 15(1), 24–34. https://doi.org/10.1080/17543266.2021.1973580
25. Csapó-Horváth, A. (2021, June 1). The Importance and Influence of Destination Advertising. European Journal of Sustainable Development, 10(2), 231. https://doi.org/10.14207/ejsd.2021.v10n2p231
26. Dalangin, J. J. G., Arthur, J. M. M., Salvador, J. B. M., & Bismonte, A. B. (2021). The impact of social media influencers purchase intention in the Philippines. Deleted Journal, 5(3), 551–568. https://doi.org/10.25139/jsk.v5i3.3708
27. Dall'Olio, F., & Vakratsas, D. (2022). The impact of Advertising Creative Strategy on advertising elasticity. Journal of Marketing, 87(1), 26–44. https://doi.org/10.1177/00222429221074960
28. Demsar, V., Sands, S., Rosengren, S., & Campbell, C. (2022). Ad creativity in a negative context: How a thanking message frame enhances purchase intention in times of crisis. Journal of Retailing and Consumer Services, 64, 102825. https://doi.org/10.1016/j.jretconser.2021.102825
29. Disastra, G. M., Hanifa, F. H., Wulandari, A., & Sastika, W. (2019). The Influence of Advertising Value on Advertising Attitude and its Impact on Purchase Intention. Proceedings of the Social and Humaniora Research Symposium (SoRes 2018). https://doi.org/10.2991/sores-18.2019.98
30. Dou, Z., & Yue-Ming, Z. (2023). The influence of short video advertisements on consumers' purchase intention. Journal of Education, Humanities and Social Sciences, 13, 269–274. https://doi.org/10.54097/ehss.v13i.7912
31. Enam, M. A., & Rahman, A. (2020). Factors Influencing Innovation in Advertisement: A study on the perspective of Bangladesh market. International Journal of Entrepreneurial Research, 3(4), 125–132. https://doi.org/10.31580/ijer.v3i4.1643
32. Faghani, V., Sahraee, M., Sarkesh, F., & Katouzian, A. R. (2024). What is Cognition? An evolutionary perspective. Authorea (Authorea). https://doi.org/10.22541/au.170665387.79875499/v1
33. Feng, T., Wang, C., & Chen, C. (2022). Creative effect of film and television advertising based on digital media interactive technology. Journal of Mobile Information Systems, 2022, 1–10. https://doi.org/10.1155/2022/6231760
34. Guo, C., Yang, Z., & Zhao, S. (2023). The impact of interactivity on live e-commerce platform on the intention of Generation Z 's consumption behavior. Lecture Notes in Education Psychology and Public Media, 3(1), 256–266. https://doi.org/10.54254/2753-7048/3/2022548

35. Hidayat, R., Menhard, M., Charli, N. C. O., Masnum, N. A., & Hartoyo, N. B. (2023). Determination purchase intention and purchase Decision: Brand Image and Promotion Analysis (Marketing Management Literature Review). Dinasti International Journal of Economics, Finance & Accounting/Dinasti International Journal of Economics, Finance and Accounting, 4(1), 152–165. https://doi.org/10.38035/dijefa.v4i1.1749
36. Hussain, A., Ting, D. H., Abbasi, A. Z., & Rehman, U. (2022). Integrating the S-O-R model to examine purchase intention based on Instagram sponsored advertising. Journal of Promotion Management, 29(1), 77–105. https://doi.org/10.1080/10496491.2022.2108185
37. Jan, M. T., Haque, A., Abdullah, K., Anis, Z., & Faisal-E-Alam, F. E. A. (2019). Elements of advertisement and their impact on buying behaviour: A study of skincare products in Malaysia. Management Science Letters, 1519–1528. https://doi.org/10.5267/j.msl.2019.5.033
38. Jiang, H., Messinger, P. R., Liu, Y., Lü, Z., Yang, S., & Li, G. (2023). Divergent versus Relevant Ads: How creative ads affect purchase intention for new products. Journal of Marketing Research. https://doi.org/10.1177/00222437231187630
39. Kahawandala, N., Peter, S., & Niwunhella, H. (2020). Profiling purchasing behavior of Generation Z. Smart Computing and Systems Engineering. https://doi.org/10.1109/scse49731.2020.9313038
40. Kala, I., & Pawar, S. (2022). What is in a Meme: Analysing the Perception of Gen Z to Memes and Virality in Modern Marketing. CARDIOMETRY, 24, 737–742. https://doi.org/10.18137/cardiometry.2022.24.737742
41. Kaur, S., Nim, D., & Jaggi, S. (2022). Relationship between humor in advertising and Attitude towards Ads : A Meta-Analysis. Indian Journal of Marketing, 52(3), 8. https://doi.org/10.17010/ijom/2022/v52/i3/160366
42. Kawulur, A., Sumakul, G., & Pandowo, A. (2022). Purchase Intention of Second-Hand: A Case Study of Generation Z. SHS Web of Conferences, 149, 02026. https://doi.org/10.1051/shsconf/202214902026
43. Kennedy, I. (2022). Sample Size Determination in Test-Retest and Cronbach Alpha Reliability Estimates. British Journal of Contemporary Education, 2(1), 17–29. https://doi.org/10.52589/bjce-fy266hk9
44. Kenton, W. (2024, September 16). What Is Brand Recognition? Why It's Important and Benefits. Investopedia. https://www.investopedia.com/terms/b/brand-recognition.asp#:~:text=While%20brand%20recognition%20refers%20to,the%20path%20to%20brand%20recognition.
45. Kim, J., Grijalva, E., Newman, D. A., & Li, Y. (2023, August 2). Entrepreneur weirdness as a double-edged sword: Effects on product creativity and investor attraction. Personnel Psychology, 77(1), 53–80. https://doi.org/10.1111/peps.12613
46. Kokins, G., Straujuma, A., & Lapiņa, I. (2021). The role of consumer and customer journeys in customer experience driven and open innovation. Journal of Open Innovation, 7(3), 185. https://doi.org/10.3390/joitmc7030185\
47. Kumar, V., & Campbell, R. (2022). Norms. In Oxford University Press eBooks (pp. 61–82). https://doi.org/10.1093/oso/9780197600122.003.0004
48. Kumari, K., & Yadav, S. (2018). Linear regression analysis study. Journal of the Practice of Cardiovascular Sciences, 4(1), 33. https://doi.org/10.4103/jpcs.jpcs_8_18
49. Levinson, J. C., & Levinson, J. (2011). The Best of Guerrilla Marketing. Entrepreneur Media, Inc. http://dspace.vnbrims.org:13000/xmlui/bitstream/handle/123456789/4913/The%20Best%20of%20Guerrilla%20Marketing%20Guerrilla%20Marketing%20Remix.pdf?sequence=1&isAllowed=y
50. Li, W., Jiang, M., & Zhan, W. (2022). Why advertise on short video platforms? Optimizing online advertising using advertisement quality. Journal of Theoretical and Applied Electronic Commerce Research, 17(3), 1057–1074. https://doi.org/10.3390/jtaer17030054
51. Ling, S., Zheng, C., Cho, D., Kim, Y., & Dong, Q. (2024). The Impact of interpersonal interaction on purchase intention in livestreaming E-Commerce: A Moderated Mediation model. Behavioral Sciences, 14(4), 320. https://doi.org/10.3390/bs14040320

52. Liu, G., Gao, P., Li, Y., & Zhang, Z. (2019). Research on the Influence of Social Media Short Video Marketing on Consumer Brand Attitude. Advances in Social Science, Education and Humanities Research, 336. https://doi.org/10.2991/icsshe-19.2019.192
53. Losa, R. (2020, November 27). RC Cola reminds us why weird Filipino commercials matter - SCOUT. Scout Magazine. https://www.scoutmag.ph/65788/rc-cola-weird-filipino-commercials-matter-nostalgia/
54. Machi, L., Nemavhidi, P., Chuchu, T., Nyagadza, B., & De Villiers, M. V. (2022). Exploring the impact of brand awareness, brand loyalty and brand attitude on purchase intention in online shopping. International Journal of Research in Business and Social Science, 11(5), 176–187. https://doi.org/10.20525/ijrbs.v11i5.1841
55. Maenhout, W. (2022). LENGTH DOES MATTER: THE EFFECTIVENESS OF LONG-FORM VERSUS SHORT-FORM VIDEO MARKETING ACCORDING TO ADVERTISING CONTEXT AND VIEWER AGE. Universiteit Gent. https://libstore.ugent.be/fulltxt/RUG01/003/117/738/RUG01-003117738_2023_0001_AC.pdf
56. Moreno, D. E., Gabatin, R. A., Agleron, M. A., Gavasan, C., Santos, E. H., & Tiongson, S. R. (2023). Penetrating through Distractions: Analyzing the Function of Storytelling Advertisements in Attracting the Focus of Gen Z Customers. Journal of Business and Management Studies, 5(3), 08–24. https://doi.org/10.32996/jbms.2023.5.3.2
57. Moussa, M. H., & Ibrahim, M. F. (2021). Studying the Relationship between Humorous Advertising and Consumer Purchasing Decision. "A Mediation Analysis of Brand Awareness." Al-Maġallah Al- ilmiyyah Lil-Iqti ād Wal Tiğārah, 51(2), 591–620. https://doi.org/10.21608/jsec.2021.158114
58. Muhammad, F. (2023, May 5). Advertising Evolution: How Personalization Has Improved over Time. Instapage. https://instapage.com/blog/advertising-evolution-how-personalization-has-improved-over-time-4/
59. Munsch, A. (2021). Millennial and generation Z digital marketing communication and advertising effectiveness: A qualitative exploration. Journal of Global Scholars of Marketing Science, 31(1), 10–29. https://doi.org/10.1080/21639159.2020.1808812
60. Murwonugroho, W., & Yudarwati, G. A. (2020). Exposure to unconventional outdoor media advertising. Pertanika Journal of Social Science & Humanities, 28(4). https://doi.org/10.47836/pjssh.28.4.50
61. Myllyoja, H., Čeh, N. B., & Said, Y. (2020). The effects of humorous and irritating advertising on brand perception in the food industry. The Ef ects of Humorous and Irritating Advertising on Brand Perception in the Food Industry. http://www.diva-portal.org/smash/record.jsf?pid=diva2:1436988
62. Nainggolan, I. S. P. (2021). The attitude of Generation Z towards YouTube skippable Ads: An Empirical Study on Lokalate Ads. Indonesian Journal of Business Analytics, 1(1), 17–32. https://doi.org/10.54259/ijba.v1i1.11
63. Neubarth, M. (2024, January 25). Bending the Mind with Weirdvertising. https://www.linkedin.com/pulse/bending-your-mind-weirdvertising-michael-neubarth-iqsme
64. Oktiani, N., Yuliantari, K., Kuspriyono, T., Haryani, H., & Kurniasari, R. (2022). The effect Promotion e- commerce toward effectiveness promotion by using (AIDA) methods. Widya Cipta, 6(1), 14–19. https://doi.org/10.31294/widyacipta.v6i1.11683
65. Othman, H. (2020, April 1). Ambient Ads as Unconventional Media Approach. International Design Journal, 10(2), 269–278. https://doi.org/10.21608/idj.2020.81142
66. Palikhe, A. (2019). Impact of Humor Advertising on Brand Purchase Strategy. the Journal of Nepalese Business Studies, 12(1), 60–70. https://doi.org/10.3126/jnbs.v12i1.28183
67. Pasharibu, Y., Soerijanto, J. A., & Jie, F. (2020, October 16). Intention to buy, interactive marketing, and online purchase decisions. Jurnal Ekonomi Dan Bisnis, 23(2), 339–356. https://doi.org/10.24914/jeb.v23i2.3398
68. Pietroń-Pyszczek, A., & Borowska, M. (2022). Generation Z need for mentor support. Zarządzanie Zasobami Ludzkimi, 149(6), 44–59. https://doi.org/10.5604/01.3001.0016.2045
69. Post, G. (2024, April 1). How to use humor in advertising to make your small business more memorable. AllBusiness.com. https://www.allbusiness.com/how-to-use-humor-in-advertising-to-make-your-small-business-more-memorable-208494-1.html

70. Reitsamer, B. F., & Becker, L. (2024). Customer journey partitioning: A customer-centric conceptualization beyond stages and touchpoints. Journal of Business Research, 181, 114745. https://doi.org/10.1016/j.jbusres.2024.114745

71. Rini, I. T. C. (2023). Factors influencing purchase intention against marketplace users in Gen Z. International Journal of Economics and Management Review, 1(2), 58–68. https://doi.org/10.58765/ijemr.v1i2.144

72. Romanova, I. D., & Smirnova, I. V. (2019). Persuasive techniques in advertising. Training, Language and Culture, 3(2), 55–70. https://doi.org/10.29366/2019tlc.3.2.4

73. Rosengren, S., Eisend, M., Koslow, S., & Dahlén, M. (2020). A Meta-Analysis of when and how advertising creativity works. Journal of Marketing, 84(6), 39–56. https://doi.org/10.1177/0022242920929288

74. Santanagopalan, S., & Raj, K. (2023). Impact of information parameters of online auto dealership website towards consumer purchase intention. International Journal of Professional Business Review, 8(2), e01073. https://doi.org/10.26668/businessreview/2023.v8i2.1073

75. Santos, S. S. S. D., Da Silva, M., & Aoki, V. C. G. (2022). Creative failure in advertising: Influencing the creative process. Revista De PsicologíA, 40(2), 957–98 https://doi.org/10.18800/psico.202202.012

76. Saputra, N. I. G. B. K., & Wardana, N. I. M. (2023). The role of brand awareness mediating the influence of social media marketing on purchase intention. International Journal of Asian Business and Management, 2(4), 559–576. https://doi.org/10.55927/ijabm.v2i4.5752

77. Schweidel, D. A., Bart, Y., Inman, J. J., Stephen, A. T., Libai, B., Andrews, M., Rosario, A. B., Chae, I., Chen, Z., Kupor, D., Longoni, C., & Thomaz, F. (2022). How consumer digital signals are reshaping the customer journey. Journal of the Academy of Marketing Science, 50(6), 1257–1276. https://doi.org/10.1007/s11747-022-00839-w

78. Selan, C. V. P., Lapian, S. L. H. V. J., & Gunawan, E. M. (2021). The Effects of Guerilla Marketing on Consumer Purchase Intention With Brand Awareness as a Mediating Variable in Pt. Solusi Transportasi Indonesia (Grab). | Jurnal EMBA : Jurnal Riset Ekonomi, Manajemen, Bisnis dan Akuntansi. ejournal.unsrat.ac.id. https://doi.org/10.35794/emba.v9i4.36297

79. Shaw, K. (2024). Grab's new spot says you can have a "No Sweat Summer" via GIGIL Philippines. Campaign Brief Asia. https://campaignbriefasia.com/2024/04/15/grabs-new-spot-says-you-can-have-a-no-sweat-summer-via-gigil-philippines/

80. Shen, W., Bai, H., Ball, L. J., Yuan, Y., & Wang, M. (2020). What makes creative advertisements memorable? The role of insight. Psychological Research, 85(7), 2538–2552. https://doi.org/10.1007/s00426-020-01439-5

81. Shen, W., Bai, H., Yuan, Y., Ball, L. J., & Fang, L. (2021). Quantifying the roles of conscious and unconscious processing in insight-related memory effectiveness within standard and creative advertising. Psychological Research, 86(5), 1410–1425. https://doi.org/10.1007/s00426-021-01572-9

82. Shen, W., Liu, Z., Ball, L. J., Huang, T., Yuan, Y., Bai, H., & Hua, M. (2020). Easy to remember, easy to forget? the memorability of creative advertisements. Creativity Research Journal, 32(3), 313–322. https://doi.org/10.1080/10400419.2020.1821568

83. Singh, P. P., & Chahal, H. S. (2019). Consumers attitude towards controversial television commercials and its impact on purchase intentions. Management and Labour Studies, 45(1), 118–141. https://doi.org/10.1177/0258042x19890242

84. Sjahruddin, H., & Adif, R. M. (2024). Gen Z Consumer Trends: Understanding the next wave of buying behavior. Management Studies and Entrepreneurship Journal, 5(1), 480–485. https://doi.org/10.37385/msej.v5i1.4085

85. Smith, K., Pickering, A. D., & Bhattacharya, J. (2022). The Creative Life: A Daily Diary study of Creativity, Affect, and Well-Being in Creative Individuals. Creativity Research Journal, 34(4), 460–479. https://doi.org/10.1080/10400419.2022.2122371

86. Sohaib, M., Safeer, A. A., & Majeed, A. (2022). Role of social media marketing activities in China's e-commerce industry: A stimulus organism response theory context. Frontiers in Psychology, 13. https://doi.org/10.3389/fpsyg.2022.941058

87. Soti, R. (2022). The impact of advertising on consumer behavior. World Journal of Advanced Research and Reviews, 14(3), 706–711. https://doi.org/10.30574/wjarr.2022.14.3.0577

88. Tabassum, S., Khwaja, M. G., & Zaman, U. (2020). Can narrative advertisement and eWOM influence Generation Z purchase intentions? Information, 11(12), 545. https://doi.org/10.3390/info11120545

89. Tanuwijaya, J., & Gunawan, L. (2021). The Influence of Advertising Value towards the Purchase Intention of MORA.id's Products. Jurnal Entrepreneur Dan Entrepreneurship/Jurnal Entrepreneur Dan Entrepreneurship, 10(1), 67–82. https://doi.org/10.37715/jee.v10i1.1727

90. Tirtayani, I. G. A., Wardana, I. M., Setiawan, P. Y., & K, I. G. N. J. A. W. (2024, March 26). Determinants of Purchase Intention in Social Media Advertising. Revista De Gestão Social E Ambiental, 18(6), e04793. https://doi.org/10.24857/rgsa.v18n6-016

91. Tsao, W. (2019). Do irritations of ads in line really matter? International Business Research, 12(5), 53. https://doi.org/10.5539/ibr.v12n5p53

92. Tumiran, M. A. (2024). HOW TO DEAL WITH INSUFFICIENT SAMPLE SIZE DUE TO NON-RESPONSE IN SURVEYS? Quantum Journal of Social Sciences and Humanities, 5(2), 70–86. https://doi.org/10.55197/qjssh.v5i2.346

93. Tuan, P. V., Chan, L. A., Ngoc Huy, D. T., Quang Anh, N., Trang, N. Q., Giang, P. C., & Uyen, N. P. (2023). Effects of Advertising Value and Attitude on the Purchase Intention Through Short-Form Video Advertising on Social Media: Empirical Research on Gen Z in Vietnam. Middle East Journal of Applied Science & Technology, 06(01), 42–59. https://doi.org/10.46431/mejast.2023.6105

94. Van Berlo, Z. M. C., Meijers, M. H. C., Eelen, J., Voorveld, H. a. M., & Eisend, M. (2023). When the medium is the message: A Meta-Analysis of Creative Media Advertising Effects. Journal of Advertising, 53(2), 278–295. https://doi.org/10.1080/00913367.2023.2186986

95. Vestil, M. (2024). Brand reputation in focus: Why it matters, what factors impact it, how to measure it—and how it drives success. Agility PR Solutions. https://www.agilitypr.com/pr-news/public-relations/brand-reputation-in-focus-why-it-matters-what-factors-impact-it-how-to-measure-it-and-how-it-drives-success/#:~:text=However%2C%20a%20negative%20brand%20reputation,to%20a%20negative%20

96. brand%20reputation.

97. Vo, T. P., Tran, M. D., & Ly, T. M. C. (2022). The relationship between new product creativity components and purchase intention and the mediating impact of perceived values: a study of consumers in Ho Chi Minh city. Cogent Business & Management, 9(1). https://doi.org/10.1080/23311975.2022.2112357

98. Wax, D. (2023, August 25). Thinking Outside the Box: 11 Tips on Shifting Perspectives. Lifehack. https://www.lifehack.org/articles/featured/11-ways-to-think-outside-the-box.html

99. Waqar, A. (2020). Impact of humorous advertising on purchase decision: In context of Pakistan's telecom industry. Global Journal of Business, Economics and Management, 10(2), 144–150. https://doi.org/10.18844/gjbem.v10i2.5089

100. Weird is Good: What Makes Absurd Ads So Ef ective? (2023, March 10). Rhythm Agency. https://rhythmagency.com/blog/march-2023/weird-is-good-what-makes-absurd-ads-so-effective#:~:text=Apparently%2C%20too%20much%20perfection%20in,more%20with%20them%20(Stackla).

101. Weismueller, J., Harrigan, P., Wang, S., & Soutar, G. N. (2020). Influencer Endorsements: How advertising disclosure and source credibility affect consumer purchase intention on social media. Australasian Marketing Journal, 28(4), 160–170. https://doi.org/10.1016/j.ausmj.2020.03.002

102. Williams, K. (2024, January 29). What is response bias and how to avoid it? A must read! SurveySparrow. https://surveysparrow.com/blog/response-bias/#:~:text=Keep%20the%20entire%20survey%20simple,use%20Likert%20scale%20questions%20correctly.

103. Wolfsteiner, E., & Garaus, M. (2023). When attention-grabbing tactics backfire during media multitasking: The detrimental effect of atypical advertising slogans on brand attitude. Telematics and Informatics, 81, 101981. https://doi.org/10.1016/j.tele.2023.101981

104. Wu, C. L., & Chen, H. C. (2019). The influence of creativity on Incongruity-Resolution and nonsense humor comprehension. Creativity Research Journal, 31(1), 110–118. https://doi.org/10.1080/10400419.2019.1577675

105. Xiao, L., Li, X., & Zhang, Y. (2023). Exploring the factors influencing consumer engagement behavior regarding short-form video advertising: A big data perspective. Journal of Retailing and Consumer Services, 70, 103170. https://doi.org/10.1016/j.jretconser.2022.103170

106. Yin, X., Li, J., Si, H., & Wu, P. (2024). Attention marketing in fragmented entertainment: How advertising embedding influences purchase decision in short-form video apps. Journal of Retailing and Consumer Services, 76, 103572. https://doi.org/10.1016/j.jretconser.2023.103572

107. Yu, Q. (2023). Research on factors affecting consumers' purchase intention. Advances in Economics, Management and Political Sciences, 37(1), 61–66. https://doi.org/10.54254/2754-1169/37/20231835

108. Yu, S., Zhang, H., Zheng, Q., Chu, D., Chen, T., & Chen, X. (2024). Consumer behavior based on the SOR model: How do short video advertisements affect furniture consumers' purchase intentions? Bioresources, 19(2), 2639–2659. https://doi.org/10.15376/biores.19.2.2639-2659

109. Zhai, L., Yin, P., Li, C., Wang, J., & Yang, M. (2022). Investigating the effects of Video-Based E-Word-of-Mouth on consumers' purchase intention: the moderating role of involvement. Sustainability, 14(15), 9522. https://doi.org/10.3390/su14159522

110. Zhang, G., Yue, X., Yan, Y., & Peng, M. Y. (2021). Understanding the impact of the psychological Cognitive process on student learning satisfaction: combination of the Social Cognitive Career Theory and SOR model. Frontiers in Psychology, 12. https://doi.org/10.3389/fpsyg.2021.712323

111. Zhou, R., & Tong, L. (2022, May 9). A Study on the Influencing Factors of Consumers' Purchase Intention During Livestreaming e-Commerce: The Mediating Effect of Emotion. Frontiers in Psychology, 13. https://doi.org/10.3389/fpsyg.2022.903023

112. Zhou, Z. (2023). User Purchase Intent on social media: A Case Study Utilizing the SOR Model on Xiaohongshu. Highlights in Business, Economics and Management, 23, 716–724. https://doi.org/10.54097/gt9ky947

The Influence of Audio Branding in YouTube Ads on the Purchase Decision of Millennials

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Abstract

This study aimed to assess how audio branding in YouTube ads affected millennials' emotional and rational decision-making processes related to purchasing, to recognize the short and long-term effects of repetitive exposure to audio branding in YouTube ads, and to determine whether audio branding influenced the buying decisions of customers. A quantitative survey with 301 respondents aged 28-43 was conducted in the BPO hub in Quezon City to gather data, utilizing a 4-point Likert scale. The findings revealed that jingles, brand music, and brand voice strongly impacted millennials' purchase decisions, while audio logos had a more moderate influence. Rational decision-making showed a stronger association with purchase decisions than emotional factors. Overall, audio branding played a significant role in shaping purchase decisions. These insights offer valuable guidance for advertisers and marketers looking to refine their strategies, while future research can further explore audio branding in other settings.

Keywords: Audio branding, YouTube advertisements, Purchase Decision, Marketing

Introduction

In the current competitive marketplace, various businesses are continuously seeking innovative ways to stand out and connect with their customers on a deeper level. This includes using eye-catching logos and emotionally appealing advertisements. Sound has also become a powerful tool for brand differentiation and fostering emotional connection. Thus, sound or audio branding has been developed —audio branding is a field that strategically uses sound elements to shape consumer perception and behavior, reinforce brand identity, and enhance brand recall (Yang et al., 2021).

Audio branding uses several auditory elements, such as audio logos, jingles, soundscapes, brand music, brand voice, user experience, interface sounds, and product sound. It represents an innovative marketing strategy that not only triggers brand recognition but also elicits emotional responses from consumers. According to a study conducted by Leicester University professors North and Hargreaves, 96% of consumers are more likely to remember a brand when music that perfectly fits with the brand identity is used. This is crucial in the digital age since attention is becoming harder to attain, and multimedia platforms are increasingly shaping consumer behavior.

Moreover, among numerous multimedia platforms, YouTube is one of the most popular. It is used for different purposes and generates a significant number of advertisements. According to Stone (2023), millennials make up one-third of YouTube's audiences. They use YouTube mainly for sponsored content, entertainment, and social commerce, while others use it for news and fan content. Additionally, the platform shows countless advertisements, which is one of the ways that content creators and YouTube generate revenue. While millennials are known as digital natives, they are also known for their critical consumer habits, as well as their preference for brands that align with their values.

Literature Review

Audio Branding of Advertisements on YouTube

Audio plays a vital role in advertising, influencing viewer attitudes and participation (Barnes, 2023). Online audio advertising, particularly in music streaming services, relies on high-quality advertisements to keep users engaged, with factors such as harmony, rhythm, and timbre influencing ad quality and user experience. Automatic assessment of audio ads employing parameters such as Long Click Rate and deep learning models improves ad quality prediction and user engagement, resulting

in more relevant and engaging commercials. The use of audio in advertising not only draws attention but also influences brand perception and audience participation (Ebrahimi, 2018).

With the growth of e-commerce and online trading, YouTube has become a powerful online marketing tool. The platform draws advertisers because of its large user base and potential for exciting content (Lu, 2023). In general, advertising on YouTube is an innovative approach that combines several factors to engage users effectively. Imperative speech is vital in public service advertisements on YouTube, performing tasks such as informing, advising, educating, inviting, and reflecting. (Bahasa & Pembelajarannya, 2023). Most products make use of YouTube's marketing strategy by inserting adverts into popular and highly viewed YouTube channel videos, either with or without a skip option; ribbon advertisements on the side of the video or pop-up advertisements are also utilized (Kumar, 2023). In addition to this, music and melodies in YouTube commercials boost consumer recall and build a relationship between the individual viewing them and the business, which significantly influences user behavior (Sadek, 2023).

Differences Between Audio Branding Elements

Audio branding, also known as sound branding, sonic branding, or acoustic branding, is defined as the use of auditory elements to facilitate brand development and brand management within the framework of brand communication (Fraile et al, 2021). The papers written by Fraile et al. (2021), Nufer and Moser (2018), Tran (2020), and Ayada and Ragab (2024) commonly stated that the elements of audio branding include, but are not limited to, audio or sound logo, jingle, brand music, and brand voice.

Audio or sound logo, according to Nufer and Moser (2018), is usually connected with the graphical corporate brand logo and has a mnemonic purpose which can help customers recognize the brand name faster. It acts as the unique signature or 'identifier' of a brand which plays a crucial role in shaping the corporate sound branding strategies of companies globally (Tran, 2020). Furthermore, Nufer and Moser stated that audio logos are often created based on intuition rather than following established advertising models and that the important characteristics that must be considered when composing an audio or sound logo are: it is a few-tone, is usually between one to three seconds, is flexible and adaptable to different surroundings and instruments, and it must be unmistakably different from the sound logos of the competitors. Examples of the audio logo are the startup chime of Apple Mac and the intro of Netflix's "Ta-dum".

Jingle, on the other hand, comprises a memorable short song or melody designed specifically for a brand (Fraile et al., 2021). It is often considered a sonic version of mnemonics since it uses techniques such as rhymes or catchy melodies to improve memory retention which consequently can boost brand recall. While jingle excels at brand identification and recall, its functions are restricted as it is unable to convey a broad range of brand attributes (ex. product quality, price, functionality, etc.) potentially leading to a weaker formation of brand impressions.

Similar to jingle, brand music or brand song, is also a musical composition. However, brand music is designed as a song that embodies the audio identity of the corporate brand. Unlike jingle, it is composed specifically for the organization to convey its values and is used for a long period of time (Fraile et al., 2021). Furthermore, according to Vidal (2018), it represents a new scenario that "aims to tie in the audience with the brand, beyond its products". There are two features a brand song can have as stated by Tran (2020), it can be a piece of music that is either specifically produced for the brand or one that is procured exclusively for the brand from an existing selection of songs. "I'd Like to Buy the World a Coke" by Hilltop is a good example of a brand song since it is a remake version of "I'd Like To Teach The World To Sing" by The New Seekers.

Finally, for the last element, brand voice pertains to the vocal aspect that is used to verbally convey the personality of a brand, which resonates with its values and identity (Fraile et at., 2021). According to Tran (2020), a brand voice encompasses both speaking and singing, which creates compelling associations and emotions solely through voice. Moreover, the tone and pace of speech should also mirror the values and character of the brand. Good examples of brand voice are from Old Spice and Dove. To conclude, audio branding is a powerful strategy that can be used to create meaningful connections with consumers. These elements work together to not only establish and reinforce brand identity but also to enhance brand recall and evoke emotional responses. By using the unique qualities of sound, brands can create memorable experiences that can resonate deeply with their target audience, which can foster greater loyalty and rapport from the customers.

Effects of Audio Branding Elements

Audio branding became a powerful tool for companies to solidify brand identity and connect with consumers. Strategic use of elements like audio logos, jingles, brand music, and brand voice can significantly enhance brand recall, recognition, and emotional association. However, simply adding sound isn't enough. The chosen audio elements must resonate with the brand's personality and target audience. Research even suggests that music tailored to a product's characteristics can positively influence purchasing behavior. Furthermore, consistency across all communication channels is also crucial for maximum impact, in order to create a cohesive brand experience that strengthens consumer memory and relationship. Finally, the most effective brand voice may differ depending on the context. Understanding these principles can help companies utilize audio branding strategically, so they can create a more memorable and impactful brand experience for their customers.

Consumer Preferences for Audio Branding Elements

The factors that influence consumer choices for audio branding elements are multifaceted. Studies reveal that audio logos have a significant impact on the feelings and opinions of consumers (Li et al., 2022). Consumers' willingness to pay for a brand, specifically, can be influenced by the amount of tones in a sound logo, with a suitable amount of tones being viewed as having more excellent value (Scott et al., 2022). Research has also indicated that the pitch levels of background music in audiovisual advertising could impact customers' assessments of brand competence, especially for individuals with holistic thinking patterns (Hong et al., 2022). Furthermore, when combined with other sensory modalities, music in sound branding has the power to change consumers' affective states, purchasing decisions, and impressions of brand personality (Rodero & Larrea, 2020). These findings highlight the necessity of carefully selecting and creating strong trademarks to increase brand recall and elicit favorable associations with the brand.

Methodology

The research design for this study used a descriptive approach and a quantitative technique, specifically the correlational method to evaluate the influence of various audio branding aspects on Millennials' purchasing decisions in Quezon City. The survey used the purposive sampling technique, and focused on millennials who are currently employed in the Business Process Outsourcing (BPO) industry. Data is gathered through online surveys, and distributed via social media sites and to guarantee a representative sample. Additionally, physical distribution of the QR codes of the survey is also implemented. The survey included structured questionnaires with Likert scale items to assess respondents' impressions and attitudes toward audio branding aspects, with a particular emphasis on jingles, brand music, brand voice, and audio logos. It also included videos for the respondents to watch, and answer questions related to it.

Analysis and Discussions

Presented in Table 10 is a comprehensive discernment of the different age ranges and their respective fraction as collected through the survey. It shows that ages 28 to 31 have the highest percentage for the age bracket accounting for 54.8%. The middle groups include ages 32-25 with 24.3% and 36-39 with 12.6%. The range 40-43 accumulated the least count with 8.3% only. Having surveyed from all different ages within millennials positively adds to the quality of this study.

Table 10. Frequency table for age

Age	Frequency	Percent
28 – 31	165	54.817
32 – 35	73	24.252
36 – 39	38	12.625
40 – 42	25	8.306
Total	**301**	**100**

Table 11 is a frequency table for sex, which depicts the gender distribution of research respondents. Of the total 301 respondents: 144 respondents (47.841%) identified as male (classified as one), 145 respondents (48.173%) identified as female (coded 2), Twelve respondents (3.987%) selected the option choose not to say (coded as three). This distribution indicates a roughly equal split among male and female responders, with both genders accounting for nearly half of the sample population.

Table 11. *Frequency table for sex*

Age	Frequency	Percent
1	144	47.841
2	145	48.173
3	12	3.987
Total	**301**	**100**

Table 12 presents the data for the awareness of audio branding of respondents. It was observed that the mean values of the three variables were close to one another, ranging from 3.0 to 3.4, indicating a similar level of agreement among respondents. The standard deviation, on the other hand, showed a low value suggesting that the data was precise. The low standard deviation implied that most of the respondents chose "Agree". From this, the researchers concluded that the majority of respondents were aware of audio branding.

Table 12. Awareness of Audio Branding

	Mean	SD	Interpretation
AOAB1	3.136	0.662	Agree
AOAB2	3.013	0.730	Agree
AOAB3	3.422	0.576	Agree
Average	**3.190**	**0.567**	**Agree**

In Table 13, the result portrayed that the mean values of the variables ranged from 3.329 to 3.575, with most values leaning towards "Strongly Agree". In order to attain the result, the mean and standard deviation were calculated. Based on the findings, question number 3 had the highest mean value of "Strongly Agree," while question number 6 received the lowest mean value.

Since the question number 3, "I usually hear music playing in the background whenever I watch an advertisement on YouTube.", received the highest mean value of M = 3.575, the researchers concluded that most respondents understood what brand music is, since they were able to identify the music or song being played in the advertisement.

On the other hand, the question "I usually hear ambient noises in the background whenever I watch an advertisement on YouTube." received the lowest mean value of M = 3.329. Although this mean value was lower, it does not signify that the respondents did not hear ambient noises but rather that the majority of respondents agreed they did hear ambient noises when watching advertisements on Youtube.

Table 13. Understanding the Different Audio Branding Elements

	Mean	SD	Interpretation
UTDABE1	3.445	0.684	Agree
UTDABE2	3.545	0.531	Strongly Agree
UTDABE3	3.575	0.534	Strongly Agree
UTDABE4	3.542	0.562	Strongly Agree
UTDABE5	3.402	0.698	Agree
UTDABE6	3.329	0.694	Agree
UTDABE7	3.568	0.559	Strongly Agree
Average	**3.486**	**0.488**	**Agree**

Table 14 affirms that jingles effectively create memorable brand associations and supports the hypothesis that jingles in YouTube ads significantly influence Millennials' purchase decisions. Respondents in the study generally agreed they recognized jingles in advertisements, with the question "Do you recognize this advertisement?" receiving the highest score (M = 3.688, SD = 0.562). Other questions, like recognizing sounds, identifying brands, and recalling previously watched ads, also showed strong agreement. While fewer respondents were aware of audio branding specifically (M = 3.498, SD = 0.651), overall, the results (average M = 3.600) confirm that jingles help make brands memorable and support the idea that jingles in YouTube ads influence Millennials' purchase decisions.

Table 14. Differences in the respondents' understanding of the audio branding element Jingle

	Mean	SD	Interpretation
2UTDABE1	3.688	0.562	Strongly Agree
2UTDABE2	3.605	0.560	Strongly Agree
2UTDABE3	3.601	0.622	Strongly Agree
2UTDABE4	3.608	0.605	Strongly Agree
2UTDABE5	3.498	0.651	Agree
Average	**3.600**	**0.068**	**Strongly Agree**

Table 15 showed that respondents generally recognized brand music in advertisements, with questions like "Do you recognize this advertisement?" (M = 3.615, SD = 0.615) and "Do you instantly recognize the brand involved?" (M = 3.654, SD = 0.510) receiving strong agreement. Other questions, such as recognizing sounds (M = 3.522, SD = 0.630) and being aware of audio branding (M = 3.468, SD = 0.656), had slightly lower agreement but were still positive. Overall, the average mean of M = 3.553 and low SD = 0.078 indicate a consistent, favorable perception of brand music. The results confirm Hypothesis 2, showing that brand music in YouTube ads significantly influences Millennials' purchasing decisions.

Table 15. Differences in the respondents' understanding of the audio branding element Jingle

	Mean	SD	Interpretation
2UTDABE6	3.615	0.615	Strongly Agree
2UTDABE7	3.522	0.360	Strongly Agree
2UTDABE8	3.654	0.510	Strongly Agree
2UTDABE9	3.508	0.671	Strongly Agree
2UTDABE10	3.468	0.656	Agree
Average	**3.553**	**0.078**	**Strongly Agree**

Table 16 revealed that respondents strongly agree that Brand Voice is among the most influential audio branding elements in purchase decisions. Using a Nescafé ad to evaluate their understanding, the question "Do you recognize this advertisement?" received one of the highest ratings (M = 3.731, SD = 0.622), followed by "Do you instantly recognize the brand involved in the advertisement?" which confirmed respondents could immediately identify the brand upon hearing the ad. However, some respondents were less aware that the sounds used in the ad constituted audio branding, as shown by the slightly lower rating for "Were you aware that the advertisement used audio branding?" (M = 3.538, SD = 0.640). Overall, Brand Voice achieved an average rating of M = 3.645 (SD = 0.076), indicating strong agreement about its effectiveness in making brands more recognizable. These findings support Hypothesis 3, confirming that Brand Voice significantly influences Millennials' purchasing decisions.

Table 16. Differences in the respondents' understanding of the audio branding element Brand Voice

	Mean	SD	Interpretation
2UTDABE11	3.731	0.527	Strongly Agree
2UTDABE12	3.638	0.546	Strongly Agree
2UTDABE13	3.704	0.499	Strongly Agree
2UTDABE14	3.615	0.615	Strongly Agree
2UTDABE15	3.538	0.640	Strongly Agree
Average	**3.645**	**0.076**	**Strongly Agree**

Table 17 showed that audio logos significantly influence Millennials' purchase decisions by serving as unique brand identifiers that enhance recognition. The question "Do you recognize this advertisement?" received the highest rating (M = 3.767, SD = 0.489), with strong agreement across other questions, such as recognizing sounds (M = 3.708, SD = 0.536) and instantly identifying the brand (M = 3.708, SD = 0.524). Although the question "Were you aware that the advertisement used audio branding?" scored slightly lower (M = 3.601, SD = 0.622), it still indicated strong agreement. The audio logo had a high overall mean (M = 3.645) and the lowest standard deviation (SD = 0.076), suggesting it elicited consistent and well-defined responses from participants. These findings confirm Hypothesis 4, demonstrating that audio logos in YouTube ads have a significant impact on Millennials' purchasing decisions.

Table 17. Differences in the respondents' understanding of the audio branding element Audio Logo

	Mean	SD	Interpretation
2UTDABE16	3.767	0.489	Strongly Agree
2UTDABE17	3.708	0.536	Strongly Agree
2UTDABE18	3.708	0.524	Strongly Agree
2UTDABE19	3.688	0.556	Strongly Agree

	Mean	SD	Interpretation
2UTDABE20	3.601	0.622	Strongly Agree
2UTDABEAVE	3.623	0.409	
Average	**3.645**	**0.076**	**Strongly Agree**

Table 18 highlights the significant influence of audio branding on Millennials' purchase decisions, aligning with research by Nufer and Moser (2018) and Fraile et al. (2021), which emphasizes the role of auditory stimuli in enhancing brand identity and memory. The results show a strong overall agreement (mean = 3.539, SD = 0.427) that audio branding positively impacts purchasing decisions, with consistent responses across participants. Statements like "The sounds used are catchy and easy to remember" (mean = 3.605, SD = 0.554) and "The advertisement is much more memorable and effective with the help of sounds" (mean = 3.658, SD = 0.509) garnered particularly strong agreement, demonstrating the effectiveness of elements like jingles and sound logos in improving brand recall. Although slightly lower ratings were observed for statements like "I will now buy from the brand in the advertisement more often" (mean = 3.266, SD = 0.768), the results still indicate a positive impact. These findings confirm that audio branding elements significantly influence Millennials' purchasing behavior.

Table 18. Influence of Audio Branding on Purchase Decision

	Mean	SD	Interpretation
IOABOPD1	3.266	0.768	Agree
IOABOPD2	3.279	0.736	Agree
IOABOPD3	3.605	0.554	Strongly Agree
IOABOPD4	3.631	0.504	Strongly Agree
IOABOPD5	3.658	0.509	Strongly Agree
IOABOPD6	3.488	0.661	Agree
IOABOPD7	3.429	0.647	Agree
IOABOPD8	3.545	0.634	Strongly Agree
IOABOPD9	3.578	0.558	Strongly Agree
IOABOPD10	3.565	0.600	Strongly Agree
IOABOPD11	3.508	0.666	Strongly Agree
IOABOPD12	3.452	0.655	Agree
IOABOPD13	3.598	0.549	Strongly Agree
IOABOPD14	3.638	0.521	Strongly Agree
IOABOPD15	3.621	0.525	Strongly Agree
IOABOPD16	3.538	0.645	Strongly Agree
IOABOPD17	3.498	0.630	Agree
IOABOPD18	3.664	0.533	Strongly Agree
IOABOPD19	3.578	0.593	Strongly Agree
IOABOPD20	3.648	0.537	Strongly Agree
Average	**3.539**	**0.427**	**Strongly Agree**

Table 19 highlights the effectiveness of repeated exposure to audio branding, with an overall average of M = 3.453 and SD = 0.447. This result indicates that respondents agreed repeated exposure to advertisements is effective in reinforcing brand recall and influencing purchase decisions. The strongest agreement was observed for "Watching the same advertisement multiple times makes me remember the brand" (M = 3.761), followed by "Hearing the same sound multiple times from an advertisement tells me what brand it is" (M = 3.701). These results suggest that repeated visual and auditory exposure helps respondents associate specific sounds with the corresponding brands or products, enhancing recall. The statement "I usually skip the advertisement if I've already watched it a couple of times" received a lower but still significant rating (M = 3.512), indicating that while respondents agree on the effectiveness of repetition, there is a tendency to skip ads after repeated viewings. On the

other hand, "I easily forget the advertisement if I don't see it for quite some time" received the lowest rating (M = 3.236), suggesting that consistent exposure is crucial for maintaining brand recall over time. These findings confirm that repeated exposure to audio branding elements positively affects memory retention and purchase decisions among Millennials.

Table 19. Differences in the respondents' evaluation of the effectiveness of repetitive exposure to advertisements

	Mean	SD	Interpretation
EOREOA1	3.761	0.443	Strongly Agree
EOREOA2	3.701	0.446	Strongly Agree
EOREOA3	3.236	0.813	Agree
EOREOA4	3.512	0.641	Strongly Agree
EOREOA5	3.429	0.647	Agree
EOREOA6	3.339	0.701	Agree
EOREOA7	3.402	0.654	Agree
EOREOA8	3.435	0.693	Agree
EOREOA9	3.452	0.623	Agree
EOREOA10	3.262	0.796	Agree
Average	**3.453**	**0.447**	**Agree**

Table 20 confirms that audio branding significantly influences Millennials' emotions, shaping their purchase decisions. Respondents strongly agreed that audio branding fosters a connection with brands, as reflected in the highest-rated statement, *"Hearing an advertisement makes me feel more connected to the brand"* (M = 3.419, SD = 0.651).

Positive emotional effects were also evident in statements like *"Hearing an advertisement uplifts my mood, making me want to purchase from that brand"* (M = 3.279, SD = 0.722). However, audio branding can evoke negative emotions, such as annoyance *("Watching the same advertisement annoys me"*, M = 3.262, SD = 0.767) and sadness (*"Some advertisements make me sad"*, M = 3.159, SD = 0.809).

The overall mean (M = 3.276, SD = 0.566) suggests a consistent agreement that audio branding elements effectively influence emotional responses, which, in turn, impact purchase decisions. These findings support the hypothesis that audio branding has a significant role in driving consumer behavior.

Table 20. Differences in the respondents' evaluation of the influence of audio branding on emotional decision-making process

	Mean	SD	Interpretation
IOABOEDMP1	3.419	0.651	Agree
IOABOEDMP2	3.262	0.753	Agree
IOABOEDMP3	3.262	0.767	Agree
IOABOEDMP4	3.279	0.722	Agree
IOABOEDMP5	3.159	0.809	Agree
Average	**3.276**	**0.566**	**Agree**

Table 21 reveals that audio branding moderately influences rational decision-making among Millennials, with an overall mean score of M = 3.324 (SD = 0.609), indicating agreement. Respondents strongly agreed with the statement, *"Hearing an advertisement makes it easier to remember brands"* (M = 3.681, SD = 0.467), highlighting audio branding's effectiveness in enhancing brand recall and supporting logical evaluations. Other items, such as *"I find myself spending more on brands that I've watched from an advertisement"* and *"Hearing an advertisement makes me want to purchase from that brand immediately,"* showed moderate agreement, with mean values ranging from 3.140 to 3.306. These findings suggest that while audio branding supports rational considerations like memory and comparison, its influence on direct purchasing behavior is more nuanced. The results support the hypothesis that audio branding elements significantly impact Millennials' purchase decisions, particularly through features like jingles and audio logos that facilitate logical evaluations of brands.

Table 21. Influence of Audio Branding on Rational Decision-Making Process

	Mean	SD	Interpretation
IOABORDMP1	3.681	0.467	Strongly Agree
IOABORDMP2	3.302	0.752	Agree
IOABORDMP3	3.140	0.857	Agree
IOABORDMP4	3.306	0.716	Agree
IOABORDMP5	3.193	0.842	Agree
Average	**3.324**	**0.609**	**Agree**

Table 22 presents the results of a chi-square test examining the relationship between audio branding elements (jingle, brand music, brand voice, and audio logo) and Millennials' purchase decisions. The results show significant relationships for all elements, with jingle ($\chi^2 = 38.318$, $p = 0.000$), brand music ($\chi^2 = 53.545$, $p = 0.000$), brand voice ($\chi^2 = 282.939$, $p = 0.000$), and audio logo ($\chi^2 = 53.678$, $p = 0.000$) all having strong influences on purchase decisions. In each case, the null hypothesis was rejected in favor of the alternative, confirming that these audio branding elements significantly affect Millennials' purchasing behavior.

Table 22. Influence of Audio Branding to Purchase Decisions

Variable		Chi-Square Test	Significant Value	Decision	Interpretation
Purchase Decision	Jingle	38.318	0.000	Reject	Significant
	Brand Music	53.545	0.000	Reject	Significant
	Brand Voice	282.939	0.000	Reject	Significant
	Audio Logo	53.678	0.000	Reject	Significant

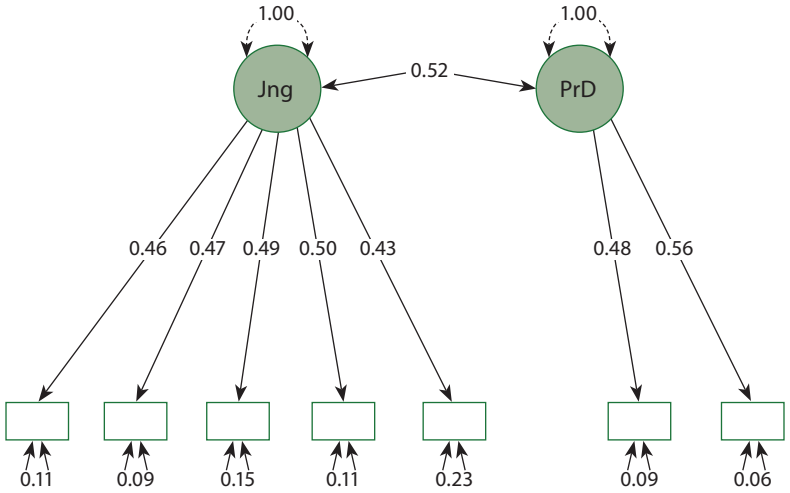
Table 23 reveals significant insights through the Comparative Fit Index (CFI) and Root Mean Square Error (RMSE) values for each audio branding element. For jingle, the CFI of 0.979 indicates a strong model fit, suggesting jingles in YouTube ads significantly influence Millennial purchasing decisions, with an RMSE of 0.080 confirming a good fit. Brand music, with a CFI of 0.964, also demonstrates a strong influence, though its RMSE of 0.102 slightly exceeds the acceptable threshold, hinting at some variability in its impact. Despite this, the high CFI supports the hypothesis that brand music affects purchase decisions. Brand voice, however, shows a weaker model fit with a CFI of 0.810, below the acceptable range, indicating less accuracy in capturing its impact on purchases. Finally, the audio logo has a CFI of 0.974, indicating an excellent fit, while its RMSE of 0.102 is slightly above the threshold, suggesting a generally acceptable model with minor variability in its effectiveness. Overall, the results support the hypothesis that audio branding elements like jingles, brand music, and audio logos significantly impact purchase decisions, though some nuances in context and frequency of use may influence their effectiveness.

Table 23. Other fit measures:

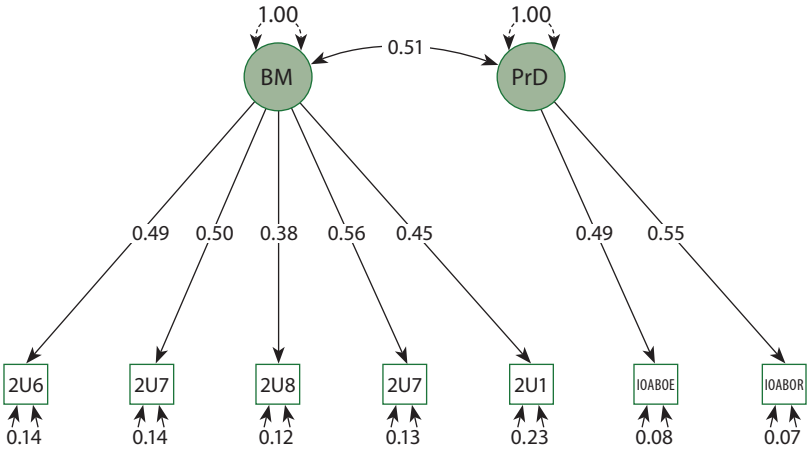
Variable		CFI	Interpretation	RMSE	Interpretation
Purchase Decision	Jingle	0.979	The Model is fit.	0.080	The Model is fit.
	Brand Music	0.964	The Model is fit.	0.102	The Model does not fit.
	Brand Voice	0.810	The Model is fit.	0.263	The Model does not fit.
	Audio Logo	0.974	The Model is fit.	0.102	The Model does not fit.

The Jingle-PD Model Plot shows a moderately strong relationship between the audio branding element "Jingle" and the "Purchase Decision" (rating of 0.52), suggesting that jingles have a noticeable but not overwhelming impact on altering Millennials' purchase decisions. Of the five elements related to Jingle, only one demonstrated a strong relationship, while four showed weak connections. However, this does not mean these elements are insignificant. Additionally, the Purchase Decision variable is

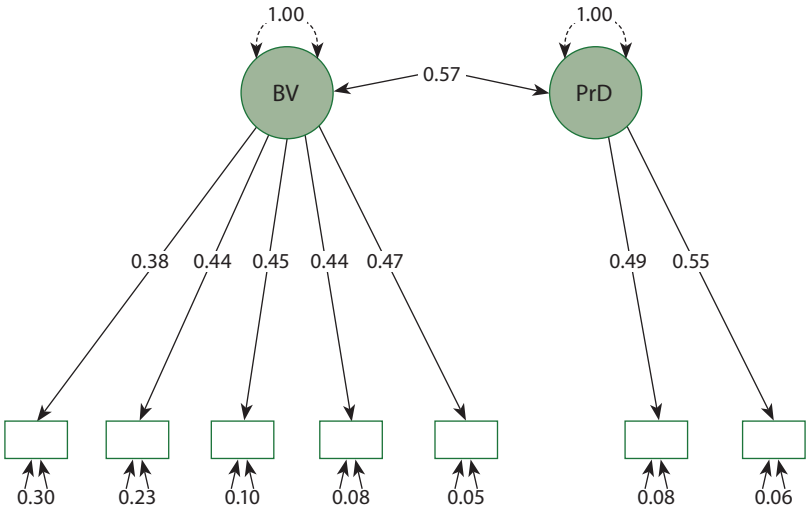
influenced by two factors: the emotional and rational decision-making processes. A stronger relationship (rating of 0.56) is observed between the rational decision-making process and the purchase decision, indicating that while jingles may help Millennials recall a brand or product, their final purchase decision is more likely to be influenced by rational considerations rather than emotional responses.



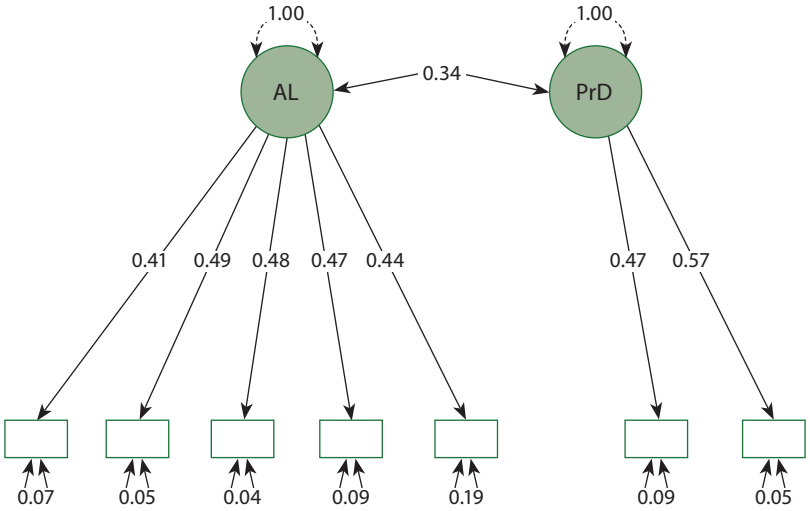
The Brand Music-PD Model Plot reveals a moderate to substantial relationship (0.51) between Brand Music (BM) and Purchase Decision (PD), indicating that brand music has a noticeable impact on purchasing decisions, though not the sole influencing factor. Within the BM construct, the component 2U9 shows a strong relationship with a factor loading of 0.56, while others like 2U6 (0.49) and 2U8 (0.38) have weaker associations. For the PD construct, the variables 2U1 and IOABOE exhibit a strong relationship with a loading of 0.55, while IOABOR (0.07) and 2U8 (0.08) show minimal influence. The moderate correlation suggests that while brand music is effective in capturing attention and fostering brand associations, its influence varies depending on musical elements like rhythm and melody, as well as exposure frequency (Miguelez-Juan, 2022). The higher factor loading of 2U9 suggests that certain musical features may have a stronger impact on consumer behavior, reinforcing the idea that emotional or positive associations from music can drive purchasing decisions (Trianto et al., 2022). Overall, the model supports the hypothesis that brand music in YouTube ads influences purchase decisions, though it is one of several contributing factors.



The Brand Voice Model Plot indicates a strong relationship between Brand Voice and Purchase Decision, suggesting that brand voice significantly influences millennials' purchasing decisions. However, none of the five elements connected to Brand Voice showed a strong relationship with the variable, implying that brand voice in audio branding is frequently used but might still be somewhat novel to millennials. These elements, associated with questions similar to those for other audio branding elements like Jingle, Brand Music, and Audio Logo, displayed weak correlations, suggesting that while brand voice is influential, its integration may still be evolving. Additionally, the correlation between Purchase Decision and the emotional decision-making process was weak, highlighting that millennials tend to rely more on rational thinking than emotional responses when making purchase decisions.



In analyzing the association of Audio Logo with various variables, it is evident that while Audio Logos can influence consumers' willingness to pay, their overall impact on purchasing decisions is relatively weak. With a value of 0.34, the correlation between Audio Logo and purchasing decisions suggests that while Audio Logos do have a positive effect, their influence is not particularly strong. This highlights the complex role of audio branding in consumer choices, where Audio Logos can contribute to brand image and quality perception but are not the sole determinants of buying decisions. Additional correlations indicate weak relationships, with values of 0.41, 0.49, and 0.48, suggesting that while some respondents recognize the audio logos in advertisements and associate them with the brand, the recognition is not instant or frequent. A value of 0.47 reveals that consumers typically encounter these advertisements only occasionally, and 0.44 indicates that musical instruments in the background are a common feature. Furthermore, Audio Logos have a weak influence on emotional decision-making, with a value of 0.47, signifying that while they can evoke emotions, they are not the dominant factor in emotional responses. On the other hand, the highest correlation of 0.57 between purchasing decisions and rational decision-making processes indicates that Audio Logos significantly influence consumers' rational thought processes, making them more likely to rely on logic when making purchasing choices.



Discussion of Results

RQ1: To determine the audio branding elements that influences the millennials' purchase decision The thesis data, supported by Fraile et al. (2021), highlights key audio branding elements that influence millennials' purchase decisions. Fraile's study found commercial songs, brand songs, and audio logos to be the most common audio branding elements in Spanish radio advertising. The thesis data shows strong correlations, with a CFI of 0.979 for jingles and 0.974 for audio logos, indicating their significant impact on purchase decisions. Additionally, the Brand Voice-PD Model Plot shows a 0.57 relationship, while the Jingle-PD Model Plot shows a 0.52 correlation, further emphasizing the influence of these elements on consumer behavior.

RQ2: To determine which audio branding element has the most influence on the millennials' purchase decision The data identifies brand voice and audio logos as the most influential audio branding elements on millennials' purchase decisions, both scoring the highest mean of 3.645. Jingles (3.600) and brand music (3.553) also impact purchasing decisions, but to a lesser extent. This aligns with Fraile et al. (2021), where commercial songs made up 44% of the audio branding elements. Additionally, Li et al. (2022) found that audio logos significantly influence consumers' emotions and decision-making, while Scott (2022) noted that specific tones in sound logos can increase consumers' willingness to pay. Tran (2020) emphasized that brand voices strengthen connections and emotions, suggesting that well-chosen audio branding enhances consumer influence and purchasing pleasure.

RQ3: To ascertain the effects of repetitive exposure to audio branding on the purchase decision of millennials According to Nufer and Moser (2018), 83% of advertisements focus on visual marketing, while only 11% rely on audio marketing. Ayada and Ragab (2024) highlight that audio branding enhances brand recall, even without visual engagement. Their study found that Netflix achieved a 94% recognition rate due to its strategic use of audio branding across its films and series. In summary, prolonged exposure to advertisements, whether seen or heard, influences millennials' purchase decisions. The sound of an advertisement can trigger immediate brand recall, potentially altering the decision-making process.

RQ4: To know the beneficial and unfavorable effects of audio branding on millennials' emotional and rational decision-making processes when it comes to purchasing Brands increasingly use audio branding to enhance recognition and influence consumer behavior. Wong (2018) suggests that consumer preferences for audio branding can be assessed by evaluating emotional and rational reactions to different brand alternatives. This study explores the impact of audio branding on millennials, focusing on how it affects emotional and rational decision-making processes in purchasing. Music, especially in YouTube ads, plays a key role in attracting consumers through emotions (Trianto et al., 2022). The right music can influence mood and, consequently, purchase decisions. Effective audio branding can strengthen emotional connections with brands.

Additionally, audio branding elements influence rational decision-making by boosting brand recall (Ayada & Ragab, 2024) and embedding the brand in memory through entertainment (Tran, 2020). In conclusion, consistent audio branding can positively impact millennials' purchasing decisions, supporting Wong's (2018) view that audio branding shapes consumers' perceptions of a brand's image.

RQ5: To discern if audio branding elements have a significant influence on the purchase decision of millennials A chi-square test was used to assess whether audio branding elements significantly influence millennials' purchase decisions. The test showed high values and consistent significant results of 0.000, leading to the rejection of the null hypothesis. This indicates that elements such as Jingle, Brand Music, Brand Voice, and Audio Logo have a significant impact on millennials' purchasing decisions. This finding supports Rodero & Larrea (2020), who noted that music in sound branding can alter consumers' emotions, purchase decisions, and perceptions of brand personality. Sadek (2023) also emphasized how music in YouTube ads enhances memory recall, fostering a connection between consumers and brands, which influences purchases.

Additionally, Ayada and Ragab (2024) highlighted that audio logos are key for brand recall and recognition, a crucial factor in purchase decisions (Wong, 2018). The survey results further validate this, showing that audio branding elements affect millennials' buying behavior. The study's findings align with the Stimulus-Organism-Response (SOR) theory, where audio branding acts as a stimulus influencing consumer responses, particularly purchasing behavior.

Conclusions

The study concluded that most respondents, aged 28 to 31, were more exposed to YouTube advertisements due to their adaptability to technology, making them more familiar with audio branding. Millennials demonstrated a strong awareness of audio branding and its elements. Among the four audio branding elements, Brand Voice and Audio Logo (both with a mean of 3.645) were frequently featured in YouTube ads, suggesting they were the most effective in reinforcing brand recall.

The study also found that audio branding significantly influenced Millennials' purchase decisions. Repeated exposure to YouTube ads impacted their emotional and rational decision-making processes, indicating that audio branding played a crucial role. It is recommended that brands incorporate various audio branding elements, beyond just Brand Voice and Audio Logo, to enhance the effectiveness of their YouTube advertisements, especially considering that audio branding is still less recognized compared to visual branding, particularly among Millennials who are less familiar with the technology.

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REFERENCES

1. 46% of Filipino millennials are shiftings toward the BPO industry. (2022, June 21). Outsource Accelerator. https://news.outsourceaccelerator.com/46-of-filipino-millennials-are-shiftings-toward-the-bpo-industry/
2. Abhishek, M., & Pradhan, V. (2022). Impact of YouTube on millennials' buying decision. CARDIOMETRY. https://doi.org/10.18137/cardiometry.2022.25.734742
3. Ajzen, I. (1991). The theory of planned behavior. Organizational Behavior and Human Decision Processes, 50(2), 179–211. https://doi.org/10.1016/0749-5978(91)90020-T
4. ARN Media, Southern Cross Austereo, & Neurolab. (2022, October). The power of sonic branding. Commercial Radio. Retrieved May 5, 2024, from https://www.commercialradio.com.au/RA/media/General/Documents/1530-The-Science-of-Audio-Sonic-Branding.pdf?ext=.pdf
5. Audio branding: The what, the why, and the ROI. (2024). The Business Journals. https://www.bizjournals.com/cincinnati/news/2024/01/20/audio-branding-the-what-the-why-and-the-roi.html
6. Ashraf, M. (2023). Theory of boundedly rational planned behavior: A new model. Zagreb International Review of Economics and Business. https://doi.org/10.2478/zireb-2023-0001
7. Ayada, W., & Ragab, D. (2024). The role of sonic logos in enhancing brand recall and recognition. International Design Journal, 14(2), 439–447. https://doi.org/10.21608/idj.2024.340785
8. Babiera, T. M., & Parado, A. J. (2023). The influence of YouTube advertisement on buying preferences of BSBA students in SLSU. figshare. https://doi.org/10.6084/m9.figshare.23726646
9. Barcelos, R. H., Dantas, D. C., & Sénécal, S. (2018). Watch your tone: How a brand's tone of voice on social media influences consumer responses. Journal of Interactive Marketing, 41, 60–80. https://doi.org/10.1016/j.intmar.2017.10.001
10. Barnes, S. J. (2023). Smooth talking and fast music: Understanding the importance of voice and music in travel and tourism ads via acoustic analytics. Journal of Travel Research. https://doi.org/10.1177/00472875231185882
11. Catic, L., & Poturak, M. (2022). Influence of brand loyalty on consumer purchase behavior. International Journal of Research in Business and Social Science, 11(8), 83–91. https://doi.org/10.20525/ijrbs.v11i8.2024
12. Chen, G., Li, Y., & Sun, Y. (2023). How YouTubers make popular marketing videos? Speech acts, move structure, and audience response in YouTube influencer marketing videos. Sage Open, 13(1). https://doi.org/10.1177/21582440231152227
13. Cher, M. F. E., & Arumugam, V. (2019). The factors affecting the effectiveness of online video advertising: A study on Malaysian consumers' perspective towards ads on YouTube. Global Business & Management Research, 11(2), 167–184.
14. Clear music on sonic branding - Clear Music. (2024, February 21). Clear Music - Music for Entertainment and Advertising. https://clearmusic.nl/what-is-sonic-branding/
15. Dili, R., Buncad, Y. J., Perez, Y. J., & Velasco, R. J. (2022). BPO industry in achieving socio-economic development inclusiveness and local governance in the Philippines. Management of Sustainable Development, 14(2), 22–26. https://doi.org/10.54989/msd-2022-0012
16. Dudziak, A., Stoma, M., & Osmólska, E. (2023). Analysis of consumer behavior in the context of the place of purchasing food products with particular emphasis on local products. International Journal of Environmental Research and Public Health, 20(3), 2413. https://doi.org/10.3390/ijerph20032413
17. Ebrahimi, S., Vahabi, H., Prockup, M., & Nieto, O. (2018). Predicting audio advertisement quality. Proceedings of the ACM International Conference on Multimedia Retrieval (pp. 153-161). https://doi.org/10.1145/3159652.3159701
18. Fraile, E. B., Jiménez, A. M. E., Veloso, M. L. B., & Payet, A. F. (2021). Sonic identity and audio branding elements in Spanish radio advertising. Analisi: Quaderns de Comunicació i Cultura, 2021(65), 103–114. https://ddd.uab.cat/pub/artpub/2021/258305/analisi_a2021n65p103.pdf

19. Galster, N. (2015). Has online audio distribution destroyed music as we know it? Medium. https://medium.com/lab-work/has-online-audio-distribution-destroyed-music-as-we-know-it-56e4e916f737
20. Global YouTube user age & gender distribution 2024. (2024, February 13). Statista. https://www.statista.com/statistics/1287137/youtube-global-users-age-gender-distribution/
21. Guy, M. J., & Hjorth, T. S. L. (2020). Audio branding: An exploratory study of audio branding as a strategic means to strengthen brand identity. Copenhagen Business School. https://research-api.cbs.dk/ws/portalfiles/portal/62174379/818161_103211_124919.pdf
22. Hogan, T. (n.d.). Audio branding creates sonic boom: 5 elements of an audio brand. Brandience. https://www.brandience.com/insights/audio-branding-creates-sonic-boom-5-elements-of-an-audio-brand
23. Hong, Y. J., Choi, B., & Lee, K. (2022). When background music in audiovisual advertisements can boost the perceived competence of the advertised brands – An empirical study from South Korea. Asia Pacific Journal of Marketing and Logistics. https://doi.org/10.1108/apjml-03-2022-0275
24. Key elements of sound branding. (2024). Sound Strategy. https://sound-strategy.com/en/scg-elements/
25. Koththagoda, K. C., & Herath, H. M. R. P. (2018). Factors influencing online purchasing intention: The mediation role of consumer attitude. Journal of Marketing and Consumer Research. https://www.iiste.org/Journals/index.php/JMCR/article/view/40727
26. Kumar, P. (2023). To study the opinion of the YouTube visitors towards advertisements and its influence among viewers. Proceedings of the International Conference on Science and Engineering, 11, 1467-1472. https://doi.org/10.52783/cienceng.v11i1.300
27. Lakens, D. (2021, January 4). Sample Size Justification. https://doi.org/10.1525/collabra.33267
28. Miguélez-Juan, B. (2022). Music branded content: The video clip as an advertising medium at the service of automotive brands. In V. S. Litovchenko (Ed.), Advances in Digital Marketing and eCommerce (pp. 99-116). https://doi.org/10.4018/978-1-6684-3971-5.ch005
29. Munidewi, I. A. B., & Pradipa, N. A. (2022). Mengisi kesenggangan ruang riset akuntansi dan auditing melalui paradigma interpretif dengan te. Proceedings of the International Conference on Science and Engineering, 11, 1467-1472.
30. Nicastro, D., & Inguanez, F. (2019). Brand usage detection via audio streams. IEEE International Conference on Consumer Electronics (ICCE-Berlin), 1-3. https://doi.org/10.1109/ICCE-BERLIN47944.2019.8966180
31. N., Z. Z., W., J., Catherine, H., & R., R. R. (2021). Measuring the impact of YouTube ads on the buying behavior of consumers in Chennai through brand recognition and motivation. Webology, 18(5), Article 53. https://www.webology.org/data-cms/articles/20220210085549pmwebology%2018%20(5)%20-%2053%20.pdf
32. Nufer, G., & Moser, H. (2018). The sound of brands. Research Papers in Economics. https://doi.org/10.15496/publikation-26654
33. Osborne, M., & Rubinstein, A. (2023). Consumer preferences. In M. Osborne & A. Rubinstein (Eds.), Introduction to Economic Theory (pp. 299-315).
34. Paramitha, S., Yatigammana, K., & Gunathunge, K. L. N. K. (2021). The impact of YouTube advertisement value on purchase intention of YouTube users in Western Province, Sri Lanka. Journal of Business and Technology, 5, 69. https://doi.org/10.4038/jbt.v5i0.54
35. Pauliene, R., Tamasevicius, V., & Gaivenis, S. (2020). Recommendations' impact on customer behavior and visiting catering organizations. Academic Journal of Interdisciplinary Studies, 9(1), 16. https://doi.org/10.36941/ajis-2020-0002
36. Porras Velázquez, N. R. (2018). Percepción y memoria en los jingles publicitarios: Reflexiones desde la psicología de la publicidad. Revista Electrónica Psyconex, 10(16), 1-19. https://revistas.udea.edu.co/index.php/Psyconex/article/view/334761

37. PR Newswire. (2022, April 12). Voices, the leading voice marketplace, calls audio branding a necessity for any successful business in 2022. PR Newswire US.
38. PR Newswire. (2022, February 10). Made Music Studio announces slate of promotions and hires to meet growth in audio and sonic branding business. PR Newswire US.
39. Rakotoarisoa, F., Laila, N., Premananto, G., Abbas, A., & Rakotoarisoa, M. H. S. (2022). Explaining littering prevention among park visitors using the theory of planned behavior and norm activation model. International Journal of Geoheritage and Parks, 11. https://doi.org/10.1016/j.ijgeop.2022.11.002
40. Ramadhany, A. K. (2023). Tuturan imperatif dalam iklan layanan masyarakat pada media YouTube (The imperatives in public service advertising on YouTube media). Jurnal Bahasa, Sastra, dan Pembelajarannya (JBSP). https://doi.org/10.20527/jbsp.v13i1.11069
41. Rathee, R., & Rajain, P. (2020). Influence of music on consumer behaviour: An experimental study. Abhigyan, 38(1), 30–39. https://doi.org/10.56401/abhigyan_38.1.2020.30-39
42. Rodero, E., & Larrea, O. (2020). Audio design in branding and advertising. In Handbook of Research on Social and Cultural Dynamics in Gaming and E-Sports (pp. 85-104). https://doi.org/10.4324/9781003009276-5
43. Sadek, S. (2023). YouTube users' behavioral responses towards musical ads: A comparative study vs voice over ads. Journal of Design Sciences and Applied Arts (Print). https://doi.org/10.21608/jdsaa.2023.182931.1244
44. Sharafat, H. (2019). Sensory marketing strategies and consumer behavior: Sensible selling using all five senses.
45. Sheer, M., Ullah, F., & Hakeem, L. (2023). An investigation of Teachers' Attitudes towards Low academic achievers: A perspective of Diagnostic Remedial Teaching at Federal Government Educational Institutions. International Journal of Multidisciplinary: Applied Business and Education Research, 4(11), 4089-4100. https://doi.org/10.11594/ijmaber.04.11.26
46. Sinaga, R. (2022). An interpretive and critical paradigm study of the "Gerakan Ekonomi Benteng" in Indonesia. Journal of Southeast Asian Economies, 1(1). https://doi.org/10.56282/sjei.v1i1.201
47. Scott, S., Sheinin, D., & Labrecque, L. (2022). Small sounds, big impact: Sonic logos and their effect on consumer attitudes, emotions, brands, and advertising placement. Journal of Product & Brand Management, 31. https://doi.org/10.1108/JPBM-06-2021-3507
48. Šostar, M., & Ristanović, V. (2023). Assessment of influencing factors on consumer behavior using the AHP model. Preprints. https://doi.org/10.20944/preprints202305.2221.v1
49. Soti, R. (2022, June 26). The impact of advertising on consumer behavior. World Journal of Advanced Research and Reviews. https://wjarr.com/content/impact-advertising-consumer-behavior
50. Stone, S. (2023, December 11). 2024 YouTube usage statistics and habits (by generation). The Shelf Full-Service Influencer Marketing. https://www.theshelf.com/the-blog/youtube-user-habits/
51. Taherdoost, H. (2022b). Designing a questionnaire for a research paper: A comprehensive guide to design and develop an effective questionnaire. Asian Journal of Managerial Science/Asian Journal of Managerial Science, 11(1), 8–16. https://doi.org/10.51983/ajms-2022.11.1.3087
52. Tran, N. (2020). How sound branding influences customer's perception: Case company: Blinkist, non-fiction book summary application. Theseus. https://www.theseus.fi/handle/10024/346600
53. Vahedi, D. (2023). The effectiveness of YouTube as a digital advertising platform: Exploring the role of YouTube in digital advertising and its impact on consumer decision making. Theseus. https://www.theseus.fi/bitstream/handle/10024/808389/Vahedi_Daniel.pdf?sequence=2&isAllowed=y
54. Vertiola, M. (2018). To what extent does YouTube marketing influence the consumer behaviour of a young target group? Theseus. https://www.theseus.fi/handle/10024/148638#:~:text=The%20results%20tell%20that%20YouTube,and%20attitudes%20of%20the%20youth.

55. Vidal, M. (2018). Branding Sonoro Sonokey: El método de impulso emocional y mnemotécnico para las marcas. Barcelona: Editorial UOC.

56. Vidal-Mestre, M., Freire Sanchez, A., Calderón-Garrido, D., Faure-Carvallo, A., & Gustems, J. (2022). Audio identity in branding and brand communication strategy: A systematic review of the literature on audio branding. El Profesional de la Información, 31, e310504. https://doi.org/10.3145/epi.2022.sep.04

57. What are some examples of successful sound branding strategies in different industries and markets? (2024). LinkedIn. https://www.linkedin.com/advice/0/what-some-examples-successful-sound-branding-strategies

58. Wong, A. (2018). Sound branding: The role of music in consumer perceptions, behaviours, and practitioner beliefs. White Rose eTheses Online. https://etheses.whiterose.ac.uk/22404/

59. Yang, Q., & Wu, X. (2022). Customer preference analysis from online reviews by a 2-additive Choquet integral-based preference disaggregation model. Technological and Economic Development of Economy, 29, 1-27. https://doi.org/10.3846/tede.2022.17972

60. Yang, Y., Yang, R., Song, X., & Tang, Y. (2021). How does music-message congruency in online video advertisements enhance consumers' behavior intentions? A serial multiple mediator model. IEEE Access, PP, 1-1. https://doi.org/10.1109/ACCESS.2021.3093353

61. Zips, S., & Riedl, J. (2019). Selected personal related and brand related predictors of the success of audio branding.

False Reviews: A Multi-Generational Study of Online Shoppers' Confidence in Assessing Review Authenticity on Selected E-commerce Platforms

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Abstract

Today, consumers rely on and contribute online reviews to be informed about their purchases. However, fake reviews have continued to plague review systems in e-commerce platforms, exposing users to fraudulent reviews, compromising their shopping experience. Utilizing the 3S model of Information Trust, through a descriptive comparative research design with a quantitative approach to compare the assessments of Generations X, Y, and Z. Data was collected from 332 respondents in Quezon City using stratified purposive sampling. The findings indicate that while younger respondents, particularly Generation Z, favor the quality and relevance of information in reviews, these preferences do not lead to statistically significant differences in assessment across generations. While slight inclinations among generations regarding the evaluation of review features, overall assessments show commonalities. Differences were found mostly in their experiences and trust judgement. The study recommends strategies to enhance protection against fake reviews, including improving reviewer validity, moderating surface features to ensure quality content, promoting awareness of fraudulent strategies, and offering insured transactions. Through these, e-commerce platforms can improve the shopping experience, increase user trust, and mitigate the effects of inauthentic reviews on consumers.

Keywords: false reviews; e-commerce platforms; online shopping; online reviews.

Introduction

In e-commerce, user reviews can play a significant role in determining the success of a product or the seller as a whole. Consumers in an online scenario rely only from information provided by sellers and other customers before making decisions about any product or service. One such way being the reviews that are posted on the items. This is crucial since it directly affects consumers' behavior towards the product or service being offered. (Mohawesh et. al, 2021). User reviews reflect the value & credibility of the product & the seller in the world of e-commerce (Kumar, 2018). However, each generation has been found to navigate digital media differently than their cohorts. Attributed to the differences in technology present in each generation's main developmental stages. (Velasco, 2022)

And as e-commerce grows, the prevalence of fake online reviews grows with it. Fake reviews tend to be more influential and complex in structure which has made it difficult to properly identify for both machines and people. And as of now there are no definite guidelines and accepted guidelines to detecting these kinds of reviews or knowing the real difference (Wu, et al. 2020). And yet despite growing public concern over disinformation in e-commerce, the industry does not seem to halt in its expansion catering to users from all ages. It is imperative to understand how the online shoppers from different generational cohorts with varying digital literacy levels approach this risk and identify their overall confidence and trust judgement of review systems on online shopping platforms.

And while there is existing literature on the digital misinformation and its effects on e-commerce and other fields, there was very little information regarding the differences between generational cohorts and their approaches in handling this issue. This study aims to identify online shoppers' confidence in assessing review authenticity and their assessment of review features on e-commerce platforms through a descriptive comparative research design with a quantitative approach to compare the assessments of Generations X, Y, and Z. To understand the pain points or areas of improvement that can benefit each generational cohort as well as detect any significant differences between them, and what recommendations can be made for improvement based on the findings.

Literature Review

E-commerce Platforms in the Philippines

E-commerce platforms have seen increased usage during the COVID-19 pandemic with consumers shifting towards online shopping for convenience and safety reasons, and extending to the post-pandemic age (Capistrano et al., 2023). Factors such as celebrity endorsements & social media influencers also played a big role in influencing the sudden spike in usage as well as influencing online shoppers' purchase behavior on these platforms (Pablo et al., 2022).

There are different e-commerce platforms available in the Philippines but the most commonly used as of today is Shopee. With study results showing that most respondents preferred Shopee over Lazada. Sitting at 86.67% of respondents stating Shopee as their primary and more preferred platform (Dulay et. al, 2022).

Shopee and Lazada are known for their monthly mega-sales, which are the crucial drivers for the platforms (Dulay et. al, 2022), with Shopee being determined to be the most prominent and best scoring online shopping platform in terms of the following factors: efficiency, fulfillment, system availability, & privacy in the National Capital Region with a score of 9.132, followed by LAZADA with a score of 6.032 (Rada et al., 2022).

Shopee stands out in the Philippines' online shopping scene thanks to its large market share and high customer involvement. Becoming one of the most frequently visited and used e-commerce site and application in Southeast Asia, having been emphasizing simplicity and security in its lifetime (Khaw, 2023). Zhang (2022), as cited in Putri et al., (2024), highlighted the growth of another online shopping platform: Tiktok that was rising in usage and popularity, attributing it to its engaging experience, as well its capability to provide sellers or influencers marketing opportunities directly providing media content for prospective buyers from the perspective of current users due to its nature as a short-form content social media platform and the live streaming features available can be used to directly interact with users. (Li et al., 2023)

Reliability of Product Information on Online Shopping Platforms

When shopping online, there is a distinct lack of physical interaction between a buyer and the product. The buyer cannot hold, inspect, or even see the product as it would appear in real life only until after they have already purchased the product and it is delivered to their doorstep. As such, buyers are entirely reliant on the information provided about the product online. Given this, consumers thoroughly research electronic products, even beyond the platform, before deciding whether to make a purchase (Dulay et al., 2022).

The results of the study of Dulay et. al (2022) and their findings however say differently; 53.3 of their respondents stated that their primary consideration when purchasing products come mostly for the Product-Related Details (Brand, Price, Quality, Advertisement) scoring the highest out of the three factors they had presented highlighting them as the most influential on their purchase decision. "While 39.7% considered User-Generated Contents (Reviews and Feedback, Electronic Word of Mouth) and only 7% (21 respondents) considered the Online Marketplaces Factors (Convenience, Delivery Time, Trust)" (Dulay et. al, 2022). Less than half stated that user-generated content and reviews from both inside & outside the platform are their primary consideration in purchasing products online. These findings are reinforced by Huang et al., (2018) as cited in Bandiling et al., (2022) stating that respondents do not particularly trust that most online opinions are persuasive nor useful.

However, this statement by Alshaer et al. (2020) argues that consumers are empowered and made confident by social media through learning the benefits and disadvantages that the product they are buying has reinforces or repels their decision to purchase, especially when they can choose from different options based on what provides the most value for the price. Further supported by Bandiling (2020) stating that online reviews expand the consumer's trust by validating or invalidating a consumer's preconceptions about the product and reinforcing the purchasing decision through confirmation bias. Concluding that product knowledge is the most important factor driving Filipino consumers to purchase a product, the more they know the more likely they are to follow through with their intention to purchase (Jiang & Regua, 2023).

Significant disparities between presented information by sellers will reduce the trust of the consumer leaving them less likely to purchase from them or online shopping platforms again; vice versa, the greater the trust in the platform and a particular seller, the more likely the customers will buy from not just the seller but also the platform itself (Davis et al., 2021) being a way for consumers to reduce the perceived risk they have when purchasing online (Chetioui, Y., Lebdaoui, H., & Chetioui, H., 2020). Given the increasing importance of E-commerce over the past years, it has become an important asset. In terms of online shopping, consumers are reliable in product information from reviews by consumers (Meng Qi et al., 2022). Even with the official product information by the seller, there is still more curiosity by the consumer if it lives by the official information. In addition to that, it has been involving consumer behavior regarding purchasing intention (Yi-Chen Lan, 2022).

Consumers generally perceive the information posted online as an "unbiased" review and knowing that, it can often cause a "make or break" success of a certain product/service provided by the seller. This can contribute to manipulating their own

online reviews by purchasing or anonymously posting fake reviews to deceive consumers and increase sales (Department for Business & Trade, Alma Economics., 2023).

Fake reviews can be favorable from the seller's product or unfavorable towards the products sold by competitors. Both intentions are made to make consumers purchase products that they might not own yet in the absence of fake reviews. Consumers might be aware though, they are consumed by the interpretations and continuous adjustment of online opinions (Zhuang, Cui and Peng 2018). It has come to the point that after reading the fake comments or reviews, that will serve as their value identification of a product. These kinds of reviews have reduced the goodwill of e-commerce platforms and harm the interests of most merchants on various platforms (Cao, 2023).

Digital Literacy

Velasco (2020) describes the three generations' digital literacy as follows; Generation X, individuals born between 1965-1980, who have witnessed the rise of digital technologies and experienced their impact on social life but remain skeptical, indifferent, or inept at their use. They were introduced to the digital world much later on in their lives already having built their lives and careers using analog technology and consequently rely less on various digital technologies in their daily lives. Pertanika J. Soc. Sci. & Hum (2021) describes them as digital immigrants as they adapt to the new digital world of the modern era from the analog world.

They then describe Generation Y or Millennials as they are commonly referred to as "the children of the transition period between the shift from analog to digital mediums; born between the years of 1981-1996. With Velasco (2020) that compared to the previous generations the millennial generation has greatly leveraged the use of social technology and a vastly greater understanding and use of digital technology and being the biggest percentage of the global workforce, they are particularly adept at using digital technology tools for their careers. Because of that, they are primarily considered to be "digital natives", a term defined by Jones and Hosein (2010) as cited by Velasco (2021) as "individuals who had been immersed in technology since birth". Atoy et al. (2020) then defines digital literacy as "the ability to use information and communication technologies to find, understand, evaluate, create, and communicate digital information, an ability that requires both cognitive and technical skills"

That is where Generation Z enters the picture; the age group was born and had developed at the peak of digital technology between 1997-2012. Çoklar & Tatli (2021) describes them as "individuals who do not know a world without the internet and digital technologies" having experienced mostly digital technology all their lives and the internet being readily available to them from birth. And while some may be familiar with pieces of analog technology, it is not something they would use in their day-to-day lives. Preferring the reliability and efficiency of digital technology.

Results from Çoklar & Tatli (2021) revealed that levels of digital literacy rose as they moved from each generation, citing differences in preferred mediums for learning; generation X preferring a more "picture and writing-oriented education process" compared to Millennials and Gen Z who are more accustomed to and actively look for different learning materials and mediums such as video, sound, image, through the use of the internet and social media. Digital technology and social media are widely embraced by Filipino youth, with 78% owning mobile phones and 59% using the Internet. Additionally, younger Filipinos are shifting their media preferences from traditional to newer forms (Velasco, 2020).

Further supported by Atoy et. al (2020) found that better online information search strategies were attributed to people with higher digital literacy. With students from younger generations being more adept at using the internet to search for information regarding daily life and casual use and pursuit of knowledge rather than direct learning activities for academic purposes.

However, findings of study on fake news assessment by Trninić et al. (2021) challenge this notion; stating that while most of their younger participants (Millennials and Gen Z) could verify the validity and reliability of information simply by checking the source, while the other half said they were neither familiar with methods for verification nor the factors that they should consider to determine if a piece of information is true or false similar to a majority of the their respondents from Generation X, many of whom admitted they only periodically check the authenticity of information they come across digitally. Their study further states that the way that younger generations are able to detect authenticity in digital media is mostly superficial, showcasing awareness but not really putting efforts in assessing the media and information they encounter due to their confidence in their intuition regarding assessment of contents' credibility. (Trninić et al., 2021).

Comparing the results of their older respondents they found that their Gen X participants take the issues a lot more seriously. They analyze negative content with complexity, even though they can't always define manipulative information clearly and intrinsically as the younger generations seemed to do. They have a deeper understanding of media bridging their experience from analogue to digital. And they concluded that while the younger generations, despite their exposure to digital technology, are not familiar with most of what they can be used for. All the while members of the Generation X, relied on

experience and a critical approach to the media, revealed a higher level of critical media literacy and information literacy that while not directly translated into digital literacy, gave them insight into how they approach information (Trninić et al., 2021).

With all of that said, how the different generations interact with the information presented on online shopping platforms may very well show a similar case. As such this study will be done to explore the difference between the different generations and how their digital literacy abilities are applied in assessing the authenticity of reviews they encounter on online shopping platforms.

Fraudulent Strategies on E-commerce Platforms

Order brushing is a tactic where retailers create fake orders to fraudulently boost sale metrics and visibility on online platforms, impacting consumer search algorithms, recommendations by the platform, and storefront engagement metrics (Jiang et al., 2023). Given the nature of ecommerce and online marketplaces, engagement with a seller and sales metrics are crucial to being recommended by the algorithm of the platform (Zhuang, Cui and Peng, 2018).

A study by He et al. (2021) navigates a similar model, although the study doesn't explicitly define the act of buying reviews as 'order brushing'. Their study investigated whether the act of buying fake reviews were done for high-quality products attempting to fix the 'cold-start' problem, trying once again to gain more interaction, or by low-quality products attempting to deceive the consumers. They stated that if the products were found to be bought by high-quality products, it might not be as harmful as it is thought to be as it does not mislead the consumers with the product's quality even though the reviews are fake. Although in that case ethical implications of the act must also be considered, this was not tackled within the scope of their study. With their study's findings, it was concluded that low-quality products were the main source of fake reviews on the platform.

With the existence of fake reviews, multiple studies have attempted to identify a uniformed criteria for detection of fake reviews from legitimate reviews through different approaches as observed. In terms of this study, the focus will be on the behavioral and contextual approach in determining review authenticity.

The difficulty of differentiating fake reviews from legitimate reviews were recognized, and so Kumar (2020) introduced this common method of identification in his study using two features: behavioral, which refers to non-verbal extracted features, and contextual for verbal features. These features were extracted from three attributes: review centric, product centric, and reviewer centric. These contextual and behavioral attributes are used to improve the classification of fake reviews and the accuracy of the assessment.

In the UK's Department of Business and Trade's research on fake online reviews (2023), their conclusions find that consumers do not change their purchasing behavior despite the exposure to fake reviews which leads them susceptible to the negative consequences of this fraudulent activity. This emphasizes the need for updated policies in place on e-commerce platforms regarding fake reviews and order brushing activities. Ananthakrishnan et al. (2020) also suggests that creating a policy showing flagged reviews and leaving them posted can increase consumer trust on the platform despite the existence of fake reviews.

The 3S Model

Theoretical Framework This study used the 3S Model of Information Trust. With the dimensions used in the study having been the following; Information Features (which itself comprises Semantic Factors, Surface Factors and Source Features) and User Characteristics (Expertise & Experience Factors) to determine their confidence in assessing review authenticity and their overall trust in online reviews on online shopping platforms.

Information Features When assessing the information they encounter products both online and in real life. Shoppers ideally look for information that is upheld by arguments they deem legitimate. They assess the quality of information within the review and how it is presented. Especially in terms of the precision, idealness, fulfillment, importance, and consistency of the information presented (Bandiling et. al, 2022).

As such, examining how information is written and presented and how language is used to convey information is a key factor when people assess reviews. Examining

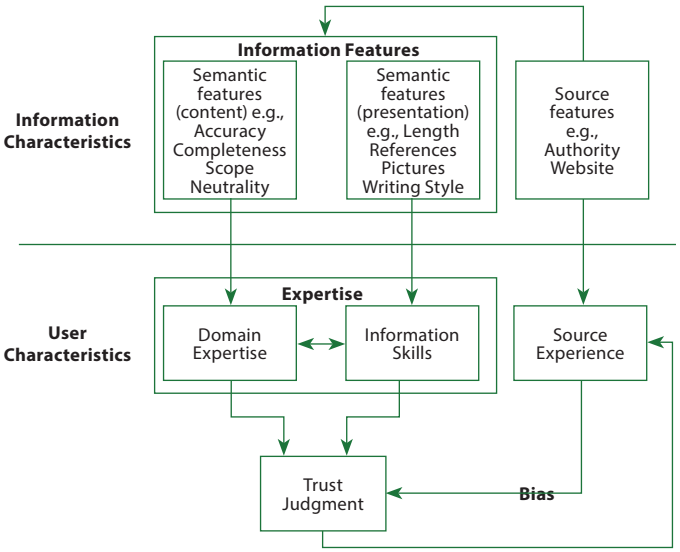


Fig. 1. 3S-model of Information Trust (Lucassen & Schraagen, 2011)

information for strong, fact-based, and realistic sentences and phrases and inversely analyzing if there is a lack of coherent and structural context in reviews. They found that ability to examine semantics also tends to be a significant boost towards the confidence of consumers in assessing authenticity of reviews (Husam, 2023). This is further supported by Azimi, Chan, & Krasnikov (2022) In their study in which they utilized formal argumentation theory to weigh semantic arguments in reviews, identifying useful and truthful content through semantic similarity and weighted semantics. Further stating that factors such as review valence, length, & readability are key to review assessment. Even for algorithms that are being designed to detect review authenticity, they utilize semantic features; with their study showing results that showcased sentiment analysis performance increased by 10.9% due to training on semantic features (Rintyarna, 2019).

Andoy et al. (2022) study showed that the textual coherence of language used, accuracy and relevance of the information, as well as how well-structured and grammatically correct the content of the reviews are; are applied by people as personal indicators of a review's authenticity; some also consider the source of the review, including the expertise and credibility of the author, to determine if the information is reliable. However, they had found that the completeness of information was not considered by shoppers to be as important because they are not sure of their authenticity in the first place.

User Characteristics Consumer personality traits and demographics also impact authenticity perception. With factors such as educational attainment playing a crucial role in shaping the digital literacy of different generations. As information skills developed from individuals' education applies also for the digital information they encounter. With respondents with higher education levels showcasing better digital literacy and inversely, the less educated individuals are likely to be less digitally literate (Nazarova & Nazarov, 2021). Additionally, lower education is linked to digital exclusion, especially in older generations (Doseděl et al., 2023). Educated and experienced users in a particular category can use their specialized knowledge to evaluate reviews more critically. The buyers' characteristics, such as brand identification and prior experiences, influence their perceptions of the credibility of reviews encountered (Román et al. 2023).

Shoppers with expertise are likely to assess reviews based on the perceived expertise of the reviewer. And compared to normal buyers are more likely to examine completeness of information presented, considering factors like product quality, seller credibility, and overall shopping experience, and spot inconsistencies within the information presented by the reviews (Yu, 2022). As such, the expertise and experience of the user in online shopping, enables them to analyze the insights of other customers and connect their own experience with different products, services, and the platform in general, and contribute to a more thorough evaluation of the review authenticity (Andoy et al. 2022). By being equipped with the knowledge of how to examine information features, such as the use of language, knowledge about the products and platform, analysis of how a review is presented, the source of the information, and possible deceptions, those users are more likely to be more confident in assessing reviews and their purchases (Barbado et al. 2019).

Methodology

The study implemented a descriptive comparative research design with a quantitative approach to assess the confidence of online shoppers in assessing the authenticity of reviews on e-commerce platforms and detect any differences in the results of Generations X, Y, and Z. Utilizing Levene's Test of Homogeneity of Variance then utilizing ANOVA or Welch's ANOVA based on the results. Quezon City was chosen as the locale of the study due to its diverse population and urban setting, as well as it being one of the biggest key hubs for e-commerce logistics. (Gupta, 2021) And utilizing stratified purposive sampling to gather respondents who fit the set criteria. First, they must be at least 18 years of age to provide their informed consent; Second is they must be residing in Quezon City; And third; they must have purchased from an online shopping platform within the past year.

To achieve this, a structured questionnaire was formulated & used in the study to systematically collect data from 332 respondents from the three ae generations. The survey questions were formulated based on the related literature & theory and validated by a panel of e-commerce & business experts. The questionnaire was divided into five (5) main parts. (1) Demographic Information, (2) Psychographic Information (3) User Characteristics Assessment. (4) Review Feature Assessment (5) Trust Judgement Assessment.

The questions in parts three to five utilized a 4-point Likert Scale (Strongly Agree, Agree, Disagree, Strongly Disagree) to measure the participants' responses. After that, the collected data was processed using SPSS v.22 & WARP-PLS v.8.

Analysis and Discussions

Discussion Of Findings

The demographic study (Table 7) indicates a primarily female sample (63.86%), with Generation Z representing the largest section at 54.22%. District 6 is the most prevalent, comprising 30.42%. Concerning educational achievement, there is a nearly similar distribution between intermediate (52.41%) and advanced (47.59%) education levels. A significant percentage of respondents engage in online shopping monthly (50.90%), with Shopee being the predominant platform (77.41%).

In Table 7, the results detected a significant presence of younger consumers (Gen Z) proficient in internet buying. The equitable depiction of educational attainment indicates that trust evaluations may be affected by differing degrees of critical thinking abilities, potentially influencing their perception of review legitimacy.

Table 7. Demographic Characteristics of the Respondents (n=135)

Demographic	Category	Frequency	Percentage
Sex	Female	212	63.86
	Male	120	36.14
Age	Generation X (1965-1980)	70	21.08
	Generation Y (1981-1996)	82	24.70
	Generation Z (1997-2012)	180	54.22
Location (District)	District 1	38	11.45
	District 2	52	15.66
	District 3	23	6.93
	District 4	24	7.23
	District 5	94	28.31
	District 6	101	30.42
Educational Attainment	Intermediate	174	52.41
	Advanced	158	47.59
Frequency of Online Shopping	At least once a week	110	33.31
	At least once a month	169	50.90
	At least once every six months	35	10.54
	At least once a year	18	5.42
Frequently Used Online Shopping Platform	Lazada	165	49.70
	Shopee	257	77.41
	Tiktok Shop	167	50.30
	Others	34	10.24

Table 8 demonstrates that respondents have a high degree of confidence when evaluating the validity of reviews, especially when it comes to their experience and experience components. The average scores for expertise (3.23) and experience (3.33) reside inside the "Strongly Agree" category, indicating a strong feeling of familiarity and analytical proficiency in review evaluation. Participants notably concur with assertions on their familiarity with rating systems and the practice of consulting external sources for verification (mean = 3.28 and 3.25, respectively).

This result implies that Quezon City consumers who shop online are generally knowledgeable about the workings of e-commerce platform review systems. Their propensity for cross-referencing reviews underscores a methodical approach to evaluating the reliability of online reviews, demonstrating an advanced comprehension of online shopping contexts which can be attributed to Quezon City being one of the biggest key hubs for e-commerce logistics in the Philippines. (Gupta, 2021)

Table 8. Level of Confidence of the Respondents based on User Characteristics

Assessment of user Characteristics	Mean	SD	Verbal Interpretation
Expertise	**3.23**	**0.64**	**Strongly Agree**
I am Familiar with how the rating system of the selected e-commerce platform/s operate/s.	3.28	0.79	Strongly Agree
I check external sources to validate the information in online reviews posted under a product in the platform.	3.25	0.85	Strongly Agree

Assessment of user Characteristics	Mean	SD	Verbal Interpretation
I have come across product reviews whose authenticity was difficult to determine.	3.24	0.79	Strongly Agree
I have knowledge of common manipulation tactics on the review systems of online shopping platforms.	3.15	0.84	Strongly Agree
Experience	**3.33**	**0.64**	**Strongly Agree**
I take note of any patterns in the complaints to identify potential recurring issues with the product.	3.28	0.81	Strongly Agree
I check if the seller provides a comprehensive overview of the product or if it's limited.	3.33	0.80	Strongly Agree
I utilize logical analysis rather than intuition in assessing review authenticity.	3.22	0.73	Strongly Agree
I read both positive and negative review to get a balanced perspective on the product.	3.49	0.77	Strongly Agree

Interpretation: 1.00 – 1.74 (strongly Disagree), 1.75 – 2.49 (Disagree), 2.50 – 3.24 (Agree), 3.25 – 4.00 (Strongly Agree)

As detailed in Table 9, respondents agree with the value of various review features, such as semantic scores with a mean of 3.09, surface with mean of 3.18, and source features with mean of 3.18.

As detailed in Table 9, respondents agree with the value of various review features, such as semantic (mean = 3.09), surface (mean = 3.18), and source features (mean = 3.18). Among these, respondents place particular emphasis on utilizing photos (mean = 3.45) and videos (mean = 3.33) as tools for verifying authenticity, indicating a preference for more tangible evidence when evaluating product reviews. The emphasis on visual aids suggests that online shoppers rely heavily on the multimedia elements of reviews for assessing authenticity. This preference may stem from a perception that visual content is less likely to be manipulated compared to textual descriptions, thus offering a more direct form of validation.

Table 9. Level of Confidence of the Respondents based on Review Features

Assessment of Review Features	Mean	SD	Verbal Interpretation
Semantic Features	**3.09**	**0.62**	**Agree**
The contents of reviews that I've viewed before have contained relevant information regarding the product.	3.22	0.74	Agree
The contents of reviews from products I've purchased before have been accurate in their assessment of the product.	3.05	0.70	Agree
The content of reviews I've seen before reliably answers my questions about the product.	3.06	0.72	Agree
The content of the majority of the reviews I've viewed before seemed authentic to me.	3.03	0.72	Agree
The contents of the reviews I have encountered have properly addressed the drawbacks or poor performance of a product.	2.97	0.77	Agree
Surface Features	**3.18**	**0.75**	**Agree**
I often examine the length of a review in evaluating the authenticity of a review.	3.08	0.83	Agree
I often utilize the attached photos in reviews in evaluating the authenticity of a review.	3.45	1.81	Strongly Agree
I often utilize the attached videos in reviews in evaluating the authenticity of a review.	3.33	0.79	Strongly Agree
I often examine the writing style of the review in evaluating the authenticity of a review.	3.07	0.88	Agree

Assessment of Review Features	Mean	SD	Verbal Interpretation
I often include the formatting of the reviews in my evaluation of the authenticity of a review.	2.98	0.91	Agree
Source Features	**3.18**	**0.60**	**Agree**
I often include seller ratings in my assessment of the trustworthiness of the product reviews posted on the selected e-commerce platforms.	3.27	0.81	Strongly Agree
I often check the profile of the reviewer when assessing the trustworthiness of the information posted in an online review.	2.87	0.98	Agree
I often take into consideration the reputation of an e-commerce platform in assessing the trustworthiness of the reviews posted in this platform.	3.24	0.74	Agree
I often take into consideration the date of reviews being posted in assessing the trustworthiness of the reviews posted.	3.13	0.83	Agree
I often take into consideration other customers' feedback to reviews in assessing the trustworthiness of the reviews posted.	3.36	0.71	Strongly Agree

Interpretation: 1.00–1.74(Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree),3.25–4.00 (Strongly Agree)

Table 10 shows that while respondents generally agree that there is trust in review systems (mean = 2.94), there is a noticeable effect of deceptive reviews, with a mean of 3.18 indicating that these evaluations have a negative influence on consumers' shopping experiences. Participants demonstrate a modest level of trust in evaluating the veracity of reviews (mean = 3.05).

This conflicting viewpoint on trust implies a tension between knowledge of review manipulation and lingering faith in the review mechanisms of large e-commerce sites. This complexity underscores a possible enhancement opportunity for e-commerce platforms in establishing and sustaining the integrity of their review systems.

Table 10. Trust Judgement of Respondents

Assessment of Trust Judgement	Mean	SD	Verbal Interpretation
Trust Judgement	**3.06**	**0.64**	**Agree**
The presence of fake reviews negatively impacts my experience on the selected e-commerce platforms.	3.18	0.85	Agree
I still have trust in the review systems of the selected e-commerce platform/s.	2.94	0.78	Agree
I am confident in my ability in assessing the authenticity of reviews that I encounter on the selected e-commerce platforms.	3.05	0.89	Agree

Interpretation: 1.00–1.74(Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)

Table 11.1 provides an analysis of how respondents from different age groups namely Generation X, Y, and Z evaluate review authenticity on selected e-commerce platforms, focusing on **expertise** and **experience** factors. The table includes mean scores, standard deviations, and statistical tests (f-values and p-values) to determine whether the differences between age groups are significant.

For Table 11, In terms of Expertise Factors, the overall mean scores indicate that Generation Z has the highest level of expertise (mean = 3.34), followed by Generation X (mean = 3.16) and Generation Y (mean = 3.05). The p-value (p = 0.002) suggests significant notable differences in expertise levels across age groups. Generation Z's higher score implies that younger respondents feel more knowledgeable about review systems and strategies for assessing review authenticity. This trend is further evident in specific items, such as checking external sources for review validation, where Generation Z has a mean score of 3.39 compared to Generation X (3.07) and Generation Y (3.09), with a p-value of 0.003 indicating statistical significance. This challenges Trninić et al. (2021) in their statement that younger generations often do not put much effort into searching for other information to verify a content's validity. Additionally, Generation Z reports greater familiarity with identifying reviews whose authenticity is difficult to determine (mean = 3.36, p = 0.007) and knowledge of manipulation tactics on e-commerce platforms (mean = 3.24, p = 0.005). These findings suggest that Generation Z is more engaged with understanding the nuances of review systems, likely due to their greater familiarity with digital platforms.

Regarding Experience Factors, results show that experience in assessing review authenticity also shows significant differences among the age groups, with Generation Z scoring the highest (mean = 3.43), followed by Generation X (mean = 3.29) and Generation Y (mean = 3.13). The p-value of 0.002 for experience factors indicates that these differences are statistically significant, suggesting that younger respondents tend to rely more on their experiences when evaluating the credibility of online reviews. Specifically, Generation Z is more likely to check if sellers provide comprehensive product overviews (mean = 3.43) compared to Generation Y (mean = 3.17), with a p-value of 0.027 highlighting this difference. Generation Z also tends to read both positive and negative reviews to form a balanced perspective (mean = 3.66), which is significantly different from the approaches of Generation X (mean = 3.41) and Generation Y (mean = 3.17), as indicated by a p-value of less than 0.001.

Table 11 discusses that in terms of user characteristics, respondents from Generation Z rated the highest out of the three age groups. As Velasco (2020) has mentioned, the younger generation grew up during the peak of digital technology and this can be attributed to their greater understanding and expertise of e-commerce platforms and its review system. However, the results show that this does not include those from Generation Y with them scoring the lowest out of the others despite being considered as a digital native. Generation X's slightly higher level of expertise could be due to their higher level of critical media literacy; although this trait does not directly translate to digital media, this can give them the insight they need to approach the information presented to them in these platforms (Trninić et al., 2021).

Unlike the results when grouped by Location, Sex, or Educational Attainment. Despite slight preferences towards specific features, they have little to no significant statistical differences that may indicate drastic differences in their assessment of their expertise and experience in authenticating reviews. Except for data that could imply a slightly greater tendency for males to rely on logical reasoning over intuition when assessing reviews.

Table 11. Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms When Grouped According to Age

Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms	Gen X n = 70		Gen Y n = 82		Gen Z n = 180		f-value	DF	p-value
	Mean	SD	Mean	SD	Mean	SD			
Expertise Factors	**3.16**	**0.58**	**3.05**	**0.75**	**3.34**	**0.6**	**6.148**	**2,329**	**0.002**
I am familiar with how the rating system of the selected e-commerce platform/operate/s.	3.16	0.8	3.2	0.83	3.36	0.76	2.175	2,329	0.115
I check external sources to validate the information in online reviews posted under a product in the platform.	3.07	0.82	3.09	0.89	3.39	0.82	5.871	2,329	0.003
I have come across product reviews whose authenticity was difficult to determine.	3.19	0.79	3.04	0.85	3.36	0.74	4.969	2,329	0.007
I have knowledge of common manipulation tactics on the review systems of online shopping platforms.	3.23	0.78	2.89	0.89	3.24	0.81	4.399	2,329	0.005
Experience Factors	**3.29**	**0.59**	**3.13**	**0.75**	**3.43**	**0.59**	**6.583**	**2,329**	**0.002**
I take note of any patterns in the complaints to identify potential recurring issues with the product.	3.3	0.82	3.11	0.85	3.35	0.77	2.554	2,329	0.079
I check if the seller provides a comprehensive overview of the product or if it's limited.	3.24	0.77	3.17	0.9	3.43	0.74	3.643	2,329	0.090
I utilize logical analysis rather than intuition in assessing review authenticity.	3.19	0.75	3.09	0.77	3.29	0.7	2.421	2,329	0.090
I read both postive and negative reviews to get a balanced pespective on the product.	3.41	0.75	3.17	0.9	3.66	0.66	12.616	2,329	0.000

Interpretation: 1.00–1.74(Strongly Disagree), 1.75–2.49 (Disagree), 2.50–.24 (Agree) ,3.25–4.00 (Strongly Agree)

Table 11.2. Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms when Grouped According to Location

Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms	District 1		District 2		District 3		District 4		District 5		District 6		f-value	DF	p-value
	n = 38		n = 52		n = 23		n = 24		n = 94		n = 101				
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD			
Expertise Factors	**3.27**	**0.61**	**3.24**	**0.7**	**3.38**	**0.46**	**3.19**	**0.63**	**3.14**	**0.71**	**3.28**	**0.59**	**0.814**	**5,326**	**0.540**
I am familiar with how the rating system of the selected e-commerce platform/operate/s.	3.21	0.78	3.23	0.85	3.35	0.78	3.33	0.56	3.23	0.86	3.35	0.74	0.361	5,326	0.875
I check external sources to validate the information in online reviews posted under a product in the platform.	3.32	0.81	3.31	0.9	3.35	0.78	3	0.98	3.16	0.88	3.32	0.79	0.909	5,326	0.475
I have come across product reviews whose authenticity was difficult to determine.	3.37	0.71	3.21	0.75	3.48	0.67	3.13	0.8	3.14	0.85	3.28	0.79	1.102	5,326	0.359
I have knowledge of common manipulation tactics on the review systems of online shopping platforms.	3.18	0.87	3.19	0.89	3.35	0.65	3.29	0.75	3.01	0.9	3.17	0.8	0.962	5,326	0.441
Experience Factors	**3.28**	**0.64**	**3.33**	**0.70**	**3.52**	**0.36**	**3.33**	**0.64**	**3.23**	**0.71**	**3.39**	**0.59**	**1.066**	**5,326**	**0.379**
I take note of any patterns in the complaints to identify potential recurring issues with the product.	3.24	0.82	3.29	0.80	3.48	0.67	3.38	0.71	3.12	0.85	3.38	0.80	1.435	5,326	0.211
I check if the seller provides a comprehensive overview of the product or if it's limited.	3.26	0.83	3.35	0.81	3.48	0.67	3.25	0.85	3.25	0.85	3.41	0.74	0.664	5,326	0.651
I utilize logical analysis rather than intuition in assessing review authenticity.	3.10	0.76	3.25	0.68	3.44	0.51	3.29	0.69	3.13	0.81	3.27	0.72	1.033	5,326	0.398
I read both postive and negative reviews to get a balanced pespective on the product.	3.50	0.76	3.44	0.83	3.70	0.47	3.42	0.72	3.45	0.84	3.52	0.76	0.487	5,326	0.786

** Interpretation: 1.00 – 1.74(Strongly Disagree), 1.75 – 2.49 (Disagree), 2.50 -3.24 (Agree),3.25 -4.00 (Strongly Agree)*

Table 11.3. Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms when Grouped According to Sex

Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms	Female		Male		f-value	DF	p-value
	n = 212		n = 120				
	Mean	SD	Mean	SD			
Expertise Factors	**3.22**	**0.64**	**3.25**	**0.65**	**0.417**	**330**	**0.677**
I am familiar with how the rating system of the selected e-commerce platform/operate/s.	3.28	0.79	3.28	0.78	0.056	330	0.955
I check external sources to validate the information in online reviews posted under a product in the platform.	3.24	0.86	3.27	0.84	0.269	330	0.788
I have come across product reviews whose authenticity was difficult to determine.	3.25	0.80	3.27	0.84	–0.278	330	0.781
I have knowledge of common manipulation tactics on the review systems of online shopping platforms.	3.11	0.85	3.23	0.81	1.218	330	0.224
Experience Factors	**3.30**	**0.65**	**3.38**	**0.63**	**0.98**	**330**	**0.328**
I take note of any patterns in the complaints to identify potential recurring issues with the product.	3.24	0.84	3.36	0.74	1.189	330	0.235
I check if the seller provides a comprehensive overview of the product or if it's limited.	3.30	0.81	3.37	0.78	0.661	330	0.509
I utilize logical analysis rather than intuition in assessing review authenticity.	3.17	0.73	3.30	0.73	1.663	330	0.097

Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms	Female		Male		f-value	DF	p-value
	n = 212		n = 120				
	Mean	SD	Mean	SD			
I read both postive and negative reviews to get a balanced pespective on the product.	3.50	0.78	3.48	0.77	–0.230	330	0.818

**Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)*

Table 11.3. Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms when Grouped Educational Attainment

Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms	Advanced		Intermediate		f-value	DF	p-value
	n = 158		n = 174				
	Mean	SD	Mean	SD			
Expertise Factors	**3.61**	**0.56**	**3.52**	**0.95**	**0.613**	**330**	**0.54**
I am familiar with how the rating system of the selected e-commerce platform/ operate/s.	3.29	0.79	3.27	0.78	0.243	330	0.808
I check external sources to validate the information in online reviews posted under a product in the platform.	3.25	0.86	3.25	0.84	0.065	330	0.948
I have come across product reviews whose authenticity was difficult to determine.	3.27	0.79	3.22	0.78	0.548	330	0.584
I have knowledge of common manipulation tactics on the review systems of online shopping platforms.	3.2	0.84	3.10	0.83	1.076	330	0.283
Experience Factors	**3.50**	**0.60**	**3.49**	**0.60**	**0.942**	**330**	**0.347**
I take note of any patterns in the complaints to identify potential recurring issues with the product.	3.32	0.78	3.24	0.83	0.819	330	0.359
I check if the seller provides a comprehensive overview of the product or if it's limited.	3.38	0.78	3.28	0.81	1.124	330	0.262
I utilize logical analysis rather than intuition in assessing review authenticity.	3.25	0.72	3.19	0.74	0.79	330	0.430
I read both postive and negative reviews to get a balanced pespective on the product.	3.50	0.76	3.48	0.79	0.271	330	0.787

**Interpretation: 1.00–1.74(Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)*

Table 12 outlines the differences in assessments of review authenticity on e-commerce platforms across Generation X, Y, and Z respondents, emphasizing semantic, surface, and source attributes.

In terms of the evaluation of review features, the analysis indicates that Generation Z assigns a slightly greater importance to semantic traits, with a mean score of 3.10, in contrast to Generation Y's 3.04 and Generation X's 2.99. Younger responders appear to favor the quality and relevancy of information in reviews more than older generations. The difference is not statistically significant ($p = 0.639$), suggesting that although variances are there, they are insufficient to imply radically different methodologies for assessing review content across age groups.

For surface feature evaluation, Generation Z exhibits a mean score of 3.43 for surface features, signifying a preference for superficial indications such as review length, photographs, and videos in assessing authenticity. Following with a mean of 3.29, Generation X is followed by Generation Y, who scores 3.13. Given the overall p-value for surface features (0.173) suggesting no significant differences among the age groups, the study uncovers a major exception. A substantial difference ($p < 0.001$) is observed when using videos as a technique to assess authenticity, indicating that Generation Z in particular perceives visual content as a more reliable source of information in online evaluations. Trninić et al. (2021), as well, states that younger generations' –Generation Z in this case– methods of assessing authenticity in digital media focus more on the surface level as they believe that they are already able to recognize contents that are not credible.

The results also partly align with the study of Çoklar & Tatli (2021) describing individuals from Generation Z and Y as those who are accustomed and actively look for other and various learning materials and mediums such as sound and image; and while the results from Generation Z support this idea with them scoring the highest mean score compared to others, Generation Y scores the lowest out of the three age groups which disproves the statement in this case. Concerning source features, Generation X respondents (mean = 3.25) exhibit a marginally greater propensity to regard elements such as seller ratings and the reviewer's profile in comparison to Generation Y (mean = 3.08) and Generation Z (mean = 3.19). This indicates that older generations may prioritize the reliability of the source when assessing reviews. As Velasco (2020) described, this

generation remains skeptical of the use of digital technologies; in this case, making sure of the legitimacy of the platform and its content. However, the disparities between the categories are not statistically significant (p = 0.228), suggesting that these preferences, although evident, do not lead to fundamentally divergent behaviors among age groups. Generation Z's increased dependence on multimedia components and Generation X's focus on source attributes underscore certain generational distinctions; yet, the uniformity in their overarching methods of review evaluation indicates common strategies between the cohorts. Similarly to the results when grouped by Location, Sex, & Educational Attainment. Despite slight preferences towards specific features, have little to no significant statistical differences that may indicate drastic differences in their approach to assessing review authenticity.

Table 12.1. Respondent's Assessment of Review Features on Selected E-commerce Platforms when Grouped According to Age

Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms	Gen X n = 70		Gen Y n = 82		Gen Z n = 180		f-value	DF	p-value
	Mean	SD	Mean	SD	Mean	SD			
Semantic Features	**2.99**	**0.54**	**3.04**	**0.58**	**3.10**	**0.60**	**1.002**	**2,329**	**0.368**
The contents of reviews that I've viewed before have contained relevant information regarding the product.	3.16	0.80	3.20	0.83	3.36	0.76	0.449	2,329	0.639
The contents of review from products I've purchased before have been accurate in their assessment of the product.	3.07	0.82	3.09	0.89	3.39	0.82	1.64	2,329	0.195
The contents of reviews I've seen before reliably answers my questions about the product.	3.19	0.79	3.04	0.85	3.36	0.74	0.729	2,329	0.483
The content of the majority of the reviews I've viewed before seem authentic to me.	3.23	0.78	2.89	0.89	3.24	0.81	0.135	2,329	0.873
The contents of the reviews I have encountered have properly addressed the drawbacks or poor performance of a product.	2.84	0.72	2.94	0.73	3.03	0.81	1.624	2,329	0.199
Surface Features	**3.29**	**0.59**	**3.13**	**0.75**	**3.43**	**0.59**	**1.765**	**2,329**	**0.173**
I often examine the length of a review in evaluating the authenticity of a review.	3.30	0.82	3.11	0.85	3.35	0.77	1.559	2,329	0.212
I often utilize the attached photos in reviews in evaluating the authenticity of a review.	3.24	0.77	3.17	0.90	3.43	0.74	1.353	2,329	0.260
I often utilize the attached videos in reviews in evaluating the authenticity of a review.	3.19	0.75	3.09	0.77	3.29	0.70	8.307	2,329	<0.001
I often examine the writing style of the review in evaluating the authenticity of a review.	3.41	0.75	3.17	0.90	3.66	0.66	0.251	2,329	0.778
I often include the formatting of the reviews in my evaluation of the authenticity of a review.	3.11	0.83	2.84	0.84	2.99	0.97	1.711	2,329	0.182
Source Features	**3.25**	**0.56**	**3.08**	**0.64**	**3.19**	**0.60**	**1.486**	**2,329**	**0.228**
I often include seller ratings in my assessment of the trustworthiness of the product reviews posted on the selected e-commerce platforms.	3.40	0.69	3.16	0.82	3.28	0.84	1.698	2,329	0.185
I often check the profile of the reviewer when assessing the trustworthiness of the information posted in an online review.	2.99	0.83	2.88	0.88	2.83	1.07	0.661	2,329	0.517
I often take into consideration the reputation of an e-commerce platform in assessing the trustworthiness of the reviews posted in this platform.	3.30	0.73	3.11	0.70	3.28	0.75	1.834	2,329	0.161
I often take into consideration the date of reviews being posted in assessing the trustworthiness of the reviews posted	3.24	0.73	3.07	0.80	3.12	0.87	0.856	2,329	0.426
I often take into consideration other customers' feedback to reviews in assessing the trustworthiness of the reviews posted.	3.30	0.71	3.23	0.73	3.44	0.69	2.848	2,329	0.059

Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)

Table 12.2. Respondent's Assessment of Review Features on Selected E-commerce Platforms when Grouped According to Location

Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms	District 1		District 2		District 3		District 4		District 5		District 6		f-value	DF	p-value
	n = 38		n = 52		n = 23		n = 24		n = 94		n = 101				
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD			
Semantic Features	**3.08**	**0.49**	**3.12**	**0.61**	**3.05**	**0.46**	**2.97**	**0.62**	**2.98**	**0.63**	**3.14**	**0.57**	**0.975**	**5,326**	**0.433**
The contents of reviews that I've viewed before have contained relevant information regarding the product.	3.37	0.63	3.21	0.78	3.30	0.47	3.08	0.78	3.05	0.81	3.33	0.72	1.919	5,326	0.091
The contents of review from products I've purchased before have been accurate in their assessment of the product.	3.05	0.61	3.17	0.71	3.04	0.56	3.00	0.72	2.95	0.79	3.09	0.65	0.82	5,326	0.536
The contents of reviews I've seen before reliably answers my questions about the product.	3.03	0.68	3.13	0.71	3.04	0.56	2.88	0.74	3.06	0.74	3.08	0.76	0.456	5,326	0.809
The content of the majority of the reviews I've viewed before seem authentic to me.	3.05	0.70	3.06	0.75	3.00	0.74	2.96	0.69	3.00	0.66	3.07	0.78	0.169	5,326	0.974
The contents of the reviews I have encountered have properly addressed the drawbacks or poor performance of a product.	2.92	0.67	3.04	0.82	2.87	0.79	2.92	0.72	2.83	0.80	3.12	0.78	1.595	5,326	0.161
Surface Features	**3.01**	**0.65**	**3.24**	**0.72**	**3.40**	**0.46**	**3.30**	**1.59**	**3.06**	**0.68**	**3.25**	**0.59**	**1.657**	**5,326**	**0.140**
I often examine the length of a review in evaluating the authenticity of a review.	3.00	0.90	4.15	0.85	3.04	0.71	3.00	0.98	3.07	0.79	3.10	0.82	0.217	5,326	0.955
I often utilize the attached photos in reviews in evaluating the authenticity of a review.	3.32	0.70	3.42	0.78	3.61	0.58	4.42	6.15	3.21	0.88	3.48	0.78	1.79	5,326	0.114
I often utilize the attached videos in reviews in evaluating the authenticity of a review.	3.18	0.83	4.47	0.79	3.78	0.52	3.29	0.91	3.19	0.85	3.41	0.70	2.642	5,326	0.023
I often examine the writing style of the review in evaluating the authenticity of a review.	2.92	0.75	3.12	0.90	3.43	0.66	2.88	1.03	2.94	0.96	3.18	0.80	2.057	5,326	0.070
I often include the formatting of the reviews in my evaluation of the authenticity of a review.	2.61	0.95	3.15	0.94	3.13	0.97	2.92	1.06	3.88	0.88	3.10	0.84	2.403	5,326	0.037
Source Features	**3.23**	**0.59**	**3.17**	**0.63**	**3.30**	**0.47**	**3.13**	**0.69**	**3.06**	**0.65**	**3.25**	**0.54**	**1.360**	**5,326**	**0.239**
I often include seller ratings in my assessment of the trustworthiness of the product reviews posted on the selected e-commerce platforms.	3.24	0.88	3.25	0.81	3.39	0.72	3.17	1.01	3.23	0.82	3.34	0.74	0.371	5,326	0.869
I often check the profile of the reviewer when assessing the trustworthiness of the information posted in an online review.	2.87	1.09	2.88	1.02	2.96	0.98	3.00	0.98	2.73	0.99	2.95	0.90	0.623	5,326	0.683
I often take into consideration the reputation of an e-commerce platform in assessing the trustworthiness of the reviews posted in this platform.	3.42	0.68	3.27	0.69	3.26	0.54	3.25	0.74	3.12	0.81	3.28	0.74	1.057	5,326	0.384
I often take into consideration the date of reviews being posted in assessing the trustworthiness of the reviews posted	3.24	0.68	3.04	0.86	3.39	0.58	3.00	0.78	2.97	0.87	3.27	0.86	2.136	5,326	0.061
I often take into consideration other customers' feedback to reviews in assessing the trustworthiness of the reviews posted.	3.37	0.54	3.40	0.63	3.52	0.51	3.25	0.85	3.26	0.78	3.43	0.73	0.961	5,326	0.442

**Interpretation: 1.00–1.74, (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)*

Table 12.3. Respondent's Assessment of Review Features on Selected E-commerce Platforms when Grouped According to Sex

Respondent's Characteristics in Assessing Review Authenticity on Selected E-commerce Platforms	Female		Male		f-value	DF	p-value
	n = 212		n = 120				
	Mean	SD	Mean	SD			
Semantic Features	**3.08**	**0.55**	**3.05**	**0.63**	**–0.409**	**330**	**0.683**
The contents of reviews that I've viewed before have contained relevant information regarding the product.	3.24	0.72	3.18	0.78	–0.773	330	0.440
The contents of review from products I've purchased before have been accurate in their assessment of the product.	3.09	0.67	2.98	0.74	–1.44	330	0.151
The contents of reviews I've seen before reliably answers my questions about the product.	3.1	0.68	2.98	0.79	–1.462	330	0.145
The content of the majority of the reviews I've viewed before seem authentic to me.	3.02	0.69	3.05	0.77	0.322	330	0.117
The contents of the reviews I have encountered have properly addressed the drawbacks or poor performance of a product.	2.92	0.73	3.06	0.83	1.572	330	0.117
Surface Features	**3.15**	**0.64**	**3.23**	**0.92**	**0.977**	**330**	**0.329**
I often examine the length of a review in evaluating the authenticity of a review.	3.10	0.79	3.04	0.89	–0.607	330	0.544
I often utilize the attached photos in reviews in evaluating the authenticity of a review.	3.36	0.78	3.61	2.83	1.184	330	0.237
I often utilize the attached videos in reviews in evaluating the authenticity of a review.	3.28	0.8	3.43	0.76	1.634	330	0.103
I often examine the writing style of the review in evaluating the authenticity of a review.	3.07	0.85	3.07	0.92	0.006	330	0.995
I often include the formatting of the reviews in my evaluation of the authenticity of a review.	2.95	0.89	3.03	0.96	0.815	330	0.416
Source Features	**3.14**	**0.56**	**3.24**	**0.67**	**1.477**	**330**	**0.141**
I often include seller ratings in my assessment of the trustworthiness of the product reviews posted on the selected e-commerce platforms.	3.25	0.77	3.32	0.87	0.722	330	0.471
I often check the profile of the reviewer when assessing the trustworthiness of the information posted in an online review.	2.79	0.97	3.02	0.97	2.022	330	0.044
I often take into consideration the reputation of an e-commerce platform in assessing the trustworthiness of the reviews posted in this platform.	3.18	0.69	3.36	0.81	2.05	330	0.04
I often take into consideration the date of reviews being posted in assessing the trustworthiness of the reviews posted	3.13	0.81	3.14	0.86	0.145	330	0.89
I often take into consideration other customers' feedback to reviews in assessing the trustworthiness of the reviews posted.	3.35	0.69	3.38	0.74	0.404	330	0.687

**Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)*

Table 12.4. Respondent's Assessment of Review Features on Selected E-commerce Platforms when Grouped According to Educational Attainment

Review Features	Post-graduate		Undergraduate		f-value	DF	p-value
	n = 158		n = 174				
	Mean	SD	Mean	SD			
Semantic Features	**3.04**	**0.52**	**3.09**	**0.63**	**0.912**	**327.183**	**0.363**
The contents of reviews that I've viewed before have contained relevant information regarding the product.	3.27	0.70	3.17	0.78	–1.294	330	0.196
The contents of review from products I've purchased before have been accurate in their assessment of the product.	2.96	0.67	3.13	0.72	2.155	330	0.031

Review Features	Post-graduate n = 158		Undergraduate n = 174		f-value	DF	p-value
	Mean	SD	Mean	SD			
The contents of reviews I've seen before reliably answers my questions about the product.	3.01	0.71	3.10	0.73	0.991	330	0.322
The content of the majority of the reviews I've viewed before seem authentic to me.	3.01	0.69	3.05	0.75	0.495	330	3.22
The contents of the reviews I have encountered have properly addressed the drawbacks or poor performance of a product.	2.91	0.74	3.02	0.80	1.315	330	0.189
Surface Features	**3.19**	**0.65**	**3.17**	**0.84**	**0.226**	**330**	**0.821**
I often examine the length of a review in evaluating the authenticity of a review.	3.04	0.86	3.11	0.80	−0.847	330	0.397
I often utilize the attached photos in reviews in evaluating the authenticity of a review.	3.37	0.79	3.52	2.39	−0.750	330	0.454
I often utilize the attached videos in reviews in evaluating the authenticity of a review.	3.37	0.73	3.30	0.83	0.788	330	0.432
I often examine the writing style of the review in evaluating the authenticity of a review.	3.13	0.81	3.01	0.93	1.195	330	0.233
I often include the formatting of the reviews in my evaluation of the authenticity of a review.	3.05	0.89	2.91	0.94	1.363	330	0.174
Source Features	**3.14**	**0.56**	**3.24**	**0.67**	**1.477**	**330**	**0.141**
I often include seller ratings in my assessment of the trustworthiness of the product reviews posted on the selected e-commerce platforms.	3.25	0.77	3.32	0.87	0.722	330	0.471
I often check the profile of the reviewer when assessing the trustworthiness of the information posted in an online review.	2.79	0.97	3.02	0.97	2.022	330	0.044
I often take into consideration the reputation of an e-commerce platform in assessing the trustworthiness of the reviews posted in this platform.	3.18	0.69	3.36	0.81	2.05	330	0.04
I often take into consideration the date of reviews being posted in assessing the trustworthiness of the reviews posted	3.13	0.81	3.14	0.86	0.145	330	0.89
I often take into consideration other customers' feedback to reviews in assessing the trustworthiness of the reviews posted.	3.35	0.69	3.38	0.74	0.404	330	0.687

**Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)*

Table 13.1. Respondents' Trust Judgement on Reviews on Selected E-commerce Platforms when Grouped According to Age

Respondent's Trust Judgement on Reviews on Selected E-commerce Platforms	Gen X n = 70		Gen Y n = 82		Gen Z n = 180		f-value	DF	p-value
	Mean	SD	Mean	SD	Mean	SD			
Trust Judgement	**2.76**	**0.65**	**2.98**	**0.71**	**3.21**	**0.56**	**13.888**	**2,329**	**0.000**
The presence of fake reviews negatively impacts may my experience on the selected e-commerce platforms.	3.06	0.88	3.01	3.30	0.82	0.82	4.177	2,329	0.016
I still have trust in the review systems of the selected e-commerce platform/s.	2.59	0.83	2.91	3.08	0.72	0.72	11.017	2,329	0.000
I am confident in my ability in assessing the authenticity of reviews that I encounter on the selected e-commerce platforms.	2.64	0.99	3.01	3.23	0.78	0.78	11.915	2,329	0.000

**Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)*

Table 13.2. Respondents' Trust Judgement on Reviews on Selected E-commerce Platforms when Grouped According to Location

Respondent's Trust Judgement on Reviews on Selected E-commerce Platforms	District 1		District 2		District 3		District 4		District 5		District 6		f-value	DF	p-value
	n = 38		n = 52		n = 23		n = 24		n = 94		n = 101				
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD			
Trust Judgement	**3.03**	**0.60**	**3.19**	**0.64**	**3.17**	**0.64**	**3.21**	**0.55**	**2.97**	**0.65**	**3.02**	**0.66**	**1.282**	**5,326**	**0.271**
The presence of fake reviews negatively impacts may my experience on the selected e-commerce platforms.	3.11	0.76	3.23	0.81	3.22	0.95	3.50	0.66	3.09	0.91	3.18	0.86	1.013	5,326	0.41
I still have trust in the review systems of the selected e-commerce platform/s.	2.97	0.72	3.06	0.85	2.96	0.71	2.96	0.69	2.88	0.79	2.9	0.79	0.405	5,326	0.845
I am confident in my ability in assessing the authenticity of reviews that I encounter on the selected e-commerce platforms.	3.00	0.77	3.27	0.82	3.35	0.71	3.17	0.82	2.95	0.92	2.97	0.97	1.678	5,326	0.139

**Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)*

Table 13.3. Respondents' Trust Judgement on Reviews on Selected E-commerce Platforms when Grouped According to Sex

Respondent's Trust Judgement on Reviews on Selected E-commerce Platforms	Female		Male		f-value	DF	p-value
	n = 212		n = 120				
	Mean	SD	Mean	SD			
Trust Judgement	**2.76**	**0.65**	**2.98**	**0.71**	**2.260**	**330**	**0.024**
The presence of fake reviews negatively impacts may my experience on the selected e-commerce platforms.	3.06	0.88	3.01	0.85	1.029	330	0.122
I still have trust in the review systems of the selected e-commerce platform/s.	2.59	0.83	2.91	0.77	1.859	330	0.064
I am confident in my ability in assessing the authenticity of reviews that I encounter on the selected e-commerce platforms.	2.64	0.99	3.01	0.91	2.256	330	0.025

**Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)*

Table 13.4. Respondents' Trust Judgement on Reviews on Selected E-commerce Platforms when Grouped According to Educational Attainment

Respondent's Trust Judgement on Reviews on Selected E-commerce Platforms	Post-graduate		Undergraduate		f-value	DF	p-value
	n = 212		n = 120				
	Mean	SD	Mean	SD			
Trust Judgement	**305**	**0.65**	**3.07**	**0.63**	**–0.266**	**330**	**0.791**
The presence of fake reviews negatively impacts may my experience on the selected e-commerce platforms.	3.18	0.86	3.17	0.85	0.119	330	0.906
I still have trust in the review systems of the selected e-commerce platform/s.	2.90	0.80	2.97	0.76	–0.849	330	0.396
I am confident in my ability in assessing the authenticity of reviews that I encounter on the selected e-commerce platforms.	3.06	0.88	3.05	0.90	0.053	330	0.957

**Interpretation: 1.00–1.74 (Strongly Disagree), 1.75–2.49 (Disagree), 2.50–3.24 (Agree), 3.25–4.00 (Strongly Agree)*

Discussion of Hypotheses

H1: Differences in Assessment in terms of Demographic Profile According to Hypothesis H1, respondents' evaluations of product reviews on e-commerce sites change significantly depending on which demographic categories they fall into. The findings in Table 12, which analyzes review assessment disparities among Generations X, Y, and Z, partially substantiate this idea.

Table 12 illustrates generational disparities in the perception of review authenticity, especially for the utilization of superficial elements such as videos. Generation Z demonstrates a greater dependence on multimedia components, particularly videos, as evidenced by a substantial p-value ($p < 0.001$) for this characteristic. This indicates that younger respondents prioritize visual elements more than older respondents when evaluating evaluations. Trninić et al. (2021) points out that younger individuals may not engage deeply in verifying information despite their technological familiarity; however the findings debunk this as nevertheless, for the majority of review assessment dimensions, including the comprehensive evaluation of semantic and source characteristics, the disparities between age groups are not statistically significant, as evidenced by p-values exceeding 0.05. This suggests that although certain distinctions occur among generations, the overall method of assessing reviews remains largely consistent throughout age groups.

Nazarova & Nazarov (2021) also claim that educated consumers are better equipped to evaluate review authenticity and information; yet the findings indicate no significant differences between respondents from different educational attainment in the approach of respondents when assessing review features. This discrepancy emphasizes that educational background alone may not predict one's ability to judge review authenticity accurately.

Thus, H1 is supported by the evidence to a limited extent. Although notable variances exist, especially regarding the dependence on video content across age groups, the majority of variations in review assessment techniques related to demographic profiles lack statistical significance. This indicates that demographic characteristics affect certain elements of the evaluation of review features, however the influence may be more complex than initially proposed.

H2: Differences in Assessment of Review Features and Confidence in Assessing Review Authenticity According to hypothesis H2, there exists a noteworthy disparity between the respondents' evaluations of review features and their level of confidence in assessing the legitimacy of such reviews. Tables 12 elucidate respondents' evaluations of reviews, while Table 8 and other portions of the study provide more comprehensive information regarding their confidence levels.

The findings reveal that participants mostly exhibit considerable confidence in their capacity to evaluate review authenticity, as evidenced by elevated mean scores for experience and expertise characteristics (Table 8). This confidence is especially robust regarding knowledge with grading systems and the capacity to utilize other sources for confirmation. The statistical studies, however, reveal no significant differences in the assessments and confidence levels among various demographic groups.

The little discrepancies noted among age groups (Table 12) indicate that although there are disparities in the evaluation of reviews, these do not significantly impact overall confidence levels. The uniformity in methods across various groups suggests that respondents' confidence in their capacity to identify genuine reviews is consistently stable, irrespective of their demographic traits.

Consequently, H2 is not entirely substantiated by the findings, as the data do not reveal a major disparity in the evaluation of review features across demographic profiles. Although respondents exhibit trust in their evaluative abilities, this confidence does not seem to fluctuate markedly when assessing review features across various groups.

However, despite the uniformity of the assessment of review features; upon analysis of the evaluations of the expertise and experience factors of the participants, notable differences had been identified between age groups. (Table 11). There are significant disparities in their assessment of factors related to expertise and experience. Additionally, the findings also reveal significant differences in their Trust Judgement. (Table 13) with data suggesting that younger respondents feel more capable of discerning authentic reviews and also higher levels of trust in review platforms. Underscoring a larger impact of experience and expertise on their confidence than initially proposed. Which supports Roman, et al (2022) & Yu (2022) findings that highlighted that experience being more effective than knowledge in evaluating information. Leading to the differences in their trust judgment and confidence.

Further supported by findings of Atoy, et al. (2020) and Velasco (2020) whose findings have indicated that the experience in digital literacy of younger generations allow them to utilize their information search skills and tools available to them. Therefore despite the uniformity in methods across demographics; how effectively they have been able to apply those strategies differ.

The data indicate that although demographic characteristics may impact specific preferences in review features evaluation (as partially evidenced in H1), they do not significantly influence respondents' confidence in their capacity to assess review authenticity, as posited in H2. However, also highlighting that individual experiences with e-commerce platforms may have more influence in shaping trust and confidence in assessing online reviews than just review features themselves.

Overall these insights offer a refined comprehension of the correlation between review evaluation and confidence levels, underscoring the need for additional research into how particular aspects of review content and experiential factors affect customer trust.

Discussion of Theory

Based on the data gathered for the study, the findings are partially in line with the theoretical framework of the 3S Model of Information Trust. As a majority of respondents were frequent users of online shopping platforms, they've developed an expertise and experience in online shopping through their consistent use. The theory posits that expert shoppers tend to prioritize the accuracy of the presented information; which is a key aspect of semantic features. (Bandiling et. al, 2022; Lucassen & Schraagen, 2011).

The findings determined that even between generational cohorts, the assessment of their utilization of Semantic Features had no significant statistical differences besides slight generational preferences, with younger generations paying slightly more attention to surface features (formatting, photos, videos) while older generations would prioritize Source Features (credibility of reviewer, store, or platform) more. Supported by the findings of Çoklar & Tatli (2021); & Lucassen & Schraagen (2011) stating that pictures & media are crucial for younger generations in judging credibility.

However, despite detecting no statistical differences between the assessment of review features between generational cohorts, the differences in the evaluation of generational cohorts in terms of User Characteristics (Expertise & Experience) & their overall trust judgment. The differences in their assessment of User Characteristics (Experience & Expertise) indicate that trust judgments are affected more by them than simply analyzing the review features only, as how they apply their experience and expertise may vary in effectiveness. Which is in line with Lucassen & Schraagen (2011) stating the possibility that users may passively rely on previous experiences rather than deeply analyze the features of the information presented. Further stating that when a user has a lot of positive (or negative) experiences with a particular source, motivation to properly assess the information they encounter. Reflecting the findings of the paper, even though the strategies utilized in assessing review features are the same, differences in their information skill, expertise, and experience in a particular platform, can lead to significant differences in the effectiveness of their assessment of information, and lead to positive or negative effects on their trust on the platform.

Ultimately highlighting that trust is not only shaped by static preferences in informational search strategies but also through a continuous learning process from individual experiences.

Discussion on the Validity of Results Given Unequal Sample Sizes Across Generational Groups

The findings in Table 8 showed a disparity in the distribution of participants between the age groups. Showcasing a majority of respondents belonging to Generation Z. However, VanVoorhis, C. R. W., & Morgan, B. L. (2007) suggests that as long as each group has sample size that meets the minimum threshold, results are generally reliable. And that for ANOVA, they recommend a minimum of 30 participants per group which the study has met. Furthermore, Field (2013) emphasizes that even when there are greater numbers of responses in other groups, achieving the minimum sample size for each group in an ANOVA is crucial to guaranteeing accurate results. It discusses the notion that if each group reaches the necessary minimum, different sample sizes can still produce accurate results. This method lessens problems with variance stability and statistical power, particularly when group variances and availability vary.

As such, despite disparities in the number of the participants between age groups and the overall total responses, the findings of this study can still yield valuable insights into the differences between the generational cohorts regarding their assessment of online review authenticity. With the larger presence of Generation Z possibly attributed to their higher digital engagement and usage of online platforms and services (Çoklar & Tatli, 2021; Velasco, 2020).

In response to possible concerns regarding the validity of the results of the study, especially given the larger sample size of Gen Z respondents compared to Gen X and Gen Y, the researchers conducted a series of statistical checks to ensure that the findings are robust and reliable. Specifically, they utilized **Levene's Test of Homogeneity of Variances** and **Welch's ANOVA** (as shown in the Robust Tests of Equality of Means table) to account for potential issues related to unequal sample sizes and variances across generational groups.

Addressing Assumptions with Levene's Test of Homogeneity of Variances Levene's Test of Homogeneity of Variances was applied to assess whether the assumption of equal variances across generational groups holds for each variable in the study. This test is crucial because ANOVA (analysis of variance) assumes that the variances among groups are roughly equal, an assumption that, if violated, could lead to misleading results.

The results from Levene's Test (see Table: Levene's Test of Homogeneity of Variances) indicated the following:

F*or most variables, Levene's Test was non-significant ($p \geq 0.05$)*, meaning that variances across Gen X, Gen Y, and Gen Z were similar. This supports the use of standard ANOVA for these items, as the assumption of homogeneity of variances is met. For instance:

- Expert1: p = .749
- Sem1: p = .525
- Sour1: p = .529

For a few variables, Levene's Test showed significant results ($p < 0.05$), indicating that variances differed significantly across groups. Notable examples include:

- Exper4: p = .001
- Trust2: p = .020
- Sour2: p = .000

In cases where the assumption of homogeneity was violated, a traditional ANOVA could produce biased results due to unequal variances across groups. The researchers applied **Welch's ANOVA** for all variables to account for these discrepancies, as detailed below.

Levene's Test of Homogeneity of Variances:0

Table 14. Test of Homogeneity of Variances

	Levene Statistic	df1	df2	Sig.
Expert1	0.289	2	329	0.749
Expert2	0.875	2	329	0.418
Expert3	0.096	2	329	0.909
Expert4	0.298	2	329	0.742
Expert	2.939	2	329	0.054
Exper1	0.116	2	329	0.890
Exper2	1.557	2	329	0.212
Exper3	0.082	2	329	0.921
Exper4	6.974	2	329	0.001
Expert	4.433	2	329	0.013
Sem1	0.646	2	329	0.525
Sem2	1.239	2	329	0.291
Sem3	1.009	2	329	0.366
Sem4	0.093	2	329	0.911
Sem5	0.253	2	329	0.777
Sur1	0.356	2	329	0.700
Sur2	2.789	2	329	0.063
Sur3	0.870	2	329	0.420
Sur4	2.843	2	329	0.060
Sur5	1.565	2	329	0.211
Sem	1.384	2	329	0.252
Sour1	0.638	2	329	0.529
Sour2	8.236	2	329	0.000
Sour3	1.490	2	329	0.227
Sour4	1.209	2	328	0.300
Sour5	0.076	2	328	0.927
Sur	0.669	2	329	0.513
Trust1	1.120	2	329	0.328
Trust2	3.949	2	329	0.020
Trust3	3.999	2	329	0.019
Trust	3.156	2	329	0.044

Using Welch's ANOVA to Address Variance Inequality and Sample Size Imbalance Welch's ANOVA is a robust statistical test that handles unequal variances and **sample sizes**. Given the more significant representation of Gen Z in the study's sample, applying Welch's ANOVA ensures that this imbalance does not disproportionately influence the results of the study.

The **Robust Tests of Equality of Means table (Welch's ANOVA)** presents the results of this analysis:

Significant Results ($p < 0.05$): Welch's ANOVA showed significant generational differences for several items, indicating that the means differ significantly across groups. Examples include:

- Expert2: p = .004
- Sur2: p = .001
- Trust2: p = .000

Non-Significant Results ($p \geq 0.05$): Welch's ANOVA yielded non-significant p-values for other items, suggesting minimal or no differences in means across generations. Examples include:

- Expert1: p = .118
- Sem1: p = .641
- Sour1: p = .147

Using Welch's ANOVA, the researchers ensured that the findings on generational differences are statistically sound, even in cases where variances differ across groups and sample sizes are unequal. Welch's ANOVA provides adjusted estimates that reduce the potential bias introduced by the larger Gen Z sample, making the observed differences more reliable.

4.4.3. Ensuring Robustness and Validity of Findings By combining Levene's Test for Homogeneity of Variances and Welch's ANOVA, we took a rigorous approach to validate study's results:

For variables with non-significant Levene's Test results (homogeneous variances): The assumption of equal variances holds, and the study's findings are not affected by variance disparities across generational groups.

For variables with significant Levene's Test results (heterogeneous variances): Welch's ANOVA, which adjusts for unequal variances and sample sizes, ensures that study's results remain accurate and reliable despite these differences.

This dual approach demonstrates that any significant generational differences we observed are not artifacts of sample size imbalance or variance inequality but represent genuine differences across Gen X, Gen Y, and Gen Z.

Hence, applying Levene's Test and Welch's ANOVA ensures that the findings of this study are robust and valid, despite the larger sample size of Gen Z respondents. The non-significant results of Levene's test for most variables support the assumption of homogeneity of variances, reinforcing the reliability of the results. For items with significant Levene's Test results, the use of Welch's ANOVA provided an effective adjustment for variances and sample size disparities, thereby ensuring that the findings on generational differences are not skewed by Gen Z's larger representation. These analyses confirm that the observed differences in respondents' assessment of reviews on e-commerce platforms across generations are statistically reliable and robust. This thorough approach strengthens the credibility of the study's findings and addresses the concerns regarding sample size imbalances.

Welch Test:

Table 14. Test of Homogeneity of Variances

		Statistic[a]	df1	df2	Sig.
Expert1	Welch	2.165	2	151.026	0.118
Expert2	Welch	5.834	2	152.180	0.004
Expert3	Welch	4.616	2	148.382	0.011
Expert4	Welch	4.865	2	154.198	0.009
Expert	Welch	5.531	2	150.179	0.005
Exper1	Welch	2.394	2	149.990	0.095
Exper2	Welch	3.418	2	148.056	0.035
Exper3	Welch	2.311	2	149.478	0.103
Exper4	Welch	10.839	2	141.622	0.000
Expert	Welch	5.631	2	147.832	0.004
Sem1	Welch	0.445	2	155.805	0.641
Sem2	Welch	1.606	2	153.597	0.204
Sem3	Welch	0.779	2	160.279	0.461

		Statistic[a]	df1	df2	Sig.
Sem4	Welch	0.139	2	157.195	0.870
Sem5	Welch	1.702	2	163.474	0.186
Sur1	Welch	1.561	2	155.191	0.213
Sur2	Welch	6.923	2	127.551	0.001
Sur3	Welch	8.270	2	151.696	0.000
Sur4	Welch	0.274	2	166.514	0.761
Sur5	Welch	2.043	2	166.213	0.133
Sem	Welch	1.816	2	144.250	0.166
Sour1	Welch	1.939	2	164.044	0.147
Sour2	Welch	0.779	2	171.320	0.460
Sour3	Welch	1.938	2	158.001	0.147
Sour4	Welch	1.026	2	163.981	0.361
Sour5	Welch	2.780	2	151.097	0.065
Sur	Welch	1.433	2	156.126	0.242
Trust1	Welch	4.146	2	151.406	0.018
Trust2	Welch	9.966	2	147.400	0.000
Trust3	Welch	10.342	2	142.130	0.000
Trust	Welch	13.722	2	142.332	0.000

a. Asymptotically F distributed.

Conclusions

The study concludes that between Generations X, Y, & Z; despite certain generational distinctions; ultimately, the overall methods of review evaluation indicate common strategies between the cohorts. The same can be said even when the respondents are grouped by District, Sex, or Educational Attainment. Therefore although notable variances exist, especially regarding the dependence on multimedia content across age groups, the majority of variations in review assessment techniques related to demographic profiles lack statistical significance. Indicating that demographic characteristics affect certain elements of review evaluation, however the influence may be more complex than initially proposed.

In terms of expertise and experience factors of Generations X, Y, & Z in assessing review authenticity; there are notable differences in expertise levels across age groups; Younger respondents tend to rely more on their experiences and expertise when evaluating the authenticity of online reviews. And when grouped according to Sex there were no significant differences in their assessment except that Male respondents had a slightly greater tendency for males to rely on logical reasoning over intuition when assessing reviews. However, when grouped according to Location/District & Educational attainment, indicating no significant differences between them.

While respondents generally still trust review systems the presence of false reviews negatively impacts their shopping experience. And when grouped by their age; significant statistical differences are found. younger respondents tend to maintain higher trust in the review systems of e-commerce platforms compared to older generations. This conflicting viewpoint on trust implies tension between knowledge of review manipulation and their continued trust in the review mechanisms of large e-commerce sites. This complexity underscores a possible enhancement opportunity for e-commerce platforms in establishing and sustaining the perceived integrity of their review systems.

When grouped according to Location, no significant differences were found. However, when grouped according to Sex, the data indicates gender influences trust & confidence in evaluating reviews, with male respondents showing slightly higher trust and self-assurance in their review assessments. And when group based on educational attainment, the overall similarity between respondents with different education levels indicates that other factors, such as individual experience with e-commerce platforms, may play a more prominent role in shaping trust and confidence than demographic background alone.

While the data indicates that although demographic characteristics may impact specific preferences in review evaluation, they do not significantly influence respondents' confidence in their capacity to assess review authenticity. These insights offer a refined comprehension of the correlation between review evaluation and confidence levels, underscoring the need for additional research into how particular aspects of review content affect customer trust.

In light of these findings, the following things can be inferred. First, despite the overall uniformity of the evaluation of confidence in assessing review features, the slight deviations in their assessments can be used in the creation of targeted solutions. In the case of Gen Z, policies and improvements to review quality and surface features will lead to better experiences for this generational cohort, meanwhile improvements to accessibility of information and validation of source features will be slightly better received for them. Second, in terms of their assessment of their User Characteristics and Trust Judgement, the data highlights younger generations as significantly more confident in assessing review authenticity and also have more trust in the review systems. However, despite the uniformity of the assessment of review features; upon analysis of the evaluations of the expertise and experience factors of the participants the significant disparities in their assessment of factors related to expertise and experience & trust judgement. Highlights a larger impact of experience and expertise on their confidence and trust than initially proposed. As such, measures should be taken to inform and educate online shoppers from older generations to improve their experience and knowledge in tackling false reviews and review systems in general. Alongside efforts to improve the systems themselves to improve their trustworthiness and the overall experience of the online shoppers.

Recommendations

By implementing the following improvements, it is expected that online shoppers' trust in the review systems of e-commerce platforms will improve and enhance the quality of information that can be found on the review systems of the platforms. Furthermore, these recommendations aspire to bridge the gap between uninformed online shoppers and the existence of fraudulent strategies growing prevalent in these platforms to protect the online shopper from falling victim to continuously evolving tactics designed to scam them. And providing more ways to validate the source of the reviews themselves (the reviewer). As well as provide motivation for legitimate buyers to leave informative & helpful reviews by incentivizing them with rewards and perks to saturate the system with more useful reviews. With the combined effort of these developmental actions, enhancement on the specific pain points of the current system regarding customer protection can be addressed and possibly improve the perceived integrity of their review systems and their platform as a whole.

Improve Reviewer Validity and Visibility. To improve the accessibility of information and the ability to validate source information, platforms can enhance user reviewer profiles to be actively engaging and helping towards the improvement of the review systems set in place. This can be done through the development of verified purchase badges, and identifying and rewarding trustworthy contributors through reviewer ranks similar to loyalty/membership tiers existing in the platforms, and also with vouchers and perks to incentivize and motivate online shoppers to leave informed and useful reviews. Additionally, stricter verification and registration requirements can be implemented before online shoppers can have the ability to leave a review for better preventive measures.

Moderate Surface Features. To protect consumers from false information or advertising, developing moderating systems for videos, images, and texts posted on these platforms for both product details and product reviews can assist in ensuring the quality and relevance of these contributions. This can include constructing specific standards regarding appropriate and quality content that can be posted on the platform such as reviewers posting photos or videos from other sources and claiming it to be theirs.

Promote Awareness of Rampant Fraudulent Strategies on the Platform. To bridge the tension between the online shoppers' knowledge of review manipulation tactics and their continued trust in the review mechanism of the platforms, focusing on launching corresponding projects to improve user awareness regarding fraudulent strategies existing on their platform can help their customers be informed and gain more confidence in assessing review authenticity. Informative campaigns and regular updates on this matter can be established in accordance with this issue.

Provide Insured Transactions against Fraud. To increase trust and lessen the vulnerability of online shoppers against scams or fraudulent transactions, platforms can provide cyber insurance covering expenses up to a certain amount in the event of fraud. Setting well-designed systems, terms, and conditions of agreement in place can aid and benefit in the protection of both the shoppers, legitimate sellers, and the platforms themselves.

Directions for Future Research

By implementing these improvements, it is expected that online shoppers' trust in the review systems of e-commerce platforms will improve and enhance the quality of information that can be found on the review systems of the platforms. Furthermore, these recommendations aspire to bridge the gap between uninformed online shoppers and the existence of fraudulent strategies growing prevalent in these platforms to protect the online shopper from falling victim to continuously evolving tactics designed to scam them. And providing more ways to validate the source of the reviews themselves (the reviewer). As well as provide motivation for legitimate buyers to leave informative & helpful reviews by incentivizing them with rewards and perks to saturate the system with more useful reviews. With the combined effort of these developmental actions, enhancement on the specific pain points of the current system regarding customer protection can be addressed and possibly improve the perceived integrity of their review systems and their platform as a whole.

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REFERENCES

1. Ananthakrishnan, U. M., Li, B., & Smith, M. D. (2020). A Tangled Web: Should Online Review Portals Display Fraudulent Reviews? SSRN Electronic Journal. https://doi.org/10.2139/ssrn.3297363
2. Asenahabi, Bostley. (2019). Basics of Research Design: A Guide to selecting appropriate research design. 6. 76 - 89.
3. Atoy, M. B., Garcia, F. R., Cadungog, R. R., Cua, J. D., Mangunay, S. C., & de Guzman, A. B. (2020). Linking Digital Literacy and online information searching strategies of Philippine University Students: The moderating role of mindfulness. Journal of Librarianship and Information Science, 52(4), 1015–1027. https://doi.org/10.1177/0961000619898213
4. Azimi, S., Chan, K., & Krasnikov, A. (2022). How fakes make it through: the role of review features versus consumer characteristics. Journal of Consumer Marketing, doi: 10.1108/jcm-04-2021-4597
5. Bandiling, V. J., Peralta, K. M., Relucio, R. M., & Etrata, A. E. (2022). Social media literacy and misinformation: The significance of consuming reliable information in relation to shopper buying behavior. Management Review: An International Journal, 17(1), 128–160. https://doi.org/10.55819/mrij.2022.17.1.128
6. Barbado, R., Araque, O., & Iglesias, C. A. (2019). A framework for fake review detection in online consumer electronics retailers. Information Processing & Management, 56(4), 1234-1244. doi:10.1016/j.ipm.2019.03.002
7. Beresford Research. (2024). Age Range by Generation. Beresford Research. https://www.beresfordresearch.com/age-range-by-generation/
8. Bhatia, M. K. (2017). Data Analysis and its Importance. International Research Journal of Advanced Engineering and Science, 2(1), 166–168.
9. Blandi, L., Sabbatucci, M., Dallagiacoma, G., Alberti, F., Bertuccio, P., & Odone, A. (2022). Digital information approach through social media among gen Z and millennials: The global scenario during the COVID-19 pandemic. Vaccines, 10(11), 1822. https://doi.org/10.3390/vaccines10111822
10. Cao, C. (2023). The Impact of Fake Reviews of Online Goods on Consumers. BCP Business & Management, 39, 420–425. https://doi.org/10.54691/bcpbm.v39i.4208
11. Capistrano, E. P., Gomez, M. M., & Isleta, A. P. (2023). Examining trust, self-efficacy, and technology acceptance in the Philippines' e-commerce sectors. Information Development, 026666692311538. https://doi.org/10.1177/02666669231153837
12. Celis, J. M., Ilagan, K. M., Mejia, P. L., & Tambungui, E. C. (2022). The Impact of the Pandemic to the Buying Behavior of the Filipino Youth. JOURNAL OF UNDERGRADUATE RESEARCH IN BUSINESS ADMINISTRATION, 1(2).
13. Chen, T., Samaranayake, P., Cen, X., Qi, M., & Lan, Y.-C. (2022, May 2). The impact of online reviews on consumers' purchasing decisions: Evidence from an eye-tracking study. Frontiers. https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2022.865702/full
14. Chetioui, Y., Lebdaoui, H., & Chetioui, H. (2020). Factors influencing consumer attitudes toward online shopping: the mediating effect of trust. EuroMed Journal of Business, ahead-of-print(ahead-of-print). doi:10.1108/emjb-05-2020-0046
15. Çoklar, A. N., & Tatli, A. (2021). Examining the Digital Nativity Levels of Digital Generations: From Generation X to Generation Z. Shamlax: International Journal of Education , 9(4), 433–444. https://doi.org/https://doi.org/10.34293/education.v9i4.4224
16. Conelly, L. M. (2021). Introduction to Analysis of Variance (ANOVA). Pitman, 30(3).
17. D. Rada, M. V., V. Santiago, M. A., Castil, F. S., Dayanghirang, N., Carmela A. Enoval, K., Gonzaga, R., Kazi, I. A., Raza, T., Rico, J. Q., & Supangco, D. P. (2022). Integrating E-S-QUAL model and analytical hierarchy process to assess consumers' preferences in selected online shopping platforms in Manila, Philippines. Proceedings of the International Conference on Industrial Engineering and Operations Management. https://doi.org/10.46254/eu05.20220025

18. Davis, F., Francis Gnanasekar, M. B., & Parayitam, S. (2021). Trust and product as moderators in online shopping behavior: Evidence from India. South Asian Journal of Marketing, 2(1), 28–50. https://doi.org/10.1108/sajm-02-2021-0017
19. Department for Business & Trade (2023). Estimating the prevalence and impact of fake online reviews. https://assets.publishing.service.gov.uk/media/6447c00c529 eda000c3b03c5/fake-online-reviews-research.pdf
20. Department of Trade and Industry. (2023). Republic Act No. 11967 or the Internet Transactions Act of 2023 | DTI ECommerce. Ecommerce.dti.gov.ph. https://ecommerce.dti.gov.ph/internet-transactions-act-of-2023/#:~:text=Signed%20by%20President%20Ferdinand%20R
21. Doseděl, T., Kafková, M. P., & Vidovićová, L. (2023). Age and gender difference in ICT literacy and Biometrics Knowledge. Sociální Studia / Social Studies, 20(1). https://doi.org/10.5817/soc2022-21291
22. Dulay, F. A., Liban, R., Rustia, J. R., Tayag, M. M., & Castaño, M. C. (2022). Generation Y and Filipino consumers' purchasing behavior on electronic products in Shopee and lazada. Journal of Business and Management Studies, 4(1), 294–305. https://doi.org/10.32996/jbms.2022.4.1.30
23. Field, A. (2013). Discovering statistics using IBM SPSS statistics (4th ed.). SAGE Publications. Galang, A. (2020, May 8). Shopee dominates e-commerce in Southeast Asia, including PH. Rappler. https://www.rappler.com/business/260791-shopee-dominates-e-commerce-southeast-asia-including-philippines#gsc.tab=0
24. Guetterman T. C. (2019). Basics of statistics for primary care research. Family medicine and community health, 7(2), e000067. https://doi.org/10.1136/fmch-2018-000067
25. Gupta, H. (2021, March). Philippines E-Commerce Logistics Market Outlook to 2025. Philippines E-Commerce Logistics Market Report, Major Players, Revenue & Trends Analysis. https://www.kenresearch.com/industry-reports/philippines-e-commerce-logistics-market
26. Gupta, S., & Chopra, C. (2020). Impact Of Social Media On Consumer Behaviour. ResearchGate; www.researchgate.net.https://www.researchgate.net/publication/342638389_Impact_Of_Social_Media_On_Consumer_Behaviour
27. Husam, M., Alawadh., Amerah, Alabrah., Talha, Meraj., Hafiz, Tayyab, Rauf. (2023). Semantic Features-Based Discourse Analysis Using Deceptive and Real Text Reviews. Information, doi: 10.3390/info14010034
28. Jiang, H., & Regua, O. U. (2023). An analysis of the purchasing behavior strategies of Philippine College graduates on online platforms. Information Systems and Economics, 4(10). https://doi.org/10.23977/infse.2023.041008
29. Jin, C., Yang, L., & Hosanagar, K. (2022). To Brush or Not to Brush: Product Rankings, Consumer Search, and Fake Orders. Information Systems Research, 34(2). https://doi.org/10.1287/isre.2022.1128
30. Khaw, J. (2023). Shopee: How Does E-commerce Affect E-consumer Perception And Satisfaction?. International Journal of Tourism and Hospitality in Asia Pacific, doi: 10.32535/ijthap.v6i1.2169
31. Kumar, J. (2020). Fake Review Detection Using Behavioral and Contextual Features. ArXiv (Cornell University). https://doi.org/10.48550/arxiv.2003.00807
32. Lago, A., & Torrano Peñaflor, A. J. (2021). THE IMPACT OF ONLINE REVIEWS SYSTEM ON FILIPINOS' ONLINE SHOPPING BEHAVIOR.
33. Li, B., Chen, S., & Zhou, Q. (2023). Empathy with influencers? the impact of the sensory advertising experience on user behavioral responses. Journal of Retailing & Consumer Services, 72, 103286. https://doi.org/10.1016/j.jretconser.2023.103286
34. Liu, Y., Jiang, M., & Wu, H. (2023). The Brushing Game in Online Marketplaces. International Journal
35. of Electronic Commerce, 27(2), 163–184. https://doi.org/10.1080/10864415.2023.2184237
36. Lucassen, Teun & Schraagen, Jan Maarten. (2011). Factual Accuracy and Trust in Information: The Role of Expertise. JASIST. 62. 1232-1242. 10.1002/asi.21545.
37. Mui, D, Kee, H. (2023). Are You Satisfied With Shopee? A Study of Customer Satisfaction Towards Shopee. International Journal of Tourism and Hospitality in Asia Pacific, doi: 10.32535/ijthap.v6i2.2342

38. Naci Çoklar, A., & Tatli, A. (2021). Examining the digital nativity levels of Digital Generations: From generation X to generation Z. Shanlax International Journal of Education, 9(4), 433–434. https://doi.org/10.34293/education.v9i4.4224

39. Nanjundeswaraswamy, T. S., & Divakar, S. (2021). Determination of sample size and sampling methods in applied research. Proceedings on Engineering Sciences, 3(1), 25–32. https://doi.org/10.24874/pes03.01.003

40. Nazarova, E. & Nazarov, A. (2021). Digital Literacy and the Development of Digital Education. KnE Social Sciences, doi: 10.18502/KSS.V5I2.8348

41. Organization for Economic Cooperation and Development. (2019). Understanding online consumer ratings and reviews. OECD Digital Economy Papers, 289, 28. https://doi.org/10.1787/eb018587-en

42. Pablo, K. I., Gallaza, G., Delos Reyes, B. E., & Al-Jarf, A. F. (2022). Insta Worthy: A study on the effects of photo editing towards Electronic Commerce in Metro Manila. Journal of Business and Management Studies, 4(2), 230–240. https://doi.org/10.32996/jbms.2022.4.2.18

43. Prasetyo, Y. T., & Dela Fuente, D. G. D. (2020). Determinant factors affecting customer satisfaction among Filipinos in Lazada online shopping during COVID-19 pandemic: A structural equation modeling approach | IEEE conference publication | IEEE xplore. IEEE Explore. https://ieeexplore.ieee.org/abstract/document/9266734

44.

45. Putri, N., Prasetya, Y., Handayani, P. W., & Fitriani, H. (2023). Tiktok Shop: How trust and privacy influence generation Z's purchasing behaviors. Cogent Social Sciences, 10(1). https://doi.org/10.1080/23311886.2023.2292759

46. Rintyarna, B. S., Sarno, R, Fatichah, C. (2019). Semantic Features for Optimizing Supervised Approach of Sentiment Analysis on Product Reviews. The first computers, doi: 10.3390/COMPUTERS8030055

47. Román, S., Riquelme, I. P., & Iacobucci, D. (2023). Fake or credible? antecedents and consequences of perceived credibility in exaggerated online reviews. Journal of Business Research, 156, 113466. https://doi.org/10.1016/j.jbusres.2022.113466

48. One way analysis of variance. StatsDirect. (n.d.). https://www.statsdirect.com/help/analysis_of_variance/one_way.htm

49. Senate of the Philippines. (2021, November 24). REVIEW OF CRIMINAL LAWS (RISE OF SOCIAL MEDIA PLATFORMS AND THE RAPID ADVANCEMENT OF TECHNOLOGY). https://legacy.senate.gov.ph/lis/bill_res.aspx?congress=18&q=SRN-953

50. Sharma H. (2022). How short or long should be a questionnaire for any research? Researchers dilemma in deciding the appropriate questionnaire length. Saudi journal of anaesthesia, 16(1), 65–68. https://doi.org/10.4103/sja.sja_163_21

51. Shopee. (2022, November 30). What is brushing? Seller.shopee.com.my; Shopee Seller Education Hub. https://seller.shopee.com.my/edu/article/9698/brushing

52. Siar, S. (2021, August). Fake news, its dangers, and how we can fight it. Policy Notes, 2021-06(2508-0865), 10. https://pidswebs.pids.gov.ph/CDN/PUBLICATIONS/pidspn2106.pdf

53. Thomas, Darrin & Zubkov, Pavel. (2023). Quantitative Research Designs.

54. Trninić, D., Kuprešanin Vukelić, A., & Bokan, J. (2021). Perception of "fake news" and potentially manipulative content in digital media—a generational approach. Societies, 12(1), 3. https://doi.org/10.3390/soc12010003

55. Uakarn, C., Chaokromthong, K., & Sintao, N. (2021). Sample Size Estimation using Yamane and Cochran and Krejcie and Morgan and Green Formulas and Cohen Statistical Power Analysis by G*Power and Comparisons. APHEIT INTERNATIONAL JOURNAL, 10(2), 76–88.

56. VanVoorhis, C. R. W., & Morgan, B. L. (2007). Understanding power and rules of thumb for determining sample sizes. Tutorials in Quantitative Methods for Psychology, 3(2), 43–50. https://doi.org/10.20982/tqmp.03.2.p043

57. Velasco, J. C. (2020). Millennials as Digital Natives: Examining the Social Media Activities of the Philippine Y-Generation. SOCIAL SCIENCES & HUMANITIES, 28(3).

58. Vetter T. R. (2017). Descriptive Statistics: Reporting the Answers to the 5 Basic Questions of Who, What, Why, When, Where, and a Sixth, So What?. Anesthesia and analgesia, 125(5), 1797–1802. https://doi.org/10.1213/ANE.0000000000002471

59. Wahpiyudin, C. A. B., Mahanani, R. K., Rahayu, I. L., Simanjuntak, M., Sumarwan, U., Yuliati, L. N., Djamaludin, M. D., Johan, I. R., & Muflikhati, I. (2022). The credibility of consumer reviews on three e-commerce in Indonesia: Mixed method approach. Jurnal Ilmu Keluarga Dan Konsumen, 15(3), 287–299. https://doi.org/10.24156/jikk.2022.15.3.287

60. Walther, M., Jakobi, T., Watson, S. J., & Stevens, G. (2023). A systematic literature review about the consumers' side of fake review detection – which cues do consumers use to determine the veracity of online user reviews? Computers in Human Behavior Reports, 10, 100278. https://doi.org/10.1016/j.chbr.2023.100278

61. Wang, N., Yang, J., Kong, X., & Gao, Y. (2022). A fake review identification framework considering the suspicion degree of reviews with time burst characteristics. Expert Systems with Applications, 190, 116207. https://doi.org/10.1016/j.eswa.2021.116207

62. Wu, Y., Ngai, E. W. T., Wu, P., & Wu, C. (2020). Fake online reviews: Literature review, synthesis, and directions for future research. Decision Support Systems, 132, 113280. https://www.sciencedirect.com/science/article/abs/pii/S016792362030035X?via%3Dihub

63. Zhang, Y., Luo, C., Wang, H., Chen, Y., & Chen, Y. (2022). 'A right to be forgotten': Retrospective privacy concerns in social networking services. Behaviour & Information Technology, 42(7), 829–850. https://doi.org/10.1080/0144929x.2022.2046162

Nostalgia Marketing: Its Influence in Purchase Intention Among Gen X Consumers

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Abstract

In the growing advertising landscape, consumer attention spans have dwindled, and nostalgia has offered a way to make its mark by tapping into consumers' memories. Nostalgia marketing leverages on memories to influence its consumers. Utilizing nostalgia in marketing endeavors can create a favorable attitude toward a brand and stimulate consumer purchasing intentions. This connection creates a way for increased purchase intention. This study explored the influence of nostalgia marketing on Generation X consumers, focusing on key elements such as nostalgia imagery, cultural references, nostalgia narratives, and brand heritage. Through a quantitative approach with purposive sampling of 395 respondents from District 3, Caloocan City, the research investigated how these elements impact purchase intentions. Through the findings, it is found that only three elements of nostalgia marketing (nostalgia imagery, nostalgia narratives, and brand heritage) significantly influenced the purchase intention among Gen X consumers. In the three commercials it was shown that the three elements were shown to be effective in influencing one's purchase intention. The results indicate that nostalgia marketing resonates among the Gen X audience. The study concludes that nostalgia marketing can be used among brands to establish a relationship and drive purchase behavior among the Gen X demographic.

Keywords: Nostalgia Marketing, Purchase Intention, Marketing, Advertising

Introduction

Through recent years, the advertising landscape has been a battleground for attention. Consumer attention spans and focus are shrinking, and advertisers are constantly working to get their message across to translate to a purchase. As Nelson (2023) mentions, effective advertising not only informs but also captures the imagination, setting aside distractions to engrave itself into our memory. Standing out in the landscape is an ever-changing strategy. The solution to this may lie in the human emotion of nostalgia. The emotional core of consumer decision making is highlighted in a by He et al. (2022), where it stated how buyers gravitate towards products that resonate with their values and preferences, often seeking a sense of comfort and familiarity. Nostalgia taps into this desire, allowing consumers to indulge in positive memories, reliving the "good old days" and past moments that shaped them. With this, by associating a certain product, service, or brand with a familiar feeling or situation, bridges an emotional connection with audiences.

Consumer nostalgia refers to the fondness for products or services associated with the past, often popular during one's youth. Nostalgia marketing, as described by Alos & Palacpac, (2024), leverages this sentiment by incorporating nostalgic elements into marketing strategies to evoke deep memories and influence consumer behavior. Using nostalgia in marketing can create a positive impression of a brand and encourage people to buy its products.This connection is what creates a way for increased purchase intention. Purchase intention is one of the first steps to purchase decision, which is why an emphasis on this increases the opportunities to grow in purchases. Research by Singh et al. (2021), emphasizes the impact of emotions in shaping consumer preferences, highlighting the role and persuasive power of nostalgia marketing.

By having brands leverage on positive memories, the emotions that are evoked through this can significantly influence purchase decisions (Cho et al., 2021). Xie et al. (2021) also highlight that nostalgia is a distinct emotion that strongly drives consumers' desire to purchase new products. The nostalgia even also acts as a coping mechanism for dealing with unpleasant present situations, having consumers seek to go back to a time of peace, indirectly influencing their intention to purchase new. This emotional connection translates into brand trust and attachment, ultimately boosting purchase desire (He et al., 2022).

This research sought to address this gap by examining how nostalgia marketing affects the purchase intentions of Gen X consumers in the Philippines. Ultimately, the study aimed to enhance the understanding of how nostalgia marketing can be effectively utilized to impact purchasing behavior among the Gen X demographic.

Literature Review

Nostalgia Marketing

As the marketing field continues to develop, marketing strategies are constantly trying to find ways to create deeper connections between consumers. Nostalgia marketing is a strategy that is becoming an increasingly prominent approach as this makes use of elements, events, ideas, which are associated in consumers' past, that then creates an avenue for brands to create emotional connections and strengthen existing relationships with consumers, which ultimately drives sales up. Nostalgia marketing proves its effectiveness for a wide audience as it evokes a sense of comfort and familiarity in older consumers reminiscing their formative years, while also attracting the younger audience who are yearning for a connection to the retro years (Alos & Palacpac, 2024).

In an article entitled "What You Should Know About Nostalgia Marketing" by Clark (2023), it was stated that nostalgia has its way of capitalizing on positive emotions by having brands associate themselves with elements that customers have fond memories of. This association could be achieved through events, songs, concepts, songs, celebrities, or products (Clark, 2023). Ultimately, brands leverage on these familiar elements from the past to create a familiar or nostalgic feeling in consumers.

Marketers have different tactics to trigger nostalgia in consumers. There are instances where it could be incorporated into advertisements, references events or people that resonate with their target audience, or leverage on events to create a sense of shared experience (Holotová et al., 2020). Nostalgia marketing extends beyond traditional marketing means. Companies can incorporate retro elements in packaging and branding to create a "sense of history" for their audience (Holotová et al., 2020).

The effectiveness of nostalgia marketing goes beyond generations. It has been shown that it encourages spending with the promise of a return to happy memories (Holotová et al., 2020). Having this emotional connection fosters brand loyalty and connection which then translates to increased sales for companies that leverage nostalgia marketing. Through a study entitled Retro Marketing - A Power of Nostalgia which Works among the Audience by Holotová et al. (2020), it was aimed to show the significance of the "retro trend" in the world of marketing to help identify the strength nostalgia holds among the generations of Generations Z, Millennials, Generation X and Baby Boomers. The results of the study the relevance of retro marketing communication and its significant impact on the generations.

A study entitled The Effect of Nostalgia Marketing on Consumers' Purchase Intention by Alkhafagi (2023), determined the outcome nostalgia marketing may have on consumer buying intent and demographic components. In the study, small sculptures that showcased the existence of early Baghdadis in the fifties and sixties of the last hundred years were displayed around shops in Babylon Mall in Baghdad, and the sample group involved in the study involved respondents who had seen these sculptures. The overall results showcase that there is a positive impact of nostalgia marketing on consumers' purchase intention. Through the results, it was recommended that nostalgia should be allocated towards product planning to further encourage consumers' purchase intention. It was also found that nostalgia marketing has a higher effect on women than on men, and also its impact varies with age, and is much stronger among widows.

A study by Veresiu et al. (2021), entitled "Marketing and Nostalgia: Unpacking the Past and Future of Marketing and Consumer Research on Nostalgia", delves into the contemporary drivers of nostalgia. Factors such as social change, economic uncertainty, and rapid technological advancements were highlighted as reasons consumers yearn for simpler times. During periods of social unrest or uncertainty of the future, consumption of nostalgia may surge (Veresiu et al., 2021).

Hartmann and Brunk (2019) further study the concept of "enchantment" in nostalgia marketing in their research entitled, "Nostalgia marketing and (re-)enchantment". It was argued here that nostalgia marketing's end goal is to take the ordinary and infuse it with special meaning, which creates a sense of magic and excitement (Hartmann & Brunk, 2019). In the study, three distinct routes to "enchantment" are identified, triggered by different nostalgia modes: Re-Instantiation, which symbolically recreating the past through retro design or imagery. Secondly, Re-Enactment, the integration of past-themed brands into present-day experiences. Lastly, Re-Appropriation, is the playful reinterpretation of the past in a new light (Hartmann & Brunk, 2019).

Nostalgia marketing stands as a powerful tool when creating relationships with consumers. By thoroughly understanding the factors that trigger nostalgia, companies can leverage this and employ effective strategies to further build brand loyalty and drive sales. As the world continues to rapidly change, nostalgia's influence in the world of marketing has a likelihood to persist, as it offers a sense of comfort and familiarity in an ever-changing world. The significance of emotions in advertising resonates deeply within the realm of consumer behavior. Poels and Dewitte's (2019) study illuminates the central role emotions play in eliciting specific responses from consumers, emphasizing the untapped potential of leveraging emotions strategically in advertising campaigns to drive desired consumer actions.

According to Poels and Dewitte's (2019), the need for marketers to adopt a nuanced understanding of emotional triggers within advertising messages. They suggest that by strategically crafting advertisements to evoke targeted emotional responses, marketers can effectively influence consumer behavior towards desired outcomes. This behavioral perspective offers useful insights for marketers aiming to enhance the effectiveness of their advertising efforts, guiding them towards leveraging emotions as a powerful tool for driving consumer actions.

Elements of Nostalgia Marketing

By incorporating familiar elements from the past there are key elements one of them is nostalgia imagery, this often uses visual elements, which evoke memories of the past.

Nostalgia Imagery refers to visual representation that evoke feelings of longing, sentimentality, and yearning of the past. Brand creates advertisements by featuring old-school graphics, or vintage photography (Hesterberg, 2023). This connection to the past can make the brand's message more memorable and connect with consumers on an extensive level.

Cultural reference holds a significant part in nostalgia marketing, brands tap into popular music, fashion, movies or TV shows and other iconic elements from previous decades to engage with their ideal audience, this gives consumers a moment from the past (Mehrishi, n.d.). The use of cultural references allows the brand to position their products or services as part of a familiar, comforting that consumers can relate to. Based on Mehrishi, n.d. This serves as a kind of shorthand that allows people within that culture to instantly recognize, relate to and connect over common points of cultural knowledge and nostalgia.

Nostalgia narratives are written stories, themes or newspapers that evoke nostalgia feelings to consumers (Hesterberg, 2023). A key element of nostalgia narratives is the use of a sentimental storytelling that invokes a longing for simpler, "better" times. By creating a narrative between characters' feelings and readers, these narratives connect to consumers' nostalgic experiences, brands can establish emotional connection, build trust, and enhance brand loyalty.

Brand heritage, it refers to a history of the company, legacy, and the emotional connection that has been with the consumers over time (Sierra, 2019), and by aligning marketing efforts with this nostalgia, brands must remain mindful of their brand heritage to ensure that the essence and emotional connection built with consumers over time are preserved (Jones, 2020). Companies can differentiate themselves in crowded markets, build customer loyalty, and position their products or services as timeless, trustworthy, and authentic.

The study finds that nostalgia marketing is an effective strategy for brands to create memorable and resonant messages with the consumers (Jones, 2020). The key elements of this strategy includes nostalgia imagery, cultural references, nostalgia narratives, and brand heritage. These elements together enable brands to create deeper emotional connections with consumers, fostering trust, loyalty and a stronger inclination to purchase.

There are several studies that have emphasized the importance of a brand's past in fostering trust with consumers. In his study "Consumer Research Insights on Brands and Branding," Keller (2020) explores the impact of brand history and established credibility greatly impact consumer views on reliability and trustworthiness. Having a long history sets a brand apart from its competitors and fosters emotional bonds with consumers, especially older ones like Gen X. This is because consumers tend to have more faith in brands with a long history, seeing them as dependable and trustworthy, which affects their purchasing choices.

Influence of Nostalgia Marketing on Consumers

Nostalgia marketing has appeared as a powerful tool for businesses to engage with consumers on an emotional level and influence their purchasing decisions, rendering an emotion that is exciting and magical, which makes it special for marketing (Hartmann & Brunk, 2019). Nostalgia marketing is known for a marketing strategy that refers to activating consumers' nostalgia, which evokes their memories deep inside and eventually promotes consumers' buying behaviour (Poels and Dewitte, 2019).

The research of Warren (2022) explores the rising use of nostalgia in marketing. In their investigation, it was discovered that nostalgic features such as characters, clothing, and decorations are utilized in advertisements that are aimed at stimulating nostalgic feelings among the audience. This is proof that nostalgia is used as a tool by marketers in an attempt to improve the purchasing intentions of consumers. In relation to this, Rana et al. (2020) also look at how nostalgia is used in the fields of marketing and business operation. Their research considered nostalgia as a powerful tool in marketing, which is very different from how nostalgia is treated in psychology. Furthermore, it was highlighted that nostalgic advertisements increase the propensity of consumers to buy from a brand.

The feelings and sentiments of consumers associated with youth were also investigated. It was discovered that nostalgia is tied to feelings of youthfulness, which can alter the views of the audience to the ads in a favorable manner. On the other hand, Singh et al. (2020) looks at how nostalgic commercials and spiritual well-being affects the purchase intentions of consumers. In employing nostalgic appeals in marketing initiatives, the study highlights how important it is to take into account the spiritual well-being of the consumers. Understanding the effect of nostalgia marketing on consumer decisions is the aim of both the current study and this study.

Pichierri (2023) provides a comprehensive exploration of nostalgia marketing and its effect on consumer behavior. Drawing from interdisciplinary literature in sociology, psychology, and consumer behavior, the paper offers real-world examples and insights into the effectiveness of nostalgia-based marketing strategies.

Nostalgia marketing has become increasingly popular, especially during times of uncertainty or crises. For instance, after 9/11 and the financial crisis of 2008, marketers turned to nostalgia as a strategy. Even during the COVID-19 pandemic, consumers searched for comfort in content related to the past, like TV shows, movies, and songs from their earlier years. Real-world examples include popular TV series like "Stranger Things," which is filled with nostalgic elements, and rebranding efforts by companies like Burger King, leveraging the power of nostalgia (Pichierri, 2023).

Purchase Intention

The research by Peña-García et al. (2020) looked at the relationship between purchase intention and the online purchasing behavior of different cultural groups. In this paper, purchase intention is discussed in relation to the widespread use of online shopping, and its importance as a prerequisite to actual purchase behavior is emphasized. Purchase intention was suggested in the study as a key element affecting consumers' online buying behavior. According to Ghorbanzadeh and Rahehagh (2021), consumers tend to develop stronger connections to brands that evoke nostalgic feelings, which can influence their purchasing intentions in favor of products associated with the memories.

In addition, Martins et al. (2019) discovered that consumer purchase intentions are shaped by a combination of factors that includes marketing value, flow experience, web design performance, and brand recognition. This is relevant to the current study as both investigate the factors that lead customers to plan a purchase. These literatures serve as a great source for the present researchers as the results of this study emphasize the role of purchase intention which is the central variable of the present study.

Xia et al. (2021) explores the relationship between nostalgia and purchasing intentions of consumers. In addition, their investigation included how the COVID-19 pandemic has impacted the feelings of nostalgia among consumers. It was revealed that COVID heightens the level of nostalgia among the respondents which then boosted their purchase intentions. In summary, they concluded that higher levels of nostalgia equate to an increase in the desire to purchase items. While this study is very similar to the present study, there are still some key differences. Xia et al. investigated nostalgia along with other external factors, while the present study aims to only focus on the nostalgia and purchase intentions connection. Furthermore, in the conducted study of Ozhan and Akkaya (2021), they discovered that nostalgia strongly influences consumers' attitudes and purchase intentions. Incorporating nostalgic elements into advertisements can positively impact attitudes towards the brand and willingness to buy.

In recent years, the concept of nostalgia has emerged as a significant role in consumer behavior, particularly in the realm of marketing strategies. Nostalgia, characterized by a yearning for the past, has been increasingly leveraged by companies to strengthen their competitive advantage in the market. Şeniz Özhan (2021).

Şeniz Özhan (2021) underscores the profound influence of nostalgia on consumer responses to advertisements. The findings, drawn from a sample of 381 consumers and analyzed through a structural equation model, reveal a robust relationship between nostalgia proneness and ad-evoked nostalgia, as well as subsequent impact on brand attitude and purchase intention.

The study suggests that tapping into consumers' sense of nostalgia through advertising can yield positive outcomes for brands. By evoking feelings of nostalgia, marketers can cultivate favorable brand attitudes and stimulate purchase intentions within their chosen audience. This highlights the importance of integrating nostalgic elements into advertising campaigns as a strategic approach to engaging consumers and fostering brand loyalty Şeniz Özhan (2021).

Social Identity

A study entitled Social Identity Theory and Marketing: A Systematic Review of the Literature by Katariina Helkkula, Chris Gummerus (2021) examines how social identity influences consumer behavior and purchase intentions. The study underscores that consumers' social identities, shaped by their interactions on social media, have a crucial role in their purchasing decisions. Brands that align their messaging and values with consumers' social identities, especially through nostalgic themes, can enhance emotional engagement and significantly increase purchase intentions.

In a study entitled The Impact of Social Media Marketing on Consumer Engagement in Sustainable Consumption: A Systematic Literature Review by Paweł Bryła, Shuvam Chatterjee, Beata Ciabiada-Bryła (2022) explores how social media marketing influences consumer involvement in maintainable use. The study highlights that social media platforms facilitate emotional connections and group affiliations, enhancing consumer engagement. By utilizing nostalgic content and aligning with social identities, brands can significantly boost purchase intentions. The review emphasizes the role of influencer marketing and content creation in fostering long-term relationships with consumers, who demonstrate higher purchase intentions when their social identity is engaged.

According to Lukas K. D. Bosman, Ruth Mugge, Jan P. L. Schoormans (2020) Nostalgia marketing taps into shared memories and cultural experiences that form a part of consumers' social identities. This connection fosters a sense of belonging and strengthens the emotional bond of the consumer and the brand, driving higher purchase intentions.

Theoretical Framework

The term "priming" is a study in psychology, referring to the phenomenon where a stimulus or action influences a subsequent response. Priming was described as the effect of a stimulus on evoking a familiar response. This concept is understood in social psychology, as the influence of events on the trigger of stored memory (Williams, 2021). Early research on priming, by Iyengar et al. (1982), suggested that most individuals rely on prior knowledge rather than in-depth understanding when making decisions, highlighting the importance of priming effects in decision-making processes.

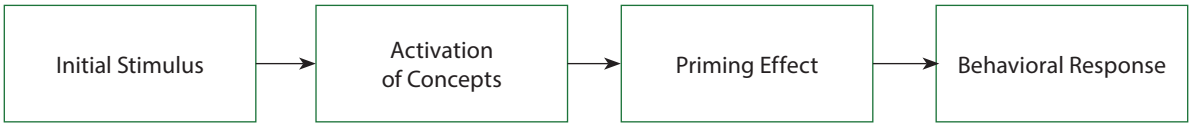


Fig. 1. Priming Theory

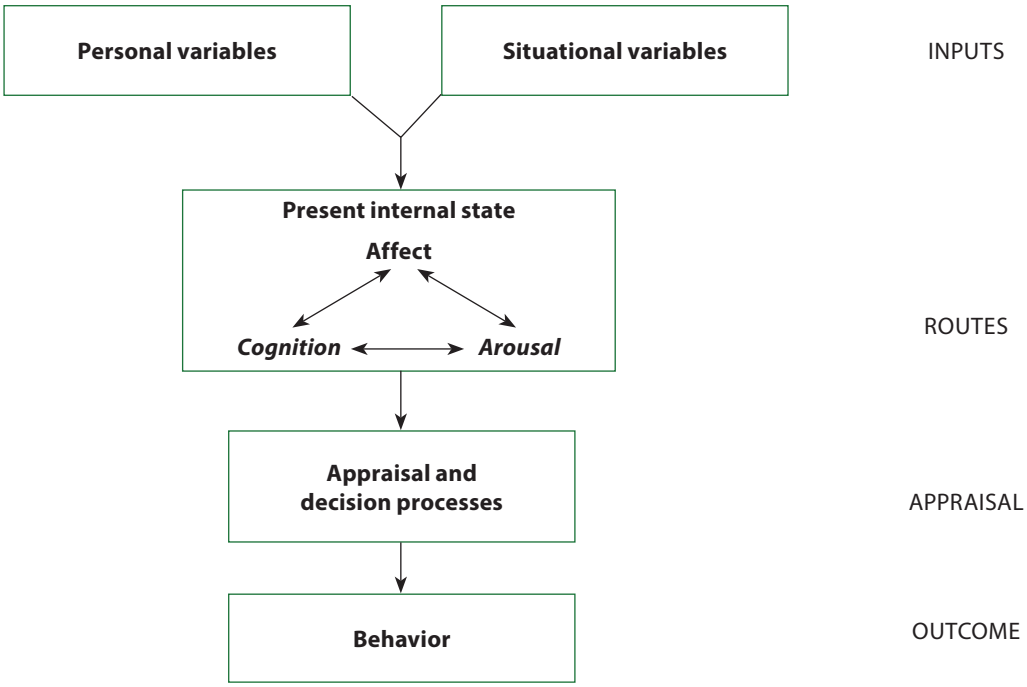


Fig. 2. The Weapon Priming Effect (Benjamin & Bushman, 2016)

Priming theory originates from social psychology and cognition and offers a unique and valuable perspective in understanding how media coverage and depiction influence individuals. (Zamith, 2022). It suggests that one's exposure to media stimuli can activate interlinked thoughts in the minds of audiences, which can then affect their subsequent thoughts and behaviors. One's reaction and perceptions to information can be influenced by constant exposure to specific themes or concepts in media content, which can strengthen associations in their minds. Additionally, the strength of the initial association, the frequency of exposure, and individual variations in memory and cognition processes can all affect the priming effects.

Methodology

The study employed a descriptive comparative research design with a quantitative approach to evaluate the influence of nostalgia marketing on the purchase intention of Generation X consumers, focusing on the effectiveness of elements such as nostalgia imagery, cultural references, narratives, and brand heritage. The research was conducted in District 3, Caloocan City, chosen for its diverse population and urban environment.

To determine the sample size, Slovin's formula was applied, resulting in 395 respondents from a target population of approximately 31,760 Gen X individuals. A combination of purposive sampling (to target relevant participants) and convenience

sampling (for practicality and accessibility) was utilized. Respondents were qualified based on their age (44–59) and residency in District 3.

Data collection involved a structured survey questionnaire, validated by experts and tested through a pilot study with 25 respondents to ensure clarity and reliability. The survey was administered both online via Google Forms and through in-person distribution. Respondents were asked to watch commercials featuring nostalgic marketing themes before answering questions. The questionnaire was divided into three parts: (1) Nostalgia Marketing Features, (2) Influence of Nostalgia Marketing Elements, and (3) Purchase Intentions. Responses were measured using a 4-point Likert Scale (Strongly Agree to Strongly Disagree).

For data analysis, descriptive statistics (mean, frequency percentage, weighted mean) and regression analysis (specifically linear regression) were employed to examine the relationships between nostalgia marketing elements and purchase intention. Regression analysis allowed the researchers to identify and quantify the predictive impact of nostalgia marketing features on consumer purchase intention, providing strong insights into the effectiveness of these strategies. This standardized and comprehensive approach enabled meaningful conclusions about consumer behavior and preferences.

Analysis and Discussions

Discussion of Findings

Respondents' Reactions to Nostalgia Marketing Features in Commercial 1: Kwentong Jollibee The commercial "Kwentong Jollibee - Hinga" effectively employs nostalgia marketing by featuring familiar scenarios that resonate with the experiences of Generation X consumers. The use of childhood memories, family gatherings, and traditional values creates an emotional narrative that invites viewers to reflect on their own past. Respondents reported strong positive reactions to the nostalgic elements presented. The portrayal of relatable situations- such as familial love and support- evoked feelings of warmth and nostalgia, reinforcing the brand's identity as staple in their lives. The commercial's imagery and storytelling align well with the values and emotions that are significant to the target demographic.

Table 1. Respondents' Reactions to Nostalgia Marketing Features in Commercial 1: Kwentong Jollibee

	Mean	SD	Interpretation
Characters			
The characters in the commercial reminds me of the past.	3.55	0.648	Strongly Agree
The commercial used characters that I felt connected to.	3.53	0.626	Strongly Agree
Total	**3.542**	**0.5822**	**Strongly Agree**
Clothing			
The clothing used in the commercial reminds me of clothing trends in the past.	3.37	0.703	Agree
The clothing used in the commercial brings back memories from the past.	3.35	0.727	Agree
Total	**3.365**	**0.6703**	**Agree**
Decorations			
The decorations (building design, old furniture, old signs, old posters) in the advertisements bring back memories from my past.	3.33	0.683	Agree
The decorations (restaurant design, old furniture, old signs, old posters) in the advertisements are relatable.	3.37	0.687	Agree
Total	**3.349**	**0.6186**	**Agree**
Overall Nostalgia Marketing Features	**3.42**	**0.5433**	**Strongly Agree**

**Interpretation: 1.00-1.50 (Strongly Disagree), 1.50-2.49 (Disagree), 2.50-3.49 (Agree), 3.50-4.00 (Strongly Agree)*

The mean scores across the different categories, it is clear that characters play the most significant impact in evoking nostalgia, with a mean score of 3.55. This contrasts with the clothing and decorations, which scored lower, reflecting a less intense connection. The clothing category, with a mean of 3.365, shows that while it resonates with viewers, it does not achieve the same level of emotional engagement as the characters.

The decorations received the lowest mean score at 3.34 (SD = 0.675), suggesting that while they contribute to the nostalgic atmosphere, their impact is less pronounced. This comparison highlights that characters are the strongest driver of emotional engagement in nostalgia marketing, while clothing and decorations, though important, are secondary in their influence. The data illustrates a hierarchy in the effectiveness of different nostalgic elements: characters lead in emotional resonance, followed by clothing, with decorations providing additional context but having a lesser impact. The findings show that characters evoke strong emotional responses, which significantly influence consumer purchasing decision. These results demonstrate the varying effectiveness of nostalgic elements, with the study by Şeniz Özhan (2021) emphasizing the recommendation for brands to focus on characters to maximize their marketing impact. By leveraging familiar and characters, brands can evoke powerful emotions that resonate deeply with consumers. This insight can guide marketers in prioritizing character-driven narratives in their advertising strategies to maximize emotional connections and consumer engagement (Alos & Palacpac, 2024).

Respondents' Influence of Nostalgia Marketing Elements in Commercial 1: Kwentong Jollibee Across the different categories analyzed for the Kwentong Jollibee Hinga commercial, table 7, portrays nostalgia imagery, cultural references, nostalgia narratives, and brand heritage, the respondents exhibited agreement that these nostalgic features effectively reminded them of the past and evoked positive emotions. The overall mean score of 3.43608 out of 5 indicates that the majority of viewers experienced nostalgia by recognizing familiar places, tunes, and stories presented in the advertisement. Each category scored above 3.4: nostalgia imagery (3.443), cultural references (3.454), nostalgia narratives (3.411), and brand heritage (3.435), suggesting a positive reception of nostalgia in the advertisement.

Table 2. Respondents' Influence of Nostalgia Marketing Features in Commercial 1: Kwentong Jollibee

	Mean	SD	Interpretation
Nostalgia Imagery			
The setting shown in the commercials remind me of my past experiences.	3.43	0.659	Agree
The overall look of the commercial reminds me of my past.	3.45	0.653	Agree
Average	**3.443**	**0.6013**	**Agree**
Cultural Reference			
The commercial uses cultural references (music, food, taglines) that I can relate to.	3.45	0.656	Agree
The commercial uses cultural references (music, food, taglines) that remind me of memories from past experiences.	3.46	0.668	Agree
Average	**3.454**	**0.6112**	**Agree**
Nostalgia Narratives			
The storytelling in the commercial reminds me of similar experiences from the past.	3.44	0.671	Agree
The story theme in the commercial makes me feel connected to the brand.	3.38	0.682	Agree
Average	**3.411**	**0.6209**	**Agree**
Brand Heritage			
Brands with rich history are more appealing to me.	3.45	0.672	Agree
A brand's reputation influences my trust in their products.	3.42	0.654	Agree
Average	**3.435**	**0.6016**	**Agree**
LEVEL	**3.43608**	**0.537708**	**Agree**

**Interpretation: 1.00–1.50 (Strongly Disagree), 1.50–2.49 (Disagree), 2.50–3.49 (Agree), 3.50–4.00 (Strongly Agree)*

The commercial strategically employs nostalgia imagery to evoke feelings of nostalgia among viewers, thereby cultivating a positive emotional connection with the Jollibee brand. By integrating cultural references, the advertisement enhances the sense of nostalgia and cultural relevance, which in turn strengthens the emotional ties viewers have with the brand. Additionally, the narrative storytelling is crafted to resonate with viewers by reminding them of similar past experiences, further deepening their emotional engagement. It builds trust and creates positive associations, thus giving the brand an emotional connection much deeper, by stating the rich brand heritage of Jollibee.

This study also supports the priming theory that exposure to certain stimuli can affect how an individual views events around him, his thoughts, feelings, and behaviors. In the case of this ad, the embedded features of nostalgia marketing primed viewers for the feelings of nostalgia to then create a positive emotive outcome with the brand. The employment of nostalgic imagery, cultural references, nostalgia narratives, and brand heritage in the ad stirred the memory and feelings of the audience, who then became more receptive to the message of the brand. Lastly, the analysis points to the huge role played by nostalgia marketing in the Jollibee commercial. This aligns with the findings of He et al. (2022), who indicated that the marketing of nostalgia evokes positive associations and feelings that somehow help to shape customers' decisions. In this case, Xia et al. (2021) concluded that as a result of advertising, nostalgia translates into high purchasing intentions by users. This contributes to the general argument that imagery, pop references, and stories elicit consumer emotional commitment and primarily determine their purchasing intent.

Respondents' Purchase Intention in Commercial 1: Kwentong Jollibee The commercial leverages nostalgia marketing by portraying narratives that resonate with Filipino cultural values, particularly focusing on themes of family, love, and togetherness. Table 3 presents the findings of the effectiveness of nostalgia as a strategic element in advertising, suggesting that when consumers encounter nostalgic elements, they are likely to connect with the brand and have this connection lead to purchase intention. This is consistent with Pascal et al. (2021), nostalgia-driven advertisements stimulate positive emotions, which, in turn, boost purchase intention by offering a sense of comfort, familiarity, and emotional engagement.

Table 3. Respondents' Purchase Intention in Commercial 1: Kwentong Jollibee

Purchasing Intentions	Mean	SD	Interpretation
The throwbacks mentioned in the commercial makes me more likely to purchase the advertised product.	3.36	0.667	Agree
The feelings I have over the commercial make me more likely to purchase the product.	3.42	0.657	Agree
The memories I connect to the commercial to, influence my purchase of the product.	3.37	0.662	Agree
The commercial makes me want to purchase the products used in the advertisement.	3.39	0.684	Agree
Throwback commercials influence me when choosing products.	3.38	0.663	Agree
Overall Purchase Intention Elements	**3.42**	**0.657**	**Agree**

**Interpretation: 1.00–1.50 (Strongly Disagree), 1.50–2.49 (Disagree), 2.50–3.49 (Agree), 3.50–4.00 (Strongly Agree)*

The findings from Table 3 indicate that mean scores for purchase intentions range from 3.36 to 3.42, demonstrating a strong consensus among respondents that nostalgia marketing positively influences their buying behavior. This aligns with Gelgile (2021), who asserts that nostalgia is an effective marketing strategy because it connects with deep-seated memories associated with joy and comfort, prompting consumers to develop favorable attitudes towards the advertised product.

The purchase intention question "The feelings I have over the commercial make me more likely to purchase the product," resulted in the highest score of 3.42. This suggests a significant correlation between nostalgic elements in advertising and increased likelihood of purchase.suggesting that nostalgia elements significantly increase the likelihood of product purchase. Furthermore, Ghorbanzadeh and Rahehagh (2021) support this argument by highlighting that consumers who experience nostalgic emotions are more likely to feel connected to brands. This emotional connection can be pivotal in driving purchasing decisions, as consumers are inclined to favor products that resonate with their personal memories. The data clearly illustrates the effectiveness of nostalgia marketing, as evidenced by the consistently favorable purchase intention score.

Overall Result for Nostalgia Marketing for Commercial 1: Kwentong Jollibee Table 4 indicates that nostalgia marketing exerts a moderately positive influence on purchase intention, as demonstrated by Regression Analysis. Through the results, it can be made clear that the nostalgia marketing elements shown in the commercial has shown an influence in the purchase intention of Gen X consumers.

Table 4. Overall Result for Nostalgia Marketing for Commercial 1: Kwentong Jollibee

Model	R value	Decision	Interpretation
Nostalgia marketing	0.696	Moderately positive influence	Significant
Model	**ANOVA**	**p-value**	**Interpretation**
Regression	170.238	<0.001	Significant

**Interpretation: 0.00 (No Influence), 0.1–0.4 (Weak or Positive Influence), 0.4–0.8 (Moderate Positive Influence), 0.8–1 (Strongly Influence)*

It is shown in the table that the R-value of 0.696 shows a moderately positive influence of nostalgia marketing, through the interpretation scale. Through the ANOVA results with a regression value of 170.238 and a p-value of <.001, it has been interpreted that the model is significant. Through the results, it is seen that nostalgia marketing significantly influences purchase intention. As Alkhafagi (2023) found, tapping into consumers' nostalgic feelings can be an effective strategy for enhancing brand engagement and building emotional connections.\

Overall Result for Nostalgia Marketing Elements for Commercial 1: Kwentong Jollibee This table summarizes the outcome of the analysis conducted by the researchers on how various elements of nostalgia marketing influence the purchasing decisions of Generation X, particularly as demonstrated by the Kwentong Jollibee commercial 1. The findings indicate that nostalgia imagery and narratives have a significant impact on purchasing choices, with P-values below .001. In contrast, cultural references and brand heritage did not show a significant effect, as evidenced by their P-values of 0.0075 and 0.076, respectively, leading to the rejection of those hypotheses.

Table 10. Overall Result for Nostalgia Marketing Elements for Commercial 1: Kwentong Jollibee

Variables	p-value	Decision	Interpretation
Nostalgia Imagery	<0.001	Accept HA1	Significant
Cultural References	0.075	RejectHA2	Not significant
Nostalgia Narratives	<0.001	Accept HA3	Significant
Brand Heritage	0.076	Reject HA4	Not significant

These results support the acceptance of alternative hypotheses HA1 and HA3, emphasizing the important roles of nostalgia imagery and narratives in shaping consumer behavior. This aligns with previous research, such as that by Alkhafagi Y. A. M. (2023), which highlights how tapping into nostalgic emotions can enhance brand loyalty and create deeper emotional connections with consumers.

On the other hand, the analysis indicates that cultural references and brand heritage did not have a statistically significant impact, with P-values of 0.075 and 0.076, both of which exceed the commonly accepted threshold of 5%. As a result, the researchers decided to reject alternative hypotheses HA2 and HA4, suggesting that these factors do not significantly influence consumer responses in this specific context. This finding contrasts with earlier studies, such as that by Picherri, M. (2020), which emphasized the importance of brand heritage in marketing strategies. Nonetheless, the results suggest that, in this context, the influence of nostalgic elements may be more significant than that of cultural references or brand heritage.

The overall findings align with the priming theory, as the stimuli presented in the commercial serve as priming mechanisms. The media stimuli, in the form of nostalgia imagery, cultural references, nostalgia narratives, and brand heritage, effectively activate and trigger stored memories from the respondents past experiences, which have then influenced their purchasing intentions.

Respondents' Reactions to Nostalgia Marketing Features in Commercial 2: Andito na si Move It Across the different categories analyzed for commercial 2 - characters, clothing, and decorations – the respondents showed agreement that all these nostalgic features reminded them of the past and brought back positive memories. The mean score of 3.36 for the "Characters" category bore the highest score which indicates that the characters used in the commercials were effective in evoking nostalgia among consumers. This finding aligns with research by Holotova et al. (2020), which stresses the importance of familiar characters in enhancing emotional engagement with brands.

Table 5. *Respondents' Reactions to Nostalgia Marketing Features in Commercial 2: Andito na si Move It*

	Mean	SD	Interpretation
Characters			
The characters in the commercial reminds me of the past.	3.36	0.736	Agree
The commercial used characters that I felt connected to.	3.36	0.722	Agree
Total	**3.365**	**0.6871**	**Agree**
Clothing			
The clothing used in the commercial reminds me of clothing trends in the past.	3.24	0.747	Agree
The clothing used in the commercial brings back memories from the past.	3.23	0.714	Agree
Total	**3.232**	**0.6909**	**Agree**

	Mean	SD	Interpretation
Decorations			
The decorations (building design, old furniture, old signs, old posters) in the advertisements bring back memories from my past.	3.18	0.732	Agree
The decorations (restaurant design, old furniture, old signs, old posters) in the advertisements are relatable.	3.23	0.735	Agree
Total	**3.204**	**0.6840**	**Agree**
Overall Nostalgia Marketing Features	**3.267**	**0.6150**	**Agree**

**Interpretation: 1.00–1.50 (Strongly Disagree), 1.50–2.49 (Disagree), 2.50–3.49 (Agree), 3.50–4.00 (Strongly Agree)*

Characters often serve as relatable figures that remind consumers of their experience. In contrast, "The clothing used in the commercial reminds me of clothing trends in the past," score 3.24, suggesting that they resonate with past trends but to a lesser extent than characters. This supports the idea that visual cues like fashion evoke emotional responses and allow consumers to connect with their cultural context, as noted by Thompson and Alaviya (2023).

The nostalgia marketing features for "The decorations (building design, old furniture, old signs, old posters) in the advertisements bring back memories from my past." consist of mean 3.18 and "The decorations (restaurant design, old furniture, old signs, old posters) in the advertisements are relatable." mean 3.23, indicating moderate effectiveness in evoking nostalgia. This suggests that while they contribute to the nostalgic atmosphere, they are secondary to characters and clothing. Overall, the average score of 3.267 shows that nostalgia marketing is generally effective across these elements, although it is slightly lower than commercial one, indicating varying levels of engagement. This aligns with findings from Holotova, et al., (2020) and Pichierri (2021), who emphasize that while decorations can enhance nostalgia, personal narratives and relatable characters often resonate more strongly with consumers in different contexts.

Respondents' Influence of Nostalgia Marketing Elements in Commercial 2: Andito na si Move It The result for commercial two, Influence of Nostalgia Marketing Elements, illustrates the effectiveness of these strategies in shaping consumer perceptions and driving purchase intention. By integrating nostalgia imagery, cultural references, nostalgia narratives and emphasizing brand heritage features. Overall, the commercial garnered an average of 3.30506, indicating a strong connection between respondents and the marketing ad, highlighting its effectiveness in eliciting positive responses.

Table 6. *Respondents' Influence of Nostalgia Marketing Elements in Commercial 2: Andito na si Move It*

	Mean	SD	Interpretation
Nostalgia Imagery			
The setting shown in the commercials remind me of my past experiences.	3.23	0.774	Agree
The overall look of the commercial reminds me of my past.	3.28	0.744	Agree
Average	**3.257**	**0.7199**	**Agree**
Cultural Reference			
The commercial uses cultural references (music, food, taglines) that I can relate to.	3.37	0.698	Agree
The storytelling in the commercial reminds me of similar experiences from past.	3.35	0.706	Agree
Average	**3.361**	**0.6546**	**Agree**
Nostalgia Narratives			
The storytelling in the commercial reminds me of similar experiences from the past.	3.26	0.732	Agree
The story theme in the commercial makes me feel connected to the brand.	3.25	0.745	Agree
Average	**3.254**	**0.6874**	**Agree**
Brand Heritage			
I prefer brands that have a rick history.	3.35	0.676	Agree
A brand's reputation influences my trust in their products.	3.34	0.659	Agree
Average	**3.348**	**0.6085**	**Agree**
Overall Level of Nostalgia Marketing Elements	**3.30506**	**0.579408**	**Agree**

**Interpretation: 1.00–1.50 (Strongly Disagree), 1.50–2.49 (Disagree), 2.50–3.49 (Agree), 3.50–4.00 (Strongly Agree)*

The result for "The setting shown in the commercials reminds me of my past experiences." scores of 3.23 and 3.28 indicate a positive response, with an average interpretation. The nostalgia imagery average, which consists of 3.257, suggests that imagery effectively evokes nostalgic feelings. This aligns with findings from Hetserberg (2023), who emphasized nostalgia imagery in advertisements, such as vintage graphics and familiar settings, can create strong emotional connections with consumers.

Under the cultural references section, the statement "The commercial uses cultural references (music, food, taglines) that I can relate to." received a mean score 3.37, while "The storytelling in the commercial reminds me of similar experiences from the past." scores 3.35. These results suggest that respondents agree these elements resonate well with their memories, averaging mean 3.361.

In the nostalgia narratives question, "The storytelling in the commercial reminds me of similar experiences from the past." scored 3.26 and "The story theme in the commercial makes me feel connected to the brand." scored 3.25. This indicates general agreement that storytelling elements contribute to emotional engagement, with an average of 3.254, and the brand heritage scores of 3.35 and 3.34 suggest a favorable view of the brand's history, with an average of 3.348, the overall average is 3.30506. This reflects that while the commercial evokes positive average scores, respondents felt a strong connection to the commercial.

The strong influence of cultural references supports findings by He et al. (2022), which emphasize that relatable elements in marketing can foster emotional ties between consumers and brands. This sentiment is further reinforced by Singh et al. (2021), who highlight nostalgia's persuasive power in shaping purchase intentions, suggesting that the integration of familiar cultural symbols can effectively enhance consumer engagement.

Respondents' Purchase Intention in Commercial 2: Andito na si Move It Table 7 shows a mean score of 3.31 for overall purchasing intentions, indicating that nostalgia marketing significantly influences consumer behavior and boosts purchase intentions. By incorporating nostalgic elements, the commercial number two effectively connects with the audience, making them more inclined to buy the advertised products.

Table 7. Respondents' Purchase Intention in Commercial 2: Andito na si Move It

Purchasing Intentions	Mean	SD	Interpretation
The throwbacks mentioned in the commercial makes me more likely to purchase the advertised product.	3.26	0.755	Agree
The feelings I have over the commercial make me more likely to purchase the product.	3.30	0.715	Agree
The memories I connect to the commercial to, influence my purchase of the product.	3.31	0.699	Agree
The commercial makes me want to purchase the products used in the advertisement.	3.30	0.721	Agree
Throwback commercials influence me when choosing products.	3.25	0.719	Agree
Overall Level of Purchase Intention Elements	**3.31**	**0.718**	**Agree**

**Interpretation: 1.00–1.50 (Strongly Disagree), 1.50–2.49 (Disagree), 2.50–3.49 (Agree), 3.50–4.00 (Strongly Agree)*

The nostalgic elements in the commercial significantly increase the chances that respondents want to buy the advertised product, with a mean score of 3.26. This supports earlier research showing that nostalgia can boost consumer preferences and purchase intentions (Liu et al., 2021).

Emotional responses triggered by the commercial also hold a significant role in influencing buying behavior, reflected in a mean score of 3.30. This suggests that nostalgia can create positive feelings, which affect how consumers make decisions (Banerjee & Shaikh, 2022).

The memories linked to the commercial are crucial in shaping respondents' intentions to purchase, as shown by a mean score of 3.31. This highlights how nostalgia can effectively bring back memories and influence preferences (Rana et al., 2022).

The commercial successfully encourages interest in the featured products, with a mean score of 3.30. This finding is consistent with existing research that shows how nostalgia can increase consumer interest in products associated with past experiences (Banerjee & Shaikh, 2022).

In summary, the overall mean score of 3.31 for Purchasing Intentions indicates that respondents generally agree that the nostalgia marketing strategies used in commercial two positively affect their intentions to purchase. This demonstrates how effective the commercial is in using nostalgia to encourage consumers to buy the advertised product.

Overall Result for Nostalgia Marketing for Commercial 2: Andito na si Move It Through Table 8, it is observed through Regression Analysis that nostalgia marketing has a moderately positive influence on consumer purchase intention. This suggests that the elements of nostalgia marketing can effectively be used to influence purchase intention.

Table 8. Overall Result for Nostalgia Marketing for Commercial 2: Andito na si Move It

Model	R value	Decision	Interpretation
Nostalgia marketing	0.758	Moderately positive influence	Significant
Model	**ANOVA**	**p-value**	**Interpretation**
Regression	203.073	<0.001	Significant

**Interpretation: 0.00 (No Influence), 0.1–0.4 (Weak or Positive Influence), 0.4–0.8 (Moderate Positive Influence), 0.8–1 (Strongly Influence)*

The table indicates that the R-value of 0.758 shows a moderately positive influence of nostalgia marketing, through the interpretation scale. The ANOVA results, with a regression value of 203.073 and a p-value of less than 0.001, have indicated that the model is statistically significant. Though the low p-value, it is confirmed that the influence observed is unlikely to be due to chance, supporting the conclusion that nostalgia marketing has an influence on purchase intention.

The findings align with the study entitled "The Effect of Nostalgia Marketing on Consumers' Purchase Intention" by Alkhafagi (2023), where it was found that nostalgia marketing has positively affected consumers' purchase intention.

Overall Result for Nostalgia Marketing Elements for Commercial 2: Andito na si Move It Table 9 presents the overall results for the nostalgia marketing elements analyzed in Commercial 2: Andito na si Move It. The findings indicate that nostalgia imagery, nostalgia narratives, and brand heritage significantly influence purchasing decisions, while cultural references do not. The results show that nostalgia imagery has a P-value of less than .001, leading to the acceptance of alternative hypothesis HA1.

Table 10. Overall Result for Nostalgia Marketing Elements for Commercial 1: Kwentong Jollibee

Variables	p-value	Decision	Interpretation
Nostalgia Imagery	<0.001	Accept HA1	Significant
Cultural References	0.444	RejectHA2	Not significant
Nostalgia Narratives	0.001	Accept HA3	Significant
Brand Heritage	<0.001	Accept HA4	Significant

This suggests that nostalgia imagery plays a vital role in influencing consumer behavior, reinforcing the idea that visual cues from the past can evoke strong emotional responses that drive purchasing decisions (Patel, 2023).

Similarly, nostalgia narratives also demonstrate significance with a P-value of .001, resulting in the acceptance of alternative hypothesis HA3. This finding emphasizes the effectiveness of storytelling in nostalgia marketing, highlighting how narratives can create emotional connections that resonate with consumers and influence their buying choices. This approach aligns with the findings of Kantar (2023), which highlight the power of nostalgia in advertising to create a deeper connection with consumers.

In contrast, cultural references exhibit a P-value of 0.444, leading to the rejection of alternative hypothesis HA2. This indicates that cultural references do not significantly impact consumer responses in this context, suggesting that while they may be relevant in other marketing strategies, they are less effective when it comes to evoking nostalgia.

Finally, brand heritage shows a P-value of less than .001, leading to the acceptance of alternative hypothesis HA4. This result underscores the importance of brand heritage in nostalgia marketing, indicating that a brand's historical background and legacy can significantly influence consumer perceptions and purchasing behavior.

These findings suggest that nostalgia imagery, narratives, and brand heritage are powerful elements in nostalgia marketing, while cultural references may not hold the same weight in this particular context. These results can be understood through the framework of priming theory, where nostalgic elements serve as stimuli that activate relevant memories in consumers. By evoking nostalgia through these marketing strategies, brands can effectively shape consumer purchase intentions.

Respondents' Nostalgia Marketing Features in Commercial 3: RC Mega Himala – RC Cola Table 10 presents which between the various nostalgia marketing feature resonate most with Gen X consumers. For the feature of characters, the statement "The characters in the commercial remind me of the past" yields a mean score of 3.50 which translates to strongly agree, which indicates that the respondents felt a strong connection between the characters and their memories. To add, the statement "The commercial used characters that I felt connected to" garnered a mean of 3.45 which translates to agree, which supports how characters play a key role in fostering familiarity and emotional resonance. A total mean score of 3.476 is computed as it indicates how the findings reflect how established characters can invoke nostalgic feelings.

The nostalgia marketing feature of clothing, the finding shows that clothing is a significant nostalgia trigger. The two statements —"The clothing used in the commercial reminds me of clothing trends in the past" and "The clothing brings back memories from the past", respectively received the mean scores of 3.45 and 3.44, which both translate to agree, garnering a total mean score of 3.447.

Lastly for the nostalgia marketing feature of decorations, the statement "The decorations in the advertisements bring back memories from my past" garnered a mean score of 3.38, while "The decorations are relatable" scored 3.45 which both translate to agree. The total mean garnered was 3.418, which suggests that the visual elements used in the commercial are tied to cultural contexts which trigger nostalgia.

Table 10. *Respondents' Nostalgia Marketing Features in Commercial 3: RC Mega Himala – RC Cola*

	Mean	SD	Interpretation
Characters			
The characters in the commercial reminds me of the past.	3.50	0.647	Strongly Agree
The commercial used characters that I felt connected to.	3.45	0.679	Agree
Total	**3.476**	**0.5961**	**Agree**
Clothing			
The clothing used in the commercial reminds me of clothing trends in the past.	3.45	0.633	Agree
The clothing used in the commercial brings back memories from the past.	3.44	0.620	Agree
Total	**3.447**	**0.5719**	**Agree**
Decorations			
The decorations (building design, old furniture, old signs, old posters) in the advertisements bring back memories from my past.	3.38	0.644	Agree
The decorations (restaurant design, old furniture, old signs, old posters) in the advertisements are relatable.	3.45	0.641	Agree
Total	**3.418**	**0.5735**	**Agree**
Overall Nostalgia Marketing Features	**3.447**	**0.5261**	**Agree**

**Interpretation: 1.00–1.50 (Strongly Disagree), 1.50–2.49 (Disagree), 2.50–3.49 (Agree), 3.50–4.00 (Strongly Agree)*

The findings show that the nostalgia marketing feature that resonates most with the respondents in this commercial were the characters, garnering a mean score of 3.476. These results are similar to the study by Alos and Palacpac (2024) where this was also suggested. The respondents' connection to the characters presented enhance consumer engagement within nostalgia marketing.

The findings for the nostalgia marketing feature of clothing reflect that the respondents identify with past clothing trends. Clothing also comes in at second as the nostalgia marketing feature that respondents resonate with, resulting in a mean score of 3.418. The results strengthen Holotoava et al. (2020)'s findings on how familiar visual cues such as fashion have the power to evoke emotional responses. Through clothing, consumers may connect their personal history and cultural context, further deepening the impact of nostalgia marketing.

Lastly, the results from the nostalgia marketing feature of decorations come as the last feature respondents resonate with, garnering a mean score of 3.418. The findings align with the research carried out by Alkhafagi (2023), where the importance of environmental details is emphasized as it enhances nostalgic connection.

Respondents' Influence of Nostalgia Marketing Elements in Commercial 3: RC Mega Himala - RC Cola Concerning the overall results of nostalgia marketing elements in the RC Mega Himala commercial, the rating had an average mean of 3.42247. The highest mean score was accorded to cultural references with a score of 3.475 meaning respondents felt connected with the type of music used in a commercial, the type of food that was advertised and specific taglines used in a commercial. The results for nostalgia imagery and nostalgia narratives were slightly lower with the mean of 3.394 and 3.392 respectively, indicating that the perceived nostalgia imagery and stories evoked a positive response among the audience but not as high as in other commercials under the study.

Table 11. Respondents' Influence of Nostalgia Marketing Elements in Commercial 3: RC Mega Himala – RC Cola

	Mean	SD	Interpretation
Nostalgia Imagery			
The setting shown in the commercials remind me of my past experiences.	3.38	0.704	Agree
The commercial used characters that I felt connected to.	3.41	0.697	Agree
Average	**3.394**	**0.6452**	**Agree**
Cultural Reference			
The commercial uses cultural references (music, food, taglines) that I can relate to.	3.48	0.614	Agree
The commercial uses cultural references (music, food, taglines) that remind me of memories from past experiences.	3.47	0.614	Agree
Average	**3.475**	**0.5627**	**Agree**
Nostalgia Narratives			
The storytelling in the commercial reminds me of similar experiences from the past.	3.40	0.670	Agree
The story theme in the commercial makes me feel connected to the brand.	3.38	0.671	Agree
Average	**3.392**	**0.6164**	**Agree**
Brand Heritage			
Brands with rich history are more appealing to me.	3.42	0.626	Agree
A brand's reputation influences my trust in their products.	3.44	0.603	Agree
Average	**3.429**	**0.5588**	**Agree**
LEVEL	**3.42247**	**0.508222**	**Agree**

**Interpretation: 1.00-1.50 (Strongly Disagree), 1.50-2.49 (Disagree), 2.50-3.49 (Agree), 3.50-4.00 (Strongly Agree)*

The data presented illustrates that nostalgia marketing elements significantly influence respondents' perceptions of the RC Mega Himala - RC Cola commercial. The average mean scores indicate a positive agreement across all categories. Notably, nostalgia imagery averages 3.394, with respondents agreeing that both the setting and the characters used in the commercial evoke past experiences. Cultural references scored slightly higher at 3.475, indicating that elements like music and food deeply resonate with viewers, reminding them of their past experiences. Nostalgia narratives and brand heritage also received favorable scores, with averages of 3.392 and 3.429, respectively, suggesting that storytelling and a brand's historical appeal play crucial roles in shaping consumer perceptions. Overall, these findings highlight the effectiveness of nostalgia as a marketing strategy in creating emotional connections with consumers.

Based on the different elements of nostalgia marketing, cultural references elicited the highest scores out of the other categories with an average score of 3.475 and therefore cultural symbols appeals to respondents. This is closely followed by nostalgia imagery with a mean of 3.394 that supported visual and character-driven elements, but less than the subcategories explored in this study. Nostalgia narratives alongside brand heritage have nearly similar results, attaining an average of 3.392 and 3.429, which implies that though storytelling has significance, it does not differ much between using the brand reputation. This comparison shows that culture share is important to establish and have influence on every element in play though it seems to show that culture has the greatest influence on the nostalgia factor and the overall emotional preferences on the brand.

The results of nostalgia marketing components on the product in the RC Mega Himala - RC Cola commercial, cultural appeals highest. This accredits the findings of Holotová et al. (2020), which point that cultural references increase appeal and appeal to feelings in adverts. However, the findings of the study regarding the role of nostalgia narratives in brand perception and importance of brand heritage confirmed the research of Keller (2020) which testifies that the brand history builds consumer trust. Altogether, these works raise an understanding of how nostalgia appeals may help brands and organizations build an emotional bond that elicits consumers' purchasing behavior.

Respondents' Purchase Intention in Commercial 3: RC Mega Himala – RC Cola Table 12 presents data for Purchase Intention. Each factor is rated on a scale, with the mean scores indicating the average level of agreement among respondents. According to Essam (2023), nostalgia appeals can create a strong emotional bond between consumers and the product. This

particularly evident in the mean score of the first question, "The throwbacks mentioned in the commercial makes me more likely to purchase the advertised product," received a score of 3.33 with a standard deviation of 0.683, indicating a positive sentiment among respondents. The second question, "The feelings I have over the commercial make me more likely to purchase the product," had a slightly higher mean score of 3.40 (SD = 0.669), supports the theory of affective advertising, suggesting that emotional connections to the advertisement positively influence purchase intentions.

The third statement, "The memories I connect the commercial to influence my purchase of the product," scored 3.40 (SD = 0.673), shows how shared experience or common reference can increase consumer relatability. Respondents also agreed with the factor "The commercial makes me want to purchase the products used in the advertisement," which received a mean score of 3.38 (SD = 0.668). Lastly, the "Throwback commercials influence me when choosing products," averaged 3.36 (SD = 0.660), indicating that nostalgic marketing is impactful in consumer decision making. The overall average for purchase intention was 3.41 (SD = 0.675), reflecting a favorable inclination among respondents towards purchasing a product.

Table 12. Respondents' Purchase Intention in Commercial 3: RC Mega Himala – RC Cola

Purchasing Intentions	Mean	SD	Interpretation
The throwbacks mentioned in the commercial makes me more likely to purchase the advertised product.	3.33	0.683	Agree
The feelings I have over the commercial make me more likely to purchase the product.	3.40	0.669	Agree
The memories I connect to the commercial to, influence my purchase of the product.	3.40	0.673	Agree
The commercial makes me want to purchase the products used in the advertisement.	3.38	0.685	Agree
Throwback commercials influence me when choosing products.	3.36	0.660	Agree
Purchase Intention Average	**3.41**	**0.675**	**Agree**

**Interpretation: 1.00–1.50 (Strongly Disagree), 1.50–2.49 (Disagree), 2.50–3.49 (Agree), 3.50–4.00 (Strongly Agree)*

The mean scores across the different statements, we can observe that the highest scores were for the second and third statements, both receiving a mean of 3.40. This suggests that the emotional connection and shared experiences or memories evoked by the commercial have the strongest influence on purchase intention. The statement with the lowest mean score (3.33) was the first one, "The throwbacks mentioned in the commercial makes me more likely to purchase the advertised product." While still indicating a positive sentiment, this score is slightly lower than the other statements. This could imply that while the nostalgic elements in the commercial are influential, the emotional and relatable aspects of the advertising may have a stronger impact on purchase intention.

Additionally, a study by Xia et al. (2021) found that nostalgia significantly enhances consumers' purchase intentions, particularly during times of crisis, reinforcing the idea that emotional connections are crucial. The respondents' favorable attitudes towards the nostalgic elements of the commercial, as reflected in the high mean scores, suggest that leveraging nostalgia can be an effective strategy for brands to connect with their target audience and influence their purchasing decisions.

Overall Result for Nostalgia Marketing for Commercial 3: RC Mega Himala – RC Cola The table presents the results of statistical analysis examining the impact of nostalgia marketing on the influence of Commercial 3: RC Mega Himala - RC Cola. The analysis utilized a regression model to assess the influence between nostalgia marketing and the commercial's performance. The "R Value" column shows the R-squared value of 179.559, indicating a strong positive influence between the variables in the model. The decision in regression shows that the p-value is less than 0.001 which is considered statistically significant. Based on the result, the decision is to accept the alternative hypothesis, meaning the model has a significant interpretation in nostalgia marketing.

Table 13. Overall Result for Nostalgia Marketing for Commercial 3: Rc Mega Himala – RC Cola

Model	R value	Decision	Interpretation
Nostalgia marketing	0.705	Moderately positive influence	Significant
Model	**ANOVA**	**p-value**	**Interpretation**
Regression	179.559	<0.001	Significant

**Interpretation: 0.00 (No Influence), 0.1–0.4 (Weak or Positive Influence), 0.4–0.8 (Moderate Positive Influence), 0.8–1 (Strongly Influence)*

The interpretation provided further clarifies that the results suggest nostalgia-based marketing strategies had a moderately positive influence on the success of this commercial. This aligns with the R-value of 0.705, which falls within the "Moderately positive influence" range specified in the interpretation guide. The analysis indicated that while the R-squared value raises some questions, the statistically significance p-value and the moderately positive R-value support the conclusion that nostalgia marketing was an effective strategy for this commercial. The results provide evidence-based insights that could inform future marketing decisions leveraging nostalgia (Alkhafagi et al., 2020).

Overall Result for Nostalgia Marketing Elements for Commercial 3: RC Mega Himala – RC Cola The table shows the statistical analysis results for the Commercial 3: RC Mega Himala. The findings reveal that both nostalgia imagery, nostalgia narratives, and brand heritage all had statistically significant effects, with p-values less than 0.00, while the Cultural references were found to be not statistically significant.

Table 14. Overall Result for Nostalgia Marketing Elements for Commercial 3: RC Mega Himala – RC Cola

Variables	p-value	Decision	Interpretation
Nostalgia Imagery	<0.001	Accept HA1	Significant
Cultural References	0.893	RejectHA2	Not significant
Nostalgia Narratives	<0.001	Accept HA3	Significant
Brand Heritage	<0.001	Accept HA4	Significant

The overall result for commercial 3: RC Mega Himala - RC Cola, nostalgia imagery has a p-value less than 0.001, leading to the decision to Accept HA1. This indicates that Nostalgia Imagery has a significant effect on the outcome. This aligns with the prior research of Şeniz Özhan (2021), who found that evoking nostalgia through advertising can enhance consumers' emotional connection to a brand. Cultural references have a p-value of 0.893, which is greater than the typical 0.05 significance level. Therefore, the decision is to Reject HA2, meaning Cultural References do not have a significant effect. This differs from the findings of Mehrishi, n.d. (2023), who emphasized the importance of cultural factors in consumer behavior. However, the current results suggest that in this variable nostalgia imagery, narratives & brand heritage have been more influential than cultural references.

Nostalgia Narratives, has a p-value less than 0.001, resulting in the decision to Accept HA3. This suggests that Nostalgia Narratives have a significant influence, consistent with the importance of evoking nostalgia found in Batcho, K. (2020) study. Brand Heritage also has a p-value less than 0.001, leading to the decision to Accept HA4. This implies that Brand Heritage is a significant factor in the model, aligning (Sierra, 2019) work on the strategic importance of brand heritage. Furthermore, these results can be interpreted through the perspective of priming theory. The nostalgic elements presented in the commercial act as stimuli that activate relevant memories stored in the minds of respondents. By evoking nostalgia through imagery, cultural references, narratives, and brand heritage, these stimuli trigger past experiences that significantly shape the consumers' purchase intentions.

Discussion of Results

Understanding the effects of nostalgia marketing on generation X consumers' purchasing intentions was applied in this study. An evaluation done on three television advertisements breakdowns; Kwentong Jollibee, Andito na si Move It, and RC Mega Himala – RC Cola offers strong supporting research to argue the efficiency of using nostalgia-driven advertising campaigns. Nostalgia Imagery: As was observed from all the analyzed advertisements of the three companies, nostalgia imagery provided a positive action stimulus on the consumers. Respondents positively correlated on the presented characters, clothes, and decorations and labeled the essence of such visualizations as evoking their own memories. Therefore, the association of nostalgia imagery with purchase intentions supports the theory that the imagery from familiar settings and of time periods positively influences consumer purchase intention.

Cultural References: Nostalgia marketing was complemented by cultural aspects such as use of music, food and taglines. The regression analysis showed that cultural references had no significant effect on purchase intentions. This again brought out that there is no relevance of using cultural appeal in the marketing communication in response to the historical appeals. Nostalgia Narratives: The "storytelling" elements that were incorporated in the body of the campaign's commercials – retrospective narratives, extended this emotional association between the buyer and the purchasing product. Originally, respondents were able to associate personal stories with the brands meaning that it would increase brand loyalty and credibility. favicon The correlation coefficients in regard to nostalgia narratives were some of the firmest, pointing out that good stories in advertisements may become key drivers of further buyer actions. Brand Heritage: Again, purpose was found to have a significantly positive impact on purchase intentions, as well as brand heritage. Consumers also became loyal with brands that could create a long standing history based on the positive response given to Jollibee and RC Cola. The brand heritage influences

positive purchase intention due to trustworthiness, credibility and genuineness (Rizma & Gunanto, 2024). Established brands need to be trusted and the consumer of Gen X tends to be more loyal and thereby more likely to purchase a particular brand.

Overall Impact on Purchase Intentions: Thus, the results of the study support the incorporation of nostalgia marketing in the current trend of adverts for the use of Generation X. Overall, respondents agreed on the sentiment that nostalgia related to imagery, cultural references, narrative, and brand heritage influenced their purchasing intentions. This accords with the findings in prior studies presenting that nostalgic elements in advertising content are not only capable of eliciting positive affect and eliciting interest in the consumers, but also enhancing consumer purchase intention. Consequently, the study points towards how nostalgia marketing indeed holds the potential to be a significant instrumental variable, for guiding the consumer buying behavior. The bonds attributed to nostalgia nurturing positive brand perception and effective manipulation of marketing messages to improve purchase decisions make nostalgia marketing an effective strategy for the brands targeting Gen X.

The study explored the various nostalgia marketing elements and its potential influence on purchase intention of consumers. The study demonstrates the theoretical framework of the priming theory, as it exists between the nostalgia marketing elements and consumer purchasing behavior. As the priming theory states that exposure to stimuli, such as nostalgia marketing elements, affects future behavior or consumer decisions. In this case, nostalgia marketing prepares the ground for consumers by triggering nostalgic memories hence tends to push them to engage with particular brands and thus make purchase decisions. This study therefore confirms the hypothesis that nostalgia marketing has an influence on consumers' purchase intentions. The influence established between nostalgia imagery, cultural references, nostalgia narratives, and brand heritage with the purchase intention confirms effective priming which are emotionally charged stimuli that conjure related feelings and perceptions, and here they affect consumer purchase intention.

Conclusions

The study demonstrates that nostalgia marketing effectively influences purchase intentions among Generation X consumers. By evoking positive memories and feelings associated with the past, brands can foster emotional connections that enhance consumer loyalty and engagement (Alos & Palacpac, 2024; He et al., 2022).

In the evaluation of nostalgia marketing features across three commercials, characters emerged as the most powerful nostalgia trigger. Clothing also played a significant role, though it was less impactful than characters. In contrast, decorations were the least effective in eliciting nostalgic feelings.

The findings suggest that the nostalgia marketing elements of nostalgia imagery, nostalgia narratives, and brand heritage, significantly impact Gen X purchase intentions. It was shown through the findings that nostalgia marketing effectively taps into the longing for the past, offering a sense of comfort and familiarity that resonates particularly with Gen X. Through a sample size of 395 respondents from District 3, Caloocan City, with the use of a quantitative research approach, it was validated that nostalgia marketing strategies significantly influence purchase intention of Gen X consumers. The findings highlight a strong influence between nostalgia elements and increased purchase intention (Xia et al., 2021).

In conclusion, as marketers seek to differentiate their brand in a competitive market, nostalgia marketing presents a powerful tool for engaging consumers on a deep emotional level. By leveraging the unique characteristics and shared experiences of Generation X, brands can create impactful campaigns that not only enhance consumer connection but also drive purchasing intention.

Recommendations

The research findings depict the positive influence that nostalgia marketing has on the buying behavior of consumers in the Gen X population. When nostalgia is incorporated, it creates bonds between the business and its clients making the business dealings more profound, and more engaging. The following recommendations are directed to marketing practitioners, advertising agencies, business managers, students, and researchers to enhance the application of nostalgia as a strategic appeal technique and to foster positive consumer responses.

Marketing Professionals. Marketing professionals should harness the power of nostalgia as a way of driving brand value. Organizing and developing the campaigns that will call such operational traditional "remember when" feelings of brand loyalty into play. The feelings fostered by nostalgia not only make for customer loyalty but also create demand for the product by old customers. It is inviting marketers to come up with new ideas and experiment with themes that appeal to the culture that is deeply synonymous with Gen X, thus making sure that each campaign is fully appreciated by this specific set of people.

Advertising Agencies. The study concludes by emphasizing that advertising agencies should particularly build on the opportunities of nostalgia marketing. As agencies carefully choose cultural symbols, visuals, and narratives that are familiar to the Gen X audience, advertisements will be more effective. Appeals in advertising are more effective because common

symbols, music, and images trigger a strong emotional response. Administrations are encouraged to use survey data to develop communications which activate the audiences' positive affect system.

Business Owners. New entrants should incorporate nostalgia in their brand creation and formation processes as should existing businesses in building relationships with customers. By using positive cues, messages can be crafted to establish and continually remind consumers of their trusted reputation. It is also a practical way to foster customer relations and build retention and repeat business.

Students. Marketing students, consumer behavior or psychology students can use the results emerging from the study to further understand nostalgia marketing. The study can stand as a starting point for them to pursue an additional analysis or study regarding the aspects of nostalgia to determine how specific aspects prompt consumer behavior.

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REFERENCES

1. Alkhafagi, Y. a. M. (2023). The effect of nostalgia marketing on consumers' purchase intention. Mağallaï Al-'ulūm Al-iqtiṣādiyyaï Wa-al-idāriyyaï, 29(136), 27–39. https://doi.org/10.33095/jeas.v29i136.2603
2. Alos, J., & Palacpac, A. (2024, January 15). Nostalgia Marketing: Statistics, Benefits, and Examples. Spiralytics. https://www.spiralytics.com/blog/nostalgia-marketing-statistics-benefits-and-examples/
3. A manual for selecting sampling techniques in Research Munich Personal REPEC Archive. (n.d.). https://mpra.ub.uni-muenchen.de/70218/
4. Banerjee, S., & Shaikh, A. (2022). Impact of brand nostalgia on intention to purchase brand extensions: moderating role of brand attachment. Journal of Product & Brand Management, 31(7), 1005–1017. https://doi.org/10.1108/jpbm-10-2020-3149
5. Batcho, K. I. (2020). When Nostalgia Tilts to Sad: Anticipatory and Personal Nostalgia. Frontiers in Psychology, 11, 1186. https://doi.org/10.3389/fpsyg.2020.01186
6. Benjamin, A., & Bushman, B. (2016). The Weapons Priming Effect. Current Opinion in Psychology, 12, 45–48. https://doi.org/10.1016/j.copsyc.2016.05.003
7. Berndt, A. E. (2020). Sampling methods. Journal of Human Lactation, 36(2), 224–226. https://doi.org/10.1177/0890334420906850
8. Bhandari, P. (2023, June 22). What Is Quantitative Research? | Definition, Uses & Methods. Scribbr. https://www.scribbr.com/methodology/quantitative-research/
9. Bhat, A. (2024, May 2). Quantitative Data Collection: Best 5 methods. QuestionPro. https://www.questionpro.com/blog/quantitative-data-collection-methods/
10. Borry, P. (2012). Medical Ethics, Use of Empirical Evidence in. In Elsevier eBooks (pp. 70–77). https://doi.org/10.1016/b978-0-12-373932-2.00371-9
11. Bosman, L. K. D., Mugge, R., & Schoormans, J. P. L. (2020). Nostalgia and consumer behavior: A review of the literature and an agenda for future research. Journal of Consumer Behavior, 19(5), 453-473. https://doi.org/10.1002/cb.1853
12. Brazil, et al. (2022). The Effect of Online Wet Markets Website Quality on the Purchase Intention of Consumers in Caloocan City. https://www.researchgate.net/publication/359541795_The_Effect_of_Online_Wet_Markets_Website_Quality_on_the_Purchase_Intention_of_Consumers_in_Caloocan_City
13. Brown, D., & Lee, S. (2022). The importance of effective time management in research: Strategies for improving productivity. https://researchleap.com/the-importance-of-effective-time-management-in-research-strategies-for-improving-productivity/
14. Bryła, P., Chatterjee, S., & Ciabiada-Bryła, B. (2022). The impact of social media marketing on consumer engagement in sustainable consumption: A systematic literature review. International Journal of Environmental Research and Public Health, 19(24), 16637. https://doi.org/10.3390/ijerph192416637
15. Caloocan City Profile – PhilAtlas. (1990, May 1). https://www.philatlas.com/luzon/ncr/caloocan.html
16. Caloocan City Profile – PhilAtlas. (2024). Www.philatlas.com. https://www.philatlas.com/luzon/ncr/caloocan.html
17. Chan, N. (2024, April 23). Filipino Online Shopping behavior: Statistics & Trends you need to know. Spiralytics. https://www.spiralytics.com/blog/filipino-online-shopping-behavior-statistics-trends/
18. Chen, T., Samaranayake, P., Xiong-Ying, C., Meng, Q., & Lan, Y. (2022). The impact of online reviews on consumers' purchasing decisions: evidence from an Eye-Tracking study. Frontiers in Psychology, 13. https://doi.org/10.3389/fpsyg.2022.865702
19. Cheung, W., Wildschut, T., Sedikides, C., Hepper, E. G., Arndt, J., & Vingerhoets, A. (2013). Back to the future. Personality & Social Psychology Bulletin, 39(11), 1484–1496. https://doi.org/10.1177/0146167213499187

20. Cho, H., Lee, H.W. and Chiu, W. (2021), "Satellite Fans: Does Sport Nostalgia Influence Purchase Intention toward Sponsors' Products?", Journal of Retailing and Consumer Services, Vol. 62, 102653. https://doi.org/10.1016/j.jretconser.2021.102653.
21. Crespo-Pereira, V., Membiela-Pollán, M., & Sánchez-Amboage, E. (2021b). Nostalgia, Retro-Marketing, and Neuromarketing: an Exploratory Review. Journal of Creative Industries and Cultural Studies, 7, 106–126. https://doi.org/10.56140/jocis-v7-5
22. Cuncic, A. (2022, May 23). What Is Nostalgia? Verywell Mind. https://www.verywellmind.com/what-is-nostalgia-5272007
23. Deshmukh, A. R. (2023). A Study on Identifying Strategies of Emotional Branding in Luxury Apparel Brands. Apeejay Journal of Management & Technology, 18(1), 14–33. https://www.researchgate.net/publication/374631749_A_Study_on_Identifying_Strategies_of_Emotional_Branding_in
24. Effect of nostalgia on customer loyalty to brand Post-Merger / Acquisition: EBSCOHost. (n.d.). https://web.p.ebscohost.com/ehost/detail/detail?vid=10&sid=d4dab5f5-4c2a-4227-bbad-6b56e6b0c49e%40redis&bdata=JnNpdGU9ZWhvc3QtbGl2ZQ%3d%3d#AN=115136418&db=bth
25. Efrat, K., & Asseraf, Y. (2019). A shift in perspective? The role of emotional branding in shaping born globals' performance. International Business Review, 28(6), 101589. https://doi.org/10.1016/j.ibusrev.2019.101589
26. Elliott, R. (2021). Likert Scale Examples and Definition. GeoPoll. https://www.geopoll.com/blog/likert-scale-examples-definition/
27. Essam, K. (2023, September 28). The Psychology of Nostalgia Marketing: Why it works and how to use it - AOK Marketing. AOK Marketing. https://aokmarketing.com/the-psychology-of-nostalgia-marketing-why-it-works-and-how-to-use-it/
28. Fleetwood, D. (2023, November 27). Sampling Methods: Guide To All Types with Examples. QuestionPro. https://www.questionpro.com/blog/types-of-sampling-for-social-research/#:~:text=Sampling%20is%20a%20technique%20of,characteristics%20of%20the%20whole%20population.
29. Fleetwood, D. (2024, February 9). Non-Probability Sampling: Types, Examples, & Advantages. QuestionPro. https://www.questionpro.com/blog/non-probability-sampling/
30. Frost, J. (2022). Cronbach's Alpha: Definition, Calculations & Example. Statistics by Jim. https://statisticsbyjim.com/basics/cronbachs-alpha/
31. Fulk, G. (2023). Descriptive statistics, an important first step. Journal of Neurologic Physical Therapy, 47(2), 63. https://doi.org/10.1097/npt.0000000000000434
32. Garcia, M., & Santos, P. (2020). Challenges in Accessing Data from Local Government Units. Journal of Research Access, 7(2),. https://files.eric.ed.gov/fulltext/EJ1285295.pdf
33. Geiger, A. (2024, April 14). Defining generations: Where Millennials end and Generation Z begins. Pew Research Center. https://www.pewresearch.org/short-reads/2019/01/17/where-millennials-end-and-generation-z-begins/
34. Geiger, A. (2024, April 14). Generation X: America's neglected 'middle child.' Pew Research Center. https://www.pewresearch.org/short-reads/2014/06/05/generation-x-americas-neglected-middle-child/
35. Gelgile, H. K. (2021). Nostalgia Marketing: Examining music retromania. https://portal.amelica.org/ameli/journal/364/3642797007/html/
36. George, T. (2023, June 22). Mixed Methods Research | Definition, Guide & Examples.
37. Ghorbanzadeh, D., & Rahehagh, A. (2021). Emotional brand attachment and brand love: the emotional bridges in the process of transition from satisfaction to loyalty. Rajagiri Management Journal, 15(1), 16–38. https://doi.org/10.1108/ramj-05-2020-0024
38. Gupta, A. (2024, June 4). Spearman's Rank Correlation: The Definitive Guide To Understand. Simplilearn.com. Retrieved October 18, 2024, from https://www.simplilearn.com/tutorials/statistics-tutorial/spearmans-rank-correlation

39. Hartmann, B. J., & Brunk, K. H. (2019). Nostalgia marketing and (re-)enchantment. International Journal of Research in Marketing, 36(4), 669–686. https://doi.org/10.1016/j.ijresmar.2019.05.002
40. Hassan, M. (2023, August 15). Purposive Sampling - Methods, types and examples. Research Method. https://researchmethod.net/purposive-sampling/
41. He, X., Zhu, L., Sun, L., & Yang, L. (2022). The influence of brand marketing on consumers' emotion in mobile social media environment. Frontiers in Psychology, 13. https://doi.org/10.3389/fpsyg.2022.962224
42. Helkkula, K., & Gummerus, C. (2021). Social identity theory and marketing: A systematic review of the literature. Journal of Business Research, 127, 57-68. https://doi.org/10.1016/j.jbusres.2021.01.027
43. Hesterberg, K. (2023, November 17). 22 Examples of nostalgia marketing that make customers feel good. blog.hubspot.com. https://blog.hubspot.com/marketing/examples-nostalgia-marketing-ads
44. Holotová, M., Kádeková, Z., & Košičiarová, I. (2020, November 1). RETRO MARKETING - A POWER OF NOSTALGIA WHICH WORKS AMONG THE AUDIENCE. | Communication Today | EBSCOhost. Openurl.ebsco.com. https://openurl.ebsco.com/EPDB%3Agcd%3A4%3A13423 08/detailv2?sid=ebsco%3Aplink%3Ascholar&id=ebsco%3Agcd%3A147726092&crl=c
45. Hurley, M., & Tenny, S. (2022). Mean. PubMed; StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK546702/#:~:text=The%20mean%2C%20which%20is%20also
46. Iyengar, S., Peters, M. D., & Kinder, D. R. (1982). Experimental Demonstrations of the "Not-So-Minimal" Consequences of Television News Programs. The American Political Science Review, 76(4), 848. https://doi.org/10.2307/1962976
47. Jones, A., Williams, B., & Taylor, C. (2023). Budgeting for Research: Strategies for Effective Resource Allocation. Journal of Budgeting Research, 15(3),. https://ojs.aaresearchindex.com/index.php/aasgbcpjmra/article/view/1049
48. Jones, C. (2020). A Hierarchy of Acceptable Changes for Audiences of Nostalgic Narrative Media Brands. Burgess, Jacqueline, Vol. 22(Issue 3/4), 151406187. https://doi.org/10.1080/14241277.2021.1922411
49. Jung, Y. M. (2019). Data analysis in quantitative research. In Springer eBooks (pp. 955–969). https://doi.org/10.1007/978-981-10-5251-4_109
50. Kalish, C. W., & Thevenow-Harrison, J. T. (2014). Descriptive and inferential problems of induction. In The Psychology of learning and motivation/The psychology of learning and motivation (pp. 1–39). https://doi.org/10.1016/b978-0-12-800283-4.00001-0
51. Kanika. (2024, April 5). Field Surveys: Methods, Survey Questions & Tools. Zonka. https://www.zonkafeedback.com/blog/field-surveys
52. Keller, L. K. (2020). Consumer Research Insights on Brands and Branding: A JRC Curation. Journal of Consumer Research, 46(5), 995-1001. https://doi.org/10.1093/jcr/ucz058
53. Kenan, J. (2024, April 10). Thanks for the memories: How to use the power of nostalgia marketing. Sprout Social. https://sproutsocial.com/insights/nostalgia-marketing/
54. Keni, D. (2019, April 16). Weighted Mean Formula | Step by Step Calculation (with Example). WallStreetMojo. https://www.wallstreetmojo.com/weighted-mean-formula/
55. Komalasari, F., Christianto, A., & Ganiarto, E. (n.d.). Factors influencing purchase intention in affecting purchase decision: A study of e-commerce customer in Greater Jakarta. UI Scholars Hub. https://scholarhub.ui.ac.id/jbb/vol28/iss1/1
56. Krause, M. (2024, April 25). How to market content to Generation X in 2024. ClearVoice. https://www.clearvoice.com/resources/generation-x-content-marketing/
57. Liu, M., Yang, L., & Zhang, W. (2021). How nostalgia affect purchase intention under the background of big data. In Advances in intelligent systems and computing (pp. 437–443). https://doi.org/10.1007/978-3-030-69999-4_59

58. Lutkevich, B. (2023, October 27). Nostalgia marketing explained: Everything you already know. WhatIs. https://www.techtarget.com/whatis/feature/Nostalgia-marketing-explained-Everything-you-already-know#:~:text=What%20is%20nostalgia%20marketing%3F,with%20the%20audience's%20fond%20memories

59. Mappesona, H., Ikhsani, K., & Ali, H. (2020). Customer purchase decision model, supply chain management and customer satisfaction: Product quality and promotion analysis. International Journal of Supply Chain Management, 9(1), 592-600 https://ojs.excelingtech.co.uk/index.php/IJSCM/article/view/4336

60. Martins, J., Costa, C., Oliveira, T., Gonçalves, R., & Branco, F. (2019). How smartphone advertising influences consumers' purchase intention. Journal of Business Research, 94(1), 378–387. https://doi.org/10.1016/j.jbusres.2017.12.047

61. Martins, J., Costa, C., Oliveira, T., Gonçalves, R., and Branco, F. (2019), "How Smartphone Advertising Influences Consumers' Purchase Intention", Journal of Business Research, Vol. 94, pp. 378-387. https://doi.org/10.1016/j.jbusres.2017.12.047.

62. MBA Skool Team. (2021). Purchase Intention Definition | Marketing Dictionary | MBA Skool-Study.Learn.Share. MBA Skool-Study.Learn.Share. https://www.mbaskool.com/business-concepts/marketing-and-strategy-terms/10976-purchase-intention.html

63. Mehrishi, M. (n.d.). Why is Nostalgia so Effective in Marketing? | OpenGrowth. https://www.blogs.opengrowth.com/why-is-nostalgia-so-effective-in-marketing

64. Mixed methods research. (n.d.). Harvard Catalyst. https://catalyst.harvard.edu/community-engagement/mmr/

65. Nelson, N. (2023, March 14). Effective Advertising Makes People Remember Your Name. https://www.wolterskluwer.com/en/expert-insights/effective-advertising-makes-people-remember-your-name

66. Ocampo, E. (2024, January 9). Gen X consumers are crucial to your ecommerce site. ClearSale. https://en.clear.sale/blog/gen-x-consumers-are-crucial-to-your-ecommerce-site

67. Othman, A. S., Rahman, I. A., Teoh, N., Omar, A. R. C., & Osman, L. H. (2022). An empirical study on the factors influencing consumers' purchase intention of cosmetic products. Malaysian Journal of Social Sciences and Humanities, 7(6), e001512. https://doi.org/10.47405/mjssh.v7i6.1512

68. Ozhan, S., & Akkaya, D. T. (2021). The Effect of Nostalgia Proneness on Ad-Evoked Nostalgia, Brand Attitude and Purchase Intention. Istanbul Business Research. https://doi.org/10.26650/ibr.2020.49.0050

69. Patel, N. (2023). Nostalgia marketing: How to leverage the past for future success. Neil Patel. https://neilpatel.com/blog/nostalgia-marketing/

70. Peña-García, N., Gil-Saura, I., Rodríguez-Orejuela, A., and Siqueira-Junior, J. R. (2020), "Purchase Intention and Purchase Behavior Online: A Cross-cultural Approach", Heliyon, Vol. 6 No. 6. https://doi.org/10.1016/j.heliyon.2020.e04284.

71. Percentage Frequency Distribution. (2023). Jove.com. https://app.jove.com/science-education/v/12584/percentage-frequency-distribution

72. Pizzi, G., & Scarpi, D. (2019). The year of establishment effect on brand heritage and attitudes. the Journal of Consumer Marketing/Journal of Consumer Marketing, 36(6), 827–834. https://doi.org/10.1108/jcm-05-2018-2665

73. Poels, K., & Dewitte, S. (2019). The Role of Emotions in Advertising: A Call to Action. Journal of Advertising, 48(2), 1–10. https://doi.org/10.1080/00913367.2019.1579688

74. Pulp Strategy. (2023, October 11). Why is nostalgia marketing the hottest trend today? https://www.linkedin.com/pulse/why-nostalgia-marketing-hottest-trend

75. Qualitative research & evaluation methods. (2024, May 15). SAGE Publications Inc. https://us.sagepub.com/en-us/nam/qualitative-research-evaluation-methods/book232962

76. Rana, S., Raut, S. K., Prashar, S., & Quttainah, M. A. (2020). The transversal of nostalgia from psychology to marketing: what does it portend for future research? International Journal of Organizational Analysis, 30(4), 899–932. https://doi.org/10.1108/ijoa-03-2020-2097

77. Rana, S., Raut, S. K., Prashar, S., & Quttainah, M. A. (2020). The transversal of nostalgia from psychology to marketing: what does it portend for future research? International Journal of Organizational Analysis, ahead-of-print(ahead-of-print). https://doi.org/10.1108/ijoa-03-2020-2097

78. Rana, S., Raut, S. K., Prashar, S., and Quttainah, M. A. (2020), "The Transversal of Nostalgia from Psychology to Marketing: What Does It Portend for Future Research?", International Journal of Organizational Analysis. https://doi.org/10.1108/IJOA-03-2020-2097.

79. Restaurant, S. J. (2023, December 15). Caloocan: a dynamic tapestry of heritage, challenges, and progress. Medium. https://medium.com/@silayushi/caloocan-a-dynamic-tapestry-of-heritage-challenges-and-progress-5677096779c8#:~:text=Caloocan%2C%20like%20many%20cities%20in,different%20provinces%20and%20indigenous%20communities.

80. Rizma, F. A. and Guganto E. (2024). The Effect of Trustworthiness to Increase Brand Trust and Purchase Intention on Social Media Promotion based on Theory of Persuasion in Generation Z. Jurnal Manajemen. 15. 61-81. https://doi.org/10.32832/jm-uika.v15i1.15766

81. Rodriguez, R., Martinez, S., & Lopez, M. (2024). Ethical Considerations in Research: Protecting Participants' Rights and Privacy. Journal of Ethics in Research, 12(1),. https://bmcmedethics.biomedcentral.com/articles/10.1186/s12910-024-01044-w

82. Satter, S. (2023, October 31). Structured questionnaire: Definition, Types + pros & cons. QuestionPro. https://www.questionpro.com/blog/structured-questionnaire/Scribbr. https://www.scribbr.com/methodology/mixed-methods-research/

83. Sd. (2023, September 19). 10+ Gen X Online Shopping Habits (Latest Data & Statistics). OnlineDasher. https://www.onlinedasher.com/gen-x-online-shopping-habits/

84. Sedikides, C., & Wildschut, T. (2018). Finding meaning in nostalgia. Review of General Psychology, 22(1), 48–61. https://doi.org/10.1037/gpr0000109

85. Sierra, O. I. &. S. M. &. J. J. S. &. V. (2019). Do customer perceptions of corporate services brand ethicality improve brand equity? Considering the roles of brand heritage, brand image, and recognition benefits. ideas.repec.org. https://ideas.repec.org/a/kap/jbuset/v154y2019i2d10.1007_s10551-017-3455-0.html

86. Simplilearn. (2021, January 5). What Is Descriptive Statistics: Definition, Type, Applications & More. Simplilearn.com. https://www.simplilearn.com/what-is-descriptive-statistics-article

87. Singh, R., Sharma, Y., & Kumar, J. (2020). A Road Less Traveled in Nostalgia marketing: Impact of Spiritual well-being on Effects of Nostalgic Advertisements. Journal of Marketing Theory and Practice, 29(3), 1–19. https://doi.org/10.1080/10696679.2020.1836975

88. Singh, R., Sharma, Y., and Kumar, J. (2021), "A Road Less Traveled in Nostalgia Marketing: Impact of Spiritual Well-being on Effects of Nostalgic Advertisements", Journal of Marketing Theory and Practice, Vol. 29 No. 3, pp. 289-307. https://doi.org/10.1080/10696679.2020.1836975.

89. Slavich, M. A., Dwyer, B., & Hungenberg, E. (2019, June 1). Taken back at the ballgame: The impact of nostalgia within the minor league baseball spectator experience. | Journal of Sport Behavior | EBSCOHost. https://openurl.ebsco.com/EPDB%3Agcd%3A7%3A17662167/detailv2?sid=ebsco%3Aplink%3Ascholar&id=ebsco%3Agcd%3A137573513&crl=c

90. Smith, J. (2021). Sample Size Considerations in Marketing Research. Journal of Marketing Studies, 8(4),. https://link.springer.com/article/10.1007/s13162-021-00219-7

91. Statistics Solutions. (2024, April 17). What is Linear Regression? – Statistics Solutions. https://www.statisticssolutions.com/free-resources/directory-of-statistical-analyses/what-is-linear-regression/

92. Stewart, L. (2024, April 8). Purposive sampling in qualitative research. ATLAS.ti. https://atlasti.com/research-hub/purposive-sampling#:~:text=Purposive%20sampling%20is%20primarily%20used,observable%20in%20the%20broader%20population.

93. Steynberg, S. (2022). Subject Guides: Research Guide: Data analysis and findings. Library.up.ac.za. https://library.up.ac.za/c.php?g=485435&p=4425510

94. Stratton, S. J. (2021). Population Research: Convenience sampling strategies. Prehospital and Disaster Medicine, 36(4), 373–374. https://doi.org/10.1017/s1049023x21000649

95. Sur, A., & Pathak, K. (2020, March). Research Publish Journals. Www.researchpublish.com. https://www.researchpublish.com/upload/book/EMOTIONAL%20BRANDING-8441.pdf

96. Take me back: P.W. Jones The power of nostalgia in advertising. (n.d.). https://www.kantar.com/inspiration/advertising-media/take-me-back-th e-power-of-nostalgia-in-advertising

97. Tenny, S., Brannan, J. M., & Brannan, G. D. (2022, September 18). Qualitative study. StatPearls - NCBI Bookshelf. https://www.ncbi.nlm.nih.gov/books/NBK470395/#:~:text=Qualitative%20research%20is%20a%20type,well%20as%20 further%20investigate%20and

98. The Impact of Brand Nostalgia on Purchase Intention. (2020). Stefanie JENSEN, Journal of Emerging Trends in Marketing and Management – Vol I, No. 1/2020. https://www.etimm.ase.ro/?p=626

99. The Impact of Brand Nostalgia on Purchase Intention among Gen Y. (2020, November). www.etimm.ase.ro. Retrieved April 22, 2024, from https://www.etimm.ase.ro/RePEc/aes/jetimm/2020/ETIMM_V01_2020_55.pdf

100. Tripathi, S. (2023, May 4). Social media has reinvented Gen X's standard of living. Times of India Blog.

101. https://timesofindia.indiatimes.com/blogs/voices/social-media-has-reinvented-gen-xs-standard-of-living/

102. Veresiu, E., Robinson, T. D., & Rosario, A. B. (2021, November 5). Marketing and Nostalgia: Unpacking the Past and Future of Marketing and Consumer Research on Nostalgia. Bristoluniversitypressdigital.com; Bristol University Press. https://bristoluniversitypressdigital.com/edcollchap/book/9781529214789/ch008.xml

103. Villegas, F. (2023, July 31). Pilot survey: Definition, importance and tips. QuestionPro. https://www.questionpro.com/blog/pilot-survey/

104. Wang, X., & Cheng, Z. (2020). Cross-Sectional studies. Chest, 158(1), S65–S71. https://doi.org/10.1016/j.chest.2020.03.012

105. Warren, S. (2022, August 26). What is Nostalgia Marketing and How Can You Use it to Sell Apparel? Scrappy Apparel. https://scrappyapparel.com/nostalgia-marketing/

106. Williams, H. (2021, September 21). Priming Theory and How it Influences PR | What's The Word. Sites.psu.edu. https://sites.psu.edu/hewilliams/2021/09/16/priming-theory/#:~:text=Within%20social%20psychology%2C%20this%20process

107. Wolff, R. (2022, March 24). How to Analyze Questionnaire Data: A Step by Step Guide. MonkeyLearn Blog. https://monkeylearn.com/blog/how-to-analyze-questionnaire-data/

108. Xia, L., Wang, J., & Santana, S. (2021). Nostalgia: Triggers and its role on new product purchase intentions. Journal of Business Research, 135, 183–194. https://doi.org/10.1016/j.jbusres.2021.06.034

109. Xia, L., Wang, J. (Feng), and Santana, S. (2021), "Nostalgia: Triggers and Its Role on New Product Purchase Intentions", Journal of Business Research, Vol, 135, pp. 183-194. https://doi.org/10.1016/j.jbusres.2021.06.034.

110. Zamith, R. (2022). Home :: The International Journalism Handbook. Ijh.rodrigo zamith.com. https://ijh.rodrigozamith.com/

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Kadiliman a Filipino-Mythology-Themed Mystery 3D PC Game

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Abstract

Kadiliman is a Filipino-mythology-themed mystery 3D PC game developed to promote Filipino culture and mythology through an immersive survival/stealth horror experience. Set within a fictionalized version of the abandoned Clark Airbase Hospital, the game follows the story of Maxine, a medical student trapped in a su-pernatural dimension filled with mythical Filipino creatures. The player must solve puzzles, avoid hostile entities, and uncover hidden truths to escape the dimension.

Developed using the Unity engine, Kadiliman incorporates traditional Filipino games and instruments into its gameplay and sound design to support cultural preservation and education. It also aligns with two United Nations Sustainable Development Goals: Quality Education and Innovation. The game was designed to accommodate varying player skill levels with multiple difficulty settings, a narrative-rich environment, and atmospheric audio elements. Feedback from beta testing highlighted its technical, visual, and thematic effectiveness. Ultimately, Kadiliman aims to inspire local game development, celebrate Filipino heritage, and offer a unique gaming experience that blends folklore, history, and horror.

Keywords: 3D PC Game, Filipino culture, Unity engine, Traditional Filipino games, Game development, Folklore, Filipino history, Quality Education, Innovation.

Introduction

There is a growing possibility of the emergence of video game companies in the Philippines, making it highly likely for game developers to choose to stay in the country rather than seeking alternatives abroad since the Western industry has always dominated the gaming development scene (Cerda, 2021) [5]. This trend could also attract foreign investors, positioning the Philippines as a potential competitor in the gaming industry. Such developments would contribute to the full recognition and growth of the gaming sector in the Philippines especially for the original Filipino games.

Based on this, the developers and designers decided to develop Kadiliman, a Survival/Stealth horror game infused with a mysterious dark atmosphere. The game incorporates elements from Filipino mythology, featur-ing entities inspired by it to serve as formidable adversaries as well as traditional Filipino games for support and further recognition within the game. In addition to this, sustainable development goals are also addressed in the game as it enlightens the youth about the less popular traditional Filipino games for educational purpos-es (Balite & Robles, 2020) [3], and to encourage innovative design thinking for medical facilities (Esposa et al., 2020) [7] based on the game's setting.

Horror games have carved their niche in the gaming industry for over four (4) decades, beginning with the first horror game titled 'Haunted House' in 1972, an early action-adventure game with simplistic graphics. Fast forward to the current releases like 'Resident Evil 7', where players traverse an abandoned house, combat an infected family, and solve puzzles (DeAngelo, 2021) [9]. These sub-genres demonstrate how horror gaming has developed; Psychological Horror, Action-Horror, Survival Horror, and Jump Scare 2 Horror. Each sub-genre gives out a distinct experience, characterizing the gameplay and the tension it evokes in the player (De S., 2020) [2]. The genre has developed throughout the years, constantly coming up with new and inventive methods to entertain and frighten players by combining sub-genres together, such as Survival/Stealth Horror.

Throughout the years, the Survival/Stealth Horror genre has consistently forged a distinct and captivating atmosphere within the realm of horror gaming. This subgenre caters to a broad spectrum of players, accom-modating both avid horror enthusiasts and those seeking thrilling experiences (Sawyer, 2022)[3]. By intertwin-ing elements of both genres together, these games not only offer an immersive ambiance but also serve as a daunting test, challenging players' abilities to survive and evade entities, thereby amplifying the thrill and engagement happening in-game (Wiggs, 2023) [4].

Kadiliman follows the story of a medical student who got trapped in another dimension of a hospital in Clark, Pampanga. The player must navigate through the halls of the abandoned hospital while defeating the creatures that brought the player into this dimension. The player will also be solving puzzles, with each one carrying a clue. Along the way, the player will be able to uncover the truth as to why they got trapped in this twisted dimension. The player is not a fighter here. The player cannot kill the creatures, but they can find ways to slow them down. The player's main objective is to escape this dimension before they

get trapped forever. Kadiliman is made for the people who enjoy dark, mystery, and thrilling PC games where the player must find a way out while avoiding the creatures that are hunting them and at the same time uncovering the truth to why they were brought into this dimension. The Filipino mythological-themed features traditional Filipino games being incorporated to be able to promote the Filipino culture within the PC gaming world. The game's setting will also help the youth appreciate local history through storytelling, as it will feature the infamous abandoned hospital located in Clark, Pampanga within the PC game.

Conceptual Framework

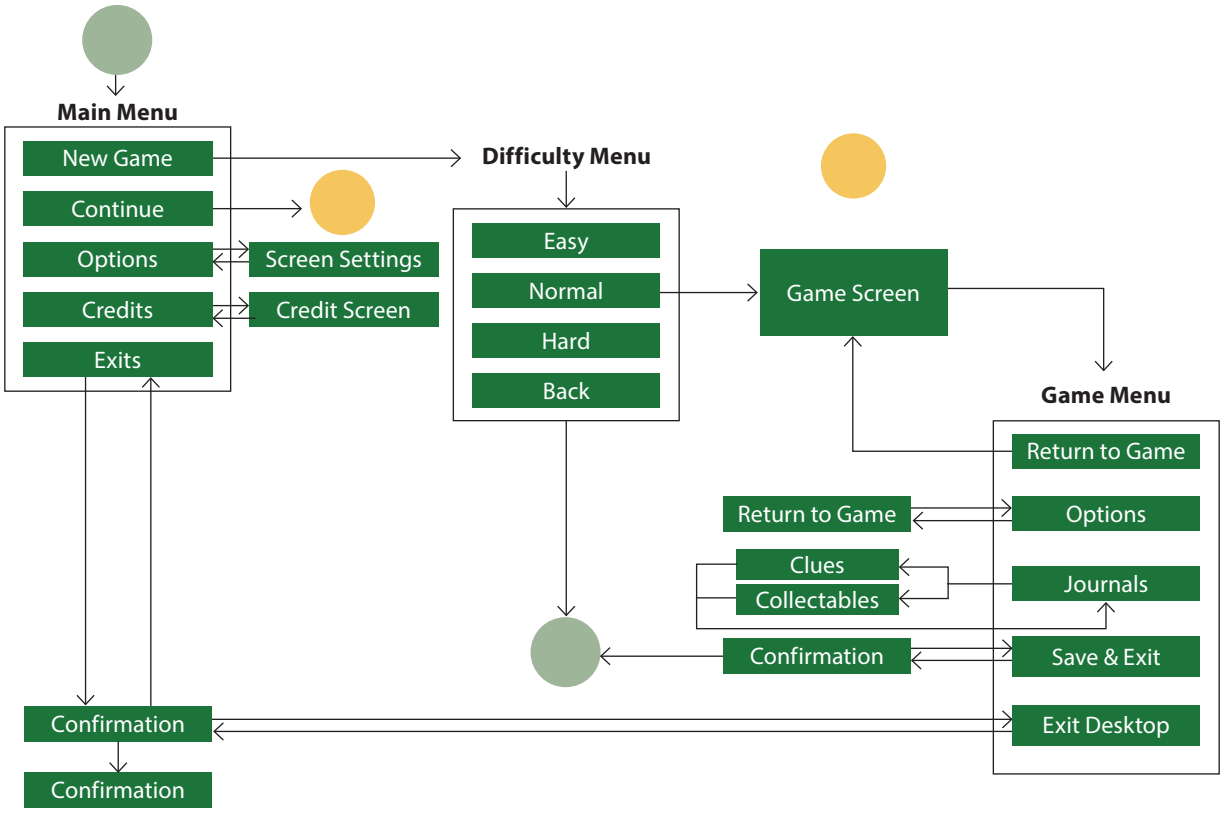


Fig. 1. Conceptual Framework of the Game

The game Kadiliman features a structured Main Menu, allowing players to start a New Game, continue an existing game, adjust settings through the Options Menu, view Credits, or exit the game. When starting a new game, players are redirected to the Difficulty Menu, where they can choose between Easy, Normal, or Hard mode, each altering UI hints, enemy damage, and puzzle complexity to cater to different playstyles. During gameplay, the Game Menu provides options to pause, modify settings, review Journals containing Clues and Collectables, save progress, return to the Main Menu, or exit directly to the Desktop. The game's intuitive UI ensures smooth navigation while allowing players to customize their experience.

Methodology

Operational Feasibility

Kadiliman is a project created by university students of FEU Diliman as a part of their school requirements. Therefore, most of the resources that the developers used are free and open source. This includes Unity 3D as the main game engine for developing the game, Photoshop for concept arts and editing, Autodesk Maya and Blender for 3D modeling the assets, and Substance Painter for texturing.

While the developers of Kadiliman were taking note of the common elements of horror and mystery games, the developers took the opportunity to integrate familiar elements into an unexplored environment. This was done by using a historically abandoned hospital in the Philippines as an inspiration for its game environment with a Filipino-mythology theme to it. Additionally, the project manager took the opportunity to distribute the tasks to each member of the group, depending on their skills and knowledge of the task given to them where they would be more efficient.

Technical Feasibility

Kadiliman is only available on Windows operating systems. After players install Kadiliman, they can play the game offline since the game is only a single-player game. However, players are required to use the internet to install the game from itch.io, and once they install the game, they must unzip it. Below are the required and minimum specifications to run the game smoothly.

Game Development Approach

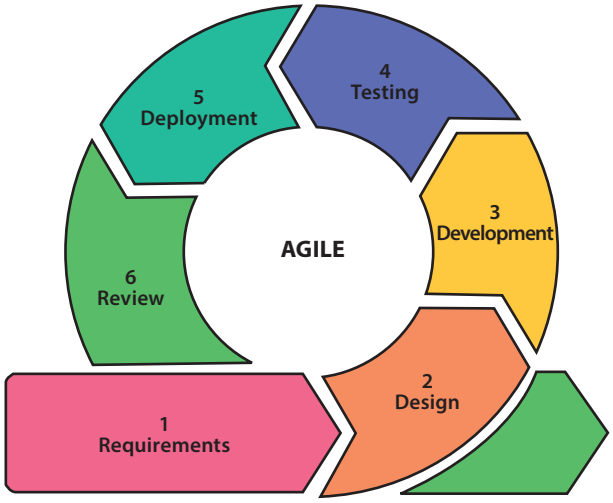


Fig. 2. Agile Approach

The developers of Kadiliman used the Agile Project Management approach. The Agile method involves plan-ning, designing, developing, testing, deployment, and reviewing. It utilizes cyclical, continuous schedules that enable the developers to improve and succeed in attaining any set goals for the project. The Agile method allows for dynamic interaction with beta testers allowing for feedback in fixing bugs and glitches. It encour-ages developers to look through the development process in its entirety, looking at every detail to ensure com-pleteness and validity, and with future game updates and patches that ensure the game is up to date and meets standards.

Results and Discussions

The survey data, website evaluation, beta test results, and results discussion are all included in this chapter. Several respondents who joined up for the beta test were sent survey forms. Survey forms were provided to 30 respondents in total.

Table 1. Demographic of the respondents

Classification	Frequency	Percentage
Gamers	25	83.3%
Game Developer	5	16.7%
Total	30	100%
Chi-Square	1667.62	1733.18

As seen in the table above, 25 of which are gamers and 5 are game developers.

Table 2. Stability of the Game

How would you rate the stability of the game?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	8	40
Good	4	4	16
Average	3	8	24
Poor	2	4	8

Sentiment Level	Value	Responses	Total
Very Poor	1	0	0
Total		30	88
			88/30 = 2.93

Most of the respondents consider the game not to be too stable and that it experiences frame drops during the gameplay.

Table 3. Good Design Practice

How well do you think the game follows good design practices (like making sure everything is easy to use, intuitive, common key mappings, and free of major issues)?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	11	55
Good	4	11	44
Average	3	7	21
Poor	2	1	2
Very Poor	1	0	0
Total		30	122
			122/30 = 4.06

Majority of the respondents consider the game to be following a good design practice such as key mappings.

Table 4. Game Performance

How satisfied are you with the game's performance?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	6	30
Good	4	11	44
Average	3	11	33
Poor	2	2	4
Very Poor	1	0	0
Total		30	111
			111/30 = 3.7

Majority of the respondents considered the performance of the game to be average due to its inconsistent frames.

Table 5. Documentation of the Game

How well is the game's documentation provided?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	14	70
Good	4	10	40
Average	3	4	12
Poor	2	2	4
Very Poor	1	0	0
Total		30	126
			126/30 = 4.2

Shows that the majority of the respondents consider the game to have a good documentation such as guides and instructions.

Production

Table 6. Quality of the Games Graphics

How would you rate the quality of the game's graphics?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	14	70
Good	4	12	48
Average	3	3	6
Poor	2	1	2
Very Poor	1	0	0
Total		30	126
			126/30 = 4.2

Show majority of the respondents rated excellent for the quality of the graphics of the game.

Table 7. Audio Quality

How do you rate the quality of the audio in the game (e.g., sound effects, music, voice acting)?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	13	65
Good	4	12	48
Average	3	4	12
Poor	2	1	2
Very Poor	1	0	0
Total		30	127
			127/30 = 4.23

Majority of the respondents consider the game to have an excellent audio quality.

Table 8. Quality of the Writing

How would you assess the quality of the writing in the game (e.g., dialogue, narrative)?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	12	60
Good	4	12	48
Average	3	5	15
Poor	2	1	2
Very Poor	1	0	0
Total		30	125
			125/30 = 4.16

Shown most of the respondents consider the game to have a good quality of writing.

Table 9. Final Version of the Game

How polished is the final version of the game?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	7	35
Good	4	12	48

Sentiment Level	Value	Responses	Total
Average	3	8	24
Poor	2	3	6
Very Poor	1	0	0
Total		30	113
			113/30 = 3.76

Most of the respondents scored the game's final version to be an average score according to the demographic.

Gameplay

Table 10. Enjoyable Upon First Play

How enjoyable is the game upon first play?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	14	70
Good	4	11	44
Average	3	4	12
Poor	2	1	2
Very Poor	1	0	0
Total		30	128
			128/30 = 4.26

Table 11. Likely to Replay the Game

How likely are you to replay the game?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	11	55
Good	4	12	48
Average	3	6	18
Poor	2	1	2
Very Poor	1	0	0
Total		30	123
			123/30 = 4.1

According to the demographic shown above majority of the respondents would likely to replay the game.

Table 12. Concept and Execution

How original do you find the game's concept and execution?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	13	65
Good	4	10	40
Average	3	6	18
Poor	2	1	2
Very Poor	1	0	0
Total		30	125
			125/30 = 4.16

Most of the respondents considered the concept and execution of the game is original and excellent.

Table 13. Accessibility of the Game

How accessible is the game to a wide range of players (e.g., difficulty options, controls)?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	13	65
Good	4	10	40
Average	3	5	16
Poor	2	2	4
Very Poor	1	0	0
Total		30	125
			125/30 = 4.16

According to the demographic, stated that the game is properly accessible; the difficulty and controls are easy to navigate for casual players.

Theme

Table 14. Elements of the Game

How well do the game's elements work together to support the theme?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	14	70
Good	4	10	40
Average	3	5	15
Poor	2	1	2
Very Poor	1	0	0
Total		30	127
			127/30 = 4.23

Based on the results, majority of the testers consider the game elements are perfectly working together to sup-port the theme.

Table 15. How Effectively the Theme is Incorporated

How effectively does the game incorporate its theme into gameplay and story?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	13	65
Good	4	11	44
Average	3	5	15
Poor	2	1	2
Very Poor	1	0	0
Total		30	126
			126/30 = 4.2

Most of the respondents consider the game successfully incorporated its theme into the gameplay and story.

Table 16. Cohesive of the Game Overall

How polished is the final version of the game?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	13	65
Good	4	10	40
Average	3	6	18
Poor	2	1	2
Very Poor	1	0	0
Total		30	125
			125/30 = 4.16

Most of the testers according to the table, consider the game is consistent with its theme, story, and mechanics.

Table 17. To What Extent is the Game Complete

To what extent do you feel the game is complete and finished?			
Sentiment Level	**Value**	**Responses**	**Total**
Excellent	5	8	40
Good	4	12	48
Average	3	8	24
Poor	2	2	4
Very Poor	1	0	0
Total		30	116
			116/30 = 3.86

Show a mixed results from the testers scoring an 3.86 average score. Most testers consider the game not to be totally complete yet.

Discussion of Results

Table 18. Overall Result

Categories	No. of Sub Categories	Total No. of Respondents	Total of Answers	Mean	Interpretation
Technical	4	120	447	3.72	Satisfactory
Production	4	120	491	4.09	Very Satisfactory
Gameplay	4	120	501	4.17	Very Satisfactory
Theme	4	120	494	4.11	Very Satisfactory

Overall, the testers' assessments, as shown in Table , point to a generally favorable experience with the game Kadiliman. The technical aspect was graded averagely, with a mean score of 3.72, indicating that although the technical performance was sufficient, there might still be space for improvement in areas like stability and optimization. With a mean score of 4.09, the production earned a solid grade, emphasizing how well-received the game's narrative, audio, and visual components were, all of which added to a polished and enjoyable expe-rience. With a good grade and a mean score of 4.17, gameplay scored the highest rating. This suggests that the testers found the main game mechanics to be especially well-accepted, making it an entertaining and repeata-ble experience. While several technological issues require improvement, the game's production and gameplay are strong overall.

Conclusion

The game effectively aligns with the quality education goal of the Sustainable Development Goals by educat-ing players about the rich history and mythology of the Philippines. Additionally, it reflects the industry, in-novation, and infrastructure objectives by incorporating the Clark Airbase Hospital as the foundation for envi-ronmental design, symbolizing the development and restoration of hospitals nationwide. The storyline is thoughtfully crafted, with cutscenes providing deeper narrative details, while scattered notes add to the game's lore. The enemies are balanced, ensuring that combat does not feel overwhelming, and the puzzles present a manageable challenge without being overly difficult, contributing to well-balanced gameplay. The user inter-face is designed effectively, offering players essential information such as battery life and damage indicators, while the controls are intuitive and well-mapped for ease of learning. The game features two distinct endings, each requiring players to make specific choices that shape the direction of the story. Background music en-hances immersion, creating a more atmospheric experience. Every character, including the monsters, plays a role in the game's progression, ensuring their presence contributes meaningfully to gameplay. Additionally, various puzzles incorporate cultural references, drawing inspiration from traditional Filipino games like Sungka and Piko, further enriching the game's unique identity.

Recommendation

Based on the poll, players have identified several key areas for improvement to enhance the gaming experience. Persistent frame drops due to graphics optimization were a major concern, which can be addressed by lowering the poly count of models and textures for smoother performance. Several participants suggested cre-ating a tutorial or guide to help new players familiarize themselves with the controls, including instructions on item usage. Additionally, a few players recommended using fonts that better align with the game's concept. The UI design requires aesthetic improvements to enhance visual appeal, while game mechanics need refining to fix bugs and glitches encountered during gameplay. Enhancing sound design can contribute to a more im-mersive atmosphere, making the experience more engaging. Lighting adjustments are necessary, as some areas are too dark, affecting player navigation and the overall ambiance; improving this will create a more harmoni-ous environment. Lastly, further puzzle refinements are needed to improve playability and make challenges more engaging for players.

References

1. A. Acero et. Al., "Hiraya An Action RPG Filipino Mythology-Themed Roguelite Mystery Video Games" B.S. In-formation Technology, FEU Diliman, Quezon City, Philippines, 2023.
2. Ancehta M.R.G. (2021). A Convergence of Filipino Worlds: An Onomastic Reading of Edgar Calabia Samar's Janus Silang Novels (2021) https://ajba.um.edu.my/ index.php/SARE/article/view/31100/13311
3. Balite, P.H., and S. Robles. (2020). Philippine Games: On the Contemporary Awareness and Involvement of Uni-versity Students (Jun 30, 2020) http://talakasaysayan.org/index.php/ talakasaysayan/article/view/38
4. Balite and Robles (2020) Philippine Games: On the Contemporary Awareness and Involvement of University Stu-dents (June 30, 2020) http://talakasaysayan.org/ index.php/talakasaysayan/article/view/38
5. Casibual (2022) Of women and monsters: A case study of Philippine creature urban legends (September 07, 2022) https://www.academia.edu/97932605/Of_women_and_monsters_A_case_study_of_Philippine_creature_urban_legends
6. Cerda (2021) Towards a "Filipino" Video Game: Teaching Filipino Culture and Identity for Video Game Develop-ment (November 18. 2021) https://archium.ateneo.edu/cgi/viewcontent.cgi?article=1077&context=filipino-faculty-pubs
7. Christofer Mitch C. Cerda. (2021). Towards a "Filipino" Video Game: Teaching Filipino Culture and Identity for Video Game Development (November 18, 2021) https://archium.ateneo.edu/filipino-faculty-pubs/78/
8. Christoffer C. (2023) The Ghosts of Empire: Asian Horror and Postcolonial Pastiche in the Filipino Visual Novel The Letter (2023) http://digra.org:9998/DiGRA_2023_CR_8882.pdf
9. Clasen, M., et al,. (2020). Horror, Personality, and threat simulation: A survey on the psychology of scary media (2020) https://psycnet.apa.org/record/2018-58515-001
10. Eslit E. (2023) Use of Mythical Creatures in Folk Literature of Southeast Asia: A Comparative Analysis (August 2, 2023) https://www.researchgate.net /publication/372823361_Use_of_Mythical_Creatures_in_Folk_Literature_of_Southeast_Asia_A_Comparative_analysis
11. Espiritu E.C. (2023). The Identity of Filipino Video Games Historical Case Study (2023) https://www.academia.edu/104267572/The_Identity_of_Filipino_Video_Games_Historical_Case_Study
12. Fawcett C.(2020). "Take a Look at Yourself": Digital Displays at Police Museums as Camouflage (2020) https://www.equalityproject.ca/wp-content/uploads/2023 /01/The_Annual_Review_of_Interdisciplinary_J.pdf#page=86
13. fivenightsatfreddys3 (n.d.) About Five Nights at Freddy's (n.d.) https://fivenight satfreddys3.com/
14. Garcia, J. Neil (2021) Myth and the Creative Imagination (April 30, 2021) https://animorepository.dlsu.edu.ph/cgi/viewcontent.cgi?article=1005&context=akda
15. Gonzalves and Lacanlale (2021) Kulintang Kultura: Danongan Kalanduyan and Gong Music of the Philippine Di-aspora (October 01 2022) https://scholarlypublishingcollective.org /uip/etm/article-abstract/66/3/542/318994/Kulintang-Kultura-Danongan-Kalanduyan-and-Gong
16. GOG (2023). Amnesia: The Bunker (June 6, 2023) https://www.gog.com/en /game/amnesia_the_bunker
17. Grand view research (n.d.). Video Game Market Size, Share & Trends Analysis Report By Device (n.d.) https://www.grandviewresearch.com/industry-analysis/video-game-market#:~:text=b.-,The%20global%20video%20game%20market%20size%20was%20estimated%20at%20USD,USD%20583.69%20billion%20by%202030.
18. Guiness world records. (1992). First 3D survival-horror videogame (1992) https://www.guinnessworldrecords.com/world-records/first-3d-survival-horror-video-game#:~:text=The%20earliest%203D%20survival%20horror,must%20fend%20off%20supernatural%20attacks.
19. IMDb (2021). Fear Therapy (2021) https://www.imdb.com/title/tt2094955/
20. Jeffrey Loeffler. (2022). History of horror games: looking back at the beginning of PC horror gaming (December 30, 2022) https://www.techradar.com/features/history-of-horror-games-looking-back-at-the-beginning-of-pc-horror-gaming

21. Klaudia J. (2022). Spatiality, Focalization and Temporality in Horror Games (May 2, 2022) https://cejsh.icm.edu.pl/cejsh/element/bwmeta1.element.desklight-d5fac478-0ea9-454e-bd2b-3c2b1f243d25
22. Marc Ariece Esposa et al. (2020). Reconstruction Proposal of Clark Airbase Hospital (March 1, 2020) https://ojs.aaresearchindex.com/index.php/aasgbcpjmra/article/view/1020
23. Maximilian Padilla-Rodriguez. (2023). Action Horror Games are an Underrated Art Form (November 2, 2023) https://www.cbr.com/action-horror-games-deserve-more-love/
24. Mikolaj Dymek. (2020). A study on the effects of horror game design theories and level design patterns on player behaviour in a horror environment (2020) https://www.diva-portal.org /smash/get/diva2:1432687/FULLTEXT01.pdf
25. Petr Hudecek (n.d.). Model Judging Criteria for Game dev Competitions (August 23, 2020) https://hudecekpetr.cz/model-judging-criteria-for-game-dev-competitions/
26. Red Barrel (2013). Outlast (September 14, 2013) https://redbarrelsgames.com/games/outlast/
27. Red Barrels (2017). Outlast 2 (April 25, 2017) https://redbarrelsgames.com/games/outlast-2/
28. R.L. Stine (n.d.) Jump scare – Wikiwand (n.d.) https://www.wikiwand.com/en/Jump_scare
29. Sarah Sawyer. (2022). 10 Best Stealth Horror Games, According to Metacritic. (April 28, 2022)https://screenrant.com/best-stealth-horror-games-metacritic/#amnesia-the-dark-descent-2010---85-8-6
30. Sarra, Graja,. Et al. (2020). Impact of visual and sound orchestration on physiological arousal and tension in a hor-ror game (June 30, 2020) https://ieeexplore.ieee.org /abstract/document/9130167/authors#authors
31. SK Desk. (2022). Horror Games (March 14, 2022) https://www.sportskeeda.com /esports/horror-games
32. Soham De. (2021). 5 Different Types of Horror Video Games, Explained. (December 7, 2021) https://www.makeuseof.com/horror-video-game-subgenres-explained/
33. Steam (2022). Inside the Backrooms (June 21, 2022) https://store.steampowered.com/ app/1987080/Inside_the_Backrooms/
34. Straight Back Games (2021). Devour (January 28, 2021) https://www.devourgame.com/
35. Švelch J. (2020) Should the Monster Play Fair?: Reception of Artificial Intelligence in Alien: Isolation (June 2020) https://www.gamestudies.org/2002/articles/jaroslav_svelch
36. Svelch J. (2023) Player vs. Monster: The Making and Breaking of Video Game Monstrosity (February 7, 2023) https://books.google.com.ph/books?id=FpxuEAAAQBAJ&printsec=frontcover#v=onepage&q&f=false
37. Szabo J. (2022) Fear and Agency in Survival Horror (July 29, 2022) https://brill.com/edcollchap/book/9783969752661/BP000010.xml
38. Tawan W. Shaller. (2020). My Genre Study on Psychological Horror (2020) https://wp.wwu.edu/twseng101/#:~:text=Psychological%20Horror%20is%20a%20very,and%20scare%20the%20audience%20simultaneously.
39. Valve. (2008). Left4Dead. (2008). https://www.l4d.com/game.html
40. Zackary Wiggs. (2023). 10 Best Stealth Horror Games. (February 18, 2023) https://www.thegamer.com/best-stealth-horror-games/#prey

Afterlife Drive: A 3D Urban Fantasy Driving Linear Adventure PC Game

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Abstract

This capstone project presents Afterlife Drive, a 3D urban fantasy driving adventure game for PC that empha-sizes narrative depth, cultural representation, and emotional impact. Developed by VAEJICO, the game cen-ters around an amnesiac angel of death driving souls to the afterlife on a tricycle, discovering each spirit's story along the way while gradually reclaiming her own lost memories.

Built using Unity and inspired by Filipino urban life and folklore, the game integrates themes of mortality, self-care, mental well-being, and personal closure within a structured, linear gameplay loop. Through stylized visuals, relatable storytelling, and thoughtful mechanics like stopovers and memory items, Afterlife Drive aspires to challenge traditional perceptions of games as mere entertainment and instead promote video games as a culturally significant and emotionally resonant art form. The accompanying promotional website, devel-oped to support distribution and feedback collection, adheres to best practices in usability and performance.

Survey data from players and developers confirm the game's strong reception, especially in storytelling and user experience, while identifying areas for improvement in cultural theme visibility and gameplay complexi-ty. Overall, Afterlife Drive not only fulfills the requirements of its academic program but also contributes meaningfully to the conversation on Filipino digital artistry and socially conscious game design.

Keywords: Urban Fantasy, Driving Game Mechanics, Narrative-Driven Gameplay, Mental Well-Being, Sustainable Development Goals (SDG), Filipino Cultural Themes, Game Design, Memory-Based Progression, Unity Game Engine, Character-Driven Storytelling

Introduction

Urban Fantasy is a genre that started out in the literary scene dating back to late 1980s and early 1990s, when writers began to incorporate existing fantasy elements with urban setting and contemporary appreciation (Crow, 2023). While it continues to flourish in the world of literary pieces, the genre found its way to other forms of art, specifically video games. Urban Fantasy in video games, as Rennick (2017) implies, became a genre known to have themes that apply staples in the fantasy genre with modern settings, one that opens a vast range of gaming possibilities, including those that have been already shown by existing games of the same genre. The versatility of this genre paved the way for it to be linked to other genres, highlighting the concept of exploration and discovery within surreal, yet familiar environments.

Drawing inspiration from the concept of Urban Fantasy, driving games also allow players to experience what it is like to navigate open-world environments. Not to be confused with other similar genres like racing games or driving simulator games—driving games, particularly known as an electronic vehicle game, is a genre in which the player controls vehicles of different kinds while being in the game's environment. By evolving this kind of genre with Urban Fantasy players can immerse themselves in expansive virtual worlds full of life, energy, and adventure.

Adventure games, meanwhile, with their emphasis on exploration, puzzle-solving, and narrative-driven game-plays, provide players with a more mentally and emotionally engaging experience. Through the use of capti-vating storylines, detailed environments, and memorable characters, games of this genre enable its players to enhance their critical and logical thinking in terms of making decisions while embarking on various quests to test their wit and resolve.

The linear genre, with regards to video games, tightens the scope of the game's progression and establishes how the player can interact and navigate through the game. Linear games set a predetermined path for the player, limiting the range of exploration and choices that the player can make. There may be additional paths or items to collect, but the overall flow of the game remains the same. The player must progress through the game in the order given by the developers.

Afterlife Drive aspires to achieve several impacts within the gaming community. These include fostering a deeper appreciation for the art of storytelling in video games, promoting significant character representation, which can inspire players to explore themes of morality, identity, and the nature of existence in meaningful ways. Incorporating Filipino themes and elements into the game also establishes another layer of cultural rich-ness and diversity to the gaming experience. By drawing

inspiration from certain Filipino beliefs and even cultural traditions, the subtle yet significant showcasing of the Philippines' expression, greater appreciation and understanding among players worldwide can foster.

Conceptual Framework

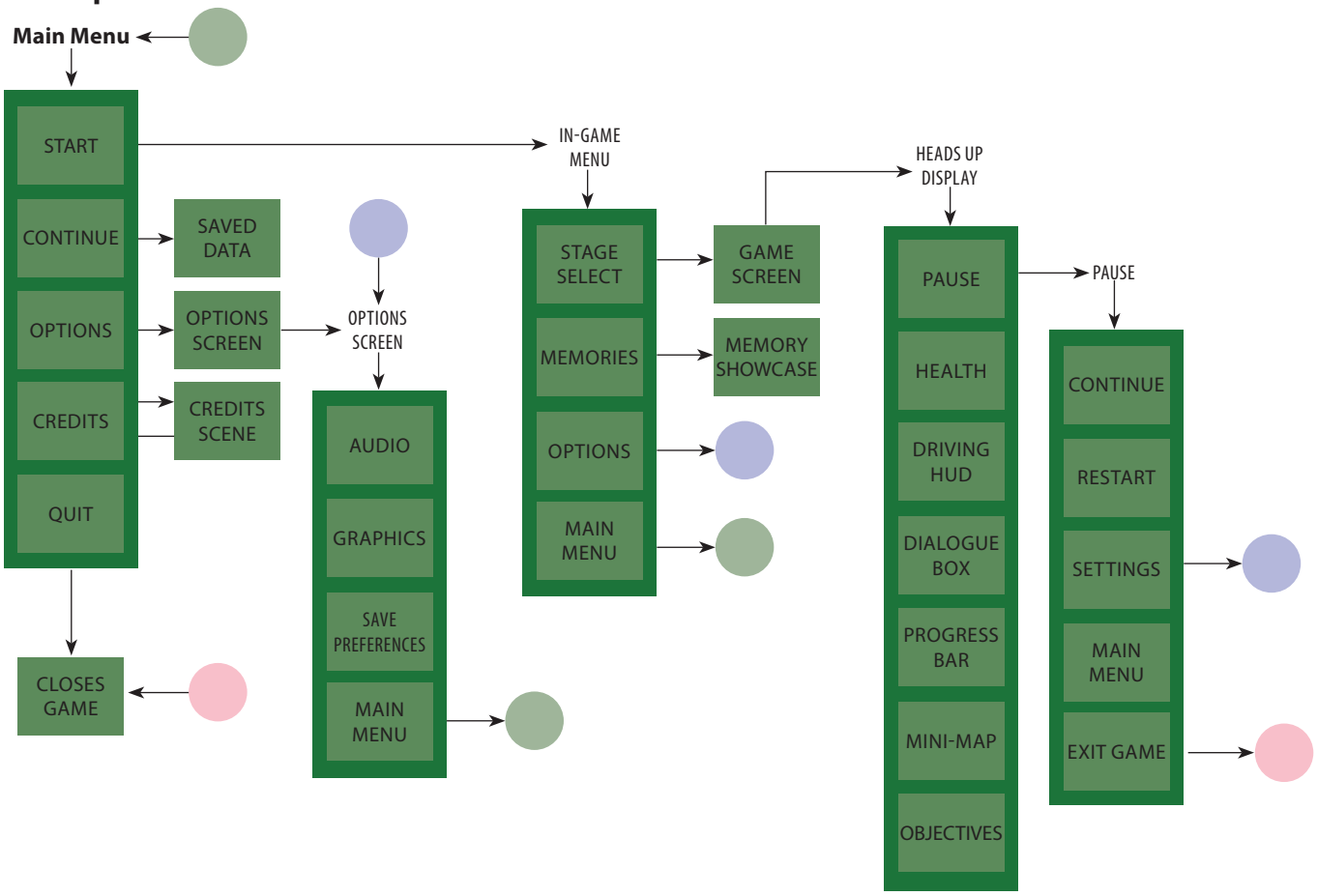


Fig. 1. Conceptual Framework of the Game

Figure 1 above showing the flow details of the game, starting from the main menu of the game that is dis-played to the player upon starting the game. The developers based the game's conceptual framework to what is known as the TAM (Technology Acceptance Model) Framework, whereas it highlights how users come to accept and use a kind of technology. It visualizes how the player navigates through the video game through the use of menus, and also details what information is displayed to the player as elements of a graphical user in-terface or heads up display during gameplay segments.

Methodology

Operational Feasibility

Afterlife Drive is a student-created capstone project. As a result, the developer' resources are limited to free and open-source software licensed for students. These resources include Autodesk Maya for modeling the game's 3D components and Unity 3D for the core game development. The developers of this project seek to incorporate elements of the urban fantasy genre into a form of an adventure driving game, thus the first goal was to bring familiar gameplay, but with a unique twist, particularly in terms of storytelling. Upon delegating tasks to be done, each team member's skills were carefully examined. The tasks assigned to each team member by the project manager were based on the task's difficulty and where each team member is both particularly effective and efficient.

Furthermore, the developers of this project intend to uphold the Sustainable Development Goals of Good Health and Well-Being and Quality Education in the experience of the video game through the narrative themes and discourse in the storytelling, by encouraging players to hold a more positive outlook on life de-spite its circumstances as well as educating players and raising awareness on the importance of maintaining one's mental health using examples given in the story of the video game.

Technical Feasibility

Afterlife Drive is a PC game that must be installed on Windows-based systems. The game does not require an internet connection to play because it is a one-player game, but it is essential for installation and update acqui-sition. C# will be utilized as the programming language for the game engine, Unity.

Game Development Approach

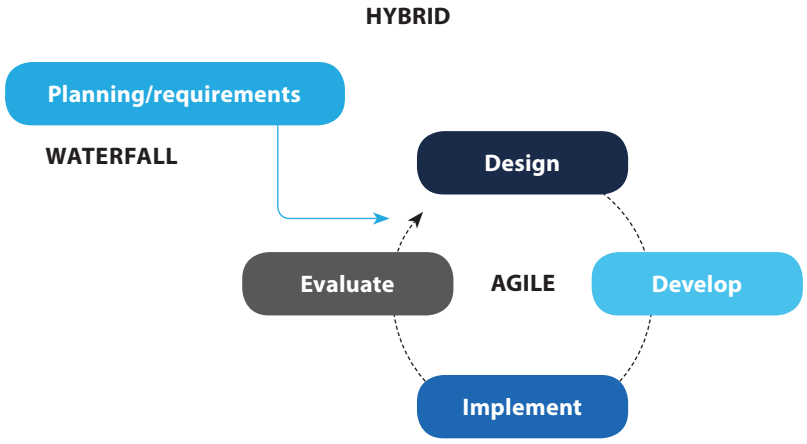


Fig. 2. Hybrid Project Management Methodology

The developers of Afterlife utilized the Hybrid Project Management methodology to produce the game. Hybrid project management combines features of both Agile and Waterfall approaches based on the project needs. This strategy emphasizes flexibility, adaptability, collaboration, and responsiveness, ensuring that all aspects of the project's development run smoothly and produce high-quality performance results.

With this knowledge, Hybrid methodology allows developers to efficiently handle the project's complexity which involves multiple areas such as 3D design, immersive gameplay mechanics, narrative-driven storytelling, and subtle cultural theme integration. It emphasizes a disciplined project management strategy while re-maining adaptable to changing requirements and objectives. It also promotes cross-functional collaboration and communication among those involved in the project.

Furthermore, the iterative nature of the Hybrid approach, as a result of its Agile elements, enables for effective risk management by highlighting strategies for dealing with inherent uncertainties and technical problems. This strategy also requires feedback from current and future software testing, since its iterative nature allows developers to actively address and respond to any modifications that must be made in alignment with the given requirements.

Overall, using a Hybrid strategy to create Afterlife Drive struck a mix between the classic Waterfall technique's formal planning and documentation and Agile's iterative, flexible nature. By using the strengths of this project management technique, the developers produced a game project that meets initial criteria while also providing an immersive gaming experience that resonates with gamers.

Results and Discussions

The survey data, website evaluation, beta test results, and results discussion are all included in this chapter. Several respondents who joined up for the beta test were sent survey forms. Survey forms were provided to 30 respondents in total.

Table 1. Likert Scale Ranges for Interpretation of Results

Scale	Range	Interpretation
5	4.21 – 5.00	The given statement is very true to the user, surpassing expectations.
4	3.41 – 4.20	The given statement is true to the user, adequately fulfilling expectations.
3	2.61 – 3.40	The user neither completely agrees nor disagrees with the given statement
2	1.81 – 2.60	The user does not believe the given statement to be true, performing below expectations.
1	1.00 – 1.80	The user believes the given statement to be very untrue, performing far below expectations.

This evaluation criteria in the evaluation instrument involved the user's experience playing the game and measured user satisfaction through a general set of responses which uniformly applied to each and every criteria category.

Each criteria category was evaluated by providing the user a statement relevant to the category and measuring how much they agree with the given statement from an ordinal scale of 1 to 5, as detailed in Table 1, which describes the 5-point Likert scale utilized in the evaluation instrument. The table also includes descriptions for each possible response that respondents provided, as well as corresponding scale ranges for the purposes of interpretation of evaluation results. These ranges were obtained by creating 5 equal divisions within the scale of 1.00 to 5.00, with a precision of up to two decimal places.

Table 2. Target Respondents' Profile and Description

Respondent Profile	No. of Respondents	Description (Must satisfy at least 1 to be Qualified)
Players	50	• Familiar of Linear Games • Familiar of Adventure Games • Familiar of Driving Games • Interested in Urban Fantasy Themes • Interested in Story-driven Gameplay • Casual or Hardcore Gamers • Either foreign or local entity
Developers	25	Currently working in the Game industry, or Has experience in Game

Afterlife Drive's developers reached out to people who satisfied at least one of the listed descriptions in Table 2 to invite them to play the game. They were instructed to evaluate the game using their extensive and expert judgments. Following the game walkthrough, an online survey-style evaluation sheet was administered.

Profile of the Respondents

Table 3. Profile of Respondents

Profile	Frequency	Percentage
Player	28	77.8
Developers	8	22.2
Total	36	100

Table 3 displays the profile demographic of the respondents of the survey, where the quantitative data was collected from to measure statistics of the game's performance and reception. A distinction was made between players and developers with respect to the difference in perspectives and experiences that each would have, which was relevant to gathering data on the game's performance and reception.

Discussion of Results

Table 4. Summary of Data and Scores for Game Evaluation

Category	No. of Questions	No. of Responses	Total Score	Mean	Interpretation
Genre & Gameplay	3	108	478	4.42	Surpassing Expectations
Themes & Art Direction	2	72	296	4.11	Within Expectations
Narrative & Storytelling	5	180	762	4.24	Surpassing Expectations
UI & UX	4	108	458	4.24	Surpassing Expectations

Table 5. Summary of Data and Scores for Website Evaluation

Category	No. of Questions	No. of Responses	Total Score	Mean	Interpretation
Genre & Gameplay	3	108	478	4.42	Surpassing Expectations
Themes & Art Direction	2	72	296	4.11	Within Expectations
Narrative & Storytelling	5	180	762	4.24	Surpassing Expectations
UI & UX	4	108	458	4.24	Surpassing Expectations

This tabular data is the basis for quantifying and interpreting the overall evaluation of the performance and reception of both the game and the promotional website that the researchers have developed. To interpret this data, Table 4 and Table 5 lists the verbal interpretation as derived in Table 1 prior.

With regards to the game's evaluation itself, the overall response is positive and indicates that it performed above expectations. The genre and gameplay aspects received a mean score of 4.42, indicating that the players had an enjoyable experience with the game overall, despite its limitations. The themes and art direction aspects received a mean score of 4.11, the lowest of the given mean scores, indicating that they were adequately received but can be improved on, especially in how the Filipino and Urban Fantasy themes could have been expressed. The narrative and storytelling received a mean score of 4.24, which is a very satisfactory result. Further emphasis is placed on this category, which also covers the lessons of mental wellness and self-care that the game intended to convey, which were received well by the players. The UI and UX received a mean score of 4.24, which meant players had a good experience with both the user interface and the user experience of the game. The difficulty of the game received a mean score of 4.28, which meant players also had a good experience with the difficulty of the game and how it meshed well with other aspects. The overall mean score for the game was 4.26, just within the range for a very satisfactory result, but leaving room for improvement, especially in conveying themes through gameplay and art direction.

Conclusion

The overall objective behind this study was to develop Afterlife Drive, a linear adventure driving game with Filipino and Urban Fantasy narrative and aesthetic themes and a heavy focus on showcasing mental well-being and self-care through character-driven storytelling. The video game tells the story where the player takes the role of an angel of death who uses a tricycle to deliver souls to heaven. In the process, the story of each soul is revealed through dialogue and gameplay, allowing the player to understand and relate to these characters as he/she progresses through the game by driving through set pieces that represent the memories of each soul bound for heaven. Afterlife Drive also intends to provide emotional impact through the storytelling, charac-ters, and gameplay of the game experience.

Recommendation

Through the experience of Afterlife Drive, the researchers of this study intended not only to deliver an enter-taining video game experience but also deliver meaningful emotional and cultural impact through the game-play and storytelling of the game. In addition, Afterlife Drive was intended to express and uphold specific Sustainable Development Goals through the gameplay and storytelling of the game. These Sustainable Devel-opment Goals are Good Health and Well-Being and Quality Education, both of which are aimed to be ex-pressed in the experience of the video game.

References

1. (2021 11) H w N f x’ ‘ ’ p f screen. RAPPLER. https://www.rappler.com/entertainment/series/transforming-trese-netflix-screen/
2. Afterlife (Stanford Encyclopedia of Philosophy). (2023, February 27). https://plato.stanford.edu/entries/afterlife/
3. Anbar, R. D. (2023, February 11). What happens after death?. Psychology Today. https://www.psychologytoday.com/us/blog/understanding-hypnosis/202212/what-happens-after-death
4. Asael. (2023, August 14). A comprehensive guide to the Philippine Tricycle. Secret Philippines. https://secret-ph.com/a-comprehensive-guide-to-the-philippine-tricycle/
5. Ashtari, H. (2022, September 29). Black Box vs. White Box Testing: Understanding 3 Key Differences. Spiceworks. https://www.spiceworks.com/tech/devops/articles/black-box-vs-white-box-testing/
6. B, K. (2023). Exploring the Cognitive Benefits of Video Games: How Gaming Can Improve Learning. Exploring the Cognitive Benefits of Video Games: How Gaming Can Improve
7. Barr, M., & Copeland-Stewart, A. (2021). Playing video games during the COVID-19 p ff p y ’ w -being. Games and Culture, 17(1), 122–139. https://doi.org/10.1177/15554120211017036
8. Cerda, C. (2021, November 18). “F p ” V : F p Culture and Identity for Video Game Development. https://archium.ateneo.edu/cgi/viewcontent.cgi?article=1077&context=filipino-faculty-pubs
9. Chaperot B. & Fyfe C. (n.d). Issues in the Physics of a Motocross Simulation. Issues in the Physics of a Motocross Simulation. https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=42cb9822c7a4298c648d5171e001c880e2cba81c
10. Crow, A. (2023, June 29). The Evolution of Urban Fantasy: From the early classics to modern masterpieces. Medi-um. https://medium.com/@authoranncrow/the-evolution-of-urban-fantasy-from-the-early-classics-to-modern-masterpieces-6a30d72c2e71
11. De, S. (2021, December 13). ’ v games? MUO. https://www.makeuseof.com/linear-non-linear-video-game-differences/
12. Finno-Ugric religion | Definition, History, Mythology, Practices, & Facts. (1999, July 26). Encyclopedia Britannica. https://www.britannica.com/topic/Finno-Ugric-religion/Institutions-and-practices#ref533392
13. French, K. (2023, November 10). Urban fantasy worldbuilding: mixing the magical & the mundane. World Anvil Blog. https://blog.worldanvil.com/worldbuilding/urban-fantasy-worldbuilding/
14. G2A.COM. (2024, January 25). What is Cel-Shading? An Overview, Examples, and Video Games. G2A News. https://www.g2a.com/news/features/what-is-cel-shading/
15. Gailloreto, C. (2021, April 19). Why urban fantasy games are more prevalent in Japan than in the US. ScreenRant. https://screenrant.com/urban-fantasy-games-persona-ff7-japan-popular-why/
16. Giglioli, I. a. C., De Juan Ripoll, C., Parra, E., & Raya, M. A. (2019). Are 3D virtual environments better than 2D interfaces in serious games performance? An explorative study for the assessment of executive functions. Applied Neuropsychology: Adult, 28(2), 148–157. https://doi.org/10.1080/23279095.2019.1607735
17. Gossett, S. (2023, January 23). The real social benefits of video games. Built In. https://builtin.com/media-gaming/online-gaming-social-benefits
18. Granic, I., Lobel, A., C, R., & Engels, M. E. (n.d.). The Benefits of Playing Video Games. The Benefits of Playing Video Games. https://www.apa.org/pubs/journals/releases/amp-a0034857.pdf
19. Granic, I., Lobel, A., & Engels, R. C. (n.d.). The benefits of playing video games. American Psychologist, 69(1), 66–78. https://doi.org/10.1037/a0034857 Learning. https://videolibrarian.com/articles/essays/exploring-the-cognitive-benefits-of-video-games/

20. Hussain, A., Shakeel, H., Hussain, F., Uddain, N., & Ghouri, T. (2020). Unity Game Development Engine: A Technical Survey. Unity Game Development Engine: A Technical Survey. https://www.researchgate.net/profile/Faizan-Hussain-2/publication/348917348_Unity_Game_Development_Engine_A_Technical_Survey/links/60167daf45851517ef2b2c0d/Unity-Game-Development-Engine-A-Technical-Survey.pdf
21. Lim, N. (2023, November 17). How much does it cost to maintain an app?. Sonamine. https://www.sonamine.com/blog/how-much-does-it-cost-to-maintain-an-app#:~:text=The%20industry%20standard%20is%20to,your%20specific%20needs%20may%20vary.
22. Massicotte, D. (2023, March 3). Unlocking the secrets behind the vital role of game mechanics in video games. https://www.linkedin.com/pulse/unlocking-secrets-behind-vital-role-game-mechanics-video-massicotte
23. McLeod, S. (2023, July 31). Likert scale questionnaire: Meaning, Examples & Analysis. Simply Psychology. https://www.simplypsychology.org/likert-scale.html
24. Mercurio, J. (n.d.). Religious Practices On Honoring The Dead: Need For Contextualized Christian Formation. Re-ligious Practices on Honoring the Dead: Need for Contextualized Christian Formation. https://iiari.org/wp-content/uploads/2021/03/V1-1-131.pdf
25. Ocampo, V. F. R. (2022, September 20). Philippine Fantasy – The Infinite Library and other stories. The Infinite Library and Other Stories. https://vrocampo.com/category/philippine-fantasy/
26. Philippine digital game: an emerging art form. (n.d.). Portia Placino. https://portiaplacino.com/2011/02/01/philippine-digital-game-an-emerging-art-form/
27. Rennick, C. (2017, November 24). The lack of urban fantasy in modern gaming. HeadStuff. https://headstuff.org/entertainment/gaming/urban-fantasy-in-modern-gaming/
28. Rönkkö, A. (2021). Psychology at play: Empathy and character relatability. Theseus. https://www.theseus.fi/handle/10024/498019
29. Sacco, M. (n.d.). Introduction to Toon Shading. https://www.occasoftware.com/blog/introduction-to-toon-shading-techniques-and-applications
30. Santos, A. K. (2021, November 3). Mythical destinations in the Philippines. Travel Up. https://www.traveling-up.com/mythical-destinations-in-the-philippines/
31. Stefyn, N. (2022, May 9). How video games are made: The game development process. CG Spectrum. https://www.cgspectrum.com/blog/game-development-process
32. Taylor-Hill, G. (2022, July 22). Non-Racing games with the best driving mechanics. Game Rant. https://gamerant.com/non-racing-games-best-driving-mechanics/#beamng-drive
33. Video game market size, share and Growth Report, 2030. Video Game Market Size, Share And Growth Report, 2030. (n.d.). https://www.grandviewresearch.com/industry-analysis/video-game-
34. Video games - Philippines: Statista Market Forecast. Statista. (n.d.). https://www.statista.com/outlook/dmo/digital-media/video-games/philippines#:~:text=This%20projection%20indicates%20an%20annual,US%242%2C595.00m%20by%202027.
35. Volle, A. (2023, April 5). Afterlife | Definition, Belief, Religion, & Facts. Encyclopedia Britannica. https://www.britannica.com/topic/afterlife-religionmar-ket#:~:text=The%20global%20video%20game%20market%20is%20expected%20to%20grow%20at,USD%20583.69%20billion%20by%202030.
36. What is a car game?. Computer Hope. (2020, August 2). https://www.computerhope.com/jargon/c/car-game.htm
37. What discount rate should be used?. Customer Lifetime Value. (n.d.). https://www.clv-calculator.com/discount-rates/discount-rate-used/#:~:text=A%20discount%20rate%20of%2010,as%20a%20%E2%80%9Churdle%E2%80%9D%20rate.